

PRINCIPLES AND PRACTICE
— OF —
AGRICULTURAL ANALYSIS.

A MANUAL FOR THE ESTIMATION OF SOILS, FERTILIZERS,
AND AGRICULTURAL PRODUCTS.

FOR THE USE OF ANALYSTS, TEACHERS, AND STUDENTS
OF AGRICULTURAL CHEMISTRY.

VOLUME I.

SOILS.

BY HARVEY W. WILEY,
CHEMIST OF THE U. S. DEPARTMENT OF AGRICULTURE.

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Title: Principles and practice of agricultural analysis. Volume 1 (of 3), Soils

Author: Harvey Washington Wiley

Release date: March 31, 2024 [eBook #73302]

Language: English

Original publication: Easton: Chemical Publishing Co., 1894

Other information and formats: www.gutenberg.org/ebooks/73302

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PREFACE TO VOLUME FIRST.

In this volume I have endeavored to place in the hands of teachers and students of Agricultural Analysis, and of analysts generally, the principles which underlie the science and art of the analysis of soils and the best approved methods of conducting it.

In the prosecution of the work I have drawn freely on the results of experience in all countries, but especially in the matter of the physical examinations of soils, of this country. Science is not delimited by geographic lines, but an author is not to be blamed in first considering favorably the work of the country in which he lives. It is only when he can see nothing of good outside of its own boundaries that he should be judged culpable. It has been my wish to give full credit to those from whose work the subject-matter of this volume has been largely taken. If, in any case, there has been neglect in this matter, it has not been due to any desire on my part to bear the honors which rightfully belong to another. With no wish to discriminate, where so many favors have been extended, especial acknowledgments should be made to Messrs. Hilgard, Osborne, Whitney, and Merrill, for assistance in reading the manuscript of chapters relating to the origin of soils, their physical properties, and mechanical analysis. With the wish that this volume may prove of benefit to the workers for whom it was written I offer it for their consideration.

H. W. WILEY.

WASHINGTON, D. C.,
Beginning of January, 1895.

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CORRECTIONS.—Page [112](#), second line from bottom, read “Fig. 14” instead of “13.”

Page [158](#), insert “and determining soluble matters therein” after “flow” in paragraph **172**, third line.

Page [468](#), paragraph **423**, read “calcium carbonate about 200 milligrams,” instead of “calcium carbonate, or gypsum fifty milligrams.”

Page [557](#), read “red-yellow” instead of “blue” in seventh line from bottom.

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PART FIRST.

INTRODUCTION.

1. Definitions.—The term soil, in its broadest sense, is used to designate that portion of the surface of the earth which has resulted from the disintegration of rocks and the decay of plants and animals, and which is suited, under proper conditions of moisture and temperature, to the growth of plants. It consists, therefore, chiefly of mineral substances, together with some products of organic life, and of certain living organisms whose activity may influence vegetable growth either favorably or otherwise. The soil also holds varying quantities of gaseous matter and of water, which are important factors in its functions.

2. Origin Of Soil.—Agriculturally considered, the soil proper is the older and more thoroughly disintegrated superficial layer of the earth, which has been longest exposed to weathering and the influences of organic life. It is usually from six to twelve inches, but occasionally several feet in depth. The subsoil, which lies directly under this, is not as a rule so thoroughly disintegrated, since it is protected in a measure by the overlying soil. It usually contains less organic matter than the soil. There is a freer circulation of air in the soil than in the subsoil, and the metallic elements usually exist therein as higher oxids. There is usually a notable difference in color between the soil and subsoil, and frequently a very sharp color line separating the two.

Geologically considered, the soil is that portion of the earth's crust which has been more or less thoroughly disintegrated by weathering and other forces from the original rock formations, or from the sedimentary rocks, or from the unconsolidated sedimentary material. The soil has, therefore, the

same essential constitution as the general mass of the earth, except that this débris has been subjected to the solvent action of water and the influence of vegetable growth.

Preliminary to the proper understanding of the methods of the analysis of soils, there should be some definite knowledge concerning the composition of the earth's crust, so that the analyst may understand more thoroughly the origin and nature of the material he has to deal with, and thereby be better equipped for his work.

3. The Chemical Elements Present in the Soil.—The chemical elements present in the soil are naturally some or all of those which were present in the original rocks. For analytical purposes relating to agriculture, it is not necessary to take into account the rare elements which may occur in the soil, but only those need be considered which are present in some quantity and which enter as an important factor into plant growth. Of the whole number of chemical elements less than twenty are of any importance in soil analysis. These elements may be grouped into two classes, the non-metals, and the metals as follows:

Non-metals.	Metals.
Oxygen,	Aluminum,
Silicon,	Calcium,
Carbon,	Magnesium,
Sulfur,	Potassium,
Hydrogen,	Sodium,
Chlorin,	Iron,
Phosphorus,	Manganese,
Nitrogen,	Barium.
Fluorin,	
Boron.	

4. Atomic Masses.—For the purpose of facilitating the calculation of results the latest revised table of atomic masses is given below. All the known elements are included in this table for the convenience of analysts who may have to study some of the rarer elements in the course of their work.

This table represents the latest and most trustworthy results reduced to a uniform basis of comparison with oxygen = 16 as starting point of the system. No decimal places representing large

uncertainties are used. When values vary, with equal probability on both sides, so far as our present knowledge goes, as in the case of cadmium (111.8 and 112.2), the mean value is given in the table.

TABLE OF ATOMIC MASSES OF THE ELEMENTS.

Revised by F. W. Clarke, Chief Chemist of the United States Geological Survey, to January 1st, 1894.

Name.	Symbol.	Atomic mass.
Aluminum	Al	27
Antimony	Sb	120
Arsenic	As	75
Barium	Ba	137.43
Bismuth	Bi	208.9
Boron	B	11
Bromin	Br	79.95
Cadmium	Cd	112
Cesium	Cs	132.9
Calcium	Ca	40
Carbon	C	12
Cerium	Ce	140.2
Chlorin	Cl	35.45
Chromium	Cr	52.1
Cobalt	Co	59
Columbium [A]	Cb[Nb]	94
Copper	Cu	63.6
Erbium	Er	166.3
Fluorin	F	19
Gadolinium	Gd	156.1
Gallium	Ga	69
Germanium	Ge	72.3
Glucinum [B]	Gl[Be]	9
Gold	Au	197.3
Hydrogen	H	1.008
Indium	In	113.7
Iodin	I	126.85
Iridium	Ir	193.1
Iron	Fe	56
Lanthanum	La	138.2
Lead	Pb	206.95
Lithium	Li	7.02
Magnesium	Mg	24.3
Manganese	Mn	55
Mercury	Hg	200
Molybdenum	Mo	96
Neodymium	Nd	140.5
Nickel	Ni	58.7
Nitrogen	N	14.03
Osmium	Os	190.8
Oxygen [C]	O	16

Palladium	Pd	106.6
Phosphorus	P	31
Platinum	Pt	195
Potassium	K	39.11
Praseodymium	Pr	143.5
Rhodium	Rh	103
Rubidium	Rb	85.5
Ruthenium	Ru	101.6
Samarium	Sm	150
Scandium	Sc	44
Selenium	Se	79
Silicon	Si	28.4
Silver	Ag	107.92
Sodium	Na	23.05
Strontium	Sr	87.6
Sulfur	S	32.06
Tantalum	Ta	182.6
Tellurium	Te	125
Terbium	Tb	160.0
Thallium	Tl	204.18
Thorium	Th	232.6
Thulium	Tu	170.7
Tin	Sn	119
Titanium	Ti	48
Tungsten	W	184
Uranium	U	239.6
Vanadium	V	51.4
Ytterbium	Yb	173
Yttrium	Yt	89.1
Zinc	Zn	65.3
Zirconium	Zr	90.6

[A.](#) Has priority over niobium.

[B.](#) Has priority over beryllium.

[C.](#) Standard or basis of the system.

PROPERTIES OF THE ELEMENTS.

Following is a brief description of the most important elements occurring in the earth's crust in respect of their relations to agriculture.

5. Oxygen exists in the free gaseous state in the atmosphere of which it constitutes about one-fifth by bulk, whilst in combination with other elements it forms nearly half the weight of the solid earth, and eight-ninths by weight of water. It enters into combination with all the other elements, except fluorine, forming what are known as oxids, and with many of the elements it unites in several proportions, forming oxids of different composition. Combined with silicon, carbon, sulfur, and phosphorus, it forms an essential part of the silicates, carbonates, sulfates, and phosphates, most of which are very abundant and all of which are very widely distributed in the earth's crust. In this form it is exceedingly stable and is rarely set free. With the exception of the oxids of silicon these oxids seldom occur uncombined with the metals as constituents of rocks or soils. The oxids of iron very commonly occur as such in rocks and soils, and play a very important part in organic life. The several oxids of iron very frequently determine the color of soils; as the iron in a soil is more or less oxidized, or as it is exposed more or less to access of air, the color of the soil changes. These oxids of iron also play an important part in the absorption capacities of soils for moisture and other physical conditions of soils, and also in the oxidation of organic matters in the soil. Many organic substances, and even the roots of growing plants when deprived of free access of air, can readily secure oxygen from the iron oxid, thus reducing the iron to a lower form of oxidation, the oxygen being used for the oxidation of the organic matter or for the needs of the growing plant; while the lower oxid of iron can more readily take up oxygen of the air and again be converted into a higher oxid, ready again to give up a part of its oxygen and thus serve as a carrier.

6. Silicon never occurs in the free state, but combined with oxygen it forms silica, which constitutes more than one-half of the earth's crust. The oxid of silicon occurs in the very common form of quartz, and likewise, as silicate of alumina, lime or magnesia. Silicon forms an essential part of many minerals, such as the feldspars, amphiboles, pyroxenes, and the micas, besides being an essential ingredient of many other minerals. Silica

is relatively very slightly affected by the ordinary forces concerned in the decay of rocks, and even after the crystals of feldspars, micas, and other common minerals occurring in rocks have been disintegrated the silica remains as hard grains of sand, forming the bulk of most soils. By far the larger part of silicon in soils is in the form of grains of quartz or silica. This form, however, is probably chemically inert in regard to plant growth, but it plays a very important part in the physical structure of soils and in the physical relation of soils to plant growth.

7. Carbon as an elementary substance occurs as diamond and graphite and in an impure form as anthracite and bituminous coals. In peats and mucks carbon is the chief constituent. This substance is also contained in the organic matters of the soil known as humus, and the relation of the carbon to nitrogen often throws important light upon the amount and character of the nitrogenous matters. In composition with oxygen it forms the chief food of growing plants, the carbon of the carbon dioxide of the air being elaborated into the tissue of the plants and the oxygen returned to the atmosphere. The content of carbon dioxide in the air is from three to five parts per thousand by volume. As carbonates this element helps to form some of the most important ingredients of the earth's crust, namely, limestones, marbles, dolomites, etc., and in an organic form it is found in the shells of the crustaceans. The calcareous matter of the soil, that is, the carbonates of the earths therein found, are of the highest importance from an agricultural point of view. The carbonates in the soil not only favor the process of converting nitrogenous bodies into forms suitable for plant food, but also exert a most potent influence on the physical state of the soil and its capacity for holding water and permitting its flow to and from the rootlets of the plant.

8. Sulfur occurs in nature in both the free and combined state. In the free state it is found in volcanic regions such as Sicily, Iceland, and the western United States. Its usual form of occurrence is in combination with the metals to form sulfids, or with oxygen and a metal to form sulfates. Sulfur and iron combine to form iron pyrites or iron disulfid (FeS_2), while sulfur, oxygen, and calcium are found in gypsum, an important fertilizing compound.

Sulfur plays an important part in the nourishment of plants, being found in them both as sulfuric acid and in organic compounds. Methods for

estimating the sulfur in both forms will be found in another part of this manual.

9. Hydrogen is a colorless, invisible gas, without taste or smell. It occurs free in small proportions in certain volcanic gases, and in natural gas, but its most common form is in combination with oxygen as water (H_2O), of which it forms 11.13 per cent by weight. It also occurs in combination with carbon to form the hydrocarbons, such as the mineral oils (petroleum, etc.) and gases. Hydrogen is of no importance to agriculture in a free state, but water is the most important of all plant foods.

10. Chlorin occurs free in nature only in limited amounts and in volcanic vents. Its most common form is in combination with hydrogen, forming hydrochloric acid, or with the metals to form chlorids. It combines with sodium to form sodium chlorid or common salt ($NaCl$), which is the most abundant mineral ingredient in sea water and which can usually be detected in rain and ordinary terrestrial waters. In this form, also, it exists as extensive beds of rock salt, which is mined for commercial purposes.

Chlorin is found uniformly in plants and must be regarded as an essential constituent thereof. Common salt applied to a soil modifies its power of attracting and holding water.

11. Phosphorus never occurs in nature in a free state but exists in combination in greater or less quantities in all soils. Its combinations are also found in large deposits of minerals known as phosphorite and apatite and as so-called pebble deposit and phosphate rock. Phosphorus in some sort of combination is one of the most essential elements in animal and plant food. In animals its compounds form almost all of the mineral matter of the bones, and in plants they are the chief constituents of the ash of seeds.

The mineral deposits of phosphorus, as well as bones, are chiefly tri-calcium phosphate, while the slag compound resulting from the basic treatment of iron ores rich in phosphorus is a tetra-calcium salt.

The pebble deposits and some rock phosphates are supposed to be of organic origin, derived from the remains of marine, terrestrial, and aerial animals.

Cereal crops remove about twenty pounds of phosphoric acid per acre from the soil annually and grass crops about twelve pounds. The total

phosphoric acid removed annually by the cereal and grass crops in the United States is nearly four billion pounds.

Gautier^[1] calls attention to the fact that the oldest phosphates are met with in the igneous rocks such as basalt, trachyte, etc., and even in granite and gneiss. It is from these inorganic sources, therefore, that all phosphatic plant food must have been drawn. In the second order in age Gautier places the phosphates of hydro-mineral origin. This class not only embraces the crystalline apatites but also those phosphates of later formation formed from hot mineral waters in the jurassic, cretaceous, and tertiary deposits. These deposits are not directly suited to nourish plants.

The third group of phosphates in order of age and assimilability embraces the true phosphorites containing generally some organic matter. They are all of organic origin. In caves where animal remains are deposited there is an accumulation of nitrates and phosphates.

Not only do the bones of animals furnish phosphates but they are also formed in considerable quantities by the decomposition of substituted glycerids such as lecithin. The ammonia produced by the nitrification of the albuminoid bodies combines with the free phosphoric acid thus produced, forming ammonium or diammonium phosphates. The presence of ammonium phosphates in guanos was first noticed by Chevreul more than half a century ago.

If such deposits overlay a pervious stratum of calcium carbonate, such as chalk, and are subject to leaching a double decomposition takes place as the lye percolates through the chalk. Acid calcium phosphate and ammonium carbonate are produced. By further nitrification the latter becomes finally converted into calcium nitrate. In like manner aluminum phosphates are formed by the action of decomposing organic matter on clay.

Davidson,^[2] explains the origin of the Florida phosphates by suggesting that they arose chiefly through the influx of animals driven southward during the glacial period. According to his supposition the waters of the ocean, during the cenozoic period contained more phosphorus than at the present time. The waters of the ocean over Florida were shallow and the shell fish existing therein may have secreted phosphate as well as carbonate of lime. This supposition is supported by an analysis of a shell of *lingula ovalis*, quoted by Dana, in which there were 85.79 per cent of lime phosphate. In these waters were also many fishes of all kinds and their

débris served to increase the amount of phosphatic material. As the land emerged from the sea came the great glacial epoch driving all terrestrial animals southward. There was, therefore, a great mammal horde in the swamps and estuaries of Florida. The bones of these animals contributed largely to the phosphatic deposits. In addition to this, the shallow sea contained innumerable sharks, manatees, whales, and other inhabitants of tropical waters, and the remains of these animals added to the phosphatic store.

While these changes were taking place in the quaternary period, the Florida Peninsula was gradually rising, and as soon as it reached a considerable height the process of denudation by the action of water commenced. Then there was a subsidence and the peninsula again passed under the sea and was covered with successive layers of sand. The limestones during this process had been leached by rain water containing an excess of carbon dioxid. In this way the limestones were gradually dissolved while the insoluble phosphate of lime was left in suspension. During this time the bones of the animals before mentioned by their decomposition added to the phosphate of lime present in the underlying strata, while some were transformed into fossils of phosphate of lime just as they are found to-day in vast quantities.

Wyatt,^[3] explains the phosphate deposits somewhat differently. According to him, during the miocene submergence there was deposited upon the upper eocene limestones, more especially in the cracks and fissures resulting from their drying up, a soft, finely disintegrated calcareous sediment or mud. The estuaries formed during this period were swarming with animal and vegetable life, and from this organic life the phosphates were formed by decomposition and metamorphism due to the gases and acids with which the waters were charged.

After the disappearance of the miocene sea there were great disturbances of the strata. Then followed the pliocene and tertiary periods and quaternary seas with their deposits and drifts of shells, sands, clays, marls, boulders, and other transported materials supervening in an era when there were great fluctuations of cold and heat.

By reason of these disturbances the masses of the phosphate deposits which had not been infiltrated in the limestones became broken up and mingled with the other débris and were thus deposited in various mounds or

depressions. The general result of the forces which have been briefly outlined, was the formation of boulders, phosphatic débris, etc. Wyatt therefore classifies the deposits as follows:

1. Original pockets or cavities in the limestone filled with hard and soft rock phosphates and débris.

2. Mounds or beaches, rolled up on the elevated points, and chiefly consisting of huge boulders of phosphate rock.

3. Drift or disintegrated rock, covering immense areas, chiefly in Polk and Hillsboro counties, and underlying Peace River and its tributaries.

Darton,^[4] ascribes the phosphate beds of Florida to the transformation of guano. According to this author two processes of decomposition have taken place. One of these is the more or less complete replacement of the carbonate by the phosphate of lime. The other is a general stalactitic coating of phosphatic material. Darton further calls attention to the relation of the distribution of the phosphate deposits as affecting the theory of their origin, but does not find any peculiar significance in the restriction of these deposits to the western ridge of the Florida peninsula.

As this region evidently constituted a long narrow peninsula during early miocene time it is a reasonably tentative hypothesis that during this period guanos were deposited from which was derived the material for the phosphatization of the limestone either at the same time or soon after.

Darton closes his paper by saying that the phosphate deposits in Florida will require careful, detailed geologic exploration before their relations and history will be fully understood.

According to Dr. N. A. Pratt the rock or boulder phosphate had its immediate origin in animal life and to his view the phosphate boulder is a true fossil. He supposes the existence of some species in former times in which the shell excreted was chiefly phosphate of lime. The fossil boulder, therefore, becomes the remains of a huge foraminifer which had identical composition in its skeleton with true bone deposits or of organic matter.

Perhaps the most complete exposition of the theory of the recovery of waste phosphates, with especial reference to their deposit in Florida, has been given by Eldridge.^[5] He calls attention to the universal presence of phosphates in sea water and to the probability that in earlier times, as during the miocene and eocene geologic periods, the waters of the ocean contained

a great deal more phosphate in solution than at the present time. He cites the observations of Bischof, which show the solubility of different phosphates in waters saturated with carbon dioxide. According to these observations apatite is the most insoluble form of lime phosphate, while artificial basic phosphate is the most soluble. Among the very soluble phosphates, however, are the bones of animals, both fresh and old. Burnt bones, however, are more soluble than bones still containing organic matter. Not only are the organic phosphates extremely soluble in water saturated with carbon dioxide, but also in water which contains common salt or chloride of ammonium. The presence of large quantities of common salt in sea water would, therefore, tend to increase its power of absorbing lime phosphates of organic origin. It is not at all incredible, therefore, to suppose that at some remote period the waters of the ocean, as indicated by these theories, were much more highly charged with phosphates in solution than at the present time.

According to Eldridge, the formation of the hard-rock and soft phosphates may be ascribed to three periods: First, that in which the primary rock was formed; second, that of secondary deposition in the cavities of the primary rock; third, that in which the deposits thus formed were broken up and the resulting fragments and comminuted material were redeposited as they now occur.

“The first of these stages began probably not later than the close of the older miocene, and within the eocene area it may have begun much earlier. Whether the primary phosphate resulted from a superficial and heavy deposit of soluble guanos, covering the limestones, or from the concentration of phosphate of lime already widely and uniformly distributed throughout the mass of the original rock, or from both, is a difficult question. In any event, the evidence indicates the effect of the percolation of surface waters, highly charged with carbonic and earth acids, and thus enabled to carry down into the mass of the limestone dissolved phosphate of lime, to be redeposited under conditions favorable to its separation. Such conditions might have been brought about by the simple interchange of bases between the phosphate and carbonate of lime thus brought together, or by the lowering of the solvent power of the waters through loss of carbonic acid. The latter would happen whenever the acid was required for the solution of additional carbonate of lime, or when, through aeration, it should escape from the water. The zone of phosphate

deposition was evidently one of double concentration, resulting from the removal of the soluble carbonate thus raising the percentage of the less soluble phosphate, and from the acquirement of additional phosphate of lime from the overlying portions of the deposits.”

“The thickness of the zone of phosphatization in the eocene area is unknown, but it is doubtful if it was over twenty feet. In the miocene area the depth has been proved from the phosphates *in situ* to have been between six and twelve feet.”

The deposits of secondary origin, according to Eldridge, are due chiefly to sedimentation, although some of them may have been due to precipitation from water. This secondary deposition was kept up for a long period, until stopped by some climatic or geologic change. The deposits of phosphates thus formed in the Florida peninsula are remarkably free from iron and aluminum, in comparison with many of the phosphates of the West Indies.

The third period in the genesis of the hard rock deposits embraces the time of formation of the original deposits and their transportation and storage as they are found at the present time. The geologic time at which this occurred is somewhat uncertain but it was probably during the last submergence of the peninsula.

In all cases the peculiar formation of the Florida limestone must be considered. This limestone is extremely porous and therefore easily penetrated by the waters of percolation. A good illustration of this is seen on the southwestern and southern edges of Lake Okeechobee. In following down the drainage canal which has been cut into the southwest shore of the lake the edge of the basin, which is composed of this porous material may be seen. The appearance of the limestone would indicate that large portions of it have already given way to the process of solution. The remaining portions are extremely friable, easily crushed, and much of it can be removed by the ordinary dredging machines. Such a limestone as this is peculiarly suited to the accumulation of phosphatic materials, due to the percolation of the water containing them. The solution of the limestone and consequent deposit of the phosphate of lime is easily understood when the character of this limestone is considered.

Shaler, as quoted by Eldridge in the work already referred to, refers to this characteristic of the limestone and says that the best conditions for the

accumulation of valuable deposits of lime phosphate in residual débris appear to occur where the phosphatic lime marls are of a rather soft character; the separate beds having no such solidity as will resist the percolation of water through innumerable incipient joints such as commonly pervade stratified materials, even when they are of a very soft nature.

Eldridge is also of the opinion that the remains of birds are not sufficient to account for the whole of the phosphatic deposits in Florida. He ascribes them to the joint action of the remains of birds, of land and marine animals and to the deposition of the phosphatic materials in the waters in the successive subsidences of the surface below the water line.

12. Nitrogen as a mineral constituent of soils, is found chiefly in the form of nitrates, but, owing to their solubility, they can not accumulate in soils exposed to heavy rain-falls. The gaseous nitrogen in the soil is also of some importance, since it is in this material that the anaerobic organisms which accumulate on the rootlets of some plants probably act in the process of the fixation of atmospheric nitrogen in a form accessible to plants. Nitrogen in the free state, it is believed, is not directly absorbed into the tissues of plants. It is necessary that it be oxidized in some way to nitric acid before it can be assimilated. The importance of nitrogen as a plant food can not be too highly estimated. It is as necessary to plant growth and development as water, phosphoric acid, lime, and potash, and far more costly. While a large quantity of nitrogen exists in the air in an uncombined state, it is, nevertheless, one of the least abundant of the elements of high importance in plant nutrition.

The conservation and increase of the stores of available nitrogen in the soil is one of the chief problems occupying the attention of agricultural chemistry. Nitrogen, which is not immediately available for the growth of plants, is conserved and restored by natural processes in various ways.

The waste nitrogen finds its way sooner or later to the sea, and is restored therefrom in many forms. Sea-weeds of all kinds are rich in recovered nitrogen. Many years ago Forchhammer^[6] pointed out the agricultural value of certain fucoids. Many other chemists have contributed important data in regard to the composition of these bodies.

Jenkins^[7] has shown from the analyses of several varieties of sea-weeds that in the green state they are quite equal in fertilizing value to stall

manure, and are sold at the rate of five cents per bushel. These data are fully corroborated by Goessmann.^[8]

Wheeler and Hartwell^[9] give the fullest and most systematic discussion which has been published of the agricultural value of sea-weeds. Sea-weed was used as a fertilizer as early as the fourth century, and its importance for this purpose has been recognized more and more in modern days, especially since chemical investigations have shown the great value of the food materials contained therein.

To show the commercial importance of sea-weed, it is only necessary to call attention to the fact that in 1885 its value as a fertilizer in the State of Rhode Island was \$65,044, while the value of all other commercial fertilizers was \$164,133. While sea-weed, in a sense, can only be successfully applied to littoral agriculture, yet the extent of agricultural lands bordering on the sea is so great as to render its commercial importance of the highest degree of interest.

A large amount of nitrogen is also recovered from the sea in fishes. It is shown by Atwater^[10] that the edible part of fishes has an unusually high percentage of protein. In round numbers, about seventy-five per cent of the water free edible parts of fish are composed of albuminoids. Some kinds of fish are taken chiefly for their oil and fertilizing value, as the menhaden. Squanto,^[11] an American Indian, first taught the early New England settlers the manurial value of fish.

Immense quantities of waste nitrogen are further secured, both from sea and land, by the various genera of birds. The well-known habit of birds in congregating in rookeries during the night and at certain seasons of the year tends to bring into a common receptacle the nitrogenous matters which they have gathered and which are deposited in their excrement and in the decay of their bodies. The feathers of birds are particularly rich in nitrogen, and the nitrogenous content of the flesh of fowls is also high. The decay of remains of birds, especially if it take place largely excluded from the leaching of water, tends to accumulate vast deposits of nitrogenous matter. If the conditions in such deposits be favorable to the processes of nitrification, the whole of the nitrogen, or at least the larger part of it, which has been collected in this débris, becomes finally converted into nitric acid and is found combined with appropriate bases as deposits of nitrates. The nitrates of the guano deposits and of the deposits in caves arise in this way.

If these deposits be subject to moderate leaching the nitrate may become infiltrated into the surrounding soil, making it very rich in this form of nitrogen. The bottoms and surrounding soils of caves are often found highly impregnated with nitrates.

While for our purpose, deposits of nitrates only are to be considered which are of sufficient value to bear transportation, yet much interest attaches to the formation of nitrates in the soil even when they are not of commercial importance.

In many of the soils of tropical regions not subject to heavy rain-falls, the accumulation of these nitrates is very great. Müntz and Marcano^[12] have investigated many of these soils to which attention was called first by Humboldt and Boussingault. They state that these soils are incomparably more rich in nitrates than the most fertile soils of Europe. The samples which they examined were collected from different parts of Venezuela and from the valleys of the Orinoco as well as on the shore of the Sea of Antilles. The nitrated soils are very abundant in this region of South America where they cover large surfaces. Their composition is variable, but in all of them carbonate and phosphate of lime are met with and organic nitrogenous material. The nitric acid is found always combined with lime. In some of the soils as high as thirty per cent of nitrate of lime have been found. Nitrification of organic material takes place very rapidly the year round in this tropical region. These nitrated soils are everywhere abundant around caves, as described by Humboldt, caves which serve as the refuge of birds and bats. The nitrogenous matters, which come from the decay of the remains of these animals, form true deposits of guano which is gradually spread around, and which, in contact with the limestone and with access of air, suffers complete nitrification with the fixation of the nitric acid by the lime.

Large quantities of this guano are also due to the débris of insects, fragments of elytra, scales of the wings of butterflies, etc., which are brought together in those places by the millions of cubic meters. The nitrification, which takes place in these deposits, has been found to extend its products to a distance of several kilometers through the soil. In some places the quantity of the nitrate of lime is so great in the soils that they are converted into a plastic paste by this deliquescent salt.

The theory of Müntz and Marcano in regard to the nitrates of soils, especially in the neighborhood of caves, is probably a correct one, but there are many objections to accepting it to explain the great deposits of nitrate of soda which occur in many parts of Chile. Another point, which must be considered also, is this: That the processes of nitrification can not now be considered as going on with the same vigor as formerly. Some moisture is necessary to nitrification, inasmuch as the nitrifying ferment does not act in perfectly dry soil, and in many localities in Chile where the nitrates are found it is too dry to suppose that any active nitrification could now take place.

The existence of these nitrate deposits has long been known.^[13] The old Indian laws originally prohibited the collection of the salt, but nevertheless it was secretly collected and sold. Up to the year 1821, soda saltpeter was not known in Europe except as a laboratory product. About this time the naturalist, Mariano de Rivero, found on the Pacific coast, in the Province of Tarapacá, immense new deposits of the salt. Later the salt was found in equal abundance in the Territory of Antofogasta and further to the south in the desert of Atacama, which forms the Department of Taltal.

At the present time the collection and export of saltpeter from Chile is a business of great importance. The largest export which has ever taken place in one year was in 1890, when the amount exported was 927,290,430 kilograms; of this quantity 642,506,985 kilograms were sent to England and 86,124,870 kilograms to the United States. Since that time the imports of this salt into the United States have largely increased.

According to Pissis^[14] these deposits are of very ancient origin. This geologist is of the opinion that the nitrate deposits are the result of the decomposition of feldspathic rocks; the bases thus produced gradually becoming united with the nitric acid provided from the air.

According to the theory of Nöllner^[15] the deposits are of more modern origin and due to the decomposition of marine vegetation. Continuous solution of soils, gives rise to the formation of great lakes of saturated water, in which occurs the development of much marine vegetation. On the evaporation of this water, due to geologic isolation, the decomposition of nitrogenous organic matter causes generation of nitric acid, which, coming in contact with the calcareous rocks, attacks them, forming nitrate of

calcium, which, in presence of sulfate of sodium, gives rise to a double decomposition into nitrate of sodium and sulfate of calcium.

The fact that iodine is found in greater or less quantity in Chile saltpeter is one of the chief supports of this hypothesis of marine origin, inasmuch as iodine is always found in sea and not in terrestrial plants. Further than this, it must be taken into consideration that these deposits of nitrate of soda contain neither shells nor fossils, nor do they contain any phosphate of lime. The theory, therefore, that they were due to animal origin is scarcely tenable.

13. Boron occurs chiefly in volcanic regions, but is much more widely distributed in the soil than formerly believed. It is a regular constituent of the ash of many plants,^[16] and is, therefore, thought to be a true plant food. It is one of the least abundant of the elements, not occurring in sufficient quantity to find a place in the table showing their relative abundance, which is to follow. Boracic acid is used to some extent as a preservative.

14. Fluorine does not occur free in nature, but it exists chiefly in combination with calcium, forming fluorspar, and traces of it are found in sea water. It occurs in bone, teeth, blood, and the milk of mammals. It is the only element that does not combine with oxygen, and it can be isolated only with the greatest difficulty. Only very small traces of it are found ordinarily and it is usually not considered in the chemical analysis of soils. Fluorine is found, however, in considerable quantities in certain phosphate deposits.

15. Aluminum is, probably, next to oxygen and silicon, the most abundant element of the earth's crust, of which it is estimated to form about one-twelfth. It has never been found, in nature, in the free state, but commonly occurs in combination with silicon and oxygen, in which form it is an abundant constituent of feldspar, mica, kaolin, clay, slate, and many other rocks and minerals.

By the weathering of feldspar, mica, and other minerals containing aluminum, kaolin or true clay is formed, which is of the greatest importance in the constitution of the soil. The compounds of aluminum are not so important as plant food as they are as the constituents of the soil, forming a large part of its bulk, and modifying in the most profound degree its physical properties. It is the custom of some authors to use the word clay to designate the fine particles of soil which have in general the same relations to moisture and tilth as the particles of weathered feldspar, etc. In a strict

chemical sense, however, the term clay is applied only to the hydrated silicate of alumina formed as indicated above. The fertility of a soil is largely dependent on the quantity of clay which it contains, its relations to moisture and amenability to culture being chiefly conditioned by its clay content. The determination of the percentage of clay in soils is an operation of the highest utility in forming an opinion of the value of a soil on analytical data alone.

16. Calcium is one of the commonest and most important elements of the earth's crust, of which it has been estimated to compose about one-sixteenth. It does not occur free in nature, but its most common form is in combination with carbon dioxide, forming the mineral calcite, marble, and the very abundant limestone rocks. In this form it is slightly soluble in water containing carbon dioxide, and hence lime has become a universal component of all soils and is very generally found in natural waters, in which it furnishes the chief ingredient necessary for the formation of the shells and skeletons of the various tribes of mollusca and corals. In combination with sulfuric acid calcium forms the rock gypsum. Lime is not only a necessary plant food, but influences in a marked degree the physical condition of the soil and the progress of nitrification. Many stiff clay soils are rendered porous and pulverulent by an application of lime, and thus made far more productive. On account of its great abundance and low price, it has not commanded the degree of attention from farmers and agricultural chemists which its merits deserve. It forms an essential ingredient of plants and animals, in the latter being collected chiefly in the bones, while in plants it is rather uniformly distributed throughout all the tissues.

17. Magnesium occurs chiefly in combination with carbon dioxide or with lime and carbon dioxide in the mineral dolomite. It is intimately associated with calcium and a trace of it is nearly always found where lime occurs in any considerable quantity. The bitter taste of sea water and some mineral waters is due to the presence of salts of magnesia. In combination with silica it forms an essential part of such rocks as serpentine, soapstone, and talc. Magnesia is not of much importance as a plant food nor as a fertilizing material.

18. Potassium combined with silica is an important element in many mineral silicates as, for instance, orthoclase. Granitic rocks usually contain considerable quantities of potassium, and on their decomposition this

becomes available for plant food. In the form of chlorid, potassium is found in small quantities in sea water, and as a nitrate it forms the valuable salt known as niter or saltpeter. Potassium, as is the case with phosphorus, is universally distributed in soils, and forms one of the great essential elements of plant food. Under the form of kainite and other minerals large quantities of potassium are used for fertilizing and for the manufacture of pure salts for commercial and pharmaceutical purposes. The ordinary potassium salts are very soluble and for this reason they can not accumulate in large quantities in soils exposed to heavy rain-fall. In the form of carbonate, potassium forms one of the chief ingredients of hard wood ashes, and in this form of combination is especially valuable for fertilizing purposes. Potash salts, being extremely soluble, are likely to be held longest in solution. Some of them, are recovered in animal and vegetable life, but the great mass of potash carried into the sea still remains unaccounted for. The recovery of the waste of potash is chiefly secured by the isolation of sea waters containing large quantities of this salt and their subsequent evaporation. Such isolation of sea waters takes place by means of geologic changes in the level of the land and sea. In the raising of an area above the water level there is almost certain to be an enclosure, of greater or less extent, of the sea water in the form of a lake. This enclosure may be complete or only partial, the enclosed water area being still in communication with the main body of the sea by means of small estuaries. If this body of water be exposed to rapid evaporation, as was doubtless the case in past geologic ages, there will be a continual influx of additional sea water through these estuaries to take the place of that evaporated. The waters may thus become more and more charged with saline constituents. Finally a point is reached in the evaporation when the less soluble of the saline constituents begin to be deposited. In this way the various formations of mineral matter, produced by the drying up of enclosed waters, take place.

The most extensive potash deposits known are those in the neighborhood of Stassfurt, in Germany. The following description probably represents the method of formation of these deposits:^[17]

“The Stassfurt salt and potash deposits had their origin, thousands of years ago, in a sea or ocean, the waters of which gradually receded, leaving near the coast, lakes which still retained communication with the great ocean by means of small channels. In that part of Europe the climate was then tropical, and the waters of these lakes rapidly evaporated but were

constantly replenished through these small channels connecting them with the main body. Decade after decade this continued, until by evaporation and crystallization, the various salts present in the sea water were deposited in solid form. The less soluble material, such as sulfate of lime or 'anhydrite,' solidified first and formed the lowest stratum. Then came common rock salt with a slowly thickening layer which ultimately reached 3000 feet, and is estimated to have been 13,000 years in formation. This rock salt deposit is interspersed with lamellar deposits of 'anhydrite,' which gradually diminish towards the top and are finally replaced by the mineral 'polyhalite,' which is composed of sulfate of lime, sulfate of potash, and sulfate of magnesia. The situation in which this polyhalite predominates is called the 'polyhalite region' and after it comes the 'kieserite region,' in which, between the rock salt strata, kieserite (sulfate of magnesia) is imbedded. Above the kieserite lies the 'potash region,' consisting mainly of deposits of carnallite, a mineral compound of muriate of potash and chlorid of magnesia. The carnallite deposit is from 50 to 130 feet thick and yields the most important of the crude potash salts and that from which are manufactured most of the concentrated articles, including muriate of potash."

"Overlying this region is a layer of impervious clay which acts as a water-tight roof to protect and preserve the very soluble potash and magnesia salts, which, had it not been for the very protection of this overlying stratum, would have been long ages ago washed away and lost by the action of the water percolating from above. Above this clay roof is a stratum, of varying thickness of anhydrite, and still above this a second salt deposit, probably formed under more recent climatic and atmospheric influences or possibly by chemical changes in dissolving and subsequent precipitation. This salt deposit contains ninety-eight per cent (often more) of pure salt, a degree of purity rarely elsewhere found. Finally, above this are strata of gypsum, tenacious clay, sand, and limestone, which crop out at the surface."

"The perpendicular distance from the lowest to the upper surface of the Stassfurt salt deposits is about 5000 feet (a little less than a mile), while the horizontal extent of the bed is from the Harz Mountains to the Elbe River in one direction, and from the city of Magdeburg to the town of Bernburg in the other."

According to Fuchs and DeLauny^[18] the saline formation near Stassfurt is situated at the bottom of a vast triassic deposit surrounding Madgeburg. The quantity of sea water which was evaporated to produce saline deposits of more than 500 meters in thickness must have been enormous and the rate of evaporation great. It appears that a temperature of 100° would have been quite necessary, acting for a long time, to produce this result.

These authors therefore admit that all the theories so far advanced to explain the magnitude of these deposits are attended with certain difficulties. What, for instance, could have caused a temperature of 100°? The most reasonable source of this high temperature must be sought for in the violent chemical action produced by the double decompositions of such vast quantities of salts of different kinds. There may also have been at the bottom of this basin some subterranean heat such as is found in certain localities where boric acid is deposited.

Whatever be the explanation of the source of the heat it will be admitted that at the end of the permian period there was thrown up to the northeast of the present saline deposits a ridge extending from Helgoland to Westphalia. This dam established throughout the whole of North Germany saline lagoons in which evaporation was at once established, and these lagoons were constantly fed from the sea.

There was then deposited by evaporation, first of all a layer of gypsum and afterwards rock salt, covering with few exceptions the whole of the area of North Germany.

But around Stassfurt there occurred at this time geologic displacements, the saline basin was permanently closed and then by continued evaporation the more deliquescent salts, such as polyhalite, kieserite, and carnallite, were deposited.

These theories account with sufficient ease for the deposition of the saline masses, but do not explain why in those days the sea water was so rich in potash and why potash is not found in other localities where vast quantities of gypsum and common salt have been deposited. It may be that the rocks composing the shores of these lagoons were exceptionally rich in potash and that this salt was, therefore, in a certain degree, a local contribution to the products of concentration.

19. Sodium is never found free in nature, but its most common form is in combination with chlorin as common salt, an important ingredient of sea

water. Combined with silica sodium is an important element in many silicates. Sodium, although closely related to potassium chemically, cannot in any case be substituted therefor in plant nutrition. In combination with nitrogen it forms soda or Chile saltpeter which is a valuable fertilizer on account of its content of nitric acid.

20. Iron is the most abundant of the heavy metals, and occurs in nature both free and combined with other elements. In the free state it is found only to a limited extent in basaltic rocks and meteorites, but in combination with oxygen it is one of the most widely diffused of metals, and forms the coloring matter of a large number of rocks and minerals. In this form, too, it exists as the valuable ores of iron known as magnetite and hematite. In combination with sulfur it forms the mineral pyrite, FeS_2 . The yellow and red colors of soils are due chiefly to iron oxids. It is an important plant food, although not taken up in any great quantity by the tissues of plants.

21. Manganese, next to iron, is the most abundant of the heavy metals. It occurs in nature only in combination with oxygen, in which form it is associated in minute quantities with iron in igneous rocks or in the forms known mineralogically as pyrolusite, psilomelane and wad. As the peroxid of manganese it occurs in concretionary forms scattered abundantly over the bottom of the deep sea. It is found in the ash of some plants but is not believed to be an essential to plant growth.

22. Barium occurs in nature combined with sulfuric acid, forming the mineral barite, or heavy spar, or with carbon dioxid forming the mineral witherite. It is of small importance from an agricultural standpoint.

23. Relative Abundance of the More Important Chemical Elements.—It will be of interest to the agricultural analyst to know as nearly as possible the relative abundance of the more important chemical elements. This subject has been carefully studied by Prof. F. W. Clarke in a paper read before the Philosophical Society of Washington.^[19] The materials considered in these calculations are the atmosphere, the water, and the solid crust of the earth to the depth of ten miles below the sea level. Of these materials the relative quantities of the three constituents named are as follows:

	Per cent.
Atmosphere	0.03
Water	7.08
Solid crust of the earth to the depth of ten miles	92.89

According to these calculations the relative abundance of the important elements composing the atmosphere, the water of the ocean and the solid crust of the earth to the depth given is as follows:

	Solid crust, ninety-three per cent.		Ocean, seven per cent.		Mean, including air.	
		per cent.		per cent.		per cent.
Oxygen	47.29		85.79		49.98	
Silicon	27.21	” ”		” ”	25.30	” ”
Aluminum	7.81	” ”		” ”	7.26	” ”
Iron	5.46	” ”		” ”	5.08	” ”
Calcium	3.77	” ”	0.05	” ”	3.51	” ”
Magnesium	2.68	” ”	0.14	” ”	2.50	” ”
Sodium	2.36	” ”	1.14	” ”	2.28	” ”
Potassium	2.40	” ”	0.04	” ”	2.23	” ”
Hydrogen	0.21	” ”	10.67	” ”	0.94	” ”
Titanium	0.33	” ”		” ”	0.30	” ”
Carbon	0.22	” ”	0.002	” ”	0.21	” ”
Chlorin	0.01	” ”	2.07 }	” ”	0.15	” ”
Bromin		” ”	0.008 }	” ”		” ”
Phosphorus	0.10	” ”		” ”	0.09	” ”
Manganese	0.08	” ”		” ”	0.07	” ”
Sulfur	0.03+	” ”	0.09	” ”	0.04+	” ”
Barium	0.03	” ”		” ”	0.03	” ”
Nitrogen		” ”		” ”	0.02	” ”
Chromium	0.01	” ”		” ”	0.01	” ”
	100.00	” ”	100.000	” ”	100.00	” ”

24. Fluorin is not mentioned in this table but it is stated that its probable percentage is 0.02 to 0.03 making it thus slightly more abundant than nitrogen.

One of the chief points of interest in connection with this table is that the nitrogen which is regarded by most persons as one of the most abundant of

the elements is almost the least abundant of those mentioned.

THE MINERALS OCCURRING IN ROCKS.

25. The Soil, as before stated, being comprised almost exclusively of decayed rocks, its characteristics would naturally be determined by the character of the minerals contained in the rocks.

A rock may be composed of a single mineral or an aggregation of several minerals.

According to the authority of the National Museum^[20] it may occur, either in the form of stratified beds, eruptive masses, sheets or dikes, or as veins and other chemical deposits of comparatively little importance as regards size and extent. The mineral composition of rocks is greatly simplified by the wide range of conditions under which the commonest minerals can be formed. Thus quartz, feldspar, mica, the minerals of the hornblende, or pyroxene group, can be formed from a mass cooling from a state of fusion; they may be crystallized from solution, or be formed from volatilized products. They are therefore the commonest of minerals and are rarely excluded from rocks of any class, since there is no process of rock formation which determines their absence.

Most of the common minerals, like the feldspars, micas, hornblendes, pyroxenes, and the alkaline carbonates possess the capacity of adapting themselves to a very considerable range of compositions. In the feldspars, for example, lime, soda, or potash may replace one another almost indefinitely, and it is now commonly assumed that true species do not exist, but all are but isomorphous admixtures passing into one another by all gradations, and the names albite, oligoclase, anorthite, etc., are to be used only as indicating convenient stopping and starting points in the series. Hornblende or pyroxene, further, may be pure silicate of lime and magnesia, or iron and manganese may partially replace these substances. Lime carbonate may be pure, or magnesia may replace the lime in any proportion.

These illustrations are sufficient to show the reason for the great simplicity of rock masses as regards their chief mineral constituents.

Whatever may be the conditions of the origin of a rock mass, the probabilities are that it will be formed essentially of one or more of a half a

dozen minerals in some of their varieties.

But however great the adaptability of these few minerals may be they are, nevertheless, subject to very definite laws of chemical equivalence. There are elements which they cannot take into their composition, and there are circumstances which retard their formation while other minerals may be crystallizing. In a mass of rock of more or less accidental composition formed under these widely varying conditions it may, therefore, be expected that other minerals will form, in considerable numbers, but minute quantities. It is customary to speak of those minerals which form the chief ingredients of any rock, and which may be regarded as characteristic of any particular variety, as the essential constituents, while those which occur in but small quantities, and whose presence or absence does not fundamentally affect its character, are called accessory constituents. The accessory mineral which predominates, and which is, as a rule, present in such quantities as to be recognizable by the unaided eye, is the characterizing accessory. Thus a biotite granite is a stone composed of the essential minerals quartz and potash feldspar, but in which the accessory mineral biotite occurs in such quantities as to give a definite character to the rock.

26. Classification Of Minerals.—The minerals of rocks may also be conveniently divided into two groups, according as they are products of the first consolidation of the mass or of subsequent changes. This is the system here adopted. We thus have:

(1) The original or primary constituents, those which formed upon its first consolidation. All the essential constituents are original, but on the other hand all the original constituents are not essential. Thus, in granite, quartz and orthoclase are both original and essential, while beryl and zircon or apatite, though original, are not essential.

(2) The secondary constituents are those which result from changes in a rock subsequent to its first consolidation, changes which are due in great part to the chemical action of percolating water. Such are the calcite, chalcedony, quartz, and zeolite deposits which form in the druses and amygdaloidal cavities, of traps and other rocks.

Below is given a list of the more common, original and secondary minerals occurring in rocks. It will be observed that the same mineral may, in certain cases, occur in both original and secondary forms. The tables following were prepared by Dr. George P. Merrill.

ORIGINAL MINERALS.

1. Quartz, SiO_2 .

2. The Feldspars:

- | | | |
|-----|--------------|---|
| 2a. | Orthoclase. | Anhydrous silicate of alumina with varying amounts of lime, |
| 2b. | Microcline. | potash, or soda and rarely barium. |
| 2c. | Albite. | |
| 2d. | Oligoclase. | |
| 2e. | Andesite. | |
| 2f. | Labradorite. | |
| 2g. | Bytownite. | |
| 2h. | Anorthite. | |

3. The Amphiboles:

- | | | |
|-----|---------------|--|
| 3a. | Hornblende. | Anhydrous silicates of lime and magnesia with iron and |
| 3b. | Tremolite. | alumina in the dark varieties. |
| 3c. | Actinolite. | |
| 3d. | Arfvedsonite. | |
| 3e. | Glaucophane. | |
| 3f. | Smaragdite. | |

4. The Monoclinic Pyroxenes:

- | | | |
|-----|-------------|---|
| 4a. | Malacolite. | Anhydrous silicates of magnesia and lime with alumina and |
| 4b. | Diallage. | iron in the dark varieties. |
| 4c. | Augite. | |
| 4d. | Acmite. | |
| 4c. | Aegerite. | |

5. The Rhombic Pyroxenes:

- | | | |
|-----|-----------------------|---------------------------------|
| 5a. | Enstatite (bronzite). | Silicates of magnesia and iron. |
| 5b. | Hypersthene. | |

6. The Micas:

- 6a. Muscovite. Anhydrous silicates of alumina with potash, soda, and iron.
- 6b. Biotite.
- 6c. Phlogopite.

- 7. Calcite,
- 8. Dolomite.
- 9. Gypsum.
- 10. Olivine.
- 11. Beryl.
- 12. Tourmaline.
- 13. Garnet, variable common form.
- 14. Vesuvianite.
- 15. Epidote.
- 16. Zoisite.
- 17. Allanite.
- 18. Andalusite.
- 19. Staurolite.
- 20. Fibrolite.
- 21. Cyanite.
- 22. Scapolite.
- 23. Apatite.
- 24. Elaeolite and Nepheline.
- 25. Leucite.
- 26. Cancrinite.
- 27. The Sodalite Group:
 - 27a. Sodalite.

27b. Häüyn (noseau).

28. Zircon.

29. Chondrodite.

30. Cordierite.

31. Topaz.

32. Corundum.

33. Titanite (sphene).

34. Rutile.

35. Menaccanite.

36. Magnetite.

37. Hematite.

38. Chromite.

39. The Spinel:

39a. Pleonast.

39b. Picotite.

40. Pyrolusite.

41. Halite (common salt).

42. Fluorite.

43. The Elements:

43a. Graphite.

43b. Carbon.

43c. Iron.

43d. Copper.

44. The Metallic Sulfids:

- 44a. Galena.
- 44b. Sphalerite.
- 44c. Pyrrhotite.
- 44d. Marcasite.
- 44e. Pyrite.
- 44f. Chalcopyrite.
- 44g. Arsenopyrite.

SECONDARY MINERALS.

- 1. Quartz:
 - 1a. Chalcedony.
 - 1b. Opal.
 - 1c. Tridymite.
- 2. Albite.
- 3. The Amphibole Group:
 - 3a. Hornblende.
 - 3b. Tremolite.
 - 3c. Actinolite.
 - 3d. Uralite.
- 4. Muscovite (sericite).
- 5. The Chlorites:
 - 5a. Jefferisite.
 - 5b. Ripidolite.
 - 5c. Penninite.
 - 5d. Prochlorite.

6. Calcite (and aragonite).
7. Wollastonite.
8. Scapolite.
9. Garnet.
10. Epidote.
11. Zoisite.
12. Serpentine.
13. Talc.
14. Kaolin.
15. The Zeolites:
 - 15a. Pectolite.
 - 15b. Laumontite.
 - 15c. Prehnite.
 - 15d. Thomsonite.
 - 15e. Natrolite.
 - 15f. Analcite.
 - 15g. Datolite.
 - 15h. Chabazite.
 - 15i. Stilbite,
 - 15k. Heulandite.
 - 15l. Harmotome.
16. Magnetite.
17. Hematite.
18. Limonite.
19. Siderite.

20. Pyrite.

21. Pyrrhotite.

ROCKS AND ROCK DECAY.

27. Types Of Rocks.—Rocks may be divided in reference to their structure into four types: First, crystalline; second, vitreous; third, colloidal; fourth, fragmental.

Of these classes there may be selected, as types of the first order, granite and crystalline limestone.

The second class is typically represented by obsidian. Rocks of this kind are confined to a volcanic origin.

The third class of rocks is completely amorphous in its structure and is less common than the others. It is found only in rocks of chemical origin. Types of this class are the siliceous sinters, opals, flint nodules, and many serpentines.

Of the fourth class of rocks, sandstone is typical, being comprised wholly of fragments of rocks pre-existing. The particles may be held together either by cohesion or by a cement composed of silica, iron oxids, carbonate of lime or clayey matter.

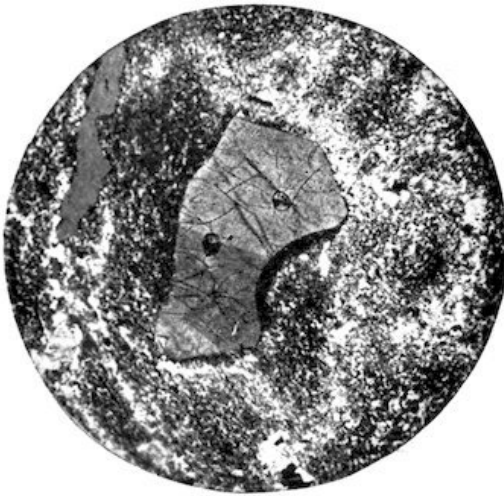
28. The Microscopical Structure of Rocks.—A great deal more light is thrown upon the nature of rock materials by microscopical study than by their study in bulk. The requisites for a microscopical study of rock are that the material should be cut into extremely thin laminae with parallel sides and polished so as to transmit the light freely. The study of the crystalline structure of the material is then conducted by means of a microscope furnished with polarizing and analyzing appliances. The light before passing through the mineral film is polarized by a Nicol prism. After passing through the film it is analyzed by a second Nicol prism. In this way the crystalline structure of the rock as affecting polarized light is distinctly brought out. The thickness of the films examined should be from $\frac{1}{500}$ to $\frac{1}{600}$ of an inch.



1



2



3



4



5



6

- FIG. 1. Microstructure of granite.
- FIG. 2. Microstructure of micropegmatite.
- FIG. 3. Microstructure of quartz porphyry.
- FIG. 4. Microstructure of porphyritic obsidian.
- FIG. 5. Microstructure of trachyte.
- FIG. 6. Microstructure of serpentine.

The method of rock study by thin microscopic sections is one of comparatively recent origin. It is scarcely more than a dozen years since the process was fairly adopted by mineralogists. The value of the method is based upon the fact that every crystalline mineral has certain definite optical properties. Therefore, when a crystalline mineral is distorted or misshapen so as to be incapable of identification by the ordinary method, it can be at once identified by its optical examination in the manner just described. In this way not only can one mineral be distinguished from another, but the crystalline system to which it belongs can be accurately pointed out. The value of the method is well summed up by Merrill,^[21] who says that it is not merely an aid in determining the mineralogical composition of a rock, but also, which is often much more important, its structure and the various changes which have taken place in it since its first consolidation. Rocks are not the definite and unchangeable mineral compounds they were once considered, but are rather ever varying aggregates of minerals which even in themselves undergo structural and chemical changes almost without number.

Another valuable result of such a study is illustrated by the discovery that the structural features of a rock are not dependent upon its chemical composition or geologic age, but upon the conditions under which it cooled from the molten magma. Portions of the same rock may vary all the way from a wholly crystalline to a pure vitreous form.

Some typical microstructures of crystalline rocks are shown in the accompanying figures 1–6.^[22]

Although this method of study has thus far been confined mainly to crystalline rocks, its efficiency is by no means limited to them. The fragmental rocks and their decomposed débris to which the name soil is given are equally worthy of study by this method. Indeed, the full value of a chemical analysis of any rock or soil can not be ascertained unless such an analysis is accompanied by a microscopic examination. It is desirable to know not merely what there is in any soil, but in what form these

compounds exist. To this latter question the chemical analysis as ordinarily made will give no clew. In Germany a beginning has been made in this line of work, and American scientists are beginning to realize its importance. An outline of this method of analysis will be given in the proper place.

29. Specific Gravity.—Much information in regard to the properties of a rock, or mineral constituent thereof, may be derived from its specific gravity.

The internal structure of a rock may have much to do with its apparent specific gravity. As an instance of this, it may be stated that an obsidian pumice will float upon water, buoyed up by the air contained in its vesicles, while a compact obsidian of the same composition will sink immediately. A careful discrimination must, therefore, be made between apparent and true specific gravity. In general it may be said that crystalline rocks have a higher specific gravity than those of a vitreous nature. The specific gravity is, therefore, largely dependent upon chemical and crystallographic properties; for instance, among siliceous rocks those which contain the largest amount of silica are the lightest, while those with a comparatively small amount, but rich in iron, lime, and magnesia, are heaviest.

30. Chemical Composition of Rocks.—Rocks are often classified with respect to the chief mineral constituent which they contain. Rocks which are composed largely of lime are termed calcareous; of silica, siliceous; of iron, ferruginous; and of clay, argillaceous. In respect of eruptive rocks, it is customary to speak of those which show above sixty per cent of silica as acidic, while those containing less than fifty per cent of silica and a correspondingly larger amount of iron, lime, and magnesia, are spoken of as basic. Illustrations of the classification of rocks on the above principles are given below.^{[\[23\]](#)}

STRATIFIED ROCKS.

Kind.	Specific Gravity.	Composition.
Calcareous:		
Compact limestone	2.6 to 2.8	Carbonate of lime.
Crystalline limestone		
Compact dolomite	2.8 to 2.95	Carbonate of lime and magnesia.
Crystalline dolomite		
Siliceous:		
Gneiss	2.6 to 2.7	Same as granite.
Siliceous sandstone	2.6	Mainly silica.
Schist	2.6 to 2.8	60 to 80 per cent silica.
Argillaceous:		
Clay slate (argillite)	2.5	Mainly silicate of alumina.

ERUPTIVE ROCKS.

	Specific Gravity.	Per cent silica.
Acidic Group:		
Granite	2.58 to 2.73	77.65 to 62.90
Liparite	2.53 to 2.70	76.06 to 67.61
Obsidian	2.26 to 2.41	82.80 to 71.19
Obsidian pumice	Floats on water.	82.80 to 71.19
Intermediate Group:		
Syenite	2.73 to 2.86	72.20 to 54.65
Trachyte	2.70 to 2.80	64.00 to 60.00
Hyalotrachyte	2.4 to 2.5	64.00 to 60.00
Andesite	2.54 to 2.79	66.75 to 54.73
Basic Group:		
Diabase	2.66 to 2.88	50.00 to 48.00
Basalt	2.90 to 3.10	50.59 to 40.74
Peridotite	3.22 to 3.29	42.65 to 33.73
Peridotite (iron rich)	3.86	23.00
Peridotite (meteorite)	3.51	37.70

31. Color Of Rocks.—The color of rocks is determined chiefly by the oxids of metals which they contain and the degree of oxidation of the mineral in each particular case. There are, however, many colors of rocks which seem to depend not upon any particular mineral ingredient which they contain, but upon some particular crystalline structure or physical condition.

The chief coloring matters in minerals are those which form colored bases such as iron, manganese, chromium, etc. The yellow, brown, and red colors, common to fragmental rocks, are due almost wholly to free oxids of iron. The gray, green, dull brown, and even black colors of crystalline rocks are due to the presence of free iron oxids or to the prevalence of silicate mineral rich in iron, as augite, hornblende, or black mica. Rarely copper and other metallic oxids than those of iron are present in sufficient abundance to impart their characteristic hues. As a rule, a white or light-gray color denotes an absence of an appreciable amount of iron in any of its forms. The bluish and black colors of many rocks, particularly the limestones and slates, are due to the presence of carbonaceous matter.

In still other cases, and particularly the feldspar-bearing rocks, the color may be due in part to the physical condition of the feldspar.

Inasmuch as the color of rocks is due so largely to metallic oxids, it is easy to see that they may undergo changes when exposed to weathering, or the degree of oxidation may change, and either, together with changes in the physical structure of the rock, may cause a distinct change in color. Luster is often considered in connection with color, and is due almost exclusively to physical conditions.

32. Kinds of Rocks.—The rocks which form any essential part of the earth's crust are grouped under four main heads, the distinction being based upon their origin and structure.^[24] Each of the main divisions may be subdivided into groups or families, the distinction being based mainly upon chemical composition, structure, and mode of occurrence. The four chief families are:

First, aqueous rocks, formed mainly through the agency of water as chemical precipitates or as sedimentary beds.

Second, aeolian rocks formed from wind-drifted materials.

Third, metamorphic rocks, changed from their original condition through dynamic or chemical agencies, and which may have been partly of aqueous and partly of igneous origin.

Fourth, igneous or eruptive rocks, which have been brought up from below in a molten condition, and which owe their present structural peculiarities to variations in composition and conditions of solidification.

33. Aqueous Rocks.—Aqueous rocks may be divided into the following general classes:

First, rocks formed as chemical precipitates.

Second, rocks formed as sedimentary deposits and fragmental in structure. The second class may again be subdivided into rocks formed by mechanical agencies and mainly of inorganic materials; and second, rocks composed mainly of the débris of plant and animal life.

In regard to the first form of aqueous rocks, namely, those formed as chemical precipitates, it may be said that while their quantity is not large they are yet of considerable importance from an agricultural point of view. They embrace those substances which, having once been in a condition of vapor or aqueous solution, have been deposited or precipitated, either by cooling or by the evaporation of the liquor holding them in solution, or by coming in contact with chemical substances capable of precipitating them. The influence of water as a solvent is perhaps not fully appreciated. Its solvent influence will be noted particularly under the head of weathering or decay of rocks. Its importance, however, in producing stratified rocks has been very great. Water, especially when under great pressure and at a high temperature, has the power of dissolving many minerals. This power is often greatly increased by the mineral matter previously in solution in the water or by the gases which it may contain. As an illustration of the latter property, the solvent action of water charged with carbon dioxid on limestone may be cited.

When mineral matters have been dissolved by the water in the ways mentioned and carried with the water beyond the condition where the solution has taken place, new conditions are found favorable to the precipitation of the dissolved matters. The water, which before may have been very hot, may reach a place where it cools, and being a supersaturated solution, the excess of the material is thrown down as the water cools.

On the other hand, if the solution be due to the presence of carbon dioxid and the water reach a place where it is exposed to the air or where the pressure under which the abundance of the gas has been due is diminished, the carbon dioxid will escape and the mineral matters which have been dissolved thereby will be precipitated.

The incrustations which often appear round the mouth of springs and the occurrence of stalagmites and stalactites in caves are illustrations of this

action.

In respect of the formation of rocks as precipitates from a state of vapor we have scarcely any illustrations excepting in volcanic regions. Rocky materials with which we are generally acquainted are practically non-volatile at the highest temperature which can be secured on the earth's surface, but it is possible that in the interior of the earth the temperature may be so high as to maintain many substances in a state of vapor.

They may, in this case, become disassociated so that the compounds or elements exist distinctly in a vaporous condition. Such a vapor transported to regions of diminished temperature would first of all on cooling permit a union of the chemical elements forming new compounds less volatile, which, of course, would be at once precipitated.

The rocks and minerals formed in this way which are of some agricultural importance may be classified as follows:

Oxids, carbonates, silicates, sulfur, sulfids, sulfates, phosphates, chlorids, and hydrocarbon compounds, the most important from an agricultural point of view being the phosphates.

The second group of rocks, namely those formed as sedimentary deposits, differ from those just described in that they are comprised mainly of fragmental materials derived from the breaking down of pre-existing rocks. The formation of fragmental rocks includes, therefore, the same processes as are active in the formation of arable soil. They are deposited from water, and are as a rule distinctly stratified.

Through the action of pressure and the heat thereby generated, or simply through the chemical action of percolating solutions, such rocks pass over into the crystalline sedimentary forms known as metamorphic. All metamorphic rocks, however, are not of a sedimentary origin. For instance, by pressure, heat, and the chemical changes thereby induced, granite may be changed into gneiss and the latter would then be a metamorphic rock.

This group of sedimentary rocks and of sedimentary material, either unchanged or metamorphosed, is of vast extent and includes materials of widely varying chemical and mineralogical nature. They form by far the greater portion of the present surface of the earth, even the mountain ranges being composed mainly of this sedimentary material. Indeed, in the whole of this country there is only a comparatively very small extent of igneous or irruptive rocks. They are of great importance from a purely scientific, as

well as agricultural standpoint, since they contain the fossil records of past geologic ages. From them it is possible to study the variations in climate, the meteorological conditions in circumstances and periods far remote, and thus form some idea of the process by which the crust of the earth has been modified by natural forces from its original form to the present time.

The sedimentary rocks may be divided, with sufficient accuracy for our purposes, into two great classes: First, rocks formed by mechanical agencies and mainly of inorganic materials. These are subdivided again as follows:

- (a) The arenaceous group.
- (b) The argillaceous group.
- (c) The volcanic group.

The second class of sedimentary rocks is formed largely, or in part at least, by mechanical agencies, but is comprised chiefly of the débris of plant and animal life. It may be subdivided as follows:

- (a) The siliceous group, such as infusorial earth.
- (b) The calcareous group, fossiliferous formations, limestone, etc.

(c) The carbonaceous group, such as peat, lignite, coals, etc. The different classes of rock described above are distinguished by special qualities represented largely by the name. The first division, the arenaceous group, is composed mainly of the siliceous or coarsely granular materials derived from the disintegration of older crystalline rocks, which have been rearranged in beds of varying thickness through the mechanical agency of water. They are, in short, consolidated or unconsolidated beds of sand and gravel. In composition and texture they vary almost indefinitely. Many of them having suffered little during the process of disintegration and transportation are composed essentially of the same materials as the rocks from which they were derived.

The sandstones, which are the type of these rocks, vary greatly in structure as well as in composition, in some the grains being rounded while in others they are sharply angular.

The microscopic structure of sandstone is shown in figure 7.^[24]

The material by which the individual grains of a sandstone are bound together is usually the material of some of the other classes. The calcareous, ferruginous, and siliceous cements being the chief ones. This cementing

substance is deposited among the granules forming the sandstone by percolating water.

The colors of sandstone are dependent usually upon iron oxids. Especially is this true of the red, brown, and yellow colors. In some of the light grey varieties, the color is that of the minerals comprising the stone. Some of the darker colored sandstones contain organic matter.



FIG. 7.

Microscopic Structure of Sandstone.

The rocks of the argillaceous group are composed essentially of a hydrous silicate of alumina, which is the basis of common clay, and varying amounts of free silica, oxids of iron and manganese, carbonates of lime and magnesia, and small quantities of organic matter. They may have originated *in situ* from the decomposition of feldspars or as deposits of fine mud or silt at the bottom of large bodies of water. The older formations of these rocks are known as shales, argillites, and slates and the fissile structure which enables this to be split into thin sheets is probably due to the conditions under which they have been formed and not to any properties of the clays themselves.

One of the purest forms of this rock is kaolin, which is almost a pure hydrous silicate of alumina formed from the decomposition of feldspathic

rocks from which the alkalis, iron oxides and other soluble constituents have been removed by water.

Under the volcanic group are included the materials ejected from volcanic vents in a more or less finely comminuted condition and which through the drifting power of atmospheric currents may be scattered over many miles of territory. Various names are applied to such products, names dependent in large part upon their state of subdivision. Volcanic dust and sand, or ashes, includes the finer dust-like or sand-like materials, and lapilli, or rapilli the coarser. The general name tuff includes the more or less compacted and stratified beds of this material, while trass, peperino, and pozzuolano are local varietal names given to similar materials occurring in European volcanic regions.

The second division, namely sedimentary rocks composed of the débris of plant and animal life includes many forms of great agricultural importance.

The first subdivision of this group is the infusorial or diatomaceous earth. It forms a fine white or yellowish pulverulent rock composed mainly of minute shells, or tests of diatoms, and is often so soft and pliable as to crumble readily between the thumb and fingers. According to Whitney the beds are of comparatively limited extent and for this reason are of little agricultural value, although the weathering of this diatomaceous material gives rise to a light yellow clay forming very fertile agricultural lands.

The second subdivision of this group includes the rocks of a calcareous nature derived from animal life; that is to say, what are properly called limestones. They vary in color, structure, and texture almost indefinitely, and include all possible grades of materials from those which can be used only as a flux, or for lime burning, through ordinary building materials to the finest marbles. These rocks are world-wide in their distribution and limited to no one particular geologic horizon, but are found in stratified beds among rocks of all ages from the most ancient to the most recent.

Owing to the fact that their chief constituent, carbonate of lime, is soluble in ordinary meteoric waters, the rocks have undergone extensive decomposition, their lime being removed, while their less soluble constituents or impurities remain to form soil. A single ton of residual soil represents not infrequently a loss of 100 tons of original rock matter. As this mass of lime carbonate is removed by solution the residual soil settles, and

as the limestone rocks are more soluble than the adjacent rock formations limestone formations usually form valley lands with ridges on either side. Caves are frequently found in such formations. Furthermore, as the lime is almost all in the form of the easily soluble lime carbonate it can be very completely removed and the fertile “limestone soils” are often very deficient in lime and respond readily to an application of burnt lime, which, not infrequently, is quarried from the same field. From an agricultural standpoint this group is of very great interest and importance.

The third subdivision of this group, namely, that of vegetable origin, includes peat, lignite, coals, etc. Rocks of this group are made up of more or less fragmental remains of plants. In many of them, as the peats and lignites, the traces of plant structure are still apparent. In others, as the anthracite coals, these structures have been wholly obliterated by metamorphisms.

Plants when decomposing on the surface of the ground give off their carbon to the atmosphere in the shape of carbon dioxide gas leaving only the strictly inorganic or mineral matter behind. When, however, they are protected from the oxidizing influence of the air, by water or by other plant growth, decomposition is greatly retarded, and a large portion of the carbonaceous and volatile matters is retained, and by this means together with pressure from the overlying mass, the material becomes slowly converted into coal. When this process goes on near the surface of the earth, and without much pressure, peat or muck is the product.

The fourth subdivision of this group, the phosphatic, forms a class of rocks limited in extent but of the greatest economic importance. Guano, coprolites, and phosphatic rocks (the phosphorites) come under this head.

34. Aeolian Rocks.—This class of rocks is of less importance than the others, either geologically or agriculturally. It is formed from materials drifted by the winds and this material has various degrees of compactness. Usually the components of these drifts form rocks or deposits of a friable texture and of a fragmental nature. The very extensive deposits of loess in China, forming their most fertile lands, are admitted now to have been formed in this way, but it is now generally admitted that similar deposits in this country are of subaqueous origin.

Chief among these rocks, are the volcanic ashes which are often carried to a long distance by the wind before they are deposited and consolidated

into rock masses. Many loose soils may be carried to great distances by the wind, the deposits forming new aggregates. This is particularly the case in arid regions.

35. Metamorphic Rocks.—This class of rocks includes all sedimentary or eruptive rocks, which, after their deposition and agglomeration, have been changed in their nature through the action of heat, pressure, or by chemical means. Sometimes these changes are so complete that no indication of the character of the original rocks remains. At other times the changes may be found in all the stages of progress, so that they can be traced from the original fragmental or irruptive to the completely metamorphosed deposit. This is especially true of rocks containing large quantities of lime. In those containing silica, the changes are less readily traced.



FIG. 8.

MICROSTRUCTURE OF CRYSTALLINE LIMESTONE.

(West Rutland, Vermont.)

The metamorphic rocks may be divided into two subclasses, namely, stratified or bedded, and foliated or schistose.

The rocks of the first class are represented by the crystalline limestones and dolomites. The microstructure of a crystalline limestone is shown in

Fig. 8.^[25] When lime and magnesia occur together in combination with carbon dioxid, the substance is known as dolomite. The chemical nature of these rocks and their soil-forming properties are the same as those of the ordinary, non-metamorphosed limestones and dolomites to which reference has already been made. The subject need not, therefore, be further dwelt upon here.

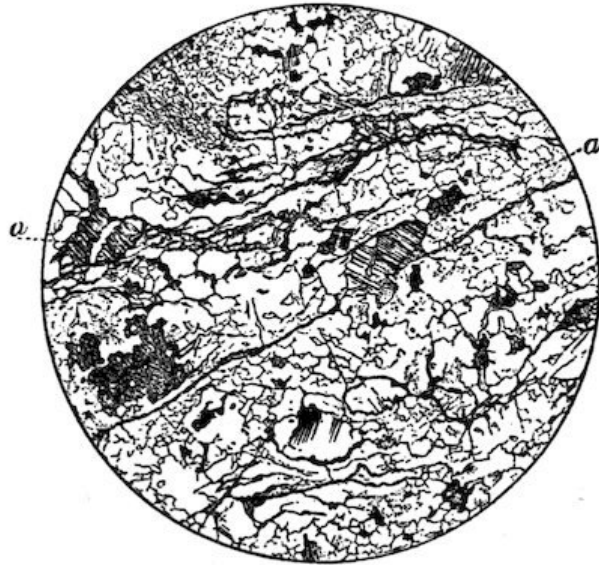


FIG. 9.

MICROSTRUCTURE OF GNEISS.

(West Andover, Massachusetts.)

At *a a* are shown plagioclase crystals broken and rounded by the sheering force producing the foliation.

The second variety of metamorphic rocks is represented by the gneisses and crystalline schists. Gneiss has essentially the same composition as granite and can frequently hardly be distinguished from it, except by a microscopic study of its sections, and even thus it is sometimes difficult to determine. Frequently a number of new minerals is formed in the metamorphic changes. The microstructure of a gneiss is shown in Fig. 9.^[26]

The schists include an extremely variable class of rocks, of which quartz is the prevailing constituent, and which, as a rule, are deficient in potash and other important ingredients.

36. Rocks Formed Through Igneous Agencies, or Eruptive Rocks.—

This group includes all those rocks, which, having been at some time in a state of igneous fusion, have been solidified into their present form by a process of cooling. It may be stated, as a general principle, that the greater the pressure under which a rock solidifies and the slower and more gradual the cooling the more perfect will be found the crystalline structure. Hence, it follows that the older and more deep-seated rocks which are forced up in the form of dikes, bosses, or intrusive sheets, into the overlying masses, and which have become exposed only through erosion and removal of the overlying rocks, are the more highly crystalline.

The eruptive rocks are divided into two main groups, *viz.*:

- (a) Intrusive or plutonic rocks, and
- (b) Effusive or volcanic rocks.

Among the more important of the first division of the plutonic form, from an agricultural point of view, are the granites. The essential constituents of granite are quartz, potash feldspar, and plagioclase. One or more minerals of the mica, amphibole or pyroxene groups are also commonly present, and in microscopic proportions apatite and particles of magnetic iron. The more valuable constituents, from an agricultural standpoint, are the minerals potash feldspar, and apatite, which furnish by their decomposition the essential potash and phosphoric acid.

In addition to the granites, which have already been mentioned, the group includes the syenites, the nepheline syenites, the diorites, the gabbros, the diabases, the theralites, the peridotites, and the pyroxenites.

The second group, the effusive or volcanic rocks, includes those igneous rocks, which, like the first group, have been forced up through the overlying rocks, but which were brought to the surface, flowing out as lavas. These, therefore, represent, in many cases, only the upper or surface portions of the first group, differing from them structurally, because they have cooled under little pressure more rapidly, and hence are not so distinctly crystalline. These comprise the following groups:

(a) Quartz porphyries. (b) Liparites. (c) Quartz-free porphyries. (d) Trachytes. (e) Phonolites. (f) Porphyrites. (g) Andesites. (h) Melaphyrs and augite porphyrites. (i) Basalts. (j) Tephrites and Basanites. (k) Picrite porphyrites. (l) Limburgites and augitites. (m) Leucite rocks. (n) Nepheline rocks. (o) Melilite rocks.

It is, in most cases, impossible to state which of the above classes is of most importance from an agricultural standpoint, since, in the process of soil formation, both chemical and physical processes are involved, whereby the character of the resultant soil is so modified as to but remotely resemble its parent rock. In most soils, the prevailing constituent is but the least soluble one of the rock mass from which it was derived. Thus a limestone soil may contain upwards of ninety per cent of silica and alumina, while the original limestone itself may not have carried more than one or two per cent of these substances. Of course, if a rock mass contains none of the constituents essential to plant growth, its resultant soil must by itself alone be quite barren. It does not absolutely follow, however, that those rocks containing the highest percentages of valuable constituents will yield the most fertile soils, since much depends on the manner in which they have been formed, the amount of leaching, etc., they may have undergone. Nevertheless, the study of the composition of the rocks in their relation to soils, is an extremely interesting and by no means unimportant one.

A comparative table of rock compositions is here given. It will be observed that there is a considerable range of variation even among rocks of the same class, a fact due to the varying abundance of their mineral constituents. The figures given are not those of actual analyses on any one particular rock, but are selected from a number of comparatively typical cases; and, it is thought, fairly well represent the composition of the class of rocks indicated.

COMPOSITION OF ROCKS.—THE FIGURES INDICATE PARTS PER HUNDRED.

	SiO ₂ .	Al ₂ O ₃ .	Fe ₂ O ₃ . FeO.	MgO.	CaO.	Na ₂ O.	K ₂ O.	P ₂ O ₅ .
Granite Quartz porphyries} Liparite	63– 78	10–15	2–3	0.3–0.5	1–2	2–3	4–5	0.05– 0.15
Syenite Orthoclase porphyries} Trachyte	55– 73	12–16	5–7	2–6	3–5	2–6	4–7	trace.
Nepheline syenites} Phonolites	54– 56	16–22	4–6	0.4– 0.88	2–4	3–7	4–6	0.15
Diorites Porphyrites} Andesites	52– 65	16–20	7–10	5–7	5–7	2–4	1–2	0.1–0.3
Gabbros Norites} Melaphyrs	48– 55	12–20	8–15	2–7	6–10	2–4	0.5– 2	0.1–0.33
Theralites Tephrites} Basanites	43– 47	15–23	9–18	1–6	6–10	5–7	2–4	trace.
Peridotites Picrite porphyrites} Limburgites	23– 43	1–10	10–15	15–45	1–4	0–4	trace.	0.0
Pyroxenites} Augitites	50– 55	0.5–4	4–10	20–25	8–15			
Leucite rocks	48– 50	15–20	7–10	1–2	5–10	3–5	5–7	0.5–2
Nepheline rocks	40– 45	8–20	10–20	1–13	4–10	4–8	1–3	0.2

37. Origin of Soils.—The soils in which crops grow and which form the subject of the analytical processes to be hereinafter described have been formed under the combined influences of rock decay and plant and organic growth. The mineral matters of soils have had their origin in the decay of rocks, while the humic and other organic constituents have been derived from living bodies. It is not the object of this treatise to discuss in detail the

processes of soil formation, but only to give such general outlines as may bear particularly on the proper conceptions of the principles of soil investigation.

38. The Decay of Rocks.—The origin and composition of rocks are fully set forth in works on geology and mineralogy. Only a brief summary of those points of interest to agriculture has been given in the preceding pages. The soil analyst should be acquainted with these principles, but for practical purposes he has only to understand the chief factors active in securing the decay of rocks and in preparing the débris for plant growth.

The following outline is based on the generally accepted theories respecting the formation of soils.[7]

The forces ordinarily concerned in the decay of rocks are:

1. Changes of temperature, including the ordinary daily and monthly changes, and the conditions of freezing and thawing.
2. Moving water or ice.
3. Chemical action of water and air.
4. Influence of vegetable and animal life:
 - (a) Shades the rock or soil surface.
 - (b) Penetrates the rock or soil material with its roots, thus admitting air.
 - (c) Solvent action of roots.
 - (d) Chemical action of decaying organic matter.
5. Earth worms.
6. Bacteria.

39. The Action of Freezing and Thawing.—In those parts of a rock stratum exposed near the surface of the earth the processes of freezing and thawing have perhaps had considerable influence in rock decay. The expansive force of freezing water is well known. Ice occupies a larger volume than the water from which it was formed. The force with which this expansion takes place is almost irresistible. The phenomenon of bursted

water pipes which have been exposed to a freezing temperature is not an uncommon one. While the increase in volume is not large, yet it is entirely sufficient to produce serious results. The way in which freezing affects exposed rock is easily understood. The effect is unnoticeable if the rock be dry. If, on the other hand, it be saturated with water, the disintegrating effect of a freeze must be of considerable magnitude. This effect becomes more pronounced if the intervals of freezing and thawing be of short duration. The whole affected portion of the rock may thus become thoroughly decayed. But even in the most favorable conditions this form of disintegration must be confined to a thin superficial area. Even in very cold climates the frost only penetrates a few feet below the surface, and therefore the action of ice cannot in any way be connected with those changes at great depths, to which attention has already been called. Nevertheless, certain building stones seem very sensitive to this sort of weathering, and the crumbling of the stone in the Houses of Parliament is due chiefly to this cause.

On the whole it appears that the action of ice in producing rock decay has been somewhat overrated, although its power must not by any means be denied. But on the other hand a freeze extending over a long time tends to preserve the rocks, and it therefore appears that the entire absence of frost would promote the process of rock decay.

At best it must be admitted that frost has affected the earth's crust only to an insignificant depth, but its influence in modifying the arable part of the soil is of the utmost importance to agriculture.

40. The Action of Glaciers.—The action of ice in soil formation is not confined alone to the processes just described. At a period not very remote geologically, a great part of our Northern States was covered with a vast field of moving ice. These seas of ice crept down upon us from more northern latitudes and swept before them every vestige of animal and vegetable life. In their movement they leveled and destroyed the crests of hills and filled the valleys with the débris. They crushed and comminuted the strata of rocks which opposed their flow and carried huge boulders of granite hundreds of miles from their homes. On melting they left vast moraines of rocks and pebbles which will mark for all time the termini of these empires of ice. When the ice of these vast glaciers finally melted the surface which they had leveled presented the appearance of an extended

plain. No estimate can be made of the enormous quantities of rock material which were ground to finest powder by these glaciers. This rock powder forms to-day no inconsiderable part of those fertile soils which are composed of glacial drift. The rich materials of these soils probably bear a more intimate relation to the rocks from which they were formed than of any other kinds of soil in the world. The rocks were literally ground into a fine powder, and this powder was intimately mixed with the soils which had already been formed *in situ*. The melting ice also left exposed to disintegrating forces large surfaces of unprotected rocks in which decay would go on much more rapidly than when covered with the *débris* which protected them before the advent of the ice. The area of glacial action extended over nearly all of New England and over the whole area of the northern tier of States. It extended southward almost to the Ohio river, and in some places crossed it. The effect of the ice age in producing and modifying our soil must never be forgotten in a study of soil genesis. It is not a part of our purpose here to study the causes which produced the age of ice. Even a brief reference to some of the more probable ones might be entirely out of place. Before the glacial period it is certain that a tropical climate extended almost, if not quite, to the North Pole. The fossil remains of tropical plants and animals which have been found in high northern latitudes are abundant proofs of this fact.

In the opinion of Sterry Hunt,^[27] rock decay has taken place largely in preglacial and pretertiary times. The decay of crystalline rocks is a process of great antiquity. It is also a universal phenomenon. The fact that the rocks of the southern part of this country seem to be covered with a deeper *débris* than those further north is probably due to the mechanical translation of the eroded particles towards the south. The decay and softening of the material were processes necessarily preceding the erosion by aqueous and glacial action.

It is possible that a climate may have existed in the earlier geologic ages more favorable to the decay of rocks than that of the present time.

41. Progress of Decay as Affected by Latitude.—Extensive investigations carried on along the Atlantic side of the country show wide differences in the rate of decay in the same kind of rocks in different latitudes. In general, the progress of decay is more marked toward the south. The same fact is observed in the great interior valleys of the country;

at least, everywhere except in the arid and semi-arid regions. Wherever there is a deficiency of water the processes of decay have been arrested. Where the rock strata have been displaced from a horizontal position the progress of decay has been more rapid. This is easily understood. The percolation of water is more easy as the displacement approaches a vertical position.

A most remarkable example of this is seen in the rocks of North Carolina. ^[28] A kind of rock known as trap is found in layers called dikes in the Newark system of rocks in that State. These dikes have been so completely displaced from the horizontal position they at first occupied as to have an almost vertical dip. The edges thus exposed vary from a few feet to nearly 100 feet in thickness. The trap rock in those localities is composed almost exclusively of the mineral dolerite, which is so hard and elastic in a fresh state as to ring like a piece of metal when struck with a hammer. In building a railroad through this region these dikes were in some places uncovered to a depth of forty feet and more. At this depth they were found completely decomposed and with no indications of having reached the lower limit of disintegration. The original hard bluish dolerite has been transformed into a yellowish clay-like mass that can be molded in the fingers and cut like putty. Similar geologic formations in New Jersey and further North do not exhibit anything like so great a degree of decomposition, thus illustrating in a marked degree the fact that freezing weather for a part of the year is a protection against rock decay. The ice of winter at least protects the rocks from surface infiltration, although it can not stop the subterranean solution which must go on continuously.

Other things being equal, therefore, it appears that as the region of winter frost is passed the decay of the rocks has been more rapid than in the North, because the chief disintegrating forces act more constantly.

42. The Solvent Action of Water.—The water of springs and wells is not pure. It contains in solution mineral matters and often a trace of organic matter. The organic matter comes from contact with vegetable matter and other organic materials near the surface of the earth. The mineral matter is derived from the solvent action of the water and its contents on the soil and rocks.

The expressions “hard” and “soft” applied to water indicate that it has much or little mineral matter in suspension, as the case may be. When

surface and spring waters are collected into streams and rivers they still contain in solution the greater part of the mineral matters which they at first carried.

When well or spring waters have more than forty grains of mineral matter per gallon they are not suitable for drinking waters. Mineral waters, so called, are those which carry large quantities of mineral matter, or which contain certain comparatively rare mineral substances which are valued for their medicinal effects.

The analysis of spring, well, or river waters will always give some indication of the character of the rocks over which they have passed.^[29] The vast quantities of mineral matters carried into oceans and seas are gradually deposited as the water is evaporated. If, however, these matters be very soluble, such as common salt, sulfate of magnesia, etc., they become concentrated, as is seen with common salt in sea waters. In small bodies of waters, such as inland seas, which have no outlet, this concentration may proceed to a much greater extent than in the ocean. As an instance of this, it may be noted that the waters of the Dead Sea and Great Salt Lake are impregnated to a far greater degree with soluble salts than the water of the ocean. The solvent action of water on rocks is greatly increased by the traces of organic or carbonic acids which it may contain. When surface water comes in contact with vegetable matter it may become partially charged with the organic acids which the growing vegetable may contain or decaying vegetable matter produce. Such acids coming in contact with limestone under pressure will set free carbon dioxid. Water charged with carbon dioxid acts vigorously on limestone and other mineral aggregates. If such solutions penetrate deeply below the surface of the earth their activity as solvents may be greatly increased by the higher temperature to which they are subjected. Hence, all these forces combine to disintegrate the rocks, and through such agencies vast deposits of original and secondary rocks have been completely decomposed.

The gradual passing of the firm rock into an arable soil is beautifully shown in Fig. 10, a print from a negative taken by Mr. Geo. P. Merrill, near Washington, D. C.

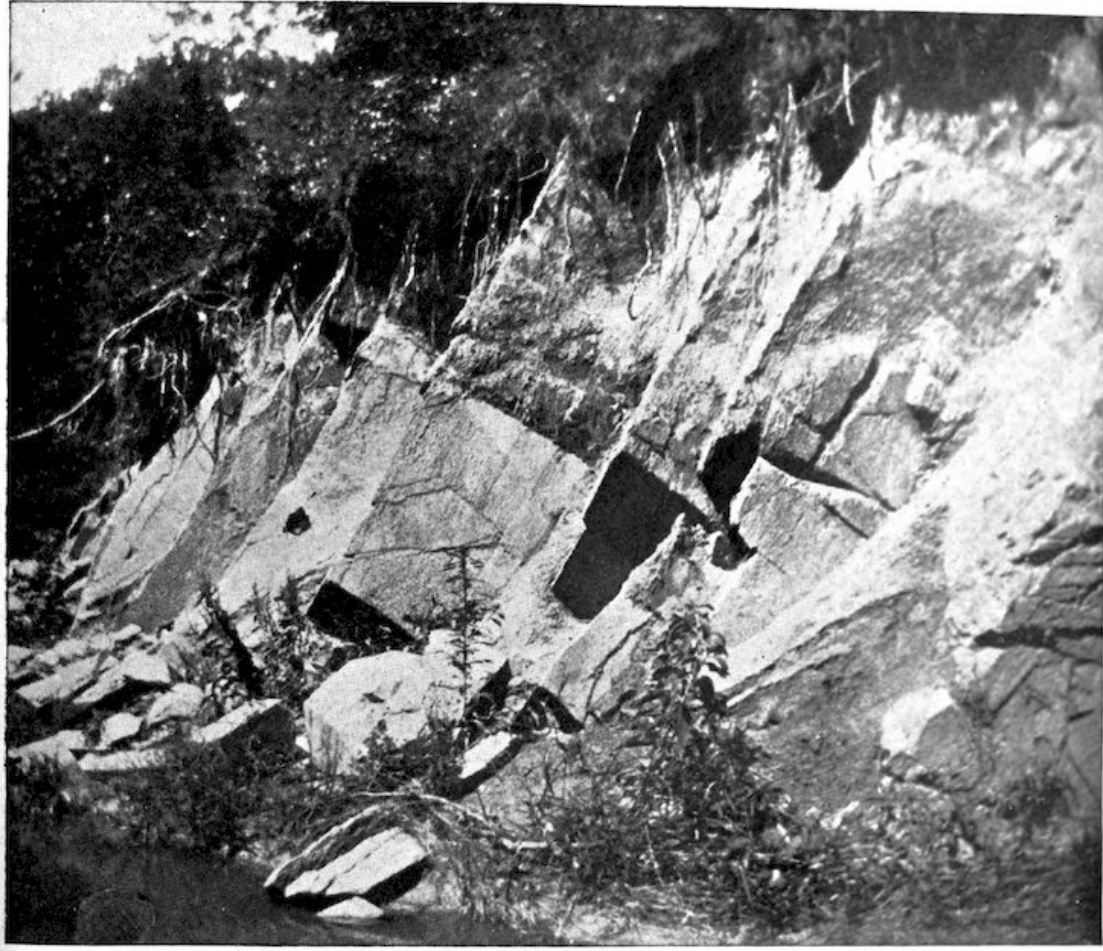


FIGURE 10.

View on the Broad Branch of Rock Creek, Washington, D. C.

The fresh but badly decomposed granitic rock is shown passing upward into material more and more decomposed until it becomes sufficiently pulverulent and soluble to support plant life. The roots showing in the upper part of the picture formerly penetrated the decomposed rock, but have been exposed through quarrying operations. Photograph by George P. Merrill, 1891.

43. Action Of Vegetable Life.—The preliminary condition to vegetation is the formation of soil, but once started, vegetation aids greatly in the decomposition of rocks. Some forms of vegetation, as the lichens, have apparently the faculty of growing on the bare surface of rocks, but the higher order requires at least a little soil. The vegetation acts by shading the surface and thus rendering the action of water more effective, by

mechanically separating the rock particles by means of its penetrating roots and by the positive solvent action of the root juices. The rootlets of plants in contact with limestone or marble dissolve large portions of these substances, and while their action on more refractory rocks is slower, it must be of considerable importance. The organic matter introduced into the soil by vegetation also promotes decay still further both directly and by the formation of acids of the humic series. This matter also furnishes a considerable portion of carbon dioxide which is carried by the water to assist in its solvent action.

44. Action of Earth-Worms.—Of animal organisms those most active in the formation of soil are earth-worms. The work of earth-worms has been exhaustively studied by Darwin.^[30] The worms not only modify the soil by bringing to the surface portions of the subsoil, but also influence its physical state by making it more porous and pulverulent. According to Darwin the intestinal content of worms has an acid reaction, and this has an effect on those portions of the soil passing through their alimentary canal. The acids, which are formed in the upper part of the digestive canal are neutralized by the carbonate of lime secreted by the calciferous glands of the worms thus neutralizing the free acid and changing the reaction to alkaline in the lower part of the canal. There is a fair presumption that the acids of the worm are of a humic nature.

The worms further modify the composition of the soil by drawing leaves and other organic matter into their holes, and leaving therein a portion of such matter which is gradually converted into humus. Stockbridge^[31] gives a striking illustration of this process due to an experiment by von Hensen. Darwin estimates that about eleven tons of organic matter per acre are annually added to the soil in regions where worms abound. A considerable portion of the ammonia in the soil at any given time may also be due to the action of worms, as much as 0.18 per cent of this substance having been found in their excrement. It is probable that nearly the whole of the vegetable matter in the soil passes sooner or later through the alimentary canals of these ceaseless soil builders, and is converted into the form of humus.

45. Action of Bacteria.—The intimate relations which have been found to subsist between certain minute organisms and the chemical reactions

which take place in the soil is a sufficient excuse for noting the effect of other similar organisms in the formation of soils.

In addition to the usual forces active in decomposing rocks Müntz^[32] has described the effects of a nitrifying bacillus contributing to the same purpose.

According to him the bare rock usually furnishes a purely mineral environment where organisms cannot be developed unless they are able to draw their nourishment directly from the air. Some nitrifying organisms belong to this class. It has been shown that these bodies can be developed by absorbing from the ambient atmosphere carbonate of ammonia and vapors of alcohol, the presence of which has been determined in the air. According to the observations of Winogradsky, they assimilate even the carbon of the carbon dioxide just as vegetables do which contain chlorophyll. Thus even in the denuded rocks of high mountains the conditions for the development of all these inferior organisms exist. In examining the particles produced by attrition, it is easily established that they are uniformly covered by a layer of organic matter evidently formed by microscopic vegetations. Thus we see, in the very first products of attrition, appearing upon the rocky particles the characteristic element of vegetable soil, viz., humus, the proportion of which increases rapidly with the products of disaggregation collected at the foot of declivities until finally they become covered with chlorophylliferous plants.

In a similar manner the presence of nitrifying organisms has been noted upon rocky particles received in sterilized tubes, and cultivated in an appropriate environment where they soon produce nitrification. The naked rocks of the Alps, the Pyrenees, the Auvergnés and the Vosges, comprise mineralogical types of the most varied nature, viz., granite, porphyry, gneiss, mica-schist volcanic rocks and limestones and all these have shown themselves to be covered with the nitrifying ferment. It is known that below a certain temperature these organisms are not active; their action upon the rock is, therefore, limited to the summer period. During the cold season their life is suspended but they do not perish, inasmuch as they have been found living and ready to resume all their activity after an indefinite sleep on the ice of the glaciers where the temperature is never elevated above zero.

The nitrifying ferment is exercised on a much larger scale in the normal conditions of the lower levels where the rock is covered with earth. This activity is not limited to the mass of rock but is continued upon the fragments of the most diverse size scattered through the soil and it gradually reduces them to a state of fine particles. It is therefore a phenomenon of the widest extension.

The action of these micro-organisms according to Müntz is not confined to the surface but extends to the most interior particles of the rocky mass. Where, however, there is nothing of a nitrogenous nature, to nitrify such an organism must live in a state of suspended animation.

When the extreme minuteness of these phenomena is considered there may be a tendency to despise their importance, but their continuity and their generality in the opinion of Müntz place them among the geologic causes to which the crust of the earth owes a part of its actual physiognomy and which particularly have contributed to the formation of the deposits of the comminuted elements constituting arable soil.

The general action of nitrifying organisms in the soil, the nature of these bodies, and the method of isolating and identifying them will be fully discussed in another part of this work.

46. Action of the Air.—The air itself takes an active part in rock decay. Wherever rocks are exposed to decay, there air is found or, at least, the active principle of air, viz., oxygen. The air not only penetrates to a great depth in the earth, but is also carried to much greater depths by water which always holds a greater or less quantity of air in solution. The oxygen of the air is thus brought into intimate contact with the disintegrating materials and in a condition to assist wherever possible in the decomposing processes.

The oxygen acts vigorously on the lower oxids of iron, converting them into peroxids, and thus tends to produce decay.

There are other constituents of rocks which oxygen affects injuriously and thus helps to their final breaking up. It is true that, as a rule, the constituents of rocks are already oxidized to nearly as high a degree as possible, and on these constituents of course the air would have no effect. But on others, especially when helped by water with the other substances it carries in solution, the air may greatly help in the work of destruction.

In a general view, the action of the air in soil formation may be regarded as of secondary importance, and to depend chiefly on the oxidation of the lower to the higher basic forms. These processes, while they seem of little value, have, nevertheless, been of considerable importance in the production of that residue of rock disintegration which constitutes the soil.

47. Classification of Soils According to Deposition.—In regard to their deposition soils are divided into five classes:

1. Those which are formed from the decomposition of crystalline or sedimentary rocks or of unconsolidated sedimentary material *in situ*.
2. Those which have been moved by water from the place of their original formation and deposited by subsidence (bottom lands, alluvial soils, lacustrine deposits, etc.).
3. Those which have been deposited as débris from moving masses of ice (glacial drift).
4. Soils formed from volcanic ashes or from materials moved by the wind and deposited in low places or in hills or ridges.
5. Those formed chiefly from the decay of vegetable matter, (tule, peat, muck).

48. Qualities of Soils.—In respect of quality, soils have been arbitrarily divided into many kinds. Some of the more important of these divisions are as follows:

1. *Sand*. Soils consisting almost exclusively of sand.
2. *Sandy Loams*. Soils containing some humus and clay but an excess of sand.
3. *Loams*. Soils inclining neither to sand nor clay and containing some considerable portions of vegetable mold, being very pulverulent and easily broken up into loose and porous masses.
4. *Clays*. Stiff soils in which the silicate of alumina and other fine mineral particles are present in large quantity.
5. *Marls*. Deposits containing an unusual proportion of carbonate of lime, with often some potash or phosphoric acid resulting from the remains of sea-animals and plants.
6. *Alkaline*. Soils containing carbonate and sulfate of soda, or an excess of these alkaline and other soluble mineral substances.

7. *Adobe*. A fine grained porous earth of peculiar properties hereinafter described.

8. *Vegetable*. Soils containing much vegetable débris in an advanced state of decomposition. When such matter predominates or exists in large proportion in a soil the term tule, peat or muck is applied to it.

With the exception of numbers six, seven and eight these types of soil are so well-known as to require no further description for analytical purposes. The alkaline, adobe and vegetable soils on the contrary demand further study.

49. Alkaline Soils.—The importance of a more extended notice of this class of soils for analytical purposes is emphasized by their large extent in the United States.

Chiefly through the researches of Hilgard attention has been called to the true character of these soils which are found throughout a large part of the Western United States and which are known by the common name of alkali. The following description of the origin of these soils is compiled chiefly from Hilgard's papers on this subject. Wherever the rain-fall is scanty, and especially where the rains do not come at any one time with sufficient force to thoroughly saturate the soil and carry down through the subsoil and off through the drainage waters the alkali contained therein, favorable conditions exist for the production of the alkaline soil mentioned above. The peculiar characteristic of this soil is the efflorescence which occurs upon its surface and which is due to the raising of soluble salts in the soil by the water rising through capillary attraction and evaporating from the surface, leaving the salts as an efflorescence.

Soils which contain a large amount of alkali are usually very rich in mineral plant food, and if the excess of soluble salt could be removed, these lands under favorable conditions of moisture would produce large crops.

The formation of the alkali may be briefly described as follows: By the decomposition of the native rocks, certain salts soluble in water are formed. These salts in the present matter are chiefly sodium and potassium sulfates, chlorids and carbonates. The salts of potash together with those of lime are more tenaciously held by the soil than the soluble salts of soda, and the result of this natural affinity of the soil for soluble potash, lime and magnesian salts is seen in the formation at the surface of the earth, by the process of evaporation above described, of a crust of alkaline material

which is chiefly composed of the soluble salts of soda. In countries which have a sufficient amount of rain-fall, these soluble salts are carried away either by the surface drainage or by the percolation of water through the soil, and the sodium chlorid is accumulated in this way in the waters of the ocean. But where a sufficient amount of rain-fall does not occur, these soluble salts carried down by each shower only to a certain depth rise again on the evaporation of the water, reinforced by any additional soluble material which may be found in the soil itself. The three most important ingredients of the alkali of the lands referred to are sodium chlorid, sulfate, and carbonate. When the latter salt, namely, sodium carbonate, is present in predominant quantity, it gives rise to what is popularly known as black alkali. This black color is due to the dark colored solution which sodium carbonate makes with the organic matters or humus of the soil. The black alkali is far more injurious to growing vegetation than the white alkali composed chiefly of sodium sulfate and chlorid.

This black alkali has been very successfully treated by Hilgard^[33] by the application of gypsum which reacting with the sodium carbonate produces calcium carbonate and sodium sulfate, thus converting the black into the white alkali and adding an ingredient in the shape of lime carbonate to stiff soils which tends to make them more pulverulent and easy of tillage.

This method of treatment, however, as can be easily seen, is only palliative, the whole amount of the alkaline substances being still left in the soil, only in a less injurious form.

The only perfect remedy for alkaline soils, as has been pointed out by Hilgard, is in the introduction of underdrainage in connection with irrigation. The partial irrigation of alkaline soils, affording enough moisture to carry the alkali down to and perhaps partially through the subsoil, can produce only a temporary alleviation of the difficulties produced by the alkali. Subsequent evaporation may thus increase the amount of surface incrustation. For this reason in many cases the practice of irrigation without underdrainage may completely ruin an otherwise fertile soil by slowly increasing the amount of alkali in the soil by the total amount of the alkaline material added in the waters of irrigation.

As Hilgard has pointed out, if a soil can be practically freed from alkali by underdrainage connected with a thorough saturation by irrigation, it may be centuries before the alkali will accumulate in that soil again when

ordinary irrigation only is practiced. It may thus become possible to reclaim large extents of alkaline soil little by little by treating them with an excess of irrigation water in connection with thorough underdrainage. The composition of the alkali on the surface of the soil due to the causes above set forth is thoroughly illustrated by the analyses of Hilgard and Weber, which follow:

TABLE SHOWING COMPOSITION OF ALKALI SALTS IN SAN JOAQUIN VALLEY.

FRESNO COUNTY.											
	Sections 13 and 24 T. 14 S. R. 19 E., 4 miles S. W. from Fresno.					Miss Austin's Ranch, Central Colony.		N.W. Cor. N ½ Sec. 20 T. 14 S. R. 21 E.		Easton.	Emigr'nt Ditch.
	Alkali soil, 1888.	Alkali Spot, 1889.				Surface soil, No. 1.	Surface soil, No. 2.	Surface soil.	Surface soil.		
		1 inch surface.	18 inches bel. surface.	26 inches bel. surface.	42 inches hardpan.						
Soluble salts in 100 parts soil		0.76	0.20	0.18	0.16	3.54	1.90	1.20	2.69		
Potassium sulfate						small	moderate				
D Potassium nitrate							small				
Potassium carbonate (Saleratus)											
Sodium sulfate (Glauber's salt)	large	small	small	very slight	very slight	large	large	much	moderate	large	
Sodium carbonate (Sal-soda)	very slight	large	small	large	large	small	chiefly	small	small		chiefly
Sodium chlorid (Common salt)	chiefly	moderate	chiefly	moderate	moderate	chiefly	large	chiefly	chiefly	large	little
D Sodium phosphate											
Calcium sulfate (Gypsum)	moderate	small	very slight	very slight	very slight	small	moderate	small	small	much	
Magnesium sulfate (Epsom salt)							small			much	
Organic matter											

[D](#). Very generally present, but not always in quantities sufficient for determination.

TULARE COUNTY.										
	Goshen	Peopl's Ditch	Near Lake Tulare	Visalia	Lemoore	Tulare Exp't Station				
	Surf'ce soil	Alkali crust	Surf'ce soil	Surf'ce soil	Alkali crust	Alkali crust				
Soluble salts in 100 parts soil	1.40		0.83	1.26						
Potassium sulfate						small				
[E]Potassium nitrate						small				
Potassium carbonate (Saleratus)				18.80						
Sodium sulfate (Glauber's salt)	44.24	1.22	31.30[F]	13.4	chiefly	32.8				
Sodium carbonate (Sal-soda)	32.98	88.09	18.2	45.3		36.16				
Sodium chlorid (Common salt)	16.74	1.00		4.4	little	31.16				
[E]Sodium phosphate	1.97		0.22	10.4						
Calcium sulfate (Gypsum)					little					
Magnesium sulfate (Epsom salt)				8.1	moderate					
Organic matter	1.59	9.21	7.5			5.37				

KERN COUNTY.										
	Alkali crusts from the Smyrna artesian belt. Townships 25 and 26 R. 23 E. W. S. W. from Delano, S. P. R. R.								Summer.	Kern Island
	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	Alkali crust	Alkali crust
Soluble salts in 100 parts soil										4.72
Potassium sulfate										
[E]Potassium nitrate										
Potassium carbonate (Saleratus)										
Sodium sulfate (Glauber's salt)	small	moderate	moderate	moderate	large	small	small	small	19.20	70.61
Sodium carbonate (Sal-soda)										14.82
Sodium chlorid (Common salt)	large	moderate	large	large	small	large	small	large	37.14	4.13
[E]Sodium phosphate										
Calcium sulfate (Gypsum)	small	small	small	small	small	small	small	small	0.96	0.08
Magnesium sulfate (Epsom salt)	small	small	small	small	small	small	small	small	18.31	
Organic matter									20.87	

[F](#). Common and Glauber's salts.

50. Adobe Soils.—In many parts of the arid regions of this country which can be recovered for agricultural purposes by irrigation the soil has peculiar characteristics.

The name adobe as commonly used applies to both the sundried bricks of the arid regions of the West and Southwest, and to the materials of which they are composed. The material is described by Russell^[34] as a fine grained porous earth, varying in color through many shades of gray and yellow, which crumbles between the fingers, but separates most readily in a vertical direction. The coherency of the material is so great that vertical scarps will stand for many years without forming a noticeable talus slope.

Distribution.—The area over which adobe forms a large part of the surface has not been accurately mapped, but enough is known to indicate that it is essentially co-extensive with the more arid portions of this country. In a very general way it may be considered as being limited to the region in which the mean annual rain-fall is less than twenty inches. It forms the surface over large portions of Colorado, New Mexico, Western Texas, Arizona, Southern California, Nevada, Utah, Southern Oregon, Southern Idaho, and Wyoming. Adobe occurs also in Mexico and may there reach a greater development than in the United States, but observations concerning it south of the Rio Grande are wanting.

In the United States it occurs from near the sea-level in Arizona, and even below the sea-level in Southern California, up to an elevation of at least six or eight thousand feet, along the eastern border of the Rocky Mountains, and in the elevated valleys of New Mexico, Colorado, and Wyoming. It occupies depressions of all sizes up to valleys having an area of hundreds of square miles. Although occurring throughout the arid region, it can be studied to best advantage in the drainless and lakeless basins in Nevada, Utah, and Arizona.

Composition.—When examined under the microscope, the adobe is seen to be composed of irregular, unsorted flakes and grains, principally quartz, but fragments of other minerals are also present. An exhaustive microscopic study has not been made, but the samples examined from widely-separated localities were very similar. The principal characteristics observed were the extreme angularity of the particles composing the deposit and the undecomposed condition of the various minerals entering into its composition. It is to be inferred from this that the material was not exposed even to a very moderate degree of friction, and had not undergone subaerial decay before being deposited. Adobe collected, at typical localities is so fine in texture that no grit can be felt when it is rubbed between the fingers; in other instances it contains angular rock fragments of appreciable size.

The composition of the material is illustrated by the following analyses:

ANALYSES OF ADOBE.

BY L. G. EAKINS.

Constituents.	No. 1. Sante Fe, New Mexico.	No. 2. Fort Wingate, New Mexico.	No. 3. Humboldt, Nevada.	No. 4. Salt Lake City, Utah.
SiO ₂	66.69	26.67	44.64	19.24
Al ₂ O ₃	14.16	0.91	13.19	3.26
Fe ₂ O ₃	4.38	0.64	5.12	1.09
MnO	0.09	trace	0.13	trace
CaO	2.49	36.40	13.91	38.94
MgO	1.28	0.51	2.96	2.75
K ₂ O	1.21	trace	1.71	trace
Na ₂ O	0.67	trace	0.59	trace
CO ₂	0.77	25.84	8.55	29.57
P ₂ O ₅	0.29	0.75	0.94	0.23
SO ₃	0.41	0.82	0.64	0.53
Cl	0.34	0.07	0.14	0.11
H ₂ O	4.94	2.26	3.84	1.67
Organic matter	2.00	5.10	3.43	2.96
	99.72	99.97	99.84	100.35

51. Vegetable Soils.—The heavy soils whose origin has been described are essentially of a mineral nature. The quantity of organic matter in such soils may vary from a mere trace to a few per cent, but they never lose their mineral predominance. When a soil on the other hand is composed almost exclusively of vegetable mold it belongs to quite another type. Such soils are called tule, peat or muck. In this country there are thousands of acres of peat or muck soils; the largest contiguous deposits being found in Southern Florida. The origin of these soils is easily understood. Whenever rank vegetation grows in such a location as to secure for the organic matter formed a slow decay there is a tendency to the accumulation of vegetable mold in shallow water or on marshy ground and where conditions are favorable to such accumulations. In Florida the muck soils have been accumulated about the margins of lakes. During the rainy season the marshes bordering these are partly covered with water, but the vegetation is

very luxuriant. The water protects the vegetable matter from being destroyed by fire. It therefore accumulates from year to year and is gradually compacted into quite a uniform mass of vegetable mold.

The composition of the muck is illustrated in the following table which shows the character of the layers at one, two and three feet in depth:^[35]

	Carbon.		Hydrogen.		Nitrogen.		Volatile matter.	
1 foot	57.67	per cent	4.48	per cent	2.24	per cent	90.60	per cent
2 feet	47.07	„	5.15	„	1.40	„	72.00	„
3 feet	8.52	„	0.53	„	0.31	„	15.00	„

In this sample, No. 3, the muck was only three feet deep, resting on pure sand. As the bottom of the deposit is approached the admixture of sand becomes greater and the percentage of organic matter less.

No reliable estimate of the time which has been required to form these deposits can be given, but in the Okeechobee region in Florida the deposit of vegetable mold in some places exceeds ten feet in depth.

The purest muck or peat soils contain only small quantities of potash and phosphoric acid, and especially is this true of the Florida mucks which have been formed of vegetable growth containing very little mineral matter.

It is not at all probable that the flora now growing on any particular area of virgin peat contains all the plants that have contributed to its formation. The principal vegetable growths now going to make up the muck soils of Florida are the following:

Common names.	Botanical names.
Saw grass	Cladium effusum
Yellow pond lily	Nymphaea flava
Maiden cane grass	Panicum Curtisii
Alligator Wampee	Pontederia cordata
Sedge	Cyperus species
Fern brake	Osmunda „
Mallow	Malva „
Broom sedge	Andropogon „
Arrow weed	Sagittaria „

The above are only the plants growing in the greatest profusion and do not include all which are now contributing to increase the store of vegetable débris.

52. Humus.—The active principle of vegetable mold is called humus, a term used to designate in general the products of the decomposition of vegetable matter as they are found in soils. In peat and muck are found a mixture of humus with undecomposed or partially decomposed vegetation.

According to Kostytchhoff^[36] vegetable matter decays under the influence of molds and bacteria. Molds alone produce the dark colored matters which give soils rich in vegetable matter, their color. One chief characteristic of humus is its richness in nitrogen. Black Russian soil contains from 4 to 6.65 per cent of nitrogen. This soil is estimated to contain sixty million organisms per gram and much of the nitrogen which it holds must be in the form of proteids. The first development in decaying vegetable matter is of bacteria and there is a tendency of the decaying matter to become acid. This causes a decay of the bacteria and the ammonia produced by this neutralizes the acid. The various kinds of mold grow when the reaction becomes neutral. Afterwards the bacteria and the molds develop together. This statement of Kostytchhoff is not a very satisfactory explanation of even our limited knowledge of the decomposition of organic matters in the soil. Ammonia and ammonia salts are formed not by the decay of some forms of bacteria but by the activities of other forms. Warington found that in nitrification there were three distinct forms of bacteria concerned in the final products of ammonia, nitrites, and nitrates. Humus always contains easily decomposable matter and consequently the rate of decay at any observed periods is nearly the same. In humus which is produced above the water-level Kostytchhoff states that all trace of the vegetable structure is destroyed by the leaves being gnawed and passed through the bodies of caterpillars and wire-worms. Under the water-level the vegetable structure is preserved and peat results. The decay of humus is most rapid in drained and open soils. For this reason the presence of clay in a soil promotes the accumulation of humus. Inferior organisms are the means of diffusing organic matter through the soil. The mycelia of fungi grow on a dead root for instance, ramify laterally and thus carry organic matter outward and succeeding organisms extend this action and the soil becomes darkened in proportion. Humic acid in black soil is almost exclusively in combination with lime.

A more common view of the difference between the formation of humus above and below the water-level is that above the water-level there is a very free access of air and even the harder parts of the leaf skeleton can be oxidized through the agency of bacteria, while under the water-level there is a very limited supply of air and this oxidation cannot proceed as rapidly. The harder parts of the leaf skeleton are preserved, and from the freer access of air humus is oxidized more readily in drained and open soils, and accumulates in clay soils where there is less circulation of air.

The real composition of humus is a matter which has never been definitely determined. Composed of many different but closely related substances it has been difficult to isolate and determine them.

Stockbridge^[37] gives the following composition of the bodies which form the larger part of humus:

Ulmic and Ulmic Acid.

Carbon	67.1	per cent	Corresponding to $C_{40}H_{28}O_{12} + H_2O$.
Hydrogen	4.2	„	
Oxygen	8.7	„	

Humin and Humic Acid.

Carbon	64.4	per cent	Corresponding to $C_{21}H_{24}O_{12} + 3H_2O$
Hydrogen	4.3	„	
Oxygen	31.3	„	

Crenic Acid.

Carbon	44.0	per cent	Corresponding to $C_{12}H_{12}O_8?$
Hydrogen	5.5	„	
Nitrogen	3.9	„	
Oxygen	46.6	„	

Apocrenic Acid.

Carbon	34.4	per cent	Corresponding to $C_{24}H_{24}O_{12}?$
Hydrogen	3.5	„	
Nitrogen	3.0	„	
Oxygen	39.1	„	

He further states that there are, aside from these humus compounds, others still less known and the action of which is not yet understood; among them xylic acid, $C_{24}H_{30}O_{17}$, saccharic acid, $C_{14}H_{18}O_{11}$, glucinic acid, $C_{12}H_{22}O_{12}$, besides a brown humus acid containing carbon, 65.8 per cent, and hydrogen, 6.25 per cent, and a black humus acid yielding carbon, 71.5 per cent, and hydrogen, 5.8 per cent.

According to Mulder humic acid has the following composition, $C_{60}H_{54}O_{27}$, while Thenard^[38] ascribes to it the formula, $C_{24}H_{10}O_{10}$.

At the present time we can only regard the various forms of humus bodies as mixtures of many substances mostly of an acid nature and resulting from a gradual decomposition of organic matter under conditions which partially exclude free access of oxygen.

For analytical purposes it is only necessary to separate these bodies by the best approved processes. A further knowledge of their composition can then be derived by determining the percentages of carbon dioxide and water which they yield on combustion.

53. Soil and Subsoil.—Many subdivisions have been made of the above varieties of soil, but they have little value for analytical purposes. For convenience in description for agricultural purposes, the soil, however, is further divided into soil and subsoil. In this sense the soil comprises that portion of the surface of the ground, usually from four to nine inches deep, containing most of the organic remains of plants and animals and in which air circulates more or less freely for the proper humification of the organic matter, which usually gives a darker color to the soil than to the subsoil. The subsoil proper lies below this, and has usually more characteristic properties, especially in respect of color and texture, as it has been less influenced by artificial conditions of cultivation and the remains of vegetation.

The subsoil extends to an indefinite depth and is limited usually by deposits of undecomposed or partly decomposed rock matter, or by layers of clay, sand or gravel.

Inasmuch, however, as the influence of the subsoil on growing crops is of little importance below the depth of eighteen inches the analysis of samples from a greater depth has more of a geologic than agricultural value.

Hilgard regards as subsoil whatever lies beneath the line of change, or below the minimum depth of six inches. But should the change of color

occur at a greater depth than twelve inches, the soil specimen should nevertheless be taken to the depth of twelve inches only, which is the limit of ordinary tillage; then another specimen from that depth down to the line of change, and then the subsoil specimens beneath that line. The depth to which the last should be taken will depend upon circumstances. It is always desirable to know what constitutes the foundation of a soil to the depth of three feet at least, since the question of drainage, resistance to drought, etc., will depend essentially upon the nature of the substratum. But in ordinary cases ten or twelve inches of subsoil will be sufficient. The sample should be taken in other respects precisely like that of the surface soil, while that of the material underlying this subsoil may be taken with less exactness, perhaps at some ditch or other easily accessible point, and should not be broken up like the other specimens.

In the method of soil sampling adopted by the Royal Agricultural College of England, the soil is regarded as that portion of the surface of the ground which is reached by ordinary tillage operations, generally being from six to nine inches deep; the subsoil is that portion which is ordinarily not touched in plowing.

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PART SECOND.

TAKING SAMPLES OF SOIL FOR ANALYSIS.

54. General Principles.—It would be unwise to attempt to give any single method of taking soil samples as the only one to be practiced in all circumstances. In the methods which follow it is believed will be found directions for every probable case. The particular method to be followed will in each case have to be determined by circumstances.

The sole object in taking a sample of soil should be to have it representative of the type of soils to which it belongs. Every precaution should be observed to have each sample measure up to that standard.

The physical and chemical analyses of soils are long and tedious processes and are entirely too costly to be applied to samples which represent nothing but themselves.

The particular place selected for taking the samples as well as the method employed are also largely determined by the point of view of the investigations. The collection of samples to illustrate the geologic or mineralogical relations of soils is quite a different matter from gathering portions to represent their agricultural possibilities. In a given area the sum of plant food in the soil would only be determined by the analyses of samples from that particular field, while samples illustrating geologic relations could or should be taken at widely distant points. Again the chemist is content with a sample of a few grams in weight while the physicist would require a much larger quantity. Much popular ignorance exists respecting the importance of the collection of soil samples. As an illustration of this may be cited a recent instance in which a sample of soil was received by the author with a request for a complete analysis and a

statement of the kinds of crops it was suited to grow. No data relating to the locality in which the sample was taken accompanied this request. The sample itself, which weighed a little less than 3.6 grams, was not a soil at all in an agricultural sense but a highly ferruginous sand.

The collector of samples who understands the purpose for which he is working will find among the approved methods which follow some one or some combination of methods, by means of which his work can be made successful. In these cases it is the collector rather than the method on which reliance must be placed to secure properly representative samples.

55. General Directions for Sampling.—The locality having been selected which presents as nearly as possible the mean composition of the field a square hole is dug with a sharp spade to the depth of eighteen inches. The walls of this hole should be smooth and perpendicular. The soil to the depth of six to nine inches is then removed from the sides of the hole in a slice about four inches thick; or the sample of soil may be taken to the depth indicated by a change of color. Any particles which fall into the bottom of the hole are carefully collected and added to the parts adhering to the spade. The whole is thrown into a suitable vessel for removal to the laboratory. The sample of soil having been thus secured, the subsoil is taken in the same way. To insure uniformity in the samples, it is well to take several of them from the same field. Where more than one sample is taken it is advisable to mix all the sub-samples in the field, remove large sticks, stones, roots, etc., and take a general sample of from three to five kilograms. The character of the *débris*, etc., removed should be carefully noted.

It is sometimes desirable to take samples of the subsoil to a greater depth than eighteen inches. A post-hole auger or large wood auger will be found very useful for this purpose. It is rarely necessary to take samples of subsoil to a greater depth than six feet. In taking samples the geologic formation and the general topography of the field should be noted, also the character of the previous crops, kind and amount of fertilizers employed, character of drainage and any other data of a nature to give a more accurate idea of the forces which have determined the physical and chemical properties of the sample.

56. Method Of Hilgard.—Hilgard^[39] recommends that samples should not be taken indiscriminately from any locality you may chance to be

interested in, but that you should consider what are the two or three chief varieties of soil which, with their intermixtures, make up the cultivable area of your region, and carefully sample these first of all.

As a rule, and whenever possible, samples should be taken only from spots that have not been cultivated, or are otherwise likely to have been changed from their original condition of virgin soils and not from ground frequently trodden over such as roadsides, cattle paths, or small pastures, squirrel holes, stumps, or even the foot of trees, or spots that have been washed by rains or streams, so as to have experienced a noticeable change, and not be a fair representative of their kind. He further suggests that the normal vegetation, trees, herbs, grass, etc., should be carefully observed and recorded, and spots showing unusual growth be avoided whether in kind or quality, as such are likely to have received some animal manure or other outside addition.

Specimens should be taken from more than one spot judged to be a fair representative of the soil intended to be examined as an additional guarantee of a fair average.

After selecting a proper spot pull up the plants growing on it, and scrape off the surface lightly with a sharp tool to remove half-decayed vegetable matter not forming part of the soil. Dig a vertical hole, like a post-hole, at least 20 inches deep. Scrape the sides clean so as to see at what depth the change of tint occurs which marks the downward limit of the surface soil, and record it. Take at least half a bushel of the earth above this limit, and on a cloth (jute bagging should not be used for this purpose, as its fibers, etc., become intermixed with the soil) or paper break it up and mix thoroughly, and put up at least a pint of it in a sack or package for examination. This specimen will, ordinarily, constitute the soil. Should the change of color occur at a less depth than six inches the fact should be noted, but the specimen taken to that depth nevertheless, since it is the least to which rational cultures can be supposed to reach.

In case the difference in the character of a shallow surface soil and its subsoil should be unusually great, as may be the case in tule or other alluvial lands or in rocky districts, a separate example of that surface soil should be taken, besides the one to the depth of six inches.

Specimens of salty or alkali soils should, as a rule, be taken only toward the end of the dry season, when they will contain the maximum amount of

the injurious ingredients which it may be necessary to neutralize.

Whatever lies beneath the line of change, or below the minimum depth of six inches, will constitute the subsoil. Should the change of color occur at a greater depth than twelve inches the soil specimen should nevertheless be taken to the depth of twelve inches only, which is the limit of ordinary tillage; then another specimen from that depth down to the line of change, and then the subsoil specimen beneath that line.

Hilgard justly calls attention to the fact that all peculiarities of the soil and subsoil, their behavior in wet and dry seasons, their location, position and every circumstance in fact, which can throw any light on their agricultural qualities or peculiarities should be carefully noted and the notes sent with the samples. Unless accompanied by such information, samples can not ordinarily be considered as justifying the amount of labor involved in their examination.

57. Whitney^[40] suggests that a geologic map of the region to be sampled should always be at hand and that all samples should be rejected from spots showing local discrepancies, washings or other disturbances.

The kind of analyses to which a sample is to be subjected also largely determines the method to be pursued in selecting it: For instance, a sample to be used for determining the size of the particles therein, may be taken in quite a different manner from that designed only for the determination of moisture.

58. In the directions collated by Richards^[41] and which have been largely followed by the correspondents of the Department of Agriculture, it is recommended to select in a field, four or five places, at least, per acre, taking care that these places have an homogeneous aspect, and represent as far as possible the general character of the whole ground. If the field, however, present notable differences, either in regard to its aspect or its fertility, the samples gathered from the different parts must be kept separate.

The sampling of arable soil should be made only after the raising of the crop and before it has received any new manure. In other soils the sample should be taken only from spots that have not been cultivated.

59. In the method of soil sampling adopted by the German Experiment Stations^[42] it is directed that the samples of soil should be taken according to the extent of the surface to be sampled, in three, five, nine, twelve or more places at equal distances from each other. They should be taken in

perpendicular sections to the depth turned by the plow; and for some studies of the subsoil to a depth of sixty to ninety centimeters. The single samples can be either examined separately or carefully mixed and an average portion of the mixture taken.

60. Method of the Official French Commission.—The official French commission^[43] emphasizes the fact that the sample of soil taken for analysis, should represent a layer of equal thickness through the total depth of its arable part. An analysis of the subsoil taken in the same way, will often be useful to complete the data of the soil study.

First of all, according to this authority, it is necessary to determine the point of view from which the sample is to be taken. If the object is a general study, having for its aim the determination of the general composition of the soils of a definite geologic formation, the sample should be taken in such a way that the different characteristics of the soil alone should enter into consideration without paying any attention to its accidental components, which have been determined by local causes, such as are produced by continued high cultivation, the application of abundant fertilizers, or the practice of a particular line of agriculture. The samples of soil therefore, with such an object in view, should be taken on parts of the earth which are beyond the reach of the causes mentioned above and which tend to modify the nature of the primitive soil. In such a case it is the soil which has not been modified, or better still, virgin soil, such as is found in the woodlands and prairies, which should be taken for a sample, choosing those places in which the geologic formation is most perfectly characterized. In such a case a soil taken in one spot corresponding to the conditions before mentioned, would be the best for the purposes in view. The sample would thus represent a true type to be studied, not one of a mean composition got by taking samples from different localities and mixing them into a homogeneous parcel. This last method of proceeding could introduce into the sample earth modified by culture or by influences purely accidental. However, it would be wise, in a region characterized by the same geologic formation, to take a certain number of samples in different localities, and examine singly each one of them in order to be assured that there is a uniformity of composition in the whole of the soils.

If, on the contrary, it is the purpose of the investigation to furnish information to the cultivator concerning the fields which are worked, it is

necessary to approach the problem from a different point of view. In this case the earth which is under cultivation should be first of all considered with all the modifications which nature causes or practical culture has caused, in it. But it often happens that upon the same farm the natural soil is variable, caused either by the washings from the adjacent soils, by the accumulation at certain points of deposits formed from standing water, or from other reasons. In such a case it would be necessary to take samples from every part of the field which exhibited any variation from the general type, in order to get a complete mean sample of the whole. It is necessary to be on guard against making a mixture of these different lots which would neither represent the different soils constituting the farm nor their mean composition. It would be better to examine each of these samples alone and then from those parts which appear to have a similar composition, to take a general sample for the mean analysis.

Most often it is necessary to confine our studies to the really important part of the farm the composition of which would have a practical interest. The aspect of the spontaneous vegetation in such a case, will often serve as a guide to determine the parts of the farms which are similar in nature. The sample should represent the arable layer, properly so-called, that is, that part of it which is stirred by the agricultural implements in use and in which the root system of the plant takes its greatest development and which is the true reservoir of the fertilizing materials.

When a trench is dug in the soil it is easy to distinguish the arable layer from the subsoil. In the first place, its color is different, generally being modified by vegetable débris which forms the supply of humus. The depth of the arable layer is variable, but it is most frequently between 200 and 300 millimeters. In the analysis the depth and layers should be indicated since the chemical composition of the earth varies according as the sample is taken to a greater or less depth. As an example of this it may be said that the quantity of nitrogen decreases in general in proportion as the depth of the layer is increased. The sample, therefore, should be limited exactly to the arable layer of soil.

61. Caldwell^[44] advises that according to the purpose of the analysis samples be taken:

a, from one or from several spots in the field, in order to subject each sample to a separate analysis; or

b, for an average representation of the soil of the whole field; in this case, several portions of earth are taken from points distributed in a regular manner over the field, all of which are most carefully mixed together, and 4–6 kilograms of the mixture, free from any large stones, are preserved as the average sample.

An excavation in the soil 30–50 centimeters deep, or through to the subsoil, and 30–50 centimeters square, with one side as nearly vertical as possible is made and a slice taken from this side of uniform thickness throughout, weighing 4–5 kilograms. If the subsoil is to be examined, a sample of it should be taken out in the same manner as directed for the upper soil, to the depth of about 60 centimeters.

If the character of the soil varies materially in different parts of the field, samples from several spots should be analyzed separately.

A small portion of the sample should be put at once in a well-stoppered bottle; the remainder may be allowed to become air-dried, by exposing it in a thin layer, in summer, to the common temperature in the shade, or, in winter, to that of a warm room, or a moderately warm drying-chamber, heated to 30°–40°; in either case it should be carefully protected from dust.

At the time of taking the sample of the soil, observations should be made in regard to the following points:

- a*. The geognostic origin of the soil.
- b*. The nature of the underlying strata, to the depth of 1–2 meters, if practicable.
- c*. The meteorology of the locality, by consulting meteorological records, if possible; otherwise, by the general opinion of the neighborhood; in this connection, the height of the locality above the level of the sea should be noted also.
- d*. The management and rotation of crops in previous years.
- e*. The character of the customary manuring.
- f*. The amount of the crops removed in the preceding year, and, if possible, the average amount of each of the more important crops yielded by the field.
- g*. The practical judgment of neighboring farmers in regard to the field.

Caldwel's method is practically identical with that of Wolff^[45] which was one of the earliest of the systematic schemes for taking soil samples.

62. Wahnschaffe^[46] insists on rather a fuller preliminary statement to accompany soil samples but gives essentially the method of Wolff with some unimportant variations which add little to the value of the process.

63. Method Of Peligot.—According to Peligot^[47] the taking of samples of soil of which the physical and chemical properties are to be determined is a delicate operation.

These samples should represent as nearly as possible both the good and bad qualities of a soil.

In the field selected are chosen a certain number of places at least four or five per hectare. The spots selected should have a homogeneous appearance—resembling as nearly as possible the general aspect of the field.

By means of a spade a few kilograms of earth are removed to the depth of the subsoil being careful to include in the sample no accidental detritus which the upper part of the soil especially may contain.

The samples should be taken immediately after the crop is harvested and before any fresh fertilizer is applied. The samples are carefully mixed and placed in a glass bottle or flask.

The sample of subsoil is obtained in the same manner. If the field presents notable differences in surface or fertility all the samples taken should be examined separately.

64. Method of Whitney.^[48]—An ordinary wood auger, 2½ inches in diameter is so arranged as to admit of additions to the stem to enable the operator to take samples at different depths. It may be fitted with a short piece of gas pipe for a handle and the several pieces of which it is composed may be taken apart and carried in a knapsack.

In taking a soil sample the boring is continued until a change in color shows that the subsoil has been reached. The auger cuts a very clean sample save in excessively sandy soil. After the soil sample is secured the hole is cleaned out and the sample of subsoil taken by the same instrument. The soil is conveniently preserved in heavy cloth bags of which the usual size is 6 by 8½ inches. Where larger samples are required the size of the bag is correspondingly increased. Each bag is to be tagged or labeled to correspond with the entry in the note book.

Samples to determine the amount of empty space in a soil are taken as follows: The sampler is a piece of brass cylinder about nine inches long and about 1½ inches in diameter. A piece of clock spring is soldered in one end and sharpened to give a good cutting edge. This arrangement permits the sample to pass into the cylinder without much friction. The area enclosed by the clock spring is accurately determined and a mark is placed in the cylinder six inches from the cutting edge. The cylinder is driven into the soil to a depth of six inches, a steel cap being used to prevent the hammer from injuring the cylinder. The earth is next removed from about the cylinder with a trowel, and the separated cylinder of earth is cut smoothly off by a sharp knife and removed together with its brass envelope. The sample is taken to the laboratory in a cloth bag, dried and weighed.

65. Taking Samples for Moisture Determination.—A number of brass tubes is provided nine inches long and ¾ inch in diameter and with a mark six inches from the bottom.

The tube is pushed down into the soil to the mark and the sample of soil removed with the tube. There is but little danger of the sample dropping out of the tube even in sandy soils. When the tube is withdrawn each end is capped with a rubber finger tip making a perfectly air tight joint. The tubes containing the samples can be kept several days with no fear of losing moisture. This method is especially useful in having samples taken by observers in different localities who can enclose the tubes in a cloth sack and send them to the laboratory by mail daily or at stated intervals. A tube of the size given holds about fifty grams of soil.

66. Taking Samples to Determine the Permeability of Soil to Water or Air.—Whitney^[49] determines the permeability of soil or subsoil to water or air in the following manner:

An excavation two feet square and eighteen inches deep is made in the soil. On one side of this hole the sample of soil or subsoil is taken by means of a narrow saw blade and a sharp carving knife. The sample of soil taken should be two inches square and 3½ to 4 inches long.

It is placed in a brass cylinder three inches long and 3¼ inches in diameter. The open space in the cylinder is filled with paraffin heated just to its melting point. As the paraffin cools the upper surface should be kept stirred to prevent the mass when set from receding from the square column of soil. Care must be taken to keep the paraffin from the ends of the soil

columns and these should be left, as far as possible in their natural condition.

The rate of percolation of the water may be determined at the time the sample is taken. For this purpose an additional section of brass tube two inches deep is secured to the one holding the sample by a rubber band. An iron rod is driven into the earth carrying a retort stand ring supporting a funnel filled with fine gravel. The lower end of the soil column in the brass cylinder is placed on this gravel. Water is next carefully poured upon the top of the sample of soil being careful not to disturb the surface. The surface of the sample may be protected with a little fine sand. The water should be poured on the paraffin thus affording an additional protection to the soil surface. When the water begins to drop from the funnel a graduated glass is set under it and the time required for a given volume to pass through under an initial pressure of two inches is noted. The volume required represents one inch in depth over the four square inches of soil surface, viz.: four cubic inches.

67. Sampling of Soil for Staple Crops.—Some variations from the usual methods are recommended by Whitney when the samples are taken from fields growing staple crops.

The immediate object of the work, for which these samples are desired, is to make a thorough study of the physical and chemical properties of a number of typical soils adapted to the different staple crops, such as grass, wheat, truck, and the different types of tobacco. They should be taken for a careful study of the texture of the soils, the relative amount and arrangement of sand and clay, the relation of the soils to moisture and heat, and the ease with which they can maintain a proper water supply for the different staple crops under existing climatic and cultural conditions. The ultimate object of such a study is to see how these conditions can be changed so as to make the soils more productive, and make them yield a better quality of crop, or to change the conditions in other soils, which differ from these, so that the culture of the different staple crops can be extended over wider areas by improved methods of cultivation and manuring.

The soil selected for sampling for these investigations should be typical, should represent fairly well a considerable area of land. It should represent either the very best type of land for the staple crop or crops of the locality,

or the very poorest lands for these same crops. Both of these extremes are desired for contrast. For example, if the staple crop of the locality is wheat or a certain type of tobacco, select the soil best adapted to this staple crop, and another soil, if possible, in the same locality, representing considerable area of land upon which this staple crop cannot be successfully grown on account of the inferior yield, quality, or the time of ripening of the crop. The soil sampled should be, or should recently have been, under actual cultivation in the crop or crops best adapted to it, so that the real agricultural value of the land can be accurately known.

The samples should be taken inside the field, some distance away from fences, roads, or trees. If there are plants growing in the field, the sample should be taken about midway between two plants. The samples should be taken where they will typify fairly well the average soil of the field and of the large area of land which they are to represent.

The samples are taken in some one of the ways described herein. Each sample should be carefully labelled at the time of taking. The following blank form will be found convenient for this purpose:

LABORATORY No.:	Locality:
No. of sack:	
Date:	Description: (virgin or cultivated).
	(a) Natural herbage:
Collector:	(b) Crops best adapted to land (grass, wheat, tobacco, truck, barren).
Depth of sample: (Soil or Subsoil?) ... in. to ... inches.	Geologic formation:

68. Method of the Royal Agricultural Society.^[50]—Have a wooden box made, six inches long and wide, and from nine to twelve inches deep, according to the depth of soil and subsoil in the field. At one of the selected places mark out a space of twelve inches square; dig around it in a slanting direction a trench, so as to leave undisturbed a block of soil, with its subsoil, from nine to twelve inches deep; trim this block to make it fit into the wooden box, invert the open box over it, press down firmly, then pass a spade under the box and lift it up and gently turn it over.

In the case of very light, sandy, and porous soils, the wooden box may be at once inverted over the soil and forced down by pressure, and then dug out.

Proceed in the same way for collecting the samples from all the selected places in the field, taking care that the subsoil is not mixed with the surface soil. The former should be sampled separately.

In preparing the plot for the gathering of the sample, take care to have it lightly scraped so as to remove any débris which may be accidentally found there.

The different samples thus procured are emptied on a clean, boarded surface, and thoroughly mixed, so as to incorporate the different samples of the same field together. The heap is then divided into four divisions, and the opposite quarters are put aside, taking care to leave the two remaining ones undisturbed; these are thoroughly mixed together, the heap divided into quarters, and the opposite ones taken away as before. This operation of mixing, dividing into quarters and taking away the opposite quarter is continued until a sample is left weighing about ten or twelve pounds.

Thus is obtained the average sample of the soil. Of course where only a single sample is taken from the field this method of quartering is not resorted to, but the bottom of the box is nailed directly on and sent to the laboratory, where the soil is to be analyzed.

69. Grandeau^[51] suggests that in taking soil samples there are two cases to be considered; first a homogeneous soil and second, a soil variable in its appearance and composition. First, if the soil is homogeneous, being of the same geologic formation it will be sufficient to take a mean sample in accordance with the following directions:

The field is first divided by diagonals or by transverse lines the direction of which need not be fixed in advance but as inspection of the form and configuration of the field may indicate. In the ordinary conditions, of homogeneity (marly, granite, argillaceous or silicious soils) it will be sufficient to select about five points per hectare from which the samples are to be taken. These points having been determined the surface is cleaned in such a way as to remove from it the detritus which may accidentally cover it; such as dry leaves, fragments of wood, foreign bodies, etc. The surface having been prepared, (five to six square meters) a hole is dug four-tenths of a meter long and as wide as the spade employed. The sides should be as

nearly vertical as possible. As to depth it varies with the usage of the country in regard to tillage. The layer of arable earth is what in effect properly constitutes the soil. It ought not to be mixed with any fragments of the subsoil. When the hole is properly cleaned the samples are secured with a spade from the sides of the excavations. About five kilograms are taken. The soil is placed in a proper receptacle as it is removed from the hole.

This operation is repeated on as many points as may be necessary to obtain a mean sample of the soil of the whole field.

All the samples are now collected on a table sufficiently large, and intimately mixed together. Two samples, each of about five kilograms, are then taken from the mixed material. One sample is immediately placed in bottles and carefully stoppered and sealed; the other is dried in the sun or on the hearth of a furnace. When sufficiently dry the second sample is also placed in bottles and well stoppered. While mixing the samples, pebbles, etc., of the size of a nut and larger are removed, the weight of the rejected matter being determined. The nature of the pebbles should also be noted; whether silicates, limestone, etc.

The sample of subsoil is taken in exactly the same manner, using the same holes from which the samples of soil were taken. The nature, the arrangement and the appearance of the strata will indicate the depth to which the subsoil should be taken. In general, a depth equal to that of the sample of soil will be sufficient. The depth to which the roots of cultivated plants reach is also a good indication in taking a sample of the subsoil. In forests the sample of subsoil should be taken from four to five-tenths of a meter below the surface.

If the soil in respect of its geologic formation, its fertility or its physical aspect presents great differences, special samples should be taken in each part in accordance with the directions given above.

70. Method of the Official Agricultural Chemists.—In the directions given by the Association of Official Agricultural Chemists^[52] it is stated that the soil selected should be as far as possible in its natural condition, not modified by recent applications of manure, or changed by the transporting action of water or wind. Surface accumulations of decaying leaves, etc., should be removed before taking the sample.

To eliminate accidental variations in the soil, select specimens from five or six places in the field which seem to be fair averages of the soil, remove

two or three pounds of the soil, taking it down to the depth of nine or ten inches^[53] so as to include the whole depth. Mix these soils intimately, remove any stones, shake out all roots and foreign matter, and dry the soil until it-becomes friable.^[54] Break down any lumps in a mortar with a wooden pestle, but avoid pulverizing any mineral fragments; pass eight to ten pounds of the soil through a sieve, having circular perforations one twenty-fifth of an inch in diameter, rejecting all pebbles and materials too coarse to pass through the sieve. Once more mix intimately the sifted soil. Expose in thin layers in a warm room till thoroughly air dry (or dry it in an air-bath at a temperature of 40°), place six to eight pounds in a clean bottle, with label of locality and date, and cork the bottle containing the soil, for analysis.

The soil is rapidly dried to arrest nitrification; it is not heated above 40° lest there should be dissipation of ammonia compounds, or a change in solubility. The normal limit to which the soil may be heated in place by the sun's rays should not be exceeded in preparing a sample for an agricultural chemical analysis.

The relative amount of fragments too coarse to pass through the sieve should be made a matter of record. They are soil material, but not yet soil, so far as agricultural purposes are concerned.

71. Method of Lawes.—In a late method of sampling proposed by Sir J. B. Lawes^[55] a steel frame ten by twelve inches, and nine inches deep open at top and bottom is driven into the earth until its upper edge is level with the surface of the soil. All above-ground vegetation is then cut off as closely as possible with scissors. The soil within the frame is then removed exactly to the depth of the frame, and immediately weighed. It is then partially dried, and mechanically separated by a series of sieves, all visible vegetable matter being at the same time picked out. The stones and roots and the remaining soil are thus separated, and the determinations of dry matter, nitrogen, etc., are made in the separated soil after being finely powdered. The loss of water at each stage of preparation and on drying the samples as analyzed is also carefully determined. This method, which requires the soil to be taken to an arbitrary depth of nine inches, could not be used when samples of strictly arable soil are to be taken.

72. In taking a sample by the French commission^[56] method it is necessary to remove from the surface, the living and dead vegetation which

covers the soil. With a spade a square hole is then dug to the depth of about 500 millimeters; in other words, to a depth considerably exceeding that of the arable layer. Afterwards on each of the four sides of the hole there is removed by the spade, a prismatic layer of the arable portion of a thickness equal to its depth. The samples thus obtained are united together and carefully mixed for the purpose of forming a sample for analysis. If there are large stones they are removed by hand and their proportion by weight determined.

In all cases it would prove useful to take a sample of the subsoil which is far from playing a secondary rôle. The rootlets bury themselves deeply in it and seek there a part of their nourishment. The subsoil, therefore, furnishes an important addition to the alimentation of the plants. For taking a sample of the subsoil a ditch is dug of sufficient depth, say one meter, and the arable soil carefully removed from the top portion. Afterwards pieces are taken from the four sides of the hole at variable depths, which should always be indicated, and which should extend in general, from six to eight-tenths of a meter below the arable soil since it is demonstrated that the roots of nearly all plants go at least to this depth. The analysis of the subsoil, however, is less important than that of the soil, properly so-called, because the agronomist does not act directly upon it and takes no thought of modifying it and enriching it as he does the layer of arable soil. But the composition of the subsoil is a source of information capable of explaining certain cultural results and capable sometimes, of leading to the correct way of improving the soil, as in cases where the subsoil can be advantageously mixed with the superficial layer.

73. Wolff^[57] suggests that a hole thirty centimeters square be dug perpendicularly and a section from one of the sides taken for the sample. To the depth of thirty centimeters the sample shall be taken as soil and to the additional depth of thirty centimeters as subsoil. The thickness of the section taken may vary according to the quantity of the sample desired. For analytical purposes, five kilograms will usually be sufficient. When culture experiments are also contemplated a larger quantity will be required.

74. Method of Wahnschaffe.—The method of sampling advised by Wahnschaffe^[58] is but little different from that of Wolff already mentioned.

A square sample hole is dug with a spade having its sides perpendicular to the horizon. The soil which is removed is thrown on a cloth and carefully

mixed. From the whole mass a convenient amount is next removed care being taken not to include any roots. In a similar manner it is directed to proceed for the sample of subsoil. At first the subsoil should be removed to a depth of two to three decimeters. The number and depth of subsequent samples will depend chiefly upon the character of the soil. Where samples are taken to the depth of two meters the use of a post-hole auger is recommended.

The samples taken should not be too small. In general from two to three kilograms should remain after all preliminary sampling is finished.

75. Method of König.—The directions given by König^[59] for taking soil samples are almost identical with those prescribed by Wahnschaffe and do not require any further illustration.

76. Special Instruments Employed in Taking Samples.—In general a sharp spade or post-hole auger is quite sufficient for all ordinary sampling but for certain special purposes other apparatus may be used.

The instrument which is used by King^[60] consists of a thin metal tube of a size and length suited to the special object in view, provided with a point which enables it to cut a core of soil smaller than the internal bore of the tube and at the same time make a hole in the ground larger than its outside diameter. Its construction is shown in figure 11, in which A B represent a soil tube intended to take samples down to a depth of four feet. A' is a cross-section of the cutting end of the tube, which is made by soldering a heavy tin collar, about three inches wide, to the outside of a large tube allowing its lower end to project about one-half an inch. Into this collar a second one is soldered with one edge projecting about one-quarter of an inch and the other abutting directly against the end of the soil tube. Still inside of this collar is a third about one-half an inch wide which projects beyond the second and forms the cutting edge of the instrument.

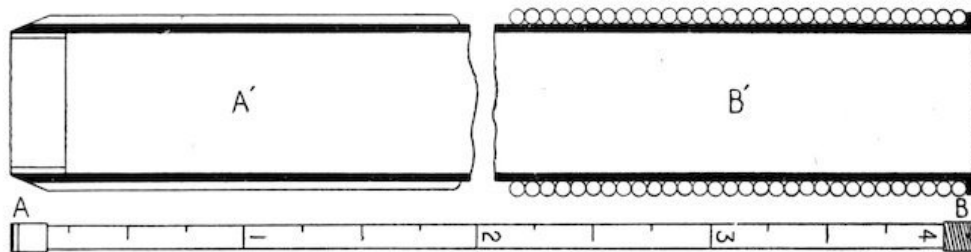


FIGURE 11.

The construction of the head of the tube is shown at B'. It is formed by turning a flange on the upper end of the tube and then wrapping it closely with thick wire for a distance of about three inches, the wire being securely fixed by soldering. The soil tube should be of as light weight as possible not to buckle when being forced into the ground, and the cutting edge thin. The brass tubing used by gas fitters in covering their pipes has been found very satisfactory for ordinary sampling. With a one inch soil tube four feet long it is possible to get a clear continuous sample of soil to that depth by simply forcing the tube into the ground with the hand and withdrawing it, or the sample may be taken in sections of any intermediate length. Later in the season when the soil becomes dryer it is necessary to use a heavy wooden mallet to force the tube, and this should be done with light blows.

The closeness with which it is possible to duplicate the samples in weight by this method will be seen below, where from each of four localities three samples were taken from the surface to a depth of four feet.

SHOWING VARIATIONS IN THE DRY WEIGHT OF TRIPPLICATE SAMPLES OF SOIL.

	A.	B.	C.
I. Surface to four feet	716.6 gms.	715.5 gms.	710.3 gms.
II. Surface to four feet	715.4 gms.	687.1 gms.	731.2 gms.
III. Surface to four feet	654.0 gms.	688.3 gms.	709.0 gms.
IV. Surface to four feet	714.0 gms.	687.8 gms.	719.3 gms.

These four series of samples were taken at the four corners of a square twelve feet on a side and serve to show how much samples may vary in that distance. The large difference shown in III, A is due to the fact that the soil tube penetrated a hole left by the decay of a rather large root as shown by the bark in the sample.

77. Auger for Taking Samples.—It has already been said that the ordinary auger used for boring fence post-holes may be used to advantage in taking soil samples. Large wood augers can also be used to advantage for the same purpose. For special purposes, however, other forms of augers may be used.

Norwacki and Borchardt^[61] have described a new auger for taking samples of soil for analytical purposes.

In figure 12, A, B and C show the general exterior and interior form of the instrument. The handle is hollow and made of iron gas pipe covered with leather. On the inside of this, in the middle, is fixed a wooden plug a,

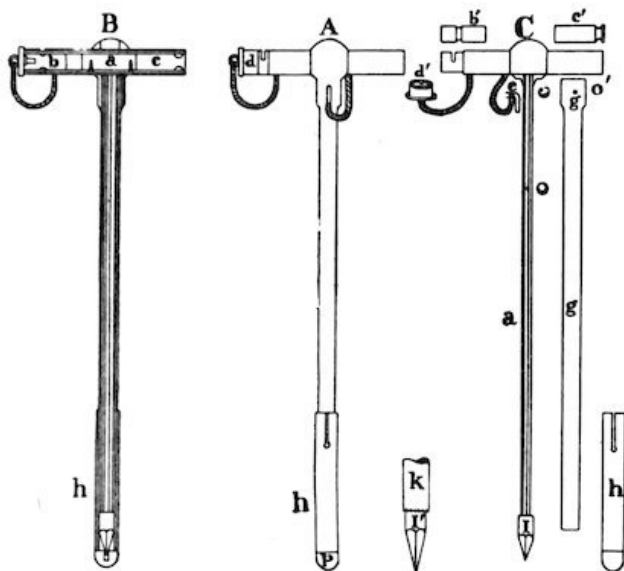


FIGURE 12.

which leaves two compartments, one in each end for holding the brass plug bb,' and the wicker lubricating wad cc.' The stem of the auger a, is heavy and made of eight-sided steel and the under end is strengthened with a heavy casting fitting into the auger guide g g. The end of the auger I I' is triangular and hardened. The auger guide g g, is made out of a single piece of drawn steel tubing. Above it is strengthened by a ring-shaped piece of iron or copper and its lower end is furnished with saw teeth as shown in K and is

hardened. The fixing key e, is bent in the form of a hook and can be passed through the two holes o o, of the auger stem and through the one hole o' in the strengthened part of the auger guide. It permits the auger guide to be fixed upon the auger stem in two different positions, higher and lower. On one end it is cut squarely across and on the other provided with a conical hole drilled into it. It fits on the one hand exactly in the auger guide and on the other loosely plays in the cavity of the handle at b, designed to hold it when not in use. The cap d' is made of heavy sheet brass and is fastened upon the end of the handle at c c' after the manner of a bayonet. The wicker cartridge is made of rolled and sewed wicker-work. At the upper end it is provided with a metallic button and before use it is saturated with paraffin oil. It fits on the one side firmly in the auger guide and on the other in the cavity, of the handle c where it is kept when not in use. The union h is made of a brass tube which below is closed with a piece of solid brass upon the inside of which a hole is bored. In this hole rests the end of the auger stem when the union is placed firmly upon the auger guide.

The auger is placed together as is shown in A B, the union h is taken off and it is driven with gentle blows, turning it back and forth, to the proper depth into the soil. After the key is loosened the auger is lifted high enough so that the second hole appears and then it is fixed in position by the key. Then the boring is continued, turning the auger to the right, by which the

auger, eating its way with its saw teeth, presses deeper into the ground and withdraws the material for analysis. After the auger guide has been filled through any desired length, say five to ten centimeters with the sample of soil, the whole auger is drawn out of the soil, the key removed, the auger stem withdrawn from the auger guide, the apparatus opened by turning the bayonet fastening of the stopper on the handle, the brass plug placed in the end and then with the smooth part forward, from above, it is allowed to fall into the auger guide until it reaches the soil. The auger stem is then put back, the point of it fitting into the hole of the plug and the sample of soil shoved out of the auger guide. The auger guide is again fixed on the auger stem by the key and then the apparatus is ready for a second operation. When the borings cease the wicker cartridge is drawn out of the handle and shoved, the soft end forward, from above, into the auger guide and the brass plug after it and pushed through with the auger stem. By this process the wicker cartridge gives up a sufficient amount of paraffin oil to completely grease the inside of the auger guide and to protect it from rust. After use the instrument should be cleaned on the outside by means of a cloth, the plug and wicker wad replaced in their proper positions, the cap fixed on the handle and the union on the point of the instrument.

The length of the whole apparatus may reach one meter or more; the internal diameter sixteen millimeters. The apparatus weighs with a length of one meter, together with all its belongings, about two kilograms. For the investigation of peat and muck soils as well as sand, instead of the steel auger guide one of brass or copper can be used. For this purpose the length of the apparatus may reach three to four meters.

In comparison with other apparatus which are used for taking samples, it appears without doubt that with the one just described a better and less mixed portion of the soil can be obtained at great depths. The apparatus is said to have many advantages over a similar one known as Fraenkel's, and is much more easy to clean. The advantages of the apparatus are said to be the following: The farmer with this piece of apparatus in a short time can go over his whole farm taking samples to the depth of ninety centimeters since a single boring does not take more than one minute. Geologists and others interested in the soil at greater depths can use an apparatus three to four meters in length and obtain unmixed samples from these lower depths. These are also interesting from a bacteriologic point of view. The entire apparatus is especially valuable for the investigation of the lower parts of

peat and muck soils. The apparatus has been tried in the collection of samples for the laboratory of the Department of Agriculture and is too complicated to be recommended for ordinary use. When however samples are to be taken at great depths as in peat soils it is highly satisfactory.

78. Soil Sampling depends for its success more on the judgment and knowledge of the collector than on the method employed and the apparatus used. One skilled in the art and having correct knowledge of the purpose of the work will be able to get a fair sample with a splinter or a jack-knife while another with the most elaborate outfit might fail entirely in collecting anything of representative value.

There are some special kinds of soil sampling, however, which cannot be left to the method of the individual and it is believed that with the descriptions given above nearly all purposes for which samples are desired may be served.

For the study of nitrifying organisms, however, special precautions are required and these will be noted in a more appropriate place.

In taking samples for moisture determinations the method of Whitney is recommended as the best. For the general physical and chemical analytical work the standard methods are all essentially the same. The principles laid down by Hilgard will be found a sufficient guide in most cases.

TREATMENT OF SAMPLE IN THE LABORATORY.

79. The Sample, or mixed sample, taken by one of the methods above described, is placed on a hard smooth board, broken up by gentle pressure into as fine particles as possible and all pieces of stone and gravel carefully removed and weighed; all roots, particles of vegetable matter, worms, etc., are also to be weighed and thrown out. This can be done very well by using a sieve of from one to two millimeter mesh. Care should be taken that the soil be made to pass through, which can be accomplished by subjecting the lumps to renewed pressure with a rubber-tipped pestle. In the above operation the soil should be dry enough to prevent sticking. The relative weights of the pebbles, roots, etc., and the soil should be determined.

80. Order of Preliminary Examination.—Hilgard^[62] commences the examination of a soil sample by washing about ten grams of it into a beaker with a water current of definite velocity, stirring meanwhile actively the part carried into the vessel. The residue not carried by the current is examined macro- and microscopically to determine the minerals which may be present, and the condition in which the fragments exist—whether sharp or rounded edges, etc.

This examination will give some general idea of the parent rocks from which the sample has been derived and of the distance the particles have been transported. Next follows the hand test, viz., rubbing the soil between the thumb and fingers first in the dry state and afterwards kneading it with water and observing its plasticity. Following this should come a test of the relations of the sample to water, viz., its capacity for absorbing and retaining moisture. Finally the separation of the soil into particles of definite hydraulic value and a chemical examination of the different classes of soil concludes the analytical work.

81. Air Drying.—The sifted soil should be thoroughly mixed and about one kilogram spread on paper and left for several days exposed in a room with free circulation of air and without artificial heat. The part of the sample to be used for the determination of nitrates should be dried more quickly as

described in another place. The sample is then placed in a clean, dry glass bottle, corked, sealed, and labeled. The label or note book should indicate the locality where the sample was taken, the kind of soil, the number of places sampled, and other information necessary to proper description and identification.

82. Caldwell^[63] directs that having taken the sample to the laboratory, the stones and larger pebbles should be separated from the finer parts by the hand, or by sifting with a very coarse sieve, and examined with reference to their mineralogical character, weight and size, making note, in this last respect, of the number that are as large as the fist or larger, the number as large as an egg, a walnut, hazel-nut, and pea, or give the percentage of each by weight.

Pulverize the air-dried soil in a mortar with a wooden pestle, and separate the fine earth by a sieve with meshes three millimeters wide; this sieve should have a tightly fitting cover of sheepskin stretched over a loop, and it should be covered in the same manner underneath, so that no dust can escape during the process of sifting.

Wash the pebbles and vegetable fibers remaining on the sieve with water, dry and weigh the residue; the water with which this gravel was washed should be evaporated to dryness at a temperature not exceeding 50° towards the close of the evaporation, and the residue mixed with what passed through the dry sieve.

The sifted fine earth is reserved for all the processes hereinafter described, and is kept in well-stoppered bottles, marked air-dried fine earth. The sieve mentioned above is too coarse for the more modern methods of analysis.

83. Wolff^[64] directs that the air-dried earth (in summer dried in thin layers at room temperature, in winter in ovens at 30° to 50°) be freed from all stones, the latter washed, dried, and weighed. The soil is next passed through a three millimeter mesh sieve, the residual pebbles and fiber washed, dried and weighed. The fine earth passing the sieve is used for all subsequent examinations. It is air-dried at moderate temperatures and preserved in stoppered glass vessels.

84. The French^[65] commission calls especial attention to the method of subsampling, and prescribes that the sample of earth which has been taken in the manner indicated, and of which the weight should be greater as the

material is less homogeneous should not be analyzed as a whole. It should be divided into two parts. The first includes the finer particles constituting the earth, properly so-called, with the elements which alone enter into play in vegetable nutrition and on which it is necessary to carry out the analysis. The second embraces the coarser particles to which only a superficial examination should be given and which may have a certain importance from a physical point of view but which cannot take any part from a chemical point of view, in the nutrition of plants. It is, however, useful to examine its mineralogical constitution and to look for the useful elements such as lime, potash, etc., which it may be able to furnish to the earth, and in proportion as it is decomposed, finer particles which may be useful in plant nutrition.

How are we to distinguish between the fine and coarse elements? All grades of fineness are observed in the soil, from the particles of hydrated silica so small that with the largest magnifying power of the microscope it is scarcely possible to distinguish them, up to grains of sand which are of palpable size and visible to the naked eye, and extending to pebbles of varying sizes. All intermediate stages are found between these and if it should be asked what is the precise limit at which it is necessary to stop in distinguishing the fine from the coarse elements of the soil, the answer is that this can only be determined by a common understanding among analysts. In general, it may be said, that the mark of distinction should be the separation which can be secured with a sieve having ten meshes per centimeter.

85. Loose Soils.—Having agreed upon a sieve of the above size, the process of separation in loose soils is as follows: The earth is exposed to the air and when the touch shows that it is sufficiently dry the conglomerated particles should be simply divided without breaking the rocky material which exists in a state of undivided fragments. There are some special precautions to be taken. Rubbing in a mortar must be forbidden since it reduces the earth to particles which are unnatural in size, by securing the breaking up of the fragments consisting of the *débris* of rocks. When it is possible the earth should be rubbed simply in the hand and after having separated that which passes the sieve, the large particles which have not passed should be again rubbed with the hand, until all the particles which can be loosened by this simple treatment have passed the sieve. The separation should be as complete as possible in order that a sample of the

particles passing the sieve should represent as nearly as possible, a correct sample of the fine particles of the soil.

In regard to the pebbles, they should be washed with water upon the sieve in order to carry through the last of the particles of earth adhering to them. They are then dried and their weight taken. The fine part of the earth is also weighed. On an aliquot part, say 100 grams, the moisture is determined and then by simple calculation the whole sample of the air-dry soil can be calculated to the dry state. The sample is then placed in a glass flask.

The pebbles are examined with a view of determining their mineralogical constitution; as for instance, on being touched with a little hydrochloric acid it can be determined whether or not they are carbonate of lime. The nature of the rock from which they have been derived is often to be determined by a simple inspection.

86. Compact Soils.—If the soils are not sufficiently loose to be treated as before described, it is necessary to have recourse to other means of division, which should not, however, be sufficiently energetic to reduce the rocky elements to fine particles. For this purpose the earth may be broken by means of a wooden mallet, striking it lightly and separating the fine elements from time to time by sifting. A wooden roller may also be used with a little pressure, for breaking up the particles or a roller made out of a large glass bottle. These methods will permit of a sufficiently fine division of the soil without breaking up any of the pebbles. Sometimes, however, a soil can not be broken up by such treatment. It is then necessary to have recourse to the following process: The soil is thoroughly moistened and afterwards rubbed up with water. The paste which is thus formed, is poured upon the sieve and washed with a stream of water until all the fine particles are removed. The wash water and the fine particles are left standing until the silt is thoroughly deposited when the supernatant water is poured off and the deposited moist earth is transferred into a large dish and dried on a sand or water-bath. In this way a firm paste is formed which can be worked up with the hand until rendered homogeneous and afterwards an aliquot portion be taken to determine moisture.

87. Method of Peligot.—The method recommended by Peligot^[66] for the preparatory treatment of the sample is essentially that already described. The sample is at first dried in the air and then in an oven at 120°. When dry

and friable 100 grams are placed in a mortar and rubbed with a wooden pestle. It is then passed through a sieve of ten meshes per centimeter. The largest particles which remain in the sieve should have about the dimensions of a pin's head. The stones are separated by hand. They should be shaken with water in order to detach any pulverulent particles adhering thereto. The turbid water resulting from this treatment is added to that which is used in separating the sand from the impalpable part of the soil.

88. Wahnschaffe prescribes^[67] in the further preparation of the sample for analysis that the coarse pieces up to the size of a walnut be separated in the field where the sample is taken and their relative weight and mineralogical character determined. The soil sample is then to be placed in linen or strong paper bags and carefully labelled. In order to avoid any danger of loss of label the description or number of the sample should be put on the cloth or paper directly.

The sample when brought to the laboratory should be spread out to dry, in a room free of dust. In the winter the room should be heated to the usual temperature. The air drying should continue until there is no sensible loss of weight. The samples then are to be placed in dry, glass-stoppered glass bottles where they are kept until ready for examination. This method of keeping the samples avoids contact with ammonia or acid fumes with which a laboratory is often contaminated.

89. The Swedish chemists^[68] direct that samples which are to be used for chemical examination in the manner described below, are most conveniently brought to such a condition of looseness and humidity that the soil feels moist when pressed between the fingers without, however, sticking to the skin. To prepare the sample in this manner, spread it in a large porcelain dish or on a glass plate in a place where it is not reached by the laboratory atmosphere; stir it frequently till it assumes the mentioned humidity (if the sample when sent is too dry, moisten it with distilled water till its condition is as indicated); then pulverize carefully between the fingers and finally sift through a sieve with five millimeter holes. In this way free the sample from stones, undecayed roots and similar parts of plants, pieces of wood, and other matter strange to the soil, which remain on the sieve; mix the sample carefully and put it into a glass bottle provided with a stopper well ground in; keep it in a cool place. Samples prepared in

this way will usually contain 20–30 per cent moisture; boggy soils 60–80 per cent and peat soils 50 per cent.

90. Petermann^[69] follows the method below in preparing samples of soil for analysis.

The soil is gently broken up by a soft pestle and all débris if of organic nature, cut fine with scissors. About 2500 grams of this soil are passed through a one millimeter mesh sieve. The organic débris is removed by forceps, washed free of adhering earth dried at 120° and weighed. The nature of the organic débris should be noted as carefully as possible.

The pebbles and mineral débris not passing the sieve are worked in a large quantity of water by decantation. They are also dried at 120° and weighed. This débris is examined mineralogically and thus some idea of the origin of the soil obtained.

91. The various methods for the preliminary treatment as practiced by the best authorities have been somewhat fully set forth in the foregoing résumé. The common object of all these procedures is to get the soil into a proper shape for further physical and chemical examination and to determine the comparative weights of foreign bodies contained therein.

The essential conditions to be observed are the proper sifting of the material and avoidance of mechanical comminution of the solid particles too large to pass the meshes of the sieve. If possible the material should be passed through a sieve of one millimeter mesh. In cases where this is impracticable a larger mesh may be used, but as small as will secure the necessary separation. Before final chemical analysis a half millimeter mesh sieve should be employed if the soil be of a nature which will permit its use. Over-heating of the sample should be avoided. Rapid drying is advisable when the samples are to be examined for nitrates.

The method recommended by the French commissions seems well adapted to the general treatment of samples, but the analyst must be guided by circumstances in any particular soil.

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PART THIRD.

PHYSICAL PROPERTIES OF SOILS.

92. The Soil as a Mass.—The soil constituted as indicated in the preceding pages, is now brought to the analyst for investigation. The properties with which he first becomes acquainted are those which impress his senses as mass characteristics. There is a perception of color, consistence, weight and other features which the soil possesses as a whole. The several constituents of the soil must first be considered as molecular and mole aggregates. In other words, the soil in its natural state is a mechanical mixture of particles which must first be considered as a whole. The physical properties of the soil, therefore, should engage the attention of the analyst before he proceeds to the investigation of the properties of its several constituents as classified by the relative size or hydraulic value of the particles of which they are composed, or to a chemical determination of the compounds or elements therein contained.

DETERMINATION OF PHYSICAL PROPERTIES.

93. Color.—The color of a soil depends chiefly upon the proportion of organic matter and iron compounds which it contains and the state of subdivision of its particles. When a soil contains a large amount of organic matter, especially when this organic matter is in an advanced state of decay, it assumes more or less a black color when moist. This black color is to be distinguished from the black alkali tint which is produced by the action of carbonate of soda on organic matter. The naturally black color of a soil containing a large amount of organic matter depends, however, either upon the action of mineral matters upon this organic matter, as in the case of the black alkali mentioned, or upon the blackish color of carbon resulting from the slow combustion of the organic matter during the period of decay.

The presence of a large amount of ferric oxid in soil gives the well-known red color so well-marked in the soils of southwestern Kentucky and other portions of the United States. The preponderance of sand in a soil tends to produce a light yellow or whitish tint, while certain kinds of clay have a bluish tint probably due to the presence of ferrous salts. The influence of the color of the soil upon the color of the vegetation is also well-marked, the black soils as a rule producing a much deeper green tint of foliage than the light colored soils. This effect should not be attributed to color alone for as a matter of fact highly colored soils are usually very close and very retentive of moisture, which is one reason, probably, for their not being more highly oxidized. Such soils will produce a more vigorous and ranker growth of vegetation, but it is the texture of the soil and the more moist condition which it maintains, rather than the color, which produce the deeper green tint of foliage.

The color of a soil is also used as an index of its fertility, the black and red soils being usually the most fertile.

It may be well to add here the probable reason as given by Whitney for this, *viz.*, that the deeper color shows that the oxids of iron and the organic compounds have less oxygen and indicate that the soils are quite retentive of moisture and rather tend to the exclusion of air, so that part of the oxygen of the iron compounds and of the organic matters has been used up in the

oxidation processes within the soil. It is known, for example, that wood oxidizes much more rapidly around a rusty nail than where it is simply exposed to the air, the iron oxid acting as a carrier between the oxygen of the air and the organic matter. In a sandy soil, on the contrary, where there is usually less moisture and much freer circulation of air, the iron compounds have more oxygen and usually have a light yellow color. If this sand is heated, however, with the exclusion of air, and especially in the presence of organic matters, part of this oxygen will be given off and there will be the same red color as in the heavier clay soils. It is frequently noticed, also, in compact clays that where air gains access through cracks or root-holes, the color is altogether modified.

94. Determination of Color.—There is no process which will give experimentally and accurately the color of a soil sample. The changes which the color of a soil undergoes in passing from a saturated to an anhydrous state are well-marked. The analyst will have to be content with giving as nearly as possible a description of the color of the sample when taken and the changes which it undergoes in air drying or on heating in a bath to 100°–110°, or in heating to redness with or without exclusion of the air. These changes in color will give some indication of the character of the organic and mineral matters present.

95. Odoriferous Matters in Soil.—It is known that the soil emits a peculiar odor which is not disagreeable except when it has been recently wet, for instance, after a short rain. Several attempts have been made to discover the nature of this odor. These researches have established the fact that the essential principle of this odor resides in an organic compound of a neutral nature of the aromatic family and which is carried by the vapor of water after the manner of a body possessing a feeble tension. The odor is penetrating, almost piquant, and analogous to that of camphorated and quite distinct from other known substances. In regard to the quantity of this substance, it is extremely minute and can be regarded as being only a few millionths of a per cent.

According to Berthelot and André^[70] this new principle is neither an acid nor an alkali nor even a normal aldehyd. It is, in a concentrated aqueous solution, precipitable by potassium carbonate with the production of a resinous substance. Heated with potash it develops a sharp odor similar to the aldehyde resin. It does not reduce the ammoniacal nitrate of silver.

Treated with potash and iodine it gives an abundant formation of iodoform, which, however, is a property common to a great number of substances. For the qualitative and quantitative estimation of the odoriferous matter the following process is employed:

About three kilograms of the soil are mixed with sand containing a small amount of carbonate of lime and some humic substance; after having freed it from all organic debris which is visible, it is placed in a glass alembic. The soil should contain from ten to twelve per cent of water at least. The alembic is placed in a sand bath and is kept at 60° for several hours. The water evaporated is condensed until about seventy-five cubic centimeters are distilled over. This distilled water is again rectified so as to obtain in all about twenty cubic centimeters. The odoriferous matter appears to be nearly all contained in this twenty cubic centimeters. The liquid thus obtained shows an alkaline reaction; it contains some ammonia and reduces ammoniacal silver nitrate. This last reaction is due to some pyridic alkali or analogue thereof, and is cause for it to be distilled anew with a trace of sulfuric acid which gives a neutral liquor deprived of all reducing action but which preserves the odor peculiar to the soil. The twenty cubic centimeters obtained as before are subjected to two additional distillations and in the final one only one cubic centimeter of liquid is distilled over.

The peculiar odor is intensified proportionately as the volume of the liquid is decreased. To this one cubic centimeter, is added some pure crystallized potassium carbonate. The liquor is immediately troubled and some hours are required for it to become clear again. Meanwhile there is formed upon its surface a resinous ring almost invisible, amounting at most to from ten to twenty milligrams of a matter which has not been identified with any known principle. The reactions described above, however, permit of its general character being known. This resinous matter contains the odoriferous principle, the composition of which is not yet definitely known.

96. Specific Gravity.—The density of a soil depends on its composition, the fineness of its particles and upon the packing which it has received. It has in other words an apparent and a real specific gravity. It is easy to see that a soil in good tilth would weigh less per cubic foot than one which had been pressed closely together, as in a road or well-pastured field. Ordinary soils in good tilth have an apparent specific gravity of about 1.2, and when entirely free from air, a real specific gravity of about 2.5. If the apparent

specific gravity of a soil sample were 1.2 and the air were removed, leaving a vacuum in the interstices of the soil, the apparent specific gravity would not be sensibly increased. The figure 1.2 is the apparent specific gravity of a mixture of soil material which is about $2\frac{1}{2}$ times heavier than water, and of an extremely small proportion by weight of air which is about 1000 times lighter than water. The figure 2.5 is about the true specific gravity of the real soil material, and shows that this material is about $2\frac{1}{2}$ times heavier than an equal volume of water.

The weights of a cubic foot of different kinds of soil as given by Schübler^[71] are as follows;

	Pounds.
Sand	110
Sand and clay	96
Common arable soil	80 to 90
Heavy clay	75
Vegetable mold	78
Peat	30 to 50

In general the specific gravity of soil decreases inversely as its content of humus.

97. Determination of Specific Gravity.—The ordinary method of proceeding to determine the true specific gravity is by means of a pyknometer. The pyknometer should have a capacity of from twenty-five to fifty cubic centimeters.

From ten to fifteen grams of earth dried to constant weight at 100° are taken, boiled for a time with a few cubic centimeters of water to remove air and poured into the pyknometer. All soil particles are washed out of the vessel in which the boiling took place into the pyknometer with freshly boiled distilled water and after cooling to the temperature at which the calibration took place, the pyknometer is filled with distilled water at the given temperature and weighed. If the soil contain materials soluble in water, alcohol of definitely known specific gravity may be employed and the number thus obtained calculated to a water basis.

The calculations when water is used are made as follows:

		Grams.
Weight of	pyknometer	13.4789
„ „	pyknometer full of distilled water at 20°	62.8934
„ „	water in pyknometer	49.4145
„ „	dry soil taken	10.0000
„ „	pyknometer + dry soil + filled with water at 20°	67.9834
„ „	soil and water	54.5045
„ „	water	44.5045
„ „	water displaced by ten grams soil	4.9100

Then specific gravity = $10.000 \div 4.9100 = 2.04$.

98. Specific Gravity of Undried Soils.—It is often desirable to determine the specific gravity of an undried portion of the soil. For this purpose a portion of the sample is dried at 100° to determine its percentage of moisture. The specific gravity is then determined on a ten gram sample of the undried soil as just given. The actual weight of soil taken is calculated from the percentage of moisture obtained in the first instance. In the case given if the percentage of moisture at 100° be ten then the actual weight of dry soil taken is nine grams. This number is therefore used in making the calculations. In all statements of specific gravity taken in the manner described the temperature at which the pyknometer is calibrated should be stated and all weighings where water is involved made at that degree.

99. Volume of Soil.—If it be desired to calculate the volume occupied by a soil it is easily done by dividing the weight of water displaced by the weight of one cubic centimeter of water of the temperature at which the determination took place.

In the case given one cubic centimeter of water at 20° weighs 0.998259. Then $4.9100 \div 0.998259 = 4.9186$ cubic centimeters = volume occupied by ten grams of dry soil excluding interstitial spaces between particles.

100. Volumetric Methods.—The water displaced by a given weight of soil may also be measured volumetrically by the method of Knop.^[72]

Place 200 grams of the soil in a flask of from three to five hundred cubic centimeters capacity. Add a measured quantity of water, and shake thoroughly to eliminate air, and fill up to the mark from a burette. The quantity of water required to complete the volume subtracted from the

number expressing the volume of the flask will give the volume of water displaced by the earth.

Another method consists in thoroughly shaking about thirty grams of the soil in a graduated cylinder with fifty cubic centimeters of water containing a little ammonium chlorid and after twenty-four hours recording the volume occupied by the whole. The increase in volume over fifty cubic centimeters shows the quantity of water displaced. This method may also be used to determine the volume occupied by a soil when saturated with water. The above methods are only to be used when approximately correct results are all that are desired.

101. Apparent Specific Gravity.—The apparent specific gravity of a soil is obtained by dividing its volume, interstitial spaces included, by the weight of an equal volume of water.

The real and apparent specific gravities of six samples of soil are given below.^[73]

Real specific gravity	2.5445,	2.6315,	2.6508,	2.6400,	2.7325,	2.6603
Apparent specific gravity of air-dried soil	1.0940,	1.1710,	1.3570,	1.2810,	1.4060,	1.2730
Apparent specific gravity of soil dried at 125°	1.0990,	1.1770,	1.3750,	1.2910,	1.4640,	1.2850

It is to be noted that in computing the apparent specific gravity of a soil dried at 125° the volume occupied by the water is assumed to occupy the same space as if it existed in a free state. The volume of this water is therefore to be subtracted from the contents of the flask before proceeding with the computations.

102. Determination of Apparent Specific Gravity.—Place in small quantity portions of the air-dried sample properly prepared, into an open glass cylinder, holding one liter, and about 170 millimeters high (if the height is exactly the mentioned one, the diameter of the cylinder will be 86.6 millimeters); pack the sample by striking the bottom of the cylinder hard against the palm of the hand after each new filling; close the cylinder thus filled by a glass plate and weigh on a balance sensitive to 0.1 gram; deduct the weight of the cylinder and glass plate, and the weight of one liter of soil in approximately similar conditions as it is found on the dry land prepared for cultivation, is thus ascertained. The weight of one liter of the

soil in grams multiplied by 2000 will give in kilograms the weight of the surface soil from a hectare (2.47 acres) of the field from which the sample is taken when the depth of this is calculated at twenty centimeters.^{[\[74\]](#)}

RELATION OF THE SOIL TO HEAT.

103. Sources of Soil Heat.—The heat of the soil comes from three sources, viz.: solar heat, as the sun's rays, heat of chemical and vital action within the soil, and the original heat of the earth's interior. The latter is sensibly a constant quantity, and of great value to plants. The heat of chemical and vital action is not great in amount except in a few special cases but is often, as in germination, of the greatest importance to plant growth. The sun, therefore, remains the greatest source of heat of practical importance in relation to the production of crops. Dark-colored soils, absorbing most and radiating the fewest rays, must attain the highest temperature. Schübler's classical researches on soil temperatures, show that there is at times a difference of over 7° in temperature between white and black soils, all other conditions being alike. Schübler's researches, being made on dry soils in the laboratory, do not, however, apply wholly to conditions in the field.

104. Influence of Specific Heat.—The heat which a soil receives and retains is largely due to the specific heat of the soil. The specific heat of a body is expressed by a number which shows the amount of heat necessary to raise a given weight of the body 1° of temperature, as compared with the amount necessary to raise the same weight of water 1° . The specific heat of the soil is usually between 0.20 and 0.25, while that of water taken as the standard is unity.

105. Influence of Moisture.—The moisture of the soil possesses great influence on the soil temperature, so much so that a dry, light-colored soil may attain a greater degree of warmth than a moist, dark-colored one. The action of water in reducing soil temperature is easily explained. In our latitude, we see the water in all its forms, solid, liquid, and gaseous, and we know that these forms are the direct result of temperature. The changing of water from the solid to the liquid or gaseous form is performed at the expense of heat; the more water evaporated from the soil the more heat must be used for the evaporation. Therefore, the more water contained in

the soil at any given time the lower must be its temperature during subsequent exposure to sun heat because of the greater evaporation. The experiments of Liebenberg, Pattner, Schübler and Dickenson have practically settled all the questions of soil temperatures. The radiation of heat from the soil, and the consequent cooling propensity of the latter, are directly proportional to the absorptive power of the soil. Two soils of like absorptive power towards heat possess, as a rule, equal radiating power.

In a general way, it can be said the greater the heating capacity and conductivity of a soil the more readily and rapidly does it give off its heat and become cooled.

106. Absorption of Solar Heat.—The quantity of heat absorbed from the sun by the earth is an important factor in the growth of vegetation. As has been established in the physics of heat, a black surface, other things being equal, will absorb a larger amount of heat than one of any other color; so, other things being equal in the physical and chemical composition of a soil, variations in the amount of organic matter producing greater or less black coloration will affect the heat absorption. Thus, black soils, in the conditions above mentioned, will absorb more heat than lighter colored soils. As a result, the vegetation in such soils gets an earlier start in the Spring and matures more rapidly. As an illustration of this it may be noted that the black prairie soils of Iowa produce uniformly crops of maize which are matured before the early frosts, while crops grown on lighter soils much farther South often suffer injury from that source.

DETERMINATION OF SPECIFIC HEAT.

107. General Principles.—The quantity of heat stored in any given weight of soil is capable of being measured and compared with the quantity stored in an equal weight of water at the same temperature. The ease, however, with which disturbing influences operate during the determination makes the manipulation somewhat difficult. The specific heat of the containing vessels must be carefully determined. Fortunately this has been done for most materials and the data thus obtained are recorded in standard works on physics. The material operated on must be protected from thermal influences from sources not controlled by the experiment and even the heat of the operator's body may often disturb the conduct of the work. The general conditions which should control the experiment as well as the details thereof are given in the following method which, however, the ingenious analyst may profitably simplify.

108. Method of Pfaundler.—The process of estimating the specific heat of soils by the method of mixture, is essentially that of Regnault and is described as follows by Pfaundler^[75].

The apparatus used is illustrated in Fig. 13.

A and A' show the heating apparatus. It consists of a vessel of sheet iron in which a test tube E is fixed by means of a cork. The test tube holds the soil whose specific heat is to be determined. The apparatus contains water, which is brought to the boiling point by means of a lamp, and the excess of steam is conducted away, as indicated in the figure, through one of the axes of the apparatus; the opposite axis is, of course, closed. It requires about thirty-five minutes boiling to bring the contents of the test tube to the temperature of the aqueous vapor. The exact temperature at which the water boils is determined by observing the barometer at the time and consulting a table of the boiling temperature of water at different barometric pressures.

The calorimeter is shown in the figures B and B'. It consists of a wooden box closed on one side by a glass plate G and on the other to the height F

by a small board on which a calorimeter of ordinary construction is placed. The cylinder of the calorimeter is seventy millimeters high and forty-seven millimeters in diameter.

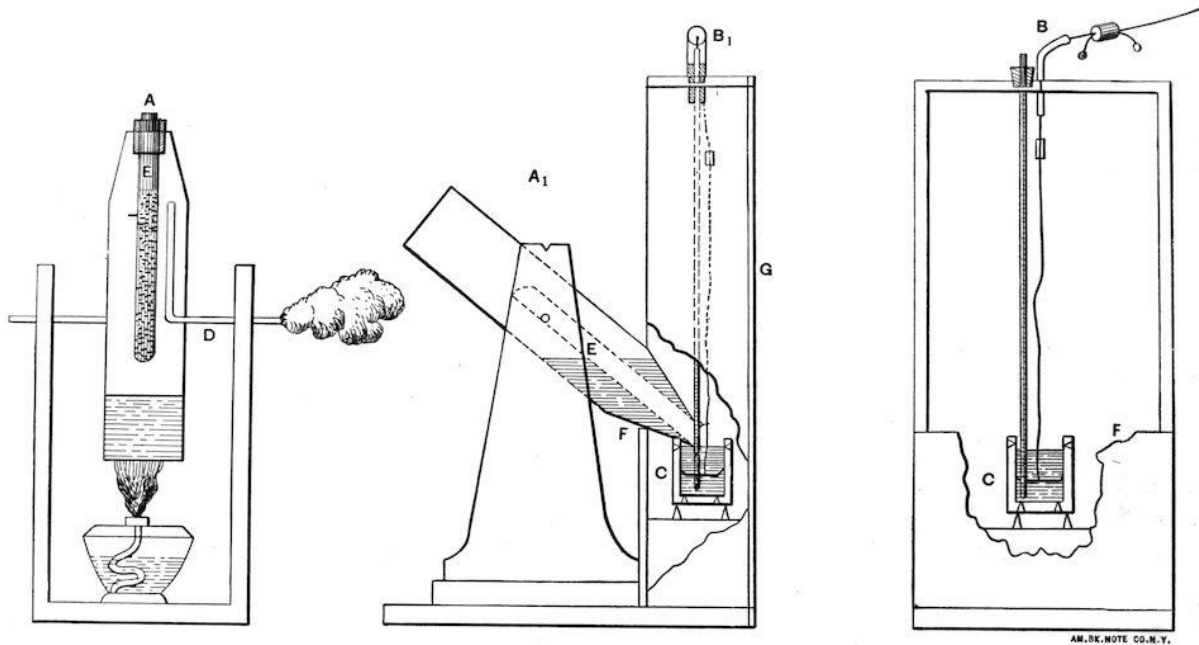


FIGURE 13.

REGNAULT'S APPARATUS FOR DETERMINING THE SPECIFIC HEAT OF SOILS.

This part of the apparatus is supported by triangular pieces of cork. A delicate thermometer is fastened to the top of the box of the calorimeter and the value of the degrees is so arranged that about twelve of them correspond to about one degree C. The scale of the instrument can be arbitrarily fixed and the temperature of any part of it determined by comparison with a delicately graduated thermometer.

Near the thermometer in the calorimeter is a stirrer made of a very thin copper disk with a bent rim. This stirrer is operated by means of a silk cord moved by appropriate machinery.

The reading of the thermometer is made through a glass plate and this should be protected from the heat of the body of the observer by a paper screen.

The test tube E is first filled with the substance, whose specific heat is to be determined, and weighed. It is then placed in the water bath until

constant weight is reached. After constant weight has been obtained the apparatus is again dried and the exact weight of the moisture lost thus determined. The test tube is then placed in the apparatus A closed with a well-fitted cork, the top covered with cotton and heated in the aqueous vapor for about one hour. The heating apparatus should be far removed from the calorimeter so that the temperature of the latter cannot be influenced thereby. Meanwhile the calorimeter is filled with water which has stood in the room for a long time until it has acquired, as nearly as possible, the room temperature.

The quantity of water is such that the water value of the whole of the calorimeter together with the immersed portions of the thermometer and stirrer shall amount to exactly 100 grams. A few minutes before bringing the substance into the calorimeter, the stirring apparatus is put in motion and the temperature observations are commenced. These should be at intervals of twenty seconds and should be continued until ten observations have been made. Meanwhile the height of the barometer is also read. A few seconds before the tenth interval the apparatus A is brought quickly to the calorimeter and its contents emptied into it at the moment of the tenth interval. The apparatus A should be removed as quickly as possible after its contents are emptied.

After the introduction of the substance and its thorough incorporation with the water of the calorimeter by the stirring apparatus, the thermometer is again read, at intervals of twenty seconds, until its maximum has been reached and as much longer thereafter as may be necessary to show that an appreciable fall of temperature has taken place. The test tube, in which the substance was heated is weighed and the exact quantity of the added substance thus determined.

In order that the sample of soil may be easily removed from the test tube in which it is heated, it is best to have it molded into appropriate forms before being placed in the heating tube. This is easily accomplished by pressing it into molds of convenient shape and of a size so that six or eight pieces (best of cylindrical shape) will be necessary to give the quantity sufficient for the experiment. Since some soils will not retain their shape after molding, the molds may be made of zinc foil whose water values in the calorimeter are previously determined and they can be placed with their contents in the calorimeter thus securing the total immersion of all the

particles of soil in the water. With very dusty materials, it is necessary that these little cylinders should be closed with pieces of foil at the ends in order to prevent the particles of dust from escaping and rising to the surface of the water.

Another source of error consists in the solution of the soluble salts which the soil may contain. This is avoided by the use of turpentine instead of water. If the cylinder containing the soil be made water-tight, this danger from the solubility of the salts in water is avoided. Another method of correcting these errors is in making a blank experiment in which a quantity of the earth taken is kept at the temperature of the water in the calorimeter until both are of the same temperature. The earth is then mixed with the water and the change of temperature produced noted. In this way the corrections made necessary by the solution of the salts in water and other causes are determined.

109. Method of Calculating Results.—Let t represent the mean temperature of the beginning period of the experiment, and v equal the loss in heat per interval. Let t' and v' represent the same values for the end period. Let $\theta_1, \theta_2, \theta_3$, etc., represent the temperature at the end of the first, second and third intervals of the middle period and θ_0 the temperature at the beginning of the middle period and θ_n the end temperature of the middle period. Let $\tau_1, \tau_2, \tau_3, \dots, \tau_n$, represent the mean temperature of the single intervals; then $\tau_1 = \frac{\theta_0 + \theta_1}{2}$; $\tau_2 = \frac{\theta_1 + \theta_2}{2}$, and $\tau_n = \frac{\theta_{n-1} + \theta_n}{2}$. The constant C represents the correction which must be applied in order to determine the true increase of temperature in the calorimetric system. The expression $\theta_n - \theta_0 + C$ represents the true temperature increase of the calorimetric system which we may represent by $\Delta\theta$ and $\theta_n + C$ represents the true maximum, that is, the end temperature, which by exclusion of external influences is reached. The correction C , as already indicated, is to be added to $\theta_n - \theta_0$ when it is positive and is to be subtracted therefrom when it is negative. The numerical value of C is usually very small, and, in the experiments indicated, varied between zero and one division of the thermometer employed, that is it seldom exceeded one degree.

110. Illustration.—The method of determining value of specific heat is best illustrated by an example:

In one determination the water value of the calorimetric system, including stirrer and thermometer was 2.50 grams, the weight of water

added was 97.50 grams and the total water value of the system 100 grams. The substance was dried at 100° and weighed in five envelopes:

Total weight	31.423	grams.
The envelopes alone weighed	10.654	„
Weight of the soil taken	20.769	„

The envelopes holding the soil were made of brass with zinc ends, the specific heat of which is 0.0939 and the water value of the whole of the envelopes was 1.0004 grams. Since, however, the ends were soldered on with zinc the true water value was somewhat smaller being equal to 0.8692 gram. The data of the observations were as follows:

Corrected height of barometer	699.6 millimeters.
Intervals between the observations	20 seconds.

	No. of Observations.	Temperature on the arbitrary scale of the thermometer.	
First Period	{ 0	162°.6	
	{10	162°.9	= θ_0 (Moment of immersion.)
	{11	185°.0	
Second Period	{12	200°.0	
	{13	206°.1	
	{14	209°.5	
	{15	210°.7	
	{16	211°.3	
	{17	211°.5	Differences.
	{18	211°.5	0
	{19	211°.5	0
	{20	211°.5	0
	{21	211°.5	0
	{22	211°.4	-0°.1
			= θ_n
Third Period			-0°.1
			-0°.1
	{23	211°.3	
	{24	211°.2	-0°.1
	{25	211°.1	-0°.1
	{26	211°.0	-0°.1
	{27	210°.9	-0°.1
	{28	210°.8	-0°.1
	{29	210°.6	-0°.2
	{30	210°.5	-0°.1

From the twenty-second interval, the regular fall of temperature begins and 211°.4 is therefore taken as θ_n . The mean temperature of the beginning period is therefore $\frac{162°.6 + 162°.9}{2} = 162°.75 = t$. The value of v is $\frac{162°.6 - 162°.9}{10} = -0°.03$. For the end period the value of t' is $\frac{211°.4 + 210°.5}{2} = 210°.95$ and the value of v' is $\frac{211°.4 - 210°.5}{8} = +0.11$. Then the sum of the observations from eleven to twenty-one inclusive = $\Sigma'_{n-1} \theta = 2280.1$

$\frac{\theta_0 + \theta_n}{2} =$	187.15
The sum =	2467.25
nt =	1953.00
Difference	514.25
This difference multiplied by $v - v' =$	0.14
gives a product equal to	71.995
This product divided by $t' - t =$	48.20
gives a quotient equal to	1.49
nv =	-0.36
The sum =	1.13 = C

Then $\Delta\theta = \theta_n - \theta_0 + C = 211^\circ.4 - 162^\circ.9 + 1^\circ.13 = 49^\circ.63$. The true end temperature = $\theta_n + C = 212^\circ.53$. The zero point of the thermometer = $24^\circ.70$, and the actual rise of temperature = $187^\circ.83$. The rise of temperature due to the proximity of the warming apparatus at the beginning was found by experiment to be equal to $0^\circ.1$ of the division of the scale. On comparing the thermometer used with a standard centigrade scale it was found that one division of the calorimetric thermometer was equal to $0^\circ.0858$. Converting these numbers into expressions of the centigrade scale we have the following summary:

The true rise of temperature, $\Delta\theta =$	$4^\circ.25$
The true end temperature, $\theta_n + C =$	$16^\circ.10$
The temperature of the steam, as determined by the height of the barometer, was equal to	$97^\circ.70$

From these data the specific heat is calculated according to the following formula:

$$\Sigma = \frac{1}{20.769} \times \left(\frac{100 \times 4.25}{97.70 - 16.10} - 0.8692 \right) = 0.2089.$$

From this formula the following rule for calculating specific heat is deduced:

Multiply the water value of the calorimetric system by the true rise in temperature in degrees Celsius and divide the product by the difference between the temperature of boiling water under the conditions of the experiment and the true end temperature. From the quotient subtract the

water value of the envelopes holding the soil sample. Divide the remainder by the weight of soil taken.

111. Variations in Specific Heat.—Different soils deport themselves very differently in respect of specific heat. In a large number of soils examined by Pfaundler, the specific heats were found to vary from 0.19 to 0.51. The highest specific heat was observed in the case of a peaty soil. Next to peaty soils came those soils which were highest in humus, and in general it was found that the specific heat varied directly with the humus content.

SOIL THERMOMETRY.

112. General Principles.—The measurement of the temperature of the soil at stated depths is often of use in analytical processes connected with agricultural chemistry and physics. The general principles on which the process rests, depend on bringing the bulb of the thermometer into as intimate contact as possible with the particles of soil at the depth required, disturbing as little as possible the normal state of the soil particles.

In the thermometer chiefly used for this purpose in this country, the stem is strong and carries the degrees figured on the glass. The whole is inclosed in a wooden case which is cut away to expose the face of the scale. The scale is about eleven inches long. The part which enters the soil is of varying lengths, according to the depth at which the temperature is desired.

113. Method of Procedure.—An excellent method of determining soil temperatures and of recording results is well illustrated by Frear.^[76]

The thermometers are set in niches cut in a trench, the earth being afterwards carefully tamped about the bulbs to secure a good contact, the trench being filled at the same time. The surface of the soil is freed from vegetation and kept in good tilth.

The depths at which observations are made are at the surface and one, three, six, twelve, and twenty-four inches. The soil tested was moderately dark and loamy to a depth of seven inches and below that a stiff clay. Solid rock existed at from five to seven feet below the surface. Readings were made three times a day.

114. Method of Stating Results.—The individual readings of the thermometers should be entered at the time they are made. At the end of each month the mean of the readings should be determined, together with the maxima and minima, and a comparison made between the mean readings of the temperature of the air and maxima and minima. As a sample of the method of stating these mean results the data are given for the month

of May, 1891, for the atmosphere, surface, and for the depths mentioned above:

MAY.**T° Fahrenheit.****ATMOSPHERE.**

Monthly mean	57.1
Maximum	85.0
Minimum	31.0
Mean daily range	22.5
Greatest daily range	32.0
Least daily range	8.0

SURFACE.

Monthly mean	56.7
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Extremes.

Maximum (10th of month)	77.0
Minimum (5th)	36.0
Mean maximum	65.2
Mean minimum	49.9

Range.

Monthly	41.0
Mean daily	14.9
Greatest daily (19th)	25.0
Least daily (21st)	4.0

ONE INCH.

Monthly mean	56.8
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Extremes.

Maximum (10th)	74.5
Minimum (5th)	36.5
Mean maximum	62.9
Mean minimum	49.5

Range.

Monthly	38.0
Mean daily	11.9
Greatest daily (10 and 19)	20.0
Least daily (23rd)	1.0

THREE INCHES.

Monthly mean	56.7
--------------	------

Extremes.

Maximum (31st)	71.0
Minimum (6th)	40.0
Mean maximum	60.9

Mean minimum	49.7
<i>Range.</i>	
Monthly	31.0
Mean daily	9.3
Greatest daily (19th)	15.5
Least daily (23rd)	1.5
SIX INCHES.	
Monthly mean	56.3
<i>Extremes.</i>	
Maximum (31st)	66.0
Minimum (6th and 7th)	43.0
Mean maximum	56.7
Mean minimum	53.2
<i>Range.</i>	
Monthly	23.0
Mean daily	4.65
Greatest daily (8 and 19)	8.5
Least Daily (5th)	1.0
TWELVE INCHES.	
Monthly mean	55.6
<i>Extremes.</i>	
Maximum (31st)	64.0
Minimum (6th and 7th)	46.0
Mean maximum	56.6
Mean minimum	54.4
<i>Range.</i>	
Monthly	18.0
Mean daily	2.18
Greatest daily (8th)	4.5
Least daily (3rd and 20th)	0.0
TWENTY-FOUR INCHES.	
Monthly mean	53.1
<i>Extremes.</i>	
Maximum (31st)	58.0
Minimum (6th and 8th)	48.0
Mean maximum	53.4
Mean minimum	52.8
<i>Range.</i>	
Monthly	10.0
Mean daily	0.48
Greatest daily (23rd)	2.0

115. Method of Whitney and Marvin.^[77]—The thermometer devised by Whitney and Marvin is shown in Fig. 14. The principle on which this modification depends is as follows:

A mercurial thermometer of the ordinary construction is liable to give wrong indications of the temperature because it is difficult to determine the temperature of the column of mercury from the bulb to the surface of the ground. To avoid this source of error the thermometer figured was constructed.

The bulb of the thermometer is made quite small and a slender portion of the stem extends into its spherical portion. The top portion of the thermometer stem does not differ in any essential respect from the stem of an ordinary thermometer.

The bulb is almost wholly filled with alcohol, which acts as the principal thermometric fluid and has the advantages of a high coefficient of expansion. The thermometer bulb and the stem of the thermometer up to a point convenient for graduation, are filled with mercury. In the drawing the mercury is represented by the heavy black marking in and just above the small bulb. The peculiar construction at this point is for the purpose of retaining the mercury about the point of the slender capillary stem inside the bulb and preventing the entrance of alcohol into the stem when the thermometer is horizontal.

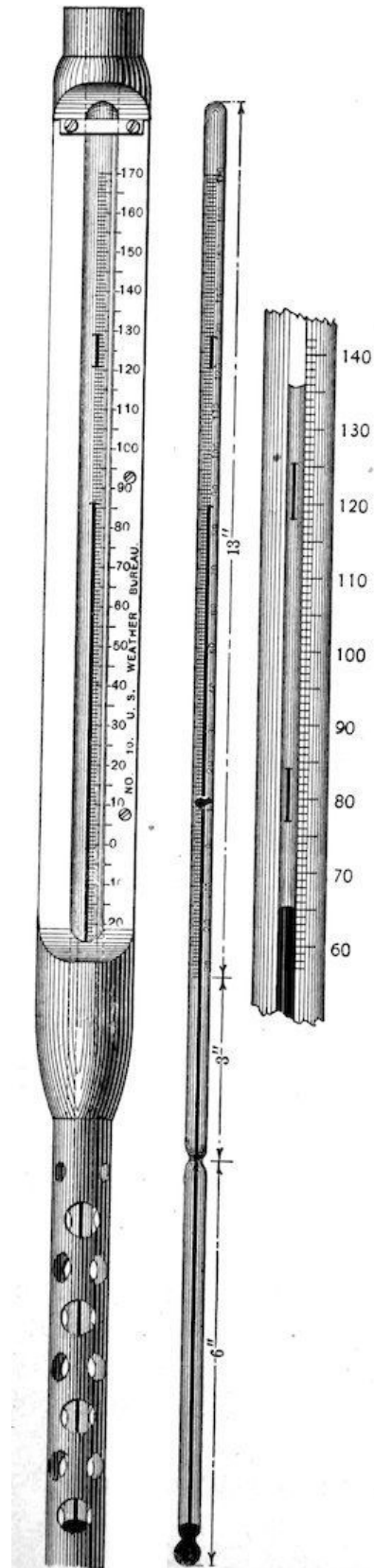
In order to register the maximum and minimum temperatures a short column of alcohol is placed in the upper portion of the stem, above the mercury, and within this are arranged two small steel indexes, so constructed that they will not slide in the tube of their own weight, but are easily pushed upward by the mercury column or pulled downward by the top meniscus of the alcohol column. The indexes are set by means of a small magnet, the one being drawn down upon the top of the mercurial column and the other raised up against the meniscus of the alcohol column.

The rise of the mercury carries its index upward, leaving it to register the highest point reached, while the alcohol meniscus withdraws the other index and leaves it at a point representing the minimum temperature. It remains only to mention that the graduations are fixed in the usual way, having reference only to the positions of the mercurial column. Beyond the

highest point supposed to be reached by the mercury, say about 120° , the graduations are extended in an arbitrary manner. The scale numbers represent temperatures by the mercurial column and are continued in regular sequence beyond the 120° . On this plan the readings for minimum temperatures are on a purely arbitrary scale and are converted into true degrees of temperature by use of a table prepared for each thermometer, which table embodies as well all the corrections for instrumental error.

The arrangement of the alcohol columns above the mercurial column and the indexes are shown enlarged at one side of the illustration. The readings of the maximum temperature are made from the bottom end of the index next to the mercurial column. The minimum temperature is the reading of the top of the uppermost index. Thus in the figure the maximum temperature indicated is 76.5° , and the minimum 125.7° , which, by reference to the table of correction for this thermometer, No. 10, is found to be 53.3° .

The use of mercury in the stem of the thermometer not only admits of the use of the index for registering the maximum temperature, but possesses the additional advantage of reducing the error due to uncertain temperature of the stem to about one-sixth what it would be if alcohol were used. Moreover, if necessary, as in the case with thermometers for greater depths than that figured, the ungraduated portion of the stem can be made of very much finer bore than the graduated portion, the effect of which is to diminish the objectionable error to a comparatively unimportant quantity.



The chief objection to thermometers of this construction is the liability of alcohol getting from the bulb into the stem during the processes of construction, graduation and subsequent handling, and the difficulty of safely shipping them.

FIG. 14. SOIL THERMOMETER—
Whitney and Marvin.

When once set up, however, there seems to be little or no possibility of derangement and the error common to mercurial thermometers due to rise of the freezing point with age does not apply owing to the high coefficient of expansion of the alcohol used in the bulb.

APPLICATIONS OF SOIL THERMOMETRY.

116. Estimation of the Absorption of Heat by Soils.—A cubical zinc box, six centimeters square, is filled with the sifted air dried soil. The box, one side of which is left open, is encased snugly in a wooden cover, exposing only the open end, and placed for a few hours in the direct rays of the sun. The temperature is then taken at a given depth. The box may be provided with thermometers at different depths, the bulbs thereof extending to the center. In this case the box should be covered with thick felt instead of wood. The temperature of the layers of soils of different depths can thus be read off directly. The air temperature directly above the box should be accurately noted while the experiment continues.

Any other kind of box well protected against all heat save the direct sunlight on the open surface of the soil will answer as well as the one described.

To determine the action of moist earth in similar conditions the soil may be previously moistened; the per cent of moisture being determined in a separate portion of the soil or the amount of water added to the air-dried soil being noted.

117. Estimation of the Conductivity of Soils for Heat.—The bulb of a thermometer is placed in the middle of a mass of fine earth which is then exposed, best in a metallic box painted with lamp black, in a warm place. The time required for the thermometer to reach a certain degree is noted. By reversing the experiment and placing the mass of earth heated to a given degree in a cool place the conductivity can be determined by the time required for the mercury in the thermometer to fall to any given point.

The experiment may also be made by packing the soil by gently jolting it in a glass tube six to eight centimeters in diameter. One end of the tube is closed with a piece of metal or fine wire gauze painted with lamp black and is exposed to the source of heat. The bulb of a thermometer is placed at a

given distance from the end of the tube and the time for the mercury to be affected observed.

COHESION AND ADHESION OF SOILS.

118. Behavior of Soil After Wetting.—The deportment of a soil when thoroughly wet in respect of its physical state on drying out is a matter of great practical concern to the agronomist. Some soils on becoming dry fall into a pulverulent state and are easily brought into proper tilth; others become hard and tenacious, breaking into clods and resisting ordinary methods of pulverization. The physical laws which determine these conditions depend largely on the principles of flocculation soon to be described. The present task is to describe briefly some of the methods of estimating the force of cohesion and adhesion.

119. General Method.—The fine earth, air-dried, is mixed with enough water to make a paste and molded into forms suitable for trial in a machine for testing strength of cement, etc. The forms most used are cakes three to five centimeters in length and one to two centimeters thick. These are used for determining crushing power. For longitudinal adhesion the paste may be molded in prismatic or cylindrical shape.^[78] The prisms should show one to two centimeters in cross section or the cylinder be one to two centimeters in diameter. Before use they are to be exposed for several days until thoroughly air-dried. The force required to separate or crush these prepared pieces will measure the adhesive or cohesive property of the sample. A great number of trials should be made and the mean taken.

120. Method of Heinrich.^[79]—This process consists in mixing the air-dried earth with water until its aqueous content is fifty per cent of the highest water capacity determined by experiment. The sample is next placed between two pieces of sheet iron of ten centimeters square, each of which in its middle point is provided with a hook. The thickness of the layer between the two pieces of iron should be about five to ten centimeters. The exuding particles of soil are cut off with a knife. The upper piece of sheet iron is next suspended by a cord in such a way that the iron piece occupies a horizontal position. A small basket is attached to the lower surface and sand added thereto, little by little, until the column of earth is separated. The sand

basket and iron plate are weighed, and the total weight gives the power necessary to separate a column of soil ten centimeters square in cross section. The iron plates may be roughened so that the adhesion thereto of the soil is greater than its cohesive force.

121. Adhesion of Soil to Wood, Iron, Etc.—The adhesive power of moist soil for wood, iron, etc., is measured by Heinrich^[80] in the following way: The soil is mixed with water, as above, until it contains just fifty per cent of its total water-holding content. It is then placed in a large vessel and the upper surface made as smooth as possible. A plate of wood, iron, etc., of ten centimeters square is then pressed on the surface until a complete contact is secured. This plate, by means of a hook and cord passing over a pulley, is then subjected to stress by weighting the cord which carries a basket for that purpose. The basket should be of the same weight as the plate in contact with the soil. The weight added to the basket necessary to separate the plate from the soil is taken to represent the cohesive force. The author of the method appears to take no account of the pressure of the air on the plate caused by the exclusion of the air from its under surface.

THE ABSORPTIVE POWER OF SOILS FOR SALTS IN SOLUTION.

122. General Principles.^[81]—It is a fact of every-day observation that soils have a particular property of absorbing certain materials with which they come in contact. If it were not for this property all our wells would soon become unwholesome from the reception of decayed animal and vegetable matter carried to them in the drainage water from the surface. It is also a well-known fact that burying dead bodies prevents the gaseous products of decomposition from reaching and vitiating the atmosphere.

Besides this well-known power of soils to absorb the decomposition products of animal and vegetable matter, they also possess a property which is of far greater importance in plant economy; that is, the power of withdrawing and retaining certain mineral constituents from their solutions.

As far back as the sixteenth century mention is made by Lord Bacon of a process for obtaining pure water on the seashore by simply digging a hole in the sand and allowing it to fill with filtered sea water, which by this means is deprived of its salt. Although certain facts were observed by some of the earlier writers in regard to soil absorption, no systematic researches were conducted with a view of demonstrating the extent and cause of this power until within a comparatively few years.

In 1850 Prof. Way published in the *Journal of the Royal Agricultural Society of England*, the results of a thorough and most excellent investigation of the subject. Since then many distinguished chemists, such as Henneberg, Stohmann, Peters, Heiden, Knop, Ullik, Pillitz, Biedermann, Tuxen, and others have given their attention to this matter.

123. Summary of Data.—If a solution of a soluble sulfate, chloride or nitrate of an alkali or an alkaline-earth metal be placed in contact with a soil, the result is that the soil takes up a part of the base but none of the acid. This absorption of base is attended with the liberation of some other base from the soil which combines with the acid of the solution. Any alkali

or alkaline earth base has the power of replacing any other such base. However, if soluble phosphates and silicates of these bases be placed in contact with the soil both the base and the acid are removed from the solution.

Peters^[82] has shown that the amount of absorption depends upon the concentration of the solution, the relation between the quantity of solution and the soil and the kind of salt used. He treated 100 grams of earth with 250 cubic centimeters of solutions of different potash salts with the following results:

Strength of solution.	$\frac{1}{10}$ Normal. Grams	$\frac{1}{20}$ Normal. Grams
Salt Used	K ₂ O absorbed.	K ₂ O absorbed.
KCl	0.3124	0.1990
K ₂ SO ₄	0.3362	0.2098
K ₂ CO ₃	0.5747	0.3154

Biedermann^[83] proves that, for phosphoric acid at least, the absorption increases with the temperature.

It has also been found that the amount of absorption depends upon the time of contact between the soil and solution. Way found that the absorption of ammonia was complete in half an hour, while Henneberg and Stohmann^[84] noticed that the phosphoric acid continued to be fixed after the expiration of twenty-four hours.

It is a very important fact that the absorption of a base is never complete; no matter how dilute the solution it will still carry a small portion of the base with it. Peters states that it requires about 28,000 parts of water to remove one part of absorbed potash and Stohmann found that it required about 10,000 parts of water to remove one part of absorbed ammonia. With phosphoric acid, the resulting compound seems to be much more insoluble.

According to Tuxen^[85] the presence of salts of soda and potash in solution decreases the power of a soil to absorb ammonia compounds and the presence of sodium salts decreases the power of a soil to absorb potash. On the other hand the presence of potassium compounds considerably increases the absorption of phosphoric acid. He further affirms that the compounds of potash, phosphoric acid, etc., formed in the soil, are decidedly more soluble in sodium salts than in pure water.

124. Cause of Absorption.—The withdrawing and fixing of phosphoric acid from solutions by the soil is not very difficult to understand as this acid forms insoluble compounds of iron, lime, and magnesium, some or all of which are present in all soils. As to the absorption of the alkalies, the explanation is far more difficult as nearly all of their ordinary compounds are readily soluble in water.

As lime is usually found combined with the acid part of an alkali salt, from which the base has been absorbed by the soil, it might naturally be supposed that the absorptive power of the soil would depend upon the amount of lime present. Way found, however, that the addition of chalk in no way influenced the absorption of ammonia by a soil which contained but a small amount of lime. This fact was also confirmed by Knop^[86] who found that chalk exerted no influence on the absorption of ammonia salts. These facts would seem to point to the conclusion that lime was present in sufficient quantity in these experiments, or that it is not essential to the phenomena of absorption. However, as any alkali or alkaline-earth base can replace any other such base, the presence of lime in the filtrate is probably more of an accidental occurrence, owing to the comparatively large amount of that substance in most soils, than a necessary condition, as any other base would doubtless answer in the absence of lime.

125. Warington^[87] has shown that hydrated oxides of iron and aluminum, and especially the former, are capable of absorbing potash and ammonia, and as more or less of these hydrates exist in nearly all soils, a part, at least, of absorptive phenomena is to be ascribed to them.

126. Way tried to determine which of the constituents of a soil exercised chiefly the absorptive power. He passed a solution of ammonia through tubes containing pure sand and found that it came through apparently unaltered from the first, while a soil treated in the same way removed the ammonia for a considerable time. He concluded from this that the absorptive power does not exist in the sand. He next oxidized the organic matter in a soil with nitric acid and then treated it with ammonia in the same way. The first portions of the filtrate showed no ammonia in any form, hence he concluded that organic matter is not essential to the act of absorption. He further showed that clay alone is capable of causing absorption phenomena, by treating powdered clay tobacco pipes with ammonia.

Having shown that clay was the main constituent in a soil which caused the absorption of alkalies, he tried next to trace out the particular compound which caused the absorption. Having tried various natural silicates he at last succeeded in producing a hydrated silicate of aluminum and soda which exhibited displacement and absorptive properties very similar to those shown by the soil.

As Way had succeeded in producing an artificial hydrated silicate possessing absorptive properties, Eichorn^[88] thought of trying natural hydrated silicates or zeolites and found that they exhibited the same power as Way's artificial preparation. It has also been shown by Biedermann,^[89] Rautlenberg,^[90] and Heiden^[91] that the absorptive power bears a close relation to the amount of soluble silicates present.

In view of these facts it is now generally accepted that the absorption of salts of the alkalies, accompanied by the change of base, is due chiefly to the presence of decomposed zeolite minerals in the soil.

Besides the purely chemical absorption of salts by the soil, we have a physical absorption of various substances similar to the action of charcoal when used as a filter.

127. Conclusions of Armsby.—The data connected with the absorption of bases by a soil have also been reviewed by Armsby.^[92] He shows that the absorption is accompanied by a chemical reaction between the salt whose base is absorbed and some constituent of the soil, and this change seems to be due particularly to certain zeolitic silicates, although Liebig and others were disposed to credit this absorption largely to physical causes.

Knop advances the idea that the soil has the power of disintegrating salts in the presence of some substances like calcium carbonate which can unite with the acid. In experiments made with hydrous silicates it was shown that the absorption resembled in all cases like phenomena in the soil; hence the supposition already advanced in regard to the influence of such silicates is doubtless true.

In respect of absorption in general, the following conclusions were reached:

1. The absorption of combined bases by the soil consists in an exchange of bases between the salt and the hydrous silicates of the soil.

2. This exchange, which is primarily chemical, is only partial, its extent varying

(a) with the concentration of the solution, and

(b) with the ratio between the volume of the solution and the quality of soil used.

3. The cause of these variations is probably the action of mass or the tendency of resulting compounds to re-form the original bodies, the absorption actually found in any case marking the point where the two forces are in equilibrium.

128. Selective Absorption of Potash.—As a rule more potash is absorbed from the sulfate than from the chlorid. This fact would seem to point to the advisability of using sulfate as a fertilizer in preference to chlorid. However, as with the exception of nitrates, the absorptive power of a soil, for the salts used as fertilizers, is many times greater than it is ever called upon to exert in fixing applied fertilizers, we need not trouble ourselves in regard to the absorption of phosphoric acid, potash or ammonia, in so far as the practical side of the matter is concerned. For example, an acre of soil to the depth of nine inches weighs about 900 tons. Now it has been found by Huston,^[93] that 100 parts of a soil experimented upon absorbed over 0.25 part of P_2O_5 , hence 900 parts would absorb over 2.25 parts of P_2O_5 ; or an acre of this soil to the depth of nine inches would absorb over two and one-fourth tons of phosphoric acid. 500 pounds per acre is a large dressing of a phosphatic fertilizer for field crops and 500 pounds of a high grade fertilizer would contain about 100 pounds of P_2O_5 ; hence the power of such a soil to absorb phosphoric acid is more than forty-five times as great as it is ever likely to be called upon to exert in fixing the phosphoric acid added to it as a fertilizer.

Huston has further shown that an acre of soil nine inches deep will absorb more than 2.7 tons of potash (K_2O) from potassium chlorid from which salt less potash is absorbed than from the sulfate. Now one-tenth ton of potassium chlorid per acre would be a large dressing of potash, hence this soil possesses the power of absorbing more than twenty-seven times as much potash as is ever likely to be applied as a fertilizer.

In like manner it may be shown that the power of an acre of soil nine inches deep to absorb ammonia from ammonium sulfate is more than thirty-

two times as great as it would be called upon to exert in fixing the ammonia from a dressing of one-quarter ton of ammonium sulfate per acre.

With sodium nitrate, however, there is no absorption; hence great care is necessary in the application of nitrogen as a nitrate, for, if it be put on in large quantities, at a season when the plant is not prepared to assimilate it, or during a period of heavy rains, there must unavoidably result loss from drainage. The best time to apply a nitrate is evidently during the active growing season.

129. Whitney^[94] places great emphasis on the surface area of soil particles in respect to their power to absorb solutions of salts. The approximate surface area of a cubic foot of each of the different typical soils of Maryland is as follows:

Pine barrens	23,940	square	feet.
Truck lands	74,130	„	„
Tobacco lands	84,850	„	„
Wheat lands	94,540	„	„
River terrace	106,260	„	„
Limestone subsoil	202,600	„	„

It will be seen that there are about 24,000 square feet of surface area in a cubic foot of the subsoil of the pine barrens, no less than 100,000 square feet or two and three-tenths acres of surface area in a cubic foot of the subsoil of the river terrace, and 200,000 square feet of surface area in a cubic foot of the limestone subsoil.

These figures seem vast, but they are probably below rather than above the true values, on account of the wide range of the diameters of the clay group. This great extent of surface and of surface attraction, which has been described as potential, gives the soil great power to absorb moisture from the air, and to absorb and hold back mineral matters from solution. A smooth surface of glass will attract and hold, by this surface attraction, an appreciable amount of moisture from the surrounding air. A cubic foot of soil, having 100,000 square feet of surface, should be able to attract and hold a considerably larger amount of moisture.

It might have been added that if the potential of the surface, separating the solution from the soil, be greater than the potential in the interior of the liquid mass, there will be a tendency to concentrate the liquid on this

surface of separation. It has been shown that certain fluids have greater density on a surface separating the fluid from a solid. On the other hand, if the potential were low there might be no tendency for this concentration, and even the reverse conditions would prevail and the soluble substance could be readily washed out of the soil.

130. Removal of Organic Matters.—It is probably largely due to this straining power that organic matters are removed from solutions in percolating through the soil. Whitney^[95] has observed that the organic matter may be coagulated and precipitated from solution by the soil constituents, and held in the soil in loose flocculent masses, while the liquid passes through nearly free of organic matter.

131. Importance of Soil Absorption.—The importance of the absorptive power of the soil can hardly be overestimated. By means of this power those mineral ingredients of plant food, of which most soils contain but little, are held too closely to allow of rapid loss by drainage, and still sufficiently available to answer the needs of vegetation, provided the store is large enough. The only important plant food liable to be deficient in the soil which does not come under the influence of absorption is nitrogen in the form of salts of nitric acid, and nature has made a wide provision for this element by binding it in the form of organic bodies which nitrify but slowly, and by supplying each year a small quantity from the atmosphere.

By means of the absorptive power of soils the farmer, if he puts on an excess of potash or phosphoric acid as a fertilizer, does not lose it but is able to reap some benefits from it in the next and even in succeeding crops. If it were not for this power the best method for applying fertilizers would be a much more complicated problem than it is at present; and it would be necessary to apply them at just the proper season and in nicely regulated amounts to insure against loss.

132. Method of Determining Absorption of Chemical Salts.—The soil which is to be used for this experiment should be treated as has been indicated and passed through a sieve the meshes of which do not exceed half a millimeter in size. From twenty-five to fifty grams of the fine earth may be used for each experiment.

The fine earth should be placed in a flask with 100 to 200 cubic centimeters of the one-tenth to one-hundredth normal solution of the substance to be absorbed. The flask should be well shaken and allowed to

stand with frequent shaking twenty-four to forty-eight hours at ordinary temperatures. The whole is then to be thrown upon a folded filter and an aliquot part of the filtrate taken for the estimation. The methods of determining the quantities of the substances used will be found in other parts of this manual. It is recommended to conduct a blank experiment with water under the same conditions in order to determine the amount of the material under consideration abstracted from the soil by the water alone. The difference in the strength of the solution as filtered from the soil, corrected by the amount indicated by the blank experiment, and the original solution will give the absorptive power of the soil for the particular substance under consideration.

If it should be desired to determine the absorptive power of the soil for all the ordinary chemical fertilizing materials at the same time, a larger quantity of the sample should be taken corresponding to the increased amount of the standard solutions used. About 500 cubic centimeters of the mixed salt solution should be shaken with 125 grams of the earth and the process carried on in general as indicated above. The absorption coefficient of an earth for any given salt according to Fesca,^[96] is the quantity of the absorbed material expressed in milligrams calculated to a unit of 100 grams of the soil.

133. Method of Pillitz and Zalomanoff.—It is recommended by Pillitz and Zalomanoff^[25] to reject the old method, viz., shaking the soil with the solution in a flask, and substitute the filtration method both because it gives a more natural process and because the results are more constant. The apparatus is shown in Fig. 15.

Two cylinders are placed vertically, one over the other. The lower cylinder is graduated in cubic centimeters, the upper cylinder is closed at each end by perforated rubber stoppers A and B through the openings of which the glass tubes *c* and *d* pass. Within the cylinder A the opening of the small tube *d* is closed with a disk of Swedish filter paper. The lower part of the small tube is *d* connected by means of a rubber tube carrying a pinch-cock C, with another small tube *e* which passes through the stopper *f*. In carrying out the process the weighed quantity of soil is placed in the upper cylinder and afterwards the measured quantity of the solution, the whole thoroughly mixed and the cylinder closed. The valve C is then opened, a given quantity of the solution, but not all, is made to drop into the lower

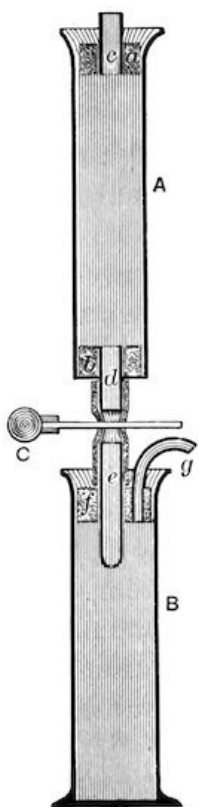


FIGURE 15.

ZALOMANOFF'S
APPARATUS FOR
DETERMINING
ABSORPTION OF
SALTS BY SOILS.

cylinder and the valve C is then closed. The liquid which has passed into the lower cylinder as well as that which remains in the upper cylinder, is thoroughly stirred and the quantity of the material remaining in both liquids determined and the absorbing power of the soil estimated from their difference. It does not appear that this method of estimation of the absorption power possesses any special advantages over the old and far simpler method of shaking in a flask.

134. Method of Müller.—The method of Müller^[97] for illustrating absorption is carried out by means of the apparatus shown in Fig. 16. A glass cylinder A about 750 centimeters long and four to five centimeters wide is closed at each end with rubber stoppers with a single perforation. The cylinder A is for the reception of the soil with which the experiment is to be made. Before using, the lower part of it is filled with glass pearls or broken glass and above this a layer of glass wool is

placed about one centimeter thick. The object of this is to prevent the soil from passing into the small tube below. As soon as the soil has all been placed in the cylinder A the upper part of the tube is also filled with glass wool. The cylinder A is connected with the pressure bottle B by means of a rubber tube and the small glass bulb tube shown in the figure. The bottle B should have a content of about two liters. It is filled with the standard solution of the material of which the absorption coefficient is to be determined. At c the rubber tube is connected with a glass T one arm of which is provided with a piece of

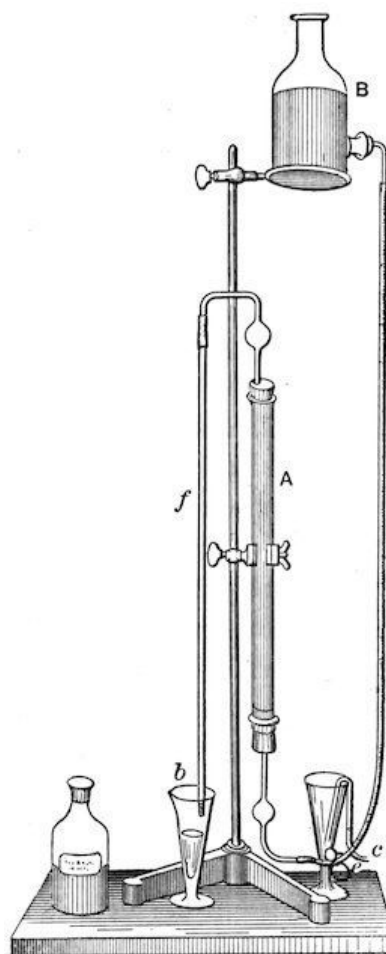


FIGURE 16.

MÜLLER'S APPARATUS TO SHOW
ABSORPTION OF SALTS BY SOILS.

rubber tubing which can be closed by means of a pinch-cock. At *c* a screw pinch-cock is placed which can be used to regulate the flow of the solution from B to A. By opening the pinch-cock at *e* on the short arm of the T piece, a sample of the original liquid can be taken and this can be compared with the part which runs to *b*. If it is desired for instance, to show that potassium carbonate has been absorbed by the soil the two bulbs shown on the small glass tubes connecting with A can be filled with red litmus paper. This paper will at once be turned blue in the lower bulb while in the upper one it will retain its original color because the liquid in passing through the soil will have lost its alkaline reaction. The solutions used should be very dilute. The apparatus is designed for lecture experiments and not for quantitative determinations.

135. Method of Knop.—For rapid determination of the absorption coefficient of the soil Knop's method may be used.^[98]

The fine earth which is employed is that which passes a sieve with meshes of half a millimeter. From 50 to 100 grams of this soil are mixed with from five to ten grams of powdered chalk and with about twice the weight of ammonium chlorid solution of known strength, viz., from 100 to 200 cubic centimeters. The ammonia solution should be of such a concentration that the ammonia by its decomposition for each cubic centimeter of the liquid evolves exactly one cubic centimeter of nitrogen. This solution is prepared by dissolving in 208 cubic centimeters of water one gram of ammonium chlorid. With frequent shaking the solution is allowed to stand in contact with the soil for forty-eight hours. The whole is now allowed to settle and the supernatant clear liquid is poured through a dry filter. From the filtrate twenty to forty cubic centimeters are removed by a pipette, and evaporated to dryness in a small porcelain dish, with the addition of a drop of pure hydrochloric acid. The ammonium chlorid remaining in the porcelain dish is washed with ten cubic centimeters of water into one of the compartments of the evolution flask of the Knop-Wagner azotometer. It is decomposed with fifty cubic centimeters of bromine lye and the nitrogen estimated volumetrically. The difference between the amount of nitrogen in this material and that of the original material will give the amount of absorption exercised by the fine earth. This number, without any further calculation, can be taken as the coefficient of absorption.

136. Method of Huston.—The salt solutions recommended by Huston^[99] are sodium phosphate (Na_2HPO_4), potassium chlorid, potassium sulfate, ammonium sulfate and sodium nitrate.

The solutions should be approximately tenth normal, the actual strength in each case being determined by analysis. The phosphorus is determined as magnesium pyrophosphate in the usual way, the potash as potassium platinochlorid, the ammonia by collecting the distillate from soda in half normal hydrochloric acid and titrating with standard alkali, and the nitrate by Warington's modification of Schlösing's method for gas analysis. The details of these methods of determination will be given later. One hundred grams of the sifted, air-dried soil are placed in a rubber stopped bottle and treated with 250 cubic centimeters of the solution to be tested. The digestion is continued for forty-eight hours in each case, the bottles being thoroughly shaken at the end of twenty-four hours. At the end of the treatment the solutions are filtered and the salts determined in aliquot portions. The details of this method are essentially those already described.

137. Statement of Results.—Duplicate analyses should be made and the tabulation of the data is illustrated in the following analyses by Huston:

Na ₂ HPO ₄ cubic centimeters filtrate taken.	Weight of Mg ₂ P ₂ O ₇ in twenty-five cubic centimeters of the solution.	Weight of Mg ₂ P ₂ O ₇ filtrate.	of in	P ₂ O ₅ absorbed by 100 grams soil.	Salt removed per cent.
(a) 25	0.1368 gram	0.0962 gram			
(b) 25		0.0963 „		0.2589 gram	29.6
	Mean	0.0963 „			
KCl cubic centimeters filtrate taken.	Weight of K ₂ PtCl ₆ in twenty-five cubic centimeters of solution.	Weight of K ₂ PtCl ₆ .	of	K ₂ O absorbed by 100 grams soil.	Salt removed per cent.
(a) 25	0.6154 gram	0.4505 gram			
(b) 25		0.4540 „		0.3161 gram	26.5
	Mean	0.4523 „			
K ₂ SO ₄ cubic centimeters filtrate taken.	Weight of K ₂ PtCl ₆ in twenty-five cubic centimeters of solution.	Weight of K ₂ PtCl ₆ .	of	K ₂ O absorbed by 100 grams soil.	Salt removed per cent.
(a) 25	0.6113 gram	0.4426 gram			
(b) 25		0.4371 „		0.3324 gram	28.0
	Mean	0.4399 „			
(NH ₄) ₂ SO ₄ cubic centimeters filtrate taken.	Number cubic centimeters one-half normal acid neutralized by fifty cubic centimeters of solution.	Half normal acid neutralized.		N absorbed by 100 grams soil.	Salt absorbed per cent.
(a) 50	10.00	7.25 grams			
(b) 50		7.25 „		0.0964 gram	27.5
	Mean	7.25 „			
NaNO ₃ cubic centimeters filtrate taken.	Number cubic centimeters N ₂ O ₂ afforded by ten cubic centimeters of solution at 0° and 1000 millimeters pressure.	Cubic centimeter N ₂ O ₂ at 0° and 1000 millimeters.		N absorbed by 100 grams soil.	Salt absorbed per cent.
(a) 10	16.63	16.77 grams			

(b) 10	16.70 „	none	00.00
Mean	16.73 „		

Upon comparing the figures it will be found that the absorption, passing from the greatest to the least, is as follows: phosphoric acid (P_2O_6), potassium sulfate, ammonium sulfate, potassium chlorid and sodium nitrate.

It will be seen that there was no absorption in the case of the nitrate, while with each of the other salts there was quite a marked absorption. It will also be noticed that the percentages of absorption are not very different, and especially is this true of the potassium and ammonium salts, the P_2O_5 being somewhat higher. Whether this fact is merely an accidental occurrence or is due to the law of combination by equivalents could hardly be predicted from the single soil experimented upon; but taking into consideration the possibility of difference in solubility of the resulting compounds in the saline solutions used, and of other varying conditions, the percentages are evidently not far enough apart to exclude the possibility of the bases uniting in equivalent proportions.

138. Preparation of Salts for Absorption.—The salts employed in the foregoing determinations are conveniently prepared, in fractional normal strength.

In grams per liter the following quantities in grams are recommended, viz., 5.35 g NH_4Cl ; 10.11 g KNO_3 ; 16.40 g $Ca(NO_3)_2$; 24.60 g $MgSO_4 + 7H_2O$; 23.4 g $CaH_4(PO_4)_2$, etc.

The ammonium chlorid, potassium nitrate and magnesium sulfate can be weighed as chemically pure salts and the standard solution be directly made up. Calcium nitrate is so hygroscopic that a stronger solution must be made up, the calcium determined and the proper volume taken and diluted to one liter.

Monocalcium phosphate is prepared as follows:

A solution of sodium phosphate is treated with glacial acetic acid and precipitated with a solution of calcium chlorid. It is then washed with water until all chlorin is removed. The fresh precipitate is saturated with pure, cold phosphoric acid of known strength. After filtering the solution is placed in a warm room and left for two or three weeks until crystallization takes place.

The crystals are pressed between blotting papers and finally dried over sulfuric acid and washed with water-free ether, and again dried. Since this salt is decomposed in strong solutions it should be used only in one hundredth normal strength, viz., 2.34 grams per liter.

POROSITY AND ITS RELATIONS TO MOISTURE.

139. Porosity.—The porosity of a soil depends upon the state of divisibility and arrangement of its particles, and upon the amount of interstitial space within the soil. If a soil be cemented together into a homogeneous mass, its porosity sinks to a minimum; if it be composed, however, of numerous fine particles, each preserving its own physical condition, the porosity of the soil will rise to a maximum. The porosity of a soil may be judged very closely by the percentage of fine particles it yields by the process of silt analysis to be described further on. In general, the more finely divided the particles of a soil, the greater its fertility. This arises from various causes; in the first place, such a soil has a high capacity for absorbing moisture and holding it; thus the dangers of excessive rain-falls are diminished, and the evil effects of prolonged drought mitigated. In the second place, a porous soil permits a freer circulation of the gases found in the soil. The influence of lime in securing the proper degree of porosity of a soil is very great, especially in alluvial deposits and other stiff soils. It prevents the impaction which will necessarily follow in a soil which is too finely divided. In general, the porosity of the soil may be said to depend on three factors, viz.: 1. Upon the state of divisibility or the number of particles per unit volume; 2. Upon the nature and arrangement of these particles; 3. Upon how much interstitial space there is in the soil.

140. Influence of Drainage.—Good underdrainage increases the porosity of a soil by removing the excess of water during wet seasons and rendering the soil more suitable to capillary attraction which will supply moisture during dry seasons. The influence of tile drainage on the production of floods has been carefully studied by Kedzie,^[100] who shows that surface ditching in conjunction with deforesting may increase floods and contribute to droughts, and that tile-draining may increase flood at the break-up in spring, when the water accumulated in the surface soil by the joint action of frost and soil capillarity during the winter, and the surface accumulations in the form of snow are suddenly set free by a rapid thaw.

He also points out that during the warm months tile-draining tends to prevent flood by enabling the soil to take up the excessive rain-fall and hold it in capillary form, keeping back the sudden flow that would pass over the surface of the soil if not absorbed by it, and it mitigates summer drought by increased capacity of the soil to hold water in capillary form and to draw upon the subsoil water supply.

141. Soil Moisture.—The capacity of a soil to absorb moisture and retain it depends on its porosity and is an important characteristic in relation to its agricultural value.

The following general principles relating to soil moisture are adapted from Stockbridge:^[101]

During dry weather plants require a soil which is absorptive and retentive of atmospheric moisture. The amount of this retention is generally in direct ratio to two factors, viz., the amount of organic matter and its state of division. The capillary water of the soil is very closely related to its percolating power, since all waters in the soil are governed in their movements by what is known as capillary force. Liebenberg has shown that this movement may be either upwards or downwards, according as the atmosphere is dry or supplies soil-saturating rain. The water absorbed by the roots passes into the plant circulation, and the greater part is evaporated from the leaves. Where the supply of water is insufficient, the plant wilts, and if the evaporation long continue in excess of the supply obtained from the soil, the plant must die. The experiments of Hellriegel have shown that any soil can supply plants with all the water they need, and as fast as they need it, so long as the moisture within the soil is not reduced below one-third of the whole amount that it can hold. The quantity of water required and evaporated by different agricultural plants during the period of growth has been found to be as follows:

One	acre	of	wheat	exhales	409,832	pounds	of	water.
„	„	„	clover	„	1,096,234	„	„	„
„	„	„	sunflowers	„	12,585,994	„	„	„
„	„	„	cabbage	„	5,049,194	„	„	„
„	„	„	grape-vines	„	730,733	„	„	„
„	„	„	hops	„	4,445,021	„	„	„

Dietrich estimates the amount of water exhaled by the foliage of plants to be from 250 to 400 times the weight of dry organic matter formed during the same time. Cultivation conserves soil moisture. It must be remembered that this water contains soil ingredients in solution. Hoffmann has estimated that the quantity of matter dissolved from the soil by water varies from 0.242 to 0.0205 per cent of the dried earth. The experiments of Humphrey and Abbott have shown that about one-sixth of the total sediment of the Mississippi river is soluble in water.

142. Determination of the Porosity of the Soil.—The porosity of the soil is fixed by the relative volume of the solid particles as compared with the interstitial space. It is most easily determined by dividing the apparent by the real specific gravity.

Let the real specific gravity of a soil be 2.5445 and the apparent specific gravity of the same soil be 1.0990.

The porosity is then calculated according to the following ratios, viz.:

$$2.5445 : 1.099 = 100 : X$$

Whence $X = 43.2$ = per cent volume occupied by the solid particles of the soil.

The per cent volume occupied by the interstitial space is therefore 56.8.

143. Method of Whitney.—The total volume of interstitial space within the soil, in which water and air can enter, is best determined by calculation from the specific gravity and the weight of a known volume of soil. To determine this in the soil in its natural position in the field, a sample is taken in the following way: A brass tube, about two inches in diameter and nine inches long, has a clock spring securely soldered into one end, and this end turned off in a lathe to give a good cutting edge of steel. The area enclosed by this steel edge is accurately determined, and a mark is placed on the side of the tube exactly six inches from the cutting edge. A steel cap fits on top of the brass cylinder to receive the blows of a heavy hammer or wooden mallet. The cylinder is driven into the ground until the six-inch mark is just level with the surface. The whole is then dug out, care being taken to slip a broad piece of steel under the cylinder before it is removed, so as to prevent the soil which it contains from falling out. The cylinder is then carefully laid over on its side, and the soil is cut off flush with the cutting edge of steel. The soil is then removed from the cylinder, carried to

the laboratory and properly dried and weighed. The object of the steel inserted in one end of the cylinder is to reduce the friction on the inside of the tube to a minimum, and thus prevent the soil inside the cylinder being forced down below the level of the surrounding earth. The volume of the soil removed with this sampler can readily be determined by calculation, as the area of the end of the tube is known and the sample is six inches deep. In a sampler, such as described here, this volume is about 300 cubic centimeters. From the weight of soil and the volume of the sample, the volume of interstitial space may be found by the following formula:

$$S = \frac{\left[V - \frac{W}{\omega} \right] \times 100}{V}$$

S is the per cent by volume of interstitial space, V is the volume of the tube in cubic centimeters, W is the weight of soil in grams, and ω is the specific gravity of the soil. The specific gravity can be determined for each soil, or the factor 2.65 can be used, which is sufficiently accurate for most work.

The per cent by volume of interstitial space in the undisturbed subsoil is found to range from about thirty-five for sandy land, to sixty-five or seventy for stiff clay lands.

For the determination of the amount of water an air-dried soil will hold, if all the space within it is completely filled with water, an eight-inch straight argand lamp chimney, with a diameter of about two inches, can be conveniently used. A mark is placed on the side of the tube, six inches from one end, and the volume of the tube up to this mark is found by covering the end with a piece of thin rubber cloth, or by pressing the chimney down firmly on a glass plate, and making a water-tight joint with paraffin or wax. Water is then poured into the tube up to the six-inch mark, and the weight or volume of water determined. The tube can then be dried, a piece of muslin tied tightly over the top and the whole then weighed. Soil is carefully poured in and the tube gently tapped on a soft support until the soil is six inches deep in the tube, and has the desired degree of compactness. The weight and volume of the soil can thus be determined, and the volume of the interstitial space from the formula already given. This can also be determined directly by introducing water from above, or by immersing the

cylinder of soil up to the six-inch mark in water, and allowing the water to enter the soil from below. With such a short depth of soil, very little water will flow out when the cylinder is suspended in the air. The amount which will flow out when the cylinder is thus suspended, will depend both upon the texture and the depth of soil. It is impossible, however, by this method, to completely remove the air or to completely fill the space within the soil with water; for as the water enters the soil, a considerable amount of air becomes entangled in the capillary spaces, and this could not be removed except by boiling and vigorous stirring, which would altogether change the texture of the soil. The amount of water held by the soil, or the amount of space within the soil into which water and air can enter, will evidently depend upon the compactness of the soil, and this is best expressed in per cent by volume of space.

144. Capacity of the Fine Soil for Holding Moisture.—The soil, as it is taken from the field, may have quite a different water coefficient from the same soil after it has been passed through a fine sieve or been dried at air temperatures or at 100° or 110°. The method of determination which depends upon adding excess of water to a given weight of fine earth, and afterwards eliminating the excess by percolation or filtration, is apt to give misleading results. If, however, the results are obtained by working on the same weight of soil, and in the same conditions, they may have value in a comparative way. The comparison between soils must be made with equal weights, in like apparatus and with the same manipulation, to have any value. These determinations, however, cannot have the same practical value as those made in the samples in a natural condition as has just been described.

145. Method of Wolff Modified by Wahnschaffe.^[102]—A cylindrical zinc tube (Fig. 17), sixteen centimeters long and four centimeters internal diameter, is used, the cubical capacity of which is 200 cubic centimeters.

The cylinder is graduated by placing the moist linen disk on the gauze and tying a piece of rubber cloth over the bottom. Water is now poured in until the level is even with the gauze bottom. Add then exactly 200 cubic centimeters of water, mark its surface on the zinc, throw out the water, and file the zinc down to the mark.

The bottom of the tube is closed with a fine nickel-wire gauze. Below this a piece of zinc tubing, of the size of the main tube, is soldered; pierced

laterally with a number of holes.

Before using, the gauze bottom of the cylinder is covered with a moist, close fitting linen disk, and the whole apparatus weighed. It is then filled with the fine earth, little by little, jolting the cylinder on a soft substance after each addition of soil to secure an even filling. When filled even full the whole is weighed, the increase in weight giving the weight of soil taken.

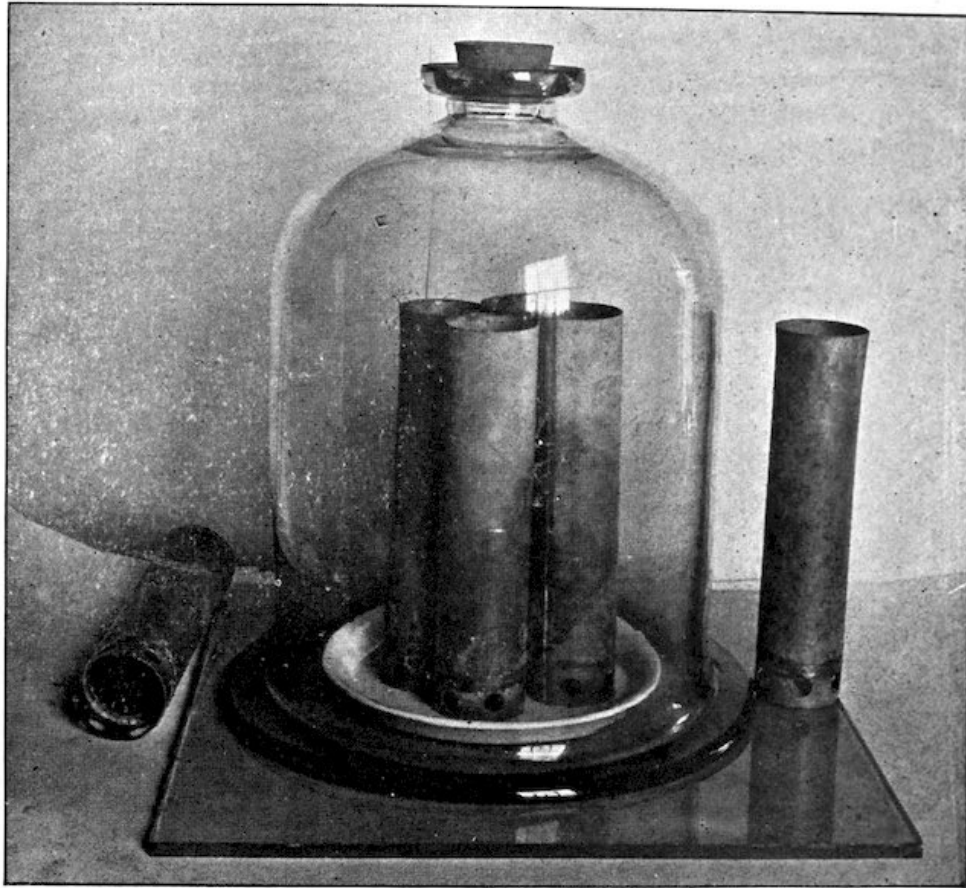


FIGURE 17.

CAPACITY OF THE FINE SOIL FOR HOLDING MOISTURE. METHOD OF WOLFF MODIFIED
BY WAHNSCHAFTE.

A large number of cylinders can be filled at once and placed in a large glass crystallizing dish containing water and covered with a bell jar (Fig. 17). The water should cover the gauze bottoms of the cylinders to the depth of five to ten millimeters. More water should be added from time to time as absorption takes place. The cylinders should be left in the water until when

weighed at intervals of an hour no appreciable increase in weight takes place. The temperature and barometer reading should be noted in connection with each determination. With increasing temperature the water coefficient is diminished.

The method of Wolff, as practiced in the laboratory of the Chemical Division of the U. S. Department of Agriculture, has given very concordant results. Five determinations were made on a sample of vegetable soil with the Wolff cylinders, which were weighed at intervals of ten, twenty, and thirty days, with the following results:

No.	1.	Water	absorbed	after	ten	days	106.25	per	cent
”	2.	”	”	”	”	”	105.68	”	”
”	3.	”	”	”	”	”	105.86	”	”
”	4.	”	”	”	”	”	106.11	”	”
”	5.	”	”	”	”	”	105.83	”	”
Mean							105.95	”	”

No.	1.	Water	absorbed	after	twenty	days	106.44	per	cent
”	2.	”	”	”	”	”	105.98	”	”
”	3.	”	”	”	”	”	106.56	”	”
”	4.	”	”	”	”	”	106.52	”	”
”	5.	”	”	”	”	”	106.38	”	”
Mean							106.38	”	”

No.	1.	Water	absorbed	after	thirty	days	108.35	per	cent
”	2.	”	”	”	”	”	107.60	”	”
”	3.	”	”	”	”	”	108.32	”	”
”	4.	”	”	”	”	”	107.86	”	”
”	5.	”	”	”	”	”	107.87	”	”
Mean							108.00	”	”

The data obtained show that there was a very slight increase in the amount of moisture absorbed after the tenth day.

As will be seen, however, from the following data, the soil within the cylinder does not contain in all parts the same percentage of moisture, the

lower portions of the cylinder containing notably larger proportions than the upper parts. The cylindrical soil column was divided into four equal parts and the moisture determined in each part. Beginning with the top quarter the percentages of moisture were as follows:

First	quarter	97.52	per	cent
Second	„	105.91	„	„
Third	„	112.83	„	„
Fourth	„	116.48	„	„

146. Method of Petermann.^[103]—The method of Wolff as practiced by the Belgian Experiment Station, at Gembloux, is essentially the same as described above.

Petermann recommends the use of tared cylinders twenty to twenty-five centimeters long and six to eight centimeters in diameter. The cylinder is to be filled with the fine earth, little by little, with gentle tapping after each addition. The bottom of the cylinder is closed with a perforated rubber stopper on which is spread a moistened disk of linen. The cylinder, thus prepared and filled, is weighed and afterwards placed in a vessel containing distilled water, to such a depth as to secure a water level about two centimeters above the lower surface of the soil in the cylinder. The level of the water is kept constant as the contents of the cylinder are moistened by capillarity. When the earth appears to be thoroughly moistened, as can be told by the appearance of the upper surface, maintain the contact with water for about five or six hours.

The cylinder is then removed, the upper surface covered to avoid evaporation, allowed to drain for a few hours, wiped and weighed. The cylinder is again placed in water to see if any increase in weight takes place. The weight of the fine earth and of the absorbed water being known, the percentage of absorption is easily calculated.

147. Method of A. Mayer.^[104]—A glass tube, one and seven-tenths centimeters in diameter, composed of two pieces, seventy-five centimeters and twenty-five centimeters in length, is united by a piece of rubber tubing. The lower free end of the seventy-five centimeter piece is closed with a piece of linen. The tube is filled, with gentle jolting, to the depth of one meter with fine earth, the earth column thus extending twenty-five

centimeters above the point of union of the two pieces. Thus prepared, a quantity of water is poured into the upper tube sufficient to temporarily saturate the whole of the soil.

During the sinking of the water in the tube there is thus effected a moistening of the material before it is wholly filled with water. After waiting until the water poured on top has disappeared the tube is separated at the rubber tube connection and a sample of the moist soil taken at that point. This is at once weighed and then dried at 100°. The loss in weight gives the water absorbed.

The number thus obtained is calculated to the standard by volume, by use of the number representing the apparent specific gravity of the fine earth.

For sand of different degrees of fineness the following numbers were found:

Degree of fineness	2	3	4
Per cent water absorbed	7.0	13.7	44.6

The numbers thus obtained are taken to represent the absolute water capacity of a mineral substance in powder.

The full water capacity, *i. e.*, the power of holding water when the powder is immersed in water, the excess of which is then allowed to flow away is much greater than the absolute number.

This difference is shown in the following data:

	Quartz, size three.		Clay, size three.	
Full water capacity	49.0	per cent	46.8	per cent
Absolute water capacity	13.7	„ „	24.5	„ „

In general the absolute is markedly inferior to the full water capacity. Only in the finest dust do the two numbers approach each other.

148. Volumetric Determination.—A convenient apparatus for this determination has been devised by Mr. J. L. Fuelling, of the Chemical Division, Department of Agriculture. It is shown in Fig. 18.

It consists of an ordinary percolator the diameter of which decreases slightly towards the lower end, a thick-wall rubber tube and an ordinary burette, divided in tenths. A rubber stopper is fitted to the mouth of the

percolator and perforated twice—in the middle and at the side, the former for a small tube provided with pinch-cock and the latter for the neck of a small funnel. The whole is supported on a convenient stand, the clamp holding the percolator being placed above that supporting the burette, both clamps arranged to slide on the stand-rod.

The method is as follows:

A mark is placed upon the projecting tube at the lower end of the percolator, and the tube at this point may be drawn out sufficiently to decrease the width of meniscus to one-eighth inch. Into the percolator is first introduced a small disk of wire gauze or perforated porcelain, with heavy wire pendant in the tube. Through the rubber stopper a small glass tube is passed and its lower end pressed firmly upon the wire or porcelain disk, its upper end being curved and supplied with a pinch-cock. Into the percolator is now poured one inch of fine shot (No. 20) and then one inch of fine sand which has been previously digested with hydrochloric acid and well cleaned of dust by washing.

The zero.—After the shot and sand have been shaken even, the burette is filled with water and raised above the level of the sand, wetting the percolator for four inches of its length. The burette is lowered and the shot and sand bed allowed to drain by opening the pinch-cock of the inner tube. The burette is raised and the shot-sand flooded repeatedly until, by lowering the burette until the zero mark of the percolator tube is reached, a uniform reading on the burette is secured.

Thus the shot-sand bed is completely charged with water. The water level is now made zero on the percolator

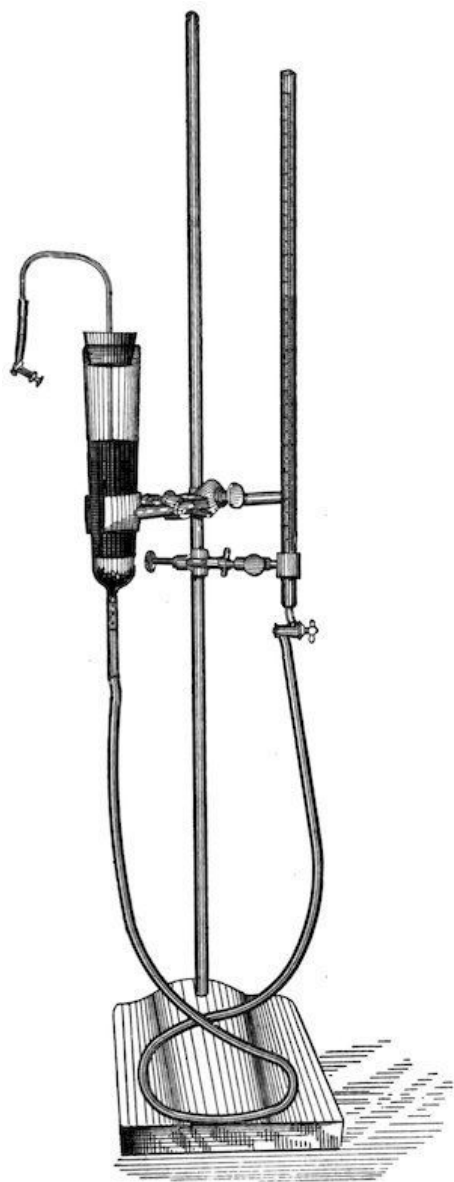


FIGURE 18.

FUELLING'S APPARATUS.

stem, the burette filled to its zero mark and the apparatus is prepared for introduction of the soil.

The Determination.—From 100 to 200 grams of soil, previously dried free of moisture, are weighed, the burette raised until the water level is three inches above the sand, and the soil gently dropped through a funnel into the water. When the soil has been introduced and wetted completely the water level is raised above the soil and allowed to remain thus two hours. The burette is then lowered and the water allowed to drain from the wetted soil. Four to six hours are usually given the draining, the reading taken on the burette after establishing the zero on the percolator stem, the volume of absorbed water thus ascertained and divided by the weight of soil multiplied by 100; the result expresses the water absorbed per hundred of soil.

Example:

Water required to saturate disk, etc.	0.50	cubic centimeter.
Weight of air-dried soil taken	20.00	grams.
Moisture at 105° therein	14.25	per cent.
Weight water in soil	2.85	grams.
Reading of burette after saturation	10.75	cubic centimeters.
Less water required for disk, etc	9.25	„ „
Temperature	20°.00	
Weight of 9.25 cubic centimeters H ₂ O at 20°	9.22	grams.
Total weight of water retained by soil	12.07	„
Per cent water retained by soil	60.35	per cent.

For general analytical work the correction for variations in the weight of water for different temperatures is of no practical importance.

149. Accuracy of Results.—A sample of soil from the beet sugar station, in Nebraska, gave the following duplicate results:

First trial	45.75	per	cent	water.
Second trial	44.85	„	„	„

Muck soils from Florida, containing varying proportions of sand, gave the following numbers:

Soil number one, 144.85 per cent, and 145.43 per cent; soil number two, 109.13 per cent, and 107.93 per cent; soil number three (very sandy), 46.86 per cent, and 46.51 per cent.

150. Method of Wollny.^[105]—A zinc tube, ninety centimeters long and four centimeters internal diameter, carries at each end, at right angles to the axis, a flattened rim 1.5 centimeters broad. The lower end of the tube is closed with a strong piece of coarse linen. The soil to be examined is then filled in little by little, with gentle tamping.

On the upper end two glass tubes are placed, each ten centimeters long and four centimeters internal diameter. These tubes are furnished at each end with cemented brass cylinders which are expanded to a circular, evenly ground rim, 1.5 centimeters wide, also at right angles to the axis of the main tube. These rims are greased and placed together, one on the other, and held together by wooden clamps.

The glass tube in immediate connection with the zinc tube is also firmly filled with the soil sample, while the second tube is only partly filled, so that any settling which may take place in the soil on the addition of water may still find the first glass tube full of the sample.

The empty part of the upper glass tube is now filled with water and additional quantities of water are added from time to time until the soil is saturated. In order to be able to observe when this takes place there is a slit at the lower end of the zinc tube which is closed with a piece of glass. This slit should be about two centimeters broad and ten centimeters long. The lower end of the zinc tube is set on a glass plate to prevent evaporation.

As soon as the water shows itself at the lower end of the zinc tube, the excess of water in the upper glass tube is at once removed by a pipette and a stopper inserted through which a glass tube passes drawn out into a fine point above. The object of this is to avoid evaporation on the upper surface. The apparatus is then left at rest for thirty-six hours.

At the end of this time the clamps are removed and the column of moist earth cut with a piece of platinum foil, and the two ends of the glass tube, next to the zinc tube, covered with glass plates. It is then weighed and the weight of moist earth determined by deducting the weight of the tube and its glass covers. The moist earth is carefully removed to a large porcelain dish and dried at 100°. Before weighing it is allowed to stand twenty-four

hours in the air. The data obtained are used to calculate the water content to volume per cent.

The volume of the glass tubes should be determined by careful calibration.

151. Method of Heinrich.^[106]—The soil to the depth turned by the plow is dug out and in the hole a lead vessel without bottom, twenty centimeters in diameter and forty centimeters high, is placed. The soil is then thrown back around and outside the lead vessel until the latter appears buried in the fragments.

The rest of the soil is passed into the lead vessel, through a sieve having four meshes to the centimeter, using for this purpose enough water to thoroughly moisten it. Care should be taken not to use enough water to cause any separation of the fine from the coarse particles.

By this process all coarse stones, sticks, etc., are separated. In sandy soils the flask is left for a few hours while in clay soils a much longer time is necessary. When the excess of water has disappeared the lead cylinder is removed, and a piece cut out of the center of it placed in a weighed drying flask and dried at 100°.

152. Effect of Pressure on Water Capacity.^[107]—The increasing capacity of soil to hold water developed by shaking or pressure, is determined by Henrici in the following way:

Into a glass cylinder of twenty millimeters internal diameter are poured twenty cubic centimeters of water. A given quantity of soil is next added, and after standing until thoroughly saturated, the residual water is measured by pouring off, or better, by graduations on the side of the tube. The increase in the volume of the clear water is also measured, after shaking, in the same way. The data of a determination made as above described follow:

Water in cylinder	30	cubic	centimeters.
Water and saturated soil	40	„	„
Volume of unsaturated soil = e =	10	„	„
Volume of saturated soil = e + w =	20.5	„	„
Water contained therein = w =	10.5	„	„

By repeated shaking the volume of e + w, the content of w therein, and the relative values of $\frac{e}{w}$ were found to be as follows:

	Cubic centimeters.	Cubic centimeters.	Cubic centimeters.	Cubic centimeters.
$e + w$	20.5	16.0	15.7	15.0
w	10.5	6.0	5.7	5.0
$\frac{w}{e}$	1.05	0.60	0.57	0.50

If e' represent the volume of the saturated soil then $e' = e + w$, and this gives the relation to the volume of dry earth represented by the equation $\frac{e'}{e} = 1 + \frac{w}{e}$. This indicates that the relative volume of the saturated soil is equal to unity increased by the relative content of water.

153. Coefficient of Evaporation.—At an ordinary room temperature in the shade, samples of soil, if they are subjected to experiment in tolerably thin layers have nearly an equal coefficient of evaporation. That is, the absolute quantity of water evaporated in a given time is almost entirely conditioned upon the magnitude of the surface exposed and the temperature of the surrounding air. Only when exposed in conditions as nearly as possible natural in thin layers to the action of the sunlight and shade do the soils show their peculiarities in respect of the evaporation of moisture. In order to see these peculiarities, samples of soil which have been previously examined must be subjected to examination at the same time with the soil whose properties are to be determined.

The zinc box, before described, should be protected with a well fitting cover of thick paper, and the different samples of soil which are to be tested placed therein. This should now be placed in a wooden box, the top of which is exactly even with the top of the zinc vessel. This box containing the vessel should be exposed to the sunlight. After twenty-four hours the zinc boxes can be taken away from position and their loss in moisture determined, and these weighings, according to the condition of the atmosphere, can be continued from fourteen days to three weeks, the temperature of the air of course being carefully determined at each time. At first, all the different soils being saturated with moisture, it will be observed that the loss of moisture is proportionately the same for all. Soon, however, the rapidity of the evaporation in the samples of soil rich in humus and clay will be decreased as compared with the sandy soils, and in general, those which possess a high capillary power capable of bringing the moisture

rapidly from the deeper layers to the surface. There soon comes a point when the difference in evaporation is at its greatest; and then there will be a gradual diminution until the samples lose no further moisture. This point, for the different soils, can be determined by frequent weighings of the vessel.

154. Determination of Capillary Attraction.—Long glass tubes graduated in centimeters may be used for this determination, or plain tubes so arranged as to admit of easy measurements with a rule. The tubes may be from one to two centimeters internal diameter and about one meter long. The fine earth should be evenly filled in little by little, with gentle jolting. The lower end of each tube, before filling, is closed with a piece of linen.

The tubes, after filling, are supported in an upright position by a frame AE, Fig. 19, in a vessel B containing water in which the linen covered ends D dip to the depth of two centimeters. The height of the water in the several tubes should be read or measured at stated intervals. The water contained in the supply vessel should be kept at a constant height by a Mariotte bottle.

The observations may be discontinued after one hundred and twenty hours, but even then the water will not have reached its maximum height.

It is recommended by some experimenters to cut the tubes, after the above determination is completed, into pieces ten centimeters in length, and to determine the per cent of water in each portion.

155. Statement of Results.—The following table illustrates a convenient method of tabulating the observed data as given by König.^[108]

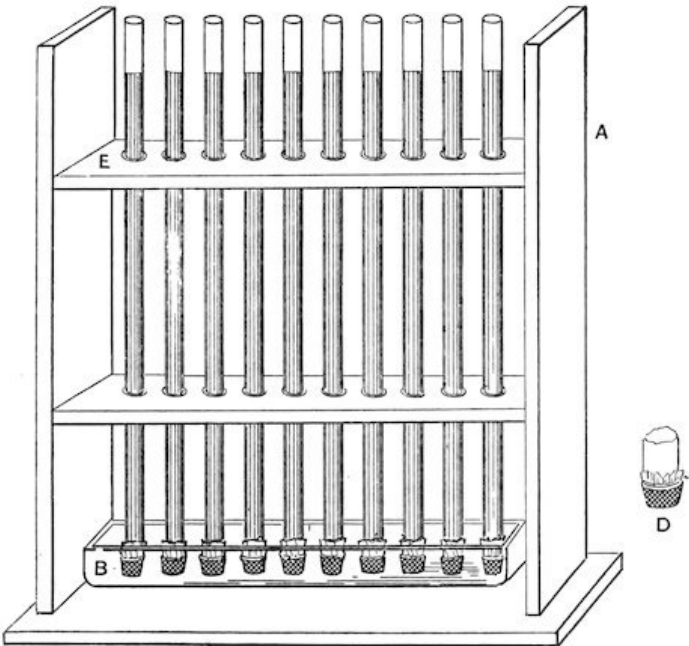


FIGURE 19.

APPARATUS TO SHOW CAPILLARY ATTRACTION OF SOILS FOR WATER.

Number of sample			1	2	3	4	5	6	
Height of moisture column after:	24	hours.	27.3	38.0	16.7	36.4	8.0	28.8	centimeters.
	48	„	35.9	50.8	24.5	49.2	11.9	40.5	„
	72	„	41.5	59.5	30.0	57.9	15.2	49.1	„
	96	„	44.4	66.2	33.5	63.8	17.5	55.2	„
	120	„	46.7	70.0	36.3	68.5	19.2	60.5	„

156. Inverse Capillarity.—In tubes filled with fine earth, as described in paragraph 154, water is quickly poured, the same quantity into each tube of the same diameter, or such quantities in tubes of different diameters as would form a water column of the same depth over the surface of the sample. The rate at which the water column descends in each tube, the time of the disappearance of the water at the surface and the final depth to which it reaches, are the data to be entered.

157. Statement of Results.—The points to be observed in the determination of inverse capillarity are the number of hours required for the total absorption of a column of water of a given height, the depth of the

moisture column at that moment, and the total depth to which the moisture column finally reaches. The data of observations with six samples with a water column four centimeters high are given by König^[109] as follows:

Number of sample	1	2	3	4	5	6	
Number of hours required for water to disappear	4.3	1.8	10.3	3.0	21.0	4.3	
Depth of moisture at time of disappearance of water	11.0	12.0	11.4	13.3	11.7	12.0	centimeters.
Total depth of moisture	13.0	18.1	13.0	19.0	12.0	16.5	„

158. Determination of the Coefficient of Evaporation.—The coefficient of evaporation is the number of milligrams of water evaporated from a square centimeter of soil surface in a given unit of time. It is evident that this number will vary with the physical state of the soil, the velocity of the wind, the saturation of the air with aqueous vapor and the temperature. In all statements of analyses these factors should appear.

The process may be carried on first (a) with soil samples kept continually saturated with water and (b) with samples in which the water is allowed to gradually dry out.

Method a.—The determination may be made in the shade or sunlight.

In the Shade.—A zinc cylinder (Z Fig. 20), fifteen centimeters in diameter and 7.5 centimeters high, with a rim one centimeter wide and one centimeter from top, is covered at one end with linen or cotton cloth and filled with fine earth, with gentle jolting, until even with the top. It is then placed in a zinc holder H, into the circular opening of which it snugly fits as in A. This holder is twenty centimeters in diameter and 7.5 centimeters deep. It has an opening at O through which water can be added until it is filled so as to wet the bottom of Z when in place. As the water is absorbed by the soil more is added and, the top being covered, the apparatus is allowed to stand for twenty-four hours. At the end of this time the soil in the zinc cylinder is saturated with water to the fullest capillary extent.

The whole apparatus, after putting a stopper in O, is now weighed on a large analytical balance and placed in an open room, with free-air circulation, for twenty-four hours. At the end of this time it is again weighed and the loss of weight calculated to milligrams per square centimeter. Where large and delicate balances can not be had, the apparatus

can be constructed on a smaller scale suitable for use with a balance of the ordinary size.

In the Sunlight.—The apparatus described above is enclosed in a wooden box having a circular opening the size of the soil-zinc cylinder. In the determination of the rate of evaporation, the apparatus, charged and weighed as above described, is exposed to the sun for a given period of time, say one hour. On the second weighing the loss represents the water evaporated. The time of year, time of day, velocity of wind and temperature, and degree of saturation of the air with aqueous vapor, should be noted. The data obtained can then be calculated to milligrams of water per square centimeter of surface for the unit of time.

Method b.—As in method a the determination may be made in the shade or in the sunlight. The rate of evaporation is, in this method, a diminishing one and depends largely on the reserve store of water in the sample at any given moment.

The same piece of apparatus may be used as in the determinations just described. After charging the sample with moisture all excess of water in the outer zinc vessel is removed and the rate of evaporation determined by exposure in an open room or in the sunlight, as is done in the operations already described.

Alternate Method.—The zinc cylinders used in determining saturation coefficient, paragraph 145, may also be employed in determining the rate of evaporation. Each cylinder should be wrapped with heavy paper or placed in a thick cardboard receptacle, and all placed in a wooden box, the cover of which is provided with circular perforations, just admitting the tops of the cylinders, which should be flush with the upper surface of the cover.

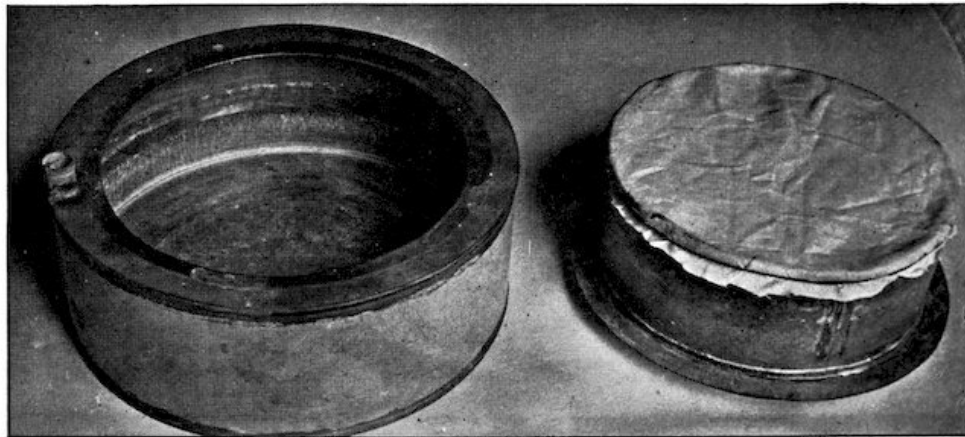
Arranged in this way the cylinders previously weighed are exposed in the shade or to direct sunlight and reweighed after a stated interval. On account of the small surface here exposed in comparison with the total quantity of soil and moisture it is recommended to weigh the cylinders once only in twenty-four hours. The weighings may be continued for a fortnight or even a month.

In soils fully saturated with water the rate of evaporation is at first nearly the same on account of the surface being practically that of water alone. As the evaporation continues, however, the rate changes markedly with the character of the soil.

159. Rapid Method of Wolff.—In order to expose a larger surface to evaporation and to secure the results in a shorter period of time, Wolff^[110] fills square boxes, having wire-gauze bottoms, with fine earth, and after saturating with moisture weighs and suspends them in the open air. The wire-gauze bottoms are previously covered with filter paper to prevent loss of soil.



A



H

Z

FIGURE 20.

APPARATUS FOR DETERMINING COEFFICIENT OF EVAPORATION.

160. Estimation of Water Given up in a Water-Free Atmosphere.—

The air-dried sample, in quantities of from five to ten grams in a thin layer on glass, is placed over a vessel containing strong sulfuric acid. It is then placed on a ground glass plate and covered with a bell jar. The sample is weighed at intervals of five days until the weight is practically constant.

This method is valuable in giving the actual hygroscopic power of a soil depending on its structure alone.

161. Estimation of the Porosity of the Soil for the Passage of Gases.—

Some further notion of the physical state of the soil known as porosity, may also be derived by a study of the rate at which it will admit of the transmission of gases. A method for estimating this has been devised by Ammon.^[111]

Air is compressed in two gas holders by means of a column of water of proper height to give the pressure required.

The tubes through which the air passes out of the gas holders are each furnished with a stop-cock and united with a glass tube having a side tube set in at right angles for carrying off the air.

The use of two holders makes it possible to carry on the experiment as long as may be desired, one holder being filled with air while the other is emptying.

The common conducting tube is joined with a meter which is capable of measuring, to 0.01, the volume of air passing through it. The pressure is regulated by means of the stop-cocks. The air passing from the meter is received in a drying tube filled with calcium chlorid.

From the drying tube the air enters a drying flask filled below with concentrated sulfuric acid and above with pumice stone saturated therewith. Next the dried air passes through a worm, eight meters long, surrounded with water at a given temperature. The dried air of known temperature next enters the experimental tube. This tube is made of sheet zinc 125 centimeters in length and five centimeters in internal diameter. It is placed in an upright position, and about six centimeters from its upper end carries a small tube at right angles to the main one for connection with a water-filled manometer.

The upper and lower ends of the tube are closed with perforated rubber stoppers carrying tubes for the entrance and exit of the air.

In the inside of the zinc tube are found two close-fitting but movable disks, of the finest brass wire gauze, between which the material to be experimented upon is held.

The layer of fine soil is held between these disks and may be of such a depth as is required for the proper progress of the experiment. With soils of firm texture opposing a great resistance to the passage of the air the column of earth tested should be shorter than with light and very permeable soils. The experimental tube is surrounded with a water jacket, which may also be made of sheet zinc, carrying small tubes directed upwards for holding thermometers. The water jacket should be kept at the same temperature as the air which is used in the experiment.

The process of filling the tube, the amount of pressure to be used and the air and soil temperature, will naturally vary in different determinations.

The volume of air at a given pressure and temperature which passes a column of soil of a given length in a unit of time will give the coefficient of permeability.

162. Determination of Permeability in the Open Field.—A method for determining the rate of transmission of a gas through the soil in the field has been devised by Heinrich.^[112]

A box C (Fig. 21) is made of strong sheet iron and has an opening below, ten centimeters square, and a height of about twenty centimeters. At exactly ten centimeters from the bottom, the box has a rim at right angles to its length so that it can be placed only ten centimeters deep in the soil. The box holds a volume of earth equal to 1,000 cubic centimeters.

The part of the box above ground is connected with the bottle B by a glass tube as indicated in the figure. The bottle B should have a capacity of about ten liters. The air in B is forced out through C by water running in from the supply A and the pressure in B is recorded by the manometer D. The experiment should be tried on a soil thoroughly moist.

In measuring the pressure in B the water pressure should be cut off by the pinch-cock between A and B, and the pressure on the manometer observed after the lapse of one to two minutes.

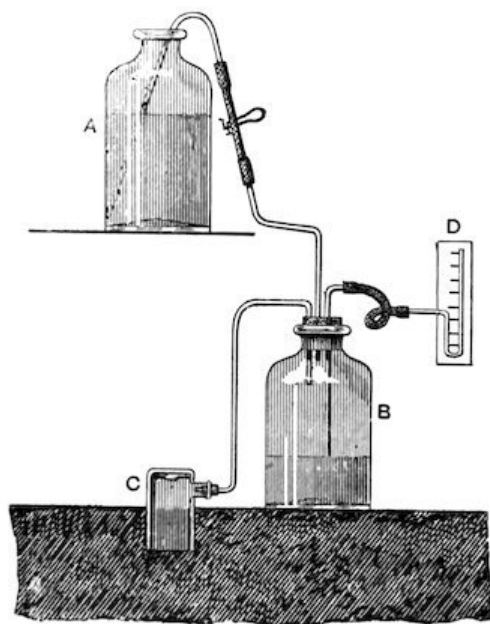


FIGURE 21.

METHOD OF HEINRICH.

MOVEMENT OF WATER THROUGH SOILS: LYSIMETRY.

163. Porosity in Relation to Water Movement.—The intimate relation which water movement in a soil bears to fertility makes highly important the analytical study of this feature of porosity. A soil deficient in plant food, in so far as chemical analysis is concerned, will produce far better crops when the flow of moisture is favorable than a highly fertile

soil in which the water may be in deficiency or excess.

Aside from the actual rain-fall the texture of the soil, in other words its porosity, is the most important factor in determining the proper supply of moisture to the rootlets of plants. Even where the rain-fall is little, a properly porous soil in contact with a moist subsoil will furnish the moisture necessary to plant growth. This fact is well illustrated by the beet fields in Chino Valley, California. In this locality most excellent crops of sugar beets are produced without irrigation and almost without rain.

164. Methods of Water Movement—The translocation of soil water is occasioned in at least two ways; namely,

1. By changing the porosity of a given stratum of soil.
2. By changing the amount of water a given stratum contains.

The following experiment by King^[113] illustrates a convenient method of studying this movement of water:

On a rich fallow ground of light clay soil, underlaid at a depth of eighteen inches by a medium-grained sand, water, to the amount of two pounds per square foot on an area of eight by eight feet, was slowly added with a sprinkler, samples of soil having been previously taken in six-inch sections

down to a depth of three feet. The samples were taken along a diagonal of the square under experiment and one foot apart. The middle sample of the line being from the center of the area. The sampling and wetting occurred between one and three p. m., on July 22, and on the evening of the 23 a corresponding series of samples was taken along a line parallel to the first but eight inches distant. The changes in the percentages of water in the soil are given in the following table, showing the translocation of water in soil due to wetting the surface:

Inches.	PER CENT OF WATER.		DIFFERENCE.	
	Before wetting.	After wetting.	In per cent.	In pounds per cu. ft.
0-6	14.00	22.23	+8.23	+2.873
6-12	15.14	15.71	+0.57	+0.199
12-18	16.23	15.75	-0.48	-0.213
18-24	17.70	16.92	-0.78	-0.347
24-30	16.76	14.41	-2.35	-1.032
30-36	15.51	15.21	-0.30	-0.132

The figures given in the last column of the table are computed from the absolute dry weights of the upper three feet of soil as determined in a locality some rods from the place of experiment, and are therefore only approximations, but the error due to this cause is certainly small. It will be seen that while only two pounds of water to the square foot were added to the surface, the upper six inches contained 2.87 pounds per square foot more than before the water was added, and the second six inches contained 0.199 pound more, and this too in the face of the fact that the evaporation per square foot from a tray sitting on a pair of scales close by, was 0.428 pound during the interval under consideration. Similar experiments were made by taking the samples of soil at 5.30 p. m. in one-foot sections down to four feet, at four equally distant places along the diagonal of a square, six by six feet, and having the ground sprinkled. At the same time four similar sets of samples were taken on lines vertical to each of the sides of the square but four feet distant from them. The amount of water the soil contained was then determined, and at 11.30 A. M., nineteen hours later, another series of samples was taken at points about four inches distant from the last and the amount of water determined with the result given below.

TRANSLATION OF WATER OCCASIONED BY WETTING THE SURFACE.

Depth of samples.	WET AREA.				AREA NOT WET.			
	Before wetting.		After wetting.		First samples.		Second samples.	
	Per cent of water.	Pounds of water per cubic foot.	Per cent of water.	Pounds of water per cubic foot.	Per cent of water.	Pounds of water per cubic foot.	Per cent of water.	Pounds of water per cubic foot.
0-12 inches	16.86	11.78	20.15	14.06	17.72	12.38	18.27	12.75
12-24 „	17.76	15.79	19.71	17.52	19.18	17.05	19.94	17.72
24-36 „	16.76	14.73	17.72	15.58	16.97	14.92	17.52	15.40
36-48 „	15.01	14.03	16.47	15.40	15.49	14.48	15.16	14.17
Averages	16.59	14.08	18.51	15.64	17.34	14.71	17.71	15.01
Total amount of water		56.33		62.56		58.83		60.04
Amount of change				+6.23				+1.21

The above data show sufficiently well the method of investigation to be pursued in studies of this kind.

165. Capillary Movement of Water.—The method of investigation proposed by King^[114] consists in taking samples of soil at intervals of one, two, three, or four feet in depth, and determining the amount of moisture in each in connection with the amount of rain-fall during the period. The quantity of water contained in a given soil, at various depths and on different dates, is shown in the following table:

Depth in feet.	Date.	Per cent water.	Pounds per cubic foot.	Increase or decrease. Pounds per cubic foot.
1	March 8th	24.33	16.98	
1	April 18th	22.37	15.61	−1.37
2	March 8th	15.80	14.05	
2	April 18th	21.64	19.24	+5.19
3	March 8th	11.16	9.81	
3	April 18th	16.24	14.27	+4.46
4	March 8th	7.87	7.36	
4	April 18th	11.19	10.46	+3.10

The rain-fall during the interval was 4.18 inches, equal to 21.77 pounds per square foot.

166. Lateral Capillary Flow.—To determine the lateral capillary flow of water in a soil the following method, used by King^[115] may be employed:

A zinc lined tray, six by six feet in area and eight inches deep, is filled with a soil well packed. In one corner of this tray a section of five inches of unglazed drainage tile, having its lower end broken and jagged, is set and the dirt well filled in round it. By means of a Mariotte bottle water is constantly maintained in the bottom of this tile, three-quarters of an inch deep, so that it will flow laterally by capillary action into the adjacent soil, the object being to determine the extent and rate of capillary flow laterally.

The water content of the soil is determined at the time of starting the experiment, on the circumferences of circles described with the tile as a center, the distance between the circles being one foot. At stated periods, usually at intervals of one day, the content of moisture is again determined at the same points. The investigations show that the lateral movement of water in the soil is not rapid enough to extend much beyond three feet in thirty-one days, for beyond that distance the soil was found to be drier than at the beginning of the experiment. A record is to be kept of the amount of

water delivered to the soil by weighing the supply bottle at intervals, and the rates given at which the soil takes up the water in grams per hour and pounds per day. Also the amount of flow per square foot of soil section together with the mean daily evaporation should be noted. The mean flow per foot of soil section is computed on the assumption that the outer face of the zone of completely saturated soil is the delivering surface. In King's work this point, as nearly as could be determined, was twelve inches from the corner of the tray and hence the figures at best can only be regarded as approximations. The method of stating results is shown in the following table:

SHOWING THE RATE OF LATERAL CAPILLARY FLOW OF WATER IN CLAY LOAM.

Date.	No. of days.	Total mean, hourly flow, grams.	Total mean, daily flow, pounds.	Mean daily flow per square foot, pounds.	Mean daily evaporation, pounds.
Jan. 28 to Feb. 2	5	70.70	3.73	2.38	
Feb. 2-7	5	85.98	4.54	2.91	
Feb. 7-12	5	79.33	4.19	2.64	
Feb. 12-17	5	79.41	4.19	2.64	0.598
Feb. 17-22	5	70.79	3.74	2.38	0.534
Feb. 22-28	6	59.89	3.16	2.01	0.451
Feb. 28 to March 6	6	60.74	3.21	2.04	0.458
Mar. 6-13	7	60.37	3.14	2.00	0.448
Means				2.38	0.498

From this table it will be seen that the flow of water in the soil varied in rate, being slower during the first five days than in the succeeding fifteen days. After twenty days the flow dropped again to the beginning rate and then fell below, but remained quite constant during the following nineteen days. For the sake of uniformity in units of measure the daily quantity of flow should be given in kilograms when the hourly flow is given in grams.

167. Causes of Water Movement in the Soil.—The movement of water in a soil as explained by Whitney^[116] is due to two forces, viz., gravitation and surface tension.

The force of gravitation in a given locality is always uniform, both in direction and magnitude per unit volume of water.

Surface tension is the tendency of any exposed water surface to pull itself together. It may act in any direction, according to circumstances, and may thus sometimes help and sometimes antagonize the force of gravitation.

According to the law of surface tension any particle of moisture tends to assume the smallest possible area. This tendency is a constant definite force per unit of surface at a given temperature. In the soil this constant strain on the free surface of water particles serves, in a high degree, to move them from place to place, in harmony with the requirements of the different portions of the field.

When a soil is only slightly moist the water clings to its grains in the form of a thin film. When these soil particles are brought together the films of water surrounding them unite, one surface being in contact with the soil particles and the other exposed to the air. If more water enter the soil the film thickens until finally, when the point of saturation is reached, all the space between the soil particles becomes filled with water, and surface tension within the soil is thus reduced to zero. Gravity then alone acts on the water and with a maximum force.

In a cubic foot of ordinary soil the total surface of the soil particles will be at least 50,000 square feet. It follows that when the soil is only slightly moist the exposed water surface of the films surrounding the soil particles approximates that of the particles themselves. If such a mass of slightly moist soil be brought in contact with a like mass saturated with water, the films of water at the point of contact will begin to thicken in the nearly dry soil at the expense of the water content of the saturated mass. The water will thus be moved in any direction.

During evaporation the surface tension near the surface of the soil is increased, and water is thus drawn from below. In like manner, when rain falls on a somewhat dry soil, the surface tension is diminished and the greater surface tension below pulls the moisture down even when gravitation would not be sufficient for that purpose.

Certain fertilizers have the faculty of modifying surface tension and thus change the power of the soil in its attraction for moisture. In this way such fertilizers act favorably on plant growth, both by providing plant food and by supplying needed moisture.

168. Surface Tension of Fertilizers.—Whitney gives the following data in respect of the surface tension of aqueous solutions of some of the more

common fertilizing materials. It is expressed in gram meters per square meter, *i. e.*, on a square meter of liquid surface there is sufficient energy to lift the given number of grams to the height of one meter.

SURFACE TENSION OF VARIOUS FERTILIZING SOLUTIONS.

Solution of—	Specific gravity.	Gram meters per square meter.
Salt	1.070	7.975
Kainite	1.053	7.900
Lime	1.002	7.696
Water	1.000	7.668
Acid phosphate	1.005	7.656
Plaster	1.000	7.638
Ammonia	0.960	6.869
Urine	1.026	6.615
Magnesium chlorid	1.1000	7.964
Basic slag	1.0012	7.890
Marl	1.0013	7.855
Potassium chlorid	1.1000	7.853
Ammonium sulfate	1.1000	7.834
Dried blood	1.0001	7.764
Ground bone	1.0007	7.749
Sodium nitrate	1.1000	7.730
Sodium sulfate	1.1000	7.730
Wood ashes	1.0038	7.674
Potassium nitrate	1.1000	7.661
Potassium sulfate	1.0830	7.658
Ammonium nitrate	1.1000	7.656
Dried fish	1.0026	7.594
Stable manure	1.0013	7.464
Cotton-seed meal	1.0054	6.534
Tankage	1.0169	4.844
Cotton seed	1.0070	4.788

SURFACE TENSION OF SOIL EXTRACTS.

Kind of Soil.	Specific gravity.	Surface tension.
Kentucky blue grass	1.000	7.244
Triassic red sandstone	1.000	7.244
Wheat soil	1.000	7.098
Garden soil	1.000	7.089

169. Method of Estimating Surface Tension.—The determination of surface tension is made by measuring the rise of the liquid in a capillary tube. A short piece of thermometer tubing is used, the diameter of the bore being determined by careful microscopic measurements with a micrometer eyepiece. The diameter of the tube should be about 0.5578 millimeter. The tube is very thoroughly cleaned after each observation, or set of observations, with a strong caustic potash solution, and, after washing, is allowed to stand for some time in a saturated solution of potassium bichromate in strong sulfuric acid. The height of the rise in the capillary tube is measured with a cathetometer.

The following formula is used for the calculation of the results:

$$T = \frac{h d \omega}{4 \cos. a}$$

Where T is the surface tension, d is the diameter of the tube in centimeters; h the height to which the liquid rises in the capillary tube in centimeters; ω is the specific gravity of the solution; and $4 \cos. a$ refers to the angle of the liquid with the sides of the glass tube. For a tube of the size given above, $5^{\circ} 24'$ is the value of this edge angle. In regard to saline solutions, Quincke^[117] says, that the edge angle appears to increase a little with augmenting concentration of the saline solution, but otherwise to differ only inconsiderably from the edge angle of pure water.

170. Effect of the Solutions on Surface Tension.—The mineral fertilizers, as a rule, increase the surface tension of water, while organic matters in solution decrease it. But it must not be forgotten in this connection that but little of the organic matter in the fertilizers employed for the experiment passes into solution. Moreover, with these substances, the accuracy of the work is impaired somewhat by the increased viscosity. In general, the results of the experiment are in harmony with the well-known effect of magnesium, sodium, and potassium chlorids, and sodium nitrate,

to make the soil more moist in dry weather, and the opposite effect produced by the application of organic matter.

171. Method of Preparing Soil Extracts.—The soil extracts used in determining the surface tension, as given in the above table, are prepared as follows:

Ten grams of the soil are rubbed up with fifteen to twenty cubic centimeters of distilled water and allowed to stand for twenty-four hours with frequent stirring. Any fine particles not removable by a filter are neglected, although they may give a turbid appearance to the solution.

172. Lysimetry.—The process of measuring the capacity of a soil to permit the passage of water and of collecting and determining the amount of flow and determining soluble matters therein is known as lysimetry. In general, the rate at which water will pass through a soil depends on the fineness and approximation of its particles. Water will pass through coarse sand almost as rapidly as through a tube, while a fine clay may be almost impervious. The study of the phenomena of filtration through soil, and the methods of quantitatively estimating them, are therefore closely related to porosity.

Two cases are to be considered, viz.: First, percolation through samples of soil prepared for analysis, and second, the passage of the water through soil *in situ*, whether it be virgin or cultivated.

The determination of the rate of flow through a soil in laboratory samples, gives valuable information in respect of its physical properties, while the same determination made on the soil *in situ*, has practical relations to the supply of moisture, to growing plants, and the waste of valuable plant food in the drainage waters. The determination of the rate of flow of water through a small sample, disturbed as little as possible in its natural condition, is classed with the first divisions of the work, inasmuch as the removal of a sample of soil from a field, and its transfer to the laboratory, subjects it to artificial conditions, even if its texture be but little disturbed by the removal.

173. Calculation of the Relative Rate of Flow of Water Through Soils.—There will evidently be one space, or opening, into the soil for every surface grain, as pointed out by Whitney,^[118] and the approximate number of grains, or of openings, on a unit area of surface may be found by the following formula:

$$N = \left(\sqrt{\frac{M \times W}{V}}\right)^{\frac{2}{3}}$$

where N is the number of grains, or openings, on one square centimeter of surface, M is the approximate number of grains in one gram of soil, W is the weight of soil, V is the total volume of the soil grains and the empty space.

If the grains are assumed to be symmetrically arranged and the spaces between them cylindrical in form, the radii of the spaces can be found by the following formula:

$$r = \sqrt{\frac{V_e}{\pi N L}}$$

where r is the radius of a single space, V is the total volume of the empty space, N is the number of grains or spaces on one square centimeter of surface, and L is the depth of the soil.

If the space within the soil is completely filled with water the relative rate of flow of water through the soil will be according to the fourth power of the radius of a single space multiplied by the number of spaces on the unit area of surface, as shown by the following formula:

$$T_1 = \frac{N(r)^4 T}{N_1(r_1)^4}$$

where $N-N_1$ are the numbers of spaces, and $r-r_1$ are the radii of single spaces in the respective soils, and $T-T_1$ the times required for a unit volume of water to flow through the soils under the same head or pressure.

The space within the soil is rarely filled with water in agricultural lands, and the most favorable amount of water for the soil to hold, as Hellriegel and others have shown, is from thirty to fifty per cent of the total amount of water the soil can hold if all the space within it were filled.

If the space within the soil be only partly filled with water, as in most arable lands, the water will move in a thin film surrounding the soil grains and according to the fourth power of the thickness of the film. The mean thickness of the film surrounding the soil grains may be theoretically determined by the following formula, which is based on the conception that the film is cylindrical and of uniform size throughout:

$$t = r \left(1 - \sqrt{\frac{s}{s+p}}\right)$$

where s is the per cent by weight of water which the soil will hold when the empty space is filled with water, p the per cent of water actually contained

in the soil, r the radius of a single space, and t the mean thickness of the film surrounding the soil grains.

The relative rate of flow of water through the soils will then be according to the following formula:

$$T_1 = \frac{N(t) \cdot T}{N_1(t_1)^4}$$

It must be remembered that these formulæ give only approximate and comparative values for comparing one soil with another. The structure of the soil is altogether too intricate to expect ever to obtain absolute values.

If the observed rate of flow varies widely from the relative rate calculated from the mechanical analysis, it will indicate a difference in the arrangement of the soil grains, or in the amount or condition of the organic matter in the soils. In the older agricultural regions of the United States, south of the influence of the glacial action, the great soil areas appear to have sensibly similar arrangements of the soil grains, and sensibly uniform conditions of organic matter, save where these have been modified by local conditions.

174. Measurement of Rate of Percolation in a Soil Sample.—In order to measure the power of the soil for permitting the passage of water, a box, about twenty-five centimeters high and having a cross section of about three centimeters square, is used. Below, this box has a funnel-shaped end with a narrow outlet tube, which at its lower end is closed with cotton, in such a way that a portion of the cotton extends through the stem of the funnel. A little coarse quartz sand is scattered over the cotton and afterwards the funnel part of the apparatus filled with it. The sand and the cotton are saturated with water and the apparatus weighed. The box is then filled with the fine sample of earth, with light tapping, until the depth of earth has reached about sixteen centimeters. The apparatus, after the addition of the air-dried earth, is again weighed to determine the amount of earth added, and the soil is then saturated by the careful addition of water. After the excess of water has run down the funnel, the total quantity of absorbed water is determined by reweighing the apparatus and the total water-holding power of the soil is determined. There is carefully added, without stirring up the surface of the soil, a column of water eight centimeters high, making in all from sixty to seventy grams. The time is observed until the water ceases to drip from the funnel. The dripping begins immediately after the water is poured on and ceases as soon as the liquid on the surface of the soil has

completely disappeared. On the repetition of this operation a longer time for the passage of the water is almost always required than at the first time. The experiment, therefore, must be tried three or four times and the mean taken.

175. Method of Welitschowsky.^[119]—The soil is placed in the vessel *a*, Fig. 22, which is cylindrical in shape and five centimeters in diameter. The lower end of the cylinder is closed with a fine wire-gauze disk and the upper end is provided with an enlargement for the reception of the tube *b*, which is connected to *a* with a wide rubber band. The lower end of the tube *b* is also closed with a wire-gauze disk. These tubes may be conveniently made of sheet zinc. The tube *b* carries on the side, at distances of ten centimeters, small tubes of fifteen millimeters diameter. On the opposite side it is provided with a glass tube set into a side tube near the bottom for the purpose of showing the height of the water. The side tube carrying the water meter is provided with a stop-cock as shown in the figure.

In conducting the experiment, after the apparatus has been arranged as described, the small lateral tubes are, with one exception, closed with stoppers. On the open one, *d*, a rubber tube is fixed for the purpose of removing the water. The required water pressure is secured by taking the lateral opening corresponding to the pressure required. Water is introduced into the apparatus slowly through the glass tube *f*.

The water rises to *d* and then any excess flows off through *e*. By a proper regulation of the water supply the pressure is kept constant at *d*. The water flowing off through *a* is collected by the funnel and delivered to graduated flasks where its quantity can be measured for any given unit of time. Since the rate of flow at first shows variations, the measurement should not be commenced until after the flow becomes constant.

In general, the experiments should last ten hours, and, beginning with a water pressure of 100 centimeters, be repeated successively with pressures of eighty, sixty, forty, and twenty, centimeters, etc. In coarse soils, or with sand, one hour is long enough for the experiment.

176. Statement Of Results.—In the following tables the results for ninety centimeters, seventy centimeters, etc., are calculated from the analytical data obtained for 100 centimeters, eighty centimeters, etc.

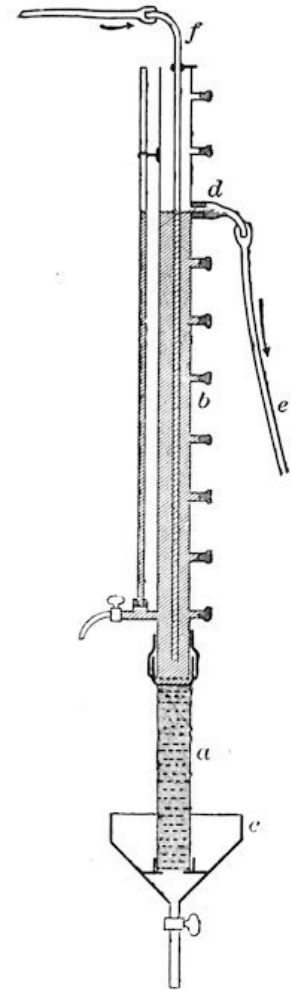


FIGURE 22.

METHOD OF
WELITSCHOWSKY.

MATERIAL—QUARTZ SAND.

No. of Exp.	Diameter of sand particles in mm.	Water pressure in cm.	LITERS OF WATER PASSING IN TEN HOURS.		
			Thickness of Soil Layer.		
			10 cm.	20 cm.	30 cm.
1.	0.01–0.71	10	0.244	0.187	0.151
		20	0.282	0.198	0.154
		30	0.320	0.209	0.158
		40	0.358	0.220	0.161
		50	0.396	0.231	0.165
		60	0.434	0.242	0.168
		70	0.472	0.253	0.172
		80	0.510	0.264	0.175
		90	0.548	0.275	0.179
		100	0.586	0.286	0.182
2.	0.071–0.114	10	2.194	1.724	1.425
		20	2.898	2.012	1.578
		30	3.602	2.300	1.731
		40	4.306	2.588	1.884
		50	5.010	2.876	2.037
		60	5.714	3.164	2.190
		70	6.418	3.452	2.343
		80	7.122	3.740	2.496
		90	7.826	4.028	2.649
		100	8.530	4.316	2.802

Similar sets of data have been collected with powdered limestone, clay and humus.

The general conclusions from the experiments are as follows:

1. Clay (kaolin) and humus (peat) are almost impermeable for water, and fine quartz and limestone dust are also very impermeable.

2. The permeability of a soil for water increases as the particles of the soil increase in size, and when particles of different sizes are mixed together the permeability approaches that of the finer particles.

3. The quantity of water passing through a given thickness of soil increases with the water pressure but is not proportional thereto, increasing less rapidly than the pressure.

4. The quantity of water passing under a given pressure is inversely proportional to the thickness of the soil layer when the particles are very

fine and the pressure high.

177. Method of Whitney.—To determine the permeability of the soil or subsoil to water or air, in its natural position in the field, the following method, due to Whitney, can be recommended:

A hole should be dug, and the soil and subsoil on one side removed to the depth at which the observation is to be made. A column of the soil or subsoil, two inches or more square, and four or five inches deep, is then to be carved out with a broad bladed knife, or a small saw can be conveniently used for cutting this out. A glass or metal frame, a little larger than the sample and three or four inches deep, is slipped over the column of soil, and melted paraffin is run in slowly to fill up the space between the soil and the frame. The soil is then struck off even with the top and bottom of the frame, preferably with a saw, or at any rate taking care not to smooth it over with a knife, which would disturb the surface and affect the rate of flow. The frame is then placed upon some coarse sand or gravel, contained in a funnel, to prevent the soil from falling out and to provide good drainage for the water to pass through. Another similar frame can then be placed on top and secured by a wide rubber band. A little coarse sand, which has been thoroughly washed and dried, is then placed on the soil, and water carefully poured on until it is level with the top of the frame. When the water begins to drop from the funnel more water must be added to the top, so as to have the initial depth of water over the soil the same in all the experiments. A graduated glass is then pushed under the funnel, and the time noted which is required for a quantity of water to pass through the soil. The quantity usually taken for measurement is equivalent to one inch in depth over the soil surface. In taking the sample, root and worm-holes are to be avoided, and these are particularly troublesome in clay lands.

178. Measurement of Percolation through the Soil in Situ.—If lateral translocation could be prevented, the measurement of the quantity of water descending in the soil through a given area would be a matter of simplicity. But to secure accurate results all lateral communication of a given body of soil with adjacent portions must be cut off. Various devices have been adopted to secure this result. An elaborate system of lysimetric measurements is illustrated by the apparatus erected by the Agricultural Experiment Station, of Indiana.

The plan and section of the apparatus are shown in Fig. 23.

Each lysimeter box, when finished, resembles somewhat a hogshhead with one head out. The sides, however, are perfectly straight inside, having a slight thickening in the center, on the outside, for making them stronger. The sides and bottom of the apparatus are constructed of oak and lined with sheet copper carefully soldered so as to be water-tight. Six inches above, and parallel to the bottom of each of the boxes, is a perforated copper tube, which extends entirely across the lysimeter, and passing through one of the sides connects the box with an underground vault in which the observations are taken. These tubes give an outlet to the drainage water, as described further on. The lysimeters are made of any required depth, the two which are shown in section being three and two-thirds and six and two-thirds feet deep, respectively.

The following method is employed for filling them with soil: There are first placed in the bottom of each lysimeter six inches of fine sand, sifted and washed, which fills them up to the level of the drainage tubes. The lysimeters are then filled with fine, sifted surface soil, to the depth of three and six feet, respectively, making a complete pair of lysimeters, and leaving two inches of the lysimeter boxes projecting above the surface of the soil so that each one will receive exactly its proper share of the rain-fall.

The lysimeters of the other pair, which are the same size as the first, are filled in a different way. The lysimeters are first constructed and placed over vertical columns of soil *in situ*, which are obtained by digging away all the surrounding soil and leaving the columns standing. The shorter lysimeter is sunk in this way to within two inches of its entire length. It is then tipped over carrying the column of soil with it. Six inches of the subsoil are then removed, when the drainage tube and sand are put in, as in the first pair, and the bottom of the tube soldered in place. The lysimeter is thus filled with the natural soil in place. The longer box is in the same way filled, as far as possible, with the soil in place, but a gravelly nature of the soil may render it impossible to do the filling with a single column unbroken, so the gravel and sand from the lower portion of the soil are to be filled in separately. The drainage tube and bottom of sand are placed in the longer lysimeter in the same way as in the shorter.

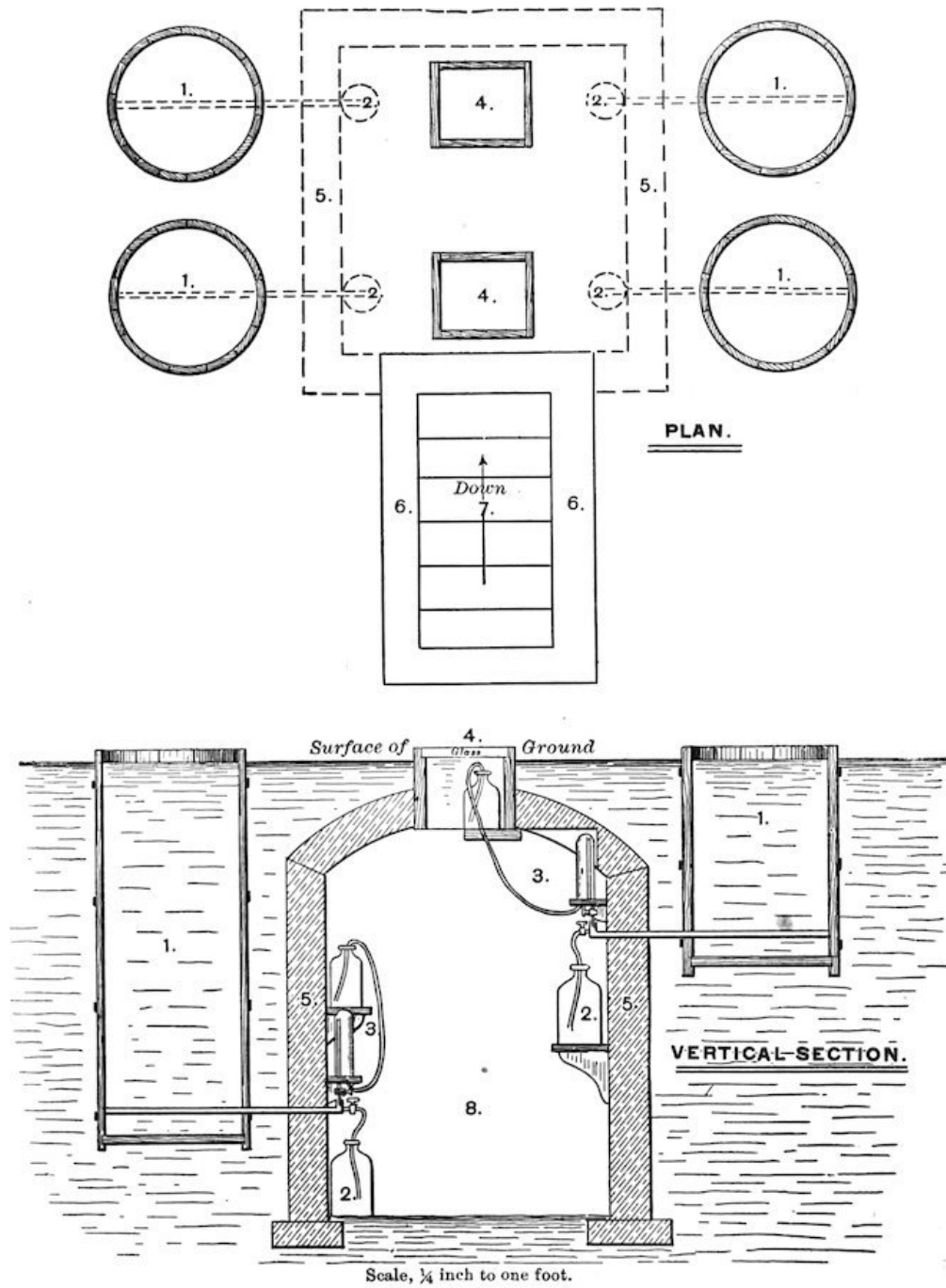


FIGURE 23.

GROUND PLAN AND VERTICAL SECTION OF LYSIMETERS AND VAULTS SHOWING POSITION OF THE APPARATUS.

1, 1, 1, 1, Lysimeters.
2, 2, 2, 2, Receiving bottles.

- 3, 3, Supplying apparatus.
- 4, 4, Skylights.
- 5, 5, 5, 5, Wall of vault.
- 6, 6, Brick walls.
- 7, Entrance Steps.
- 8, Vault.

The purpose of placing sand at the bottom of each lysimeter is to offer a porous stratum in which free water may collect and rise to the level of the perforated copper tube, which would prevent any further rise by conveying the surplus above into the vault as drainage water. The soil above the tube will therefore be constantly drained and the sand below constantly saturated, unless the water be drawn up by the capillary action of the soil as the result of evaporation from the surface.

By means of a proper arrangement within the vault, of a kind of Mariotte's bottle, the water may be caused to flow back through the drainage tube into the lysimeter to take the place of that lost by evaporation, and thus maintain the level of free water just below the drainage tube. The water flowing back to the lysimeter, and the amount of drainage water, are carefully measured by a system of graduated tubes.

The lysimeters thus constructed represent tile-drained land; in one case the tile being three feet below the surface and in the other six feet below. The drainage waters collected in the receiving bottles can be measured and analyzed from time to time, as occasion may require, to determine the amount of plant food which is removed.

179. Improved Method of Deherain.^[120]—Deherain's earlier experiments were made in pots containing about sixty kilos of soil. These vases serve very well for some kinds of plants, but there are other kinds which do not grow at all normally when their roots are imprisoned. For instance, in pots, even of the largest size, wheat is always poor, beets irregular, maize never acquires its full development, and the conclusions which can be drawn from the experiments can not be predicated of the action of the plant under conditions entirely normal.

It is necessary therefore to carry on the work in an entirely different way, and to construct boxes so large as to make the conditions of growth entirely normal. The arrangement of these boxes is shown in Fig. 24.

They are placed in a large trench, two meters wide, one meter deep, and forty meters long. There are twenty boxes in this trench, the upper surface of each containing four square meters area. The boxes are one meter deep, and therefore can contain four cubic meters of soil. The sides and bottoms of the boxes are made of iron lattice work, covered with a cement which renders them impervious to water.

The bottom inclines from the sides towards the middle, and from the back to the front, thus forming a gutter which permits of the easy collection of the drainage. The drainage water is conveyed, by means of a pipe and a funnel, into a demijohn placed in the ditch in front of the apparatus, as shown in the figure. These receptacles stand in niches under the front of the cases, and are separated by the brick foundations. Access to them is gained by means of the inclined plane shown in the figure, and this plane permits the demijohns in which the drainage water is collected, to be removed with a wheelbarrow for the purpose of weighing. This apparatus is especially suitable for a study of the distribution of the nitrogen to the crop, the soil and the drainage waters. The loss in drainage waters of potash and phosphoric acid is insignificant in comparison with the loss in nitrogen.

The cases having been placed in position they are filled with the natural soil, which is taken to the depth of one meter, in such a way that the relative positions of the soil and subsoil are not changed.

While the soil is transferring to the cases it is carefully sampled in order to have a portion representing accurately the composition of both the soil and subsoil. These samples are subjected to analysis and the quantities of nitrogen, phosphoric acid, and potash contained therein carefully noted.

One or two cases should be left without crop or fertilizer to determine the relations of the soil and subsoil to the rain-fall. Three or four cases should be kept free of vegetation and receive treatment with different fertilizer, in order to determine the influences of these on the deportment of the soil to rain-fall. The rest of the cases should be seeded with plants representing the predominant field culture of the locality, and some of them should be fertilized with the usual manures used in farm culture.

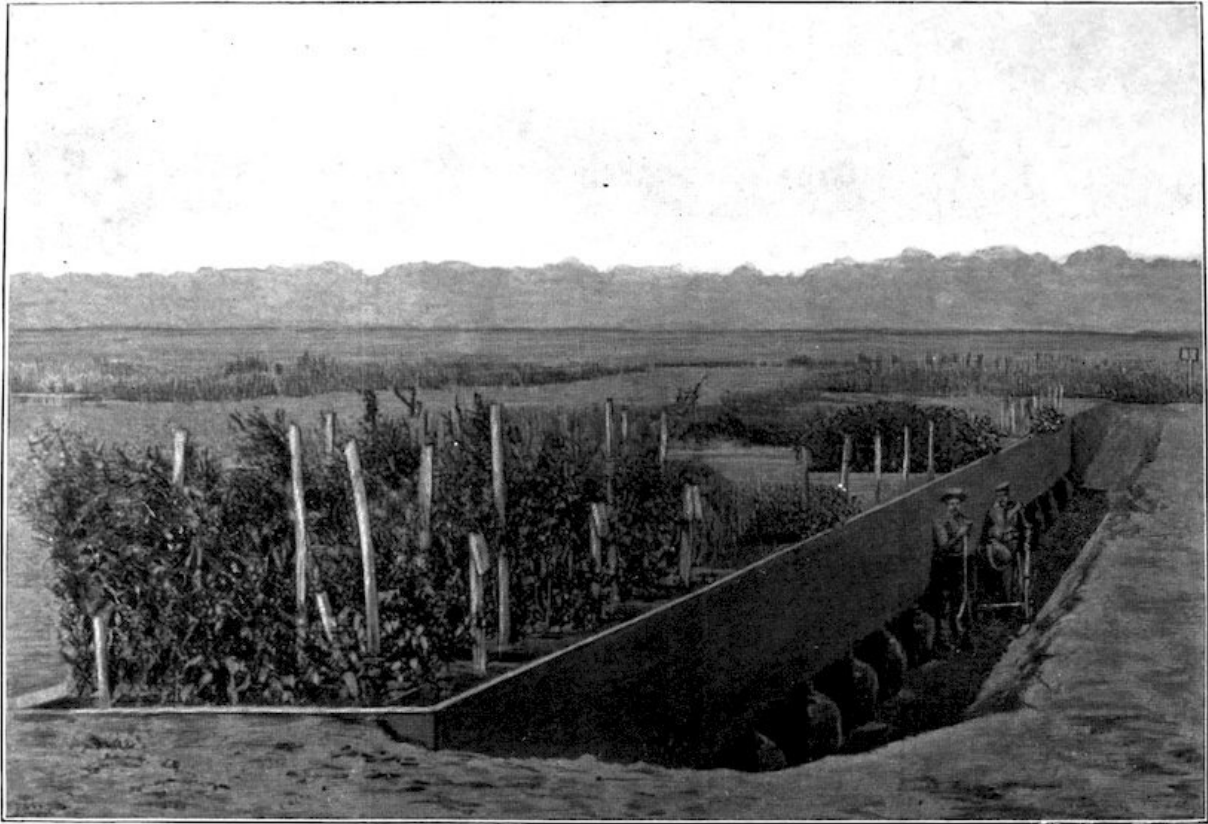


FIGURE 24.

DEHERAIN'S APPARATUS FOR COLLECTING DRAINAGE WATER.

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PART FOURTH.

MECHANICAL ANALYSIS OF SOILS.

THE FLOCCULATION OF SOIL PARTICLES.

180. Relation of Flocculation to Mechanical Analysis.—The tendency of the fine particles of silt to form aggregates, which act as distinct particles of matter, is the chief difficulty connected with the separation of the soil into portions of equal hydraulic value by the silt method of analysis. This tendency has been discussed fully by Johnson^[121] and Hilgard.^[122]

181. Illustration of Flocculation.—A sediment, consisting of particles of a hydraulic value, equal to one millimeter per second, is introduced into an ordinary conical elutriating tube placed vertically, in which the current of water entering below performs all the stirring which the particles receive.

A current of water corresponding to a velocity below one millimeter per second will, of course, not carry any of the particles out at the top of the cylindrical tube, but will keep them moving through the conical portion of the tube. If now the current be increased until its velocity is greater than one millimeter per second after having run at the slower velocity for fifteen or twenty minutes, very little of the sediment will pass over, although theoretically the whole of it should. Even at a velocity of five millimeters per second, much of the sediment will remain in the tube. This, of course, is due to the coagulation of the particles into molecular aggregates having a higher hydraulic value even than five millimeters per second. These

aggregates can be broken up by violent stirring or moderate boiling, and the sediment reduced again to its proper value. The conclusions which Hilgard derives from a study of the above phenomena are as follows:

1. The tendency to coagulation is, roughly, in an inverse ratio to the size of the particles. With quartz grains it practically ceases when their diameter exceeds about two-tenths of a millimeter having a hydraulic value of eight millimeters per second. The size of the aggregates formed follows practically the same law as above. Sediment of 0.25 millimeter hydraulic value will sometimes form large masses like snow-flakes on the sides of the elutriator tube.

2. The degree of agitation which will resolve the aggregates into single grains is inversely as the size of the particles; or, more properly perhaps, inversely as their hydraulic value.

3. The tendency to flocculation varies inversely as the temperature. So much so is this the case that Hilgard at one time contemplated the use of water at the boiling point in the mechanical analysis of soils, in place of mechanical stirring.

4. The presence of alcohol, ether, and of caustic or carbonated alkalies, diminishes the tendency to flocculation, while the presence of acids and neutral salts increases it.

5. As between sediments of equal hydraulic value, but different densities, the tendency to flocculation seems to be greater with the less dense particles.

In regard to the mechanical actions which take place between the particles, Hilgard considers them as irregular spheroids, each of which can at best come in contact at three points with any other particle. The cause of aggregation cannot therefore be mere surface adhesion independent of the liquid, and the particles being submerged there is no meniscus to create an adhesive tension.

Since experiment shows that the flocculative tendency is measurably increased by the cohesion coefficient of the liquid, it seems necessary to assume that capillary films of the latter interposed between the surfaces of solids create a considerable adhesive tension even in the absence of a meniscus.

182. Effect of Potential of Surface Particles.—Whitney suggests that this is due to the potential of the surface particles of solids and liquids.^[123] The potential of a single water particle is the work which would be required to pull it away from the surrounding water particles and remove it beyond their sphere of attraction. For simplicity, it may be described as the total force of attraction between a single particle and all other particles which surround it. With this definition, it will be seen that the potential of a particle on an exposed surface of water is only one-half of the potential in the interior of the mass, as half of the particles which formerly surrounded and attracted it were removed when the other exposed surface of water was separated from it. A particle on an exposed surface of water, being under a low potential, will therefore tend to move toward the center of the mass where the potential, *i. e.*, the total attraction, is greater, and the surface will tend to contract so as to leave the fewest possible number of particles on the surface. This is surface tension.

If, instead of air, there is a solid substance in contact with the water, the potential will be greater than on an exposed surface of the liquid, for the much greater number of solid particles will have a greater attraction for the water particles than the air particles had. They may have so great an attraction that the water particle on this surface, separating the solid and liquid, may be under greater potential than prevails in the interior of the liquid mass. Then the surface will tend to expand as much as possible, for the particles in the interior of the mass of liquid will try to get out on the surface. This is the reverse of surface tension. It is surface pressure, which may exist on a surface separating a solid and liquid.

Muddy water may remain turbid for an indefinite time, but if a trace of lime or salt be added to the water the grains of clay flocculate, that is, they come together in loose, light flocks, like curdled milk, and settle quickly to the bottom, leaving the liquid above them clear. Ammonia and some other substances tend to prevent this and to keep the grains apart if flocculation has already taken place.

If two small grains of clay, suspended in water, come close together they may be attracted to each other or not, according to the potential of the water particles on the surface of the clay. If the potential of the surface particle of water is less than that of the particle in the interior of the mass of liquid, there will be surface tension, and the two grains will come together and be

held with some force, as their close contact will diminish the number of surface particles in the liquid. If, on the other hand, the potential of the particle on the surface of the liquid is greater than of the particle in the interior of the mass, the water surface around the grains will tend to enlarge, as there will be greater attraction for the water particles there than in the interior of the mass of liquid, and the grains of clay will not come close together and will even be held apart, as their close contact would diminish the number of surface particles in the liquid around them.

183. Influence of Surface Tension.—Hilgard supposes that the surface tension which is assumed to exist between two liquid surfaces must exert a corresponding influence between the surfaces of solids and liquids, apart from any meniscal action.

It is then to be expected that the adhesion of the particles constituting one of these floccules will be very materially increased whenever the formation of menisci between them becomes possible by the removal of the general liquid mass. Suppose one of the floccules to be stranded, it will, in the first place, remain immersed in a sensibly spherical drop of liquid. As this liquid evaporates, the spherical surface will become pitted with menisci forming between the single projecting particles, and as these menisci diminish their radius by still further evaporation, the force with which they hold the particles together will increase until it reaches a maximum. As the evaporation progresses beyond this point of maximum, the adhesion of the constituent particles must diminish by reason of the disappearance of the smaller menisci, and when finally the point is reached when liquid water ceases to exist between the surfaces, the slightest touch, or sometimes even the weight of the particles themselves, will cause a complete dissolution of the floccule, which then flattens down into a pile of single granules.

In regard to natural deposits from water, Hilgard supposes that they are always precipitated in a flocculated state. The particles of less than two-tenths millimeter diameter are carried down with those of a larger diameter having much higher hydraulic value. Thus the deposition of a pure clay can take place under only very exceptionable circumstances.

Whitney, on the other hand, suggests that grains of sand and clay carry down mechanically the particles of fine silt and clay as they settle in a turbid liquid in a beaker; and it is often difficult to wash out a trace of fine

material from a large amount of coarse particles, for this reason, although there may be no trace whatever of flocculation.

184. Destruction of Floccules.—The destruction of the natural floccules is seen in the ordinary process of puddling earth or clay. It is also the result of violent agitation of water or of kneading or boiling, or, finally, to a certain extent, of freezing. All these agencies are employed by the workers in clay for the purpose of increasing the plasticity which depends essentially upon the finest possible condition of the material to be worked. As an illustration of this, Hilgard cites the fact that any clay or soil which is worked into a plastic paste with water, and dried, will form a mass of almost stony hardness. If, however, to such a substance one-half per cent of caustic lime be added, a substance which possesses in an eminent degree the property of coagulating clay, the diminution of plasticity will be obvious at once, even when in a wet condition. If now the mass be dried, as in the previous case, it is easily pulverized. This is an illustration of the effect of lime upon stiff lands, rendering them more readily pulverulent and tillable. The conversion of the lime into a carbonate in the above experiment by passing bubbles of carbonic acid through the mass while still suspended in water does not restore the original plasticity, thus illustrating experimentally the fact known to all farmers that the effect of lime on stiff soil lasts for many years, although the whole of the lime in that time has been converted into carbonate.

185. Practical Applications.—The practical application of this is, according to Hilgard, that the loosely flocculated aggregation of the soil particles is what constitutes good tilth. For this reason the perfect rest of a soil, if it is protected from the tamping influence of rains and the tramping of cattle, may produce a condition of tilth which cannot be secured by any mechanical cultivation. As an illustration of this, the pulverulent condition of virgin soils protected in a forest by the heavy coating of leaves may be cited. On the contrary, as pointed out by Hilgard, there are some kinds of soil in which a condition of rest may produce the same effect as tamping. These are soils which consist of siliceous silt without enough clay to maintain them in position after drying. In such a case, the masses of floccules collapse by their own weight or by the least shaking, and fall closely together, producing an impaction of the soil. This takes place in some river sediment soils in which the curious phenomenon is presented of injurious effects produced by plowing when too dry, which is the direct

opposite of soils containing a sufficient amount of clay and which are injured by plowing too wet.

It is further observed that the longer a soil has been maintained in good tilth, the less it is injured by wet plowing. This is doubtless, according to Hilgard, due to the gradual cementation of the floccules by the soil water which fixes them more or less permanently.

Whitney believes that the arrangement of the grains, or the condition of flocculation in the soil, or the distance apart of the soil grains, is determined, to a large extent, by the potential on the surface of the grains; and he suggests that by changing this the exceedingly fine grains of silt and clay can be pulled together or can be pushed further apart, and so alter the whole texture of the land.

The action of alkaline carbonates in preventing flocculation, and thus rendering tillage difficult or impossible, is pointed out by Hilgard in the case of certain alkali soils of California. The soils which are impregnated with alkaline carbonates are recognized by their extreme compactness. The suggestion of Hilgard to use gypsum on such soils has been followed by the happiest results. This gypsum renders any phosphates present insoluble, and thus prevents loss by drainage, and yet leaves the plant food in a sufficiently fine state as to be perfectly available for vegetation.

186. Suspension of Clay in Water.—The suspension of clay in water and the methods of producing or retarding flocculation and precipitation have also been studied by Durham.^[124] His experiment is made as follows:

In a number of tall glass jars fine clay is stirred with water, and the results of precipitation watched. In all cases it will be noticed that the clay rapidly separates into two portions, the greater part quickly settling down to the bottom of the jars, and the smaller part remaining suspended for a greater or less length of time.

The power which water possesses of sustaining clay is gradually destroyed by the addition of an acid or salt; a very small quantity, for instance, of sulfuric acid, is sufficient to precipitate the clay with great rapidity. In solutions of sulfuric acid and sodium chlorid of varying strengths, suspended clay is precipitated in the order of the specific gravity of the solutions, the densest solutions being the last to clear up. This may be due to the greater viscosity of the denser liquids.

The power which water possesses of sustaining clay is gradually decreased by the addition of small quantities of certain salts and of lime.

187. Effect of Chemical Action—Brewer^[125] emphasizes the importance of chemical action in the flocculation of clays. As expressed by him the chemical aspects of the phenomena of sedimentation have either been lightly considered or entirely ignored. Brewer is led to believe that the action of clay thus suspended is analogous to that of a colloidal body. Like a colloid, when diffused in water, the bulk of the mass is very great, shrinking enormously on drying. He therefore concludes that clays probably exist in suspension as a series of hydrous silicates feebly holding different proportions of water in combination and having different properties so far as their behavior to water is concerned.

Some of them he supposes swell up in water much as boiled starch does, and are diffusible in it with different degrees of facility, and that the strata observed on long standing of jars of suspended clay represent different members of this series of chemical compounds which hold their different proportions of combined water very feebly and are stable under a very limited range of conditions.

These compounds are probably destroyed or changed in the presence of acids, salts and various other substances, and are stable only under certain conditions of temperature, those which exist at one temperature being destroyed or changed to other compounds at a different temperature.

188. Theory of Barus.—Brewer's hypothesis, however, is not in harmony with the demonstration of Barus, who proves that a given particle of clay has the same density in ether as in water.

The physical and mathematical aspects of sedimentation have also been carefully studied by Barus.^[126] The mathematical conditions of a fine particle suspended in a liquid and free from the influences of flocculation are described by Barus in the following equations.

If P be the resistance encountered by a solid spherule of radius r , moving through a viscous liquid at the rate x , and if k be the frictional coefficient, then $P = 6\pi krx$. Again, the effective part of the weight of the particle is $P' = \frac{4}{3}\pi r^3 (\rho - \rho')g$, where g is the acceleration of gravity and ρ and ρ' the density of solid particle and liquid, respectively. In case of uniform motion $P = P'$. Hence $x = \frac{2}{9k}r^2 (\rho - \rho')g \dots (1)$.

In any given case of thoroughly triturated material the particles vary in size from a very small to a relatively large value; but by far the greater number approach a certain mean figure and dimension. An example of this condition of things may be formulated. To avoid mathematical entanglement let $y = Ax^{\frac{3}{2}}e^{-x^2} \dots$ (2) where y is the probable occurrence of the rate of subsidence x . If now the turbidity of the liquid (avoiding optical considerations) be defined as proportional to the mass of solid material particles suspended in unit of volume of liquid, then the degree of turbidity which the given ydx particles add to the liquid is, *caeteris paribus*, proportional to r^3ydx , where r is the mean radius. Hence the turbidity, T , at the outset of the experiment (immediately after shaking), is $T = T_0 \int_0^\infty r^3 y dx = T_0$, where equations (1) and (2) have been incorporated.

If the plane at a depth d below the surface of the liquid be regarded, then at a time after shaking the residual turbidity is

$$(3) \dots T_d = T_0 \int_0^d r^2 y dx = T_0 \left(1 - \left(1 + \frac{d^2}{t^2} e^{-\frac{d^2}{t^2}} \right) \right)$$

The equation describes the observed occurrences fairly well.

The phenomena of stratification observed by Brewer are explained by Barus from the above formula: In proportion as the time of subsidence is greater, the tube shows opacity at the bottom, shading off gradually upward, through translucency, into clearness at the top. If, instead of equation (2), there be introduced the condition of a more abrupt maximum, if, in other words, the particles be very nearly of the same size, then subsidence must take place in unbroken column capped by a plane surface which at the time zero coincided with the free surface of the liquid. Again, suppose one-half of the particles of this column differ in some way uniformly from the other half. Then at the outset there are two continuous columns coinciding, or, as it were, interpenetrating throughout their extent. But the rate of subsidence of these two columns is necessarily different, since the particles, each for each, differ in density, radius and frictional qualities, by given fixed amounts. Hence the two surfaces of demarcation at the time zero coincided with the free surface. In general, if there be n groups of particles uniformly distributed, then at the time zero n continuous columns interpenetrate and coincide throughout their extent. At the time t , the free surface will be

represented by n consecutive surfaces of demarcation below it, each of which caps a column, the particles of which form a distinct group.

From a further discussion of the mathematical condition under which the subsidence of the particles takes place, Barus is of the opinion that Durham's theory of suspension being only a lower limit of solution is rapidly gaining ground, yet without being attended with concise experimental evidence which will account for the differences in the rate of subsidence. On the contrary, Brewer's hypothesis of colloidal hydrates is more easily subjected to experimental proof. The test shows that the particles retain their normal density, no matter how they are suspended or circumstanced.

Further, in the explanation of the phenomenon of sedimentation, the following principle may be regarded as determined; namely, if particles of a comminuted solid are shaken up in a liquid, the distribution of parts after shaking will tend to take place in such a way that the potential energy of the system of solid particles and liquid, at every stage of subsidence, is the minimum compatible with the given conditions.

According to Barus it is necessary, in order to pass judgment on the validity of any of the given hypotheses, to have in hand better statistics of the size of the particles relatively to the water molecule, than are now available. Inasmuch as the particles in pure water are individualized and granular, it is apparently at once permissible to infer the size of the particles from the observed rates of subsidence. His observations show that the said rate decreases in marked degree with the turbidity of the mixture. Hence the known formulæ for single particles are not rigorously applicable, though it cannot be asserted whether the cause of discrepancy is physical or mathematical in kind. It follows that special deductions must be made for the subsidence of stated groups of particles before an estimate of their mean size can fairly be obtained.

Rowland^[127] reaches a closer approximation for the fall of a single particle by showing that the liquid, even at a large distance from the particle, is not at rest.

In the case of water, however, it is noticed that despite the large surface energy of the liquid, subsidence takes place in such a way that for a given mass of suspended sediment the surfaces of separation are a maximum. On the other hand, in case of subsidence in ether or in salt solutions, the solid

particles behave much like the capillary spherules of a heavy liquid shaken up in a lighter liquid with which it does not mix. In other words, the tendency here is to reduce surfaces of separation to the least possible value, large particles growing in mass and bulk mechanically at the expense of smaller particles; in other words, exhibiting the phenomenon of flocculation.

189. Physical Explanation of Subsidence.—Whitney^[128] thinks that the phenomena of the suspension of clay in water may be explained on purely physical principles, and that neither the partial solution nor hydration hypotheses are necessary, or will explain the suspension of clay in water, for the solution, or hydrated substance, would still have a higher specific gravity than the surrounding liquid. He calls attention in the first place to the fact, that in a turbid liquid, which has been standing for weeks and which is only faintly opalescent, the grains in suspension are still of measurable size, if properly stained as in bacteriological examinations and viewed through an oil emersion objective. He gives a value of 0.0001 millimeter, as the lower limit of the diameters of these particles of “clay,” which are usually met with in agricultural soils. He refers to the fact that fine dust and ashes, and even filings of metals, may remain in suspension in the air for days and even months in very apparent clouds, or haze, although they may be a thousand times heavier than the surrounding air. Particles of clay, no smaller than the limits which have been assigned, should remain in suspension in the much heavier fluid, water, for an indefinite time, for the volume or weight of the particles $(\frac{4}{3})(\pi r^3)$ decreases so much more rapidly in proportion than the surface $(4\pi r^2)$, that there is, relatively, a larger amount of surface area in these fine clay particles, and a great deal of surface friction in their movement through a medium, and they would settle very slowly. Under ordinary conditions, however, the mean daily range of temperature is about twenty degrees, the mean monthly range is fifty degrees, and the yearly range 100° F., and the ordinary convection currents, induced by the normal change of temperature, would be sufficient of itself to keep these fine particles in suspension in the liquid for an indefinite time, as it is known that currents of air keep fine particles of dust and ashes in suspension. If the volume or weight of a fine gravel, having a diameter of one and five-tenth millimeters, be taken as unity, then for a particle, having a diameter of 0.00255 millimeter, which is the mean diameter for Whitney’s

clay group, the volume decreases in the ratio 1:0.000000004853, and the surface decreases only in the ratio 1:0.000286.

190. Practical Applications.—The action of mineral substances in promoting flocculence has been taken advantage of in later times in the construction of filters for purifying waters holding silt in solution. In these filters the introduction of a small quantity of alum, or some similar substance, into the water usually precedes the mechanical separation of the flocculent material. In the same way the action of iron and other salts on sewage waters has been made use of in their purification and in the collection of the sewage material for fertilizing purposes.

191. Separation of the Soil Into Particles of Standard Size.—The agronomic value of a soil depends largely on the relative size of the particles composing it. The finer the particles, within a certain limit, the better the soil. The size of the particles may be estimated in three ways: (1) by passing through sieves of different degrees of fineness; (2) by allowing them to subside for a given time in water at rest; (3) by separating them in water moving at a given rate of speed. The first method is a crude one and is used to prepare in a rough way, the material for the second and third processes.

192. Separation in a Sieve.—The soil should be dry enough to avoid sticking to the fingers or to prevent agglutination into masses when subjected to pressure. It should not, however, be too dry to prevent the easy separation of any agglutinated particles under the pressure of the thumb or of a rubber pestle.

The sieve should have circular holes punched in a sheet of metal of convenient thickness to give it the requisite degree of strength. Sieves made of wire gauze are not so desirable but it is difficult to get the finer meshes as circular perforations. Such sieves cannot give a uniform product on account of the greater diagonal diameter of the meshes and the ease with which the separating wires can be displaced. It is convenient to have the sieves arranged *en batterie*; say in sets of three. Such a set should have the holes in the three sieves of the following dimensions; viz.,

1st	sieve	2	millimeters	diameter.
2nd	„	1	millimeter	„
3rd	„	0.5	„	„

Coarser single sieves may be used to separate the fragments above two millimeters diameter if such a further classification be desired. Each sieve fits into the next finer one and the separation of a sample into three classes of particles may be effected by a single operation. In most cases, however, it is better to conduct each operation separately in order to promote the passage of agglutinated particles by gentle pressure with the thumb or with a rubber pestle. In no case should a hard pestle be used and the pressure should never be violent enough to disintegrate mineral particles.

There is much difference of opinion concerning the smallest size of particles which should be obtained by the sieve.

Most analytical processes prescribe particles passing a sieve of one millimeter mesh ($\frac{1}{25}$ inch). There is little doubt, however, of the fact that a finer particle would be better fitted for subsequent analysis by the hydraulic method.

For this purpose a sieve of 0.5 millimeter circular mesh is preferred.

193. Sifting with Water.—In soils where the particles adhere firmly the sifting should be done with the help of water. In such cases the soil is gently rubbed with a soft steeple or the finger in water. It is then transferred to the sieve or battery of sieves which are held in the water, and rubbed through each of the sieves successively until the separation is complete. After the filtrate has stood for a few minutes the supernatant muddy liquor is poured off, the part remaining on the sieve is added to it and the process repeated until only clean particles larger than 0.5 millimeter are left on the sieve. These particles can be dried and weighed and entered on the note book as sand. The filtrate should be evaporated to dryness at a gentle temperature and when sufficiently dry be rubbed up into a homogeneous mass by a rubber pestle.

The sieve recommended by the Association of Official Agricultural Chemists^[129] for the preparation of fine earth for chemical analysis has circular openings $\frac{1}{25}$ inch (one millimeter) in diameter.

Wahnschaffe^[130] directs that a sieve of two millimeters mesh be used in preparing the sample for silt analysis and that the residue after the silt analysis is finished, which has not been carried over by a velocity of twenty-five millimeters per second, be separated in sieves of one millimeter and 0.5 millimeter meshes respectively.

Hilgard objects to leaving this coarse material in the sample during the process of churn elutriation on account of the attrition which it exerts and therefore directs that it be separated by sieve analysis before the elutriation begins.

194. Method of the German Experiment Stations.^[131]—In the method recommended for the German Agricultural Stations an attempt is made to secure even a finer sieve separation than that already mentioned.

Sieves having the following dimensions are employed; sieve No. 1, square meshes 0.09 millimeter in size, diagonal measure 0.11 millimeter; sieve No. 2, square meshes 0.14 to 0.17 millimeter in diameter, diagonal measure 0.22 to 0.24 millimeter; sieve No. 3, square meshes 0.35 to 0.39 millimeter in diameter, diagonal measure 0.45 to 0.50 millimeter; finally a series of sieves one, two and three millimeters circular perforations.

Five hundred grams of the soil (in the Halle Station only 250) are placed in a porcelain dish with about one liter of water and allowed to stand for some time with frequent stirring, on a water bath. After about two hours, when the soil is sufficiently softened so that with the help of a pestle it can be washed through the sieves, the process of sifting is undertaken in the following manner: Sieve No. 3 is placed over a dish containing water, the moistened soil placed therein and the sieve depressed a few centimeters under the water and the soil stirred by means of a pestle until particles no longer pass through. After the operation is ended the residue in the sieve is washed with pure water and dried. The part passing the sieve is thoroughly stirred and then washed with water into sieve No. 2 and treated as before. The product obtained in this way is brought into sieve No. 1 and carefully washed. All the products remaining on each of the sieves are dried at 100° and weighed. The portion passing sieve No. 1 is either dried with its wash water or estimated by loss by deducting from the total weight taken, the sum of the other weights obtained. If a more perfect separation of the first sieve residue be desired it can be obtained by passing it through sieves of the last series which may have meshes varying in size, viz.: one, two, or three millimeters in diameter. Each sieve of the same class should have holes uniformly of the same size.

The sieve products are characterized as follows: The part passing a three millimeter sieve is called fine earth, while the part remaining is called gravel. The fine earth is separated into the following products: The part that

passes through the three millimeters opening and is left by the two millimeters opening is called *steinkies*. The product from the two millimeters opening and the residue from the one millimeter opening is called *grobkies*. The product from the one millimeter opening and the residue on the sieve No. 3 is called *feinkies*. The product from the sieve No. 3 and the residue from the sieve No. 2 is called coarse sand. The product from sieve No. 2 and the residue from sieve No. 1 is called fine sand. The product from sieve No. 1 is called dust. The dust can be further separated into sand, dust, and clay. For the examination of the clay the Kühn silt cylinder as modified by Wagner, is recommended. The cylinder has a diameter of eight centimeters and a height of thirty centimeters, and is furnished with a movable exit tube reaching to its bottom.

195. General Classification of the Soil by Sieve Analysis.—The classification recommended by the German chemists is satisfactory but the following one is more simple. All pebbles, pieces of rock, etc., should first be separated by a two millimeters circular mesh sieve, dried at 105° and weighed. The result should be entered as pebbles and coarse sand.

The finer sand may be separated with a sieve of one millimeter circular openings.

The still finer sand is next separated with the sieve of 0.5 millimeter circular openings as indicated above.

The sample may now be classified as follows:

1. Coarse pebbles, sticks, roots, etc., separated by hand.
2. Pebbles and coarse sand not passing a two millimeters sieve.
3. Sand not passing a one millimeter sieve.
4. Fine sand not passing a 0.5 millimeter sieve.
5. Fine earth passing a 0.5 millimeter sieve.

196. Classification of Orth.^[132]—As fine silt are reckoned those particles which range from 0.02 to 0.05 millimeter; as fine sand the groups from 0.05 to 0.2 millimeter; as medium sized sand those ranging from 0.2 to 0.5 millimeter and for large grained sand those particles ranging from 0.5 to 2 millimeters in diameter. Particles over two millimeters form the last classification.

SEPARATION OF THE EARTH PARTICLES BY A LIQUID.

197. Methods of Silt Analysis.—The further classification of the particles of a soil passing a fine sieve can best be effected by separation in water. The velocity with which the current moves or with which the particles subside will cause a separation of the particles into varying sizes. The slower the velocity the smaller the particles which are separated. There is, however, a large and important constituent of a soil which remains suspended in water, or in a state of seeming solution. This suspended matter would still be carried over by a current of water moving at a rate so slow as to make a subclassification of it impossible. This suspended matter passing off at a given velocity may be classed as clay, and it consists in fact chiefly of the hydrated silicate of alumina, or other particles of equal fineness. The laws which govern its deposit have already been discussed.

The apparatus which have been used for silt analysis may be grouped into four classes.

(1) Apparatus depending on the rate of descent of the particles of a soil through water at rest. The apparatus for decanting from a cylinder or a beaker belong to this class.

(2) Apparatus which determine the rate of flow by passing the liquid through a vessel of conical shape. The system of Nöbel is a good illustration of this kind of apparatus.

(3) Apparatus in which the elutriating vessel is cylindrical and the rate of flow determined by a stop-cock or pressure feed apparatus. The system of Schöne represents this type.

(4) Apparatus in which the above system is combined with a device for mechanically separating the particles and bringing them in a free state into the elutriating current. The system of Hilgard is the type of this kind of apparatus.

In practice the use of cylindrical apparatus with or without mechanical stirring and the method by decantation have proved to be the most reliable and satisfactory procedures. Between the beaker and churn methods, of separation there is little choice in regard to accuracy. Which is the superior method, is a question on which the opinions of experienced analysts are

divided. The various processes will be described in the order already mentioned.

198. Methods Depending on Subsidence of Soil Particles.—The simplest method of effecting the further separation of the soil particles is without doubt that process which permits them to fall freely in a liquid sensibly at rest. The practical difficulties of this method consist in the trouble of securing a perfect separation of the particles, in preventing flocculation after division and in avoiding currents in the liquid of separation.

For the separation of the soil particles for this method boiling and wet pestling are the only means employed. The flocculation of the separated particles may be partially prevented by adding a little ammonia to the water employed. The author has also tried dilute alcohol as the separating liquid but the results of this method are not yet sufficiently definite to find a place in this manual. Evidently the practical impossibility of avoiding convection currents prevents the use of water at a high temperature for this separation, although the tendency to flocculation almost disappears as the temperature approaches 100°. The general method of avoiding the errors due to flocculation in the subsidence method consists in repestling the deposited particles and thus subjecting them as often as may be necessary to resedimentation. These principles are well set forth by Osborne,^[133] who states that when a soil is completely suspended in water by vigorous agitation, particles of all the sizes present are to be found throughout the entire mass of liquid. When subsidence takes place, the larger particles will go down more rapidly than the smaller ones, but some of the small particles that are near the bottom will be deposited sooner than some of the larger ones which have a much greater distance to travel. Thus, independently of the fact that the larger particles in their descent are somewhat impeded by the smaller, the smaller being at the same time somewhat hastened by the larger, the sediment that reaches the bottom at any moment is a more or less complex mixture of all the mechanical elements of the soil. The liquid, however, above this sediment at the same moment will have completely deposited all particles exceeding certain dimensions of hydraulic value, determined mainly by the time of subsidence.

If now the aforesaid first sediment be suspended in pure water, and allowed to subside for the same time as before, the larger part of it will be

again deposited, but some will remain in suspension, consisting of a considerable part of the finer matter of the first sediment. By pouring off these suspended particles with the water and agitating the sediment again with clear water as before, another portion of fine particles will be suspended and may be decanted from it. On continuing this process of repeated decantations it will soon be found that the soil has been separated into two grades.

It is evident that in this way a separation can be made, but it is perhaps not so clear that such a separation would be sharp enough for the purposes of a mechanical soil analysis. If, for instance, the separation is to be made at 0.05 millimeter diameter, it is evident that by repeated decantations all below 0.01 millimeter can be washed out of that above 0.05 millimeter, but it may not appear so probable that all below 0.045 millimeter can be removed without removing some above 0.055 millimeter.

Such a result may be easily attained, however, if the following principle be adhered to:

Make the duration of the subsidence such that the liquid decanted the first few times shall contain nothing larger than the desired diameter. Then decant into another vessel, timing the subsidence so that the sediment shall contain nothing smaller than the chosen diameter. This can not be done without decanting much that is larger than the chosen diameter, but the greater part of the particles greater and less than the chosen diameter can be removed and an intermediate product obtained, the diameters of whose particles are not very far from that desired.

If this intermediate portion be again subjected to the same process, two fractions may be separated from it, one containing particles larger than the chosen diameter and another containing particles smaller than this diameter, while a new intermediate product will remain which is less in amount than that resulting from the first operation. By frequent repetitions of this process this intermediate product can be reduced to a very small amount of substance the particles of which have diameters lying close to the chosen limit and may then be divided between the two fractions.

The principles of the separation described by Osborne set forth with sufficient clearness the purposes to be achieved by the analysis. The chief methods of manipulation practiced will be found below.

199. Kühn's Silt Cylinder.—A simple form of apparatus for the determination of silt by the sedimentation process is the one described by Kühn.^[134]

The cylinder should be about twenty-eight centimeters high with a diameter of 8.5 centimeters. At the lower end of the cylinder five centimeters from the bottom it carries a tube 1.5 centimeter in diameter furnished with a pinch-cock and held in position by a rubber stopper.

In carrying out the process thirty grams of sifted soil (two millimeters mesh sieve) are boiled with water for an hour and after cooling the soil and water are washed into the separating cylinder. The cylinder is then filled with water with constant shaking.

After standing for ten minutes the stop-cock is opened and the water with its suspended matter allowed to flow into a porcelain dish.

The cylinder is then again filled with water and the process is continued until the water drawn off is practically clear.

The fine particles having been separated in this way the next coarser grade of particles is separated by repeating the process at intervals of five minutes.

By these two operations it is considered that the clay is entirely removed. The residue remaining in the cylinder is dried and weighed. The relative proportions of clay and residue in the sample are thus determined.

The residue is then separated into two portions by sieves of one millimeter and 0.5 millimeter mesh.

The soil is thus separated into the following parts:

1. By the first sifting coarse quartz larger than two millimeters diameter.
2. Fine quartz two millimeters, to one millimeter diameter.
3. Coarse sand one millimeter, to 0.5 millimeter diameter.
4. Fine sand finer than 0.5 millimeter diameter.
5. Silt, clay, humus, etc., separated by the water.

200. Knop's Silt Cylinder.—The cylinder recommended by Knop^[135] is essentially that of Kühn being furnished with four lateral tubes instead of one (Fig. 25).

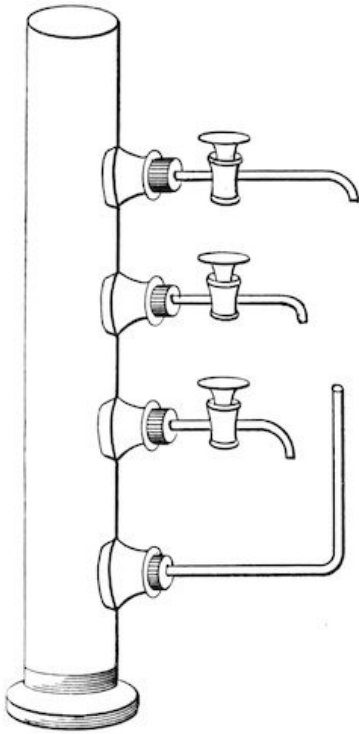


FIGURE 25.

KNOP'S SILT CYLINDER.

The sample of soil, twenty-five to thirty grams, after passing a two millimeters mesh sieve, and long boiling, is washed through a series of sieves of the following diameters of mesh respectively; viz., one millimeter, 0.5 millimeter, 0.25 millimeter, and 0.1 millimeter. The part which passes the finest sieve is placed in a Knop's cylinder, the cylinder filled with water one decimeter above upper tube and well shaken. The cylinder is allowed to rest for five minutes when the upper cock is opened and the water drawn off. After five minutes more the next tube is opened and so on with equal intervals for the three upper tubes. The operation is repeated with fresh water until the water drawn off is clear. Finally the lowest tube is opened and all the water poured off of the sandy residue. The space between each tube is one decimeter. The dust remaining is dried and weighed and the weight of material carried over as silt determined by difference.

201. Siphon Silt Cylinder.—Instead of the tubulated cylinder one furnished with a siphon can be employed^[136] (Fig. 26).

It should be about forty centimeters high and six centimeters internal diameter. The cylinder first receives twenty-five to thirty grams of the well boiled fine earth and then water until there is but a small space between it and the stopper when the latter is inserted.

The cylinder is marked exactly 200 millimeters below the surface of the water with a narrow strip of paper at *a*, stoppered, inverted and well shaken. The cylinder being again placed in normal position the soil particles under the influence of gravity tend to sink with greater or less rapidity according to their size. The siphon *a b c* is filled with water, the cock at *c* closed and the opened end *a* placed in the cylinder A just at the mark 200 millimeters below the surface of the water, and the water thus transferred to B when desired. If the suspended matter is allowed to stand for 100 seconds the particles of more than two millimeters hydraulic value will have fallen below the open end of the siphon. If allowed to stand 1,000 seconds the silt

value of the particles will be 0.2 millimeter per second. Whatever the number of seconds may be, the operation is continued until the water removed is practically clear. The open end of the siphon *a* should be bent upwards so that no disturbing current may bring the particles below the line into the liquid discharged into B.

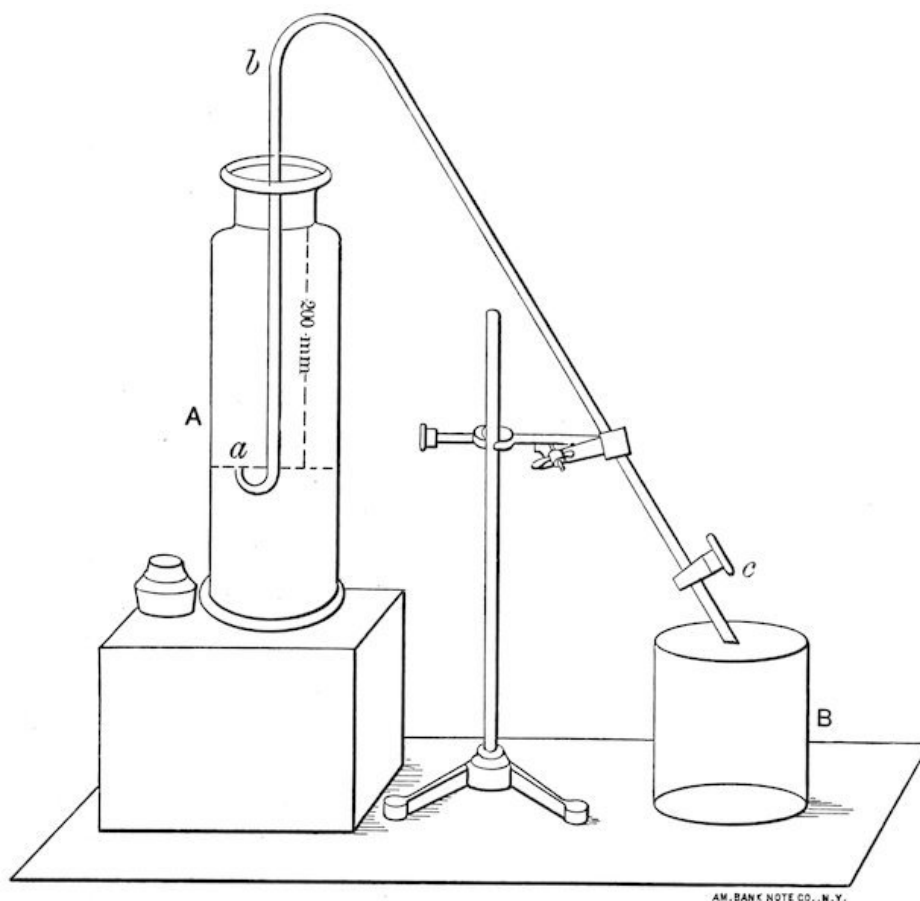


FIGURE 26.

SIPHON CYLINDER FOR SILT ANALYSIS.

While the results obtained by this method are satisfactory as compared with other similar processes it cannot be highly recommended because of the time and trouble required to get a complete separation and by reason of the difficulty of collecting the separated silt.

202. Wolff's Method.^[137]—As modified by Wolff the Knop process is conducted as follows: Fifty grams of fine earth are boiled with water and then the entire mixture is passed through three sieves with openings of one

millimeter, 0.5 millimeter, and 0.25 millimeter in diameter, respectively. The finest part is mixed with water to a height of eighteen centimeters in a bottle twenty centimeters high and having a capacity of one liter and thoroughly agitated, after which it is left to rest, and finally the turbid liquid is drawn off with a siphon, the bottle refilled with water, agitated, and left to rest, and the process repeated as long as the water carries any suspended matter after a definite time.

Wolff proposes for the first three periods of rest one hour, for the second three, a half an hour, for the third three, a quarter of an hour, and for the fourth three, five minutes.

203. Moore's Modification of Knop's Method.^[138]—The sample of soil is first passed through a sieve having round perforations three millimeters in diameter. The weight of the particles remaining on the sieve is then determined, and likewise that of the portion passing through, which is known as fine earth. The last named portion constitutes the material for all subsequent operations of mechanical and chemical analysis.

Thirty grams of the fine earth are boiled rapidly with water until the lumps are disintegrated and clayey portions separated from the sand. The material is then successively washed through perforated metal sieves, the holes of which are respectively 1, 0.5, and 0.25 millimeter in diameter. The portions retained on the sieves are severally dried, ignited and weighed, and the finest portion, or that passing through the 0.25 millimeter sieve, is then submitted to the following process of separation:

The sediment and water passing through the 0.25 millimeter sieve are placed in a glass cylinder fifty centimeters long and thirty-seven millimeters in internal diameter. The cylinder is closed at the bottom and is provided with a lateral tube inserted six centimeters above the bottom. Three other lateral tubes are inserted at intervals of ten centimeters above the first tube, and a ring is etched into the cylinder ten centimeters above the uppermost tube. The lateral tubes are closed with rubber tubes compressed by spring clips. The sediment being placed in the cylinder, water is added to the mark or ring, the cylinder closed with a rubber stopper, and vigorously shaken until the contents are thoroughly mixed. It is then placed upright, the stopper removed, and after standing undisturbed for five minutes the clip on the uppermost tube is opened and the water allowed to flow into a beaker. After five minutes further standing, the second clip is opened and the water

drawn off into the same beaker; in the same manner the water is drawn off from the other tubes at intervals of five minutes until the level of the lowest tube is reached. The cylinder is then refilled with water to the mark, thoroughly shaken after inserting the stopper and the water again drawn off at intervals of five minutes, as before; the operation being repeated until the water drawn off is almost free from turbidity. The sediment remaining in the cylinder from this process of washing by subsidence is termed by Knop, fine sand, the material flowing off in suspension in the wash waters, dust, and the process of separation by Knop's original method ends here.

In order to remedy the imperfect separation into definite particles secured by the above method, Moore proposes the following device:

The fine sand from the first series of subsidences is placed in a separate vessel, the washings are allowed to remain undisturbed for twelve hours, the turbid liquid decanted and the sediment returned to the cylinder. Water is then added to the mark, the whole shaken, and the liquid drawn off at intervals of five minutes, as in the first series. The sediment from this operation is placed in a separate beaker, the washings returned to the cylinder, and again allowed to subside as before; the sediment from this second subsidence is added to that from the preceding operation, and the washings again returned to the cylinder, the operation being repeated as long as any sediment can be obtained from renewed treatment of the washings; the final washings are then placed in a separate vessel for subsequent microscopic measurements.

The collective sediments from the last series of operations are then returned to the cylinder and allowed to subside with fresh additions of water, as in the case of the first series; the fine sand thus obtained being added to that from the first series, and the washings being collected in a large beaker. The latter are left at rest for twelve hours, and the sediment returned to the cylinder and treated as before until no further separation can be effected. The fine sand resulting from all of these operations is then dried, ignited and weighed; the weight of the portion removed by the washing being determined by difference, as it is, owing to its excessively slow rate of subsidence, found impracticable to collect it for direct weighing. The size of the particles of fine sand is then determined by micrometric measurement. Similar measurements are made on the material obtained by long subsidence from the washings from the foregoing

operations. The average diameter of the largest particles should not exceed 0.01 millimeter.

204. Statement of Results.—The results of the analyses on three soils from the localities indicated in the table, and the method of stating them, are given in the following table:

	New Milford, Conn., per cent.	Clarksville, Tenn., per cent.	Granville, N. C., per cent.
Particles larger in diameter than 3.0 millimeters	8.55	0.32	0.23
Particles of diameter from 3.0 millimeters to 1.0 millimeter	4.96	0.45	15.04
Particles of diameter from 1.0 millimeter to 0.5 millimeter	4.43	0.96	33.43
Particles of diameter from 0.5 millimeter to 0.25 millimeter	11.86	1.25	18.82
Particles of diameter from 0.25 millimeter to 0.01 millimeter	60.54	61.58	23.59
Particles smaller in diameter than 0.01 millimeter	9.66	35.44	8.89
Total	100.00	100.00	100.00

205. Method of Bannigsen.—The silt flasks recommended by Bannigsen^[139] are shown in Fig. 27.

The glass flask *b* carries a long cylindrical neck *a* the upper part of which is graduated in cubic centimeters. Ten grams of the fine soil are shaken with water in the flask, the neck of which is closed with a rubber stopper. The flask is then inverted bringing the soil and water into the neck. The flask is hung up and sedimentation is assisted by imparting a pendulous motion to the neck for ten minutes. After an hour the soil particles have separated into a coarse layer below and a fine layer above. The relative volumes of the two layers are then read off in cubic centimeters. While this method may be useful in helping to form a speedy judgment concerning the character of a soil it can lay no claim to being an accurate method of silt separation.

206. Method of Gasparin.—The method of Gasparin only gives a very primitive separation of the various components of the earth according to their fineness. It is conducted as follows:

Ten grams of sifted earth are put into a beaker, water is added and strongly agitated; after five minutes the water is decanted into another

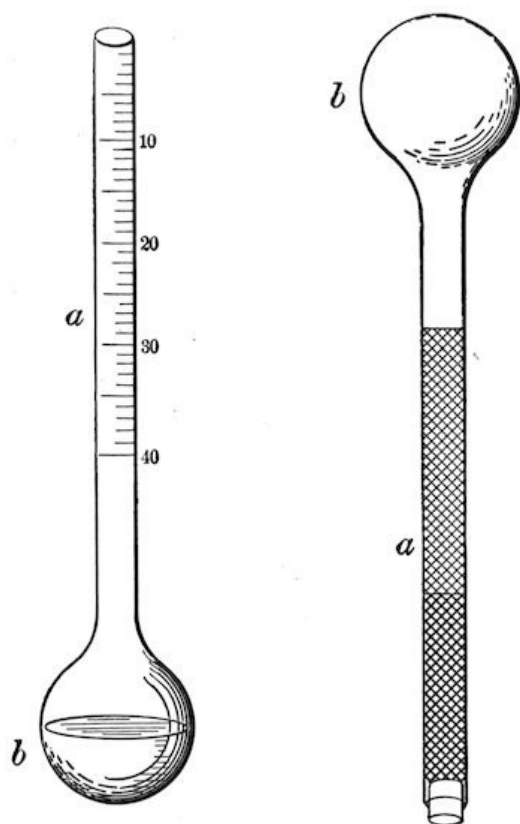


FIGURE 27.

BENNIGSEN'S SILT FLASKS.

vessel, the first vessel is filled anew with water, agitated, decanted, and this process is repeated until the liquid remains perfectly clear. Only two portions are weighed, *i. e.*, the pebbles which remain in the sieve and the coarse sand which remains in the beaker; while the argillaceous portion drawn off with the water is determined by the difference.

207. The Italian Method.—The following modification of Gasparin's process is practiced by the Italian chemists.^[140]

Twenty grams of earth are passed through a sieve having openings of one millimeter in diameter, then the sifted part is mixed with 100 cubic centimeters of water in a 200 cubic centimeters beaker and left to rest for some hours, then strongly agitated and after ten seconds the turbid liquid is

poured into another vessel of half a liter capacity. This manipulation is repeated until the liquid is clear.

The decanted liquid is thoroughly agitated, then left to stand until the movement shall be completely arrested, after which the supernatant liquid is poured into another vessel holding two liters. To the residuum is added more water; it is agitated, decanted, and this process is repeated until the water is no longer turbid.

208. Method of Osborne.—In the foregoing paragraphs the methods of silt separation by subsidence as practiced in different countries have been outlined. The good points of the various methods are combined in the process as carried out by Osborne.^[141] The details of this method will be given with sufficient minuteness to make its practice possible by all analysts.

Selecting the Sample.—Several pounds of air-dried, fine earth are secured by passing the soil through a sieve, the holes of which are three millimeters in diameter.

Sifting.—Thirty grams of the above fine earth are stirred with 300 to 400 cubic centimeters of water and then thrown successively upon sieves with circular holes of 1, 0.5, and 0.25 millimeter diameter respectively. By means of successive additions of water and the use of a camel's hair brush, all the fine material is made to pass through the sieves and these at the last are agitated under water in a shallow dish in such a way that the soil is immersed. The finest sieve should be well wet with water on its lower surface just before using. The finest particles which render the water turbid are easily washed through. The turbid water is kept separated from the clear water which comes off with the last portions that pass the sieves. The turbid water usually does not amount to more than one liter.

Elutriation.—The elutriation should be carried on so as to secure three grades of silt; the diameters of the particles ranging in the first grades from 0.25 to 0.05 millimeter, in the second grade from 0.05 to 0.01 millimeter, and in the third grade from 0.01 millimeter to the impalpable powder. The term sand is applied to the first grade, silt to the second, and dust to the third. After the turbid liquid from the sifting has stood a short time it is decanted from the sediment and after standing until a slight deposit is formed, is again decanted and the sediment examined with a microscope. If sand be present, the subsidence of the turbid liquid is continued until no more sand is deposited. As the sand subsides rapidly there is no difficulty in altogether freeing the liquid first decanted from this grade of particles. The sediment thus obtained contains all the sand, a part of the dust and much silt. As only dust and the finest silt render the water turbid the sediment is stirred a few times with a fresh quantity of water and decanted after standing long enough to let all the sand settle. When the water decanted is free from turbidity, the last portions of the soil passing through the sieve with clear water are added to the sediment and the decantations continued so as to remove most of the silt. When no more silt can be easily removed from the sediment without decanting sand, the decantations are made into a different vessel and the subsidences so timed as to remove as much of the silt as possible. By using a little care, at least three-quarters of the sand are thus obtained free from silt. The rest of the sand is mixed with the greater part of the silt which has been decanted into the second vessel. The size of

the smallest particles in this vessel is determined with the microscope, to make sure that its contents are free from dust as they usually will be if, after settling for a few moments, they leave the water free from turbidity.

The soil is thus separated into three portions, one containing sand, one sand and silt, and the other silt, dust, and clay. The sand and silt are separated from each other by repeating the subsidences and decantations in the manner just described.

In this way there is removed from the sediment, on the one hand, a portion of silt free from sand and dust, and on the other hand a portion of sand free from silt. Thus is obtained a second intermediate portion consisting of sand and silt, but less in amount than the first and containing particles of diameters much more nearly approaching 0.05 millimeter. By repeating this process a few times, this intermediate portion will be reduced to particles whose diameters are very near 0.05 millimeter and which may be divided between sand and silt, according to judgment. The amount of this is usually very small. As soon as portions are separated, which the microscope shows to be pure sand or pure silt, they are added to the chief portions of these grades already obtained.

The same process is applied to the separation of silt from dust. When all the silt has been removed from the dust and clay, the turbid water containing the dust and clay is set aside and allowed to settle in a cylindrical vessel for twenty-four hours. The vessel is filled to a height of 200 millimeters. According to Hilgard, the separation of the dust from clay during a subsidence of twenty-four hours, will give results of sufficient accuracy, although the clay then remaining suspended will not be entirely free from measurable fine particles up to 0.001 or 0.002 millimeter diameter.

Small beakers and small quantities of distilled water are used at first for the decantations, as thus the duration of subsidence is less and more decantations can be made in a given time than when larger quantities of water are employed. Beakers of about 100 cubic centimeters capacity are convenient for the coarser grades, but it is necessary to use larger vessels for the fine sediments from which turbid water accumulates that cannot be thrown away, as may be done with the clear water, from which the coarse sediments settle out completely in a short time.

It is best to keep the amount of water as small as possible in working out the dust since loss is incurred in using too large quantities.

It is also necessary in most cases to subject the various fractions obtained during elutriation, to careful kneading with a soft rubber pestle so that the fine lumps of clay may be broken up and caused to remain suspended in the water. This treatment with the pestle should be done in such a way as to avoid as far as possible all grinding of the particles, the object being merely to pulverize the minute aggregations of clay and extremely fine particles which always form on drying a sample of soil after removing it from the ground.

Measurement of the Particles.—To determine the size of particles in suspension, a small glass tube is applied to the surface of the liquid in such a way as to take up a single drop which is transferred to a glass slide. This drop will contain the smallest particles in the liquid.

To obtain a sample of the coarsest particles the liquid is allowed to stand long enough to form a very slight sediment and a portion of this sediment is collected with a glass tube.

To determine the diameter of the particles in a sediment it is stirred vigorously with a little water and the pipette at once applied to the surface of the water. On decanting the greater part of the sediment, the large particles remain at the bottom of the beaker and may be easily examined.

Time.—The time required to make the separations, above described, is about two hours for each, so that an analysis including the sittings, is made in five or six hours, exclusive of the time necessary for collecting the dust and separating the clay, for which a subsidence of twenty-four hours is allowed.

Weighing the Sediments.—The sediments are prepared for weighing by allowing them to subside completely, decanting the clear water as far as possible, rinsing them into a weighed platinum dish and igniting. The dish is cooled in a desiccator.

Effect of Boiling.—The analyses show a very decided increase in the particles smaller than 0.01 millimeter diameter at the expense of coarser particles as the result of boiling. The surfaces of the coarser particles are seen to be polished and of a lighter color than those not boiled. The surfaces of the unboiled particles are coated with a film of fine material probably cemented to them by clay. When these coarse particles which have not been

boiled, are violently stirred with water for a short time, no fine particles are detached from them; and a careful examination under the microscope fails to reveal in any of the sediments more than an occasional grain exceeding the 0.05 millimeter limit by so much as 0.01 millimeter, or the 0.01 limit by as much as 0.005 millimeter. It would, therefore, appear that these small particles thus set free by long boiling are really a part of the larger ones and should be treated as such in a mechanical analysis of these soils.

209. The French Method.—The Schloesing method^[142] as practiced by the French agricultural chemists^[143] differs essentially from those already described in attempting to first free the silt from carbonates and organic matter. It is conducted as follows:

One kilogram of the soil previously dried in the air, is taken and passed through a sieve of which the meshes are five millimeters. The agglomerated particles of earth are broken up by the hand. The pebbles are also taken out and weighed. The pebbles are then treated with hydrochloric acid until all effervescence is over. The insoluble part is dried and again weighed. The difference in weight gives the quantity of calcium carbonate contained on the external surface of the pebbles. The earth which passes the sieve of five millimeters mesh is next passed through a sieve having ten meshes to the centimeter. The masses on the sieve are broken up with the hand or with a pestle, in such a manner as to separate the fine agglomerated particles. The material which remains upon the sieve after being dried at 100°, is weighed. This gives the coarse sand. This is treated with hydrochloric acid as were the pebbles before, washed and the residue dried and weighed. The difference in weight gives the quantity of calcium carbonate adhering to the surface of the coarse sand.

The mechanical analysis is continued with the matter which has passed the sieve with ten meshes to the centimeter and which consists of the soil, properly so-called. Ten grams of this are taken, dried at 100° until no further loss takes place and the moisture thus determined. Another ten grams are taken and placed in a capsule with a flat bottom, and from nine to ten centimeters in diameter. This is moistened with a small quantity of water in such a way as to make a paste. This paste is rubbed with the finger in fifteen cubic centimeters of water. Ten seconds after the stirring is completed the supernatant liquid is poured into a precipitating jar of about 250 cubic centimeters capacity, taking great care not to allow any particles

to pass over which have been deposited during that time. This operation is repeated in the same way waiting about ten seconds each time before decanting, until the decanted liquor is almost perfectly clear. In this way the particles of different fineness are separated. The decanted portions contain the fine sand and clay. The remaining portion contains the sand and particles of medium fineness. This last part is dried, being kept at 100° until it has a constant weight. It is afterwards treated with dilute nitric acid to dissolve the calcium carbonate. When the carbonate is abundant, it is sufficient to determine it by difference which is done by washing the material, drying and weighing. But when the proportion of carbonate is very small and in consequence when its exact determination acquires a greater importance, it is better to determine the lime directly. For this purpose the part soluble in dilute nitric acid is collected, treated with ammonia and acetic acid and precipitated with ammonium oxalate. Details of this operation will be given in another part of this manual.

In regard to the matter which is insoluble in nitric acid, it is composed chiefly of silica or silicates, and sometimes also of vegetable débris. The vegetable matter is determined by the incineration of the material which has been previously dried. The loss of weight gives the proportion of vegetable or organic débris contained in the soil and of combined water.

The portion which has been decanted, the volume of which should not exceed 500 cubic centimeters, is treated with nitric acid until effervescence ceases. It is then left to digest for some time, in order to permit the whole of the carbonate to dissolve. It is next thrown upon a smooth filter about one decimeter in diameter. After filtration it is washed to secure the complete elimination of the soluble lime salts. The lime is determined in the filtered liquid.

The insoluble portion contains the fine sand, the clay and humus bodies. In order to separate the three elements the precipitate which was received upon the filter, is rubbed with water, the filter is broken and all its contents washed through. The volume of wash water is made up to 200 cubic centimeters; two or three cubic centimeters of ammonia are added and the whole left to digest for two or three hours. The volume of the liquid is then made up to one liter with distilled water, vigorously shaking in such a way as to put all the matter in suspension. It is then left to settle for twenty-four hours. At the end of this time the supernatant liquid is decanted by the aid

of a siphon. To the residue are added two cubic centimeters of ammonia and one liter of water. The matter is again brought into suspension and allowed to settle for twenty-four hours. The supernatant liquid is again decanted with a siphon, and added to the liquid previously removed. For ordinary soils two decantations are generally sufficient but when the soils contain a large quantity of clay it is convenient to decant three or four times. By an examination of the supernatant liquid it is easy to tell if the washings have been sufficiently prolonged. The decanted liquors contain the organic matter and that which it is convenient to call clay, which is constituted of very fine particles of sand and colloidal clay which play, in arable soil, a rôle somewhat like that of cement.

These matters are estimated in the following manner: The liquor is first treated with nitric acid and the clay and the humic matters are precipitated together. They are collected upon a smooth filter one decimeter in diameter and washed with water. By means of a washing bottle all the solid matters which have stuck to the sides of the filter are finally collected in the bottom of it. Since the last washings pass the filter very slowly, they can be removed after the complete deposition of the matter they contain, by means of a pipette. When all the liquid is removed the filter is placed upon blotting paper, great care being taken to avoid desiccation, having in view only the elimination of the excess of humidity. The folds in the filter are then carefully smoothed out with the finger. The matter which has collected upon the filter is then removed completely with a washing bottle, placed in a dish and dried at 100° and weighed. After weighing, the mass is incinerated in a muffle in order to destroy the humic bodies. The difference in weight before and after incineration, gives the total weight of the humic bodies and since the diminution in weight comprises not only the weight of the humic bodies, but also the weight of the combined water which is lost during the process of incineration, there should be subtracted from the total loss of weight ten per cent of the weight of the residual mineral matter, which represents the water of composition of the hydrated silicate.

210. Statement of the Analysis.—Schloesing in his original paper^[144] recommends that the analysis be commenced with 1,000 grams of soil. The data of the analysis and the method of arrangement are illustrated by the following example.

The physical examination of the earth having been completed as above, the results can be tabulated as follows: taken, 1,000 grams of dry earth, digested in water, thoroughly worked by hand, sifted, and passed through the meshes of the sieve by a stream of water, the meshes having a diameter of one millimeter.

Dry residue, fifty-five grams, contains	Pebbles	21	grams.
	Gravel	33	„
	Organic débris	1	„
Sifted earth by difference, 1000 – 55 =		945	„
		—	
		1000	grams.

Humidity of the homogeneous paste, twenty-seven per cent. Then 945 grams of the dry sifted earth correspond to $\frac{945}{1.00 - .27} = 1294.5$ of paste.

The analysis, therefore, should be carried on upon this weight or some aliquot part say 0.01 thereof; viz., 12.945 grams.

12.945 grams of the paste after successive kneadings and decantations furnish dry:	1st.—Coarse sand dry giving by treatment with acid and ignition.	Noncalcareous sand	3.05	grams.
		Calcareous sand	1.19	„
		Organic débris	0.08	„
	2nd.—Fine elements decanted with the water, their weight calculated by difference, 9.45 – 4.32 = 5.13 grams.			

Treatment of the Fine Elements.—Treated by nitric acid until a complete decomposition of the calcareous matter is secured, filtered, washed, the residual matter collected upon a filter, and the liquid received in a two-liter flask, a little ammonia added, allowed to digest, the flask filled with distilled water, left for twenty-four hours at repose, and decanted:

The decantation furnishes	1st.—A deposit of fine calcareous sand weighing dry	3.14	grams.
	2nd.—Clayey liquid giving after coagulation by acid, filtration and drying 0.85 grams of clay.		
Then: Total fine elements			5.13 grams.
Fine elements determined directly.	Fine calcareous sand	3.14	3.99 „
	Clay	0.85	
Fine calcareous sand by difference			1.14 „

Calculating these results to the original quantity of 1,000 grams the following data are obtained:

RÉSUMÉ.

One thousand grams of dry earth contain:

Pebbles	21	grams.
Gravel	33	„
Organic débris	1	gram.
Fine earth	945	grams.
Total	1000	„

945 grams of fine earth contain:	Coarse sand 432 gms.		Noncalcareous sand	305	gms.
			Calcareous sand	119	„
			Organic débris	8	„
	Fine elements 513 gms.		Fine, noncalcareous sand	314	gms.
			Clay	85	„
			Fine calcareous sand	114	„
			Total	1000	„

There are counted as clay all the elements which have remained in suspension in the water after a period of repose of twenty-four hours. In fact, these elements comprise a notable proportion of very fine sand which is not deposited during that time. In order that the liquid should become entirely freed from this sand it would be necessary to wait several weeks and even several months. Such a prolongation of the analysis is evidently inadmissible. The period of twenty-four hours of repose therefore has been adopted. This is merely conventional, in the same way that the period of ten seconds adopted for the precipitation of the gravel is conventional. But this convention is justified by the fact that the substance which is called clay presents, when it has a proper degree of humidity and cohesion, a plasticity entirely analogous to that property of natural clay. Moreover, as has already been said, that which is chiefly important in these analyses is the employment of processes always comparable among themselves in their results and generally followed.

211. The Belgian Method.—The method of estimating the percentage of sand and clay practiced at the Gembloux Station^[145] is essentially that recommended by Schloesing with a few minor modifications.

With the ball of the thumb or with the finger, 100 grams of fine earth are rubbed with water in a porcelain capsule or mortar with a capacity of about

250 cubic centimeters. The suspended particles are poured off with the wash water and the process repeated five or six times, using in all about 200 cubic centimeters of water.

The water containing the sediment is rendered slightly acid (hydrochloric acid) adding the acid in minute particles with constant stirring for about an hour in order to dissolve all the carbonate and to separate the organic acids from the bases with which they are combined.

The liquid is allowed to remain at rest for five or six hours and a part of the liquor decanted to remove any supernatant particles of organic matter which may have passed the sieve in the original preparations of the sample. Filter through a smooth filter about twelve centimeters in diameter, wash until the chlorin has disappeared, and throw the filtrate away.

Break the filter paper over the vessel in which the soil was treated with hydrochloric acid and wash all the contents of the filter into this vessel with as little water as possible (about 100 cubic centimeters), add five cubic centimeters of strong ammonia water, allow to stand for three hours, shaking from time to time and with distilled water make the volume up to 250 cubic centimeters. Stir vigorously with a glass rod or spatula, take this out and wash any adhering particles back, leave at rest for twenty-four hours, siphon the turbid liquid into a two-liter vessel. Make the volume up again to 250 cubic centimeters and treat as above described and repeat the operation until the water becomes clear after standing for twenty-four hours. Usually eight or ten washings are necessary. Wash the residual sand into a weighed dish, evaporate to dryness, ignite and weigh. The weight obtained divided by the weight of the original sample gives the per cent of sand. The sand is separated by sieves of varying fineness into coarse, fine, and pulverulent sand.

Add to the ammoniacal liquor collected in the two-liter flask some powdered potassium chlorid (five grams per liter) to hasten the coagulation and rapid deposit of the clay.

After twenty-four hours siphon the clear liquor, collect the deposited clay in a smaller vessel, allow to remain at rest and decant as much of the clear liquor as possible. Pass through a plain tared filter about nine centimeters in diameter, dry at 150° and weigh the clay.

212. The Italian Method.—Schloesing's method as carried out by the Italian chemists^[146] is as follows:

A kilo of earth dried in the air is passed through a sieve the threads of which are separated a distance of five millimeters; and with this the small pebbles are separated.

With another sieve having spaces of one millimeter, the coarse sand is separated. The pebbles and sand are dried, weighed, treated with hydrochloric acid and again weighed in order to find the quantity of calcareous matter contained in them. In ten grams of this fine earth the humidity is determined by drying at 100°.

Ten grams are mixed in a capsule with fifteen to twenty cubic centimeters of water and after eight to ten seconds the supernatant liquid is poured into a beaker having a capacity of 250 cubic centimeters. The same operation is repeated until there are contained in the beaker the fine sand and the clay, while the coarser sand remains in the capsule.

This last is then dried and weighed and the quantity of calcium carbonate determined by treating it with diluted nitric acid. By means of calcination the organic matter is determined. The liquid decanted in the beaker, the volume of which must not surpass 200 to 250 cubic centimeters, is treated with nitric acid, filtered after some time, washed and the calcium is directly determined by precipitating the solution with ammonium oxalate.

The part in the filter which contains the fine sand, the clay, and the humus material is mixed with water to a volume of about 200 cubic centimeters; there are then added to it two to three cubic centimeters of ammonia and after two or three hours it is diluted to a liter and strongly agitated.

After twenty-four hours of rest it is decanted and the residuum is treated a second time with diluted ammonia, decanting after twenty-four hours. Ordinarily these two treatments suffice, if, however, the earth is very argillaceous, this operation should be repeated three and even four times.

The clay which is found in the liquid suspended in colloidal form coagulates and is precipitated by adding thirty to forty cubic centimeters of a saturated solution of potassium chlorid, while the humus substance, under the influence of the ammonia remains dissolved.

Sestini found that the method of Schloesing was the only one which indicated exactly the quantity of clay in the soil. He modified this method by reducing the time of rest from twenty-four hours, as proposed by

Schloesing, to only twelve hours, a reduction which in his opinion does not in the least impair the exactness of the method.

Sestini also proposes twelve treatments instead of six.

SEPARATION OF THE SOIL PARTICLES BY A LIQUID IN MOTION.

213. General Principles.—The laws, already discussed, applying to the subsidence of a solid particle in a liquid, are equally applicable to the separation of the particle by imparting a motion to the liquid at a given rate. If a solid particle subside in a given liquid at the rate of one millimeter per second it follows that this particle will remain at rest if the liquid be set in motion upward with a like velocity. If the velocity be greater the particle will be carried upward and eventually out of the containing vessel. Such a particle is said to have a hydraulic value of one millimeter per second. If there be a perfect separation of a soil into its constituent particles and no subsequent flocculation, all the particles of one millimeter hydraulic value and less will be separated by a current of the velocity mentioned.

The general principles on which the separation rests, therefore, are the securing of the proper granulation of the sample and the maintenance of a fixed velocity of the current until the separation is finished. The separation must be commenced with a period of subsidence so as to remove first of all the suspended clay or impalpable particles. The velocity can then be increased in a certain fixed ratio to secure a separation into particles of any required hydraulic value.

214. Nöbel's Apparatus.—One of the earliest methods of separating the soil particles by a moving liquid is that of Nöbel.^[147] The apparatus is shown in Fig. 28. The four separating vessels 1, 2, 3, 4 are of glass, pear shaped, and have a relative capacity of 1^3 , 2^3 , 3^3 , 4^3 , or 1 : 8 : 27 : 64. No. 4 has an outlet tube leading to the beaker B, of such a capacity as to allow the passage of just nine liters of water in forty minutes, constant pressure being maintained by means of a Mariotte's bottle or of the constant level apparatus A, *a*, *b*, which is connected with the main water supply through the tube *a* by means of a rubber hose. The reservoir C should hold about ten liters. The sample of soil to be separated should be previously boiled and passed through a sieve having circular openings one millimeter in diameter. The flask in which the sample is boiled is allowed to stand for some time when the muddy supernatant liquid is poured into elutriator No. 2 and the

remaining sediment washed into No. 1. No. 1 is filled with water by connecting it with the water supply and opening the pinch-cock *p*. The water is carefully admitted until the air is all driven out and Nos. 1 and 2 connected. The cock *p* is then opened and the vessels all filled, and the water allowed to run into B for forty minutes, the level being maintained uniformly at A.

Of the water used, four liters are found in the elutriating vessels and nine liters in the receiving vessel No. 5. The apparatus is left standing for an hour until the liquid in the elutriators is clear and the portions in each vessel are received on weighed filters dried at 125°, and the weight of each portion determined.

It is recommended that the loss on ignition of each part be also determined. The separated particles thus secured are classified as follows:

No. 1. Débris and gravel. No. 2. Coarse sand. No. 3. Fine sand. No. 4. Clayey sand. No. 5. Finest parts or clay.

Although the method of Nöbel has been much used, the results which it gives are entirely misleading. The convection currents produced in the conical vessels by the passing water and the flocculation of the soil particles prevent any sharp separation into classes of distinct hydraulic value. The process may be useful for a qualitative test, but its chief claim to a place in this manual is in its historic interest arising from its use in the first attempts at silt analysis.

215. Method of Dietrich.^[148]—The difficulties attending the silt separation by the Nöbel method, led Dietrich to construct an apparatus in which the sides of the elutriating vessels were parallel, but these vessels, with the exception of the first, were not set in an upright position.

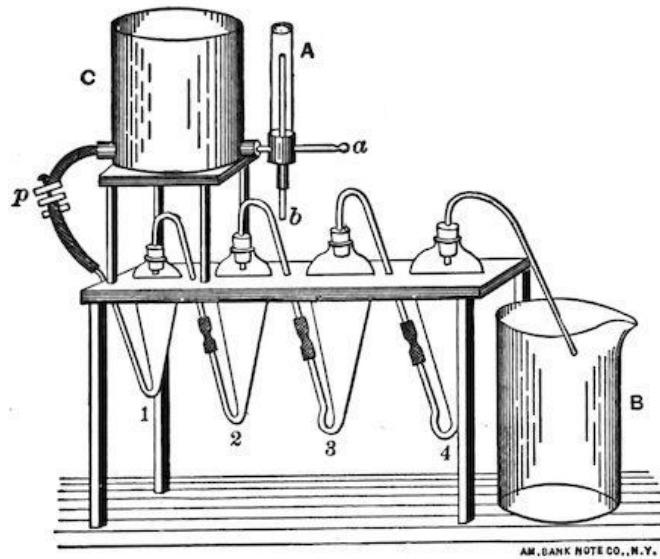


FIGURE 28.

NÖBEL'S ELUTRIATOR.

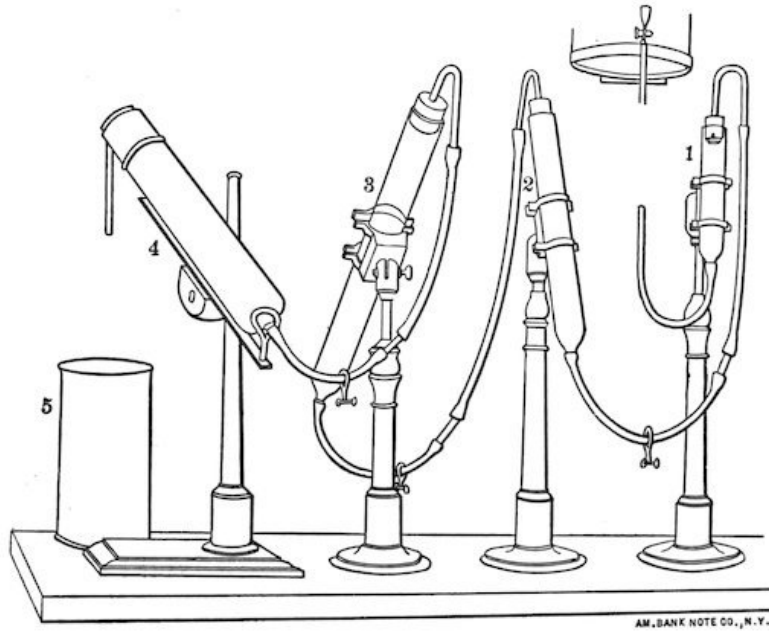


FIGURE 29.

DIETRICH'S ELUTRIATOR.

The apparatus (Fig. 29) consists of a series of cylindrical vessels connected by rubber tubing.

The elutriators are of the following dimensions:

- No. 1. Seventeen centimeters long, 2.8 centimeters in diameter, position upright.
- No. 2. Thirty-four centimeters long, four centimeters in diameter, inclined $67^{\circ}.5$.
- No. 3. Fifty-one centimeters long, 5.2 centimeters in diameter, inclined 45° .
- No. 4. Sixty-eight centimeters long, 6.4 centimeters in diameter, inclined $22^{\circ}.5$.

The rubber tubes passing from one vessel to the other are furnished with pinch-cocks so that each one of the elutriating vessels can be shut off from the others and independently removed from the circuit.

The stream of water is made to pass through the apparatus under a constant pressure of one meter.

Only the fine earth, boiled with water or hydrochloric acid, is to be placed in the apparatus. The part coming through a sieve with a mesh 0.67 millimeters is to be used and placed in No. 1. About thirty grams of soil, are employed for each elutriation. Before adding the soil, the air is completely

removed from all parts of the apparatus by connecting it with the water supply and allowing it to be filled with water.

The rate of flow is controlled by the orifice of the last effluent tube and the analyst is directed to continue the operation until the effluent water collected in the beaker glass (5) is clear. The particles then remaining in each of the vessels are collected separately.

The author of the method claims that in respect of likeness of particles the results are especially gratifying and that duplicate analyses give results fully comparable. The process, however, has not commended itself to analysts, but it marks a distinct progress toward the principles of later investigators. Had each of the elutriating vessels been placed upright and the rate of flow determined, the apparatus of Dietrich would have served, to a certain extent, for the more rigid investigations of his successors.

216. Method of Measure.—The sifted earth, from ten to fifteen grams, is carefully mixed with 200 cubic centimeters of water. It is then introduced into a doubly conical elutriator B, Fig. 30, of about 250 cubic centimeters capacity. A current of distilled water is allowed to flow from a Mariotte's bottle, A, which secures a regular and constant flow. The bottle A is joined to the elutriator B by means of a rubber tube and the vertical glass tube D, the top of which is expanded into a funnel for the purpose of receiving the water from the Mariotte flask. The current of water flowing upward through the elutriator B carries in suspension the most finely divided particles of clay, and these are collected with the emergent water in the receiver C. The sand and coarser particles of clay remain in the elutriator. The water flows out by the tube F, the diameter of which should be less than that of D. When the emergent water becomes limpid the operation is terminated. After the apparatus is disconnected, the water is decanted from the sand in the elutriator, and the whole residue is weighed after drying for two hours at 110°.

The fine soil collected in C may also be separated and weighed, for control, after drying as above.

The pebbles and coarse sand separated by the sieves should also be weighed. By this process the soil is separated into four portions; viz.,

(1) Pebbles. (2) Coarse sand. (3) Fine sand and other materials not carried off by the current of water. (4) Fine soil, carried into the receiver C.

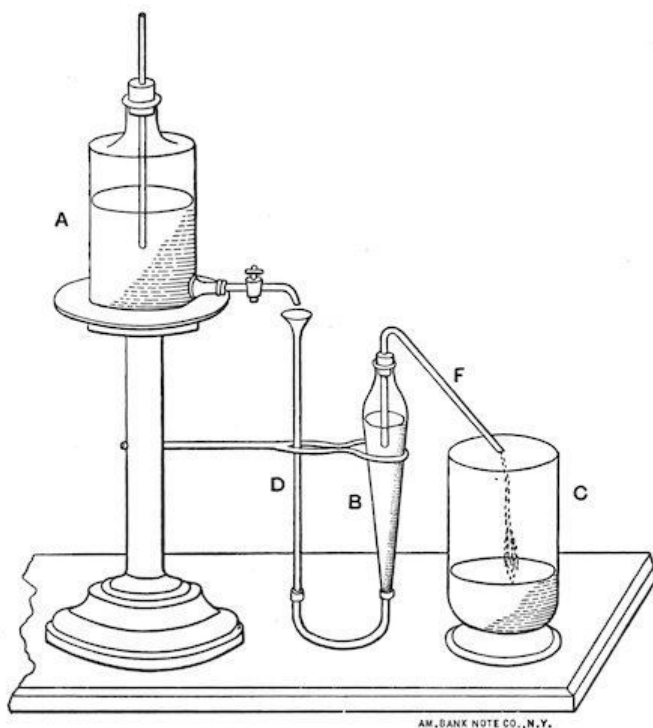


FIGURE 30.

MASURE'S SILT APPARATUS.

217. Method of Schöne.—

The method of Schöne^[149] is based on the combination of a cylindrical and conical separatory tube through which the flow of water is regulated by a piezometer.

If, in the process of silt separation, the water move perpendicularly upward with a given velocity, *e.g.* = *v* the separation is dependent:

(1) On the volume of the silt particles, (2) On their specific gravity, and, (3) On their state of disintegration.

If it be assumed that the silt particle is a sphere with a diameter = *d*, then according to Newton's law of gravity, the

following formula would be applied: $d = v^2 \left(\frac{3Z}{4g(S-1)} \right)$.

In the above formula *Z* = a coefficient which depends on the condition of the surface against which the hydraulic pressure or resistance works, in this case a sphere; *g* = the acceleration of gravity equivalent to 9.81 meter; and *S* = the specific gravity of the particle.

This expression signifies that in a given case, the velocity of the current in the apparatus is just sufficient to counteract the tendency of a given particle to sink. All particles of a smaller diameter, in such a case, will be carried on by the current, while all of a greater diameter would separate by sedimentation. These theoretical conditions are not met with in practice where silt particles of all shapes and degrees of aggregation abound. These particles, whatever their shape, may be said to have the same hydraulic value when carried by the same current. It is necessary, therefore, to secure some uniform standard of expression to assume a normal form of particle and a normal specific gravity. For the form, a sphere is evidently the normal which must be considered and for specific gravity that of quartz is taken;

viz., 2.65. The mean coefficient for Z may also be placed at 0.55, although slightly different values are ascribed to it. Substituting these values in the formula, it is reduced to the expression; $d = v^2 \times 0.0000255$ millimeters. It can, therefore, be said that by this or that velocity of the current, silt particles will be removed of this or that diameter, it being understood that all particles of equal hydraulic value to spherules of quartz of the given diameter are included in each class. In order to have the theoretical formula agree with the results of analysis it is necessary to modify it empirically to read $d = v^{7/11} \times 0.0314$ millimeters.

This formula is found to agree well with the results obtained for all velocities between 0.1 millimeter and twelve millimeters per second, the ordinary limits of silt separation.

The Apparatus.—The conic-cylindrical elutriating vessel A, B, C, D, E, F, G, Fig. 31, is of glass. The part B, C, is cylindrical, ten centimeters in length and as nearly as possible five centimeters in diameter.

The conical part C, D, is fifty centimeters in length. Its inner diameter at D must not be greater than five centimeters nor smaller than four centimeters.

The bend, D, E, F, should have the same diameter; viz., four to five centimeters.

The part A, B, C, D, and D, E, F, G, may be made of separate parts and joined by a rubber tube.

Outflow Tube and Piezometer.—The outflow tube and piezometer, H, J, K, L, is constructed as shown in Fig. 32. It should be made of barometer tubing having an internal diameter of about three millimeters. The tube is bent at J at an angle of forty to forty-five degrees. The knee J must be as acute as possible not to interfere with the inner diameter. The form and especially the magnitude of the outlet are of great importance. It must be circular and nearly 1.5 millimeter in diameter. It must not be larger than 1.67 millimeter nor smaller than 1.5 millimeter. The opening should be so made as to direct the stream of outflowing liquid in the direction shown by the arrow.



FIGURE 31.

SCHÖNE'S
ELUTRIATOR.

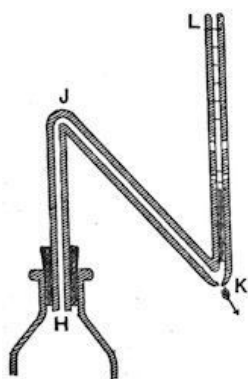


FIGURE 32.

SCHÖNE'S
ELUTRIATOR,
OUTFLOW TUBE.

The piezometer L, K is parallel to the arm H, J, of the delivery tube. Its graduation has its zero point in the center of the outlet K. It commences with the one centimeter mark. From one to five centimeters it is divided into millimeters, from five to ten centimeters into one-fourth centimeter, from ten to fifty centimeters into one-half centimeter, and from fifty to 100 centimeters into centimeters. The dimensions given are those required for ordinary soils and for velocities ranging from two-tenths millimeter to four millimeters per second.

For greater velocities, a delivery tube with a larger outlet must be used and the piezometer must be of greater internal diameter than indicated.

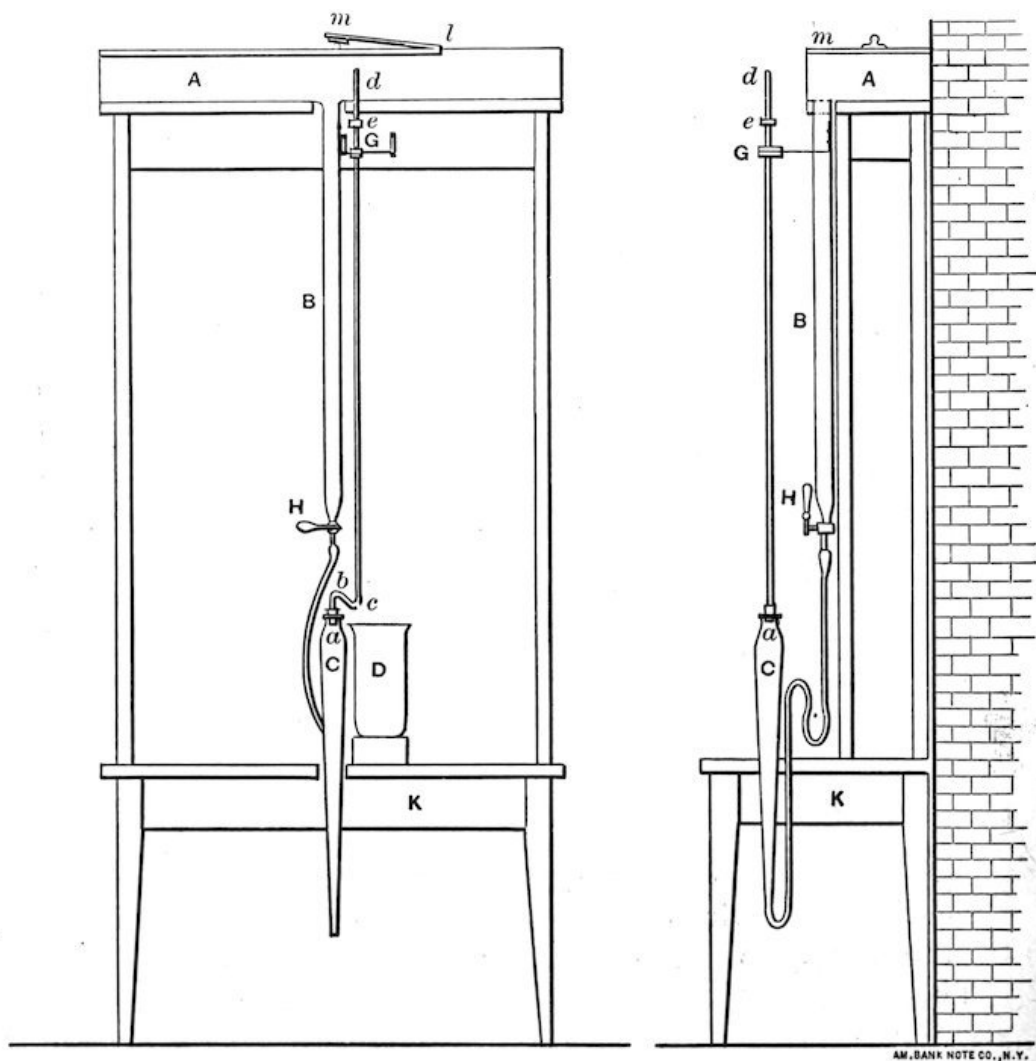


FIGURE 33.

SCHÖNE'S ELUTRIATOR, ARRANGEMENT OF APPARATUS.

Arrangement of the Apparatus.—The apparatus is conveniently mounted as shown in Fig. 33, giving front and side views of all parts of apparatus in position ready for use. When numerous analyses are to be made much time is saved by having a number of apparatus arranged *en batterie*.

The Sieve.—The soil, before being subjected to elutriation, should be passed through a sieve of which the meshes are 0.2 millimeter square.

The Process.—To measure the diameter of the cylinder, two marks are made with a diamond upon the glass which are distant from each other a certain space, for instance, h centimeters. The space between these two

marks is filled with water exactly measured. Suppose that a cubic centimeters were used, then the diameter is determined by the formula:

$$D = \sqrt{\frac{4a}{\pi h}} \text{ centimeters.}$$

In order to determine that the elutriating cylinder is strictly comparable in all its parts this measurement should be made upon several parts thereof.

The apparatus should now be tested in regard to the quantity of liquid which it will deliver under a given pressure in the piezometer. By means of the stop-cock H the flow of water is so regulated that the outflow at *c* can be measured at a given height of the water in the piezometer. Suppose that *a* cubic centimeters of water flow in *t* seconds, then the quantity which would flow in one second is determined by the formula, $Q = \frac{a}{t}$ cubic centimeters. Since according to the law of hydraulic outflow the quantities are proportional to the square root of the height of the column it is easy to compute from any given height the quantity which will flow from any other one desired. For the retardation due to capillary attraction, it is sufficient, in general, to take it in a constant quantity; if this constant quantity be represented by *C*, the observed height of the water in the piezometer by *h*, and the quantity of water flowing out by *Q*, the data required for any given velocity can be calculated from the following proportion:

$$\sqrt{h_1 - C} : \sqrt{h_2 - C} = Q_1 : Q_2.$$

It is necessary to compute the magnitude of this constant *C* which is to be subtracted. This is accomplished by measuring the quantity of water which flows out at two different heights of the column in the piezometer. From the foregoing proportion, the value of *C* is as follows:

$$C = \frac{Q_1^2 h_2 - Q_2^2 h_1}{Q_1^2 - Q_2^2} \text{ centimeters.}$$

The value of *C* can be the more exactly determined as *h*₁ is greater and *h*₂ smaller. It is best to choose the lowest height from which an exact reading can be made; that is, by which the regular rise and fall of the level of the water in the piezometer (in consequence of the formation of drops) just begins to disappear. This usually takes place when *h*₂ = 1.5 centimeter to 1.7 centimeter. For the higher value *h*₁ it is best to take about 100 centimeters. Suppose, for example, the following results are obtained:

Observed height.		Observed quantity of outflow.		Height of column to be subtracted due to capillary attraction.
h_2 centimeters.	h_1 centimeters.	Q_1 cubic centimeters.	Q_2 cubic centimeters.	centimeters.
80	1.6	5.53	0.406	1.21
100	1.6	6.13	0.484	1.17
80	1.8	5.53	0.406	1.19
100	1.8	6.13	0.484	1.19

The same quantity of water which flows out in a unit of time passes also at the same time over a cross section of the elutriating cylinder. The diameter of this cylinder being D the equation is derived

$$v = Q \frac{4}{\pi D^2} \text{ centimeters.}$$

Since the velocity in the elutriating cylinder v is directly as the quantity of water overflowing so is

$$v : v_n = \sqrt{h - C} : \sqrt{h_n - C};$$

$$v_n = \sqrt{(h_n - C) \frac{v}{\sqrt{h - C}}}$$

$$\text{and } h_n = v_n^2 \frac{h - C}{v^2} + C. \text{ The constant } \frac{h - C}{v^2}$$

is obtained from the means of a number of estimations; for example as illustrated in the following data:

Observed height, centimeters.	Observed quantity of outflow, cubic centimeters.	Corresponding velocity in elutriating cylinder of 4.489 centimeters diameter, millimeters.	Constant. $\frac{h - C}{v^2}$
1.6	0.406	0.0257	621
1.8	0.484	0.0306	652
80.0	5.530	0.3490	647
100.0	6.130	0.3870	660
Mean			645

Then are obtained the following values of h_n and v_n :

$$h_n = 645(V_n^2) + 1.19 \text{ centimeters.}$$

$$\text{and } v_n = \sqrt{h_n - 1.19} \times 0.0394 \text{ centimeters.}$$

In order to be able easily and rapidly to judge under what pressure the outflow has taken place in any particular instance, a larger number of values are computed with the help of the formula given and placed together in tabular form. As an example the following table may serve which was computed for one of the apparatus used. Usually it will be sufficient to test the apparatus for four different heights and then to interpolate the values for all the others. The numbers marked with a star in the table are those which were determined by experiment; the others were calculated.

Height of column in piezometer. h	Velocity in the elutriating cylinder of 4.489 centimeters diameter. v		Corresponding diameter of silt particles. $d = v(\frac{7}{11})^{0.0314}$ millimeters.
centimeters.	Observed, millimeters.	Calculated, millimeters.	millimeters.
1.5	0.222	0.220	0.0120
1.6	0.257*	0.252	0.0131
1.7	0.284	0.281	0.0140
1.8	0.306*	0.307	0.0148
1.9	0.323	0.332	0.0155
2.0	0.346	0.355	0.0162
2.5	0.427	0.451	0.0185
3.0	0.531	0.530	0.0210
3.5	0.577	0.599	0.0227
4.0	0.650	0.660	0.0236
4.5	0.694	0.717	0.0254
5.0	0.751	0.769	0.0265
6.0	0.850	0.864	0.0286
7.0	0.942	0.950	0.0304
8.0	1.050	1.028	0.0320
9.0	1.120	1.101	0.0334
10.0	1.170	1.169	0.0347
15.0	1.490	1.460	0.0400
20.0	1.730	1.710	0.0441
25.0	1.940	1.920	0.0476
30.0	2.100	2.110	0.0506
35.0	2.310	2.290	0.0532
40.0	2.460	2.450	0.0556
45.0	2.610	2.610	0.0578
50.0	2.770	2.750	0.0598
60.0	3.030	3.020	0.0635
70.0	3.290	3.270	0.0667
80.0	3.490*	3.500	0.0697
90.0	3.710	3.710	0.0724
100.0	3.870*	3.920	0.0749

Suppose the problem is by means of the apparatus tested above, to separate into a number of groups a mixture of silt particles, whose hydraulic values are found between the following diameters: 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07 millimeters. The table will show at once under what

pressure of water the piezometer must be placed in order to give the values; viz., 1.4, 2.8, 7.0, 15.0, 29.0, 53.0, and 83.0 centimeters respectively.

The apparatus described above, is adapted for velocities in the elutriating cylinder varying from two-tenths millimeter to four millimeters per second. The largest silt particles which can be separated by the velocities given above, have approximately a diameter of 0.08 millimeter. For the separation of larger particles a sieve can take the place of the silt apparatus. If, however, it be desired to subject larger particles to silt analysis, the dimensions of the elutriating cylinder and of the outlet of the delivery tube must be changed accordingly.

Preparation of Sample.—The conduct of silt analysis of natural soils must, in certain cases, be preceded by a special treatment of the sample. If the latter be rich in humus the organic substance must previously be separated as completely as possible. With sandy soils this can be accomplished by ignition. With clayey soils, on the contrary, it is to be performed by boiling the soils at least one hour with water which contains from one to two per cent of free alkali. Soils which contain lime must also be subjected to treatment with dilute hydrochloric acid, and the hydrochloric acid must be as carefully removed, as possible before the sample is subjected to elutriation; afterward follows the boiling of the sample in the ordinary way with water. This, of course, can be omitted when it has already been treated with boiling dilute alkali. It is also important to remove the larger particles by a sieve before the elutriation begins. It is well to pass a sample through a sieve after it has been boiled, by which all particles of a larger diameter than 0.2 millimeter are removed. This will usually require about one liter of water and this water should be allowed to rest from one to two hours and poured off with the suspended material which it contains. Only what subsides should be brought into the apparatus. In rinsing the sample as much water must be used as will fill the apparatus up to its cylindrical portion.

After the sample has been placed in the apparatus, the water is allowed slowly to enter, being careful to avoid reaching more than the lowest required velocity, until the outflow begins. The water then is so regulated by the stop-cock as to bring it to the desired height in the piezometer. This being accomplished, the different velocities which have been decided upon for separating the particles of silt are used one after the other, as soon as all

the silt which can be removed at each given velocity, has been secured. From three to five liters of water will be required for the separation of each class of particles. Sometimes the reading of the height of the water in the piezometer is difficult; as, for instance, when foam or bubbles accumulate therein. These bubbles can be removed by simply blowing into the tube, or dropping into it a little ether. The outflow of water can be received in vessels, beaker glasses, or cylinders, in which it is allowed to subside. The finest particles which remain in suspension in the water are best determined by difference. If it be desired to weigh them directly, the water can be treated with ammonium bicarbonate until it contains from one to two per cent thereof. The precipitation then takes place in a few hours.

The collection and weighing of silt particles are accomplished in the usual way. That which finally remains in the elutriating vessel is taken out after the end of the operation by closing the stop-cock, removing the stoppers with the piezometer tube, pouring the contents of the elutriating vessel into a beaker glass and rinsing out carefully all adhering particles. Examples of the working of the apparatus follow:

The soil was taken from the Imperial Russian Agricultural Experimental Institute at Gorki. It was a fine clay sand and was carefully treated with hydrochloric acid. The results of the analysis are given in the following table:

Velocities employed in millimeters.	Largest diameter of the collected particles in millimeters.	Percentage of silt product obtained in repeated elutriations.		
0.25	0.012	13.4	12.6	11.9
0.5	0.020	9.1	8.7	9.5
1	0.032	21.0	21.4	20.8
2	0.050	30.4	29.8	31.7
3	0.063	16.7	16.1	15.5
4	0.076	5.3	5.5	5.5
	Residue	4.2	4.9	3.8
	Total	100.0	99.0	98.7

Holthof modifies the apparatus of Schöne by putting into the lower mouth of the elutriator a little mercury so that the particles of earth are

deposited upon its surface and are thus better agitated and washed by the current of water.

218. Mayer's Modification of Schöne's Method.—An improvement of Schöne's apparatus in the direction of greater simplicity has been tested by Mayer^[150] with satisfactory results:

The apparatus, (Fig. 34), consists of a glass vessel having a glass stop-cock at the bottom for admitting the water. For a distance of twenty centimeters the sides of the tube are parallel and the diameter about one centimeter. Next for a distance of fifty centimeters the tube is conical expanding at a regular rate until the internal diameter reaches five centimeters. For a distance of ten centimeters the vessel is again strictly cylindrical and it is in this cylindrical portion that the separation of the different constituents takes place. The vessel is then rapidly narrowed until it carries the stopper A two centimeters in diameter. This stopper carries two glass tubes, one F bent downward to conduct the overflow into the receiving vessels, and one H for the purpose of regulating the rate of overflow by the height of the column of water therein. The orifice of the overflow tube F should be so regulated that with a pressure of five centimeters water in H, one liter shall pass over in ten minutes.

If the separation be conducted in an apparatus thus mounted and graduated with a pressure of two centimeters in H all that portion of the soil which can properly be called clay will pass over. The fine earth, that is, earth in which all coarse particles have been removed by proper sifting, is used in ten-gram lots for each experiment. The residue, after the separation is complete, consists of pure sand or at least pure sand mixed with humus. Before the fine earth is placed in the apparatus, the calcium carbonate therein is removed with hydrochloric acid. The treatment with hydrochloric acid, however, is not to be recommended in soil containing many undecomposed particles of calcium carbonate or dolomite for then large additions to the silt output might be made from these particles, which could not be regarded as coming from the soil as it actually exists. For alluvial soil, however, previous treatment with hydrochloric acid is recommended unconditionally.

219. Schöne's Method as Practiced by Osborne.—The apparatus used by Osborne^[151] was obtained from Germany and was similar to that described by Schöne in his original paper, except that it was furnished with

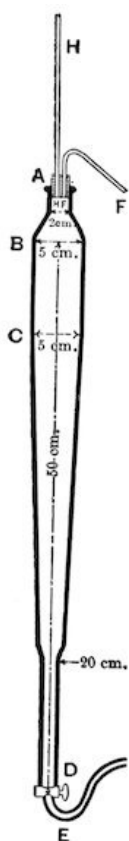


FIGURE 34.

SCHÖNE'S
APPARATUS
FOR SILT
ANALYSIS,
MODIFIED
BY MAYER.

a second elutriating tube as suggested by Orth. The modification made by Orth consists, essentially, of a second elutriating tube with straight sides into which the bulk of the soil is introduced, only the final part being carried over into the Schöne's tube proper. Water is supplied to the apparatus under constant pressure by means of a Mariotte's bottle.

The preliminary treatment recommended by Schöne is omitted, as these steps have been shown to be undesirable, on account of affecting the accuracy of the results.

Twenty grams of the air-dried soil are passed, under water, through a sieve of one-fourth millimeter mesh. That part of the soil which remains in suspension after being sifted is placed at once in the Schöne's tube of the apparatus, the coarser portion being rinsed into the Orth tube. The current is regulated so that the largest particles of quartz carried off have an average diameter of 0.01 millimeter. When all is carried off that can be removed at this rate the current is increased until the largest quartz grains passing off have a diameter of 0.05 millimeter.

As noticed by Hilgard with Schulze's apparatus, secondary currents are formed during the process of elutriation which descend along the walls of the conical portion of the Schöne's tube and some distance along the sides of the cylindrical portion. The tendency of these currents is to produce globular aggregates of particles which fall to the bottom. They are broken up from time to time by increasing the velocity of the current but even this method fails to disintegrate a considerable quantity of them.

220. Statement Of Results.—Two samples of soil from the garden of the experiment station analyzed by Schöne's method gave the following proportions of sediment.

In the table the term clay is used to designate all that part of the soil which has diameters less than 0.01 millimeter and which remains suspended after twenty-four hours standing in water having a depth of 200 millimeters.

SOIL, FROM GARDEN OF THE EXPERIMENT STATION.—NOT BOILED.

Analyses with the Schöne-Orth Elutriator.

	A.	B.	C.
Above 0.25 millimeter	48.82	48.82	48.82
0.25–0.05	27.36	29.94	22.37
0.05–0.01 millimeter	8.63	6.07	13.70
0.01 millimeter and less depos.	7.36	7.31	7.20
Clay (by difference)	1.00	1.03	1.08
Loss on ignition	6.83	6.83	6.83
	100.00	100.00	100.00

The last column *C* represents the average of three direct beaker elutriations according to the method of Osborne.

The differences which these figures show are found to be due to imperfect separation of the finer grades from the coarser and even when the various fractions separated by the Schöne method are subjected to beaker elutriation and the portions separated from them added to the grades to which they properly belong the Schöne elutriator was found to effect far less exact separations than the beaker method.

Samples of prairie soil from Mercer County, Ill., not boiled, were examined by the two methods with the following results:

	Schöne-Orth elutriation.	Beaker method.
Above 0.25 millimeter	0.76	0.62
0.25–0.05 millimeter	11.25	2.42
0.05–0.01 millimeter	52.65	43.58
0.01 millimeter and less deposited	14.84	31.58
Clay	4.44	5.81
Loss on ignition	14.49	14.49
	98.43	98.50

In this case it is seen that Schöne's method varies considerably from the beaker method and if the beaker method be regarded as correct the Schöne method is evidently less reliable.

In the next table are given the data of the examination of brick clay from North Haven, Conn., by the two methods.

BRICK CLAY FROM NORTH HAVEN, CONN.		
	Schöne-Orth elutriation.	Beaker method.
Above 0.25 millimeter	1.02	1.02
0.25–0.05 millimeter	3.91	0.76
0.05–0.01 millimeter	29.63	20.95
0.01 millimeter and less	58.58	71.01
Loss on ignition	6.60	6.60
	99.74	100.34

The failure of the Schöne method to give the results obtained by the beaker method is ascribed to the fact that it is impossible for the current of the strength used to disintegrate the clay and further that the particles after they are once separated tend to coalesce by the currents produced by the elutriating process.

221. The Berlin-Schöne Method.—Osborne has also made a study of the Schöne method as modified by the Bodenkulturlaboratorium of Berlin. The directions for the analysis by this laboratory method are as follows:

Five hundred grams of the soil are sifted through a sieve with circular holes two millimeters in diameter. Of the earth passing the sieve from 30 to 100 grams are boiled in water with constant stirring from one-half to one hour or longer, according to the character of the soil. The finer the texture of the soil the smaller the quantity taken and the longer the time of boiling. Treatment with acids or alkalies is not practiced.

The finer portion of the soil remaining suspended in the water, after boiling, is poured into the Schöne tube, the remaining coarse part is rinsed into the Orth tube. The clay, together with the finest sand, is collected in a separate vessel, the water in which it is suspended is evaporated and the residue after drying in the air is weighed. The rest of the operation is carried out as previously described except that the products of elutriation are not ignited but weighed air dried, in order that they may be further examined, chemically if desired. By proceeding in this manner the following results were obtained:

SOIL, FROM GARDEN OF THE EXPERIMENT STATION, BOILED FORTY-FIVE MINUTES.

Separations by the Berlin-Schöne method.

	Air-dried.	Ignited.
Above 0.05 millimeter	72.63	71.76
0.05–0.01 millimeter	14.17	12.53
0.01 millimeter and less	12.97	9.38
Loss on ignition		6.83
	99.77	99.50

For the sake of comparing the mechanical separation attainable by this procedure with those yielded by other methods, the air-dried products were ignited and again weighed and examined.

By subtracting from the ignited portion above 0.05 millimeter, 49.37 per cent, the amount of this soil that remained on a 0.25 millimeter sieve, the fraction between 0.25 millimeter and 0.05 millimeter is found, and the separations in this analysis may be compared with those previously obtained by the beaker method as follows:

SOIL FROM GARDEN OF EXPERIMENT STATION.

			Beaker Method.		
	Berlin-Schöne, boiled forty-five minutes.		Boiled twenty-three hours, average of four analyses.	Pestled, not boiled.	Not boiled nor pestled, average of three analyses.
Above 0.25 millimeter	49.37		47.77	48.82	48.82
0.25–0.05 millimeter	21.39		20.75	22.44	22.37
0.05–0.01 millimeter	12.53		11.18	12.55	13.70
<0.01 clay included	9.38		13.47	9.36	8.28
Loss on ignition	6.83		6.83	6.83	6.83
	99.50		100.00	100.00	100.00

Osborne concludes from the above facts that the Berlin-Schöne method, while showing close agreement with the beaker method, does not give results which are identical with that method. On subjecting portions separated by the Berlin-Schöne method to the beaker analysis additional separations were secured. In the case of heavy loams the inability of the Berlin-Schöne method to effect even a rough or approximate separation of the several grades becomes very conspicuous.

222. Method of Hilgard.—Two important principles lie at the foundation of this method; viz., 1, the use only of separating vessels of true cylindrical shape and 2, the employment of a mechanical stirrer to break up the floccules formed during the process of separation. The points in the apparatus to be considered are uniformity of the cross section of the elutriator at every point, exact perpendicularity of position, careful control of the rate of flow and continuous operation of the mechanical stirrer. According to Hilgard's observations the stirring due to the current of water alone is not sufficient to break up the floccules unavoidably formed during the separation, while any inclination of the sides of the elutriating vessel from the perpendicular due either to a conical shape or false position favors in the highest degree the formation of floccules due to reflex currents formed in the body of the liquid.

In order to carry out the idea suggested by Türschmidt of substituting for the accidental and indefinite products usually appearing in the statements of silt analyses sediments of known and definite hydraulic value a constant head of water is used, secured by means of a Mariotte's bottle connecting with the tube delivering the current through a cock provided with an arm moving on a graduated arc.

According to Hilgard the separation of sediments by the method of subsidence does not possess the analytical accuracy of the moving liquid method, especially when the latter is combined with mechanical stirring. The subsidence method requires close and continuous attention and in the case of fine sediments tending to flocculation the difficulties of the method are greatly increased. The views of Hilgard in respect of the laboriousness of the subsidence method lose, however, some of their force since the modifications of Osborne have come into use. The simplicity and cheapness of the apparatus required for subsidence give it at the start many advantages over the more elaborate process with a churn elutriator. For rigid scientific

investigation, however, the method of Hilgard is commended as a standard of comparison in all cases.

223. The Elutriator.—The instrument devised by Hilgard^[152] for the purpose of breaking up these flocculent aggregates is shown in figure 35, together with the simpler form, a Schöne's elutriator, figure 36, which can serve for grain sizes above eight millimeters hydraulic value. The latter is conveniently selected so as to have half the cross-section of the former, so that with the same position of the index lever the velocity will be just doubled. The cylindrical glass tube, of about forty-five millimeters inside diameter at its mouth, and 290 to 300 millimeters high, has attached to its base a rotary churn consisting of a brass cup, shaped like an egg with point down, so as to slope rather steeply at base, and triply perforated; viz., at the bottom for connection with the relay reservoir, and at the sides for the passage of a horizontal axis bearing four grated wings. This axis, of course, passes through stuffing boxes, provided with good thick leather washers, saturated with mutton tallow. These washers, if the axis runs true, will bear a million or more revolutions without material leakage. When a beginning is noted additional washers may be slipped on without emptying the instrument, until the analysis is finished.

For the finest sediments, from five to six hundred revolutions per minute is a proper velocity, which may be secured by clock work, turbine or electric power. The driving pulley should not be directly connected with the axis, both because it is liable to cause leakage, and because it is necessary to be able to handle the elutriator quickly and independently. This is accomplished by the use of "dogs" on the pulley and churn axis. For the grain sizes of one to eight millimeters hydraulic value lower velocities are sufficient; too low a velocity causes an indefinite duration of the operation and may be recognized by the increase of turbidity as the velocity is increased.

As the whirling agitation caused by the rotation of the dasher would gradually communicate itself to the whole column of water and cause irregularities, a wire screen of 0.8 millimeter aperture is cemented to the lower base of the cylinder.

The relay vessel should be a thick, conical test glass with foot; its object is to serve as a reservoir for the heavy sediments not concerned at the velocity used in the elutriator tube, and whose presence in the latter or in its base, the churn, would only cause abrasion of the grains and changes of current velocity, such as occur in the apparatus of Schöne, and compel the current measurement of the water delivered. It is connected above with the churn by a brass tube about ten millimeters in clear diameter, so as to facilitate the descent of the superfluous sediments, which the operator, knowing the proportion of area between the connecting tube and elutriator, can carry to any desired extent; thus avoiding the disturbance of the gauged current velocities, as well as all material abrasion.

A glass delivery tube should extend quite half way down the sides of the relay vessel, to insure a full stirring up of the coarse sediments when required. By means of a rubber hose, not less than twenty inches in length, this delivery tube connects with the siphon carrying the water from near the bottom of the Mariotte's bottle, a ten-gallon acid carboy. A stop-cock provided with a long, stiff index lever, moving on an empirically graduated arc, regulates the delivery of water through the siphon. Knowing the area of the cross section of the elutriator tube, the number of cubic centimeters of water which should pass through it in one minute, at one millimeter velocity, is easily calculated, and from this the lever positions corresponding to other velocities are quickly determined and marked on the graduated arc. The receiving bottle for the sediments, also shown in the figure, must be wide and tall, so as to allow the sediment to settle while the water flows from the top into the waste pipe. The receiving funnel tube must dip nearly to the bottom of the bottle. Thus arranged, the instrument works very satisfactorily, and by its aid soils and clays may readily be separated into sediments of any hydraulic value desired. But in order to insure correct and concordant results, it is necessary to observe some precautions; viz.,

(1) The tube of the instrument must be as nearly cylindrical as possible and must be placed and maintained in a truly vertical position. A very slight variation from the vertical at once causes the formation of return currents, and hence of molecular aggregates on the lower side.

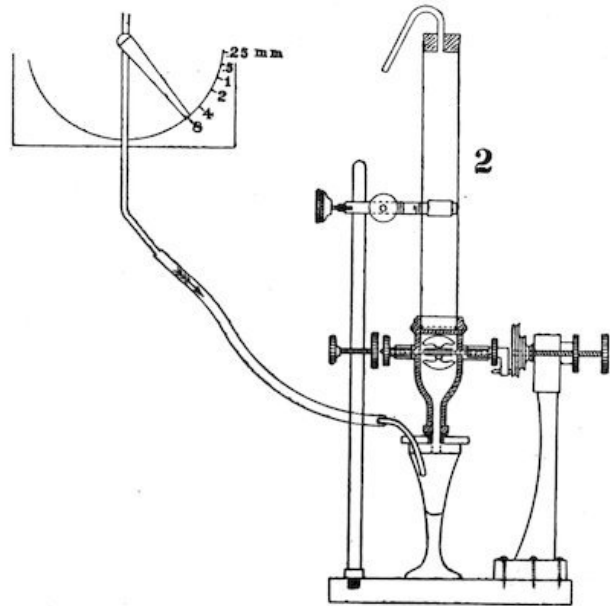


FIGURE 35. HILGARD'S CHURN ELUTRIATOR.

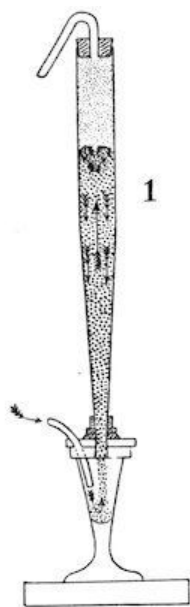


FIGURE 36.

IMPROVED SCHÖNE'S
APPARATUS
WITH
RELAY.

(2) Sunshine, or the proximity of any other source of heat, must be carefully excluded. The currents formed when the instrument is exposed to sunshine will vitiate the results.

(3) The Mariotte's bottle should be frequently cleansed, and the water used be as free from foreign matters as possible. For ordinary purposes it is scarcely necessary to use distilled water. The quantities used are so large as to render it difficult to maintain an adequate supply, and the errors resulting from the use of any water fit for drinking purposes are too slight to be perceptible, so long as no considerable development of the animal and vegetable germs is allowed. Water containing the slimy filaments of fungoid growths and moss protonema, algae, vorticellae, etc., will not only cause errors by obstructing the stop-cock at low velocities, but these organisms will cause a coalescence of sediments that defies any ordinary churning, and completely vitiates the operation.

(4) The amount of sediment discharged at any time must not exceed that producing a moderate turbidity. Whenever the discharge becomes so copious as to render the moving column opaque, the sediments assume a mixed character, coarse grains being, apparently, upborne by the multitude of light ones whose hydraulic value lies considerably below the velocity used, while the churner also fails to resolve the molecular aggregates which must be perpetually reforming where contact is so close and frequent. This difficulty is especially apt to occur when too large a quantity of material has been used for analysis, or when one sediment constitutes an unusually large portion of it. Within certain limits the smaller the quantity employed the more concordant are the results. Between ten and fifteen grams is the proper amount for an instrument of the dimensions given above.

224. Preparation of the Sample.^[153]—In some cases simple sifting will be sufficient to prepare the air-dried soil for the elutriator. In most cases, however, some mechanical aid must be invoked to secure particles of sufficient fineness. Nothing harder than a rubber pestle should be used and care must be taken not to break up any calcareous or ferruginous masses which the particles of fine soil may contain. The use of water in this mechanical attrition should be avoided, if possible, but in some heavy clay and adobe soils wetting becomes necessary. In this case the parts separated by the sieve are collected separately and the turbid mass removed by water and dried for further examination.

A sieve of 0.5 millimeter mesh is recommended as the best because that is almost exactly the diameter of the particles passing off at the maximum velocity of sixty-four millimeters per second to which the elutriator is adapted. The particles passing the 0.5 millimeter mesh are called fine earth.

225. Preparation by Boiling.—The method of preparation by boiling may be applied to all samples of fine earth. The fact pointed out by Osborne, that diffusibility of some clays is diminished by long boiling, renders it important to restrict the time of this operation as much as possible. With most soils from eight to fifteen hours will be long enough, occasionally extending to even twenty-four hours. A thin long-necked flask of about one-liter capacity should be used; filled three-quarters full with distilled water and the sample of soil added. The flask is supported over the lamp on a piece of wire gauze at an angle of 45°. It carries a cork with a long condensing tube. At first the boiling goes on smoothly, but after a time violent bumping may supervene, endangering the flask but promoting the object in view.

The contents of the flask are transferred to a beaker and diluted with distilled water to one and a half liters, shaken and allowed to settle for a time necessary to allow all particles of 0.25 millimeter hydraulic value to reach the bottom. The supernatant turbid liquid is decanted and the process repeated with smaller quantities of water until no further turbidity is produced. The united decantations, of which there will be from four to eight liters, are well shaken and a proper time allowed for the 0.25 millimeter hydraulic value sediments to fall. This last step is necessary to remove any such sediments which may have been carried over mechanically in the first separation. The dilution being very great, a fairly perfect separation is thus secured and the sediments are then ready for the elutriator.

226. Separation of Clay and Finest Silt.—The property which pure clay possesses, of remaining suspended almost indefinitely in pure water, affords a ready means of separation from the silt particles of less than 0.25 millimeter hydraulic value. But the finest silt particles subside so slowly that this method of separation is too long to become practically applicable to secure a perfect demarcation between the finest silt and so-called colloidal clay.

Hilgard recommends the following procedure: The clay water from the previous separation is placed in a cylindrical vessel of such a diameter as to allow the column of water to be 200 millimeters high where it is allowed

to settle for twenty-four hours. When the clay is very abundant a longer time may be allowed; viz., from forty to sixty hours. The line of separation between the dark silt below and the translucent clay above is sharply defined. Finally the clay water is decanted and the remaining liquid poured off leaving the sediment as sharply defined as possible. The sediment is rubbed with a rubber pestle and a few drops of ammonia water added. Distilled water is added, the beaker well shaken or stirred to break up the floccules that may have formed and subsidence permitted as before. This operation is repeated from six to nine times until the water remains quite clear after subsidence or the decanted turbid water fails to be precipitated by brine showing the suspended matter to be fine silt and not clay.

The diameter of the particles of silt thus obtained is from 0.001 to 0.02 millimeter, and it is impossible to obtain it quite free from any admixture with clay.

227. Estimation of the Colloidal Clay.—The importance of the colloidal constituent of the clay is such as to make its direct determination desirable. The volume of the clay waters at this stage of the analysis may amount to twenty liters. One method of determination consists in evaporating an aliquot portion and this method will yield good results if the sample be free from soluble salts and the quantity taken be not too small. At least 500 cubic centimeters should be used for this purpose. A better method consists in precipitating the clay by means of a saline solution. A saturated solution of salt is recommended for this purpose of which fifty cubic centimeters are sufficient to precipitate the clay from one liter of the clay water. The precipitation is hastened by heating. Each portion of the clay water should be precipitated as soon as obtained, the total volume of the precipitate at the end of twenty-four hours is thus reduced to a minimum. The clay water from the succeeding separations of the same analysis can be mixed with the precipitate which diffuses therein, thus promoting the precipitation of the rest of the clay inasmuch as the separation takes place more readily where more clay is present. When all the clay is thus collected it can be gathered on a tared filter and washed with weak brine. Pure water may not be used because of the diffusibility of clay therein. After drying at 100° and weighing it is washed with a weak solution of ammonium chlorid until all sodium is removed. The filtrate is evaporated to dryness, ignited at low redness, and weighed. The weight of the sodium chlorid thus obtained plus the weight of the filter deducted from the total weight gives the weight of the clay precipitate. Whenever the clay collected as above will not diffuse in water it may be washed with water and its weight directly obtained. An excess of iron in clay will usually allow of the above treatment.

228. Properties of Pure Clay.—The percentage of pure clay as obtained by the procedure described is about seventy-five in the finest natural clays, forty-five in heavy clay soils, and fifteen in ordinary loamy soils. When freshly precipitated by brine it is gelatinous resembling a mixed precipitate of iron and aluminum oxids. Its volume greatly contracts on drying, clinging tenaciously to the filter, from which it may be freed by moistening. On drying, it becomes hard, infriable, and often resonant. It usually possesses a dark brown tint due to iron oxid. Under the action of water it swells up like glue, the more slowly as the percentage of iron is greater. In the dry state it adheres to the tongue with great tenacity. According to Whitney the finest particles of colloidal clay have a diameter of 0.0001 millimeter. With a magnifying power of 350 diameters, however, Hilgard states that no particles can be discerned.

229. Chemical Nature of the Fine Clay.—The fine particles separated as above consist essentially of hydrous aluminum silicate or kaolinite. It doubtless contains, however, other colloids or hydrogels whose absorptive powers are similar to those of clay. It appears also to contain sometimes free aluminum hydroxid, and colloidal ferric hydroxid, and amorphous zeolitic compounds.

While the most careful mechanical separation can give at best only approximately the really plastic kaolinite substance, yet it is far closer than that attained by determination of total alumina with boiling sulfuric acid. By the latter treatment all the lime-kaolinite particles are decomposed and the method does not lead to even an approximate estimate of the soil's plasticity.

230. Separation of the Fine Sediments.—The sediments remaining after the separation of the clay and fine silt are ready for separation in the churn elutriator. The apparatus mounted, as already described, is brought into use by beginning with a low velocity of the water in the upright tube. The rate of flow should be set at from 0.25 millimeter to 0.50 millimeter per second, and the churn put in motion.

When the elutriating tube is partly full of water the sediments should be poured in from a small beaker which is perfectly cleaned by means of a washing flask. The stopper and delivery tube of the elutriator are then put in place. The rate of flow should be so regulated that the sediments shall have had a few seconds of subsidence before the water is within thirty millimeters of the top. At this point the required velocity for the first sedimentation should be turned on; viz., 0.25 millimeter per second. At first the sediment passes off rapidly and the water in the elutriator is distinctly turbid. This excess of turbidity ceases in a few hours and then some attention is necessary in order to determine when the process is complete. In fact it never is completely finished, but where no more than one

milligram of silt comes off with one liter of water it may be said to be practically done. The time required for the first operation varies from fifteen to ninety hours. Downward currents in the elutriator are likely to form in spite of all precautions, and floccules of silt adhere to its walls. These should be detached from time to time with a feather in order to bring them again in contact with the churn.

Hilgard has found that, practically, 0.25 millimeter per second is about the lowest velocity available within reasonable limits of time, and that by successively doubling the velocities up to sixty-four millimeters a desirable ascending series of sediments is obtained; provided always, that a proper previous preparation has been given to the soil or clay. It would seem that according to the prescription given above for the preliminary sedimentation, no sediment corresponding to 0.25 millimeter velocity should remain with the coarser portion. That such is nevertheless always the case, often to a large percentage, emphasizes the difficulty, or rather impossibility, of entirely preventing or dissolving the coalescence of these fine grain sizes by hand stirring, as in beaker elutriation. It is only by such energetic motion as is above prescribed that this can be fully accomplished, and the delivery of 0.25 and 0.50 millimeter hydraulic value really exhausted.

It is desirable to run off the upper third of the column at intervals of fifteen to twenty minutes by temporarily increasing the velocity. Recent sediments, river alluvium, etc., are more easily separated than soils of more ancient formation. The second, third, etc., separations are naturally accomplished in much less time than the first. The respective velocities of the separations should be 0.25 millimeter, 0.50 millimeter, one millimeter, two millimeters, four millimeters, eight millimeters, sixteen millimeters, thirty-two millimeters, and sixty-four millimeters a second. Below a velocity of four millimeters a second the mechanical stirrer is indispensable. Above this velocity the current of water in the conical base will be sufficient to bring the desired particles into the ascending column. At this velocity also a smaller elutriating tube having one-half or one-quarter the cross-section of the first may be employed to hasten the operation and diminish the quantity of water required. The quantity of water required for a complete separation is from 100 to 120 liters. Any soft water free of organic matter may be used, but distilled water is best. Hard water should be avoided.

The mean time required for the different separations is as follows: 0.25 millimeter hydraulic value, thirty-five hours; 0.50 millimeter hydraulic value, twenty hours; one millimeter hydraulic value, seven and a half hours; two to sixty-four millimeters hydraulic value, eight hours. With proper arrangements for night work, an analysis may be finished in three or four days not counting the time required for the previous separation of the clay.

231. Weighing the Sediments.—The sediments should be dried at the same temperature used for drying the soils. Hilgard dries both at 100°. Great care should be used in weighing the exceedingly hygroscopic clay sediments. In the case of the sediment of 0.25 millimeter hydraulic value it is allowed to subside as much as possible and after removing the supernatant water the residue, twenty-five to fifty cubic centimeters, is evaporated in a platinum dish and weighed therein. The water can be completely decanted from the other sediments, and they can be dried and weighed without any unusual precautions.

The loss in the separation of clays and subsoils containing but little organic matter is usually from 1.5 to 2 per cent. This loss is partly due to the fine silt which comes off during the whole of the process and which is lost in the decanted waters of the sediments of 0.25 millimeters hydraulic value and above. The procedures indicated above are not strictly applicable to soils rich in humus and other organic matters, but the destruction of these matters by ignition leaves the residual soil in a condition wholly unfit for sedimentary separation.

232. Classification of Results.—A convenient method of stating the results of an analysis may be seen from the following classification. The percentage obtained for each of the classes is to be entered in the column provided for that purpose.

	No.	Names of Silt Classes.	Diameter of grains in millimeters.	Velocity of current millimeters hydraulic value.	Per cent.
Sieves.	1.	Grits	1–3		
	2.	Fine grits	0.5–1		2.07
Elutriator without churn.	3.	Coarse sand	0.50	64	
	4.	Medium sand	0.30	32	0.55
	5.	Fine sand	0.16	16	
Elutriator with churn.	6.	Finest sand	0.12	8	0.21
	7.	Coarse silt	0.072	4	1.21
	8.	Large silt	0.047	2	2.92
	9.	Medium silt	0.036	1	7.36
	10.	Silt	0.025	0.5	8.86
	11.	Fine silt separated in elutriator	0.016	0.25	7.85
Beaker sedimentation.	12.	Fine silt separated from clay water	0.010	<0.25	35.22
	13.	Clay	0.0001	<0.0023	33.16
Total					99.36

The measurements of diameters in the above table is of the best formed quartz grains in each class. Naturally the actual size of the particles may vary in each class within the extreme limits of the diameter next above and below. It is not easy to indicate in popular language distinctions not popularly made but the grades of particles designated by the names grits, sand and silt, may serve, at least, to establish uniformity of expression. The term grits is thus applied to all grains above one millimeter in diameter up to gravel. Below one millimeter down to 0.1 millimeter may be called sand and below that silt may designate the particles down to an impalpable powder.

233. Influence of Size of Tube.—The diameter of the elutriating tube exerts a sensible influence on the character of the sediments. The friction against the sides of a small tube is comparatively greater than in a large tube. Strictly speaking, no class of sediments strictly corresponds to the hydraulic value calculated from the cross section of the tube and the quantity of water supplied thereto. The sediments correspond actually to higher velocities, due to the fact that the lateral friction causes a more rapid flow in the center of the water column. This may be demonstrated by slightly diminishing the velocity while a sediment is copiously discharging. The turbid column then remains stationary while clear water is running off.

234. Statement of Results.—A complete silt analysis of a soil, conducted by the method of Hilgard, depends largely for its practical value on an intelligible tabulation. The method of collating results is illustrated in the table of analyses of Mississippi soils shown on page 237.

The character of the soils entering into the given analyses is as follows:

Nos. 248, 206, 209, 397, 219, belong to the end of the drift period.

No. 230 is one of the two chief varieties of soils occurring in what is known as the flat-woods, a level surface bordering on the cretaceous area, having lower tertiary clays near the surface.

No. 165 is a light soil which occurs in the former in irregular strips and patches, is easily tilled, absorbs rain water readily, but is subject to drought and does not hold manure.

SILT ANALYSES OF MISSISSIPPI SOILS AND SUBSOILS.																			
Designation of Materials.	Diameter. Millimeters.	Velocity (Hydraulic value). Millimeters per second.	DRIFT	UPLAND.										MISSISSIPPI BOTTOM.					
				YELLOW LOAM.										Champlain.			MODERN.		
				SANDY.					LOAM.					Tertiary.			Swamp River.		
				238 White pipeclay. Tishomingo Co.	248 Tallahoma subsoil. Jasper Co.	165 L.L. Flatwoods soil. Chickasaw Co.	206 Pine Hill soil. Smith Co.	209 Pine Hill subsoil. Smith Co.	397 Oxford subsoil. Lafayette Co.	219 Table Land subsoil. Benton Co.	173 Prairie subsoil. Monroe Co.	230 High Flatwoods soil. Pontotoc Co.	246 Red Hills subsoil. Attala Co.	196 Hog Wallow subsoil. Jasper Co.	390 Buckshot soil. Issaquena Co.	237 Loess. Claiborne Co.	365 Tallahatchie soil. Panola Co.	377 Frontland subsoil. Sunflower Co.	395 Dogwood ridge soil. Coahoma Co.
1 Coarse grits	1.0 to 3.0				6.94	2.90	0.36	0.36		0.23	2.10	0.33	1.97	0.83	0.09	0.24	0.09		
2 Fine „	0.5 to 1.0				17.65	6.96	2.98	0.83				0.35		1.19	0.05				
3 Coarse sand	0.40	6	0.06		18.81	2.81	6.62	6.21	0.79	1.47	0.62		0.72	1.96		0.37	0.04	0.32	0.15
4 Medium „	0.30	32			10.16	4.41	7.75	3.38		2.33			2.32	1.64	0.36	0.61	0.05		
5 Fine „	0.16	16			2.66	3.13	3.01	3.85		1.17			2.09	0.88		0.93	0.21	2.97	0.47
6 Finest „	0.12	8	0.08		1.66	2.02	1.59	1.49	0.18	0.78	0.20	0.23	0.70	0.26	0.31	1.65	1.30	2.41	3.74
7 Dust „	0.072	4	0.02		1.02	2.23	1.19	0.64	0.78	0.76	1.26	0.18	1.29	0.19	0.27	1.95	2.68	16.90	12.38
8 Coarsest silt	0.047	2	0.04		0.88	5.06	3.56	2.63	3.56	9.79	2.92	1.61	1.81	2.49	1.56	14.25	9.38	19.79	13.27
9 Coarse „	0.036	1	0.08		1.96	9.67	6.50	5.40	13.12	7.26	7.36	2.66	3.60	3.67	2.23	16.20	9.88	13.90	15.87
10 Medium „	0.025	0.5	0.08		7.89	14.18	13.97	7.77	16.64	13.14	8.81	9.13	2.73	5.39	3.68	20.08	20.37	4.27	9.93
11 Fine „	0.015	0.25	2.00		8.40	22.03	14.20	16.65	27.28	15.07	7.85	26.64	13.30	10.31	8.97	5.59	19.79	1.89	7.26
12 Finest „	0.008	<0.25	21.15		15.53	15.62	29.36	37.75	18.87	26.50	35.22	32.35	25.33	24.18	38.19	33.38	25.30	30.08	19.67
13 Clay	0.0001	<0.0023	74.65		8.63	7.86	4.58	10.70	17.23	19.19	33.16	25.48	40.25	47.03	44.30	2.51	9.64	5.51	10.35
Compaction (resistance to tillage)				98.16	99.28	98.68	95.67	97.77	98.35	97.65	99.50	97.87	96.11	100.00	100.01	97.74	98.73	98.04	96.58
Porosity				0.36	32.56	45.33	48.14	45.10	63.38	60.82	69.77	84.47	78.88	81.52	89.46	41.48	54.63	37.48	28.57
Hygrosopic Moisture (+7° to +21°)				59.55	40.40	37.89	47.13	20.23	26.04	17.04	6.40	39.18	10.12	4.87	38.44	23.63	58.25	61.50	49.20
Ferric Oxide				9.09	1.80	3.36	2.48	7.69	8.79	7.21	11.35	9.33	18.60	14.48	14.31	4.19	6.12	5.68	3.95
				0.13	1.10	1.45[G]	1.25	4.15	2.53	5.11	5.42	5.90[G]	10.50	4.00	5.82[G]	3.27	2.58	2.31	2.69

G. Bog ore.

No. 248 is from a soil stratum three feet thick. The soil is so light that the finer particles of it are carried away by high winds.

Nos. 206 and 209 are typical of the soils producing the long-leaf pine. This soil is much improved by an admixture of the subsoil No. 209, which enables it to hold manure.

No. 219 is a cotton upland soil of the best quality, found in Western Mississippi and Tennessee.

No. 397 is the same soil of a second rate quality. These lands are easily washed into gullies on account of their lack of perviousness to water. They also easily swell up in contact with water, and become thereby readily diffused. The denudations produced by heavy rains are rapidly destroying the lands covered by these soils.

No. 173 is a sedimentary or residual subsoil of the cretaceous prairies of Northeastern Mississippi, forming a stratum from three to seven feet thick.

No. 230 is a residual soil which is formed by the disintegration of the old tertiary clays. It yields good crops only in very favorable years, and is easily injured both by wet and dry seasons.

No. 246 is a soil of the same origin, but is more easily tilled than the foregoing, does not crack, but becomes very hard when dried slowly. Its superiority to the former soil as regards tillage consists in the presence of the large amount of iron and lime.

No. 196 is a typical heavy clay soil; is better suited for the potter than the farmer. It cracks on drying, whence its popular name. On the accession of rain the edges of these cracks crumble and fall, until finally the lumpy surface is produced which is locally known as hog wallows.

No. 390, the richest soil of the Yazoo Bottom, seems to have a physical composition like the preceding one. Its superiority is due not only to the increased quantity of plant food which it contains, but to its property of crumbling on rapid drying. Even when plowed wet, on drying each clod crumbles into a loose pile resembling buck-shot; whence its name. It is strongly calcareous.

As comparative data, are added the soils 365, 377, and 395, representing alluvial deposits, and two deposits from the Delta of the Mississippi.

235. Comparison of Osborne's Method with Hilgard's Method.^[154]—The comparative results obtained by Osborne's method, beaker elutriation, and Hilgard's method, churn elutriation, are given in the following tables:

SOIL FROM EXPERIMENT STATION GARDEN, NEW HAVEN, CONN. SURFACE SOIL, BOILED TWENTY-THREE HOURS.

Diameter in millimeters.	Churn elutriation.		Beaker elutriation. Average of four analyses.	
	per cent.	per cent.	per cent.	
Removed by sieves	47.77	47.77		47.77
0.25–0.05	22.06	21.95		20.75
0.05–0.01	11.20	11.62		11.18
<0.01	9.82	9.14		10.72
Clay (difference)	2.32	2.69		2.75
Loss on ignition	6.83	6.83		6.83
	100.00	100.00		100.00

SUBSOIL, BOILED TWENTY-THREE HOURS.

Diameter in millimeters.	Churn elutriation.		Beaker elutriation.	
	per cent.	per cent.	per cent.	per cent.
Removed by sieve	39.33	39.33	39.33	39.33
0.25–0.05	33.61	30.83	32.35	32.95
0.05–0.01	10.91	12.25	10.32	10.37
<0.01	7.05	8.11	8.29	7.64
Clay	5.02	5.40	5.63	5.63
Loss on ignition	4.08	4.08	4.08	4.08
	100.00	100.00	100.00	100.00

These analyses agree quite as well as could be expected from two such different methods.

Elutriation of Clayey Soils.—Hilgard found that by churn elutriation no satisfactory results could be obtained on clay without long boiling and subsequent kneading of the finer sediments. Osborne examined a sample of clay by his method after previous boiling for twenty-three hours. When the sediments were examined by the microscope they were found to contain many aggregations of particles which broke into dust under the pressure of the thin glass slide-cover. These sediments were then gently crushed in the beaker with the help of a soft rubber stopper with a glass rod for a handle, the grinding together of the particles being, as much as possible, avoided. This pestling was continued with clear water as long as it occasioned turbidity. Comparison of the analyses shows that practically identical results were obtained on this soil whether it was boiled or not and indicates that the sediments are reduced to their elements by gentle pestling alone. For such soils, therefore, it is demonstrated that pestling is a much safer treatment than boiling. The same remark may be applied to the fertile prairie soil of Mercer County, Illinois, where boiling proved quite insufficient and in which the pestling process proved completely successful. The general conclusions arrived at from the results obtained by Osborne are as follows:

1. On sands and silts of pure quartz or similar resistant material Hilgard's method and beaker elutriation give practically identical results.
2. With coarse sands and silts upon whose grains finer matter has been cemented by silicates, etc., and with soils containing soft slaty detritus, the churn elutriator with preliminary boiling may give results too low for the coarse and too high for the finer grades. In these cases beaker elutriation with pestling yields more correct figures.
3. Some loamy soils containing no large amount of clay or of extremely fine silt, as well as prairie soils rich in humus, cannot be suitably disintegrated by twenty-four hours' boiling, but are readily reduced by pestling.
4. Beaker elutriation preceded by sifting, gives results in five or six hours with use of two to three gallons of pure water, which, in churn elutriation, require several days and consume eight to ten gallons of pure water.
5. Hilgard found that practically 0.25 millimeter is about the lowest velocity of water current per second available within reasonable limits of time in his elutriator. Such a current carries over particles up to 0.015 millimeter diameter and hence the silts of less dimensions cannot be conveniently separated by churn elutriation. In beaker elutriation there is no difficulty in making good separations at 0.01 millimeter and at 0.005 millimeter.
6. Beaker elutriation requires no tedious boiling or preliminary treatment and with careful pestling of the sediments gives, we believe, as nearly as possible, a good separation of adhering particles and at every stage of the process carries with it, in the constant use of the microscope, the means of testing the accuracy of its work and of observing every visible peculiarity of the soil. It is not claimed that pestling may not easily go too far, but in any case a good judgment may be formed of its effects and of the extent to which it is desirable to carry it.
7. In beaker elutriation the flocculation of particles occasions little inconvenience and does not impair the accuracy of the results.

236. Comparison of the Osborne with the Schloesing Method.—Schloesing's method has been compared by Osborne^[155] with the beaker method of elutriation with the following results:

SCHLOESING'S METHOD.

	Per cent.
Calcium carbonate	4.20
Sand	64.91
Clay	22.65
Humus	none
Loss on ignition	6.60

BY THE DIRECT BEAKER METHOD.

	Per cent.
Above 0.25 millimeter diameter	1.02
0.25–0.05 millimeter diameter	0.76
0.05–0.01 millimeter diameter	20.95
Below 0.01 millimeter diameter	71.01
Loss on ignition	6.60

It is seen by the above that there is little agreement between the results of the two methods.

With the prairie soil from Mercer County, Ill., the following results were obtained working on the original sample and the sand separated by the Schloesing process:

SCHLOESING'S METHOD.

	Per cent.
Calcium carbonate	0.88
Humus	1.57
Loss at 150° C.	4.42
Sand	82.86
Clay	7.86
	97.59

BEAKER METHOD.

	Schloesing's sand.		Original soil.
	Dried at 150°C., per cent.	Ignited, per cent.	Ignited, per cent.
Above 0.25 millimeter diameter	0.12	0.10	0.92
0.25–0.05 millimeter diameter	3.58	3.55	2.89
0.05–0.01 millimeter diameter	42.69	41.87	42.86
0.01–0 millimeter diameter	23.66	20.47 }	31.44
Clay	12.81	10.14 }	
Clay			7.40
Loss on ignition		6.73	14.49
	82.86	82.86	100.00

Osborne says the above figures indicate that the treatment with acid has disintegrated the particles of less than 0.01 millimeter diameter so that one-third of this portion appears as clay, according to the Hilgard method of estimating clay, which is the one employed.

As to the humus it may be noted that loss in the analysis by Schloesing's method; viz., 2.41 per cent, plus loss at 150° = 4.42 per cent, plus humus found = 1.57 per cent, plus carbon dioxid ($\frac{44}{56}$ of 0.88 =) 0.69 per cent amounts to 9.09 per cent, while the loss on ignition which represents humus, carbon dioxid and water is 14.49 Per cent. The 5.40 per cent difference must evidently be, for the most part, humus which has escaped estimation by the Schloesing method, having been distributed among the sand and clay.

237. The Mechanical Determination of Clay.—Schloesing's method for the separation of the clay as stated by Osborne^[156] is essentially one of subsidence for twenty hours from a volume of from 200 to 250 cubic centimeters of water, but of no specified height. Hilgard's conventional method requires the same time and a height of solution of 200 millimeters.

Such methods of separation assume, first, that most of the sand and, second, that little of the clay shall settle within the fixed time. That both of these assumptions are fallacious, the following experiments show. The clay obtained by twenty hours subsidence from thirty grams of brick clay is suspended in four liters of distilled water and allowed to settle out completely, which requires several days. The water is then decanted so as to remove all soluble matters, the jar again filled with distilled water, and the clay and fine sand allowed to settle again for several days. The upper three-quarters of the liquid are then decanted and made up to a volume of four liters, and this is allowed to stand several days, when a considerable sediment forms. A decantation is again made as before. The operations are repeated until the clay water has been so far freed from the clay as to become opalescent; then it first ceases to deposit any appreciable sediment. A microscopic examination of the several sediments thus collected shows them all to contain particles of sand. It appears, therefore, that only after the liquid containing the clay has become opalescent does it cease to deposit fine particles of sand as well as of clay.

Furthermore, the character of the true clay itself is so changed under certain conditions that it loses the property of remaining in prolonged suspension in water.

A sample of clay which has been freed from particles of sand exceeding 0.005 millimeter diameter is suspended in water and precipitated from it by freezing. It is then washed by decantation with alcohol and dried in the air. A portion of this clay is shaken with water and allowed to stand a few hours, during which time the greater part of it has settled. After decanting the water and suspended clay and repeating this process a few times, a very considerable part of the clay is left which will subside completely through 100 millimeters in a few hours. After standing under water for several months, only a small part of the clay has regained the quality of prolonged suspension. It has been found, however, that if this clay be pestled, this quality of prolonged suspension is restored to it to a very considerable degree.

It is evident, therefore, that conventional methods depending on simple subsidence can give no accurate results because the ever varying amounts of finest sand and clay in different soils yield variable mixtures of the two when subjected to any simple course of treatment by elutriation and subsidence.

The method of persistent pestling and repeated subsidences and decantations continued until no further separation can be effected, although extremely tedious, is the only one which has so far yielded even approximately good separations on any of the clayey soils examined by Osborne.

A single subsidence of the clay water for twenty-four hours will free it from all particles of sand having a diameter greater than 0.005 millimeter, but in many cases a considerable amount of finer sand will remain in suspension for many hours or days.

On the other hand, the sediment formed during the twenty-four hours subsidence will not be free from clay, as may be easily seen by suspending it in water a second time and allowing it to stand again for twenty-four hours. Both Hilgard and Schloesing direct attention to these defects, but assume that they do not usually influence the results to a sufficient extent to deprive them of value. In many cases this is undoubtedly true, as, for example, in such soils as that from the garden of the Experiment Station, at New Haven, in which there is but little clay and fine sand; but in soils of the opposite character, as in the North Haven brick clay where exact separations are most desirable, a very considerable error is thus inevitably encountered.

238. Effect of Boiling on the Texture of Clayey Soils.—Most investigators who have worked upon mechanical soil analysis advise boiling with water in order to detach clay and sand from each other and make a good separation of the several mechanical elements practicable or possible. In general, however, the instructions as to the time and manner of boiling are rather indefinite, and no definite research as to the effects of this treatment has been undertaken.

The practice of Hilgard, to boil twenty-four hours or even longer in case of adhesive clays, according to Osborne^[157] appears to be objectionable in view of the dehydration and change of physical properties known to occur in case of many hydroxids, especially those of iron and aluminum, which may be present in the soil. It is a familiar fact that the hydroxids above named and many other amorphous substances when precipitated from cold solutions are more bulky and less easily washed upon a filter than when thrown down hot. It is also well known that their properties are considerably changed by warming or boiling with water. Heating with water to boiling for some hours or days gradually converts the bulky brown-red ferric hydroxid, which when precipitated cold and air-dried for eighteen days, contains thirty-eight per cent of water, into a much denser, bright red substance containing

but two per cent of water. St. Gilles has also observed the partial dehydration of aluminum hydroxid from $\text{Al}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ to $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ by prolonged boiling.

The hydrate of silica and the highly hydrated silicates are most probably affected in a similar manner, and if such be the case, boiling would evidently change the constitution of clay in a very essential degree.

The following experiments throw light on this subject:

Ten grams of North Haven brick clay were boiled continuously for nine days with about 700 cubic centimeters of distilled water, in a glass flask of one liter capacity and furnished with a reflux condenser. Fifteen grams were boiled in the same manner for eight and one-half days. When the boiling was concluded, the soil was found to have assumed a granular condition, the clay and fine sand being collected into a mass of small grains resembling coarse sand and settling rapidly. One portion thus boiled was elutriated by the beaker method, the other by Hilgard's. The pestle was not used on either of those portions as it was desired to determine simply the effect of prolonged boiling. The separations thus accomplished are here compared with the elutriations of the same soil boiled twenty-three hours and of the pestled but unboiled soil.

Diameter of particles.	Hilgard elutriation.		Beaker elutriation.	
	Not pestled.	Not pestled.	Not Pestled.	Pestled.
	Boiled twenty-three hours, per cent.	Boiled nine days, per cent.	Boiled eight and a half days. per cent.	Not boiled. per cent.
By sieves	3.36	3.24	3.63	3.49
0.25–0.05	1.21	1.11	1.91	1.29
0.05–0.01	28.27	33.04	33.61	27.02
0.01–0	56.29	48.85	54.78	52.21
Clay	4.92	3.05	1.97	10.15
Loss on ignition	5.95	5.95	5.95	5.95
	100.00	95.24	101.85	100.11

Here it is observed that the eight to nine days boiling diminished the clay as determined by Hilgard's conventional method by seven to eight per cent, increasing the dust by two to three per cent, and the silt by about six per cent.

Under the microscope small, rounded, opaque, brown granules were seen in large numbers, which when pressed under the cover glass, broke up into a multitude of very fine particles.

From these experiments it would appear to be conclusively proved that too long boiling precipitates clay and thereby defeats the very object of the operation.

In these experiments the time of boiling was prolonged in order to bring out unmistakably the effects of this operation. If ebullition for eight or nine days reduces clay from ten to two per cent, increasing the 0.05–0.01 millimeter diameter grades by six per cent, it is evident that boiling for one day or a shorter time becomes a questionable treatment.

Further experiments^[158] made by boiling clay in a platinum vessel with a platinum condenser showed that this precipitation of the clay was largely if not wholly due to the salts extracted from the soil.

When the clay has once been converted into the granular condition, considerable difficulty is experienced in restoring it to the state in which it is capable of prolonged suspension in water.

The results of the studies herewith reported may be summed up as follows:

1. The Berlin-Schöne method of elutriation gives fairly correct separations with sandy soils containing little clay or matters finer than 0.01 millimeter diameter, but on soils of fine texture, as loams rich in humus and clays, it gives results which are grossly inaccurate, the error on single grades amounting to from eight to fourteen per cent.
2. In respect of rapidity, economy of time, and ease of operation, the Schöne elutriation has no advantage over the beaker method.
3. Schloesing's method on its mechanical side makes no satisfactory separations, and the chemical treatment it employs is liable to alter seriously the texture of the soil.
4. The determination of clay from a single subsidence from any conventional depth or volume of water, or for any conventional time, is not a process certain to effect even a roughly approximate separation of the finest quartz grains from true clay.

5. Boiling with water must be rejected as a treatment preliminary to mechanical analysis, because it not only abrades and reduces the coarser sediments, but may dehydrate and coagulate the true clay and thus alter essentially the texture and grain of the soil.

239. General Conclusions.—The methods of Hilgard and Osborne have been given in detail and largely in the descriptive language used by the authors. The other methods of elutriation in use in other countries have also been described. For practical use the methods of Hilgard and Osborne are to be preferred to all others. For simplicity and speed the Osborne method has the preference over the Hilgard. For rigid control of the work the Hilgard method is to be preferred. The effect of long boiling on clay pointed out by Osborne would suggest that the boiling process preliminary to the Hilgard method be made as short as possible. It would seem that the churn attrition in the Hilgard method might well be regarded as a substitute for the soft pestling of the Osborne process, and any prolonged boiling in the former method might be safely omitted. When carefully carried out, the results of the Hilgard and Osborne method are fairly comparable.

240. Distribution of Soil Ingredients.—The determination of the distribution of the soil ingredients in the sediments obtained in silt analysis is illustrated by the following table:[^{\[159\]}](#)

Hydraulic value Per cent in soil	Clay 21.64		<0.25mm 23.56		0.25mm. 12.54		0.5mm. 13.67		1.0mm. 13.11		Other sediments.	Total sediments.	Original soil.
	A	B	A	B	A	B	A	B	A	B			
Insoluble residue	15.96	4.35	73.17	17.29	87.96	11.03	94.13	12.72	96.52	12.74	13.76	71.89	70.53
Soluble silica	33.10	7.17	9.95	2.34	4.27	0.53	2.35	0.32				10.36	12.30
Potash	1.47	0.32	0.53	0.12	0.29	0.04	0.12	0.01				0.49	0.63
Soda	(1.70)		0.24	0.06	0.28	0.04	0.21	0.02				0.12	0.09
Lime	0.09	0.03	0.13	0.03	0.18	0.02	0.09	0.01				0.09	0.27
Magnesia	1.33	0.29	0.46	0.11	0.26	0.03	0.10	0.01				0.44	0.45
Manganese	0.30	0.06	0.00	0.00	0.00	0.00	0.00	0.00				0.06	0.06
Ferric oxid	18.76	4.06	4.76	1.11	2.34	0.29	1.03	0.14				5.60	5.11
Alumina	18.19	3.97	4.32	1.04	2.64	0.33	1.21	0.17				5.51	8.09
Phosphoric acid	0.18	0.04	0.11	0.02	0.03	0.00	0.02	0.00				0.06	0.21
Sulfuric acid	0.06	0.01	0.02	0.01	0.03	0.00	0.03	0.00				0.02	0.02
Volatile matter	9.00	1.33	5.61	1.43	1.72	0.23	0.92	0.29	0.36			3.64	3.14
Total	100.14	21.64	99.30	23.56	100.00	12.54	100.21	13.67	13.10			98.28	100.63
Total soluble matter.	75.18		20.52		10.32		5.16						
" " bases	41.84		10.44		5.99		2.76						
Soluble silica in crude substance.	0.38	0.01											0.19

A. Calculated on the amount of sediment. B. Calculated on the amount of soil.

It is seen from the above analyses that the clay is by far the richest in mineral constituents, of all the ingredients separated in silt analysis, the amount in the clay being more than twice that of all the others combined. Its volatile matter is also the largest. The large amount of soda, however, is probably in part due to the sodium chlorid used in the precipitation of the diffused clay. The following points in regard to the distribution of the different ingredients are instructive:

1. The iron and alumina exist in almost identical relative proportions in each sediment, making it probable that they are in some way definitely correlated.

2. Potash and magnesia also exist in almost the same quantities, and their ratio to each other in all the sediments being almost constant seems to indicate that they occur combined, perhaps in some zeolitic silicate which may be a source of supply to plants.

3. Manganese exists only in the clay, a mere trace being found in the next sediment.

4. The lime appears to have disappeared in the clay, having probably been largely dissolved in the form of carbonate by the large quantity of water used in elutriation. Its increase in the coarser portions may be owing to its existence in a crystallized form not so readily soluble.

5. In a summary of the ingredients, it is seen that there is a loss in potash, magnesia and lime in the sediments as compared with the original soil; and this loss is doubtless due to the solution of these bodies in the water of elutriation.

A noteworthy fact shown in this table is the rapid decrease of acid-soluble matter in the coarser sediments; even what is dissolved from so fine a sediment as 1.0 millimeter hydraulic value, equal to a diameter of 0.04 millimeter, is in this case a negligible quantity. This suggests forcibly the inutility of introducing into chemical soil analysis, grains of as large a size as will pass a sieve of one millimeter aperture. The hydraulic value of these grains would be somewhere between 150 and 200 millimeters per second. While the exact results of the above analysis may not be applicable to all soils, yet the range is so wide that the systematic exclusion from chemical analysis of inert material, by means of preliminary mechanical separation, seems likely to lead to important improvements in the interpretation of the results.

241. Percentage of Silt Classes in Different Soils.—The adaptation of a soil to different crops depends largely on the sizes of the particles composing it and consequently on the relative percentages of the silt classes.

The following table gives the mechanical analysis of some markedly different types of subsoils:^[160]

Diameter, millimeters.	Conventional names.	Early truck.	Truck and small fruit.	Tobacco.	Wheat.	Grass and wheat.	Limestone.
		1	2	3	4	5	6
2–1	Fine gravel	0.49	0.04	1.53	0.00	0.00	1.34
1–0.5	Coarse sand	4.96	1.97	5.67	0.40	0.23	0.33
0.5–0.25	Medium sand	40.19	28.64	13.25	0.57	1.29	1.08
0.25–0.1	Fine sand	27.59	39.68	8.39	22.64	4.03	1.02
0.1–0.05	Very fine sand	12.10	11.43	14.95	30.55	11.57	6.94
0.05–0.01	Silt	7.74	4.95	28.86	13.98	38.97	29.05
0.01–0.005	Fine silt	2.23	2.02	7.84	4.08	8.84	11.03
0.005–0.0001	Clay	4.40	8.79	14.55	21.98	32.70	43.44
		99.70	97.52	95.04	94.20	97.63	94.23
Org. matter, water and loss		0.30	2.48	4.96	5.80	2.37	5.77

242. Description of the Soils.—Number one represents the very early truck lands of southern Maryland. It is a light yellow sand, belonging to the Columbia terrace formation. Under an intense system of cultivation and heavy manuring with organic matter, good crops of garden vegetables are produced which mature very early, at least ten days or two weeks before the crops from any other part of the state. Under the prevailing meteorological and cultural conditions this soil maintains about five or six per cent of moisture, while a heavier wheat and grass soil maintains from twelve to twenty per cent. The truck soil is so loose and open in texture that the rain-fall passes through it very readily, and it is undoubtedly owing to this drier soil that the plant is forced to the early maturity which secures it from competition from other parts of the State and insures a good market price.

Number two represents the later truck and fruit lands of southern Maryland. These lands contain rather more clay than those just described; they are somewhat heavier and closer in texture, and are rather more retentive of moisture. This land gives a larger yield per acre than the one just described, and in every way crops make a more vigorous growth and development, but the crop is about a week or ten days later in maturing, and for this reason it brings a lower price in the market. It is much better land than number one for small fruit and peaches. These lands are altogether too light in texture for the profitable production of wheat, and it would cost altogether too much to improve them so that even a moderate yield of wheat could be obtained.

Number three is a tobacco land of southern Maryland. The finest tobacco lands of this locality come between the truck and wheat lands in texture, and contain from ten to twenty per cent of clay. The lighter the texture of the soil and the less clay it contains, the less tobacco it will yield per acre, but the finer the texture of the leaf. The tobacco yields more per acre on the heavier wheat soils, but the leaf is coarse and sappy and cures green and does not take on color. It brings a very low price in the market and does not pay for cultivation. The crop on the lighter lands is of much finer quality; there is a smaller yield per acre but the leaf takes on a fine color in curing, and brings a much better price per pound. Wheat is commonly raised on these tobacco lands to get advantage of the high manuring, and because the rotation is better for the land than where tobacco is grown continuously on the same soil. The finest tobacco lands are, however, too light in texture for the profitable production of wheat. These lands belong to the neocene formation.

Number four is a type of the wheat lands of southern Maryland. These lands represent soil of about the lightest texture upon which wheat can be economically produced under the climatic conditions which there prevail. They contain from eighteen to twenty-five per cent of clay, and are much more retentive of moisture than the best tobacco lands. This type is about the limit of profitable wheat production. These soils will maintain about twelve per cent of water during the dry season. Garden truck is so late in maturing on these lands that there is often a glut in the market when the crop matures, and the crops often do not pay the cost of transportation. The lands are too light in texture for a permanent grass sod. They belong to the neocene formation.

Number five represents the heavier wheat lands of southern Maryland, belonging probably to a different horizon of the neocene formation and containing about thirty per cent of clay. This soil is much more retentive of moisture and produces very much larger crops of wheat than the last sample. It is strong enough and sufficiently retentive of moisture to make good grass lands. It is too close in texture and too retentive of moisture for the production of a high grade of tobacco, or to be profitable for market truck.

Number six is from a heavy limestone soil of lower Helderberg formation. It is a strong and fertile wheat and grass land.

243. Interpretation of Silt Analysis.—The primary conceptions upon which the interpretation of the mechanical analysis is based may be briefly stated as follows:^[161] The circulation of water in the soil is due to gravity, or the weight of water, acting with a constant force to pull the water downward, and also to surface tension, or the contracting power of the free surface of water (water-air surface), which tends to move the water either up or down, or in any direction, according to circumstances. There is a large amount of space between the grains in all soils in which water may be held, ranging from about thirty per cent in light sandy lands to sixty-five or seventy per cent in stiff clay soils. The relative rate of movement of water through a given depth of soil will depend upon how much space there is in the soil; upon how much this space is divided up, *i. e.*, upon how many grains there are per unit volume of soil; upon the arrangement of the grains of sand and clay; and upon how this skeleton structure is filled in and modified with organic matter. It also appears that the ordinary manures and fertilizers change this surface tension, or pulling power of water; that they also change the arrangement of the grains, and consequently the texture or structure of the soil may be changed and the relation of the soil to water, through the effect of the ordinary manures and fertilizers in causing flocculation or the reverse.

244. Number of Particles in a Given Weight of Soil.—The approximate number of particles in the soil can be calculated from the results of the mechanical analysis by the following formula:^[162]

$$\frac{\frac{a}{\pi(d)^3 \omega}}{6} \div A$$

Where a is the weight of each group of particles, d the mean diameter of the particles in the several groups in centimeters, ω is the specific gravity of the soil, and A is the total weight of soil. For the specific gravity of ordinary soils, the constant 2.65 may be used.

In using the formula the per cents are expressed as grams. Thus, if there were twenty per cent of silt, this would be taken as twenty grams, and if the results of the analysis added up ninety-seven per cent the whole weight of soil would be taken as ninety-seven grams. The diameter d is taken as the mean for the extreme diameters taken for any group, for instance, for the silt this would be 0.003 centimeter, which is assumed to be the diameter of the particles in that group. This formula can only give approximate values, as the number of separations in a silt analysis must necessarily be small, amounting usually to not more than eight or ten grades, on account of the time and labor required for closer separations. There is relatively rather a wide range in the diameters of grains within any one of these grades, and absolute values could not be expected without a vast number of separations, so that all the grains in each group would be almost exactly of the same size.

The clay group has relatively the widest limits, which is unfortunate, as this is the most important of all the groups on account of the exceedingly small size of the particles. The figure 0.0001 millimeter is taken as the lowest limit of the diameter of the clay particles. These particles have been heretofore assumed to be ultra-microscopic, but by the use of a microscope of high power with oil-immersion objective and staining fluids, it has been possible to define the clay particles in a turbid liquid which has stood so long as to be only faintly opalescent.

Pending more exact measurements, the figure 0.00255 millimeter has been used as the diameter of the average sized particle in the clay group.

The following table gives the approximate number of grains per gram in the different types of subsoils calculated from the mechanical analysis of the typical soils already given:

NUMBER OF PARTICLES OF EACH CLASS IN ONE GRAM OF SOIL.			
Silt classes. Diameter (d) in centimeters.	No. 1 Early truck.	No. 2 Truck and small fruit.	No. 3 Tobacco.
0.15	0	0	3
0.075	85	34	102
0.0375	5,511	4,011	1,900
0.0175	37,230	54,610	11,890
0.0075	207,500	199,700	267,900
0.003	2,073,000	1,355,000	8,092,000
0.00075	38,210,000	35,360,000	140,900,000
0.000255	1,915,000,000	3,918,000,000	6,637,000,000
	1,955,000,000	3,954,973,355	6,786,273,795
Silt classes. Diameter (d) in centimeters.	No. 4 Wheat.	No. 5 Grass and wheat.	No. 6 Limestone.
0.15	0	0	12
0.075	726	4	60
0.0375	8,273	181	157
0.0175	32,340	5,556	1,456
0.0075	554,100	202,600	125,900
0.003	3,962,000	10,670,000	8,231,000
0.00075	73,990,000	154,900,000	199,900,000
0.000255	10,150,000,000	14,570,000,000	19,430,000,000
	10,228,547,439	14,735,778,341	19,638,258,585

245. Estimation of the Surface Area of Soil Particles.—The approximate extent of surface area of the soil grains in one gram of soil can be calculated from the foregoing by the following formula:^[163]

$$\pi(d)^2n$$

in which d is the mean of the diameters of any group in centimeters, and n is the number of particles in the group.

The following table gives the approximate extent of surface area of the particles in one gram of soil calculated from the preceding table:

APPROXIMATE EXTENT IN SQUARE CENTIMETERS, OF SURFACE AREA IN ONE GRAM OF SOIL.

Diameter, millimeters.	Soil number.					
	1	2	3	4	5	6
1.5	0.0	0.0	0.4	0.0	0.0	0.1
0.75	1.8	0.6	1.8	12.8	0.1	0.1
0.375	24.3	17.7	8.4	36.5	31.0	0.7
0.175	35.8	52.6	11.4	31.1	5.3	1.4
0.075	21.3	35.3	47.3	97.9	35.8	22.2
0.03	218.8	38.3	228.9	112.0	301.4	232.7
0.0075	67.4	62.5	248.9	130.8	273.5	353.4
0.00255	390.8	800.5	1355.0	2072.0	2976.0	3965.0
Total	760.2	1007.5	1902.1	2493.1	3593.1	4575.3

246. Logarithmic Constants.—The following logarithmic constants have been used in the calculation of the approximate number of grains per gram and of the surface area, using 2.65 in all cases as the specific gravity of the soil.

Diameter. (<i>d</i>)	Approximate number of grains. $\log. \frac{\pi(d)^2 w}{6}$	Surface area. $\log.(d)^2 \pi$
0.15 centimeters	3.6703	2.8493
0.075 „	4.7674	2.2473
0.0375 „	5.8641	3.6451
0.0175 „	6.8711	4.9831
0.0075 „	7.7674	4.2473
0.003 „	8.5734	5.4513
0.00075 „	10.7674	6.2473
0.000255 „	11.3616	7.3101

247. Mineralogical Examination of the Particles of Soil Obtained by Mechanical Analysis.—The principal object of the mechanical analysis of soils as has already been set forth is the separation of the soil into portions, the particles of which have the same hydraulic value. It is evident without illustration that particles of the same hydraulic value do not necessarily have the same size. The rate of flow of a liquid carrying certain definite particles does not imply that these particles are of the same dimensions. Of two particles of the same size and shape, that one which has the lower specific gravity, will be carried off at the lower rate of flow. At the end of the operation, therefore, the several portions of the soil obtained will be found composed of particles of sizes varying within certain limits, and of these particles the larger ones will tend to be composed of minerals of lower specific gravity, and the smaller ones of minerals of higher specific gravity. Of the same mineral substance, the particles which are most irregular, exposing for a given weight the largest surface will be found to pass over at a lower velocity than those of a more nearly spherical shape. The same law holds good for particles falling through a liquid at rest, *i. e.*, the heavier and more spherical particles, weight for weight, will sooner reach the bottom of the containing vessel. To complete the value of a mechanical analysis, it becomes necessary to submit the several portions of soil obtained not only to a chemical but also to a mineralogical examination. Only the outlines of the methods of examining silt separates for mineral constituents can be given here and special works in petrography must be consulted for greater details.^[164]

It is evident that the methods of separation and examination from a mineralogical point of view about to be described can only be applied to silts of the largest size. The finer silts can not be separated into portions of different specific gravities by separating liquids of varying densities on account of the slowness with which they subside, thus tending to adhere to the sides of the separating vessels and to form floccules which are not all composed of the same kind of mineral particles. While, therefore, these processes are more appropriately described in connection with the silts obtained by hydraulic elutriation, they can be applied with greater success to the fine particles passing the different sieves used in the preparation of the soil for analysis or to the finely pulverized soil as a whole.

The minerals which have contributed to soil formation, moreover, are better preserved in the larger silt particles and therefore more easily identified. While the desirability of securing like determinations in the finer silts is not to be denied, in the present state of the art the analyst must be content with the examination of the larger particles.

248. Methods of Investigation.—The chief points to be observed in the examination of the fine particles of soil are the following: (1) the size and shape of the particles; (2) measurement of crystal angles; (3) separation into classes of approximately the same specific gravity; (4) separation by means of the magnet; (5) determination of color and transparency; (6) determination of refractive index; (7) examination with polarized light; (8) examination after coloring; (9) chemical separation. For many of the optical studies above noted, it is first necessary to prepare thin laminae of the mineral particles and properly mount them for examination. For the purposes of this manual only those processes will be described which are essentially connected with a proper understanding of the nature of the soil particles. For the more elaborate methods of research the analyst will consult the standard works on mineralogy and petrography.

249. Microscopical Examination.—The direct examination of the silt particles with the microscope should attend the progress of separation. Unless the particles obtained have the same general appearance, the separation is not properly carried on. Especially is the microscope useful to determine that the value of the silt separation is not impaired by flocculation. Unless flocculation be practically prevented during the separation of the finest particles, many of these will be left as aggregates to be brought over subsequently with particles of far different properties. No special directions are necessary in the use of the microscope. The silt particles are removed with a few drops of water by means of a pipette, a drop of the liquid with the suspended particles is placed on the glass, covered and examined with a convenient magnification. A micrometer scale should be employed in order that the approximate sizes of the particles may be determined. A *camera lucida* may also be conveniently used for the purpose of delineating the form of particles of peculiar interest.

250. Petrographic Microscope.—Any good microscope furnished with polarizing apparatus may be used for the examination of the silt particles and sections. For directions in manipulating microscopes the reader is referred to works on that subject. A special form of microscope for petrographic work is made by Bausch and Lomb of Rochester. The stand of this instrument is shown in Fig. 37. The base, upright pillars and arm are made of japanned iron. The stage is made in two forms, first, plain revolving, having silvered graduates at right angles and second, a mechanical stage with silvered graduations on the edge with vernier and graduations for the rectangular movements. The mirror bar is adjustable and graduated and the mirror is of large size, plane and concave. The double chambered box in the main tube carries the upper Nicol prism (analyzer). The lower Nicol prism (polarizer) is mounted in a cylindrical box beneath the stage to which it is held by a swinging arm. It is adjustable also up or down and is provided with a compound lens for securing converged polarized light. In revolving the prism a distinct click shows the position of the crossed Nicols.

251. Form and Dimensions of the Particles.—In order to study the contour of the fine silt particles, it is well to suspend them in a liquid whose refractive index is markedly lower than that of the particles themselves, and for this purpose pure water is commonly used. Care must be taken that not too many particles are found in the drop of water which is to be placed on the object holder and protected with a thin, even glass. The tendency to flocculation in these fine particles will make the study of their form difficult if they are allowed to come too close together. The size of the particles, or linear diameter, is to be determined by means of an eye-micrometer. This consists of a glass plate on which a millimeter scale is engraved with a diamond, or photographed. The millimeter scale is the one usually employed, each millimeter being divided into tenths. On microscopes designed especially for photographic work the micrometer is fastened to the eyepiece, and so adjusted as to read from left to right, or at right angles thereto. Sometimes an eyepiece-micrometer has two scales at right angles so that dimensions may be read in two directions without change. With an eyepiece-micrometer, not the dimensions of the object, but those of its magnified image are read, and the degree of magnification being known, the actual size of the object is easily calculated. The actual measurements may also be obtained by placing in the field of vision, a stage-micrometer and determining directly the relation between that and the eyepiece-scale. If, for example, the stage-micrometer is ruled to 0.01 millimeter, and the eye-micrometer to 0.1 millimeter, and one division of the stage-rule should cover three divisions of the eye-rule, then the one division of the eye-micrometer would correspond to an actual linear distance of 0.0033 millimeter in the object. If the two lines of division in the two micrometers do not fall absolutely together, the calculation may be made as follows: suppose that six divisions, 0.6 millimeter, in the eyepiece correspond to nearly five divisions, 0.25 millimeter, in the stage piece. To get at the exact comparison, take ninety-six divisions of the eye-scale and they will be found to be somewhat longer than eighty-one and somewhat shorter than eighty-two divisions of the stage-scale. It follows therefore that



FIGURE 37.

one	division	of	the	eye-scale	>0.008438	millimeter,	and
"	"	"	"	"	<0.008541	"	;

and, hence, one division of the eye-scale corresponds almost exactly to 0.008489 linear measure.

252. Illustrations of Silt Classes.—In figure 38 are shown the relative sizes and usual forms of a series of silt separates made by the Osborne beaker method. The photomicrographs were made by Dr. G. L. Spencer from specimens furnished by Prof. M. Whitney.

The soil represented by the separates is from a truck farm near Norfolk, Virginia.

The particles represented in each class are not all strictly within the limits of size described. For instance, in the largest size (No. 1) are two particles at least which show a diameter of more than one millimeter. The particles in general, however, are within the limits of the class; viz., one-half to one millimeter, and this general observation is true of all the classes. In the case of the finer particles, especially of clay, the tendency to flocculation could not be overcome in the preparation of the slides for the photographic apparatus. The clay particles are so fine as to present but little more than a haze at 150 diameters of magnification. The particles seen are clearly, in most cases, aggregates of the finer clay particles. The larger particles show the rounded appearance due to attrition and weathering. It would have been more instructive to have had the particles of the different classes all photographed on the same scale, but this is manifestly impossible. The lowest power which shows any of the clay particles to

advantage is at least 150 diameters, and with the larger particles such a magnification would have been impracticable.

253. Measurement of Crystal Angles.—The fine silt particles rarely retain sufficient crystalline shape to permit of the measurement of angles and the determination of crystalline form thereby. The rolling and attrition to which the silt particles have been subjected have, in most cases, given to the fragments rounded or irregular forms which render, even in the largest silts, the measurement of angles impossible. For the methods of mounting minute crystals and the measurement of microscopic angles, the analyst is referred to standard works on mineralogy and petrography.

254. Determination of the Refractive Index.—For a study of the theory of refraction, works on optics should be consulted. The general principles of this phenomenon which concern the determination of the refractive power of fine earth particles are as follows: if a transparent solid particle is observed in the microscope imbedded in a medium of approximately the same refractive power and color, its outlines will not be clearly defined, but the imbedded particle will show in all of its extent the highest possible translucency. If, therefore, the form or perimeter of the particle is to be studied with as much definiteness as possible, it should be held in a medium differing as widely from it as possible in refractive power. For minerals, water is usually the best immersion material. On the other hand, when the internal structure of the particles is the object of the examination, it should be imbedded in oil, resin (Canada balsam), etc., or in some of the liquids mentioned below.

If particles of different refractive powers and the same character of surface be studied in the same medium, they will not all appear equally smooth on the field of the microscope. Some of the surfaces will seem smooth and even, others will appear rough and wrinkled. Those particles whose refractive index is equal to or less than that of the liquid appear smooth, because all the emergent light therefrom can pass at once into the environing medium. On the other hand, the surfaces of those particles which have a higher refractive power than the medium will appear roughened, because, on account of the unavoidable irregularities on the surface, many of the emergent rays of light must strike at the critical angle and so suffer total reflection, and consequently those portions of the surface will be less illuminated, producing the phenomenon of apparent roughness above noted. In the case of any given particle, liquids of increasing refractive power can be successively applied until the change in the appearance of the surface of the particle is noticed. The refractive index of the liquid being known, that of the particle is in this way approximately to be determined.

The following liquids, having the indexes mentioned, are commonly employed:

Substance.	Refractive index.
Water	1.333
Alcohol	1.365
Glycerol	1.460
Olive oil	1.470
Canada balsam	1.540
Oil of cinnamon	1.580
Oil of bitter almonds	1.600
Oil of Cassia	1.606
Concentrated solution of potassium and mercuric iodid	1.733
Concentrated solution of barium and mercuric iodid	1.775

The solution of potassium and mercuric iodid may also be used for all refractive indexes from 1.733 to 1.334 by proper dilution with water.

The mineral particle may also be imbedded in Canada balsam and over it a drop of a liquid of known refractive power placed. By a few trials one of the liquids will be found having practically the refractive index of the particle under examination.

255. Examination with Polarized Light.—The internal structure of a mineral particle can often be determined by its deportment with polarized light. The theory of polarization is fully set forth in works on optics and will not be discussed here. The principle on which the utility of polarized light in the examination of soil particles rests is found in the information it may give in respect of crystalline structure. The structure of mineral particles which make up the bulk of an ordinary soil is, as a rule, so thoroughly disintegrated that all trace of its original form is lost. Some particles may exist, however, in which there is no determinable element of shape and which yet possess an internal crystalline structure which the microscope with polarized light may be able to reveal.

256. Staining Silt Particles.—The finer silts and clays before microscopic examination should be colored or stained. The methods used in staining bacteria may be employed for the clay particles.

Evaporation to dryness with a solution of magenta will often impart a color to the clay particles which is not removed by subsequent suspension in water. The harder and larger silt particles are not easily stained, especially if they be firm and undecomposed. On the other hand, if the particles be broken and seamed, and well decomposed, the stain will be taken up and held firmly in the capillary fissures. Valuable indications are thus obtained respecting the nature of the silt particles. Particles of mica, chlorite and talc are easily distinguished in this way from the firmer and less decomposed quartz grains.

The staining of the particles after ignition and treatment with acids gives better results than the direct treatment. Particles of carbonate which are stained with difficulty before ignition take the stain easily afterwards on account of the decomposition produced by the loss of carbon dioxide. This is the case also with particles containing water of composition or crystallization.

257. Cleavage of Soil Particles.—A microscopic examination of the cleavage of soil particles may be useful in determining their mineral origin. The course followed by cleavage lines and their mutual position is dependent on the direction in which the separation of the mineral fragment takes place. The character of the microscopic fragments produced by crushing a soil particle is determined primarily by the system of crystallization to which it belongs. Perhaps the most distinguishing cleavage marks in soil particles will be found in fragments of mica and orthoclase. These characteristic forms are shown in Figs. 39 and 40. The first (Fig. 39) shows the pinacoidal cleavage in a fragment of mica. Fig. 40 illustrates the appearance of the cleavage lines in a fragment of orthoclase. Figs. 41 and 42 show the characteristic cleavage lines in fragments of epidote and titanite.

258. Microchemical Examination of Silt.—The methods of quantitative chemical examination of silts will be given in another part of this manual. Certain qualitative and microchemical tests, however, are useful in identifying silt particles. For instance, any soluble iron mineral will be detected, even in minute quantity, by the blue coloration of the solution produced by the addition of potassium ferrocyanide. Manganese will be revealed by fusion with soda and saltpeter on platinum foil, in the oxidizing flame, producing the well-known green coloration due to the sodium manganate formed.

More valuable indications of the character of the fragments examined are obtained by microchemical processes. The best method of decomposing the silt particles for this purpose is by treatment with hydrofluosilicic acid. When the particles are composed of silicates, pure hydrofluoric acid is to be preferred.

The method of treatment is essentially that of Boricky.^[165] The slide used is protected by a film of Canada balsam, and a few of the silt particles are placed thereon, and fixed in place by slightly warming the balsam. Each particle is then treated with a drop of hydrofluosilicic acid, care being taken not to let the drops flow together. The acid must be pure, leaving no residue on evaporation. The acid should be prepared by the analyst from a mixture of barium fluoride, sulfuric acid and quartz powder, or the commercial article should be purified by distillation before using. The acid should be kept in ceresin or gutta-percha bottles and must be applied with a ceresin or gutta-percha rod. Each particle should be as completely dissolved as possible by the acid, and the rate of solution may be hastened by gentle warming, provided the heat is not great enough to remove the balsam and allow the acid to attack the glass. The bases present in the silt particles crystallize on drying as fluosilicates. In case of a too rapid crystallization, the mass may be dissolved in a drop of water or of very dilute hydrofluosilicic acid, and allowed to evaporate more slowly. Some fragments need more than one treatment with acid to secure complete solution, and particles of mica may even resist repeated applications. In such a case the decomposition may be made in a platinum crucible with hydrofluoric acid, adding afterwards an excess of hydrofluosilicic acid and evaporating to dryness. The crystals may then be dissolved in a little water and a drop of the solution allowed to crystallize on the slide.

259. Special Reactions.—The number of microchemical reactions is very great, but there will be given here only some of the more important for silt identification.

Sodium.—Sodium mineral fragments dissolved in hydrofluosilicic acid and dried give the combinations shown in Fig. 43. With sodium and aluminum the forms shown in Figs. 44 and 45 are obtained. With an increasing amount of lime in the mineral, the crystals tend to become longer. For microscopic work it is not advisable to try to produce the tetrahedral crystals of the double uranium sodium acetate because the commercial uranium acetate often contains sodium and even the pure article will often take up sodium from the bottles.

Potassium.—Fragments containing potash give isotropic clear cubes, or octahedra of low refracting power, or combinations of these forms with each other and with rhombic dodecahedra. These crystals have the composition K_2SiF_6 . Their forms are shown^[166] in Figs. 46 and 47. In case much sodium be present, the first crystals obtained

may be strongly double refractive rhombohedra, but on dissolving in water and allowing to recrystallize, the normal forms will be obtained. If the crystals be dissolved in hydrochloric or sulfuric acids, and treated with platinum chlorid, the characteristic yellow octahedral crystals of K_2PtCl_6 will be obtained. Ammonium and cesium compounds also give this reaction.

Lithium.—When fragments containing lithium are treated with the solvent mentioned, monoclinic crystals are produced on drying. These crystals dissolved in sulfuric acid and freed from calcium sulfate by treatment with potassium carbonate give aggregates of lithium carbonate resembling a snowflake. At a high temperature lithium solutions treated with sodium phosphate give spindle-shaped crystals of lithium phosphate. The double lithium aluminum silicofluorid is shown in Fig. 48. The ease with which traces of lithium may be detected by the spectroscope renders unnecessary any further description of its microchemical reactions.

Calcium.—Nearly all mineral particles, save quartz grains, contain calcium. When these particles are dissolved by treatment with hydrofluosilicic acid, they form on drying hydrated monoclinic crystals of calcium silicofluorid ($CaSiF_6 + 2H_2O$). These crystals assume many forms, some of which are shown in Figs. 49 and 50. These crystals are easily decomposed by sulfuric acid, the well-known long prismatic crystals of gypsum taking their place. On treatment of silt particles containing lime with hydrofluoric and sulfuric acids, only a part of the lime passes into solution if the content thereof be large. Where but little lime is present and the sulfuric acid is in large excess, all the lime passes into solution and the characteristic gypsum crystals appear as in Fig. 51.

Magnesium.—Rhombohedral crystals of magnesium silicofluorid separate from the solution of particles containing magnesium in hydrofluosilicic acid. They have the composition $MgSiF_6 \cdot 6H_2O$ and their common forms are shown in Fig. 52. Quite characteristic also are the crystals of struvite ($NH_4MgPO_4 + 6H_2O$), which are produced in a very dilute solution of the magnesium compound first obtained by carefully adding ammonium hydroxid and chlorid until a faint alkaline reaction is produced, and then placing a drop of dilute sodium phosphate at the edge of the solution. The crystals should be allowed to form slowly in the cold. Their form is shown in Fig. 54.

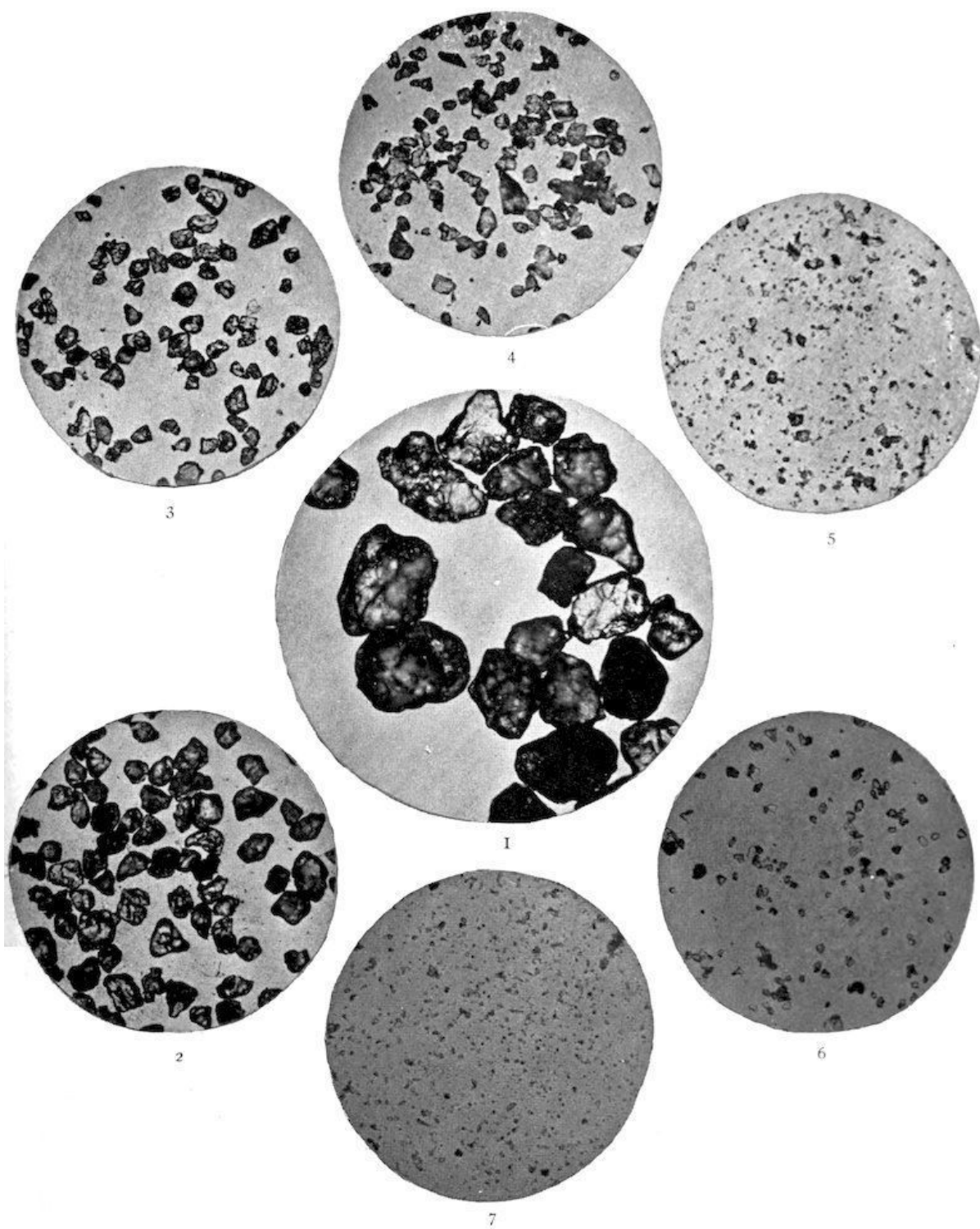


FIGURE 38. PHOTOMICROGRAPHS OF SILT PARTICLES.

No.	Diameter in mm.	Name.	Magnification. Diameters.
1	1.0–0.5	coarse sand	×10
2	0.5–0.25	medium sand	×10
3	0.25–0.1	fine sand	×10
4	0.1–0.05	very fine sand	×30
5	0.05–0.01	silt	×30
6	0.01–0.005	fine silt	×150
7	0.005–0.0001	clay	×150

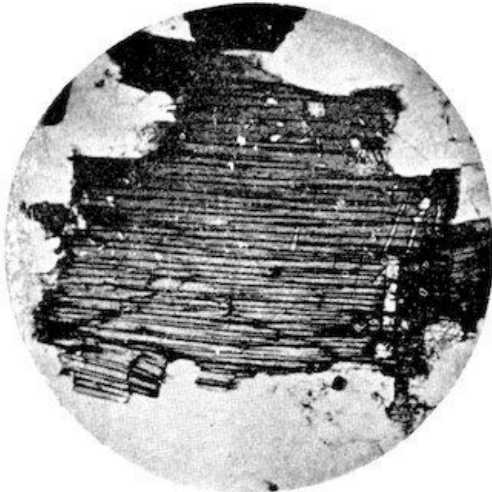


FIGURE 39.



FIGURE 40.



FIGURE 41.

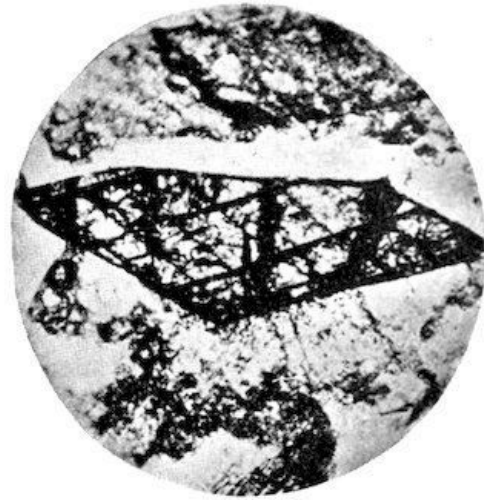


FIGURE 42.

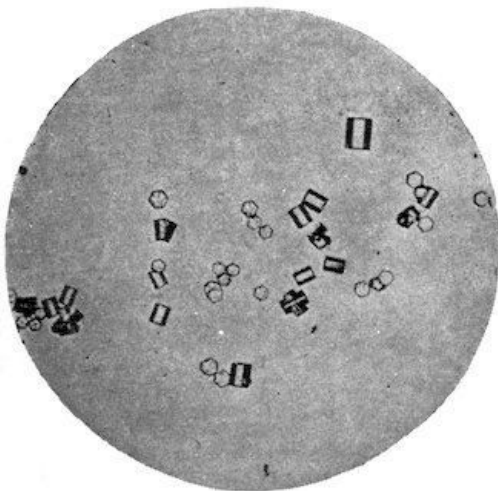


FIGURE 43.

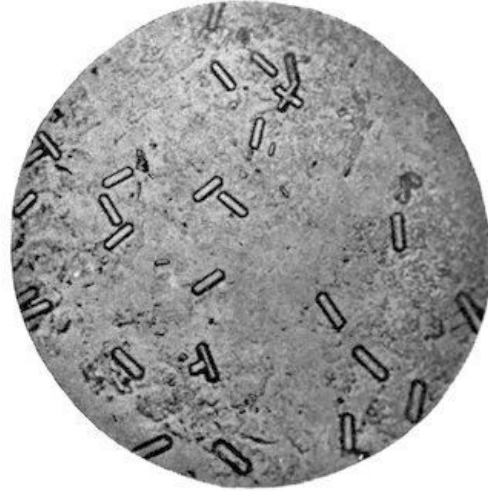


FIGURE 44.

Figures 39-42, show examples of the various degrees of perfection and relative positions of cleavage lines.
 Figure 39, illustrates pinacoidal cleavage in mica from granite. Magnified thirty diameters.
 Figure 40. A cleavage of orthoclase from augite syenite magnified twenty-seven diameters.

Figure 41. Cleavage of epidote magnified sixty diameters.
Figure 42. Cleavage of titanite magnified seventy-five diameters.
Figure 43. Sodium fluosilicate crystals magnified seventy-two diameters.
Figure 44. The same with aluminum fluosilicate magnified twenty-seven diameters.

Taken from Rosenbusch, Mikroskopische Physiographie.

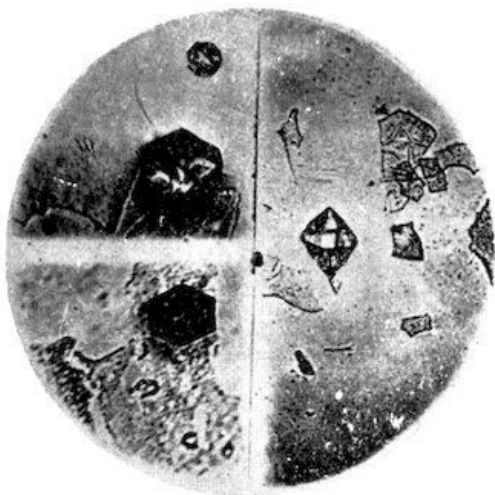


FIGURE 45.

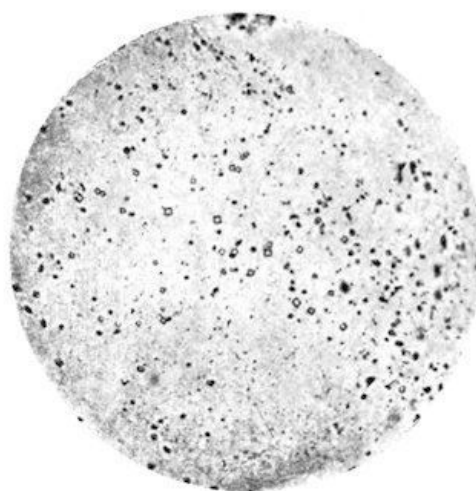


FIGURE 46.

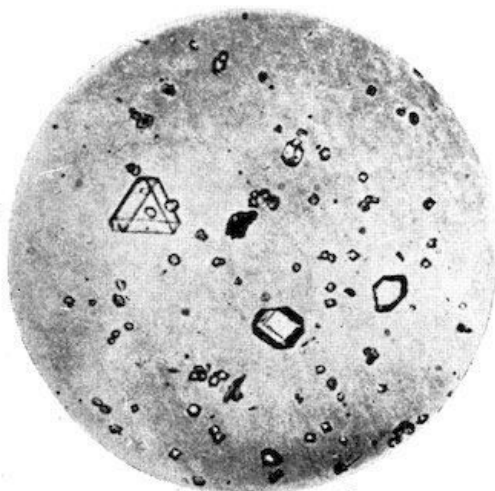


FIGURE 47.

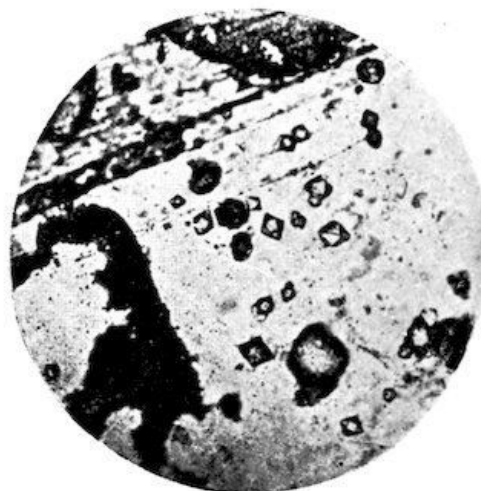


FIGURE 48.



FIGURE 49.



FIGURE 50.

Figure 45. Sodium and aluminum silicofluoride crystals magnified 100, 140 and 160 diameters.
 Figure 46. Potassium silicofluoride crystals magnified 130 diameters.
 Figure 47. Another preparation of the same magnified 140 diameters.

Figure 48. Lithium and aluminum silicofluorid crystals magnified 100 diameters.

Figure 49. Calcium silicofluorid crystals magnified 45 diameters.

Figure 50. Another preparation of the same magnified 42 diameters.



FIGURE 52.

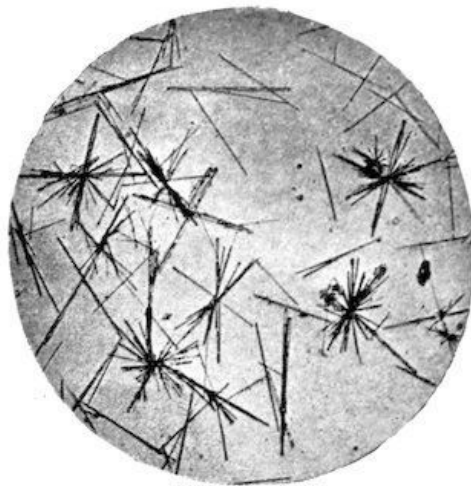


FIGURE 51.

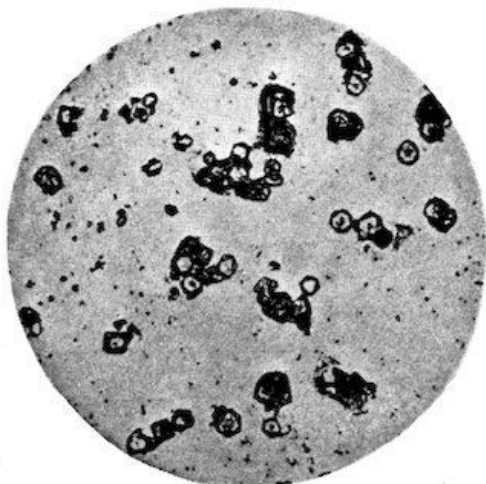


FIGURE 53.

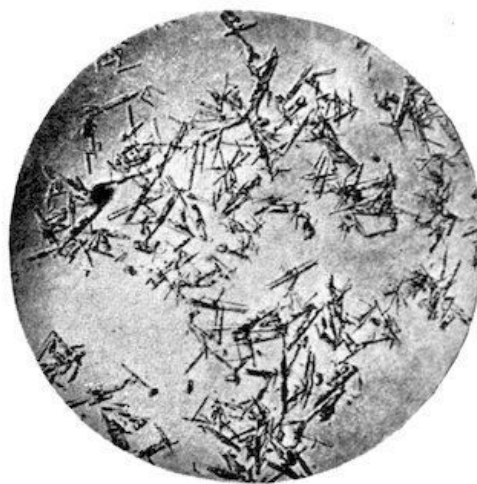


FIGURE 54.

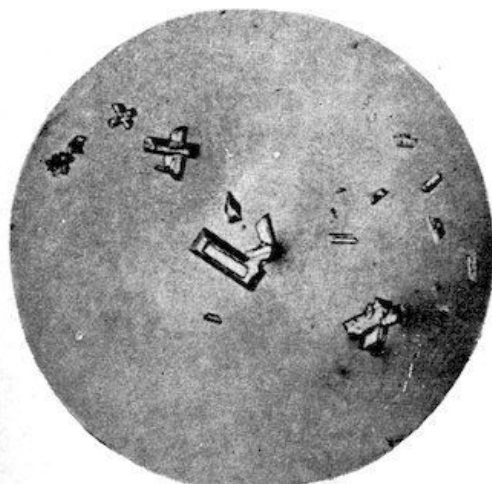


FIGURE 55.

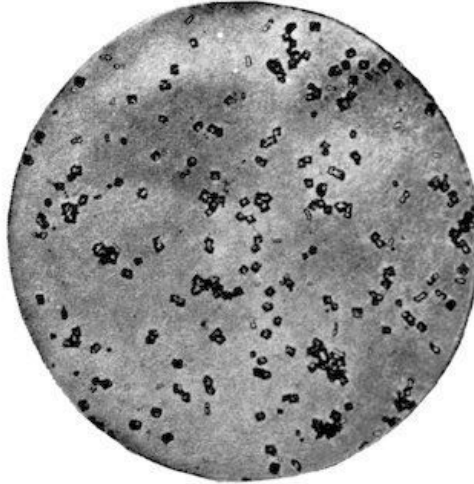


FIGURE 56.

Figure 51. Calcium sulfate crystals magnified twenty diameters.
 Figure 52. Magnesium silicofluoride crystals magnified thirty diameters.
 Figure 53. Cesium aluminum sulfate crystals magnified twenty diameters.
 Figure 54. Ammonium magnesium phosphate crystals magnified ten diameters.

Figure 55. The same crystallized from dilute solution magnified thirty diameters.

Figure 56. Ammonium phosphomolybdate crystals magnified 140 diameters.

Barium.—From solution of barium bearing minerals in hydrofluosilicic acid fragments, no characteristic crystals, are obtained. Treated with hydrofluoric and sulfuric acids the barium is left as sulfate. If this salt be dissolved in boiling oil of vitriol and a drop of the solution placed on the slide, a mixture of rectangular tablets and St. Andrew's cross-shaped growths will be separated before any crystals of gypsum which may be present appear. When strontium is present, the barium sulfate residue obtained by treatment with hydrofluoric and sulfuric acids should be fused with sodium and potassium carbonate, washed with water until the sulfuric acid is removed, the residue dissolved in hydrochloric or nitric acids, and the solution treated with potassium chromate. Pale yellow crystals of barium chromate are thus obtained, which resemble in form those secured by dissolving the barium sulfate in oil of vitriol. Strontium is not precipitated by this treatment. If potassium ferrocyanid be used instead of barium chromate with the hydrochloric acid solution, crystals of barium potassium ferrocyanid are formed of a bright yellow color and rhombohedric shape.

Strontium.—From a hydrofluosilicic acid solution, strontium crystallizes in columns or tablets of the monoclinic system as strontium silicofluorid, SrSiF_6 . On treating these with sulfuric acid, rhombic plates of strontium sulfate are formed, which serve to distinguish this element from calcium. On treatment of the particles of the original mineral with hydrofluoric and sulfuric acids, the strontium remains in the insoluble residue. When this residue is treated with boiling oil of vitriol, rhombic plates of celestine are separated. If the residues above mentioned be dissolved by fusion with the alkaline carbonates, washed with water, dissolved in hydrochloric acid and treated with oxalic acid, octahedral crystals of strontium oxalate are formed.

Iron.—Mineral particles containing iron give crystals, when treated as is first described above, which are fully isomorphous with those obtained from magnesium. By moistening the crystalline mass with potassium ferrocyanid, the presence of iron is at once revealed by the blue coloration produced.

Aluminum.—No crystals containing aluminum are formed from the mineral particles containing this substance when dissolved in the solvent already mentioned. If, however, the gelatinous mass be dissolved in a little sulfuric acid and a fragment of a cesium salt added, beautiful crystals of cesium alum are obtained, illustrated in Fig. 53.

Phosphorus.—When a mineral fragment containing phosphorus is treated according to the usual analytical methods for securing the ammonium magnesium phosphate, crystals are obtained of the form shown in Figs. 54 and 55. A phosphatic fragment of silt may be identified when soluble by treatment with nitric acid and ammonium molybdate. On slowly drying, rhombohedral crystals are produced, yellow by reflected, and green by transmitted light. Their form is shown in Fig. 56.

260. Petrographic Examination of Silt Particles.—The larger silt particles and the minute fragments of minerals in the soil can best be studied in thin sections. For this purpose the following plan, proposed by Thoulet, may be used. Mix the soil minerals in considerable proportion—Thoulet recommends ten per cent, but a greater percentage is often better—with zinc oxid and make into a paste with sodium silicate. The paste should be worked to the consistence of putty and then rolled into little tablets about one-eighth of an inch thick and an inch in diameter. After drying a day or two without heating, the tablets become hard enough to mount and grind like rock sections. These tablets are mounted in Canada balsam on glass slides and ground as thin as possible with fine emery on the turn-table or glass plate, as rock sections are treated. As these tablets are not as strong as rock sections usually are, they require care in this treatment. Some of the grains also are apt to be torn out in the process of grinding and to compensate for this loss a number of slides should be prepared with each lot of soil minerals. When this operation has been successful, the optical properties of the various minerals can be studied as in rock sections.

As the iron oxid contained in the soils obscures the transparency of the minerals, it is well to treat a portion of the material under examination with hot hydrochloric acid for a short time to remove this oxid and then prepare slides with the cleansed material and compare results with the untreated. As the acid will dissolve phosphates and carbonates, and will partly or wholly decompose some other minerals, the operator must be guided by his judgment in its use.

261. Machine for Making Mineral Sections.—A convenient apparatus for this purpose has been described by Williams^[167] and is represented in Fig. 57. It is supported on a substantial table provided underneath with electric batteries and a motor for driving the cutting disks seen on the top. The table is three feet six inches square and two feet nine inches high.

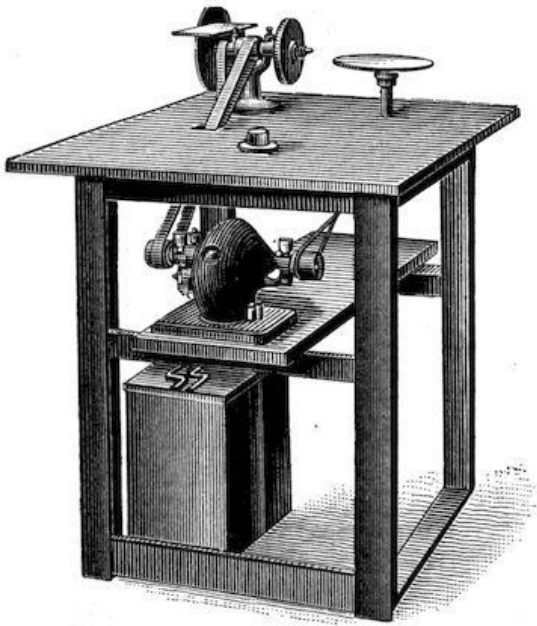


FIGURE 57.

MACHINE FOR MAKING MINERAL SECTIONS.

The grinding apparatus consists of two circular disks of solid copper, nine inches in diameter, and three-eighths inch thick, which may be used alternately as different grades of emery are required. They are attached either by a screw or square socket to a vertical iron spindle which revolves smoothly in a conical bearing. The grinding disk is surrounded when in use by a large cylindrical pan of tin, which is not shown in the cut, which has an opening in its center to allow of the passage of the spindle.

The sawing apparatus consists of a horizontal countershaft placed on a different part of the table and connected with the motor by a separate belt. It carries at one end a vertical wheel of solid emery, and at the other an attachment, level-table and guide for the diamond-saw. A small water-can with spout, not shown in the cut, is suspended over the edge of the table to keep the saw wet when it is in use.

The machine is very conveniently driven by a storage battery when street circuits cannot be drawn on.

For the details of making mineral sections, the works on petrography may be consulted.

262. Separation of Silt Particles by Specific Gravity Solutions.—In silt separates the specific gravity of the different mineral particles present may vary from graphite (1.9–2.3) to hematite (5.2–5.3).

The following list gives the specific gravities of some of the more common minerals which may be met with in soils:

Gypsum	2.31
Albite	2.56–2.63
Quartz	2.65
Talc	2.74
Chlorite	2.78
Muscovite	2.85
Calcite	2.5–2.78
Dolomite	2.90
Tourmaline	2.94–3.3
Biotite	3.01
Apatite	3.16
Pyroxenes	3.22–3.5
Epidote	3.39
Titanium Minerals	3.48–4.75
Iron oxids	5.2–5.3

The finest particles of silt are separated by gravity with great difficulty, inasmuch as they tend to remain suspended in the solutions for an indefinite period. With the coarser silts, however, useful data are often obtained by this method. The separation is preceded by extraction of the particles with hydrochloric acid to remove encrusted soluble matter, and by ignition to destroy any traces of organic matter. Those mineral matters which are soluble in acid or are changed by ignition must, of course, be sought for in separate portions of the silt,

263. Thoulet's Solution.^[168]—The standard solution is of such a density that particles of 2.65 specific gravity will just float thereon, using for this purpose a solution of about 2.7 specific gravity. The solution from which the above standard is prepared is made as follows:

One part of potassium iodid is weighed and placed in a beaker and one and one-quarter part of mercuric iodid is placed on top of it. Then water is added in the proportion of ten cubic centimeters to 100 grams of the mixture, and after some time (twelve to twenty-four hours), with occasional stirring, the salts will nearly completely dissolve. Filter from the undissolved residue and evaporate in a porcelain dish until crystals form on the surface of the

liquid. Allow to cool, pour off the liquid from the crystals and evaporate the liquid for another crop. The first solution, after cooling, has a specific gravity between 3.10 and 3.20, the second a specific gravity of 3.28, practically the limit of density of the solution. The solution of 2.7 specific gravity and other densities are made by cautiously adding a few drops of water at a time and ascertaining the specific gravity by the Westphal balance or other convenient method.

The strong solution, according to Goldschmidt,^[169] may be prepared directly by using potassium iodid and mercuric iodid in the ratio of 1 : 1.24. Twenty-five cubic centimeters of water, 210 grams of potassium iodid, and 280 grams of mercuric iodid afford a solution of 3.196 specific gravity at 15°, on which fluorspar fragments will float.

264. Klein's Separating Liquid.—A solution of cadmium borotungstate, of the composition $2\text{H}_2\text{O}, 2\text{CdO}, \text{B}_2\text{O}_3, 9\text{WO}_3 + 16\text{H}_2\text{O}$, has been proposed by Klein^[170] for separating silt particles. This salt is obtained by dissolving pure sodium tungstate in five times its weight of water, adding one and a half parts of boric acid and boiling until, complete solution takes place. On cooling; the borax is separated in crystalline form. The mother-liquor after the removal of the crystals is carefully concentrated by boiling. By stirring the cold solution, there is a further separation of sodium borate and polyborate. This operation is continued until glass will float on the mother-liquor. The salt in solution then has the following composition: $4\text{Na}_2\text{O}, 12\text{WO}_3, \text{B}_2\text{O}_3$. To this boiling concentrated solution, is added a boiling saturated solution of barium chlorid, in the proportion of one part of the chlorid to three parts of the original double tungstate. An abundant pulverulent precipitate is formed, making the whole mass mushy. The mass is filtered under pressure and well-washed with hot water. The residue is then suspended in hot water containing one part in ten of hydrochloric acid of 1.18 specific gravity. It is then evaporated to dryness in the presence of an excess of hydrochloric acid and decomposed, by which process hydrated tungstic acid is separated. The boiling mass is taken up with water and the boiling continued for two hours with occasional addition of water to take the place of that evaporated, and the tungstic acid separated by filtration.

From the solution, beautiful quadratic crystals separate having the composition $9\text{WO}_3, \text{B}_2\text{O}_3, 2\text{BaO}_2\text{H}_2 + 18\text{H}_2\text{O}$. These are purified by several recrystallizations and freed from any scales of boric acid by washing with alcohol. Any reducing action, revealed by a violet coloration of the crystals, can be avoided by adding a few drops of nitric acid. From a boiling solution of these crystals, the cadmium salt desired is obtained by treatment with the proper amount of cadmium sulfate solution to precipitate the barium. The barium sulfate is separated by filtration. The cadmium borotungstate is soluble in less than ten parts by weight of water. From this solution it is obtained in pure form by evaporation under a vacuum, or by carefully concentrating on a water-bath and cooling. A saturated solution of these crystals at 15° has a bright yellow color and a specific gravity of 3.28.

If a dilute solution of the above salt be carefully evaporated on a water-bath, any violet color which may be present disappears when the specific gravity reaches 2.7. If the evaporation be continued until a crystal of augite will float on the hot liquid, crystals may be obtained on cooling which, dissolved in as little water as possible, make a solution which will almost support olivine. If the two solutions be united, the specific gravity of the mixture is 3.30–3.36. The highest attainable specific gravity; viz., 3.6, is produced by continuing the evaporation on a water-bath until the liquid will support olivine, and then allowing to stand in a closed place for twenty-four hours. The crystals of cadmium borotungstate thus obtained are freed as much as possible from the mother-liquor by drainage and then melted at about 75° in their own water of crystallization. A liquid is thus obtained on which spinel will float. The same concentration may also be obtained by careful heating on a water-bath. At its highest specific gravity this solution has an oily consistence and this renders its practical use in the separation of fine particles somewhat restricted. By filtering the liquor when a crystalline crust begins to form during evaporation, a cold solution of 3.360–3.365 specific gravity is obtained which is found practically useful. It has a higher specific gravity than Thoulet's mixture, is not injurious to any of the mineral particles, not even of iron with which it is brought into contact, but the trouble of preparing it is far greater than that of the mixture of mercuric and potassium iodids.

265. Rohrbach's Solution.—The solution of barium mercuric iodid recommended by Rohrbach^[171] for this purpose was originally prepared by Suchsin. The solution must be rapidly prepared on account of the tendency of the barium salt to decomposition. The solution is prepared by weighing rapidly 100 grams of barium and 130 grams of mercuric iodid, mixing the two salts well in a dry flask and adding twenty cubic centimeters of water. The mixture is raised to a temperature of 150°–200° on an oil-bath. The formation and solution of the double salt are promoted by constant stirring.

After solution, the liquor is boiled for a few minutes and then evaporated on a water-bath until it will bear a crystal of epidote. On cooling, a small quantity of a yellow double salt is separated by crystallization and the resulting mother-liquor is dense enough to carry a fragment of topaz. Inasmuch as the liquor is filtered with

difficulty, the clear mother-liquor should be separated by decantation after standing for several days. This solution has the disadvantage of not being dilutable with water, the addition of which causes a separation of red mercuric iodid. Were this solution not so easily decomposed, it would prove of high value in silt separation.

266. Braun's Separating Liquid.—In many respects the separatory solution proposed by Braun^[172] is superior to those already mentioned. It is the commercial methylene iodid, CH_2I_2 , which has at 16° a specific gravity of 3.32, at 5° of 3.35, and at 25° of 3.31. It is a strongly refractive liquid having a refractive index of 1.7466 for the yellow ray.

As a separating medium the liquid is open to two objections; viz., first, it cannot be diluted with water and, second, it turns brown on heating or on long exposure to the sunlight.

When dilution is necessary, it should be accomplished with benzene or xylene. To bring the diluted liquor again to its maximum density, the benzene must be removed by evaporation, which causes a considerable loss in the liquid. When this substance becomes opaque, the transparency may be restored by removing the separated iodine by shaking with potash lye, washing with pure water, drying by the addition of pieces of calcium chlorid and filtering. The same result may also be reached by freezing and separating the liquid portion. The frozen portion on melting will have the density of the original liquid.

267. Method of Bréon.—Instead of a solution of a salt, Bréon^[173] has proposed to use salts in a fused state for separating mineral particles. Lead and zinc chlorids may be used for this purpose in a melted state, having the specific gravities of 5.0 and 2.4, respectively. By mixing the molten salts in different proportions, any desired specific gravity between the extremes mentioned may be secured. The fusion is accomplished at 400° in a test-tube. The silt is added gradually with constant stirring until a sharp separation is secured between the sinking and floating particles. After cooling, the tube is broken, the two parts separated, and the silt recovered by dissolving the mixed salts in hot water containing a little nitric acid. Only the coarser silts can be separated by this method. Fused silver nitrate, melting point 198° , specific gravity 4.1, has also been used for separation.

268. The Separation.—Forty cubic centimeters of the solution in the Thoulet process are placed in the separatory tube A, Fig. 58, together with from one to two grams of the silt and the stopper F inserted. The tube G is connected with a vacuum apparatus by means of which any air particles adhering to the mineral fragments are removed. The silt which sinks in the solution is removed after G has been disconnected by opening the cock C and sucking through B at I. The cock C is closed and the separated particles washed into a beaker at H after opening D. Water is next added to the materials left in A in quantities previously determined to secure a given specific gravity and thus a second, a third, etc., separation secured. An intimate mixture of the solutions in A can be effected by closing D, opening C, and blowing through B in such a way that no liquid is allowed to pass through C.

The quantity of water to be added in each case to secure a given specific gravity is determined by the formula $v_1 = \frac{v(D-d)}{d-1}$, in which v is the volume of the solution, D its specific gravity, and d and v_1 the specific gravity desired and volume of the water to be added.

Example.—Let the specific gravity of the original solution be 3.2, its volume thirty cubic centimeters, and the desired specific gravity of the new solution 2.85.

Then $v_1 = \frac{30(3.2-2.85)}{2.85-1} = 5.68$.

The desired specific gravity is therefore secured by adding 5.68 cubic centimeters of water, which is easily accomplished by means of the graduations on the tube.

According to Rosenbusch,^[174] the calculated specific gravity as made above is not wholly reliable on account of the contraction which takes place. An empirical process is rather to be commended which consists in introducing a fragment of mineral of known or desired specific gravity and then adding water drop by drop until the fragment remains suspended in the mixture. Should too much water be added, the necessary increase in density can be secured by adding a little of the strong solution.

269. Method of Packard.—A separatory funnel, according to Packard,^[175] may be safely used to hold the solution while separation is going on. As the lighter minerals form the bulk of soils, the heavier constituting only a small percentage, it is well to use a wide funnel holding as much as one-half liter for quantitative separations, because a large quantity of soil, say 100 grams, is necessary from which to recover the small quantity of heavy particles satisfactorily. The soil is introduced into the solution contained in the funnel, agitated, stirred with a glass rod, and allowed to stand some time. This operation may be repeated as often as desired. Separation is not absolute by this operation, the heavy and light particles being sometimes so united that they sink or float together according as one or the other preponderates. There are also particles having so nearly the same specific gravity as the solution that they remain indifferent to its action in any position. After separation has been effected, the heavy portion is drawn off

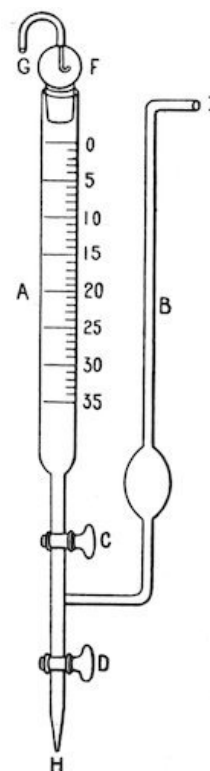


FIG. 58.

THOULET'S
SEPARATING
APPARATUS.

through the stop-cock of the funnel and the lighter is skimmed off the top. Both must be thoroughly washed from the adhering heavy solution for further examination with the microscope, and by chemical, microchemical, and blow-pipe tests. One who has familiarized himself with the appearance of minerals in minute fragments under the microscope, in ordinary and polarized light, will be able to determine some minerals in that way. But for certain identification it is necessary to ascertain their optical properties as is done in the case of the minerals in thin sections of rocks.

Illustration.—The following example from the work of Packard will serve to illustrate the results of separating a soil by the specific gravity method:

One hundred grams of soil, residual clay from the Trenton limestone, were placed in the Thoulet's solution contained in the large separatory funnel. The heavy portion, after washing and drying, weighed 0.6886 gram, or 0.69 per cent. Of this, the magnet removed 0.1635 gram, or 0.16 per cent. This heavy material consisted of rounded yellowish and brown grains up to twenty-five millimeters in diameter, mingled with lustrous angular black grains which were seen under the microscope to be cubes with striated faces, cubes penetrating each other and aggregations of cubes. Combinations of cubes with octahedra and instances of the pentagonal dodecahedron were also observed. These forms, characteristic of pyrites, were also seen in the fine sand obtained as a residue on elutriating the same soil. As these crystals dissolved in hydrochloric acid, giving a strong iron solution, they were regarded as pseudomorphs of iron oxid after pyrites. The yellowish grains on treatment with acid left a grayish residue which contained some grains of quartz but was not wholly quartz. The lighter portion of the soil, over ninety-nine per cent, which floated in the Thoulet's solution of 2.8 was next examined. It was colored red by the iron oxid which coated and adhered to the other minerals. It contained all the quartz, the feldspars if present, and the other minerals whose specific gravity is less than 2.8. It was examined by the microscope and found to consist largely of irregular grains of a mineral which acted on polarized light, obscured somewhat by the iron oxid, and was apparently quartz; and another mineral which was yellowish-brown in color and seemed to be dull and not transparent. Besides there was a large quantity of indistinguishable amorphous material. To clean these minerals the material was treated with hydrochloric acid to remove the iron oxid and other matter soluble in acid,

when the quartz grains appeared transparent and gave interference colors in polarized light. But mingled with these were grains of the other mineral which now appeared grayish, dull, and without action on polarized light. The character of this mineral substance could only be determined by chemical analysis.

270. Harada's Apparatus.—Although it has been affirmed by some analysts that in the subsidence of small particles it is advisable that the containing vessels have parallel sides, yet in the method just given, and in those about to be described, good results are obtained in a funnel or pear-shaped holder.

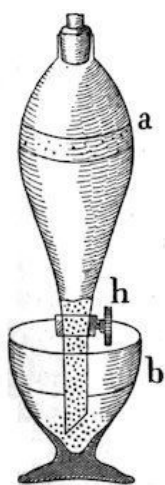


FIGURE 59.

HARADA'S
APPARATUS.

In the apparatus of Harada,^[176] Fig. 59, the separating vessel *a* is made of thick glass furnished with a glass stopper above and a glass stop-cock *h* below. The separating liquid and silt are placed in the pear-shaped vessel *a*, the stopper inserted, and the whole well-shaken. As soon as a ring of clear liquid is seen between the sinking and floating silt, the lower end of the apparatus is brought near the bottom of a conical glass *b*, the cock *h* opened and the heavy silt allowed to fall out. Very little of the liquor will flow out because of the air pressure. Should an air bubble enter the apparatus and be held at the stop-cock, it should be made to ascend by gently tapping. When all the heavy silt has passed into the conical glass, the cock *h* is closed and some water poured over the solution and silt in *b*. The separatory apparatus is now raised until the beveled end of it is in the water layer, when the water at once rises to *h* and thus washes all the silt particles adhering to the glass into *b*. The liquid in *a* may then be diluted by inverting the apparatus, adding the required amount of water through *h*, again shaken after closing *h*, and another separation secured as before.

This apparatus is somewhat easier to manipulate than Thoulet's but does not admit of the same quantitative dilution of the separating liquid.

271. Apparatus of Brögger.—All silt separations in narrow tubes are open to the objection of permitting more or less flocculation. Some of the lighter particles are thus carried down by the heavier, and, on the other hand, some of the heavier float with the lighter. This disturbing action Brögger^[177] seeks to avoid by the following device, Fig. 60, *a*, *b*, *c*. The

length of the apparatus is forty-six centimeters, and its greatest diameter 3.5 centimeters. The opening in the large stop-cock A is the same diameter as that of the apparatus at that point. The cubical content of the apparatus with A open and B closed is about seventy-five cubic centimeters. In conducting the separation the cock B is closed, the separating liquid and silt introduced, A being open, the stopper

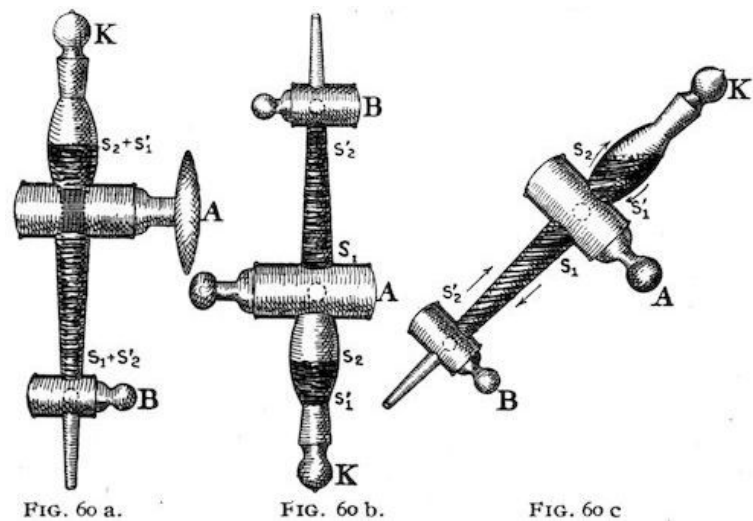


FIG. 60 a. FIG. 60 b. FIG. 60 c.

BRÖGGER'S APPARATUS.

K inserted and the whole well-shaken. In the first separation, the silt S , lying over B is contaminated with some of the lighter particles S'_2 , while the lighter particles above A, S_2 , are mixed with some of the heavier particles, S'_1 . After closing A the apparatus is again well-shaken and inverted as in Fig. 60 b. The two parts of the silt will now undergo another separation as indicated. The apparatus is now carefully inclined as in c, when the various grades of silt will flow in the directions indicated by the arrows, but without mixing, passing each other on opposite sides of the apparatus. When the movement is complete, A is carefully opened, the apparatus still being held as in c, and the light silt formerly between A and B will flow above A, while the heavy silt above A will flow down and join the silt collected over B. This operation may be repeated until a perfect separation is effected. Finally B is opened and the heavy silt collected in a beaker, and the lighter silt then removed from the upper part of the apparatus.

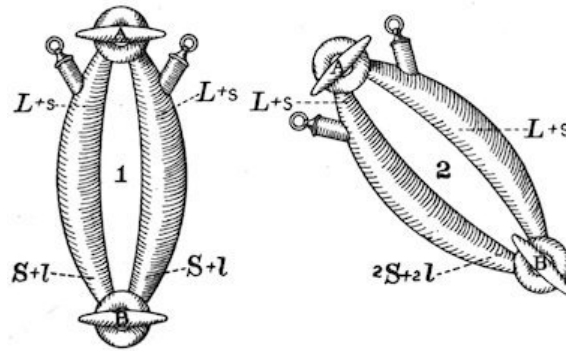


FIGURE 61.

APPARATUS OF WÜLFING.

272. Method of Wülfig.—A somewhat more convenient method of purifying the silt segregates and freeing them of mechanically occluded particles of differing specific gravities has been proposed by Wülfig.^[178] An elliptical ring of heavy glass tubing carries glass stop-cocks A and B, Fig. 61, at the two extremities of the ellipse, each arm of which is provided with a lateral glass-stoppered neck. The perforation in the stop-cocks has the same diameter as the sides of the ellipse. The apparatus has an interior cubical content of about forty cubic centimeters. Thirty cubic centimeters of the separating fluid are introduced through one of the lateral apertures and brought to the same height in the two arms by opening the cock B. The silt is then introduced in equal quantities into each of the arms. The stoppers having been inserted, the whole is well-shaken. At the beginning of the separation, the apparatus being held in position 1, the lighter soil above and the heavier soil below are somewhat mixed by reason of flocculation and mechanical entanglement. At this point B is opened and the apparatus placed in the inclined position 2. The heavier particles $S + l$, on the right arm, are thus united with the same class of particles in the left arm making $2S + 2l$. This operation is hastened by opening A and allowing the higher column of liquid in the right arm to pass into the left. The liquid in the left arm is allowed to rise to A. After all of $S + l$ in the right arm has passed into the left B is closed, the apparatus then placed back in position 1 and inclined in the opposite direction until $L + s$ in the top of the left arm has been transferred to the $L + s$ in the top of the right, and the same quantity of liquid is found in each arm. The operation is then repeated and this

continued until all S + s is found in the bottom of the left arm and all L + l in the top of the right arm.

273. Separation with a Magnet.—Particles of magnetic iron oxid are easily separated from the fine soil particles by means of a magnet. A strong bar or horseshoe magnet may be used. Electro-magnets are rarely necessary except for the separation of particles of feeble magnetic power. Particles of iron which may be found would owe their origin to the mortars in which the soil had been pulverized, or they might come from a recently crushed meteorite. Some minerals, as limonite, after ignition are attracted by the magnet and it is advisable to subject a part of the sample to this treatment. The best method of separation consists in spreading the particles evenly on paper and gradually bringing the magnetic particles to one side by moving the magnet underneath.

274. Color and Transparency.—But little can be learned from the color and transparency of the smallest silt particles, but these properties in the larger grains have considerable diagnostic value. Many minerals of distinct color appear wholly colorless in petrographic sections or in silt particles, as for instance, highly-colored quartz. On the other hand, even the smallest particle of chlorite will show its distinctive tint. The colors in some minerals are due to occluded matter not essential to their structure, and these foreign bodies would naturally escape when the crystal mass is reduced to an almost impalpable powder.

275. Value of Silt Analyses.—As in the case of chemical analyses a silt analysis of a soil which is not typical or representative has little value. On the other hand, a systematic separation of soils into classes of particles can not fail to reveal a definite correspondence of mechanical composition to soil properties. The production of a crop is the result of certain functions, chief among which are temperature, moisture, and plant food. In a given soil the temperature is markedly affected by its physical state. It has been demonstrated in previous paragraphs that the circulation of moisture in the soil and its capacity to be held therein are chiefly functions of the state of aggregation of the soil itself. The availability of plant food in a soil is not measured by its quantity alone, but rather by its state of subdivision. It is not therefore a matter of surprise that the fertility of a soil is found, *caetèris paribus*, to be commensurate to a certain limit with the percentage of fine silt and clay which it contains. It is true that two soils quite different in

fertility, may have approximately the same silt percentages, but in such a case it is demonstrable that even in the poorer soil the measure of fertility is largely the percentage of fine particles and not its actual content of plant food. In other words, almost all soils, even the poorest, have still large quantities of plant food, but these stores, owing to certain physical conditions, are not accessible to the rootlets of plants. An illustration of this is seen in the use of concentrated fertilizers. It might seem absurd to suppose that the addition of 100 pounds of sodium nitrate would prove useful to a plat containing already many tons of nitrogen; but the nitrate is at once available and its beneficial influences are easily seen.

The full value of silt analysis will only be appreciated when many typical soils from widely separated areas are carefully studied in respect of their chemical and physical constitution and the character of the crops which they produce.

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PART FIFTH.

ESTIMATION OF GASES HELD IN SOILS.

276. Relation of Soil Composition to Gases.—The power of a soil for occluding gases rests primarily on its composition as determined by silt analysis. The discussion of this part of the subject is so nearly related to that of the physical properties of the soil that it might properly have been included in that part of the work. Since, however, we deal in this part more with the determination of the gas constituents of the soil, it was deemed preferable to place it after the silt analysis and as introductory to the general estimation by more strictly analytical processes of the chemical constituents of the soil.

277. Occurrence of Carbon Dioxid.—The amount of organic matter in the soil, according to Wollny,^[179] is no indication of the quantity of carbon dioxid when the organic matter is in excess. The percentage of carbon dioxid is only proportional to the amount of organic matter when this is in small quantities. Large quantities of organic matter do increase the amount of carbon dioxid, but the increase is not a proportional one, since a larger quantity of this gas in the air of a soil reduces the activity of the organisms which produce oxidation. Water and temperature have a greater influence on the oxidation, and act in an opposite direction to that of the organic matter. The amount of free gas in the soil affords no indication either of the intensity of the action of oxidation or of the amount of organic matter.

The addition of liquid manure to the soil results in a reduction of the decomposition of the organic matter when the quantity of the salts therein contained is greater than that already present in the soil. But if the liquid

manure is dilute, and the absorptive power of the soil for salts is great, then the decomposition is promoted.

278. Absorption of Aqueous Vapor.—The power of a soil to resist drought depends largely upon its coefficient of absorption for aqueous vapor. Hilgard has shown^[180] that at temperatures between 7° and 21°, the amount of aqueous vapor absorbed by a thin layer of a clay or soil not unusually rich in humus, in a saturated atmosphere, is sensibly constant. In general, clay soils are more absorbent than sandy ones, yet there is no direct connection between the amount of clay present and the absorbent power of the soil. Evidently the hygroscopic coefficient is largely controlled by the presence with the clay of the powdery ingredients which determine its looseness of texture, and it is found that the finer silts themselves possess a considerable absorbing power. According to Whitney this is largely dependent upon the extent of the surface area of the soil grains and upon the size and arrangement of these grains. Again, the presence of hydrated ferric oxid materially influences this power, so that the amount of iron present must always be taken into consideration.

279. Methods of Study.—The study of the deportment of a soil with vapors or gases may be divided into two general classes. The first depends on the subjection of a sample of soil to the saturating influence of a given vapor or gas and measuring the amount thereof absorbed, either directly by increase of weight, or by the diminution in the amount of gas originally supplied. The maximum absorbent capacity of a soil under given conditions for a gas or vapor is in this way determined.

In the second class the determination consists in accurately estimating the amount of gas which is absorbed by a soil in natural conditions or *in situ*, thus giving the natural percentages of the gaseous constituents of the soil.

In the first case in general, the principle of the method depends upon the exposure of the soil for a given time under given conditions, to an atmosphere of the gas to be absorbed. The principle of the second class of determinations depends upon the extraction, usually by means of suction, from a given mass of soil of the gaseous matters therein contained. The general details of the methods of procedure for the first class are found in the following directions for manipulation:

280. Determination of the Maximum Hygroscopic Coefficient.—The fine earth, in Hilgard's method, is exposed to an atmosphere saturated with

moisture for about twelve hours at the ordinary temperature (60° F.) of the cellar in which the box should be kept. The soil is sifted in a layer of about one millimeter thickness upon glazed paper, on a wooden table, and placed in a small water-tight covered box, twelve by nine by eight inches, in which there is about an inch of water; the interior sides and cover of the box should be lined with blotting paper, kept saturated with water, to insure the saturation of the air.

Air-dried soil yields results varying from day to day to the extent of as much as thirty to fifty per cent, nor have we any corrective formula that would reduce such observations to absolute measure. Knop's law, that the absorption varies directly as the temperature, while applicable to low percentages of saturation, is wide of the truth when saturation is approached. The ordinary temperature of cellars will serve well in these determinations without material correction.

After eight to twelve hours the earth is transferred as quickly as possible, in the cellar, to a weighed drying tube and weighed. The tube is then placed in a paraffin bath; the temperature gradually raised to 200° C. and kept there twenty to thirty minutes, a current of dry air passing continually through the tube. It is then weighed again and the loss in weight gives the hygroscopic moisture in saturated air.

The reason for adopting 200° C. as the temperature for drying instead of 100° is that water will continue to come off from most soils at the latter temperature for an indefinite time, a week or more, before an approach to constancy of weight is attained; and that up to 200° only an arbitrary limit can be assigned for the expulsion of hygroscopic moisture. Moreover, the great majority of soils, especially those poor in humus, will reabsorb moisture from a saturated atmosphere to the full extent of that driven off at 200° C.

281. Estimation of the Absorption Power of Soils for Aqueous Vapors.^[181]—*Method A.*—The fine earth, ten to twenty grams, is spread out on a surface of about twenty-five square centimeters, and left for several days with the observation of the temperature of the air and the loss of weight determined from time to time. This evaporation is continued until the weight remains practically constant. Afterwards by drying the sample at 100° the amount of hygroscopic moisture is determined. A similar result can be reached if the sample is first dried at 100°, or over sulfuric acid at

ordinary temperatures, and then the increase in weight observed which the sample acquires on being exposed for several days to the atmosphere under ordinary conditions. Soils with about the same content of humus show variations in the power to absorb aqueous vapors which are almost proportional to the amount of clay which they contain. With the increase of humus substance, the power of the soil for absorbing moisture is increased, so that a sandy soil which is rich in humus often will retain as much moisture in an air-dried state as a clay soil which is poor in humus. If the experiment is carried on by drying over sulfuric acid instead of at 100°, the sample should be left from four to seven days in order that a constant weight may be reached. Even after this time the loss in weight is 0.2 to 1.5 per cent less than when the sample is dried at 100°.

Method B.—In order to determine the amount of aqueous vapor which a soil will absorb in an atmosphere saturated with the vapor the following method is used:

The sample of air-dried soil in a flat dish of given surface; viz., about twenty grams of soil to twenty-five square centimeters surface is placed in a vessel over water without contact with the water, and the whole of the apparatus is covered with a glass bell-jar. The sample is weighed at intervals of six or eight hours until no appreciable increase of weight is observed. An empty vessel of the same size and character as that containing the soil is kept under the bell-jar, also in the same conditions, so that any increase in weight by the deposition of moisture on this vessel may be determined. This increase in weight is to be deducted from the total increase in weight of the vessel and the soil. Sandy and loamy soils become saturated in this manner in the course of the first twenty-four hours and remain after that unchanged in weight. Very clayey soils, and also those which are very rich in humus, require a much longer time, three or four days even. In this case it is better to take a smaller sample of the soil; viz., ten grams. The temperature of the air within the glass vessel, of course, must be taken into consideration.

Method C.—The same flat dish and the same quantity of soil as in the other methods are taken in this one. The sample is left out over night where it can be fully saturated with dew. The amount of dew which appears on the bushes should be noted and also the temperature of the air and the percentage of clouds in the sky. An experiment should also be made on

spots of earth which are entirely free from vegetation in order that the difference in the amount of water absorbed in places practically devoid of dew and in places where the dew is abundant may be observed.

Method D.—Deeper flat dishes should be used for this determination so that the depth of soil contained in them shall be from one to three, or even six centimeters. The sample of soil should be completely air-dried and in a state of fine subdivision. The vessels containing the soil should be placed in a locality saturated with aqueous vapor or in the open air during the night where they are subjected to the influence of the cooling of the atmosphere and the deposition of dew. Note should be made of the different amounts of moisture absorbed by the layers of earth of different thicknesses in a given time. Observation should also be made of the depth to which the moisture sinks in the sample of soil under consideration.

282. Estimation of the Absorption Power of the Soil for Oxygen and Atmospheric Air.^[182]—From fifty to one hundred grams of air-dried soil are placed in a glass vessel of about 500 cubic centimeters capacity, and the flask closed with a stopper after the addition of enough water to make the percentage of moisture in the soil about twenty. After from eight to fourteen days the air contained in the vessel is analyzed for oxygen, nitrogen, and carbon dioxid, with special reference to the determination of how much oxygen has disappeared and how much the carbon dioxid has been increased. As an alternative method, twenty-five grams of the soil may be moistened with tolerably concentrated potash lye in a small glass vessel, which is itself joined with air-tight connections to an azotometer in which a known volume of air is confined by quicksilver. The glass vessel is frequently shaken during the progress of the experiment. The diminution of the volume of air in the apparatus after from one to four days gives approximately the quantity of oxygen absorbed.

283. General Method of Determining Absorption.—This method, due to Freiherrn von Dobeneck,^[183] is as follows: The soil, in a state of fine powder, is dried at 100° to 105° to a constant weight. It is then placed in an absorption tube of the following construction:

The absorption tube consists of a shaped wide glass tube, both ends of which are supplied with small glass tubes sealed upon the end of the tube, and those are furnished with tightly-ground glass stop-cocks. Above these stop-cocks these small tubes are bent in opposite directions at right

angles. On the bend of the is sealed another tube which is furnished with a ground glass stopper. Through this opening the tube can be filled with the sample of soil. When the tube is filled, the glass stopper inserted, and the two stop-cocks on the small tubes closed, the contents of the tube are completely excluded from the external atmosphere. Many of these tubes can be used at once so as to hasten the progress of the work.

The tubes after being filled are placed in a drying oven with the stop-cocks open. The stop-cocks are then closed before the tubes are removed, when they are placed in a desiccator for cooling preparatory to weighing. The weighed tubes are held in a tin box which can be placed in a water-bath which is kept at a given temperature by means of a thermostat. The top of the tin box should be hinged and made of a thick non-conducting material so as to prevent any rapid change of temperature within. On the inner side of the box a small thin-walled glass tube is carried around four times. One end of this tube passes through an opening in the side of the box by means of which it can be connected with the gas apparatus outside. The other end of it is connected directly with the absorption tubes.

The absorption tubes are so connected among themselves that when ammonia or carbon dioxid is employed the gas passes through one of the tubes before it can reach the next, and so on. For experiments with water-gas, however, that is, air charged with aqueous vapor, the arrangement must be different. While in the case of ammonia and carbon dioxid the composition of the gas is not changed by passing through the samples of soil, the case is quite different when air charged with aqueous vapor passes through. In the latter case the amount of aqueous vapor in the air would be notably lessened in passing from sample to sample on account of the retention of a part of the aqueous vapor by the soil. In this case, therefore, the saturated air, after it has passed through the glass tube around the inside of the box in order to reach the proper temperature, is conducted into a receptacle of glass which has a number of connections equal to the number of absorption tubes so that the saturated air can pass directly into each one of them.

The gases which are to be used for the experiments are prepared in proper apparatus and are forced through the samples of soil, either by pressure as in the case of ammonia or carbon dioxid, or by means of aspirators as in the case of air saturated with aqueous vapor.

The carbon dioxide employed is purified by passing over sodium carbonate and calcium chloride.

The ammonia is prepared by the action of finely powdered lime on ammonium chloride, and is dried by passing over lime and sticks of potassium hydroxide.

The air which is to be saturated with aqueous vapor, in order to purify it from dust, carbon dioxide, and ammonia, is passed through two flasks in which are contained respectively, diluted sulfuric acid and potash lye. It is afterwards thoroughly saturated with aqueous vapor at the temperature desired.

Various kinds of soil material may be employed as follows:

(1) Pure quartz sand.—Freed from all fine particles by subjection to silt analysis, afterwards boiled with hydrochloric acid and washed with water to free it from all clayey materials. The sand prepared in this way should be passed through different sieves in order to prepare it in different states of fineness.

(2) Quartz powder.—Prepared from pure quartz crystals by grinding in an iron mortar.

(3) Kaolin.—Material such as is used in the manufacture of the finest porcelain which, after being freed of all foreign matter, is rubbed to a fine powder in a porcelain mortar.

(4) Humus.—Washed with ether and alcohol, boiled with hydrochloric acid, washed, dried and reduced to a state of fine powder.

(5) Iron oxide.

(6) Calcium carbonate.—Precipitated, washed, and dried.

(7) Soil mixtures.—Prepared artificially by mixing the kaolin, quartz, and humus, above mentioned.

The quantity of gas absorbed by each of these materials is determined by filling the tubes, as above mentioned, with the dried material. The content of each tube is previously determined by filling with mercury and weighing. Having determined the weight of the substance to the exclusion of the air contained within its pores, it is treated with the gas in the apparatus described above and weighed from time to time until no further increase of weight takes place.

The method of calculating the results is shown in the following scheme:

V = content of the absorption tube obtained by filling with mercury and weighing.

P' = weight of the empty tube filled with air at 100°.

pl = weight of the air in the tube (pl = V × specific gravity of the air at 100°).

pt' = weight of the tube (pt' = P' - pl).

P² (second weighing) = weight of the tube filled with the substance with the included air at 100°.

v^s = volume of the substance calculated according to the formula

$$v^s = \frac{P^2 - P'}{s^s - \text{specific gravity of air}}.$$

s^s = specific gravity of the substance.

vl = volume of the air in the flask filled with the substance (vl = V - v^s).

pl' (weight of this included air) = vl × specific gravity.

p^s = weight of the substance (pl = p² - pt' - pl)

P³ = weight of the apparatus at the end of the experiment.

sg = specific gravity of the gas employed for saturation.

pg (weight of the gas remaining over the substance) = vl × sg.

pa (weight of the absorbed gas) = P³ - pt' - p^s - pg.

p^s gram of substance absorbs pa gram of the gas and 100 grams of substance would absorb $\frac{100 \times pa}{p^s}$ grams.

The specific gravities of the gases employed are calculated from the tables given by Landolt and Börnstein in "Physical and Chemical Tables," page 5.

The specific gravity of the quartz sand employed was 2.639; of the quartz powder, 2.622; of the kaolin, 2.503; of the humus, 1.462; of the iron hydroxid, 3.728; and of the calcium carbonate, 2.678.

One liter of ammonia, at a pressure of 760 millimeters of mercury and a temperature of 0°, weighs 0.7616 gram; one liter of carbon dioxide, 1.9781 grams; one liter of aqueous vapor, 0.8064 gram; and one liter of dried air, 1.2931 grams.

At a pressure of 720 millimeters, and at 20° temperature, a liter of air saturated with aqueous vapor at 0° weighs 1.1383 grams; saturated at 8.6°, 1.1362 grams; saturated at 10°, 1.1358 grams; saturated at 14°, 1.1340 grams; saturated at 18.2°, 1.1330 grams; saturated at 20°, 1.1321 grams; saturated at 30°, 1.1313 grams.

The general results of the experiments are as follows:

ABSORPTION AT 0°.

		Aqueous vapor from saturated air.		Ammonia.		Carbon dioxid.	
		Grams.	Cubic cm. [H]	Grams.	Cubic cm. [H]	Grams.	Cubic cm. [H]
100 grams	quartz	0.159	197	0.107	145	0.023	12
„ „	kaolin	2.558	3,172	0.721	947	0.329	166
„ „	humus	15.904	19,722	18.452	24,228	2.501	1,263
„ „	Fe ₂ (OH) ₆	15.512	19,236	4.004	5,275	6.975	3,526
„ „	CaCO ₃	0.224	278	0.256	320	0.028	14

[H](#). Reduced to 0° and 760 millimeters pressure mercury.

The foregoing methods will suffice to show the procedures to be followed in estimating the maximum amount of any common gas or vapor a given quantity of soil may be made to absorb. We pass next to consider the quantities of gases or vapor soils *in situ* may hold.

284. Method of Boussingault and Lewey.^{[\[184\]](#)}—This method is the oldest and most simple procedure for estimating the nature of the gases held in a soil *in situ*.

For the purpose of collecting the sample of gas from the soil a hole, thirty to forty centimeters in depth, is dug, and a tube placed in it in a vertical position, having on its lower extremity a bulb perforated with fine holes. The hole is filled and the earth closely packed around the tube which is left for twenty-four hours. At the end of that time the tube is slowly aspirated until a volume of gas approaching from five to ten liters is obtained.

Estimation of Carbon Dioxid.—The carbon dioxid in the sample of gas is estimated by allowing it to bubble through a solution of barium hydroxid.

Estimation of the Oxygen.—The oxygen is estimated in a separate sample of the gas by means of potassium pyrogallate.

The chief objection to this simple process is the uncertainty of being able to obtain an average sample of the occluded gas. In digging the hole and refilling, there must evidently be a considerable disturbance of the original distribution of the gas or vapor.

The methods of Pettenkofer^[185] and Aubry^[186] are essentially like that just described. Pettenkofer found the largest quantities of carbon dioxide in the earth gases in July, August, and September, and the smallest quantities in the winter months.

No greater detail concerning these methods of the direct aspiration of the air is considered necessary inasmuch as the methods about to be described, while more elaborate, are superior in accuracy to the older methods mentioned. In general, in these experiments, it is deemed sufficient to determine the carbon dioxide only.

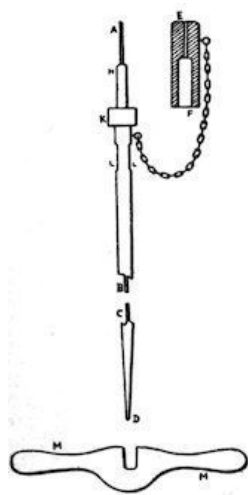


FIGURE 62.

SCHLOESING'S SOIL-TUBE FOR COLLECTING GASES.

285. Method of Schloesing.—The apparatus used by Schloesing^[187] in the collection of the soil gases consists of a steel tube (Fig. 62) a little over one meter in length, ten millimeters in external diameter, and one and one-half to two millimeters in internal diameter. The end which penetrates the soil is made slightly conical for a distance of twenty-five to thirty centimeters. By reason of the shape of the tube, when it is driven into the soil all connection between the orifice in the point of the tube and the external air is prevented. The obstruction of the internal canal of the tube is prevented by introducing a thread of steel which penetrates the whole length of the tube. This thread, represented by A, B, C, D, is flush with the interior extremity of the tube at D. It extends for about three centimeters above the upper end of the tube in order to be easily handled when it is to be removed.

For the purpose of driving the tube into the soil its upper part is covered with a cylindrical piece of steel, EF, in the interior of which are freely engaged H and A. This head piece rests upon a ring of steel, K. This ring is fastened solidly into the tube. On striking the piece EF the tube and the steel wire in the center are driven together into the soil. The tube is

flattened at L and L' in order to be embraced by the key MM, the employment of which is necessary in order to revolve the tube around its axis when it is being driven into the soil. When the tube has been driven to the depth desired, the steel wire is withdrawn and it is immediately connected at H with the rubber tube N (Fig. 63) belonging to the system PQT, and furnished with a pinch-cock X. The system PQT comprises the following elements: PQT made of a capillary glass tube in the form of a T. The lower end of the tube P is closed by the larger glass tube O, sealing the end of P with a little mercury. O is held to P by the cork S, which is attached firmly enough to prevent O from dropping off, but is furnished with a canal in order to allow the air to flow in or out freely. This system is connected with the system UV by the rubber connection T. U is a glass vessel having the constrictions as indicated in its stem above and below the bulb. V is a glass vessel of convenient size connected with U by the rubber tubing as indicated. The capacity of the cylindrical portion of U should be from fifteen to eighteen cubic centimeters.

To take a sample of soil gas, V is lifted above U. The air is driven from U and escapes through O, which acts as a true valve. When the mercury has completely filled U the pinch-cock X is opened and V depressed gradually. The gas coming from the soil is thus collected in U. A few cubic centimeters of the soil gas are collected in this way, the pinch-cock X is again closed and V is raised in order to drive the whole of the contents of U again through O. In this way the whole of the air which the capillary vessel originally contained is removed and all parts of it remain filled with soil gas. Two or three operations, using from five to ten centimeters of soil gas in all, will be sufficient to completely free the apparatus from its original content of air. U is then entirely filled by depressing V, and it is then hermetically sealed at the two constricted points by means of an alcohol lamp. The sealed tube can then be transported to the laboratory and its contents subjected to eudiometric analysis.

Without displacing the tube from the soil, several samples of gas can be taken from the same spot. A sufficient number of the bulbs V should be at

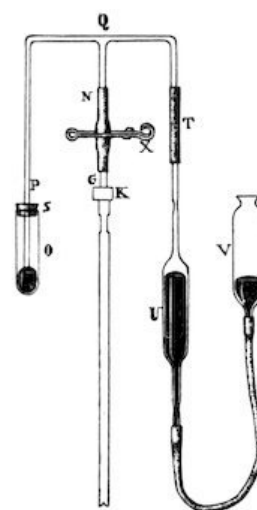


FIGURE 63.

SCHLOESING'S
APPARATUS FOR
COLLECTING GASES
FROM SOIL.

hand to hold the required number of samples. Instead of submitting the sample to eudiometric analysis it is usually sufficient to determine the quantity of carbon dioxide which it contains, inasmuch as numerous experiments have shown that in 100 parts of soil gas the oxygen and carbon dioxide together constitute twenty-one parts. No appreciable trace of marsh gas, or other combustible gas, has yet been detected in ordinary arable soils. These gases have only been found in special soils from marshes, in the neighborhood of gas wells, etc., and not in arable soils.

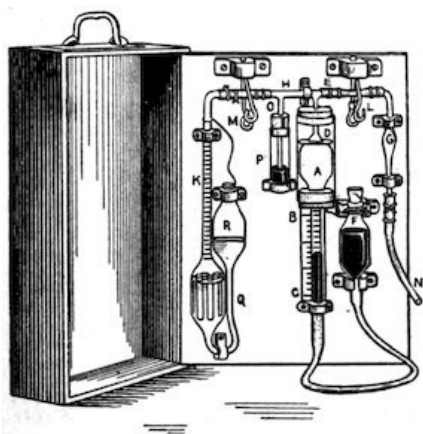


FIGURE 64.

SCHLOESING'S APPARATUS FOR
DETERMINATION OF CARBON DIOXID.

286. Apparatus for Estimating the Carbon Dioxid.—The apparatus used for determining the carbon dioxide in Schloesing's work consists of the apparatus shown in Fig. 64. A represents a glass vessel surrounded by a jacket of glass, full of water, and sealed on its lower part to the tube BC of about six millimeters internal diameter. On its upper part it is sealed to the capillary tube D. The tube BC is graduated from C in hundredths of the volume of DAC, which volume is about twelve cubic centimeters. On its lower part it is connected by a rubber tube with a reservoir F which is capable of being raised or lowered. GHK are capillary tubes connected together by the rubber tubes L and M, which are furnished with pinch-cocks. The tube G is connected to a vacuum by the rubber tube N. The rubber tube should be of very small internal diameter and from forty to fifty centimeters in length. To the tube H are sealed, at right angles, the branch D and another branch O. This last dips into a little mercury which the tube P contains. It serves as a valve, permitting the exit of the gases but not their entrance. The tube K carries some lines engraved on its inferior part and is sealed to the system of the two bulbs Q and R. The bulb Q contains a concentrated solution of potash. It carries a number of pieces of glass tubing for the purpose of increasing the surface of the potash solution.

All the parts of the apparatus are fixed upon a rectangular board, nineteen centimeters broad by twenty centimeters long. This forms one of the faces of a wooden box to which it is hinged and which serves for the transportation of the apparatus in a vertical position. The graduation of the

tube BC is recorded behind this tube upon a card fixed upon the board. By means of these two graduations, the height of the mercury in the tube BC is most easily read, even when the tube is not perfectly vertical. Each one of the pinch-cocks L and M, on its upper part is fixed in a sort of guard which prevents it from being displaced laterally during the processes of the manipulation, thus avoiding all danger of breakage.

After the operation is finished a little air is sent into Q in such a manner as to sensibly lower the level of the solution of potash, and the upper extremity of R is closed with a rubber stopper. Afterward, the apparatus can be transported without any danger of the potash becoming engaged in the tube K and reaching the measuring tank A.

To proceed to the analysis, a stake is driven into the soil to which all of the apparatus can be fixed. At the side of the stake the apparatus for taking the sample, already described, is driven into the soil and this apparatus is connected by the tube N with the apparatus for determining the carbon dioxid. The pinch-cocks L and M being closed, F is lifted until the mercury which runs from it fills A and approaches D. During this time the air which the apparatus contains has been driven out through O. The tube NGD is freed from air by opening the pinch-cock L, lowering F and drawing into A the gas coming from the soil; afterward closing L and driving out the gas through O. After two or three rinsings of this kind, which employ altogether only ten to twelve cubic centimeters, the gas which is to be analyzed is sucked into A. For this purpose F is lowered until the mercury in the tube BC is very near C. The pinch-cock L is closed and M opened. The reservoir F is displaced little by little by pressing lightly against the rectangular board in order to give it greater firmness in such a way as to fix the level of the mercury exactly at C, and the line is noticed where the solution of potash in K stands. The gas contained in the apparatus is under a pressure, the difference of which from the external pressure is represented by the column of the potash solution between the mark just noticed and the level of the same solution in the bulb R. In order to absorb the carbon dioxid, F is lifted until the mercury stands between D and E. The gas thus passes from A into Q. It gives up immediately its carbon dioxid to the potash solution. It is then made to come again into A, and afterward a second time into Q in order to free it from the last trace of dioxid. Finally it is made to return to A and F is kept at such a height that the potash solution maintains in the tube K the same level as at the commencement of the operation. The gas is then at the

same pressure to which it was subjected before absorption. The level of the mercury is then read on BC. At the time the apparatus is used, the measuring tube A should be slightly moist. If it is not so, a small quantity of water should be introduced which is afterward rejected, but which leaves a sufficient quantity of moisture upon the internal walls of A. In this way the gas will always, before or after absorption of carbon dioxide, be saturated with vapor of water, and the figure read in the last place upon the tube BC represents the percentage of carbon dioxide in 100 parts of the gas extracted from the soil supposed to be saturated with vapor at the temperature of the experiment.

During the course of the analysis, the temperature of the measuring flask, which is almost entirely surrounded with water, does not vary sensibly, but in a series of experiments which are executed at different times, the temperature of the measuring apparatus, which is that of the ambient air, may change much. It may oscillate between 10° to 25° , and exceptionally between 0° and 30° , whence there are notable variations in the tension of the vapor of the gas measured. If it should be desired to calculate to 100 parts of dry gas the observations made at 30° upon 100 parts of saturated gas, it would be necessary to increase the percentage of carbon dioxide by about $\frac{1}{25}$ of its value. It is noticed that with the apparatus described above, the gas upon which the estimation is really conducted comprises not only that which the measuring apparatus contains from E to C before the absorption of the carbon dioxide, but also the small quantity which remains in the capillary tube KME at the moment when closing the pinch-cock M, after the second rinsing, the gas from the soil is aspired into EAC. On the other hand, there is left in the same tube KME, when the final reading is made, some gas which belongs to that which has been measured at the end. These two small gaseous portions which we consider in the tube KME to be sensibly equal, do not contain any carbon dioxide and may be left out of consideration. That is why the volume of the measuring apparatus is limited to E and the graduation of the tube BC is in hundredths of the volume comprised from E to C. In reality the two portions are not absolutely equal because the two successive levels of the potash solution, which limit them in the tube K, are not absolutely identical. These two levels can differ in such a manner as to correspond to a volume of about $\frac{1}{1000}$ of the measuring apparatus. Thus the estimation is really made upon a volume of gas which may be greater or less by $\frac{1}{1000}$ than the volume of EAC; whence there

might result an error of $\frac{1}{1000}$ in the estimation of the carbon dioxide, an error which is wholly negligible.

As a result of numerous analyses it is concluded, first, that the oxygen exists normally in the atmosphere of soils in large proportion; second, very probably the gaseous atmosphere of arable soils, to a depth of sixty centimeters, contains scarcely one per cent of carbon dioxide and about twenty per cent of oxygen; third, the highest percentages of carbon dioxide correspond to epochs of highest temperature and periods of greatest calm; fourth, the proportion of carbon dioxide increases ordinarily with the depth at which the samples are taken. This disposition of the carbon dioxide would appear almost necessary, since near the surface the internal atmosphere is almost constantly diluted by external air by virtue of diffusion. Fifth, from one epoch to another the composition of the atmosphere of the soil can undergo considerable variation.

287. Determination of Diffusion of Carbon Dioxide in Soil.—The method proposed by Hannén^[188] is a convenient one to use in studying the rate of diffusion of carbon dioxide in soils. A large Woulff's bottle with three necks serves for the reception of the gas. The two smaller outer necks of the bottle carry two glass tubes bent outwards and provided with stop-cocks. One of these passes to near the bottom of the bottle and the other just through the stopper. The middle tubule of the bottle is of a size to give in section an area of about twenty-two square centimeters. It is made with a heavy rim two centimeters wide and plane ground. This rim carries a plane-ground glass plate with a circular perforation in one-half of it, of the size of the opening in the central tubule of the bottle. A glass cylinder, carrying a fine wire-gauze diaphragm near the lower end, fits with a ground-glass edge air-tight, over this aperture, being held in position by a brass clamp. The ground-glass plate moves air-tight between the cylinder and the bottle, so that the cylinder can be brought into connection with the bottle or cut off therefrom without in any way opening the bottle to the air. The plate and all ground movable surfaces should be well lubricated with vaseline.

The experiment is carried on as follows: The glass cylinder is filled with the soil to be tested, closed above with a rubber stopper carrying a gas tube, and then by moving the perforated-glass plate brought into connection with the bottle. The side tube, with short arm inside the bottle, is then closed, and

carbon dioxid introduced through the other lateral tube until the gas passing from the tube at the top of the cylinder is pure carbon dioxid.

The lateral tube is then closed and the bottle is placed in a water-bath and kept at a constant temperature of 20°. When the temperature within and without the apparatus is the same the reading of the barometer is made, the stopper removed from the top of the cylinder, and the process of diffusion allowed to begin. After from six to ten hours the glass plate is moved so as to break the connection between the cylinder and bottle. The carbon dioxid remaining in the bottle is driven out by a stream of dry, pure air. The air is allowed to pass through the apparatus for about ten hours. The carbon dioxid driven out is collected in an absorption apparatus and weighed. The absorption apparatus should consist of a series of Geissler potash absorption bulbs and finally a form soda-lime tube. In front of the absorption apparatus is placed a drying bulb containing sulfuric acid. Inasmuch as the temperature and pressure can be readily determined, the weight of carbon dioxid obtained is easily calculated to volume.

The weight of 1,000 cubic centimeters of carbon dioxid at 0° and 760 millimeters pressure is 1.96503 grams. Therefore one milligram is equivalent to 0.5089 cubic centimeter of the gas. The volume of the bottle should be carefully determined by calibration with water. The results should be calculated to cubic centimeters per square centimeter of exposed surface in ten hours. The depth of the soil layer is conveniently taken at twenty centimeters.

288. Statement of Results.—

THE SOIL PACKED LOOSELY IN THE DIFFUSION TUBE.

DIFFUSION TIME, TEN HOURS.

Diameter of soil particles, millimeters.	Weight of soil taken, grams.	Pure carbon dioxid at beginning of experiment, cubic cm.	Carbon dioxid at end of experiment, cubic cm.	Cubic centimeters of carbon dioxid diffused for each square cm.
0.01–0.071	520	2549.4	1230.3	59.9
0.071–0.114	550	2545.9	1269.2	58.0
0.114–0.171	590	2556.4	1354.2	54.6
0.171–0.250	620	2538.9	1336.1	54.6
0.250–0.500	660	2532.0	1374.5	52.6
0.500–1.000	680	2528.2	1440.2	49.5
1.000–2.000	690	2496.6	1396.9	50.0
Mixture of the above	720	2514.3	1572.5	42.8

In greater detail the calculation and statement of the results may be illustrated by the following data:

In the first experiment given in the above table the diameter of the soil particles varied from 0.010 to 0.071 millimeter. The weight of soil in the diffusion tube was 520 grams. The volume of gas, at 0° and 760 millimeters, before the diffusion began was 2549.4 cubic centimeters. The volume of carbon dioxid under standard conditions remaining after ten hours of diffusion was 1230.3 cubic centimeters. This volume is calculated from the weight of carbon dioxid obtained in the potash bulbs, each milligram being equal to 0.5089 cubic centimeter of carbon dioxid. The volume of carbon dioxid diffused is therefore $2549.4 - 1230.3 = 1319.1$ cubic centimeters. The per cent of carbon dioxid diffused is $1319.1 \div 2549.4 = 51.74$. The volume of carbon dioxid diffused for each square centimeter of cross section of the diffusion tube is $1319.1 \div 22 = 59.9$ cubic centimeters.

The carbon dioxid should be passed long enough to secure complete expulsion of the air before the determination is commenced.

289. General Conclusions.—The general results of the experiments with the diffusion apparatus to determine the effect of the physical condition of the soil upon the rate of diffusion are as follows:

1. The diffusion of carbon dioxid through the soil is, at a constant temperature, chiefly dependent upon the pores in the cross section of the column of soil. Therefore, the absolute quantity of the diffused gas is greater the larger the total volume of the pores and *vice versa*.

2. Every diminution of the volume of the pores, whether secured by pressure of the soil or by an increase in the moisture thereof, is followed by a decrease in the volume of diffused gas. The giving up of the carbon dioxid present in the soil atmosphere to the upper atmosphere by the method of diffusion is therefore the less the finer the soil is, the more compressed the soil particles are, and the larger the water capacity of the sample and *vice versa*.

3. The quantity of diffused carbon dioxid is diminished according to the measure of compression to which the soil is subjected but is not strictly proportional to the height of the soil layer.

4. In soils in which rain water percolates slowly the diffusion of the carbon dioxid on account of this property is depressed to a greater or less extent.

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PART SIXTH.

290. Preliminary Considerations.—The sample of soil intended for chemical analysis should consist of the fine earth which has passed at least a one-millimeter mesh sieve and subsequently been completely air-dried. According to Petermann the air-drying of a soil should continue for about four days for an ordinary arable soil, and about six days for one very rich in organic matter. With peat and muck soils I have found that ten or twelve days with frequent stirring, even when in thin layers, are necessary to attain approximately a constant weight.

The soil is conveniently spread on a zinc or other metal sheet of sufficient area so that the layer will be only one or two centimeters in thickness. The weight before and after desiccation will give the percentage of moisture lost on air-drying, which, of course, will depend chiefly on the degree of saturation of the sample when taken and the atmospheric conditions prevailing during drying.

If samples of soil are taken in very dry times it is often necessary to moisten them with distilled water in order to prepare them properly for air-drying.

The quantity of hygroscopic water which the sample loses at 100° – 105° should be determined, and all subsequent calculations of the percentages of the various constituents be based on the water-free material. When a soil which has been dried at 100° – 105° to a constant weight is heated to 140° – 150° it loses additional weight not due to loss of water of constitution. A part of this loss may be due to hygroscopic moisture which is not given off at 100° – 105° , and a part may be hydrocarbons, or other easily volatile organic or inorganic bodies. Before estimating the total loss on ignition it is recommended by most chemists to dry at 140° – 150° . The samples of soil, however, intended for chemical examination should never be dried beyond

the point which is reached by exposure in thin layers at ordinary room temperatures. The state of aggregation, degree of solubility, and general properties of a soil, may be so changed by absolute desiccation as to render the subsequent results of chemical investigation misleading. In the methods which follow the actual processes employed have been given, which in some instances transgress the general principle stated above, but in all cases standard and approved methods are given in detail, even if some of their provisions seem unnecessary or imperfect.

291. Order of Examination.—First of all in a chemical study of the soil should be determined, its reaction (with litmus), its water-holding power in the air-dried state (hygroscopicity), its content of combined water (giving hydrous silicates of alumina), its organic matter (humus and organic nitrogen), its content of carbon dioxide (carbonates of the alkaline earths), and the part of it soluble in acids. A determination of these values gives the analyst a general view of the type of soil with which he is engaged, and leads him to adopt such a method of more extended analysis as the circumstances of the case may demand.

For this reason those operations which relate to the above determinations are placed first in the processes to be performed, while the estimation of the more particular ingredients of the soil is left for subsequent elaboration.

Next follows a description of the standard methods of estimating the more important elements passing into solution on treatment of a soil sample with an acid. The method of treating the insoluble residue, and the detection and estimation of rare or unimportant soil constituents, closes the analytical study of the soils.

With respect to the determination of nitrogen as nitric or nitrous acid in the soil and drainage waters, it has been thought proper to collect all standard methods relating particularly thereto into one group, and they will appear separate from the methods under nitrogen analysis in fertilizers.

The question of the utility of chemical soil analyses is one which has been the subject of vigorous discussion, a discussion which finds no proper place in a work of this character. Unless, however, intelligent soil analysis be productive of some good it would be a thankless task to collect and arrange the details of the processes employed. An accurate determination of the constituents of a soil may not enable the chemist to recommend a proper course of treatment, but it will help in many ways to develop a rational soil

diagnosis which will permit the physician in charge of the case, who last of all is the farmer, to follow a rational treatment which in the end will be productive of good.

The analyst will find in the methods given all that are approved by bodies of official or affiliated chemists, or by individual experience, and among them some method may be found which, it is hoped, will be suited, in the light of our present knowledge, to each case which may arise.

292. Reaction of the Soil.—In soils rich in decaying vegetable matter the excess of acid is often great enough to produce a distinct acid reaction.

On the contrary, in arid regions the accumulation of salts near the surface may produce the opposite effect.

The reaction of the soil may be determined with a large number of indicators among which, for convenience, sensitive litmus paper, both red and blue, stand in the front rank. A sample of the soil, from fifteen to thirty grams, is mixed with water to a paste and allowed to settle. The litmus paper is then dipped into the supernatant liquid.

293. Determination of Water in Soil.—The following problems are presented:

(a) *The Determination of Water in Fresh Samples taken in situ.*—The content of water in this case varies with the date and amount of rain-fall, the capacity of the soil for holding water, the temperature and degree of saturation of the atmosphere, and many other conditions, all of which should be noted at the time the samples are taken.

(b) *The Determination of Water in Air-Dried Samples.*—In this case the soil is allowed to remain in thin layers, and exposed to the air until it ceases to lose weight. The quantity of water left is dependent on the capacity of the soil to hold hygroscopic water and to the temperature and degree of saturation of the air.

(c) *The Determination of the Total Water by Ignition.*—This process not only gives the free and hygroscopic moisture, but also combined water present in the hydrous silicates and otherwise. The estimation is complicated by the presence of carbonates and organic matter.

294. Determination of Water in Fresh Samples.—This determination requires that the sample, when taken in the field, should be so secured as to be weighed before any loss of moisture can take place. For this purpose it

can be sealed up in tubes or bottles and preserved for examination in the laboratory.

According to Whitney, the relations of soils to moisture and heat are such prominent factors in the distribution and development of agricultural crops, that the determination of the actual moisture content of soils in the fields should be considered a necessary part of the meteorological observations, and of far more importance, indeed, or having far more meaning to the agriculturist than the simple record of the rain-fall.

In order to determine the relation of the soil to moisture, uninfluenced by the varying conditions of cultivation and of the different size of crop, he recommends that a small plot of ground be reserved at each station, adjacent to the soil thermometers, where the samples may be taken for the moisture determinations. No crops should be allowed to grow on this area and the soil is not to be disturbed, except that weeds and grass are carefully removed by hand when necessary. Samples of the soil should be taken every morning at 8 o'clock, by correspondents in the principal soil formations from the different parts of the area under observation, and sent by mail to the laboratory.

The samples should be taken as described in paragraph 65. The locality and date are written on a label attached to the tube. The tube contains about sixty or seventy grams of soil, and the moisture determination is made on this in the laboratory in the usual way.

It would be desirable to have this sample represent a depth of from six to nine inches, thus rejecting the surface three inches which are more liable to sudden and accidental changes. These tubes are very inexpensive, and a sufficient number should be purchased to keep each station supplied. The sample represents a definite depth, and it does not have to be subsampled or even transferred in the field. This record of the moisture of the soil will show the amount of moisture which the different soils can maintain at the disposal of the plants, which, together with the temperature of the soil, is believed to be a most important factor in crop distribution and development.

295. Method of Berthelot and André.—The estimation of the water according to Berthelot and André^[189] should be made under three forms; viz.,

1. Water eliminated spontaneously at ordinary temperatures.
2. Water eliminated by drying to constant weight at 110°.

3. Water eliminated at a red heat.

The water may be determined directly on a sample weighed at the time of taking and afterwards dried in the open air, and finally, if necessary, in a desiccator. For a general idea the desiccation should be made on a sample of 100 grams, for exact work on ten grams. The dish in which the drying takes place should be shallow, and during the time the sample should be frequently stirred and thoroughly pulverized with a spatula which is weighed with the dish. The drying in the air should continue several days. The data obtained are not fixed since they depend on the temperature and the degree of saturation of the air with aqueous vapor. The variations due to these causes, however, are not very wide. The process may be regarded as practically finished when successive weights sensibly constant are obtained. In this state the soils contain very little water eliminable at 110°.

296. Estimation of Water Remaining after Air-Drying.—The sifted sample is placed in quantities of five or ten grams in a flat-bottomed dish and dried at 110° to constant weight. This treatment not only removes the moisture, but all matters volatile at that temperature.

Petermann,^[190] in the Agricultural Station, at Gembloux, practices drying the sample to constant weight at 150°.

It is further recommended by Petermann to determine total volatile and combustible matters by igniting to incipient redness, allowing to cool, moistening with distilled water, and drying at 150°.

The German experiment stations^[3] estimate hygroscopic moisture for analytical calculations by drying to constant weight at 100°. In determining loss on ignition, however, the preliminary drying is made at 140°, with the exception of peaty samples where so high a temperature is not admissible.

The Official Agricultural Chemists^[191] place five grams of air-dried soil in a flat-bottomed and tared platinum dish; heat in an air-bath to 110° for eight hours; cool in a desiccator, and weigh; repeat the heating, cooling, and weighing, at intervals of an hour till constant weight is found, and estimate the hygroscopic moisture by the loss of weight. Weigh rapidly to avoid absorption of moisture from the air.

In the German laboratories, according to König,^[192] from ten to twenty grams of the fine earth, properly prepared by air-drying and sifting, for analysis, are heated at 100° to constant weight. For control, five grams are placed in a desiccator over sulfuric acid for two or three days.

Wolff directs that a small portion of the well-mixed earth, for example, twenty grams, be spread out on a flat zinc plate, and its changes in weight observed through several days. These observations are continued until the variations are so slight that the means can be determined with sufficient exactness from the last weighings. The soil is then dried at 125° in a hot air-chamber. The loss in weight will give the mean hygroscopic moisture in the soil under the conditions in which the experiment is made.

297. Drying in a Desiccator.—The sample dried as indicated previously by the method of Berthelot and André is placed in a desiccator over sulfuric acid. It is better to have the sample traversed by a current of perfectly dry air, and in this case it should be placed in a tube, which is closed while weighing, to prevent absorption of moisture. Much time is also required for this operation, and it does not possess the practical value of the method of drying in the free air.

298. Water Set Free at 110° .—This is determined by Berthelot and André on a weight of five to ten grams of soil. The sample which has been employed for the preceding determination may be used. While this is going on in an air-bath heated to 110° , about ten times as much soil should be dried for the same time at the same temperature, and this should be preserved in a well-stoppered flask. All subsequent determinations are to be made with the soil dried at 110° .

The loss of weight in a soil increases with the temperature to which it is exposed. The apparent quantity of water, therefore, determined at 140° or 180° is always greater than that obtained at 110° . But when the temperature exceeds 110° there is danger of decomposing organic bodies with the loss of a part of their constituent elements. Carbon dioxid and ammonia may also be lost, as well as acetic acid and other volatile bodies.

299. Loss on Ignition.—The loss on ignition represents any hygroscopic moisture not removed by previous drying, all water in combination with mineral matters as water of constitution, all organic acids and ammoniacal compounds, all organic matter when the ignition is continued until the carbon is burned away, all or nearly all of the carbon dioxid present in carbonates, and, finally, some of the chlorids of the alkalis, if the temperature have been carried too high or been continued too long.

The loss of carbon dioxid in carbonates may be mostly restored by moistening the ignited mass two or three times with ammonium carbonate,

followed by gentle ignition for a few minutes to incipient redness, to remove excess of the reagent. The apportionment of the rest of the loss justly among the remaining volatile constituents of the original sample is a matter of some difficulty but may be approximately effected by the methods to be submitted.

300. Determination of Loss on Ignition.—*Method of the Official Agricultural Chemists.* The platinum crucible and five grams of soil used to determine the hygroscopic moisture may be employed to determine the volatile matter. Heat the crucible and dry soil to low redness. The heating should be prolonged till all organic material is burned away, but below the temperature at which alkaline chlorids volatilize. Moisten the cold mass with a few drops of a saturated solution of ammonium carbonate, dry, and heat to 150° to expel excess of ammonia. The loss in weight of the sample represents organic matter, water of combination, salts of ammonia, etc.

According to Knop^[193] the total loss on ignition is determined as follows: About two grams of the fine earth are carefully ignited until all organic matter is consumed. The sample is then mixed with an equal volume of finely powdered, pure oxalic acid, and again heated until all the oxalic acid is decomposed. After cooling, the sample is weighed, again mixed with oxalic acid, ignited, cooled, weighed, and the process continued until the weight is constant.

The method recommended by König consists in igniting about ten grams of the fine earth at the lowest possible temperature until all the humus is destroyed. Thereafter the sample is repeatedly moistened with a solution of ammonium carbonate and ignited after drying at 100° , until constant weight is obtained. In soils rich in carbonates some carbon dioxid may be lost by the above process. For this a proper correction can be made by estimating the carbon dioxid in the sample, both before and after the execution of the above described process.

The method described by Frühling as much used in the German laboratories, consists in igniting ten grams of the fine earth, previously dried at 140° in a crucible placed obliquely on its support and with the cover so adjusted over its mouth as to give a draft within the body of the crucible. The ignition, at a gentle heat is continued until on stirring with a platinum wire no evidence of unconsumed carbon is found. The moistening with solution of ammonium carbonate, should not take place until the

contents of the crucible are cool. Subsequent ignition, at a low heat for a short time, will remove the excess of ammonium salt.

301. Method of Berthelot and André.^[194]—The earth dried at 110° contains still a greater or less quantity of combined water. This is the water united with alumina, silica and certain salts, but not the water of constitution belonging to organic bodies. The exact estimation of this water offers many difficulties. The determination of loss obtained at a red heat embraces:

(1) The water combined with zeolitic silicates, with alumina and with organic compounds.

(2) The water produced by the combustion of the organic compounds.

(3) The carbon dioxid resulting from the partial decomposition of the calcium and magnesium carbonates.

(4) The carbon burned and the nitrogen lost during ignition. The measure of the loss of weight in an earth heated to redness in contact with the air is not therefore, an exact process of estimating water or even volatile matters.

A better defined result is obtained in carefully burning a known weight of earth either in a current of free oxygen, or with lead chromate. The water produced in such a combustion is secured in a tube filled with pumice stone saturated with sulfuric acid, the carbon dioxid being absorbed afterwards in potash bulbs and by solid potash. The weight of earth burned is chosen so as to furnish a convenient weight of both water and carbon dioxid. In general about five grams are sufficient. When the combustion is made with oxygen, the soil is contained in a boat and the products of the combustion are carried over a long column of copper oxid heated to redness. The residue left in the boat is weighed at the end of the operation, and in this residue it is advisable to determine any undecomposed carbonate. Should the sample burn badly and be mixed with carbonaceous matter at the end of the operation, it will be necessary to substitute the lead chromate method. In this case, of course, the residue left after combustion is not weighed. Whichever method is employed gives a quantity of water originally combined with the soil, plus the quantity arising from the combustion of the hydrogen of the organic matter. The details of the processes for organic combustion, will be given in a subsequent part of this manual. It is not possible to divide the water between these two sources

directly, but this can be done by calculation, which gives results lying within the limits of probability. The method follows:

The organic nitrogen, determined separately, by soda-lime, the method of Kjeldahl, or volumetrically, is derived from proteid principles resembling albuminoids containing about one-sixteenth of their weight of nitrogen. The nitrates contained in the earth are in such feeble proportion, as to be negligible in this calculation. The total weight of these nitrogenous principles in the soil is therefore easily calculated. The carbon contained in the proteids is then calculated on a basis of 53 per cent of their total weight, and the hydrogen on a basis of 7.2 per cent. From the weight of the total organic carbon (determined as described further on) is subtracted the carbon present in the proteids. The remainder corresponds to the organic carbon present as carbohydrates, (ligneous principles) containing 44.4 per cent carbon and 6.2 per cent hydrogen. By adding together the weight of the hydrogen contained in the ligneous principles, and the hydrogen contained in the proteids, and multiplying the sum by 9, the weight of water formed by the combustion of all the organic matter in the sample is obtained. This is subtracted from the weight of the total water obtained by direct determination as described above. The difference represents the weight of water combined with the silicates, etc., as well as with organic matters.

302. Method of Von Bemmélén.^[195]—According to the view of Von Bemmélén, the soil contains colloidal humus and colloidal silicate, which complicate the determination of water. The colloids retain water in varying quantities, depending upon the following conditions:

- (1) Upon their composition and state of molecular equilibrium.
- (2) Upon the pressure of the aqueous vapor of the room.
- (3) Upon the temperature.

At each degree of temperature, the quantity of absorbed water which a colloid can retain in a room saturated with aqueous vapor, is different. The quantity of water which air-dried earth gives off at 100°, has therefore, no special significance unless all conditions are known.

In addition to the estimation of the quantity of water which soils, in their natural condition, are capable of taking up and holding, at ordinary temperatures, the estimation of the quantity of water which they can take up in different temperatures in rooms saturated with aqueous vapor should be of interest. It follows, therefore, that there is no special value in data

obtained by drying earth at 100° or 110°. For the purpose of comparison, he prefers to select that point at which the soil is dried over sulfuric acid, the point at which the tension of the water vapor in the earth, at a temperature of plus or minus 15°, approaches zero. The water which still remains in the earth under these conditions is characterized as firmly combined water.

Von Bemmelen truly observes that only in soils which contain no carbonates and no chlorids and sulfids, can the loss on ignition be regarded as the sum of the humus and water content. By moistening with ammonium carbonate, the correction for lime or carbon dioxide cannot be correctly made as has been the custom up to the present time. In the first place, ignited magnesia, when it has lost its carbon dioxide, does not take this up completely on moistening with ammonium carbonate; in the second place, reactions with the chlorids may take place; and in the third place, the lime which is in the humus will be converted into calcium carbonate. Chlorids on ignition may be volatilized or oxidized. The sulfuric acid formed from the sulfids, on ignition, can expel carbon dioxide; further than this the iron of pyrites takes up oxygen on ignition. All these influences make the numbers obtained from loss on ignition extremely variable.

With sea soils, Von Bemmelen has weighed the soil after the elementary analysis and estimated, in addition to the carbon dioxide, both chlorine and sulfuric acid therein. The comparison of these estimations with those of CO_2 , Cl, SO_3 and S, made in the original soil, gave the necessary corrections; viz., for the increase in the weight through oxidation of sulfur and iron, and for the decrease in weight through the volatilization of sodium chlorid, sulfur, and carbon dioxide. A trace of chlorine was evolved as ferric chlorid, nevertheless, the molecular weight of sodium chlorid, 58.5, is scarcely different from the equivalent quantity of ferric chlorid 54.1. For this reason the estimation of loss of water, on ignition, of sea soils is less exact than that of soils which are free from carbonates and sulfids and which, as is usually the case with tillable soils, contain only small quantities of chlorids and sulfates.

The Strongly Combined Water.—Water which, at a temperature of plus or minus 15°, in a dry room, still remains in the soil, is chiefly combined according to Von Bemmelen with the colloidal bodies therein. Its estimation, presents, naturally, difficulties and is not capable of any great exactness. The quantity of strongly combined water, on the one hand is

determined from the difference between the loss on ignition and the quantity of humus present, calculated from the content of carbon; on the other hand, from the difference between the water obtained by elementary analysis and the water which corresponds to the calculated quantity of humus. If the hydrogen content of humus is correctly taken and no appreciable error is introduced through the factor 1.724, both of these differences must agree. On the other hand the hydrogen content of the humus can be computed from the difference between the water found and the calculated content of the firmly combined water.

The hydrogen content of humus bodies, dried at 100°, varies between four and five per cent. Eggertz has found the content from 4.3 to 6.6 per cent of hydrogen in thirteen soils which he first treated with dilute hydrochloric acid then extracted with ammonia or potash lye and precipitated this alkaline extract with acid. The method of applying these principles to soil analysis is indicated in the following scheme:

A volcanic earth from Deli gave, on elementary analysis:

	Per cent.
Carbon	2.94
Water	14.78
Nitrogen	0.28
Loss on ignition	17.54

FIRST CALCULATION.

	Per cent.
Loss on ignition	17.54
Humus = carbon, $2.94 \times 1.724 =$	5.07
Difference = firmly combined water	12.47

Assuming that a humus dried over sulfuric acid contains five per cent of hydrogen, the second calculation is made as follows.

SECOND CALCULATION.

5.00 humus $\times$ 5 per cent = 0.25 per cent of hydrogen in humus corresponding to 2.28 per cent of water.

	Per cent.
Water found	14.79
Difference = firmly combined water	12.51

THIRD CALCULATION.

	Per cent.
Firmly combined water	12.47
Water from the hydrogen in humus	2.28
Total water	14.75
Found	14.79

In this way, in three other volcanic earths and in an ordinary alluvial clay from Rembang, there were found by analysis and by calculation the following percentages of water:

		1.	2.	3.	4.	5.
Percentage of water	calculated	14.75	7.74	8.06	4.90	6.01
„ „ „	found	14.97	7.63	8.05	4.70	6.00

On the contrary, when the calculation is made from sea-slime taken from under the water a higher content of hydrogen must be assumed; viz., about six per cent. In two samples of sea-slime calculated in this way the following numbers were obtained:

Percentage of water	calculated	8.61	3.71
„ „ „	found	8.53	3.57

It is, therefore, quite evident that the organic compounds of soil taken from under the sea-water are richer in hydrogen than those exposed to the air or in cultivation.

303. General Conclusions.—In the foregoing paragraphs have been collected the most widely practiced methods of determining moisture in soil

in both a free and combined state. The following conclusions may serve to guide the analyst who endeavors to determine the water in any or all of its conditions:

(1) In determining water in fresh samples the method of Whitney is satisfactory. Although the samples taken by this method are small they may be easily secured in great numbers over widely scattered areas, and can be easily transported without change. These samples should be dried at 100° to 110° for rapid work, or where time can be spared may be air-dried.

(2) For a simple determination of the water left in the soil after air-drying (hygroscopic water) the method of the Association of Official Agricultural Chemists may be followed. There is much difference of opinion in respect of the proper temperature at which this moisture is to be determined. Much here depends on the nature of the soil. An almost purely mineral soil may safely be dried at 140° or 150°. A peaty soil, on the contrary, should not be exposed to a temperature above 100°. For general purposes the temperature chosen by the official chemists is to be recommended.

(3) Water of composition can only be determined by ignition. As has been fully shown, this process not only eliminates the water, but also destroys organic matter, decomposes carbonates and sulfids, and, to some extent, chlorids. Subsequent repeated treatment with ammonium carbonate may restore the loss due to carbon dioxide, but in many cases not entirely. The water which comes from organic matter may be approximately calculated from the humus content of the sample, but as will be seen further on the methods of estimating humus itself are only approximate. Nevertheless, in distributing the losses on ignition properly to the several compounds of the soil there is no better method now known than that of taking into consideration the humus content and carbonates present. The principles of procedure established by Berthelot and André, and Von Bemmelen, are to be applied in all such cases, modified as circumstances may arise according to the judgment of the analyst.

304. Estimation of the Organic Matter of the Soil.—The organic matter in the soil may be divided into two classes. First, the undecayed roots and other remains of plant and animal life, and the living organisms existing in the soil. The study of the organisms which are active in the condition of plant growth will be the subject of a special chapter. Second, the decayed or partially decayed remnants of organic matter in the soil

known as humus. Such matter may be present in only minute traces, as in barren sand soils, or it may form the great mass of the soil under examination, as in the case of peat, muck, and vegetable mold. It is with the investigation of the second class of matter that the analyst has chiefly to do at present. The problems which are to be elucidated by the analytical study of such bodies are the following: (1) The total quantity of such matter in the soil. (2) The determination of the organic carbon and hydrogen therein. (3) The determination of total nitrogen. (4) The determination of the availability of the nitrogen for plant growth. (5) The estimation of the humic bodies (humus, humic acid, ulmic acid, etc.).

The importance of humus in the promotion of plant growth is sufficient excuse for the somewhat extended study of the principles which underlie the analytical methods, and the methods themselves, which follow.

305. Total Quantity of Organic Matter.—The total approximate quantity of organic matter in the soil can be determined by simple ignition, in the manner noted in paragraphs 294 and 295. The proper correction for free and combined water being applied by the further copper oxid or lead chromate combustion of the sample, and for carbonates and volatile chlorids, the approximate total of the organic matter of all kinds is obtained.

306. Estimation of the Organic Carbon.—To estimate the organic carbon in an earth the sample may be burned in a current of oxygen, or after mixing with lead chromate.

In a Current of Oxygen.—When burned in a current of oxygen the sample is held in a boat and the gases arising from the combustion directed over copper oxid at a red heat. The carbon thus disappears as carbon dioxid and is absorbed and weighed in the usual way.

With Lead Chromate.—The lead chromate employed should be previously tested since it often contains other compounds, especially lead acetate and nitrate, furnishing in the one case both carbon dioxid and water, and in the other hyponitric acid.

From two to ten grams of earth are employed, according to its richness in organic matter. The total carbon dioxid is obtained in this process both from carbonates and organic bodies. The water and carbon dioxid are secured and weighed in the usual manner.

The oxygen method should be used in all cases possible. Although it does not always give the whole of the carbon dioxid present as carbonates, the

rest can be easily estimated by treating the residue in the boat with hydrochloric acid, in an apparatus for estimating that gas.

Calculation of Results.—The whole of the carbon dioxid is determined either by direct combustion with lead chromate, or by taking the sum of the amounts by burning in a stream of oxygen and treating the residue in a carbon dioxid apparatus.

The carbon dioxid contained in the original carbonates should be determined by direct treatment of the sample in the usual way.

The carbon in organic compounds is determined by subtracting the carbon present as carbonates from the total.

From the organic carbon contained in the soil the humus is calculated by Wolff on the supposition that it contains fifty-eight per cent of carbon. It is, therefore, only necessary to multiply the percentage of carbon found by 1.724, or the carbon dioxid found by 0.471, to determine the quantity of humus in the dried soil.

307. Details of the Direct Estimation of Carbon in Soils by Various Methods.—(1) *Oxidation by Chromic Acid.*—The method of Wolff by oxidation with chromic acid has been worked out in detail by Warington and Peake.^[196] It consists in treating the soil with sulfuric acid and potassium bichromate, or by preference with a mixture of sulfuric and chromic acids, the carbon dioxid evolved being estimated in the usual way. This method is recommended by Fresenius as an alternative to a combustion of the soil with copper oxid or lead chromate. It is apparently the method which has been most generally employed in agricultural investigations.

Ten grams of the finely powdered soil are placed in a flask of about 250 cubic centimeters capacity, provided with a caoutchouc stopper, through which pass two tubes, one for the supply of liquids, the other for the delivery of gas. The soil is treated with twenty cubic centimeters of water and thirty cubic centimeters of oil of vitriol; and the whole, after being thoroughly mixed, is heated for a short time in a water-bath, the object in view being the decomposition of any carbonates existing in the soil. Air is next drawn through the flask to remove any carbon dioxid which has been evolved. The stopper is next removed, and coarsely powdered potassium bichromate introduced. In the case of a soil containing three per cent of carbon, six grams of bichromate will be found sufficient, a portion

remaining undissolved at the end of the experiment. The stopper is then replaced, its supply-tube closed by a clamp, and the delivery-tube connected with a series of absorbents contained in tubes. The first of these tubes contains solid calcium chlorid; the second, fragments of glass moistened with oil of vitriol; the third and fourth are nearly filled with soda-lime, a little calcium chlorid being placed on the top of the soda-lime at each extremity. The last named tubes are for the absorption of carbon dioxid, and have been previously weighed. The series is closed by a guard-tube containing soda-lime, with calcium chlorid at the two ends.

The flask containing the soil and bichromate is now gradually heated in a water-bath, the contents of the flask being from time to time mixed by agitation. A brisk reaction occurs, carbon dioxid being evolved in proportion as the soil is rich in organic matter. The temperature of the water-bath is slowly raised to boiling as the action becomes weaker, and is maintained at that point till all action ceases. As bubbles of gas are slowly evolved for some time, it has been usual in these experiments to prolong the digestion for four or five hours. When the operation is concluded the source of heat is removed, an aspirator is attached to the guard-tube at the end of the absorbent vessels, and air freed from carbon dioxid is drawn through the flask and through the whole series of tubes. The tubes filled with soda-lime are finally weighed, the increase in weight showing the amount of carbon dioxid produced. The object of the calcium chlorid placed on the surface of the soda-lime is to retain the water which is freely given up when the soda-lime absorbs carbon dioxid. The second tube filled with soda-lime does not gain in weight till the first is nearly saturated; it thus serves to indicate when the first tube requires refilling. The same tubes may be used several times in succession.

No increase in the carbon dioxid evolved is obtained by substituting chromic acid for potassium bichromate.

The organic matter of the soil appears to the eye to be completely destroyed by the digestion with sulfuric acid and potassium bichromate; the residue of soil remaining in the flask when washed with water is perfectly white, or the dark particles, if any, are found to be unaltered by ignition, and therefore to be inorganic in their nature. Under these circumstances considerable confidence has naturally been felt in this method. The complete destruction of the humic matter of the soil does not, however,

necessarily imply that the carbon has been entirely converted into carbon dioxid as has been pointed out by Wanklyn. According to his demonstration of the action of chromic acid on organic matter the oxidation frequently stops short of the production of carbon dioxid. While oxidation with chromic acid apparently leads to a complete reaction when the carbon is in the form of graphite, it would probably yield other products than carbon dioxid when the carbon exists as a carbohydrate. The doubt thus raised as to the correctness of the results yielded by the chromate method makes it desirable to check the work by the use of other methods for the determination of carbon. For this purpose Warington and Peake recommend:

(2) *Oxidation with Potassium Permanganate*.—In the trials with this method ten grams of soil are digested in a closed flask with a measured quantity of solution of caustic potash containing five grams of potash for each twenty cubic centimeters, and crystals of potassium permanganate. Seven grams of the permanganate are found to be sufficient for a soil containing 3.3 per cent of carbon. The flask is heated for half an hour in boiling water, and then for one hour in a salt-bath. The flask during this digestion is connected with a small receiver containing a little potash solution, to preserve an atmosphere free from carbon dioxid; distillation to a limited extent is allowed during the digestion in the salt-bath.

The first part of the operation being completed a rubber stopper, carrying a delivery and supply-tube, is fitted to the flask, which is then connected with the system of tubes already described. Dilute sulfuric acid is then poured down the supply-tube, a water-bath surrounding the flask is brought to boiling, and maintained thus for one hour, after which air, free from carbon dioxid, is drawn through the apparatus, the tubes containing soda-lime being finally disconnected and weighed.

In the first stage of this method the carbon of the organic matter is converted into carbonate, and probably also into potassium oxalate.^[197] In the second stage the oxalate is decomposed by the sulfuric acid and permanganate, and the carbon existing, both as oxalate and carbonate, is evolved as carbon dioxid, and absorbed by the weighed soda-lime tubes. Both F. Schulze and Wanklyn have employed potassium permanganate for the determination of organic carbon, but they have preferred to calculate the amount of carbon from the quantity of permanganate consumed, as,

however, by so doing everything oxidizable by permanganate is reckoned as carbon, it seems better to make a direct determination of the carbon dioxide formed.

From the amount of carbon dioxide found, is to be subtracted that existing as carbonates in the soil, and in the solution of potash used. For this purpose an experiment is made with the same quantities of soil and potash previously employed, but without permanganate, and the carbon dioxide obtained is deducted from that yielded in the experiment with permanganate. If the potash used contains organic matter two blank experiments will be necessary, one with potash and permanganate, and one with soil alone.

A further difficulty arises from the presence of chlorides in the materials, which occasions an evolution of free chlorine when the permanganate solution is heated with sulfuric acid. This error occurs also with the chromic acid method, but in that case the quantity of chloride is merely that contained in the soil, which is usually very small; in the permanganate method we have also the chloride present in the caustic potash, and this is often considerable. Corrections for chlorine by blank experiments are unsatisfactory, the amount of chlorine which reaches the soda-lime tubes depending in part on the degree to which the calcium chloride tube has become saturated with chlorine. It is better therefore to remove the chlorine in every experiment by the plan which Perkin has suggested, by inserting a tube containing silver foil, maintained at a low red heat, between the flask and the absorbent tubes.

The amount of carbon dioxide yielded by oxidation with potassium permanganate is found to be considerably in excess of that obtained by oxidation with chromic acid; to ascertain whether these higher results really represented the whole of the carbon present in the soil, trials were next made by actual combustion of the soil in oxygen.

(3) *Combustion in Oxygen*.—The most convenient mode of carrying out the combustion of soil is to place the soil in a platinum boat, and ignite it in a current of oxygen in a combustion tube partly filled with cupric oxide. A wide combustion tube is employed, about twenty inches long, and drawn out at one end; the front of the tube is filled for eight inches with coarse cupric oxide, the hind part is left empty to receive the platinum boat. The drawn out end of the combustion tube is connected with a series of

absorbent tubes, quite similar to those employed for the estimation of carbon dioxide in the chromic acid method. Between these absorbent vessels and the combustion tube is placed a three-bulbed Geissler tube filled with oil of vitriol. The oil of vitriol is quite effective in retaining nitrous fumes. The wide end of the combustion tube is connected with a gas-holder of oxygen; the oxygen gas is made to pass through a tube of soda-lime before entering the combustion tube, to remove any possible contamination of carbon dioxide.

In starting a combustion the part of the combustion tube containing the cupric oxide is brought to a red heat, and oxygen is passed for some time through the apparatus. Ten grams of soil, previously dried, are placed in a large platinum boat, which is next introduced at the wide end of the combustion tube. The combustion is conducted in the usual manner, a current of oxygen being maintained throughout the whole operation. It is very useful to terminate the whole series of absorbent vessels with a glass tube dipping into water; the rate at which the gas is seen to bubble, serves as a guide to the supply of oxygen from the gas-holder, the consumption of oxygen varying, of course, with different soils, and at different stages of the combustion. At the close of the combustion, oxygen, or air freed from carbon dioxide, is passed for some time through the apparatus to drive all carbon dioxide into the absorbent vessels. One experiment can be followed by another as soon as the hind part of the combustion tube has cooled sufficiently to admit a second platinum boat. The same combustion tube can be employed for several days, if packed in the usual manner in asbestos.

The presence of carbonates in the soil occasions some difficulty in working the combustion method, as a part of this carbon dioxide will, of course, be given up on ignition, and be reckoned as carbon. The simplest mode of meeting this difficulty is to expel the carbon dioxide belonging to the carbonates before the combustion commences. The method of Manning; namely, treatment with a strong solution of sulfurous acid, may be employed for this purpose. The ten grams of soil taken for combustion are placed in a flat-bottomed basin, covered with a thin layer of sulfurous acid, and frequently stirred. After a time the action is assisted by a gentle heat. When the carbonates have been completely decomposed the contents of the basin are evaporated to dryness on a water-bath; the dry mass is then pulverized, and removed to the platinum boat for combustion in oxygen.

For the action of the sulfurous acid to be complete it is essential that the carbonates should be in very fine powder, since even chalk is but imperfectly attacked when present in coarse particles.

308. Comparison of Methods.—A considerable number of soils analyzed by the chromic acid method and by the combustion, method, by Warington and Peake, with the assistance of Cathcart, shows the following comparisons:

PERCENTAGE OF CARBON FOUND BY TWO METHODS IN SOILS DRIED AT 100°.

No.	Kind of soil.	Chromic acid method.			Combustion method.			Per cent. yielded by chromic acid.
		Exp. 1.	Exp. 2.	Mean.	Exp. 1.	Exp. 2.	Mean.	
1.	Old pasture	2.85	2.79	2.82	3.58	3.55	3.57	79.0
2.	„ „	2.83	2.79	2.81	3.57	3.53	3.55	79.1
3.	„ „	2.76	2.76	2.76	3.46	3.46	3.46	79.7
4.	„ „	2.74	2.76	2.75	3.37	3.38	3.38	81.4
5.	„ „	2.64	2.54	2.59	3.31	3.36	3.34	77.5
6.	„ „	2.51	2.43	2.47	3.15	3.15	3.15	78.4
7.	„ „	2.40	2.44	2.42	3.09	3.13	3.11	77.8
8.	New pasture	1.92	1.93	1.93	2.41	2.40	2.41	80.1
9.	„ „	1.66	1.81	1.74	2.39	2.43	2.41	72.2
10.	Arable soil	1.78	1.78	1.78	2.14	2.13	2.14	83.2
11.	„ „	1.21	1.14	1.18	1.40	1.43	1.42	83.1
12.	Subsoil	0.28	0.27	0.28	0.37	0.38	0.38	73.7

Of the above soils the arable soils, Nos. 10 and 11, were the only ones containing carbonates in any quantity exceeding a minute trace. The two soils in question were treated with sulfurous acid before combustion, the others not.

All the determinations by the chromic acid method were made by Mr. P. H. Cathcart, with the exception of Nos. 9 and 12, which were executed by another experimenter, and are seen to give distinctly lower results. Excluding these two analyses the relation of the carbon found by the two methods is tolerably constant, the average being 79.9 of carbon found by oxidation with chromic acid for 100 yielded by combustion in oxygen. The

results obtained by the chromic acid method thus appear to be very considerably below the truth.

Four typical soils were analyzed by the permanganate, as well as by the chromic acid and combustion methods. The results obtained were as follows:

PERCENTAGE OF CARBON FOUND BY THREE METHODS IN SOILS DRIED AT 100°.

Kind of soil.	Combustion method. Mean. Per cent.	Chromic acid method. Mean. Per cent.	Permanganate method.			Yielded by permanganate if carbon by combustion = 100. Per cent.
			Exp. 1. Per cent.	Exp. 2. Per cent.	Mean. Per cent.	
Old pasture	3.55	2.81	3.26	3.30	3.28	92.4
New pasture	2.41	1.93	2.29	2.30	2.30	95.4
Arable soil	1.42	1.18	1.28	1.33	1.31	92.3
Subsoil	0.38	0.28	0.34	0.34	0.34	89.5

Oxidation by permanganate thus gives a much higher result than oxidation with chromic acid; but even the permanganate fails to convert the whole of the carbon into carbon dioxide, the product with permanganate being on an average of the four soils 92.4 per cent of that yielded by combustion in oxygen.

Wanklyn states that a temperature of 160°–180° is necessary in some cases to effect complete oxidation with permanganate and caustic potash. Such a temperature is found impracticable when dealing with soil, from the action of the potash on the silicates present; hence possibly the low results obtained.

Combustion in oxygen appears from these experiments to be the most satisfactory method for determining carbon in soil, nor is this method, on the whole, longer or more troublesome than the other methods investigated.

Warington and Peake have further determined the loss on ignition of the four soils mentioned above, with the view of comparing this loss with the amount of organic matter calculated from the carbon actually present. In making this calculation they have taken as the amount of carbon in the soil,

that found by combustion in oxygen, and have assumed with Schulze, Wolff, and Fresenius, that fifty-eight per cent of carbon will be present in the organic matter of soils. The four soils were heated successively at 100°, 120°, and 150°, till they ceased to lose weight; the loss on ignition in each of these stages of dryness is shown in the following table:

PERCENTAGE LOSS ON IGNITION COMPARED WITH ORGANIC MATTER CALCULATED FROM CARBON.

Kind of soil.	Between 100° and ignition.	Between 120° and ignition.	Between 150° and ignition.	Organic matter at fifty-eight per cent carbon.
	Per cent.	Per cent.	Per cent.	Per cent.
Old pasture	9.27	9.06	8.50	6.12
New pasture	7.07	6.88	6.55	4.16
Arable soil	5.95	5.70	5.61	2.44
Clay subsoil	5.82	5.39	4.76	0.65

The loss on ignition is seen to be in all cases very considerably in excess of the organic matter calculated from the carbon, even when the soil has been dried at as high a temperature as 150°. The error of the ignition method is least in soils rich in organic matter, as, for instance, the old pasture soil in the above table. The error reaches its maximum in the case of the clay subsoil, which contains very little carbonaceous matter, but is naturally rich in hydrated silicates, which part with their water only at a very high temperature.

The above methods of Warington and Peake have been given in detail, and in almost the verbiage of the authors for the reason that the working directions are clearly set forth, and may serve, therefore, as guides to the previous methods where only general indications of manipulation have been given.

309. Estimation of Organic Hydrogen.—The estimation of the total hydrogen is made without difficulty either by burning the sample in a current of oxygen or with lead chromate, and weighing the water produced. This water comes from two sources, the pre-existing water and organic

hydrogen. There is no direct method of distinguishing one from the other. They may, however, be estimated indirectly. The method of calculating the organic hydrogen has already been given (paragraph 299). Experience shows that the hydrogen thus calculated is a little greater than is necessary to form water with the whole of the oxygen found in the organic matters.

310. Estimation of Organic Oxygen.—The determination of this oxygen cannot be made directly. It is obtained by calculation, according to Berthelot and André,^[198] from the oxygen in the proteid and ligneous matters.

Let p represent the weight of the proteid bodies in a sample of soil.

$$\text{Then } O = \frac{p \times 33.5}{100}$$

Let p' = weight of ligneous bodies.

$$\text{Then } O' = \frac{p' \times 49.4}{100}$$

The total oxygen = $O + O'$.

An approximate result is thus obtained, very useful to have when account is taken of the oxidizing processes which go on in the soil during agricultural operations.

311. Estimation of Humus (Matière Noire).—The original method of determining this substance is due to Grandeau.^[199] It is carried on as follows: Ten grams of the fine earth are mixed with coarse sand previously washed with acids and ignited. The mixture is placed in a small funnel, the bottom of which is filled with fragments of glass or porcelain. The mass is moistened with ammonia diluted with an equal volume of distilled water, and allowed to digest for three or four hours. The ammonia dissolves the dark matter without attacking the silica. The ammoniacal solution is displaced by treating the mass with pure water, or water to which some ammonia has been added, and the whole of the dark matter is thus obtained in a volume of twenty to fifty cubic centimeters of filtrate. The filtrate is evaporated to dryness in a weighed platinum dish, and the weight of residue is determined and the percentage of *matière noire* calculated therefrom. The residue is incinerated, and when in sufficient quantity the phosphoric acid is determined in the ash. In soils poor in humus a larger quantity than ten grams may be taken. If the soil be previously treated with hydrochloric acid, Grandeau recommends that the phosphoric acid be determined always in the ash of the dark matter.

The method has undergone various modifications and, as given by Hilgard, is now practiced as follows:

About ten grams of soil are weighed into a prepared filter. The soil should be covered with a piece of paper (a filter) so as to prevent it from packing when solvents are poured on it. It is now treated with hydrochloric acid from five-tenths per cent to one per cent strong (twenty-five and one-third cubic centimeters of strong acid and 808 cubic centimeters of water) to dissolve the lime and magnesia which prevent the humus from dissolving in the ammonia. Treat with the acid until there is no reaction for lime; then wash out the acid with water to neutral reaction. Dissolve the humus with weak ammonia water, prepared by diluting common saturated ammonia water (178 cubic centimeters of ammonia to 422 cubic centimeters of water). Evaporate the humus solution to dryness in a weighed platinum dish at 100°; weigh, then ignite; the loss of weight gives the weight of humus.

The residue from ignition is carbonated with carbon dioxide, heated and weighed, thus giving the ash. It is then moistened with nitric acid and evaporated to dryness. The residue is treated with nitric acid and water, allowed to stand a few hours, and the solution filtered from the insoluble residue, which is ignited and weighed, giving the silica.

The soluble phosphoric acid is determined in the solution by the usual method, as magnesium pyrophosphate. It usually amounts to a fraction, varying from one-half to as little as one-tenth of the total in the soil. While the phosphoric acid so determined is manifestly more soluble and more available to vegetation than the rest of that found by extraction with stronger acid, it is clearly not as available as that which, when introduced in the form of superphosphates, exerts such striking effects even though forming a much smaller percentage of the whole soil. Nevertheless, very striking agreement with actual practice is often found in making this determination.

The estimation of humus by combustion, in any form, of the total organic matter in the soil, gives results varying according to the season, and having no direct relation to the active humus of the soil. The same objection lies against extraction with strong caustic lye.

312. Modification of Grandeau's Method for Determining Humus in Soils.—According to Huston and McBride^[200] the function of the vegetable matter in the soil has long been a matter of contention among those

interested in the science of agriculture. Two factors have contributed to the uncertainty existing in this matter: First, the very complex and varying nature of the compounds resulting from the decomposition of vegetable matter in the soils; and second, the lack of uniformity in the methods of determining either the total amount of organic matter present in a soil, or the amount that has been so far decomposed as to be of any immediate agricultural value. Prominent among these methods are the methods in which a combustion is resorted to, the substance being either burned in air or in a combustion tube with some agent supplying oxygen. The loss on ignition is no measure of the amount of organic matter present since it is practically impossible to remove all the water from the soil previous to ignition, and neither of the methods gives information regarding the extent of the decomposition of the organic matter. Pure cellulose and the black matter of a fertile soil are of very different agricultural value.

Determinations of carbon in soils by oxidation with chromic and sulfuric acid, and with alkaline permanganate have been used. The method with alkaline permanganate agrees fairly well with combustion with copper oxid or lead chromate, but the chromic sulfuric acid method gives only about eighty per cent of the carbon found by combustion processes. However valuable these processes may be for determining the total carbon in the soil, they furnish no information regarding the condition of the carbonaceous soil constituents, and as the determination is really one of carbon, the organic matter must be calculated by using an arbitrary factor. Generally the organic matter of the soil is considered to have fifty-eight per cent carbon; yet different values are given from forty to seventy-two per cent.

There is a general opinion that the black or dark brown material of the soil, resulting from the decay of vegetable matter, has a much higher agricultural value than the undecomposed vegetable matter. No very sharp dividing line can be drawn, for changes in the soil are continually going on, and material may be found in almost every stage between pure cellulose and carbon dioxid. The character of the intermediate products will vary according to the conditions of tillage and the supply of air and water.

For agricultural purposes some means of determining the amount of decomposed matter is very desirable. Several solvents have been tried for this purpose. The earlier attempts were made by treating the soil with successive quantities of boiling half-saturated solution of sodium carbonate

until the soil appeared to yield no more coloring matter to the solvent. The solutions were then united, rendered acid with HCl, which precipitated the humic acid, which was then washed, dried, and weighed. This was considered the more soluble portion of the humic acid. The soil was afterward treated with caustic potash solution in the same manner, and the humus thus extracted was called insoluble humus. This last process was really more in the nature of manufacturing humus, for sawdust treated with caustic potash yields humic acid, and the inert organic matter in the soil was decomposed to some extent by the caustic alkali. Neither of the processes provided for the separation of the humic acid from the lime, magnesia, alumina, and iron with which it is usually combined in the soil.

In case results of different workers are to be compared, it is of the greatest importance that methods should be used that are of such a nature that errors resulting from difference of manipulation, and from difficulty of reproducing duplicate work can be reduced to a minimum.

Hence, a simple modification of the Grandeau method has been tried which has the advantage of keeping a definite amount of the soil in contact with a definite volume of ammonia for a fixed time, the strength of the ammonia remaining constant.

The process is as follows: The soil is washed with acid and water as usual. It is then washed into a 500 cubic centimeter cylinder with ammonia, the cylinder closed and well shaken and allowed to remain for a definite time, usually thirty-six hours. The material is shaken at regular intervals. The cylinder is left inclined as much as possible without having the fluid touch the glass stopper, thus allowing the soil to settle on the side of the cylinder and exposing a very large surface to the action of the ammonia. During the last twelve hours the cylinder is placed in a vertical position to allow the soil to settle well before taking out the aliquot part of the solution.

The process of washing the soil with hydrochloric acid, water and ammonia, is very tedious when performed in the usual way with the wash-bottle. A simple automatic washing apparatus was devised by which a fixed volume of the washing fluid can be delivered at regular intervals, giving ample time for the thorough draining between each addition of the fluid, and requiring no attention. By this apparatus work can be continued day and night. Instead of washing on the usual form of filter paper in funnels, it is preferable with this apparatus to hold the soils on a disk of filter paper

resting on a perforated porcelain disk in the bottom of the funnel. This removes the necessity of washing out the filter papers, does not permit of the accumulation of humus on the edge of the filter paper when the Grandeau process is used, and insures that all the washing fluids pass through the soil and not around it. This form of apparatus reduces the labor to a minimum and permits many determinations to be carried on at once.

This form of apparatus was only lately devised and has only been used long enough to test it and to show its advantages. The reported results were obtained by the ordinary methods of washing.

In all the work reported, five grams were used, as the soils contained so much humus that this amount gave enough humus for good work in the final weighings. The results obtained so far appear in the following tables:

TABLE I.

COMPARISON OF METHOD OF GRANDEAU WITH HUSTON'S MODIFICATION AND OF INFLUENCE OF STRENGTH OF AMMONIA SOLUTION. TIME OF DIGESTION IN MODIFIED METHOD THIRTY-SIX HOURS.

		Two per cent NH ₃ .		Four per cent NH ₃ .		Seven per cent NH ₃ .		Eight per cent NH ₃ .	
		Grandeau.	Huston.	Grandeau.	Huston.	Grandeau.	Huston.	Grandeau.	Huston.
1.	Peat soil,			16.40	20.06				
	Bogus „			13.98	20.80				
	„ „			17.43					
	Mean			15.94	20.43				
2.	Peat subsoil,			13.98	19.38				
	Bogus „			13.85	20.30				
	Mean			13.92	19.84				
3.	Peat soil,	9.05	15.60	14.71	21.24	19.77	21.70	16.05	21.42
	Good „	10.27	15.88	15.34	20.20	19.85	21.90	15.40	21.80
	Mean	9.61	15.74	15.03	20.72	19.81	21.80	15.73	21.61
4.	Peat subsoil,			16.75	24.34				
	Good „			18.60	23.52				
	Mean			17.68	23.93				
5.	Black soil, A	3.90	6.90	(1.86)	7.42				
	„ „ „			(1.67)	6.98				
	„ „ B	3.88	7.00	4.42					
	„ „ „			4.20					
	Mean	3.99	6.95	(3.05)	7.20				
	„			4.31					
6.	Clay loam,	1.86	4.20	2.40	4.26	2.14	4.02	1.85	4.12
	West side, A				4.28				
	„ „ B	1.76	4.36	2.48	(3.40)	2.13	4.48	1.90	4.40
	„ „ „				(3.10)				
	Mean	1.81	4.28	2.44	(3.76)	2.14	4.25	1.88	4.26
	„				4.27				
7.	Clay loam, A	1.90	4.12	(1.60)	(4.59)				
	Lysimeter soil, B	1.61	4.22	(1.41)	(4.58)				
	„ „ C			1.80	4.12				
	„ „ D			1.95	4.04				
	„ „ E			1.92	3.85				
	„ „ F			1.95	4.08				
	„ „ G			1.90	3.93				
	„ „ H			1.90	3.80				
	Mean	1.76	4.17	(1.80)	(4.12)				
	„			1.90	3.97				

NOTE.—Numbers in parentheses indicate results, generally the earliest ones, which the authors do not consider strictly comparable with the rest of the work. They are given solely for the purpose of exhibiting all the work that has been done to date. When a mean is included in parentheses it

indicates that it is calculated from all the results obtained, including those not considered strictly comparable. Bogus is a name given to a peaty soil which is very sterile.

TABLE II.

INFLUENCE OF TIME OF DIGESTION. FOUR PER CENT OF AMMONIA USED THROUGHOUT. HUSTON'S METHOD.

	Thirty-six hours.	Forty-eight hours.	Sixty-eight hours.	Ninety-eight hours.
Peat Soil,	21.24	22.28	24.04	
Good „	20.20	21.70	23.94	
Mean	20.72	21.99	23.99	
Clay loam,	4.28	4.00		4.40
„	4.26	4.01		4.85
West side	(3.40)			
„ „	(3.05)			
Mean	4.27	4.01		4.63

TABLE III.

INFLUENCE OF TIME OF EXTRACTION. TIME, TEN DAYS. GRANDEAU'S METHOD, FOUR PER CENT AMMONIA. PEAT SOIL.

		A. Per cent.	B. Per cent.	Mean. Per cent.	Remarks.
1st extraction,	750 cc	16.90	18.96	17.93	
2nd „	250	2.80	2.38	2.59	
	„				
3rd „	250	1.77	1.10	1.44	
	„				
4th „	250	1.34	1.30	1.32	Stood over night.
	„				
5th „	250	0.89	0.85	0.87	
	„				
6th „	250	1.41	1.65	1.53	Stood overnight.
	„				
7th „	250	2.10	1.80	1.95	Washed again with HCl for Ca. Trace found. HCl washed out, but trace of chlorids found in ash. Probably HCl absorbed from air as humus showed small quantity of a white volatile solid on evaporation.
	„				
8th „	250	0.67	0.65	0.66	
	„				
9th „	250	0.57	0.50	0.53	
	„				
Total	2750	28.45	29.19	28.82	
	„				

313. Summary of Results.—1. The modified method gives much higher results than the original method of Grandeau.

2. In the Grandeau method marked irregularities follow a change in the strength of the ammonia solution. These differences in results bear no relation to the strength of the solution used. They seem to be errors due to the difficulty of securing uniform and complete washing of the soil by the ammonia solution.

In the modified method the change in the strength of the ammonia solution makes practically no difference in the amount of humus extracted, except in the case of the peat soil where two per cent ammonia failed to extract all the humus. But the results show no considerable increase when the strength is increased to over four per cent.

3. The factor of time has not been fully investigated, but the results so far obtained indicate that the time exerts less influence in the modified than in the Grandeau method.

4. Table III shows that considerable quantities of the peat soil are still passing into solution in the Grandeau method at the end of ten days. With ordinary soils this is not true; but in the case of soil No. 5, a black soil, the solutions were colored at the end of a week. On the peat soil the modified method extracted from ten to fifty per cent more than the Grandeau, and on the ordinary soil from two to three times as much humus.

5. In comparing duplicate results by both methods it is found that with soil No. 3, peat soil, the following differences appear calculated to percentage of the total amount involved in the determination:

	Per cent.	Per cent.	Per cent.	Per cent.
Strength of ammonia	2.	4.	7.	8.
Modified	1.7	5.	1.0	1.8
Grandeau	13.0	4.3	0.5	3.4

Special attention was paid to this point in case of soil No. 7, an ordinary soil; taking all results into consideration the greatest difference in percentage of total amount involved was, by the modified method, nineteen per cent, and by the Grandeau, thirty per cent. In the set of six special determinations made by both methods to test this point and which are strictly comparable with each other, the maximum range was by the modified method 7.8 per cent and by the Grandeau 8.3 per cent of the total amount involved in the determination. From which it appears that the modified method is on the whole capable of yielding rather more concordant results than the Grandeau.

314. Estimation of Free Humic Acids.—This process, due to Müntz^[201] is essentially that of Huston and McBride. Twenty grams of the soil are reduced to a fine powder and saturated with fifty cubic centimeters of

concentrated ammonia and allowed to digest two or three days in a warm place. The volume is then made up to one liter with water, well shaken, and set aside for one day in order to permit the subsidence of the solid matter. At the end of this time 500 cubic centimeters of the supernatant liquor are taken and acidified with hydrochloric acid in order to precipitate the humic bodies. The humus is collected on a filter, dried and weighed. It is then ignited and the weight of ash deducted from the first weight thus giving the actual weight of the humus obtained, free from mineral matter. This process gives the free humic acids. By previous treatment of the sample with hydrochloric acid as in the process of Huston and McBride, the total humus is obtained. The estimation of the free humic acids is of importance in determining the quantity of lime or marl which should be added to acid lands.

315. Humus Method of Von Bemmélén.^[202]—Von Bemmélén obtains the content of humus by the multiplication of the content of carbon in the soil by the factor of Wolff; viz., 1.724. The estimation of carbon, water, and of the loss on ignition is conducted in combustion tubes in a current of oxygen. The nitrogen estimation is carried on according to the method of Dumas.

In soils containing calcium carbonate the carbon content is derived from the carbon dioxid taken up by the potash bulbs during combustion (a); from other carbonates not decomposed on ignition and which are subsequently determined in the residue by treatment with hydrochloric acid in a carbon dioxid apparatus, (b) and the total carbon dioxid derived from the carbonates in the soil (c).

For each estimation from three to five grams of the soil are taken, because with smaller quantities the errors of analysis too strongly influence the results. The carbon is then calculated according to the formula:

$$\text{Carbon} = \frac{3}{11} (a + b - c).$$

The Carbon Dioxid of Carbonates.—It is necessary to expel the carbon dioxid at ordinary temperatures, because on heating to boiling, carbon dioxid would be formed from the humus. In a flask, as small as possible, the soil is treated at ordinary temperature, with dilute sulfuric or citric acid, the escaping gas dried over sulfuric acid and taken up with soda-lime. Behind the soda-lime is a small tube filled with pieces of glass and moistened with sulfuric acid, which retains any moisture taken out of the soda-lime. A

stream of about one liter of air, free from carbon dioxid, is sufficient to drive out all of the carbon dioxid when the estimation is made at ordinary temperatures.

A volcanic earth from Deli, which contained five per cent of humus, gave, at a temperature plus or minus 15° , 0.01 per cent CO_2 . At boiling temperature two analyses gave 0.54 and 0.56 CO_2 . This soil contained no carbonate, and the carbon dioxid found at the boiling temperature, must have come from the humus substances under the influence of the dilute acids.

A heavy clay containing 6.9 per cent of humus gave, at plus or minus 15° , 3.60 per cent CO_2 ; at 100° without boiling, it gave an additional 0.53 per cent, and with boiling an additional 0.11 per cent, or a total of 4.24 per cent CO_2 . A light clay containing 3.2 per cent of humus, gave, at 15° , 5.09 per cent CO_2 ; at a boiling temperature an additional 0.43 per cent, and by continued boiling an additional 0.27 per cent.

316. Estimation of Humus by the German Method.—The German experiment stations follow the method of Loges,^[203] depending on the oxidation of the humic bodies with copper oxid after evaporation of the sample with phosphoric acid. The object of the preliminary evaporation is to set the humic acids free in order that they may be better and more easily oxidized than when burned in the combined state.

The sample of soil is placed in a Hoffmeister dish (Schälchen), moistened with dilute phosphoric acid and evaporated to complete dryness. The dish and its contents are rubbed up with pulverized copper oxid and placed in a combustion tube of sixty centimeters in length, open at both ends. There is then placed in the tube, and held in place by asbestos plugs, granular copper oxid to a length of twenty centimeters.

The combustion tube is placed in a proper furnace and one end connected with two washing-flasks, the first containing potash lye, and the other a solution of barium hydroxid. These flasks are to free the aspirated air from carbon dioxid. The other end of the combustion tube is connected with an appropriate apparatus for absorbing the carbon dioxid. Loges recommends the Pettenkofer absorption tube and a Fresenius drying cylinder.

Between the absorption apparatus and the aspirator, is also placed a washing-flask containing barium hydroxid solution, serving to detect any unabsorbed carbon dioxid. The layer of granular copper oxid is first heated,

the air being slowly aspirated through the apparatus meanwhile, but not through the absorption bulbs. All the carbon dioxide is thus removed from the apparatus.

The absorption system being connected, the tube is heated slowly from the front, backwards, and after the tube is well heated a slow current of air is drawn through and continued until the combustion is complete, which is usually in about three-quarters of an hour.

After the tube is cool the powdered copper oxide and residue of combustion are removed, and for this reason the tube is stopped with a cork at both ends instead of being drawn out and sealed at one end. The tube can thus be refilled without disturbing the granular layer of copper oxide.

The drying cylinder used between the combustion tube and the absorption system has its upper part filled with cotton to avoid the deleterious effects of the nitric oxide produced in the combustion. With this arrangement the use of metallic copper in the combustion tube to reduce the nitric oxide can be dispensed with, the moist cotton holding back the acid fumes. The per cent of humus is obtained by multiplying the per cent of carbon found by 1.724.

317. Method of Raulin for the Estimation of Humus.^[204]—The volumetric estimation of humus in soil by a solution of potassium permanganate would be convenient and practical if the combustion of the organic matter were complete, and if the browning of the liquor did not render the end of the reaction uncertain. The process of Schmidt, modified as below, has given satisfactory results.

In a small flask, with flat bottom, containing about 250 cubic centimeters, are introduced ten cubic centimeters of a solution of manganese sulfate containing sixteen grams of the anhydrous salt per liter, and ten cubic centimeters of a ten per cent solution of potassium permanganate. The solution is heated for a few minutes, the liquor is decolorized and manganese dioxide is precipitated. One hundred cubic centimeters of water are added, and four cubic centimeters of sulfuric acid containing 150 cubic centimeters of monohydrated acid per liter. There is now added an exactly measured volume of the humic liquid properly prepared, so that in oxidizing completely it destroys at most only half of the manganese dioxide. The mixture is submitted to gentle ebullition for eight hours, the water being kept at a constant volume. The excess of manganese

dioxid remaining is dissolved hot by a measured portion of decinormal oxalic acid in slight excess, and the excess of oxalic acid is removed by a solution of potassium permanganate containing one gram per liter. The volume of oxalic acid not destroyed by manganese dioxid is calculated from the amount of permanganate consumed. The volume of oxalic acid, which corresponds to the same quantity of dioxid as the introduced humus, is also calculated by taking the difference between the volume of oxalic acid necessary to destroy all the dioxid formed by ten cubic centimeters of the ten per cent permanganate solution, and the volume of the oxalic acid which has destroyed the dioxid remaining after the action of the humus. The first volume of oxalic acid, that is to say, that which destroys the dioxid formed by ten cubic centimeters of ten per cent permanganate is determined in a preliminary titration.

In regard to the humic liquor, it is prepared by treating ten grams of earth with soda solution in the usual manner. It will be easy to calculate the volume of the oxalic solution equivalent to the total volume of the humic solution, of which a determined fraction has been assayed, and consequently the volume of oxalic solution equivalent to the humus in ten grams of the dry earth. This number of cubic centimeters of the decinormal oxalic solution multiplied by 0.8 will express in milligrams the weight of oxygen necessary to burn the humus from ten grams of dry earth. Humus not being a definite compound, but a residue of complex organic matters partially oxidized, it will require as much more oxygen to complete the combustion as the previous oxidation has been less pronounced. This weight of oxygen necessary to burn the humus from ten grams of dry earth may serve to detect the total value as well as the weight of the humus itself. However, if we wish to have directly the weight of the humus, resource can be had to a table which, without being rigorous, can be regarded as sufficiently exact when the variability of the constitution of humus is taken into account.

Volume of decinormal oxalic acid for ten grams of earth. Cubic centimeters.	Corresponding humus, directly determined. Milligrams.
50	80
100	150
200	280
300	400
400	510
500	610
600	705
700	790
800	885
900	975
1,000	1,060
1,200	1,225
1,400	1,390
1,600	1,560
1,800	1,720
2,000	1,890
2,500	2,315
3,000	2,735
3,500	3,170
4,000	3,605
4,500	4,035
5,000	4,460
5,500	4,890
6,000	5,310
6,500	5,745

318. Pasturel's Method.—According to Pasturel^[205] the process of Raulin does not furnish figures that are rigorously exact only with soil of which the humus contains forty-five per cent of carbon. When the richness in organic carbon is less, the results of the estimation are too high. Pasturel modifies the process as follows:

Manganese Sulfate.—Dissolve sixteen grams of the pure anhydrous manganese sulfate in distilled water and make the solution up to one liter.

Potassium Permanganate.—Make a solution of ten grams of potassium permanganate in one liter of water; 100 cubic centimeters of the liquor just

mentioned are diluted to one liter and constitute the potassium permanganate solution one to ten.

Oxalic and Sulfuric Acids.—A solution of oxalic acid is prepared containing 6.3 grams of the acid in one liter of water, and a dilute solution of sulfuric acid, by dissolving 150 grams of the monohydrated acid in one liter of water.

Humus Solution.—The solution of humus is prepared by the following process: Ten grams of fine earth are freed from all their carbonates by dilute hydrochloric acid. After washing, the filter is broken and the dirt is washed into a small flask. Not more than twenty or thirty cubic centimeters of water should be employed for this purpose. Twenty cubic centimeters of a liquor containing two grams of caustic soda are added, and the flask is placed upon a sand-bath and maintained at a boiling temperature for six hours. It is then diluted with water, filtered and washed as long as the waters are colored. The liquor is treated with dilute sulfuric acid until almost the whole of the soda is saturated. It is indispensable, however, to maintain a slight alkalinity in order that the organic matter may rest totally dissolved. The precipitation of silica which is almost always produced is without inconvenience. Afterward the volume is completed to 500 cubic centimeters and the humus solution is then ready for use.

Estimation of the Humus.—Ten cubic centimeters of the manganese sulfate are placed in a flask and ten cubic centimeters of the permanganate added, and the whole is then slightly heated, and afterward 100 cubic centimeters of water and four cubic centimeters of sulfuric acid are added. The humic liquor is now introduced in such proportion that the humus which it contains dissolves at the greatest, a half of the precipitated manganese and the rest of the process is continued as described by Raulin.

319. Estimation of Carbonates in Arable Soil.—The principle of the determination depends on the liberation of the carbon dioxid from its compounds in the soil by acting on them with strong acid, and the desiccation, absorption, and weighing of the evolved gas. Any of the ordinary forms of apparatus for estimating carbon dioxid may be used in this determination.

The apparatus of Knorr^[206] has been used with satisfaction for many years in the laboratory of the Department of Agriculture.

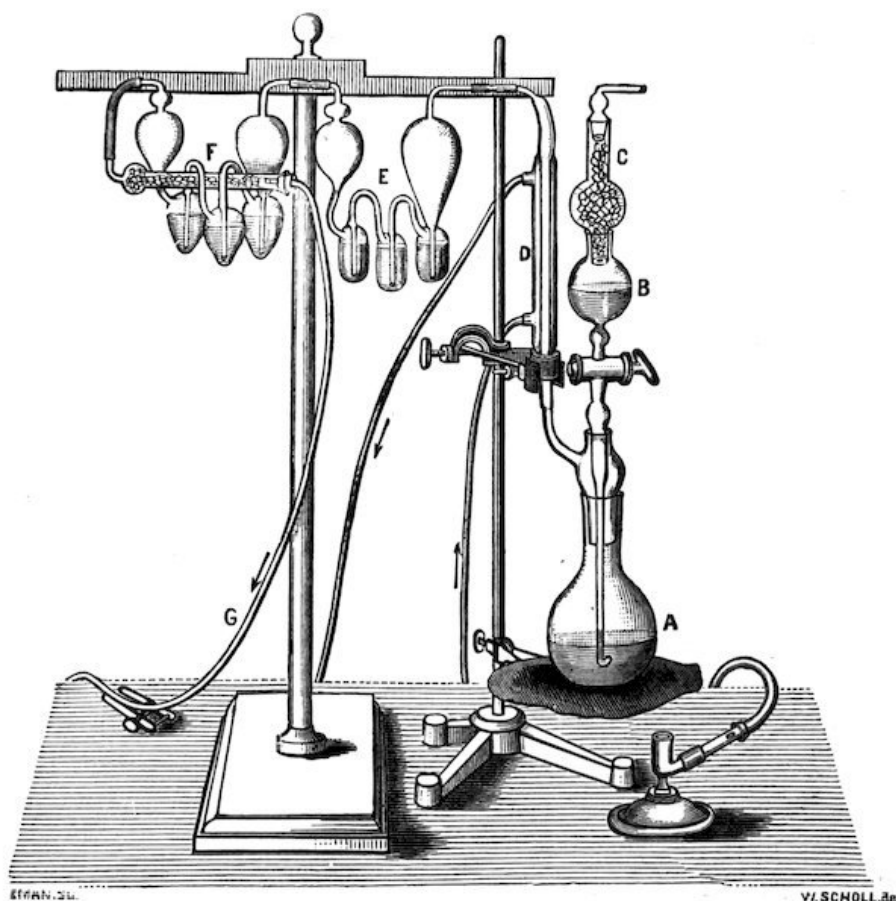


FIGURE 65.

KNORR'S APPARATUS FOR THE DETERMINATION OF CARBON DIOXID.

The apparatus consists of a flask A, Fig. 65, in which the carbon dioxide in the soil is liberated. A condenser, D, fits by means of a ground-glass joint into the neck of the flask in which the liberated gas, together with any air or aqueous vapor which may be carried forward, is cooled. This prevents any excess of vapor of water from entering the absorbing bulbs, which could easily happen at the end of the experiments when the contents of A are raised to the boiling point. The bulb B contains the acid, usually hydrochloric, which is employed for decomposing the carbonates. It is provided with a guard bulb-tube, C, which serves to absorb any carbon dioxide which might enter the apparatus with the air during aspiration at the close of the determination. The carbon dioxide is dried in the bulb-tube, E, in oil of vitriol, and absorbed in the potash solution in F. It is advisable to aspirate a slow current of air through the apparatus by means of the tube G during the whole of the operation. The quantity of the sample to be taken

depends on its richness in carbonates. Many soils are so poor in carbonates as to render any attempt at exact determination nugatory. On the other hand, a comparatively small sample of marls will be sufficient. A preliminary qualitative test will indicate, in a general way, the quantity of the sample to be taken. The sample of soil, five to fifty grams, having been transferred to A, which should be perfectly dry, is made into a batter with freshly boiled distilled water. When all the parts of the apparatus are properly connected gas-tight, the cock between B and A is slowly opened and the hydrochloric (nitric) acid in B allowed to flow into A at such a rate as will secure a moderate evolution of gas.

When the carbonate is entirely decomposed, a lamp is brought under A and its contents gradually raised to the boiling point. The aspiration of air, free from carbon dioxide, is meanwhile continued until all the liberated gas has been absorbed in F. Usually about fifteen minutes will be sufficient to accomplish this purpose.

320. Bernard's Calcimeter.—For a rapid and approximately accurate method of determining the amount of carbonate in the soil, estimated as calcium carbonate, Bernard makes use of the well-known method of the volumetric estimation of carbon dioxide. The sample to be examined should not be powdered in any way. The sample in a natural state, but well air-dried, is gently broken up by the fingers and passed through a sieve having ten meshes to the centimeter. Of the fine earth thus obtained, one gram is taken, for the determination. If the percentage of carbonate in the soil exceeds fifty then only half a gram is taken.

The apparatus employed is one well known. The small erlenmeyer C is fitted with a rubber stopper carrying an exit tube for the gas and a small thermometer. This flask is connected by means of a rubber tube and small glass tube to the measuring burette B. This burette is graduated from 0 to 100 cubic centimeters. Below, by means of a rubber tube, it is connected with the open bulb A, which, by means of a cord about its neck, can be suspended by the hook as shown in the figure. The measuring tube is filled with water through A until the level of the liquid in B is slightly above the zero mark. Meanwhile the one gram of earth has been placed in C, together with the tube D three-fourths filled with an equal mixture of water and strong hydrochloric acid. The greatest care must be taken that no part of the acid be spilled.

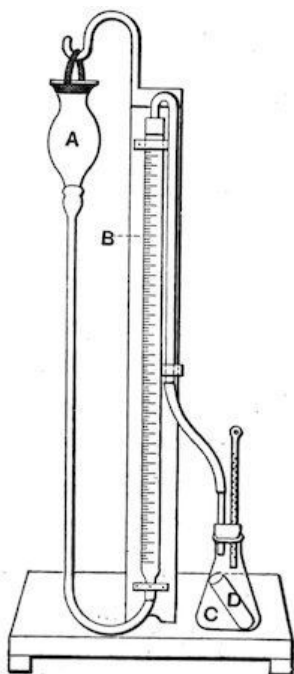


FIGURE 66.

BERNARD'S CALCIMETER.

The rubber stopper is now forced into C until the level of the water in B is just at the zero mark. Grasping C in the right hand and A in the left, the operator inclines C until the contents of D are emptied. Meanwhile as the gas is evolved, A is lowered at such a rate as to always keep the level of the water in B and A on the same plane. In a few moments the evolution of gas is complete, and the volume given off is read at once without correction. This volume multiplied by 0.4 gives the percentage of carbonate in the sample examined. It is understood that the determination is made at ordinary temperatures; viz., 17° to 22°. Example:

One gram of a soil treated as above, gave of carbon dioxide (uncorrected) 65 cubic centimeters. $65 \times 0.4 = 26.00 =$ per cent calcium carbonate in sample.

The above method is useful in the classification of soils and in determining approximately the quantity of calcium carbonate which they contain. The practical use of this method is of great value in determining the character of fertilizer to be applied. It is well to know the percentage of carbonate in selecting mineral fertilizers.

321. Soils Deficient in Carbonates.—When a soil contains but a small quantity of carbonates, Müller^[207] has called attention to the fact that the carbon dioxide absorbed by the water in which the soil is rubbed up may vitiate the result. Instead of water a titrated solution of sodium carbonate is employed. The apparatus is composed of a flask containing the mixture of the sodium carbonate and the soil on which the hydrochloric acid is to act. The hydrochloric acid is contained in a small tube, as in Scheibler's apparatus. The gas is received in a rubber tube 1.5 meters long and three to four millimeters interior diameter, and connected with a burette, the open mouth of which dips into the water of a cylinder of proper length. The volume of gas is read when the burette is raised or lowered in the cylinder until the liquid within and without stands at the same level.

During the action of the acid on the carbonates the flask is constantly shaken.

Several readings of the volume of gas are made, the evolution flask being vigorously shaken before each one. Finally, in order to allow for the variations in temperature and pressure of the exterior air which may take place between the beginning and the end of the reaction, a second flask containing air is placed by the side of the evolution flask and communicating with a narrow tube half filled with water. Any variations in the volume of the air in the flask will be shown by variations in the height of the liquid in the two arms of the tube, and the volume of the variation can be easily determined by having the tube calibrated.

If now a equals the volume per cent of carbon dioxide in the atmosphere of the evolution flask at the end of the reaction, v the volume of gas disengaged, and V the volume of the atmosphere in the evolution flask, the per cent of carbon dioxide contained in a given length of the rubber tube will be equal to $\frac{a}{2}$. This arises from the fact that the first gas which passes into the rubber tube is composed solely of air, while the last contains a per cent of carbon dioxide. By reason of the shaking of the flask the mean richness of the contents of the tube in carbon dioxide, will be sensibly $\frac{a}{2}$.

From the above data the following equations are derived:

$$1. v\frac{a}{2} + Va = v.$$

$$2. a = \frac{v}{\frac{v}{2} + V}$$

If the weight of the carbon dioxide dissolved in V' cubic centimeters of the liquid in the evolution flask be represented by q , the coefficient of the solubility of pure carbon dioxide in this liquid will be, according to the law of the solubility of a gas, equal to $k = \frac{q}{V'a}$

The volume of k has been determined for various strengths of the sodium carbonate solution, using five cubic centimeters of hydrochloric acid containing 1.6 grams pure hydrochloric acid. For solutions disengaging from five to fifty milligrams of carbon dioxide, the mean value of k was found to be 1.8 milligrams in the absence of calcium chloride. When calcium chloride was present in quantities varying from 0.03 to 0.07 gram per cubic

centimeter of liquid in the evolution flask, the value of k was 1.4 milligrams.

By adopting, according to circumstances, the one or the other of the above numbers and multiplying it by Va , as determined by experiment, results are obtained differing only 0.2 to 0.3 milligram from those secured by direct weighing of the evolved gas.

Dietrich^[208] has called attention to the necessity of adding the volume of the dissolved gas to the measured volume in such determinations, and this volume or weight is easily determined by the above formulas.

322. Belgian Method.—The method pursued at the Gembloux Station^[209] consists in taking from five to fifty grams of the sample of soil, according to its content in carbonate, rubbing it up in a porcelain dish with distilled water in order to make a thin paste. The mass is worked to drive out all the air, the whole washed into a flask of 300 cubic centimeters capacity, and the amount of carbon dioxid estimated by setting free with an acid, and collecting the carbon dioxid evolved in potash bulbs.

DIGESTION OF SOILS WITH SOLVENTS.

323. General Considerations.—There are two points in connection with the determination of mineral matters in the soil which must always be kept in view; *viz.*, first, the estimation of the total quantities of material in the soil, and second, the study of those materials which are more easily brought into solution and thus made available for the food of plants.

It is well understood that the soil particles do not give up entirely to the plant the food materials which they contain. The practical value therefore of an analysis of a soil depends more upon the exact determination of the plant food available than upon its total quantity. From a mineral and geological point of view, on the other hand, an idea of the total composition of the soil is the object to be attained.

For the determination of the available plant food, various solvents have been proposed, none of which, perhaps, imitates very accurately the natural solvent action of organic life and moisture on the soil materials. A description of the standard methods of preparing soil extracts will be the subject of a few succeeding paragraphs.

324. Estimation of the Quantity of Materials Soluble in Water.^[210]—Five hundred grams of the air-dried soil are treated in a flask with 1,500 cubic centimeters of water, less the quantity of water already contained in the air-dried soil, which is volatile at 125°. The mass is frequently shaken and, after seventy-two hours, 750 cubic centimeters of the liquid filtered. The filtrate is evaporated to dryness in a platinum dish, dried at 120° and weighed. This is then incinerated and, after treatment with ammonium carbonate and gentle ignition, is again weighed. The further examination of the residue for acids and bases is made by some of the methods hereafter described.

325. Treatment with Water Saturated with Carbon Dioxid.—Two thousand five hundred grams of the air-dried soil are treated with 8000 cubic centimeters of distilled, and afterwards with 2000 cubic centimeters of water, which have previously, at room temperature, been saturated with carbon dioxid. The mixture is left in a closed flask for seven days, frequently shaken, after which 7,500 cubic centimeters of the liquid are

filtered. The clear filtrate, after treatment with a little hydrochloric acid and a few drops of nitric acid, is evaporated to dryness. After the separation of the silica the traces of iron, alumina, lime, sulfuric acid, magnesia, potash, and soda, are estimated in the liquid in the manner hereinafter to be described. Phosphoric acid is always present in such a case, in such small quantities as to make its estimation unnecessary.

326. Treatment with Water Containing Ammonium Chlorid.—In the flask containing the residue from the last experiment; viz., the soil with 2,500 cubic centimeters of liquid, are added 1,500 cubic centimeters of water saturated with carbon dioxid, and 8,000 cubic centimeters of pure water in which five grams of ammonium chlorid are dissolved. The mixture is then left for seven days, with frequent shaking, and 7,500 cubic centimeters of the liquid are then filtered, and the substances dissolved, determined in the filtrate. In addition to the usual quantities of lime and magnesia, from two to four times as much alkali is dissolved by this treatment as is found in the solution from the water containing carbon dioxid alone.

327. Treatment with Water Containing Acetic Acid.—The acetic acid should be of such a strength that after it has fully acted on the soil it should still contain twenty per cent of free acid. 1000 grams of the soil dried at 100° are taken and the acid added in proper proportions and treated in the manner to be described for determining the solvent action of hydrochloric acid.

328. Treatment with Citric Acid Solution.—In ascertaining the quantities of soil materials soluble in a solution of citric acid, Dyer^[211] recommends the use of a carefully prepared citric acid solution. The digestion is carried on as follows: Place in a flask or bottle, holding about three liters, 200 grams of air-dried soil and two liters of distilled water, in which are dissolved twenty grams of pure citric acid. The soil is left, at room temperature, in contact with the one per cent acid for seven days, with thorough shaking several times a day. At the end of the digestion the solution is filtered and 500 cubic centimeters of the filtrate, corresponding to fifty grams of the soil, are taken for analysis for each ingredient to be determined.

The digestion in citric acid is especially recommended by Dyer because of its supposed near resemblance to the methods of solution of plant food

practiced by the rootlets of plants. It is evident, however, that this process is in no sense an imitation of natural methods. The solution is to be used exclusively for the estimation of potash and phosphoric acid. Dyer concludes, from a comparison of the action of a solution of citric acid on soils of known fertility, that when as little as 0.01 per cent of phosphoric acid is dissolved from a soil by this treatment it is justifiable to assume that it stands in immediate need of phosphatic manure. The methods used by Dyer to determine the phosphoric acid and potash in the citric acid solution will be given in their appropriate place.

329. Treatment with Hydrochloric Acid.—The solutions of soils usually subjected to chemical analysis are those obtained by long treatment with hot mineral acids, among which the most common is hydrochloric.

It has long been assumed by soil analysts, perhaps not with justness, that such treatment removed from the soil, all those elements of plant food which could possibly be available for the needs of the growing crop. In this connection, however, the analyst must not forget that nature, in a series of years, with her own methods may easily accomplish what he in five days, even with the help of a hot mineral acid, may not be able to secure. Since, however, this method of solution has been so long practiced it is not the place here to throw doubt on its effectiveness without being able to suggest a better way. Of the mineral acids available no one possesses solvent powers for soils in a higher degree than hydrochloric. A somewhat detailed description will therefore be given of the methods of its use.

330. Strength of Acid to be Employed.—The fact that hydrochloric acid of nearly constant strength; *viz.*, specific gravity 1.115, equivalent to 22.9 per cent hydrochloric acid, may be obtained by distillation, led Owen to use acid of this density in his classic work on soil analysis. Hilgard has lately reviewed the conditions of constant strength in the solvent with results confirming the statements of Owen.^[212] He evaporated on a steam-bath, to one-half its bulk, fifty cubic centimeters of hydrochloric acid, specific gravity 1.116, obtained by using the distillate from a stronger acid after rejecting the first third. The same operation was conducted with similar acid diluted with ten per cent of water. The acid used contained 22.96 per cent hydrochloric acid. The residual acid contained 21.49 per cent hydrochloric acid. These results lead Hilgard to believe that the changes arising from evaporation in hydrochloric acid during soil digestion are insignificant,

compared with those due to its action on the soluble matters, and that evaporation during digestion is effective in maintaining a definite strength in the solvent. For this reason it is contended that evaporation in a porcelain beaker covered by a watch-glass is more effective in constancy of conditions than digestion in a closed flask under pressure.

331. Influence of Time of Digestion and Strength of Acid.—Loughridge has made an interesting study of the influence of the strength of acid and time of digestion on the extraction of soils.^[213] The method of preparing the soil for the determination of the above points is as follows:

The soil, having been passed through the appropriate number of sieves to obtain the fine earth is pulverized with a wooden pestle and thoroughly mixed. The hygroscopic moisture is determined, after exposing it in a place saturated with vapor, in a layer not exceeding one millimeter in thickness for twelve hours, and subsequently drying at 200° in a paraffin-bath. Of this dried substance, from two to three grams are used in the general analysis, the methods employed being in general those adopted by Peter.^[214]

The quantities of materials dissolved by acids of different densities are shown below. The determinations were made by methods hereafter to be described.

Ingredients.	Specific gravity of acid.		
	1.00	1.115	1.160
Insoluble residue	71.88	70.53	74.15
Soluble silica	11.38	12.30	9.42
Potash	0.60	0.63	0.48
Soda	0.13	0.09	0.35
Lime	0.27	0.27	0.23
Magnesia	0.45	0.45	0.45
Manganese oxid	0.06	0.06	0.06
Ferric oxid	5.15	5.11	5.04
Alumina	6.84	8.09	6.22
Sulfuric acid	0.02	0.02	0.02
Volatile matter	3.14	3.14	3.14
Total	100.02	100.69	99.29
Amount of soluble matter	24.00	27.02	22.27
„ „ „ bases	13.50	14.70	12.83

From the above table it is seen that the strongest acid exerts the least soluble effect upon the substances present in the soil, while the greatest degree of solution was obtained by the acid of 1.115 specific gravity. This result indicates that while lime and magnesia are probably present chiefly as carbonates, potash as well as alumina, and to some extent lime, are present as silicates, and for that reason are not as fully extracted by acid of low strength as by that of medium concentration.

In regard to the influence of the time of digestion, the acid of specific gravity 1.115 being used, the data obtained are given in the following table:

Ingredients.	Number of days digested.				
	1.	3.	4.	5.	10.
Insoluble residue	76.97	72.66	71.86	70.53	71.79
Soluble silica	8.60	11.18	11.64	12.30	10.96
Potash	0.35	0.44	0.57	0.63	0.62
Soda	0.06	0.06	0.03	0.09	0.28
Lime	0.26	0.29	0.28	0.27	0.27
Magnesia	0.42	0.44	0.47	0.45	0.44
Manganese oxid	0.04	.06	0.06	0.06	0.06
Ferric oxid	4.77	5.01	5.43	5.11	4.85
Alumina	5.15	7.38	7.07	7.88	7.16
Phosphoric acid				0.21	0.21
Sulfuric acid	0.02	0.02	0.02	0.02	0.02
Volatile matter	3.14	3.14	3.14	3.14	3.14
Total	99.63	100.68	100.55	100.69	99.80
Amount of soluble matter	19.67	24.88	25.57	27.02	24.87
„ „ „ bases	11.05	13.68	13.91	14.49	13.68

From this table it appears that the amount of dissolved ingredients increases up to the fifth day, the increase becoming, however, very slow as that limit is approached. It is also found that the ingredients offering the greatest resistance to this action are the same as those whose amounts were sensibly affected by the strength of the acid; namely, silica, potash, and alumina.

In regard to lime and magnesia, one day's digestion not being sufficient for full extraction, it is evident that they do not exist in the soil as carbonates or hydric oxids only, as has been supposed, but also as silicates. A comparison of the results of the five and ten days' digestion shows that the solvent action of the acid has substantially ceased at the end of five days, there being no further increase of the amount of dissolved matter.

332. Digestion Vessels.—Hilgard prescribes that the digestion of the sample of soil with acid be conducted in a small porcelain beaker covered with a watch-glass.^[215] Kedzie, however, prefers beakers of bohemian glass, and shows that hydrochloric acid attacks the porcelain with greater energy

than the glass.^[216] Platinum would be the ideal material for the digestive vessels, but its great cost would exclude its general use. In most cases it will be found that the error introduced into the analysis by the use of porcelain or bohemian glass beakers is quite small and not likely to affect the quantitative estimation of soluble soil ingredients to any extent.

In this laboratory some comparative tests made by Mr. W. D. Bigelow have shown that vessels of hard glass of special manufacture are less soluble in hot hydrochloric acid of 1.115 specific gravity than porcelain, thus confirming the observation of Kedzie. Following are the data showing the weights of material dissolved in fifty hours:

Berlin porcelain	2.8	milligrams
Bohemian glass	1.7	„
Kaehler and Martini glass	1.2	„

In each case twenty-five cubic centimeters of the acid were used. The vessels all had approximately a capacity of 200 cubic centimeters.

333. Processes Employed—Hilgard's Method.—The sample of soil sifted through a 0.5 millimeter mesh sieve and thoroughly air-dried, is conveniently preserved in weighing tubes. The actual content of hygroscopic and combined moisture may be previously made on a separate sample of soil.

In determining the amount of material to be employed for the general analysis regard must be had to the nature of the soil. This is necessary because of the impracticability of handling successfully such large precipitates of alumina as would result from the employment of as much as five grams in the case of calcareous clay soils; while in the case of very sandy soils even that quantity might require to be doubled in order to obtain weighable amounts of certain ingredients. For soils in which the insoluble portion ranges from sixty to eighty per cent, two and a half to three grams are about the right measure for general analysis, while for the phosphoric acid determination not less than three grams should be employed in any case. It has been alleged that larger quantities must be taken for analysis in order to secure average results. It is difficult to see why this should be true for soils and not for ores, in which the results affect directly the money value, while in the case of soils the interpretation of results allows much wider limits in the percentages. Correct sampling must be presupposed to

make any analysis useful; but with modern balances and methods it is difficult to see why five grams should be employed instead of half that amount, which in some cases is still too much for convenient manipulation of certain precipitates.

The weighed quantity, usually of two to two and a half grams, is brought into a small porcelain beaker, covered with a watch-glass, treated with eight to ten times its bulk of hydrochloric acid of 1.115 specific gravity, and two or three drops of nitric acid, and digested for five days over the laboratory steam-bath. At the end of this time it is evaporated to dryness, first on the water-bath and then on the sand-bath. By this treatment all the silica set free is rendered insoluble.

334. Provisional Method of the Official Agricultural Chemists.—Place ten grams of the air-dried soil in a round bottom 150 to 200 cubic centimeter bohemian flask, add 100 cubic centimeters of pure hydrochloric acid of specific gravity 1.115, insert the stopper, wire it securely, place in a steam-bath, and digest for thirty-six hours at the temperature of boiling water. Pour the contents of the flask into a small beaker, wash with distilled water, add the washings to the contents of the beaker and filter through a washed filter. The residue is the amount insoluble in hydrochloric acid. Add a few drops of nitric acid to the filtrate, and evaporate to dryness on the water-bath; take up with hot water and a few drops of hydrochloric acid, and again evaporate to complete dryness. Take up as before, and filter into a liter-flask, washing with hot water. Cool and make up to the mark. This is solution A. The residue represents the silica originally dissolved. In comparing the two preceding methods it is found that the former; *viz.*, digestion in flasks covered only with a watch-glass gives a larger quantity of dissolved matter in five days than the digestion under pressure does in thirty-six hours. In comparative tests in this laboratory made by Mr. W. D. Bigelow the respective quantities of soluble and insoluble matter obtained by the two methods in two soils are as follows:

Method of Digestion.	Soil No. 1.		Soil No. 2.	
	Per cent.		Per cent.	
	Insoluble.	Soluble.	Insoluble.	Soluble.
Open flask	75.62	24.38	79.62	20.38
Closed flask	76.81	23.19	80.48	19.52

335. The German Station Method.—The method recommended by the German Stations^[217] is greatly different from that described above, both in temperature and time of digestion. To one part of the soil are added two parts by volume of a twenty-five per cent hydrochloric acid solution, the quantity being increased to correspond to any excess of carbonates. The mixture is left for forty-eight hours with frequent shaking. As an alternate method, one part of soil is treated with two parts by volume of ten per cent hydrochloric acid, and heated on the water-bath, with frequent shaking, for three hours.

The soluble materials are determined in the filtrate by some of the methods usually employed.

336. The Gembloux Method.—The method of making the acid extract of the soil at the Gembloux Station does not differ greatly from some of those already described.

The quantity of air-dried material taken is such that it may weigh exactly 300 grams exclusive of the moisture which it contains. It is dried at 150° for at least six hours. The drying is necessary in order to obtain an extract in hydrochloric acid of exactly 1.18 specific gravity. The dry earth is placed in a flask of two or three liters capacity to which one liter of hydrochloric acid of 1.18 specific gravity is added, being careful to take precautions to prevent frothing if much carbonate be present. The acid is allowed to act for twenty-four hours, it being frequently shaken meanwhile. After settling it is decanted and filtered upon a double folded filter, the apex of which rests upon a small funnel covered with a plain filter of strong paper. Five hundred cubic centimeters of the filtrate are taken for the estimation, and in this filtrate are estimated the silica, phosphoric and sulfuric acids, potash, soda, iron, alumina, lime, and magnesia.

The filtrate is evaporated to dryness in a porcelain capsule, a few drops of nitric acid added and the liquid kept well stirred. The residue should be taken up with water, and if not perfectly bright a second and even a third evaporation with nitric acid should take place, until all the organic matter is destroyed, which will be indicated by the clear yellow or reddish-yellow color of the liquid, caused by the iron oxid. After the last evaporation the material is dried in a drying oven one hour at 110°.

337. Treatment with Cold Hydrochloric Acid.—According to the digestion method of Wolff^[218] the soil sample is treated with cold

concentrated hydrochloric acid. The process is as follows:

Four hundred and fifty grams of the soil dried at 100° are placed in a glass flask and treated with 1,500 cubic centimeters of hydrochloric acid of 1.15 specific gravity, corresponding to thirty per cent of gaseous hydrochloric acid. For every five per cent of calcium carbonate which the soil may contain, an additional fifty cubic centimeters of hydrochloric acid are added. With frequent stirring, the soil is left in contact with the acid for forty-eight hours and then 1,000 cubic centimeters of liquid, as clear as possible, are poured off, which corresponds to 300 grams of the soil. After dilution with water it is filtered and the filtrate treated with a few drops of nitric acid and evaporated to dryness. After the separation of the silica the solution is again made up with water to 1,000 cubic centimeters.

Two hundred cubic centimeters of this solution, corresponding to sixty grams of the soil, are taken for the estimation of iron, alumina, lime, manganese, and magnesia.

Four hundred cubic centimeters of the solution, corresponding to 120 grams of the soil, are left for the estimation of sulfuric acid and alkalies. This method gives from five to six times less alkalies and a much smaller quantity of iron than the treatment with hot acid. In the use of hot acid, therefore, Wolff reduces the quantity of soil acted on to 150 grams.

338. Treatment with Nitric Acid.—For the purpose of estimating phosphoric acid Grandeau^[219] directs that the soil be extracted with nitric acid. For this purpose 100 grams of the air-dried fine earth are placed in a bohemian flask and treated cautiously with nitric acid in small quantities at a time. If the soil be calcareous in its nature it should be previously moistened with water, and the acid so added as to avoid undue effervescence, the flask being inclined during the operation. Sufficient acid is added to strongly saturate the sample and it is then digested on the sand-bath for two hours; or at least until the organic matters are destroyed, which will be indicated by the cessation of evolution of nitrous vapors. When the supernatant liquid has become clear it is decanted. The residue is washed with distilled water and separated on a filter, and washed until the wash-water is colorless. The decanted portion is united with the filtrate and the whole made up to a volume of one liter. The determinations are made in portions of 200 cubic centimeters each.

339. Digestion with Hydrofluoric and Sulfuric Acids.—When a complete disintegration of the siliceous substances in soils is desired as in analysis in bulk, the decomposition is easily accomplished by digestion with the above named acids in a platinum dish. The fine earth is saturated with a concentrated aqueous solution of hydrofluoric acid to which a few drops of sulfuric acid are added. It is then digested until nearly dry. If any undecomposed particles remain, the treatment is continued until complete decomposition is secured. The silica is thus all volatilized as hydrofluosilicic acid and the bases pre-existing in the soil are left as sulfates. This method of treatment is especially recommended when it is desired to estimate the whole quantity of any of the soil constituents with the exception of silica. The silica may, however, be determined in the distillate. Instead of using the solution of hydrofluoric acid, ammonium fluorid may be employed. In this process the sample of earth reduced to an impalpable powder by grinding in an agate mortar is mixed with four or five times its weight of the ammonium fluorid in a platinum dish and thoroughly moistened with sulfuric acid and allowed to stand at room temperature for several hours. It is then gently heated until all fumes of hydrofluosilicic acid have been driven off, but is not raised to a red heat. If any undecomposed particles remain, the above treatment is repeated.

DETERMINATION OF THE QUANTITY OF DISSOLVED MATTER.

340. Substances in Solution.—By treatment with solvents as indicated in the preceding paragraphs, greater or less quantities of the original constituents of soil are brought into solution. The total quantity of dissolved matters is determined by drying and weighing the insoluble residue and the percentages of soluble and insoluble matters should be noted; and each portion saved for further examination. In this country the common practice of soil analysis is to digest the sample with hydrochloric acid. The following paragraphs, therefore, will be devoted to the general methods of determining the matters dissolved by that treatment, leaving for later consideration the special methods of analysis. The fundamental principle on which the treatment with hydrochloric acid rests is based on the belief that such treatment practically extracts from the soil all those elements which are likely to become, immediately or in the near future, available for plant food.

341. Provisional Methods of the Official Agricultural Chemists.^[220]—

(1) *The Analytical Operations* are conducted with solution A, paragraph 334.

(2) *Ferric Oxid, Alumina, and Phosphoric Acid.*—To 100 or 200 cubic centimeters, according to the probable amount of iron present, of the solution A, add ammonium hydroxid to alkaline reaction to precipitate ferric and aluminum oxids and phosphates. Expel the excess of ammonia by boiling, allow to settle, decant the clear solution through a filter; add to the flask fifty cubic centimeters of hot distilled water, boil, settle, and decant as before. After pouring off all the clear solution possible, dissolve the residue with a few drops of warm hydrochloric acid and add just enough ammonium hydroxid to precipitate the oxids. Wash by decantation with fifty cubic centimeters of distilled water, and then transfer all the precipitate to the filter and wash with hot distilled water till the filtrate becomes free from chlorids. Save the filtrate and washings which form solution B. Dry the filter and precipitate at 110°, transfer the precipitate to a tared platinum crucible, burn the filter and add the ash to the precipitate, heat the whole red

hot, cool in a desiccator, and weigh. The increase of weight, minus the ash of filter and the phosphoric acid (found in a separate process), represents the weight of the ferric and aluminum oxids.

(3) *Ferric Oxid*.—Precipitate 100 cubic centimeters of solution A, as under (2), except that only one precipitation is made; wash with hot water; dissolve in dilute sulfuric acid; reduce with zinc and estimate as ferrous oxid by a standard solution of potassium permanganate.

To prepare the potassium permanganate solution, dissolve 3.156 grams of pure crystallized potassium permanganate in 1,000 cubic centimeters of distilled water, and preserve in a glass-stoppered bottle, shielded from the light. Standardize this solution with pure ferrous sulfate, ammonium ferrous sulfate or oxalic acid.

(4) *Alumina*.—The calculated weight of ferric oxid deducted from that of ferric oxid and alumina with corrections for filter ash and phosphoric acid, will give the weight of alumina in two grams of air-dried soil.

(5) *Phosphoric Acid*.—This may be estimated in the above iron solution, if the soil is sufficiently rich, by the molybdate method, given under fertilizers; or if the quantity of soil represented in the iron solution is not sufficient, a fresh portion of solution A may be taken, and the phosphoric acid determined directly by the molybdate method.

(6) *Manganese*.—Concentrate the filtrate and washings (solution B) to 200 cubic centimeters or less; add ammonium hydroxid to alkalinity; add bromin water and heat to boiling, keeping the beaker covered with a watch-glass; as the bromin escapes, the beaker is allowed to cool somewhat, ammonia and bromin water again added, and heated as before.

This process is continued until the manganese is completely precipitated, which requires from thirty to sixty minutes, and the solution filtered while still warm; the precipitate is washed, dried, ignited and weighed; estimate as manganese protosesquioxid.

(7) *Lime*.—If no manganese is precipitated, add to solution B, or the filtrate and washings from (6) twenty cubic centimeters of a strong solution of ammonium chlorid and forty cubic centimeters of saturated solution of ammonium oxalate to completely precipitate all the lime as oxalate and convert the magnesia into soluble magnesium oxalate. Heat to boiling and let stand for six hours till the calcium oxalate settles clear, decant the clear solution on a filter, pour fifty cubic centimeters of hot distilled water on the

precipitate and again decant the clear solution on the filter, transfer the precipitate to the filter, and wash it free from all traces of oxalates and chlorids. Dry and ignite the precipitate over the blast-lamp until it ceases to lose weight, weigh and estimate as calcium oxid; carefully moisten with sulfuric acid, heat the inclined covered crucible gently to avoid loss, then intensely, and weigh as calcium sulfate.

(8) *Magnesia*.—Concentrate the filtrate and washings (from 7) to 200 cubic centimeters, place in a half-liter erlenmeyer, add thirty cubic centimeters of a saturated solution of sodium phosphate and twenty cubic centimeters of concentrated ammonium hydroxid, cork the flask, and shake violently at intervals of a few minutes till crystals form, then set the flask in a cool place for twelve hours. Filter the clear liquid through a tared gooch, transfer the precipitate to the filter, and wash with dilute ammonium hydroxid (1 : 3) till the filtrate is free from phosphates; dry and ignite the crucible, at first gently and then intensely, to form magnesium pyrophosphate. The increase of weight $\times 0.36024 = \text{MgO}$. By using an erlenmeyer free from scratches and marks, and shaking violently instead of stirring with a glass rod, the danger is almost entirely avoided of crystals adhering to the sides of the vessel; but if crystals do adhere they are readily removed by a rubber-tipped glass rod.

(9) *Sulfuric Acid*.—Evaporate 200 cubic centimeters of solution A (1) nearly to dryness on a water-bath to expel the excess of acid; then add 100 cubic centimeters of distilled water, heat to boiling and add ten cubic centimeters of a solution of barium chlorid, and continue the boiling for five minutes. When the precipitate has settled, pour the clear liquid on a tared gooch, treat the precipitate with fifty cubic centimeters of boiling water, and transfer the precipitate to the filter and wash with boiling water till the filtrate is free from chlorids. Dry the filter and ignite strongly. The increase in weight is barium sulfate, which multiplied by $0.34331 = \text{SO}_3$ in two grams of air-dried soil.

(10) *Potash and Soda*.—To another portion of 200 cubic centimeters of solution A, add barium chlorid in slight excess, and make alkaline with ammonia to precipitate sulfuric and phosphoric acids, ferric oxid, etc. Then precipitate the calcium and barium by ammonium oxalate. Evaporate the filtrate and washings to dryness, heat to a low red heat to decompose oxalates and expel ammonia salts, dissolve in twenty-five cubic centimeters

of distilled water, filter and wash the precipitate; add to the filtrate and washings ten cubic centimeters of baryta water, and digest for an hour. Filter and wash the precipitate, add ammonium carbonate to the filtrate to complete precipitation of baryta, filter and wash this precipitate. Evaporate the filtrate and washings in a tared platinum dish, gently ignite the residue to expel ammonia salts, cool and weigh. The increase of weight represents the potassium and sodium chlorids in two grams of air-dried soil.

342. Hilgard's Methods.^[221]—(1) *Soluble Silica*.—The acid filtrate obtained by the process given in paragraph 333 is employed for the following determinations. After the solution obtained has been evaporated to dryness to render silica insoluble, it is moistened with strong hydrochloric acid and two or three drops of nitric acid. The mass is warmed, and after allowing to stand for a few hours on a steam-bath is taken up with distilled water. After clearing, it is filtered from the insoluble residue, which is strongly ignited and weighed. If the filtrate should be turbid the insoluble residue which has gone through the filter can be recovered in the iron and alumina determination.

The insoluble residue is next boiled for fifteen or twenty minutes in a concentrated solution of sodium carbonate, to which a few drops of caustic lye should then be added to prevent reprecipitation of the dissolved silica. The solution must be filtered hot. The difference between the weight of the total residue and that of undissolved sand and mineral powder is recorded as soluble silica, being the aggregate of that set free by the acid treatment and that previously existing in the soil. The latter, however, rarely reaches five per cent.

(2) *Destruction of Organic Matter*.—The acid filtrate from the total insoluble residue is evaporated to a convenient bulk. In case the filtrate should indicate by its color, the presence of any organic matter, it should be oxidized by aqua regia, otherwise there will be difficulty in separating alumina.

(3) *Precipitation of Iron and Alumina*.—The filtrate thus prepared is now brought to boiling and treated sparingly with ammonia, whereby iron and alumina are precipitated. It is kept boiling until the excess of ammonia is driven off, and then filtered hot. (Filtrate A.) The previous addition of ammonium chlorid is usually unnecessary. If the boiling is continued too long, filtration becomes very difficult and a part of the precipitate may

redissolve in washing. Filtration may be begun as soon as the nose fails to note the presence of free ammonia; test paper is too delicate. Failure to boil long enough involves the contamination of the iron-alumina precipitate with lime and manganese.

(4) *Estimation of Iron and Alumina.*—The iron and alumina precipitate with filter of (3) is dissolved in a mixture of about five cubic centimeters of hydrochloric acid and twenty cubic centimeters of water. Then filter and make up to 150 cubic centimeters. Take fifty cubic centimeters for the determination of iron and alumina together by precipitation with ammonia, after oxidizing the organic matter (filter) with aqua regia; also fifty cubic centimeters for iron alone; keep fifty cubic centimeters in reserve. Determine the iron by means of a standard solution of potassium permanganate after reduction; this latter is done by evaporating the fifty cubic centimeters almost to dryness with strong sulfuric acid, adding water and transferring the solution to a flask, and then reducing by means of pure metallic zinc in the usual way. The alumina is then determined by difference. This method of determining the two oxids in their intermixture is in several respects more satisfactory than the separation with alkaline lye, which, however, has served for most determinations made, until within the last ten years. It is, however, much more liable to miscarry in unpracticed hands than the other.

(5) *Estimation of Lime.*—The filtrate A from iron and alumina is acidified slightly with hydrochloric acid, and if too bulky is evaporated to about twenty-five cubic centimeters, unless the soil is a very calcareous one, and the lime is precipitated from it by neutralizing with ammonia and adding ammonium oxalate. The precipitation of the lime should be done in the hot solution, as the precipitate settles much more easily. It is allowed to stand for twelve hours, then filtered (filtrate B), washed with cold water, and dried. By ignition the lime precipitate is partially converted into the oxid. It is then heated with excess of powdered ammonium carbonate, moistened with water, and exposed to a gentle heat (50° – 80°) until all the ammonia is expelled. It is then dried below red heat and weighed as calcium carbonate. When the amount of lime is at all considerable, the treatment with ammonium carbonate must be repeated till a constant weight is obtained.

(6) *Estimation of Sulfuric Acid.*—The filtrate B from the calcium oxalate is put into a bohemian flask, boiled down over the sand-bath, and the

ammoniacal salts destroyed with aqua regia. From the flask it is removed to a small beaker and evaporated to dryness with excess of nitric acid. This process usually occupies four to five hours. The residue should be crystalline-granular; if white-opaque, ammonium nitrate remains and must be destroyed by hydrochloric acid. The dry residue is now moistened with nitric acid, and the floccules of silica usually present separated by filtration from the filtrate, which should not amount to more than ten or fifteen cubic centimeters; sulfur trioxid is then precipitated by treatment with a few drops of barium nitrate, both the solution and the reagent being heated to boiling. If the quantity of sulfuric acid is large it may be filtered after the lapse of four or five hours (filtrate C). If very small let it stand twelve hours. The precipitate is washed with boiling water, dried, ignited, and weighed. Care should be taken in adding the barium nitrate to use only the least possible excess, because in such a small concentrated acid solution the excess of barium nitrate may crystallize and will not readily dissolve in hot water. Care must also be taken not to leave in the beaker the large heavy crystals of barium sulfate, of which a few sometimes constitute the entire precipitate, rarely exceeding a few milligrams. Should the ignited precipitate show an alkaline reaction on moistening with water, it must be treated with a drop of hydrochloric acid, refiltered and weighed. The use of barium acetate involves unnecessary trouble in this determination.

(7) *Estimation of Sodium and Potassium*.—Filtrate C is now evaporated to dryness in a platinum dish; the residue is treated with an excess of crystallized oxalic acid, moistened with water, and exposed to gentle heat. It is then strongly ignited to change the oxalates to carbonates. This treatment with oxalic acid must be made in a vessel which can be kept well covered, otherwise there is danger of loss through spattering. As little water as possible should be used, as otherwise loss from evolution of carbon dioxide is difficult to avoid. Spatters on the cover should not be washed back into the basin until after the excess of oxalic acid has been volatilized. The ignited mass should have a slightly blackish tinge to prove the conversion of the nitrates into carbonates. White portions may be locally retreated with oxalic acid. The ignited mass is treated with a small amount of water, which dissolves the alkaline carbonates and leaves the magnesium carbonate, manganese protoxyd, and the excess of barium carbonate behind. The alkalies are separated by filtration into a small platinum dish (filtrate D), and the residue is well but sparingly washed with water on a small filter.

When the filtrate exceeds ten cubic centimeters it may, on evaporation, show so much turbidity from dissolved earthy carbonates as to render refiltration on a small filter necessary, since otherwise the soda percentage will be found too large and magnesia too small. If, on dissolving the ignited mass, the solution should appear greenish from the formation of alkaline manganates, add a few drops of alcohol to reduce the manganese to insoluble dioxid. The residue of barium, magnesium, and manganese compounds is treated on the filter with hydrochloric acid, and the platinum dish is washed with warm nitric acid (not hydrochloric, for the platinum dish may be attacked by chlorine from the manganese oxid) dissolving any small traces of precipitate that may have been left behind.

The filtrate D, which should not be more than ten or fifteen cubic centimeters, containing the carbonates of the alkalies, is evaporated to dryness and gently fused, so as to render insoluble any magnesium carbonate that may have gone through; then redissolved and filtered into a small weighed platinum dish containing a few drops of dilute hydrochloric acid, to change the carbonates into chlorids; evaporated to dryness, exposed to a gradually rising temperature (below red heat), by which the chlorids are thoroughly dried and freed from moisture, so as to prevent the decrepitation that would otherwise occur on ignition. Then, holding the platinum basin firmly by forceps grasping the clean edge, pass it carefully over a very low bunsen flame, so as to cause, successively, every portion of the scaly or powdery residue to collapse, without fully fusing. There is thus no loss from volatilization, and no difficulty in obtaining an accurate, constant weight. The weighed chlorids are washed by means of a little water into a small beaker or porcelain dish, treated with a sufficient quantity of platinum chlorid, and evaporated to dryness over the water-bath. The dried residue is treated with a mixture of three parts absolute alcohol and one part ether, leaving the potassium platinochlorid undissolved. This is put on a filter, and washed with ether-alcohol. When dried, the precipitate and filter are put into a small platinum crucible and exposed to a heat sufficiently intense to reduce the platinum chlorid to metallic platinum and to volatilize the greater part of the potassium chlorid. This is easily accomplished in a small crucible, which is roughened by being constantly used for the same purpose (and no other), the spongy metal causing a ready evolution of the gases. The reduced platinum is now first washed in the crucible with hot acidulated water, then with pure water; then all moisture is driven off and it is

weighed. From the weight of the platinum, is calculated the potassium chlorid and the oxid corresponding; the difference between the weights of the total alkaline chlorids and potassium chlorid gives the sodium chlorid, from which may be calculated the sodium oxid. When the heating of the platinum precipitate has not been sufficient in time or intensity, instead of being in a solid spongy mass of the color of the crucible itself, small black particles of metallic platinum will obstinately float on the surface of the water in the crucible, and it becomes difficult to wash without loss.

(8) *Estimation of Manganese.*—The solution containing the magnesium and manganese chlorids is freed from barium salts by hot precipitation with sulfuric acid, and the barium sulfate, after settling a few hours, is separated by filtration. The filtrate is neutralized with ammonia, any resulting small precipitate (of iron) is filtered, and the manganese precipitated with ammonium sulfid, let stand twelve hours and filtered (filtrate E); wash with cold water, dry, ignite, and weigh as manganese protosquioxid, Mn_3O_4 . If preferred the manganese may be precipitated with chlorin or bromin water as dioxid; but the process requires a rather longer time and may fail in inexpert hands more readily than the other.

(9) *Estimation of Magnesium.*—The filtrate E from the manganese is now freed from sulfur by acidulating with hydrochloric acid, evaporating, if necessary, and filtering. From the filtrate the magnesia is precipitated by adding an equal bulk of ammonia water and then sodium phosphate. After standing at least twenty-four hours, the magnesium salt may be filtered, washed with ammoniacal water, dried, ignited, and weighed as magnesium pyrophosphate.

343. Examination of Acid Extract by the Methods of Petermann.—*Estimation of the Silica.*—The Gembloux method of estimating silica consists in taking up the dry extract obtained from the treatment of the earth, in the manner described in paragraph 336, with water and a few drops of hydrochloric acid, heating for a short time on a sand-bath to facilitate the solution, and filtering, washing, drying, igniting, and weighing the residue obtained as silica.

Estimation of the Sulfuric Acid.—The method employed consists in heating the filtrate obtained in the estimation of silica for half an hour with a few drops of nitric acid and making the volume up to 500 cubic centimeters. One hundred cubic centimeters of this are precipitated with

barium chlorid, diluted to double its volume, heated for some time, the precipitate of barium sulfate collected and weighed, and the quantity of sulfuric acid calculated therefrom.

Potash and Soda.—Potash and soda are estimated at the Gembloux Station by heating the filtrate obtained in the estimation of the sulfuric acid and precipitating the excess of barium in the hot solution after the addition of ammonia by ammonium oxalate and carbonate. The whole is allowed to digest for six hours at a gentle heat and then allowed to remain at rest for twenty-four hours, filtered, washed, and the filtrate evaporated to dryness in a large platinum dish and the ammoniacal salts driven off at a low temperature. At the end, the temperature is carried a little higher until it reaches low redness. The residue is taken up by distilled water, filtered into a weighed platinum dish, a few drops of hydrochloric acid added, evaporated, dried, heated with great care and the sodium and potassium chlorids obtained weighed together. The respective quantities of potash and soda in the earth are estimated in the usual way by precipitating the potash with platinum chlorid.

Estimation of the Iron and Aluminum Oxids.—The iron and aluminum oxids are estimated by taking twenty-five cubic centimeters of the primitive solution obtained with hydrochloric acid and adding ammonium carbonate almost to complete neutralization, that is to say until the precipitate formed is just redissolved in the feeble excess of hydrochloric acid which remains. Dilute with distilled water and precipitate with a little excess of ammonium acetate, and boil for a moment; after boiling, the basic iron and aluminum acetate and the small quantity of iron and aluminum phosphate present are easily deposited, and the supernatant liquid should be completely limpid and colorless. Wash the precipitate by decantation, boiling each time, filter, wash the filter with boiling water to which a little ammonium acetate has been added, dry, ignite, and weigh. The material obtained consists of ferric oxid, aluminum oxid, and iron and aluminum phosphates. Deduct from the whole, the phosphoric acid determined in another portion. The residue will be the sum of the iron and aluminum oxids.

Estimation of the Lime.—The filtrate from the portion used for the estimation of the iron and alumina is treated with ammonium oxalate. The mixture is kept at a low temperature for at least twelve hours, after which it

is filtered, washed with hot water, dried, and ignited over a blast-lamp to constant weight and weighed as calcium oxid.

Estimation of the Magnesia.—For the estimation of the magnesia the filtrate obtained in the estimation of lime is evaporated to dryness in a platinum dish, the ammoniacal salts driven off, the residue taken up with water slightly acidified with hydrochloric acid, filtered, the filtrate saturated with ammonia and heated some time to the boiling point to precipitate any traces of iron and alumina which may have remained in solution. Filter, wash, allow to cool and precipitate the magnesia by the addition of sodium phosphate. It is then allowed to stand for twelve hours, collected on a filter, ignited, and weighed as pyrophosphate, and the quantity of magnesia calculated from the weight of salt obtained.

Estimation of the Phosphoric Acid.—The phosphoric acid is estimated by taking 100 cubic centimeters of the original solution obtained by the treatment of the soil with hydrochloric acid and evaporating it to dryness on the water-bath. The residue is taken up with water to which a few drops of nitric acid have been added and filtered. The total phosphoric acid is then obtained by precipitation with ammonium molybdate in the usual way.

344. Analysis of the Insoluble Residue.—The insoluble residue left after digestion with hydrochloric acid is not without interest from an agricultural and analytical point of view. While it is true that the plant food, therein contained, is not immediately available, yet it must not be forgotten that the method of the chemist may not fix a limit to nature's method of collecting nutriment for plants. In however refractory a state they may exist, it is possible that all nutritive elements may eventually become available for assimilation. For the completion of an estimate of the total nutritive power of a soil, therefore a further examination of the insoluble residue should be made. The methods of securing this are essentially those of making a bulk analysis of the soil.

The principle of the method depends on the reduction of the sample to an impalpable powder and the subsequent decomposition of the insoluble portions by treatment with hydrofluoric and sulfuric acids or by fusion with the alkalis.

345. Method of Wolff for Treating Residue Insoluble in Hot Acid.—The well-washed residue is dried with the filter, then separated therefrom, the filter burned and the ash weighed with the whole of the residue. About

eight grams of the residue are ignited and serve for the estimation of the insoluble mineral matter. Another portion of ten grams of the dried, but not ignited, residue is boiled with a concentrated solution of sodium carbonate with the addition of caustic soda, and the quantity of dissolved silicic acid estimated. A third portion of about fifteen grams is treated with about five times its weight of pure concentrated sulfuric acid, and is evaporated until the mass has taken the form of a dry powder. After moistening with concentrated hydrochloric acid the mass is boiled with water, filtered, and the filtrate examined according to the ordinary methods for silicic acid, alumina, iron, lime, magnesia, and alkalies. The residue after treatment with concentrated sulfuric acid is dried, but not ignited, and boiled with a concentrated solution of sodium carbonate with the addition of a little caustic soda, filtered, heated, and the silicic acid separated from the solution. After thorough washing, the residue, after ignition, is weighed and represents the material insoluble in concentrated hydrochloric and sulfuric acids. The silicic acid found as before, together with the small quantity dissolved in the hydrochloric acid extract, gives, in connection with the alumina contained in the sulfuric acid extract, approximately the quantity of pure water-free clay contained in the soil.

In six samples of soils of very different compositions which were examined by the above process, it was found that the clay had the following mean composition: Silicic acid, 55.1 to 61.5 per cent, alumina, 38.6 to 44.9 per cent; as a mean 58.05 per cent silicic acid and 41.95 per cent alumina.

Finally, four or five grams of the residue, after treatment with sulfuric acid and sodium carbonate, are rubbed up in an agate mortar and completely separated into silt by water. The silt mass is dried, lightly ignited, and three grams of it spread in a flat platinum dish moistened with sulfuric acid, and subjected to the action of hydrofluoric acid in a lead oven at 60°, until a complete decomposition of the material is accomplished. In the solution all the different bases can be determined.

346. Method of the Belgian Chemists.—The method employed by Petermann^[222] at the Gembloux Station in the examination of the part of the soil insoluble in hydrochloric acid consists in washing the insoluble portion by decantation with distilled water until all acid reaction is removed. Place the contents of the flask and of the filter in a porcelain dish and dry. After a careful mixing of the mass take out about fifty grams and wash upon the

filter until all reaction for chlorin has disappeared, dry, detach the mass from the filter, and incinerate. Place in a platinum crucible two grams of the ground and ignited residue and mix it, using a platinum stirring rod, with twelve grams of ammonium fluorid; heat slightly over a bunsen burner in a muffle with a good draught and regulate the flame in such a way that the operation shall continue for about one hour. After complete decomposition add about two cubic centimeters of sulfuric acid in such a way as to moisten completely the residue, drive off the sulfuric acid carefully at a low red heat and take up the residue with water slightly acidulated with hydrochloric acid and wash the whole into a flask of 500 cubic centimeters capacity. Oxidize by heating for an hour with nitric acid, make up to the mark and filter. The percentages of potash, soda, lime, magnesia, and the silicates are determined exactly as in the hydrochloric acid extract.

347. Bulk Analysis.—It is frequently desirable to determine the total composition of a soil sample as well as the nature of that part of it soluble in any of the solvents usually employed. The latest methods for this purpose have been well studied by Packard^[223] who finds that the variations which occur between duplicates are probably due to the small quantities of material taken for analysis, it being difficult to obtain average samples of a material which is not very finely powdered when small quantities are taken. Moreover, as it is likely to become of importance to know whether the proportions of lime and magnesia vary by as much as one-tenth per cent, and such small variations are within the limits of error of an analysis, and as the total proportion of lime and magnesia in highly siliceous soils, probably does not exceed one-tenth per cent, it is deemed best to take a large quantity of soil for the bulk analysis in each case. The amount adopted for the highly siliceous soils containing much quartz is ten grams. This quantity, taken after quartering down the entire sample, is ground to an impalpable powder and used for the determination of the lime, magnesia, and alkalies, the silica, iron oxid, alumina, and loss on ignition, being determined in one gram samples. The ten grams are decomposed by hydrofluoric and sulfuric acids in a large platinum dish, the solution evaporated, at first on the water-bath until all water is removed and then at a higher temperature until all the free sulfuric acid is driven off, when the residue is heated in a muffle at a low red heat for several hours. At this temperature the sulfuric acid combined with the iron oxid and alumina is driven off, leaving the remaining sulfates unchanged and the iron oxid and alumina are in the form

of a powder of no great volume which is easily and quickly washed. This operation is usually successful at first but in some cases the decomposition is not complete as is shown by the appearance of a precipitate on adding ammonia to the filtrate from the aluminum and iron oxids. In such cases the precipitate is dissolved in hydrochloric acid, reprecipitated by ammonia and removed by filtration. In the filtrate from the thoroughly washed aluminum and iron oxids, lime is precipitated as oxalate and separated by filtration; the filtrate is evaporated to dryness and the ammonia salts driven off by heat; the magnesia in the unfiltered watery extract of this residue is precipitated by baryta water, which also removes the sulfuric acid with which the bases had been combined. In the filtrate from this precipitate baryta is precipitated by ammonium carbonate and removed by filtration, leaving the alkalies to be determined in the usual way after conversion into chlorids. The mixed precipitate of magnesia and barium sulfate is treated with hydrochloric acid, filtered, the baryta present removed as sulfate, and the magnesia precipitated in the filtrate from the latter as phosphate. The advantages of this method are that the large quantity of material employed gives some assurance that an average sample has been operated on, and all the bases present in small proportions are estimated in the same sample. The objection to it is the time consumed both in grinding the samples and in determining all the bases in one solution. As a small quantity of material is generally used for determining the silica, iron oxid, alumina, and loss by ignition, and a larger quantity for the remaining bases, slight differences in the unground samples are unavoidable, especially when the quartz grains are somewhat large, it being practically impossible to take two small samples of such a soil which would have the same number of quartz grains. Consequently tedious grinding of large quantities of the soils for the bulk analysis is necessary. This objection does not apply to the official analysis or assay of soils in which considerable quantities are extracted by acid and the solution analyzed, and silica is not determined. In any case, it may be said, when it becomes an object to know whether a soil contains a total of 0.1 or 0.2 per cent of lime or magnesia, of 0.7 or 0.5 per cent of potash, one analysis even of the large quantity of ten grams would be insufficient to decide the point, and at least the mean of two determinations should be taken.

SPECIAL METHODS OF DETERMINATION OF SOIL CONSTITUENTS.

348. Preliminary Considerations.—In the foregoing paragraphs the general outline of the chemical methods of soil examination have been given. There are often occasions, however, which demand a special study of some particular soil constituent. It has been thought proper, therefore, to add here some of the best approved methods of special determinations which have been approved in this and other countries.

In the main, the final determination of any particular element of the soil, and its previous separation from accompanying elements, are based on the general processes already given. The variations in many instances, however, seem to require special mention.

349. Condition of Potash in Soils.—Potash exists in the soil in very different states. That part of it which is combined with the humus material, or with the hydrated silicates, is easily set free from its combinations and is to be regarded as the more assimilable portion.

The potash in the soil is found chiefly in combination with silicates, and particularly with the hydrated aluminum silicates, forming clay. As the particles with which it is combined are found in a state of greater or less fineness, the potash itself is set free under the influences of the agents which are active in the soil, with greater or less rapidity, passing into a form in which it can be utilized by plants. In silicates which are very finely divided, such as clay, the potash becomes active in a relatively short time, while in the débris of rocks in a less advanced state of decomposition it may rest for an indefinite period in an inert state. The estimation of the potash which is assimilable in the clay is quite as important for agricultural purposes as to determine that which may be present in the soil in firmer combination. Treating the sample of soil with water does not furnish any useful information in regard to the potash which it contains. Indeed, the absorbing properties of the soil tend to prevent the elimination of the potash in this way, even when it is found in the soluble state. It is therefore, necessary to employ an acid to set the potash free, but variable results are

obtained, according to the employment of acids of greater or less concentration and for longer or shorter periods of contact.

350. Estimation of the Potash Soluble in Concentrated Acids.—In the method of the French agricultural chemists^[224] twenty grams of the earth are placed in a dish with a flat bottom, eleven centimeters in diameter, and rubbed up with twenty to thirty cubic centimeters of water. There is added carefully, and in small quantities, some nitric acid of 36° Baumé until all effervescence has ceased, the mass meanwhile being thoroughly stirred. When the carbonates have been decomposed, which can be told by the cessation of the effervescence, twenty cubic centimeters more of the same acid are added. The dish is heated on the sand-bath for five hours, regulating the heating in such a way that there still remains some acid at the end of the operation and the mass is not thoroughly dry. The acid mass is then taken up with hot water, filtered, and washed with hot water until the amount of filtrate is about 300 cubic centimeters. The filtrate should be received in a flask of about one liter capacity. The filtrate will contain the dissolved potash, soda, magnesia, lime, iron and aluminum oxids, and traces of sulfuric and phosphoric acids. For the elimination of the other substances, with the exception of potash, soda and magnesia, a few drops of barium nitrate are added, afterwards sufficient ammonia to render the solution alkaline, and finally an excess of ammonium carbonate in powder added in small portions. These materials are added successively and the whole is left to stand for twenty-four hours. By this operation the sulfuric acid is separated in the form of barium sulfate; the iron and aluminum oxids are precipitated, carrying down with them the phosphoric acid, and the lime is thrown down in the form of carbonate. The mass is now filtered and washed several times with hot water. The filtrate contains in addition to potash, soda, magnesia, and the ammoniacal salts which have been introduced. The ammonium salts are destroyed by adding aqua regia and evaporating the liquid to a very small volume, as described in the method for the estimation of magnesia. The mass is now evaporated in a porcelain dish with a flat bottom, of about seven centimeters diameter, and an excess of perchloric acid added. The evaporation is carried to dryness on a sand-bath, and the heating prolonged until the last white fumes of perchloric acid are disengaged. The mass is now left to cool. There are then added five cubic centimeters of alcohol, of 90° strength. The mass is triturated by a stirring rod, the extremity of which is flattened, in such a manner as to

reduce it all to an impalpable powder. It is then left to settle and the supernatant liquid is decanted upon a small filter. The treatment with alcohol of the kind, quantity, and strength described, is continued four or five times. Afterwards, as there may still remain a trace of the sodium and magnesium perchlorates in the interior of the crystals of potassium perchlorate, there are added to the capsule in which all of the alkaline residue has been collected, two or three cubic centimeters of water, and it is evaporated again to dryness and taken up twice with small quantities of alcohol. There are thus removed the traces of sodium and magnesium perchlorates. By means of a jet of boiling water the stirring rod and the filter, which contains the small quantities of potassium perchlorate, are washed, and the liquid passing through is received in the capsule which contains the larger part of the salt. It is then evaporated to dryness and weighed.

When there is very little magnesia present, as is generally the case, the estimation of the potash is made without any difficulty by the process just mentioned, but when the proportion of magnesia is high it is found useful to separate it before the transformation into perchlorates. The magnesia is separated by carbonating the residue as indicated in the method for the estimation of magnesia, by treatment with oxalic acid and ignition. By extracting the carbonates formed with very small quantities of water, and filtering, the alkalies are obtained free from magnesia.

It is advisable to test the purity of the potassium perchlorate formed which sometimes contains a little silica. For this purpose it is dissolved in boiling water, and any residue which remains is weighed, and that weight deducted from the total weight of perchlorate. By multiplying the weight of potassium perchlorate found by the coefficient 0.339, the quantity of potash contained in the twenty grams of earth submitted to analysis is obtained.

Estimation of the Potash Soluble in Cold, Dilute Acids.—(Method of Schloesing.) Introduce 100 grams of the soil into a one or one and a half liter flask with 600 to 800 cubic centimeters of water. A little nitric acid, of 30° Baumé, is added until the carbonate is decomposed and a slight acid reaction is obtained. Afterwards five cubic centimeters of the same acid are added and it is left to digest for six hours, shaking every fifteen minutes. Instead of taking the whole of the wash-water for the examination, it is

better to extract only a portion of it and so dispense with washing. This process is conducted in the following manner:

The weight P of the full flask having been determined, as much as possible of the solution, is decanted by means of a very small siphon, of which the flow is moderated by fixing a rubber tube with a pinch-cock to its lower extremity. After the decantation is complete, the flask is again weighed, giving the weight of P' ; the weight of liquid taken, therefore, is equal to $P - P'$. To determine the total weight of the liquid, throw upon a filter the earthy residue insoluble in the acid, and after washing and drying it determine its weight r . The weight of the empty dry flask p is also determined. The total weight of the soil will be, therefore, $P - r - p$. The part of the liquid which was extracted from the flask, and upon which the analytical operation is to be conducted is represented by the fraction $\frac{P - P'}{P - r - p}$. This method avoids washing and evaporation which would be of very long duration. It rests upon the supposition that the solid matter from which the liquor is separated has no affinity for the dissolved substances, and that the total of these substances has passed into the liquor, and that the solution is homogeneous.

In the liquor first decanted as described before, the potash is estimated. This liquor contains in addition to potash, soda, lime, magnesia, iron and aluminum oxids, as well as phosphoric, sulfuric, and hydrochloric acids. There is first added to it a little barium chlorid to precipitate the sulfuric acid. It is then heated to about 40° in a glass flask and some ammonium carbonate added in a solution containing an excess of ammonium hydroxid. By this process the lime and baryta are precipitated in the form of carbonates; the alumina and iron as oxids, and the phosphoric acid in combination with the last two bases. The magnesium carbonate is not precipitated because it is soluble in the ammonium carbonate with which it forms a double salt.

The employment of a gentle heat favors the formation of the precipitate of calcium carbonate in a granular form which lends itself easily to filtration. The contents of the flask are now thrown upon a filter and the insoluble residue washed. The filtrate contains the potash, soda, magnesia, ammonia, and nitric and hydrochloric acids. It is concentrated as rapidly as possible by heating in a flask, and afterwards the ammoniacal salts are destroyed by weak aqua regia and the whole is then transferred to a

porcelain dish and evaporated to dryness. There is thus obtained a mixture of potassium, sodium, and magnesium nitrates, from which the potash is separated by means of perchloric acid in the manner already described.

Estimation of the Total Potash.—Beside the potash which can be dissolved by the boiling concentrated acids the soil contains potash combined with silicates, which becomes useful for plant life with extreme slowness. It is often of great interest to estimate the total potash contained in a soil, that is to say, the reserve for the future. In this case it is necessary to free entirely this base from its combinations by means of hydrofluoric acid. The operation is conducted upon two grams of earth previously ignited and reduced to an impalpable powder. The decomposition is conducted in a platinum capsule by sprinkling the sample with a few cubic centimeters of hydrofluoric acid, or solution of ammonium fluorid, and adding a few drops of sulfuric acid. It is then evaporated to dryness and dissolved in boiling hydrochloric acid. The part which remains insoluble is treated a second time by hydrofluoric and sulfuric and afterwards by hydrochloric acid. All of the potash is thus brought into solution. The estimation of the potash, after having obtained it in a soluble state, is conducted in the manner previously described.

Estimation of the Potash as Platinochlorid.—Instead of estimating the potash as perchlorate it can also be transformed into platinochlorid. This process gives as good results as the preceding one but it is necessary in all cases, to separate the magnesia. After having treated the soil as indicated in the case of the estimation of the potash as perchlorate, the separation of the sulfuric and phosphoric acids, of alumina and iron, of magnesia, and the destruction of the ammoniacal salts in the manner already described, there are finally left the alkalies potash and soda in the form of carbonates. These are transformed into chlorids by adding hydrochloric acid; afterwards they are evaporated to dryness and the mixture of the two chlorids weighed in order to determine what quantity of platinum chlorid it is necessary to add, in order that it be in excess. The quantity of chlorid to be added is calculated so as to be in sufficient quantity to saturate the whole of the chlorids weighed, whether they may be composed wholly of sodium or potassium. In this way there is a certainty of having an excess of platinum. The solution of platinum chlorid used should contain in 100 cubic centimeters seventeen grams of platinum. Each cubic centimeter of this solution will be sufficient for a decigram of the sodium and potassium

double chlorids. After the addition of the platinum chlorid the mixture is evaporated in a capsule with a flat bottom, on a water-bath. It is important that the temperature should not exceed 100°. If the temperature should go above this there would be a tendency to form some platinum subchlorids insoluble in alcohol.

The evaporation is continued until the contents of the dish are in a pasty condition and form a rather solid mass on cooling. It is necessary to avoid a complete desiccation. After cooling, the residue is taken up by alcohol of 95° strength. It is allowed to digest with alcohol of this strength for some time, after having been thoroughly mixed and shaken therewith in order to obtain a complete precipitation of the platinochlorid. This digestion should take place under a small bell-jar resting upon a piece of ground glass. The evaporation of the alcohol is thus prevented. The mass is then washed by means of alcohol of the same strength and the liquors decanted upon a small filter placed within another filter of identical weight, which serves as a tare for it on the balance. The washing is prolonged until the filtrate becomes colorless. All of the particles in the dish should be brought upon the filter by means of a hair-brush. The filters are now dried at a temperature not exceeding 95° and the platinochlorid received upon the interior filter is weighed. The precipitate may also be washed from the small filter into the capsule in which it was formed by means of a jet of alcohol. The alcohol is evaporated and the precipitate weighed in the capsule. The weighing should be made rapidly on account of the hygroscopicity of the material. The weight obtained multiplied by 0.193 gives the corresponding quantity of potash in the soil.

Purification of the Oxalic Acid.—The commercial oxalic acid used in separating the magnesia, often contains lime, magnesia, and potash. When this reagent is used in a sufficiently large quantity in the estimation of the above substances, it is indispensable to free it entirely from them. This is secured by submitting the oxalic acid to successive recrystallizations which are obtained by dissolving it in warm water, filtering and leaving to cool. The mother waters are thrown away. After two or three successive crystallizations the traces of potash and magnesia have disappeared and the oxalic acid obtained after ignition leaves no trace of residue.

The purification may also be secured in the following manner: At a temperature of 60° a saturated solution of oxalic acid is made; the liquid is

decanted, carried to the boiling point and filtered. Five per cent of nitric acid are added and it is allowed to cool. The crystals which are deposited are collected upon a funnel in which a plug of cotton has been placed, and are washed with a little cold water.

Purity of the Ammonium Carbonate.—The ammonium carbonate employed should not leave any residue whatever after volatilization. In general, it may be said of all the reagents employed in analyses and especially of those employed in large quantities, that it is indispensable to be sure that they contain no traces of the substances which are to be estimated. The acids, ammonia, etc., should always be examined with this point in view.

Estimation of the Soda.—It is often of interest to estimate the soda in the soil, not that it is an element of any great fertility but rather because it is hurtful when in excess. It is determined in the residue obtained in the estimation of potash and is estimated by difference. The weight of the mixture of sodium and potassium chlorids being known when the potash is determined, the weight of its chlorid is to be deducted from the weight of the two chlorids and thus the direct weight of the sodium chlorid is obtained.

A better way is to make a direct estimation. The soda is found entirely dissolved in the alcoholic solution obtained by washing the potash salt as before described, for the separation of the potassium platinochlorid. This alcoholic liquor is evaporated to dryness on a water-bath, in a bohemian flask of about 100 cubic centimeters capacity. The residue obtained consists of sodium platinochlorid and a little platinum chlorid. There is now fitted to the bohemian flask a cork stopper carrying two tubes. The apparatus is placed upon a water-bath and kept at about 100°. Through the tube which reaches to the bottom of the bohemian flask, a current of pure hydrogen is passed. The hydrogen passes off through the second tube. The hydrogen completely reduces the salts of platinum. In order that the decomposition may go on more rapidly a few drops of water are added. When the whole mass in the flask has become black owing to the separation of the platinum, it is shaken, evaporated to dryness and hydrogen passed through a second time. This operation is repeated three or four times, being stopped when the water no longer shows a yellow color. There is then in the flask only a mixture of reduced platinum and sodium chlorid. No trace of sodium

chlorid has been lost because the temperature has never exceeded 100°. The sodium chlorid is dissolved by washing with water and filtered. The liquor, which must be absolutely colorless, is evaporated to dryness in a platinum capsule and weighed. There is thus obtained the weight of the sodium chlorid. For verification, the sum of the weight of potassium chlorid calculated from the platinochlorid and the weight of the sodium chlorid should be equal to the initial weight of the mixture of the two chlorids.

351. Potash Methods of the German Experiment Stations.^[225]—*a.* To one volume of air-dried fine earth which is obtained by sifting through a three millimeter sieve, two volumes of twenty-five per cent hydrochloric acid are added, or more if the soil contains much carbonate. The acid is allowed to act with frequent stirring for forty-eight hours at room temperature.

b. To one volume of the soil, as above prepared, are added two volumes of hydrochloric acid and allowed to stand for three hours with frequent shaking, at the temperature of boiling water.

c. (Halle method.) One hundred grams of the fine earth are treated with 500 cubic centimeters of forty per cent hydrochloric acid, made up to one liter with water and allowed to stand for forty-eight hours with frequent shaking. After filtering, a large aliquot part of the filtrate is evaporated for the estimation of the potash. The evaporated residue is washed into a half-liter flask in which the sulfuric acid is precipitated with barium hydroxid the flask filled to the mark and an aliquot part of the filtrate in a half-liter flask, treated with ammonium carbonate, filtered and the potash estimated as platinochlorid by the usual method.

352. Method of Raulin for the Estimation of Potash in Soils.^[226]—The process rests upon the very feeble solubility in aqueous solution of potassium phosphomolybdate, while sodium, magnesium, calcium, iron, and aluminum phosphomolybdates are more or less soluble. The process does not require complicated separation and permits of the treatment of a small quantity of soil, since the weight of the phosphomolybdate obtained is equivalent to nineteen times that of the potash.

The reagent is prepared by dissolving 100 grams of crystallized ammonium molybdate in as little water as possible and adding six and a half grams of neutral crystallized ammonium phosphate dissolved in a little water. Aqua regia is now added cold and some ammonium

phosphomolybdate is precipitated. The mixture is heated, adding a little aqua regia from time to time, until the solution of the precipitate is accomplished. The whole is then evaporated to dryness, the final temperature of evaporation not being carried above 70°. Four hundred cubic centimeters of water are now added and five cubic centimeters of nitric acid, and the contents of the dish heated and filtered. The reagent is then ready for use.

The liquid to be used for washing the potassium phosphomolybdate is prepared by dissolving twenty grams of sodium nitrate in one liter of water, two cubic centimeters of pure nitric acid, and a mixture of about twenty cubic centimeters of the phosphomolybdic reagent and one and a half cubic centimeters of a solution of potassium nitrate containing eighty grams per liter, slightly heated in order to saturate the liquid with potassium phosphomolybdate. The solution is shaken, allowed to rest, and the liquid decanted.

For the preparation of the solution in which the potash is to be estimated, a sample of soil is carefully weighed of such magnitude as to contain about fifteen milligrams of anhydrous potash. The potash salts are dissolved by the usual processes and are separated from the largest part of the calcium, iron, and aluminum salts, and converted into nitrates. The solution is reduced to a volume of a few cubic centimeters and slightly acidulated with nitric acid. Four cubic centimeters of the phosphomolybdic reagent are added for every ten milligrams of anhydrous potash supposed to be present. The solution is evaporated to dryness at 50° and immediately brought upon very small weighed filters, of which each one is double, by using sixty cubic centimeters of the washing liquor mentioned above. The tared filter is likewise washed with the same liquid at 50° and weighed. The weight multiplied by 0.052 gives the anhydrous potash. This method for a direct precipitation of the potash salts does not have the merits of the perchlorate process and both are inferior in accuracy to the usual platinochlorid procedure.

353. Russian Method for Estimating Potash in Soils.^[227]—Ten grams of the soil are digested with 100 cubic centimeters of ten per cent hydrochloric acid on a steam-bath for twenty-four hours. After adding five cubic centimeters of nitric acid to the filtrate it is evaporated to dryness, taken up with dilute hydrochloric acid, filtered, the filtrate saturated with

ammonia, the excess of ammonia driven off, again filtered, and the lime separated by ammonium oxalate.

The filtrate is treated with a little barium chlorid for the removal of sulfuric acid and afterwards with ammonium carbonate in excess, and digested for twenty-four hours. After filtering, the solution is evaporated in a platinum dish, the excess of ammonia driven off, the residue taken up with water, filtered, treated with hydrochloric acid, evaporated to dryness, and ignited at low heat. The residue is again dissolved in water, filtered, and the potash precipitated with platinum chlorid and estimated in the usual way.

354. Potash Method of the Italian Stations.^[228]—The potash in the soil should be determined in three forms; viz.,

1. Assimilable potash.
2. Potash soluble in concentrated acid.
3. Total potash.

For determination of the first, 100 grams of earth are put into a retort holding a liter and digested with dilute nitric acid.

For the analysis, an aliquot portion of the clear liquid is taken or weighed, and the determination of the potash is made by the common methods.

For an alternate method, from twenty to fifty grams of earth are put into a retort of 500 cubic centimeters, moistened with water, and nitric acid is gradually added. After one or two hours there are added from 200 to 300 cubic centimeters of water; the liquid is poured without filtering into a retort and the residue washed by decantation.

In the liquid, after the elimination of the other substances with barium chlorid, ammonium carbonate, etc., the potash is determined by the ordinary methods.

In the second case, by using warm concentrated acids, a portion of the insoluble silica is decomposed, but this decomposition is always partial and the quantity of the potash extracted depends upon the temperature, upon the concentration, upon the duration of the action, and upon the nature of the acid.

The method of moistening twenty to fifty grams of earth with water and adding, thereto, concentrated nitric acid of 1.20 density, in such a manner

that the earth shall be completely saturated, may also be employed. Then the temperature is kept at 100° during two hours. In the solution, the potash is determined as usual.

In the third case the soil is to be decomposed by hydrofluoric and sulfuric acids, or by fusion with alkaline carbonates, and the total potash determined by one of the standard methods.

If it is desired to adopt a general method for the determination of the potash the following points must be carefully considered:

1. The quantity of the earth to be examined.
2. The state of humidity or dryness of the same.
3. The quantity, nature, and concentration of the acid.
4. The quantity of the water.
5. The duration of the treatment.

355. Method of J. Lawrence Smith for Potash.—This method, designed especially for mineral analysis, has been fully approved by the general experience of analysts.

The principle of the method^[229] depends upon the decomposition of silicates on ignition with calcium carbonate and ammonium chlorid. The object of this mixture is to bring into contact with the mineral, caustic lime in a nascent state at a red heat, the caustic lime being soluble to some extent in calcium chlorid at a high temperature. Pure calcium carbonate, made by precipitation of marble, should be used.

The ammonium chlorid should be prepared by taking crystals of pure, sublimed sal ammoniac, dissolving in water, and filtering, and evaporating the solution until small crystals are deposited, the solution being well-stirred until one-half or two-thirds of the whole has crystallized. The mother-liquor is poured off while still hot, and the crystals dried on an asbestos filter at ordinary room temperature.

A special platinum crucible should be used in the Smith method, but the common crucible, especially if very deep, can be employed. The special crucible is of about double the usual length. Smith recommends a crucible ninety-five millimeters in length, diameter at top twenty-two millimeters, at bottom sixteen millimeters, and weighing thirty-five to forty grams. The object of the long crucible is to have the part of the bottom containing the

silicate subjected to a high heat, while the top of the crucible is at a much lower temperature, thus preventing the loss of alkalies by volatilization.

Method of Analysis.—The samples of soil or silicate containing the alkalies are well pulverized in an agate mortar, and from one-half to one gram of the finely pulverized material taken for analysis. This is carefully mixed with the same weight of finely powdered sal ammoniac and the mineral and sal ammoniac rubbed well together in a mortar. Eight parts by weight of calcium carbonate are next added in three or four portions, and the whole intimately mixed after each addition. The contents of the mortar are emptied on a piece of glazed paper and then introduced into the crucible, which is tapped gently upon the table until the contents are well settled. It is then fixed in the furnace which is used for heating, and a small bunsen burner is placed beneath the crucible, and the heat applied just about at the top of the mixture and gradually carried toward the lower part until the sal ammoniac is completely decomposed, which requires from four to five minutes. The heat is then applied by means of a blast-lamp and the crucible kept at a bright red heat for from forty to sixty minutes. The crucible is allowed to cool, the contents detached and placed in a platinum or porcelain dish of about 150 cubic centimeters capacity, and sixty to eighty cubic centimeters of distilled water added. The solution of the flux may be hastened by heating the water to the boiling point. The crucible and its cover are also well washed with hot water until all matter adhering to them is dissolved. After the slaking of the mass it is best to continue the digestion with hot water for six or eight hours, although this is not absolutely necessary. The contents of the crucible are filtered and washed well with about 200 cubic centimeters of water. The filtrate contains in solution all the alkalies of the mineral, or soil, together with calcium chlorid and caustic lime. A solution of pure ammonium carbonate containing about one and one-half grams of the pure salt is added to the filtrate. This precipitates the lime as carbonate. The dish containing the material is placed on a water-bath and its contents evaporated to about forty cubic centimeters. Two additional drops of ammonium carbonate are added, and a few drops of caustic ammonia, to precipitate any lime which may be redissolved by the action of the ammonium chlorid solution on the calcium carbonate. Filter on a small filter and wash with as little water as possible and collect the filtrate in a small beaker. The filtrate contains all the alkalies as chlorids, together with a little ammonium chlorid. Add a drop of ammonium

carbonate solution to be sure all the lime is precipitated, evaporate on a water-bath in a deep platinum dish, in which the alkalies are to be weighed. The dish should have from thirty to sixty cubic centimeters capacity, and during the evaporation should never be more than two-thirds filled. After the evaporation has been completed the dish is slowly heated and then gently ignited over a gas-flame to drive off any ammonium chlorid which may be present. During this process the platinum dish may be covered with a thin piece of platinum to prevent any possible loss by the spitting of the salt after the ammonium chlorid has been driven off. The heat should be gradually increased until it is brought to a point a little below redness, leaving the cover off. The platinum dish is again covered, and when sufficiently cooled placed on a balance and weighed.

If lithium chlorid be present it is necessary to weigh it quickly as the salt being very deliquescent takes up moisture rapidly. The alkalies may now be separated in the usual way.

If the sample under examination contains magnesia the residue in the capsule should be dissolved in a little water and sufficient pure lime-water added to render the solution alkaline. It should then be boiled and filtered. The magnesia will, in this way, be completely separated from the alkalies. The solution which has passed through the filter is treated with ammonium carbonate in the manner first described, and the process continued and completed as above mentioned.

If it be suspected that the whole of the alkalies have not been obtained by the first fusion, the residue upon the filter can be rubbed up in a mortar with an amount of ammonium chlorid equal to one-half the weight of the mineral, mixed with fresh portions of calcium carbonate and treated exactly as in the first instance. Any trace of alkali remaining from the first fusion is thus recovered in the second one.

Method of Heating the Crucible.—The apparatus used by Smith for igniting the crucible is shown in Fig. 67. It consists of an iron filter-stand HG, a clamp, ED, carrying the muffle NC, attached by the supports AB, and heated by the lamp F. The muffle NC is a chimney of sheet iron, eight to nine centimeters long, ten centimeters high, the width at the bottom being about four centimeters on one side and three centimeters on the other. It is made with the sides straight for about four centimeters and then inclining toward the top so as to leave the opening at the top about one centimeter in width. A piece is cut out of the front of the chimney of the width of the diameter of the hole in the iron support and about four centimeters in length, being semi-circular at the top, fitting over the platinum crucible. Just above this part of the chimney, is riveted a piece of sheet iron in the form of a flattened hook, N, which holds the chimney in place by being slipped over the top of the crucible support; it serves as a protection to the crucible against the cooling effects of the currents of air.

356. International Method for Assimilable and Total Potash.—In the International Congress of Chemists held in Paris in 1889,^[230] the discrimination between the assimilable and total potash was declared to be of prime importance. Unfortunately no method is known by which the potash which is present in the soil in a state suited to the wants of plants can

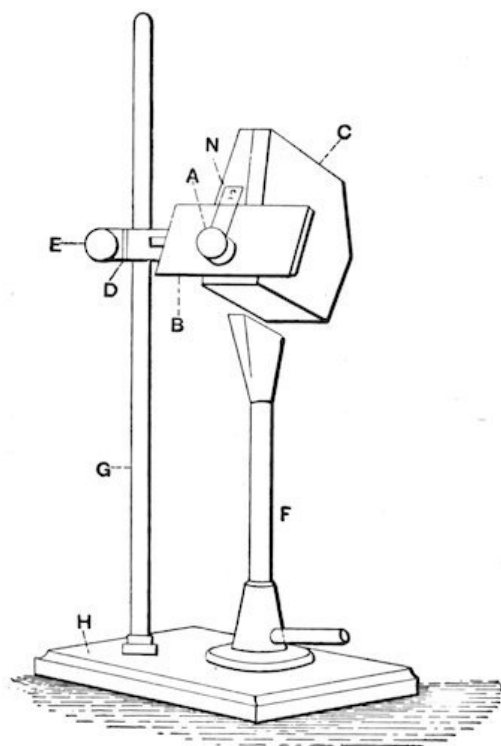


FIGURE 67.

SMITH'S MUFFLE FOR DECOMPOSITION OF
SILICATES.

be determined with approximate accuracy. In general, that portion which is given up to weak acids may be assumed to be available. In the treatment of soils with weak acid, as pointed out in the Congress, it is demonstrable that with a 0.05 to 0.1 per cent nitric acid solution, the quantity of potash which goes into solution increases by continued stirring of the mixture with the time of action of the acid up to a certain maximum which is reached in from three to four hours, and after that, it is not changed even when the strength of the acid mixture is increased to two per cent. From this time on, concentrated acids withdraw from the soil which has already been exhausted by the weak acid, a new quantity of potash. The soils which have been exhausted by concentrated acids yield also an additional quantity of potash when they are treated with hydrofluoric acid, or

melted with barium or sodium carbonate. Potash, therefore, appears to exist in the soil in various forms.

First. In the form of indecomposable silicates which have, agriculturally perhaps, very little interest.

Second. In the form of silicates which are more basic than those just mentioned. These silicates are attacked by strong acids and give up probably every year a portion of their potash to vegetation.

Third. In a form which is easily soluble in weak acids and consequently directly assimilable by plants.

In view of the fact that it would be of interest to chemists and agronomists to establish certain methods of investigation so as to be able to obtain comparative results, it was decided to adopt the original method recommended by Gasparin for the estimation of the potash decomposable

by concentrated acids. This method consists in the treatment of the soil with boiling aqua regia until the sand which is not decomposed, is white.

Determination of the Fineness of the Earth which is Used for Analysis.—For the estimation of potash, the soil should be divided as finely as possible, and passed through a sieve of thirty meshes to the centimeter. The decomposition is then completed in two hours, while if a sieve of only ten perforations per centimeter is used, the acid must be allowed to work for twelve hours.

The determination of the potash after solution, is accomplished by any of the standard methods.

357. Method of Tatlock as Used by Dyer.—Attention was called in paragraph 328 to the estimation of the total plant food in the soil by extraction of the sample with citric acid. Dyer first determines the total potash by Tatlock's method which is as follows:

To determine potash ten grams of fine dry soil are treated with ten cubic centimeters of hydrochloric acid and evaporated to dryness on the water-bath, the residue taken up with another ten cubic centimeters of acid, warmed, diluted with water, boiled, filtered, and washed. The filtrate and washings are concentrated and gently incinerated to get rid of organic matter, and the residue redissolved in hydrochloric acid, and evaporated slowly with a considerable quantity of platinum chlorid. If the evaporation be conducted slowly, the potassium platinochlorid settles out well, despite the iron, aluminum, and calcium salts, and is easily washed with some more platinum chlorid solution, followed by alcohol. The application of this modification of the platinum chlorid process to solutions containing comparatively minute quantities of potash amid an overwhelming excess of iron, aluminum, and calcium salts is probably new to many chemists. It works admirably, and obviates the necessity for removing iron, aluminum, calcium, magnesium, etc., with the necessary use of ammonia, and the tedious processes of concentration and final volatilization of the ammonium salts; but, of course, the process cannot be employed if soda also is to be determined.

The potash, soluble in hydrochloric acid, having been thus determined, the undissolved siliceous matter is incinerated, weighed, and finely ground in an agate mortar. A weighed portion of it is then, as in the Smith method, mixed with a large bulk of pure calcium carbonate and a little ammonium

chlorid and heated, beginning with a low temperature, rising slowly to bright redness. The mass is then boiled with water, washed, incinerated, reground, mixed with some more ammonium chlorid, and again heated, boiled, and washed. The process is repeated and the filtrates from all the treatments concentrated, the calcium being removed as carbonate, and the potash determined in the filtrate, after evaporation and incineration at a low temperature, by means of platinum chlorid.

Five hundred cubic centimeters of the citric acid solution of the soil, made as described in paragraph 328, corresponding to fifty grams of soil, are evaporated to dryness in a platinum dish and ignited at a low temperature. The residue is dissolved in hydrochloric acid filtered and washed, and the filtrate again evaporated to dryness and treated again as just described. The potash is then determined as above.

358. Estimation of Total Alkalies and Alkaline Earths.—To properly determine the exact amount of these substances in a sample of soil it is necessary first to remove the silica. This is accomplished in the process of Berthelot and André^[231] by intimately incorporating with the sample, in a state of very fine powder, four or five times its weight of ammonium fluorid. The mixture, in a platinum dish, is moistened with strong sulfuric acid and allowed to stand for a few hours. It is then gently heated until all fumes of hydrofluosilicic acid have disappeared, but the mass is not raised to a red heat. If there is any doubt about the complete decomposition of the silica the treatment is repeated.

At the end of the operation there remain only sulfates without excess of sulfuric acid. The sulfates likely to be present are of potash, soda, lime, magnesia, alumina, and iron. The separation of these bodies is conducted in the ordinary manner.

Fusing the soil with potash does not give reliable results but it can be used in certain cases for the rapid estimation of alumina and iron. In this case after the separation of the silica in the ordinary way the iron can be determined as ferric oxid.

The iron can also be directly determined by reducing to the ferrous state and titrating with potassium permanganate.

Comparison of Fluorin Method with Common Methods.—To establish the difference in the data obtained by the old and new processes samples of

the same earth were treated by Berthelot and André by different methods with the following results:

	By the fluorin method.	By the cold dilute hydrochloric acid method.	By the concentrated hydrochloric acid method.	By incineration and subsequent treatment with boiling hydrochloric acid method.
	Per cent.	Per cent.	Per cent.	Per cent.
Potash	0.886	0.021	0.149	0.176
Soda	0.211	0.024	0.033	0.042
Magnesia	0.087	0.033	0.033	0.067
Lime	1.160	0.879	1.120	1.060
Alumina	3.950	0.102	1.009	2.631
Ferric oxid	2.150	0.296	1.401	1.678

The impossibility of getting all the alkalies and oxids into solution by even the prolonged action of a boiling acid is clearly set forth in the above table. Boiling sulfuric acid might do a little better but would not give correct results. Lime alone of the elements in the soil can be correctly determined by solution in boiling hydrochloric acid, a circumstance due to the fact that lime is found chiefly as carbonate, sulfate, and phosphate in the soil, and these compounds are easily soluble in hot hydrochloric acid with the exception of the sulfate. Even lime could not be thus determined in soils containing silicates rich in lime. The other mineral elements cannot be determined by the wet method. This is due to the forms in which they occur, being mostly silicates of different composition, with excess of silica.

As to the silicates they may be divided into two groups. The first of these are the hydrated silicates, resembling the zeolites, capable of being completely decomposed by boiling acids. The first group of silicates is doubtless of greater importance to vegetable life than the second since it would, doubtless, give up its alkalies with greater ease. This distinction is, however, arbitrary. It is, in fact, impossible to place on one side the soluble and on the other the insoluble silicates. This distinction represents only the unequal degrees in the speed of decomposition of the different silicates contained in the primitive rocks under the influence of atmospheric agents,

the soil being nothing more than the products of the decomposition of these rocks with vegetable mold. The second group is insoluble in acids.

That part of the silicates least decomposed at any given moment will be attacked more easily by acids, while that portion whose decomposition has been pushed furthest will be more slowly attacked. The action of the acid will grow more feeble as the time of contact is prolonged, and after a time a point is apparently reached where the results are nearly constant. But it is evident that this distinction is purely conventional and bears no necessary or even probable connection with the quantity of alkali really assimilable by plants.

Vegetables, moreover, exert on a soil, for the extraction of its alkalies and other matters, chemical reactions peculiar to themselves, altogether distinct from the tardy action of atmospheric agents and still more distinct from the rapid action of mineral acids.

It is well known with what energy, it ought to be said with what admirable instinct, plants take from the soil the least traces of phosphorus, of sulfur, of potash, of iron, and other substances necessary to their sustenance.

These specific actions of vegetables on the soil merit, in the highest degree, the attention of analysts and agronomists. Their intervention plays a most important part in the restitution to the soil, by means of complementary fertilizers, the mineral elements removed by vegetable growth.

359. Estimation of Lime by the French Method.—The quantity of lime contained in the soil varies within wide limits. Sometimes this base is entirely absent to such a degree that it is even impossible to discover feeble traces of it. Sometimes it composes almost the whole of the earthy mass. Lime is found in the soil principally in the state of carbonate. It is also found combined with organic matter under the form of humates, with sulfuric acid, etc. It is customary to estimate the lime as a whole, without distinguishing between the different states in which it exists. The quantity of material which is used in the French method^[232] varies in proportion to the amount of calcareous matter contained in it. For a soil which contains a large amount of lime, one or two grams would be sufficient for the analysis. For a soil which is poor in calcareous matter ten or even twenty grams must be taken. The quantity of lime dissolved differs according to the strength of

the acids employed and length of contact of the acid with the soil. The calcium carbonate, the sulfate, the nitrate, and the humate rapidly pass into solution when treated with acid as above, but this is not the case with calcium silicates which are attacked much more slowly. Sometimes the silicates give only an insignificant increase in the amount of lime, and in this case it is immaterial what process of solution is employed. For simplicity it is best to adopt the method of solution in boiling concentrated nitric acid, prolonging the boiling for a period of five hours. This method of operation is sufficient to bring into solution at one treatment, not only the lime, but also the potash and magnesia. After having heated with acid for the necessary time there are added in the capsule in which the solution took place ten cubic centimeters of nitric acid and fifty cubic centimeters of water. The mixture is heated, collected upon a filter and the residue washed. To the filtrate, the volume of which should be from 400 to 500 cubic centimeters, a sufficient quantity of ammonia is added to render it slightly alkaline. There is formed a precipitate of alumina and of iron oxid containing phosphoric acid and also sometimes a trace of the lime combined with the same acid. In order to keep the whole of the lime in solution it is necessary to add a little acetic acid, about ten cubic centimeters more than is necessary to neutralize the ammonia which has been added in excess. If the liquid is turbid on account of the presence of the iron and aluminum phosphates it is necessary to filter it. There is afterwards added a slight excess of ammonium oxalate in solution, and the whole is left for twenty-four hours in order that the calcium oxalate may deposit. Indeed, the complete precipitation is not always immediate, and especially in the presence of magnesia it takes place with slowness. The calcium oxalate is collected upon a filter and washed with hot water. To determine the quantity of the lime the best procedure consists in transforming the oxalate into carbonate by a careful ignition, and afterwards heating in a Schloesing or Leclerc furnace for four or five minutes. The oxalate for this purpose should be contained in a covered platinum crucible. By this method the calcium carbonate is transformed into calcium oxid, in which form it is weighed rapidly to avoid absorption of moisture.

In laboratories which have no means of securing so high a temperature as is mentioned before, the lime may be weighed as sulfate. For this purpose the calcium oxalate is transformed into carbonate by ignition in a platinum crucible. Afterwards it is treated with nitric acid until the carbon dioxid is

completely driven off. The platinum crucible is now covered with a funnel which is afterwards washed in order to bring back into the dish the small drops which have been projected in the process of boiling. An excess of sulfuric acid is added and evaporated to dryness on a sand-bath. Afterwards, in a muffle, the temperature is carried to a feeble redness until the vapors of sulfuric acid are all driven off. The lime is weighed in the form of sulfate, and the weight multiplied by 0.412 gives the lime contained in the quantity of earth analyzed.

In special researches in which it is desired to avoid attacking the siliceous pebbles of the soil, the concentrated nitric acid is replaced by dilute nitric acid in slight excess, and heated for a few moments only. The calcium carbonate is then dissolved with the other calcareous salts not combined with silica in the rock products. The analysis is continued in other respects as just described.

360. Estimation of the Actual Calcium Carbonate.—The lime which is found in the state of carbonate plays one of the most important rôles in the chemical phenomena which take place in the soil. It is often of great importance to determine it. The most certain process is to estimate the carbon dioxid which is disengaged from the carbonate under the influence of an acid and to receive this gas in a jar graduated to measure it by volume. The flask recommended by the French Commission for this purpose contains about 300 cubic centimeters. The neck of the flask is connected with a condensing tube of about one centimeter interior diameter, which is cooled by a current of water.

According to the presumed richness in calcium carbonate varying quantities of earth are taken for analysis, from as little as half a gram for soils which are rich in carbonate, up to five or even ten grams for soils which are poor in carbonate. The apparatus is connected with a mercury pump for the purpose of exhausting the air as completely as possible therefrom. For this purpose the flask in which the carbonate is disengaged is made in the shape of a tubulated retort. Through the opening into the retort, a narrow tube is introduced and connected with a small funnel by means of a rubber tube supplied with a pinch-cock. When the retort has been connected with the mercury pump a slight vacuum is produced and the pinch-cock is opened and forty cubic centimeters of distilled water allowed to enter. The pinch-cock is closed soon enough to retain a portion of the

water in the funnel. The retort is then heated and a vacuum partially produced by means of the pump. When the flask is boiling, the steam drives out the air. A refrigerating jacket is connected with the tube leading from the retort to the pump by means of which the steam is condensed and falls back into the flask. After some minutes of boiling, a vacuum is produced; the lamp is then taken away and a cylinder, graduated at 100 cubic centimeters and filled with mercury, is placed over the lower orifice of the pump, and there is introduced into the apparatus, by the funnel above described, some hydrochloric acid in small quantities, but sufficient only to saturate the whole of the carbonate in the sample of soil taken. Usually three or four cubic centimeters will be sufficient. The acid should be added in such quantities as to prevent the production of any large amount of foam. If frothing should be excessive a little oil can be added to the flask. The whole of the carbon dioxide produced in the reaction is withdrawn by means of the mercury pump and collected in the graduated jar. Towards the end of the operation the flask is heated anew in order to produce an ebullition which is continued for some time. The volume of gas collected is measured after making the proper corrections for pressure and temperature. Afterwards the carbon dioxide which has been produced is absorbed by two or three cubic centimeters of a solution of potash of 42° baumé. This potash is introduced into the graduated jar by means of a pipette bent into the form of a  in the lower portion. If the whole of the gas is not absorbed the volume which remains is read, and this is subtracted from the original volume after having made the proper corrections for pressure and temperature. The difference gives the quantity of carbon dioxide contained in the amount of earth employed. From this the actual weight of the calcium carbonate is computed. This official French method does not appear to possess any advantage in accuracy to the usual absorption method and is far more complicated.

361. Estimation of the Active Calcareous Matter in Soils.—Like other soil elements, the calcium carbonate exists in different degrees of fineness and availability in the soil. It must be admitted that the fine particles play the most important rôle. The calcium carbonate, which exists in large fragments, presents only a circumscribed surface and remains almost inactive, although it is easily corroded by the rootlets of plants. It is possible to estimate in a rapid way, the quantity of fine carbonate in the soil, considering that in a time relatively short, feeble acids act upon calcareous

matter proportionally to the surface which it presents, and that it attacks, therefore, especially the finest particles. By measuring the amount of carbon dioxide set free under the action of dilute acids it is possible to estimate the content of available calcareous matter in the soil.

The apparatus of Mondesir is used for this purpose by the French chemists. It is composed of a tubulated flask of about 600 cubic centimeters capacity. The interior tubulature carries a manometer fixed by means of a stopper. This is formed of a rubber tube, terminated by a glass tube, whose extremity is united to a little rubber bag, very flexible, placed in the interior of the flask.

Graduation of the Apparatus.—If the apparatus is new it is necessary to begin by graduating it. The rubber bag is filled with water, the air being carefully excluded, in such a way that the level of the water comes just a little above the bend in the tube. There are placed in the flask 125 cubic centimeters of water and it is shaken for a few seconds. The flask and the manometer are then unstoppered and the level of the water in the manometer is made to equal the level of the water in the flask. With a rubber ring the level of the water in the manometer tube is marked. The manometer is then stoppered. There are then added to the flask two-tenths gram of pure calcium carbonate. The flask is closed and shaken for a minute. There are then added, enclosed in a little piece of filter paper, six-tenths of a gram of pulverized tartaric acid and the flask immediately closed and shaken several times. The manometer tube is then uncorked and moved until the level of the water reaches the point marked before. The difference in level after the height of the water remains constant is then read. The depression in the level observed, corresponds to two-tenths gram of pure calcium carbonate.

362. Estimation of the Available Calcareous Matter in the Soil.—There is introduced into the flask of the apparatus a quantity of soil varying in amount in accordance with the content of carbonate which it is supposed to contain. There are added 125 cubic centimeters of water and the flask is shaken for a minute. As in the test given before, the level of the water in the manometer is then made to correspond to that of the water in the flask. The level in the manometer is marked as before with a rubber band, and the manometer is then closed. There are then added, contained in a piece of filter paper, two grams of pulverized tartaric acid and the operation is

finished as described before. The amount of tartaric acid added, in general, should be three times as much as the amount of calcium carbonate supposed to be contained in the earth. The pressure in the manometer being proportional to the quantity of carbon dioxide disengaged, it is easy to calculate the quantity of calcium carbonate in a state of fine division contained in the soil taken for the test.

In order to fill the rubber bag it is necessary to put it in its proper place in the apparatus. The flask is filled with water in order to flatten the rubber bag and expel the air from it. It is then closed with a cork. Afterwards, with the aid of a small funnel and with a copper wire placed in the tube, the lower extremity of which descends just to the elbow, the air in the tube is replaced by water. The operation is finished by uncorking the flask and inclining it or shaking it after a partial vacuum has been established. It is useless to attempt to drive off the last particles of the air. The rubber bag should have a content of about double the volume of the whole of the interior of the manometric tube. In the washing which is necessary between two successive operations, it is well to fill the flask entirely with water in order to expel all the carbon dioxide which it may contain.

The same remark may be made of this method of determination as was made of the last one. In the present case, however, the operation is not quite so complicated. When the apparatus is once arranged, it will admit of rapid determinations.

363. Lime Method at the Riga Station.—Ten grams of the non-ignited sample of the fine earth are digested with 100 cubic centimeters of ten per cent hydrochloric acid, in a 250 cubic centimeter erlenmeyer for twenty-four hours on the steam-bath, with frequent shaking. The filtrate, with washings after the addition of five cubic centimeters strong hydrochloric acid, is evaporated to dryness in a porcelain dish and the residue taken up with dilute hydrochloric acid. After filtering, ammonia is added in excess, the excess removed by evaporation, and the mass is again filtered. In the filtrate, the lime is thrown down with ammonium oxalate, filtered, ignited, and weighed as calcium oxide.

The above method cannot give exact results chiefly because more or less lime may be carried down with the phosphoric acid. Also if manganese be present it will be thrown down with the lime. These errors are compensatory, but only by chance could the compensation lead to

exactness. It would be better in all cases to remove the iron and alumina in such a way as would avoid loss of time.

364. Estimation of Assimilable Lime.—In the determination of the total lime in soils or even of that part present as carbonate, it is not to be assumed that the quantity assimilable by plants is known; particles of lime minerals in soils are corroded only superficially by the rootlets of plants and any process which would attack only the superficies of the lime particles would thus more nearly resemble the activity of the solvent forces of plant growth. Oxalic acid is a reagent of this kind, attacking only the surfaces of lime particles. Reverdin and de la Harpe guided by this fact have based a method for determining the amount of lime present in the soil in an available state on the solvent action of oxalic acid.^[233] After the total lime content has been determined, twenty grams of the soil sample are covered with 200 cubic centimeters of a solution containing in molecular proportion a known quantity of sodium oxalate and carbonate. The mixture is digested on the water-bath for one hour. By this treatment all lime minerals are converted superficially into oxalate while particles containing magnesia are not affected. After filtering and washing well, the filtrate and wash-waters are acidulated with hydrochloric acid. If any precipitate of organic matter be produced separate it by filtration. Treat the filtrate with a slight excess of sodium acetate by which process the excess of hydrochloric acid is replaced with acetic after which the oxalic acid may be separated by treatment with calcium chlorid and subsequently titrated with potassium permanganate in presence of excess of sulfuric acid. The oxalic acid obtained, deducted from the quantity originally present will give the amount consumed on the surfaces of the lime particles and consequently the amount of lime corresponding thereto which may be considered as available for plant growth.

365. Method of the Halle Station for Lime.^[234]—a. *In Phosphates, Limestones, etc.*—Four grams of the prepared substance are heated with fifty cubic centimeters of hydrochloric acid and five cubic centimeters of nitric acid, in a porcelain dish on the water-bath to dryness, and left for a few hours at 105° for the purpose of separating the silicic acid. The dry residue is moistened with hot water and a few drops of hydrochloric acid, and allowed to stand for some time with frequent stirring. The contents of the dish are then washed into a half-liter flask, filled up to the mark and the

separated silicic acid removed by filtration. If the silicic acid is not taken into account, the solution can be made directly in a half-liter flask.

After filtration, an aliquot part of the filtrate is neutralized in a 500 or 250 cubic centimeter flask with ammonia, again acidified with a few drops of hydrochloric acid and allowed to stand six hours at least, in the cold, with ammonium acetate. For each four grams of the substance fifty cubic centimeters of an ammonium acetate solution are used, made by dissolving in one liter of water 100 grams of ammonium acetate. If phosphoric acid is present in excess, iron and aluminum oxids are precipitated completely as phosphates. If iron and aluminum oxids are in excess, the excess must be precipitated by ammonia. If it is feared that in the subsequent precipitation of the lime by ammonium oxalate there may be still some phosphoric acid in solution, before precipitation with ammonium acetate the proper amount of ferric chlorid is added and the iron is afterwards precipitated with ammonia. It is certain that in the presence of oxalic acid and phosphoric acid the lime is precipitated as oxalate, but should it be feared that traces of calcium phosphate are precipitated with the iron and aluminum phosphates the precipitate of iron and aluminum phosphates may be dissolved in hydrochloric acid, neutralized with ammonia, again acidified and a second time precipitated with ammonium acetate and the filtrate added to that first obtained.

For the further estimation the filtrates are united and a quantity corresponding to a given part of the original sample, and being in volume from fifty to one hundred cubic centimeters is made slightly acid with acetic acid and while hot precipitated with dilute ammonium oxalate. The filtrate must contain acetic acid since calcium oxalate is best precipitated from a slightly acetic acid solution. The filtering of the calcium oxalate should not take place until from six to twelve hours after precipitation, and during this time it should stand in a warm place. Filter paper of the best quality should be used for the purpose.

The dried precipitate is brought into a platinum crucible together with the filter; the filter is first incinerated over an ordinary bunsen and the calcium oxalate converted into calcium oxid by ignition for fifteen minutes over the blast. It is then cooled in a well-closed desiccator and weighed as oxid. If in the precipitation of the iron and aluminum phosphates sodium acetate be

employed instead of ammonium acetate, the precipitation must take place hot and filtration also be accomplished on a hot filter.

b. *Estimation of Lime in Soils.*—For the estimation of lime in soils there may be used either the acid soil-extract, prepared as under the direction for the estimation of potash, or twenty grams of the soil may be treated with hydrochloric acid and a few drops of nitric acid, and evaporated to dryness in a porcelain dish and the silicic acid separated as described for the estimation of lime in phosphates and limestones. In the case of soils, iron and aluminum oxids can be precipitated directly with ammonia since the small quantity of phosphoric acid usually contained in soils is not sufficient to influence in any way the estimation of the lime. For example suppose there is 0.10 per cent of phosphoric acid contained in a soil. In case the whole of this phosphoric acid is taken down with the lime it would only amount to about 0.10 per cent of calcium oxid precipitated as phosphate. This case, however, is very improbable since it is much more likely that the iron and aluminum phosphates will be precipitated and the whole of the phosphoric acid be carried down with them instead of being precipitated with the lime.

The precipitation of the lime and its subsequent treatment are to be conducted as just described.

366. Estimation of the Magnesia.—Magnesia is a much more rare element in the soil than lime. It is usually necessary to operate upon considerable quantities of earth in order to determine the magnesia with any degree of precision. From ten to twenty grams of the soil are taken. The decomposition is accomplished as in the case of lime. A few drops of barium nitrate are added for the purpose of precipitating any sulfuric acid present. Some ammonia and ammonium carbonate are added to precipitate the iron and aluminum oxids, the lime and the excess of barium introduced, as well as the phosphoric acid. The operation is best conducted on a dilute solution having a volume of from 400 to 500 cubic centimeters. The solution from which the lime has been precipitated, contains with the magnesia, large quantities of ammoniacal salts which it is necessary to destroy. For this purpose the solution is concentrated in a flask until its volume is about ten cubic centimeters. About ten cubic centimeters of nitric acid are added and the whole brought to the boiling point. Afterwards a few drops of hydrochloric acid are added. Continuing the heating, hydrochloric

acid is added in small portions and, from time to time, some nitric acid until the bubbles indicating the setting free of gaseous nitrogen, resulting from the action of the nascent chlorine upon the ammonia, have completely ceased to appear. The whole is then evaporated on a sand-bath in a porcelain dish in order to separate the silica. The residue is taken up by water containing a few drops of nitric acid. It is filtered and evaporated to dryness in a covered porcelain dish. Upon the residue four or five grams of oxalic acid, in a state of powder, are placed. A little water is added in such a way that the moist mass covers entirely the matter in the dish. In order to avoid all losses there is placed upon the dish a funnel which serves as a cover. The dish is heated on a sand-bath, but when the film which is formed begins to break there are added from time to time, a little more oxalic acid and water until there is no longer any disengagement of the vapor of nitric acid. Afterwards it is evaporated to dryness and the heat raised to a low redness. The magnesia is found in a free state or mixed with alkalies. It is washed with a small quantity of water and collected upon a very small filter paper. The filter paper is dried, burned, the ignition carried to redness and afterwards cooled and weighed. In order to test the purity of the magnesia it is transformed into sulfate by the addition of a few drops of sulfuric acid. The excess of sulfuric acid is driven off by heating moderately by means of a gas-burner moving it in a circular manner round the bottom of the capsule and lifting the cover from time to time in order to allow the vapors of sulfuric acid to escape. The weight of the magnesium sulfate should correspond to that of the magnesia from which it was formed.

Magnesia exists most often in the soil in the state of carbonate or silicate. In this last state it is especially abundant in some soils, such as those which are derived from mica schists, serpentines, etc. In treating earth of this last quality with concentrated, nitric acid there is dissolved also a notable part of the magnesia of the silicates. If, however, it is treated for some minutes only with dilute hydrochloric acid the amount of magnesia present as carbonate alone can be estimated separately.

367. Estimation of Magnesia in Soils.—*Method of the Halle Station.*—For the estimation of magnesia the sample of soil or fertilizer is brought into solution in the same way as is given for the estimation of lime. After the separation of the silicic acid, the iron and alumina are precipitated with sodium acetate. In the case of phosphoric fertilizers, ferric chlorid should first be added in order that the excess of phosphoric acid shall be in all

cases certainly combined with the iron. After this the lime is separated as usual with ammonium oxalate. After the precipitation of the lime, the magnesia is precipitated in an ammoniacal solution with sodium phosphate and the ammonium magnesium phosphate estimated exactly as in the case with the estimation of phosphoric acid, as magnesium pyrophosphate.

A simpler method for the estimation of magnesia consists in precipitating it as ammonium magnesium phosphate in the presence of a solution of ammonium citrate, the other bases remaining in solution. In this case the operation is carried on in an inverse way as described under the estimation of phosphoric acid, the proper quantity of the acid solution being neutralized with ammonia and after the addition of sodium phosphate, the required quantity of citrate solution added and a further excess of ammonia supplied.

368. Estimation of Manganese.—The estimation of manganese in the presence of Fe_2O_3 , Al_2O_3 , CaO , etc., presents peculiar difficulties. In ordinary alluvial clays the quantity of manganese is proportionately small and its estimation may be neglected. In volcanic clays the quantity of manganese, in proportion to the lime and magnesia, is much larger. The method used for estimating manganese is that of Carnot.^[235] The hydrochloric acid extract of the soil is evaporated to dryness and heated with potassium bisulfate in order to destroy the organic substance, the neutralized solution of the residue precipitated with twenty cubic centimeters of hydrogen peroxid solution and thirty cubic centimeters of ammonia. The colorless filtrate gives, with nitric acid and bismuth peroxid, no trace of reaction for manganese. The precipitate, washed by decantation, is carried into a carbon dioxid apparatus and treated with oxalic acid and dilute sulfuric acid. From the amount of carbon dioxid obtained, the quantity of manganese is calculated on the supposition that the precipitate corresponds to the formula Mn_6O_{11} .

369. Estimation of the Manganese by the French Method.—Manganese exists in all plants and its presence in small quantities seems necessary to vegetation. The method of estimation adopted by the French Commission is the one proposed by Leclerc and is applicable even when the base exists in small quantities.

Twenty grams of the soil are taken and the organic matter destroyed by incineration. In a flask of 200 cubic centimeters capacity, are placed thirty

cubic centimeters of water and, little by little, some hydrochloric acid for the purpose of decomposing the calcium carbonate. When effervescence has ceased ten cubic centimeters of the same acid are added and boiled for half an hour, filtered, washed, and the wash-water and filtrate evaporated to dryness in a porcelain dish. Afterwards there are added twenty cubic centimeters of nitric acid of one and two-tenths density, and ten cubic centimeters of water. The liquor is boiled with constant shaking. Afterwards there are thrown in, in two or three portions, ten grams of lead dioxid. The boiling is stopped just at the moment when all the lead oxid is introduced into the liquor and the mixture is then shaken vigorously. The manganese is transformed by this treatment into a highly oxygenized compound having a deep rose coloration. It is transferred immediately afterwards to a graduated cylinder of 100 cubic centimeters capacity, with the wash-waters the volume is completed to 100 cubic centimeters and it is vigorously stirred with a rod, flattened at its extremity, in order to obtain a complete homogeneity of the liquid. The stirring rod is withdrawn and the liquid left to settle. At the end of some minutes the principal part of the liquid is clear, and it is decanted by means of a pipette graduated at fifty cubic centimeters, and this quantity of the clear liquid is poured into a small glass precipitating jar to which is added immediately, with constant stirring, a solution of mercurous nitrate from a graduated burette. The addition of the nitrate is arrested at the moment when the rose color of the liquor disappears, and the volume of the mercurous nitrate employed is read from the burette.

It is now necessary to determine the strength of the mercurous nitrate, that is the quantity necessary to decolorize one milligram of manganese. For this purpose dissolve by means of five cubic centimeters of hydrochloric acid 150 milligrams of manganese dioxid, which is prepared perfectly pure by means of precipitation. When the solution is complete evaporate to dryness, add one cubic centimeter of sulfuric acid and heat on a sand-bath until white fumes of sulfuric acid appear. Redissolve in water and make the volume up to 100 cubic centimeters. Each cubic centimeter of this solution should contain one milligram of manganese. Take five cubic centimeters of this solution, equivalent to five milligrams of manganese, treat in a capsule with twenty cubic centimeters of nitric acid and ten cubic centimeters of water, afterwards add ten grams of lead dioxid, carrying on the operation exactly as described above. Fifty cubic centimeters, taken as before described, are then decolorized by the solution of mercurous nitrate,

and thus it is easy to calculate the quantity of manganese which corresponds to one cubic centimeter of the mercurous nitrate employed. By a simple proportion the quantity of manganese contained in the twenty grams of earth to be analyzed is calculated.

The mercurous nitrate is prepared by dissolving five grams of crystallized mercurous nitrate in one liter of water; it is allowed to repose for some time and is preserved in a well-stoppered flask.

370. Estimation of Iron.—Iron, in general, is quite abundant in the soil where it is met with, principally in the state of anhydrous sesquioxid or the hydrated sesquioxid of silicates. Some soils, however, only contain iron in small proportions and it can happen that the introduction of iron as a fertilizing element may be useful. Plants assimilate iron only in small quantities, but it appears to be indispensable to their development and to the proper functional activity of their assimilating faculties. The method of estimation which is recommended is based upon the decoloration of potassium permanganate by iron in the ferrous state. The following description, based on the method proposed by the French Commission, will illustrate the process to be followed.

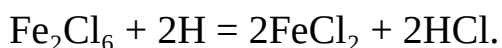
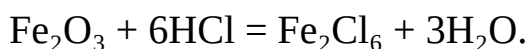
Ten grams of the soil are ignited in a porcelain capsule until all organic matter is destroyed. The ignited mass is then introduced into a flask of 100 cubic centimeters capacity with thirty cubic centimeters of hydrochloric acid and fifteen cubic centimeters of water. It is boiled for about half an hour. The iron oxid is dissolved and is found in solution in the form of ferric chlorid. After filtering and washing, the volume of the filtrate is reduced by evaporation to about twenty-five cubic centimeters. The liquor is afterwards placed in a flask of from 100 to 150 cubic centimeters capacity, which is closed by a stopper carrying a tube furnished with a valve destined to prevent the re-entrance of the air. Ten cubic centimeters of dilute sulfuric acid are added from a mixture containing twenty cubic centimeters of strong acid and eighty cubic centimeters of water. Afterwards the iron is reduced to the ferrous state by introducing into the flask, in quantities of about five decigrams, metallic zinc and waiting after each addition until the portion last added is dissolved before adding another. This addition of zinc is continued until the iron is all reduced. When this point is reached and the last portion of zinc added is dissolved, the contents of the flask are transferred rapidly to a precipitating glass of about one liter capacity, in

which there has been placed a little lately boiled but cold water. The flask is washed several times with cold water, previously boiled, to remove from it all traces of oxygen. The volume is made up to 500 cubic centimeters, and afterwards, without any loss of time, by means of a graduated burette and with constant stirring, a solution of potassium permanganate is added which is stopped exactly at the moment when the liquor begins taking on a light rose tint. The quantity of permanganate employed is read from the burette and is proportional to the amount of iron contained in the soil. A blank operation is made for the purpose of detecting traces of iron which the zinc may contain. If, as often happens, the soil contains a large amount of iron it is advisable to use only one gram of it for this operation. The aspect of the earth will indicate in general if it be very ferruginous.

Preparation and Standardization of the Permanganate Liquor.—In one liter of water are dissolved ten grams of crystallized potassium permanganate and the quantity of iron which corresponds to one cubic centimeter of this liquor is determined. It may be well enough to remark that this liquor does not remain constant and it is necessary to titrate it from time to time. For this purpose pure iron is taken. Piano wire may be used, being almost pure iron. One-tenth of a gram of this wire is dissolved in a flask in the manner recommended for treating the soil and with the same quantities of acid and water. When the solution is complete it is transferred to the flask to be estimated. It is made up to one liter and permanganate added, just as in the case before mentioned, until the rose color persists. There is thus determined the quantity of iron which corresponds to each cubic centimeter of the permanganate, and by a simple proportion, the quantity of iron contained in the soil analyzed is determined.

The Italian agricultural chemists proceed essentially in the same manner in determining the iron in soils, first igniting the sample and afterwards extracting the iron in the ferric state with boiling hydrochloric acid, reducing with hydrogen, and titrating with potassium permanganate.

The following are the reactions which take place:



Sulfuric may take the place of hydrochloric acid in the above reactions.

371. Method of Sachsse and Becker.

[236]—Ferric oxid (not as silicate) in soils can be estimated by reducing with hydrogen, and measuring the hydrogen which is evolved by the action of the reduced iron on an acid. The sample of soil is weighed in a platinum boat, the boat put into a wide glass tube and heated in a stream of dry hydrogen. While this is going on, water is boiled in the flask A (see Fig. 68) from which the stopper has been removed, to drive out the air. When the reduction of the ferric oxid is complete, the boat is slipped out of the tube into the flask without interrupting the hydrogen evolution. In order to accomplish this without allowing the reduced iron to come in contact with the air the flask is inclined, the end of the glass tube inserted until it is covered with water and the boat is then dropped beneath the water. The flask is closed with a cork provided with a funnel tube, B, and a delivery tube C; the tap *a* is opened, and tube *b* connected with a carbon dioxid apparatus from which the gas is passed into A until all the air is displaced. This point is determined by filling the burette D with potash-lye by aspiration at C and allowing the escaping gas from C to enter the burette as indicated in the figure. Any residual gas in D is removed by aspiration at C and allowing the potash-lye in *e* to enter in its place. The end of the tube C is now placed under the measuring tube D, and the clamp *f* opened and the tap *a* closed. The funnel is filled with dilute, boiled sulfuric acid, the cork of *b* replaced and connected with the carbon dioxid apparatus. The burner under A is lighted and acid let in. By continued boiling, all the hydrogen is driven into D, the carbon dioxid being absorbed. The measuring tube is then placed in a tall cylinder of water, the volume of gas read and reduced to 0° and 760 millimeters barometric pressure. To be certain that all carbon dioxid is absorbed, some fresh potash-lye may be introduced into D by carefully opening *d*. The iron is then computed from the volume and weight of the hydrogen by the formula (1) $\text{Fe} + \text{H}_2\text{SO}_4 = 2\text{H} + \text{FeSO}_4$.

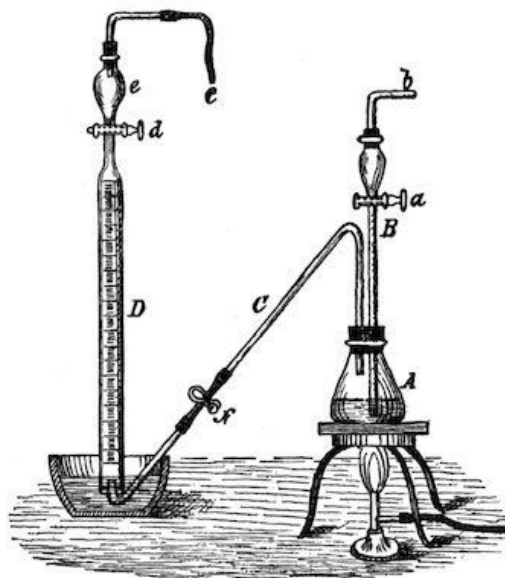


FIGURE 68.

APPARATUS BY SACHSSE AND BECKER.

If the substance analyzed contains iron silicates, these may be partly decomposed with formation of ferrous sulfate, according to the reaction (2) $2\text{Fe} + \text{Fe}_2(\text{SO}_4)_3 = 3\text{FeSO}_4$. This will redissolve a part of the metallic iron and yield ferrous oxid. In this case the contents of the flask are cooled in an atmosphere of carbon dioxid, made up to 500 cubic centimeters, of which 250 cubic centimeters are quickly filtered and titrated with permanganate. In order to properly distribute the iron in harmony with its previously existing states the following computations may be made:

Represent the	ferrous oxid corresponding to formula	(1) by x
and that	„ „ „	(2) by z
and that	found by titration with permanganate	by a.

We have then the equation $x + z = a$. Since seventy-two parts by weight of ferrous oxid formed by formula (1) are equivalent to two parts by weight of hydrogen, x parts of ferrous oxid would set free $\frac{x}{32}$ parts of hydrogen; and this corresponds to the hydrogen found in *D*; viz., b.

If $a = \frac{1}{36}$ then by solving the equations: $Z = a - 36b$ and $X = 36b$. The ferrous oxid arising according to formula (2), however, is derived in such a way that only one-third of it corresponds to metallic iron. Then: $X + (\frac{1}{3})z = (\frac{1}{3})a + 24b$. For computing the total ferric oxid reduced by hydrogen there must, therefore, be added twenty-four parts by weight of hydrogen for one-third of the ferrous oxid found by titration with permanganate, and this quantity of ferrous calculated to ferric oxid. Some silicates, such as the micas, give ferrous oxid with hot dilute sulfuric acid. A correction for this is made by making one or more determinations without previously reducing with hydrogen.

The method of procedure above described appears to be capable of giving in an easily attainable manner some valuable indications of the state in which iron exists in a soil. While plants do not use any notable quantity of iron during their growth nevertheless its physiological importance is unquestioned. The chief points of difficulty to be considered are found in the changes which the iron may undergo even while heating in a stream of hydrogen, and the practical difficulties of obtaining carbon dioxid entirely free of air. The latter difficulty may be overcome by making blank experiments with carbon dioxid alone and estimating the volume of residual

gas. The total volume of hydrogen obtained is then to be diminished by the ascertained amount.

In regard to the second point it is known that both ferrous and ferric oxides when ignited with hydrated silicates partly decompose and form new silicates. Care should therefore be taken not to carry the temperature too high during the process of ignition.

372. Carnot's Method for Estimating Phosphoric Acid in Soils.—

Carnot^[237] proposes the following procedure for the estimation of phosphoric acid in soils. The principle of this method depends upon the isolation of silica by the double precipitation of phosphomolybdate.

Ten grams of the sifted soil, dried at 100°, are charred if organic matter be present. The charred mass is next moistened with water and afterwards with nitric acid, until the carbonates are decomposed. Afterwards the mass is digested with ten cubic centimeters of nitric acid for two hours at about 100°, with frequent stirring and the addition of fresh acid, from time to time, to replace that which has been evaporated. After filtering and washing with hot water the filtrate is evaporated to a volume of fifty cubic centimeters and treated with five cubic centimeters of concentrated nitric acid and half a gram of crystals of chromic acid. After covering the dish with a funnel to return condensed vapors its contents are heated to the boiling point for half an hour. At the end of this time five grams of ammonium nitrate are added and afterwards fifty cubic centimeters of molybdate solution and the mixture kept at a temperature of about 100° for an hour. The precipitate obtained is washed twice by decantation with water containing one-fifth of its volume of ammonium molybdate solution. It is then dissolved in thirty cubic centimeters of ammonia diluted with an equal bulk of warm water. The solution and the washings should measure eighty cubic centimeters and the ammonia therein is neutralized with nitric acid, keeping the temperature below 40°. When the yellow precipitate formed ceases to redissolve on stirring, a mixture of three cubic centimeters of pure nitric acid and five cubic centimeters of water is added, together with the same quantity of molybdate solution. The precipitate is brought upon a filter, washed first with water containing one per cent of nitric acid and finally with a little pure water, and dried at 100° and weighed. The weight of the precipitate multiplied by the factor 0.0373 gives the quantity of phosphoric acid. The object of the second precipitation is to relieve the

process of the necessity of rendering the silica insoluble, as the presence of silica in the solution as above treated does not interfere with the complete precipitation of the phosphate. This was proved by the author, by the introduction of considerable quantities of sodium silicate and these were found not to interfere with the accuracy of the operation.

The results are as accurate as those obtained by the methods of the consulting committee of the agricultural stations. The coefficient employed; viz., 0.0373, is not the same as that recommended by the committee; viz., 0.043. The committee, however, itself has recognized the inaccuracy of the latter number. The composition of the compound obtained by double precipitation according to Carnot is $\text{P}_2\text{O}_5 \cdot 24\text{MoO}_3 \cdot 3(\text{NH}_4)_2\text{O} + 3\text{H}_2\text{O}$.

373. Method of the Halle Experiment Station.—The available or easily soluble phosphoric acid in soils is estimated by Maercker and Gerlach, as follows:^[238]

Sixty grams of the air-dried soil as prepared for analysis, are placed in an erlenmeyer with 300 cubic centimeters of two per cent citric acid solution and digested for twenty-four hours in the cold. It is necessary in this time to shake the flask four or five times and to put the stoppers in loosely in order to allow the escape of any evolved carbon dioxide. Of this mixture 200 cubic centimeters are filtered and evaporated in a 300 cubic centimeter dish to dryness. There remains, in most cases, a sirupy-like mass from which even by strong heating the silica is not completely separated. In order to reach this result the residue is treated with twenty cubic centimeters of concentrated sulfuric and five cubic centimeters of fuming nitric acid and heated over a bunsen. As soon as the appearance of foam denotes the beginning of the reaction the lamp must be removed. With strong foaming and the evolution of red-brown vapors the citric acid is completely oxidized. After the reaction is ended the contents of the dish are heated for about fifteen minutes over a small flame so that a continuous, yet not too violent evolution of sulfuric acid fumes takes place. After the silicic acid and the greater part of the lime have been separated in this way the contents of the dish are diluted with water, stirred with a glass rod, washed into a 200 cubic centimeter flask, cooled, filled up to the mark, and filtered. From the filtrate 100 cubic centimeters, corresponding to twenty grams of the earth, are taken, made slightly alkaline with ammonia, acidified by a few drops of hydrochloric acid, and after cooling treated with fifty cubic

centimeters of the citrate solution and twenty-five cubic centimeters of the magnesia mixture. The complete separation of the precipitate requires about forty-eight hours and shaking of the precipitate is not necessary.

374. Estimation of total Phosphoric Acid in Soils.—In the method used at the Halle Station^[239] twenty-five grams of the soil sample are boiled with twenty cubic centimeters of nitric acid and fifty cubic centimeters of concentrated sulfuric acid for half an hour. With very clayey soils only half the quantity of the sample mentioned above is used in order to avoid the too great accumulation of soluble alumina. The oxidation of the organic substances of the soils must be carried on at a moderate heat to avoid foaming. During the boiling, the flask is to be often shaken to prevent the soil constituents from accumulating too firmly at the bottom. The total volume is finally made up to 500 cubic centimeters.

For the estimation, 100 cubic centimeters of the solution, corresponding to five (or two and a half) grams of the soil, are taken. In order to nearly completely saturate the acid, the solution is treated with twenty cubic centimeters of twenty-four per cent ammonia, care being taken that the precipitate of iron and alumina which is formed is again completely dissolved. The solution is cooled and treated with fifty cubic centimeters of the citrate solution, and then with twenty cubic centimeters of ammonia of above strength, and precipitated with the magnesia mixture. The filtration of the precipitate should not be made for at least forty-eight hours, during which time the flask should be often shaken to prevent the attachment of ammonium magnesium phosphate to its sides and bottom.

A detailed description of the citrate method for estimating phosphoric acid will be found in the chapter devoted to this subject under fertilizers.

375. French Method for Phosphoric Acid.—Phosphoric acid is found in the soil principally in combination with alumina and iron oxid, with organic matters, or with lime and magnesia. Whatever may be the state in which it is found all the phosphoric acid, with the exception of that which enters into the constitution of insoluble mineral particles, can be brought into solution by acids and determined by some of the approved methods. This method of solution, therefore, is capable of determining very accurately the total proportion of phosphoric acid in the soil, but it is incapable of rendering account to us of the state in which the phosphorus is found and of its aptitude to be utilized by plants.

The estimation of soil phosphorus, as recommended by the French Commission, is carried on in the following way:^[240] Twenty grams of the earth are submitted to ignition in a muffle heated to the temperature of redness but not higher. This calcination eliminates the organic materials, whose intervention in subsequent reactions might be able to prevent the precipitation of a part of the phosphoric acid. The calcined earth is placed in a capsule of about eleven centimeters diameter and saturated with water. There is then added in small quantities, as long as effervescence is produced, nitric acid of 36° baumé. When the effervescence has ceased, after thorough shaking and the addition of a new quantity of acid, it will be found that the whole of the calcium carbonate in the soil has been decomposed. It is necessary then to proceed to the solution of the phosphoric acid by adding twenty cubic centimeters of nitric acid and heating on the steam-bath for five hours, shaking from time to time, and avoiding complete desiccation. At the end of this time the whole of the phosphoric acid has entered into solution. It is taken up by warm water, filtered, and the insoluble residue washed with small quantities of boiling water. But from the solution obtained, which holds in addition to phosphoric acid, some oxid of iron, alumina, lime, magnesia, etc., it is necessary to separate the silica which has passed into solution. For this purpose the mass is evaporated to dryness on a sand-bath, heating toward the end of the operation with precaution and not allowing the temperature to pass beyond 110°–120°. In these conditions there is obtained a magma which sometimes remains quite sirupy when the earth is very highly impregnated with calcium carbonate, but in which the silica is insoluble. It is indispensable that it be eliminated wholly because it would introduce grave errors into the results, as will be seen later on. If the temperature be carried too high during the desiccation this silica would react upon the earthy salts and alkaline earths forming silicates and it would be found ultimately again in solution. The application of a too high temperature would also render somewhat insoluble in nitric acid the iron and aluminum oxids, and these would retain small quantities of phosphoric acid. The desiccation, therefore, requires to be conducted with great precaution. When it is accomplished there are placed in the capsule five cubic centimeters of nitric acid and five cubic centimeters of water, and the whole heated on the sand-bath until the entire amount of iron oxid is dissolved, that is to say, until there is no ferruginous deposit persisting in the liquid. The solution is

then filtered and washed with small quantities of boiling water in such a way that the total volume of the filtrate will not exceed twenty-five to thirty-five cubic centimeters. Afterwards there are added twenty cubic centimeters of ammonium nitromolybdate and the whole is left at rest for twelve hours at the ordinary temperature. At the end of this time the whole of the phosphoric acid is precipitated in the form of ammonium phosphomolybdate. In order to be certain that an excess of nitromolybdate has been used in the precipitation, which is indispensable to the total precipitation of the phosphoric acid, a few cubic centimeters of the filtrate are removed by means of a pipette and are mixed with their own volume of the ammonium nitromolybdate. If, at the end of an hour or two, no precipitate is formed the operation can be regarded as terminated.

In order to collect and weigh the ammonium phosphomolybdate some precautions are necessary. Two smooth filters are used, one of which serves as a counter-weight for the other on the balance. One of these filters is placed within the other and the phosphomolybdate is collected upon the inner filter. The part of the phosphomolybdate adhering to the precipitating jar is detached by the aid of a stirring rod, one of the ends of which is covered with a piece of rubber tubing. The washing is accomplished with very small quantities of water containing five per cent of its volume of nitric acid. When all of the precipitate is collected upon the filter and the washing is terminated a few drops of water are thrown upon the upper borders of the filters; this is done to displace the acid liquor which has been used in washing. The filters are then carried to the drying oven where they are dried at a temperature not exceeding 90°. The application of a higher temperature would decompose the ammonium phosphomolybdate and lead to results which would be too low. After the drying is completed the two filters are separated and placed upon the arms of the balance and the increase in weight corresponds to the ammonium phosphomolybdate. This, multiplied by the coefficient 0.043, (see page 404) gives the quantity of phosphoric acid contained in the weight of the soil which has been employed. The ammonium phosphomolybdate is pure if all the silica has been eliminated, but if a part of that has remained in solution it would furnish an ammonium silicomolybdate whose weight would be added to that of the phosphomolybdate. The elimination of the silica, therefore, should be made with the greatest care.

Different processes have been proposed in order to determine the forms under which phosphoric acid should be regarded as most assimilable. Deherain has proposed acetic acid as a solvent for this purpose. Other scientists, oxalic or citric acid, and the ammonium oxalate or citrate. The solubility of phosphoric acid in these different reagents gives some information in regard to its state, but the relations which exist between this solubility and the assimilability of the acid have not yet been fixed.

Preparation of the Ammonium Nitromolybdate.—One hundred grams of molybdic acid are dissolved in 400 grams of ammonia with a density of ninety-five. The mixture is filtered, and the filtered liquor is received drop by drop in 1,500 grams of nitric acid of one and two-tenths density, constantly stirring. This mixture is left standing for some days in an unexposed locality, during which time a deposit is formed. The clear part is decanted and used.

The above method of the French chemists unfortunately attempts to determine the phosphorus content of the soil by weighing the yellow precipitate and using an empirical factor for the calculation, a factor which is probably too high.

Experience has shown that at this point it is far more accurate to continue the process by dissolving the yellow precipitate, and subsequently obtaining the phosphoric acid in combination with ammonia and magnesia, or according to the process of Pemberton the content of phosphoric acid in the yellow precipitate might be determined by titration. In regard to the latter method which will be given in full under fertilizers, it may be said that it has been found quite accurate by several analysts, although it is difficult to see how a precipitate which is so variable in its constitution as to be estimated with little safety by weight may yet be capable of rather exact determination by titration.

376. Petermann's Method for the Estimation of the Phosphoric Acid Soluble in Alkaline Ammonium Citrate.^[241]—From twenty-five to fifty grams of the sample of soil are triturated with 100 cubic centimeters of alkaline ammonium citrate and placed in a flask of 250 cubic centimeters capacity, and allowed to digest for one hour at a temperature of from 35°–40°. After cooling, make up to the mark, filter, take 200 cubic centimeters of the filtrate, evaporate to dryness on a sand-bath in a platinum dish, heat lightly at first, and afterwards to a higher temperature. Take up the residue

of the incineration with water and about two cubic centimeters of nitric acid, heat a few minutes gently, filter into a bohemian flask and precipitate with fifty cubic centimeters of ammonium molybdate solution, and estimate the phosphoric acid in the usual way.

377. Method of Dyer for Total and Assimilable Phosphoric Acid.—For the determination of phosphoric acid, soluble in citric acid, secured as described in paragraph 328, 500 cubic centimeters of the filtrate obtained, corresponding to fifty grams of the soil, are evaporated to dryness in a platinum dish, gently ignited, extracted with hydrochloric acid, again evaporated, ignited and extracted, and the phosphoric acid determined as below in the method applied to the hydrochloric acid extract of the soil itself.

The total phosphoric acid is determined in each case in ten grains of the dried soil and also in twenty-five grams, the mean of the two results being taken. The numbers obtained in each case are, however, all but identical, the difference in the duplicate percentages being in most cases only a small one in the third place of decimals.

The soil is incinerated and digested with hydrochloric acid, and evaporated to dryness, redigested with acid, filtered, and washed. The filtrate and washings are concentrated to a small bulk, and treated, in the cold, with excess of a solution of ammonium molybdate in nitric acid. After standing forty-eight hours, the liquor is decanted through a filter, the precipitate washed several times by decantation, first with dilute acid, then with pure water in very small doses, and finally transferred to the filter and washed free from excess of acid. The ammonium phosphomolybdate is then dissolved in ammonia, evaporated to dryness in a platinum capsule, and dried to constant weight at 100°. The residue contains three and one-half per cent of its weight of phosphoric acid. This is the method of Hehner; and for determining small quantities of phosphoric acid, such as occur in soils or in solutions of iron and steel, is in the opinion of Dyer very much to be preferred to the old-fashioned method of conversion into magnesium ammonium phosphate. The solubility of the yellow precipitate in the small quantity of wash-water used is in most cases negligible. As a matter of fact, the quantity of wash-water used in these analyses was found capable of dissolving only 0.005 gram of precipitate, of which only 0.00017 is phosphoric acid, making an error of 0.0017 per cent on the soil if ten grams

be used, or of only 0.0006 if twenty-five grams be used. In the citric acid experiments the solution from fifty grams of soil is used, when the error due to solubility of precipitate shrinks to 0.0003 per cent. The correction for this solubility is, however, made in each case.

It may be observed that the method of *Hehner* is not applicable if the molybdic solution be added to a hot liquid, since, in that case, some molybdic acid is sure to crystallize with the yellow precipitate. Moderate and careful warming to about 35° hastens precipitation, but it is preferable, when speed is not a special object, to precipitate cold, and leave the beaker standing at the laboratory temperature over night, or longer if the quantity to be determined is very minute.

378. Methods of Berthelot and André.—The phosphorus in the soil may be found under three forms; *viz.*,

1. Phosphoric acid in phosphates.
2. Phosphoric acid in ethers which alkalies decompose slowly and oxidizing agents destroy with regeneration of phosphoric acid.
3. Organic compounds or mineral compounds of phosphorus which are resolved by alkaline solutions with formation of phosphoric acid and which are not reduced to this state by the reagents employed in the wet way of decomposition except after a contact of indefinite length and uncertainty.

It is, therefore, seen that the employment of oxidizing agents for the valuation of phosphoric acid in soils and vegetables is not a very reliable procedure. The same is true after incineration by which more or less phosphorus may be lost or rendered insoluble in acids. The methods used by *Berthelot* and *André* for the estimation of these forms of phosphorus are as follows:^[242]

Total Phosphorus.—The sample is at first oxidized by a current of air near a red heat and the vapors are conducted over a column of sodium or potassium carbonate at the same temperature. The combustion is finished in a current of pure oxygen. All phosphorus compounds, even those which are volatile, are by this treatment converted into phosphoric acid. The part of the acid held by the carbonate is to be determined with the non-volatile portions.

A less certain method of oxidation consists in mixing the material with potassium nitrate and carefully throwing it little by little into a red hot

platinum crucible.

Estimation of the Phosphoric Acid Pre-existing as Phosphates.—The sample is treated with a cold dilute acid incapable of exercising an oxidizing or decomposing effect on the ethers. The dissolved acid is precipitated and weighed in the usual way. The precipitate first obtained should be ignited and the phosphoric acid taken up and reprecipitated. This is necessary to remove any organic matter or silica which the first precipitate may contain.

Estimation of Ethereal Phosphoric Acid.—The sample is boiled for some time with a non-oxidizing acid or with a concentrated solution of potash. The phosphoric acid dissolved represents that which was present as phosphates and as ethers. From this, deduct that portion pre-existing as phosphates and the remainder represents the part derived from the ethereal compounds.

Estimation of Phosphorus in Organic Compounds and Special Minerals.—From the total phosphoric acid deduct that found as phosphates and ethers. The difference represents the quantity combined as noted in the caption. Illustration.

A sample of soil contained:

Total phosphoric acid	0.292	per cent.
Of this, pre-existing as phosphoric acid	0.109	„
As ethereal phosphoric acid	0.074	„
As organic phosphoric acid	0.109	„
Sum	0.292	„

379. Method Used at the Riga Station.—In the method pursued at the experimental station at Riga^[243] the organic matter in the fine earth is first destroyed by igniting twenty-five grams in a muffle. The ignited residue is placed in a 250 cubic centimeter erlenmeyer and digested with 150 cubic centimeters of ten per cent hydrochloric acid. The digestion is continued for forty-eight hours with frequent shaking. The filtrate is evaporated to dryness in a porcelain dish to separate any dissolved silica. The residue is taken up with dilute hot nitric acid, filtered, and the phosphoric acid

precipitated with ammonium molybdate. The final weighing is made as magnesium pyrophosphate, following the usual procedure in respect of precipitation and washing. Experiments show that approximately ninety-five per cent of the phosphoric acid are obtained by one extraction and five per cent by a second, conducted exactly as the first. Thoms draws the following conclusions from a long series of determinations:

(1) For the simple purpose of determining the need of a soil for phosphorus fertilizer a single extraction with ten per cent hydrochloric acid is sufficient. The difference between the first and second extraction; *viz.*, five to six per cent is too small to be of any value from a practical point of view.

(2) A soil which has been ignited until organic matter is destroyed gives up to the hydrochloric acid solvent about fourteen per cent more phosphoric acid than a non-ignited sample would.

(3) The mean temperature in the flask during extraction on a steam-bath is 74°.

380. Method of Hilgard. ^[244]—The weighed quantity of soil (usually from three to five grams) is ignited in a platinum crucible, care being taken to avoid all loss. The loss of weight after full ignition gives the amount of chemically combined water and volatile and combustible matter.

The ignited soil is now removed to a porcelain or glass beaker, treated with four or five times its bulk of strong nitric acid, digested for two days, evaporated to dryness, first over the water-bath and then over the sand-bath, moistened with nitric acid, heated and treated with water. After standing a few hours on the water-bath it is filtered and the filtrate is evaporated to a very small bulk (ten cubic centimeters) and treated with about twice its bulk of the usual ammonium molybdate solution, thus precipitating the phosphoric acid. After standing at least twelve hours, first at a temperature of about 50°, it is filtered and washed with a solution of ammonium nitrate acidified with nitric acid. The washed precipitate is dissolved on the filter with dilute ammonia water. After washing the filter carefully, the ammoniacal solution is treated with magnesia mixture, by which the phosphoric acid is precipitated. After allowing it to stand twenty-four hours it is filtered, washed in the usual way, dried, ignited, and weighed as magnesium pyrophosphate, from which the phosphoric acid is calculated. When a gelatinous residue remains on the filter after dissolving the

phosphomolybdate with ammonia it may consist either of silica not rendered fully insoluble in the first evaporation, or, more rarely, of alumina containing phosphate. It should be treated with strong nitric acid, and the filtrate with ammonium molybdate; any precipitate formed is, of course, added to the main quantity before precipitating with magnesia solution.

381. Separation of Phosphoric Acid From Iron and Alumina.—The following methods are suggested by Wolff^[245] for the complete separation of the phosphoric acid from the iron and alumina in soil analysis, where large quantities of these bases are found in solution:

1. After the separation of the greater part of the iron and alumina the phosphoric acid is precipitated from the solution in nitric acid by molybdic acid. The process is carried on as follows:

The acid extract is heated in a flask to boiling and the iron oxid completely reduced by the gradual addition of small particles of sodium sulfite. While still warm the free acid is neutralized with soda-lye, and ammonia added until the ferrous hydroxid and the aluminum hydroxid are completely separated. Acetic acid is now added in excess and until about four-fifths of the whole precipitate have passed again into solution. Then, after boiling for a moment, the whole is quickly filtered through a large filter with a cover, and the contents of the filter finally washed slightly. All the phosphoric acid is thus obtained in combination with some alumina and a very little iron. Nearly the whole of the iron and the larger part of the alumina, by this precipitation, are found in the filtrate and therefore cannot disturb the estimation of phosphoric acid in succeeding portions. The filter is now filled with boiling water and a little nitric acid added. The precipitate is dissolved and received in a beaker. The precipitation of the phosphoric acid is then accomplished by ammonium molybdate in the presence of nitric acid. After twenty-four hours all the phosphoric acid is thus precipitated and the precipitate is free from iron.

2. By the method of Schulze^[246] the iron is completely, and the alumina, with the exception of a small quantity, separated, and the precipitation of the phosphoric acid is accomplished either by the addition of a small quantity of tartaric acid and afterwards magnesium sulfate, or directly by means of ammonium molybdate. The principle of the separation depends on the fact that when the hydrochloric acid is nearly neutralized with soda or ammonia, and the solution boiled after treatment with ammonium formate,

the greater part of the alumina remains in solution. The precipitate is quickly filtered, washed with hot water, dried, taken from the filter and fused in a silver crucible with pure caustic alkali, either soda or potash. On solution and boiling with water, the iron is completely separated from the phosphoric acid, and from the small quantity of the alumina present the precipitation of the phosphoric acid can now be accomplished, either by saturation of the alkaline solution with hydrochloric acid and the direct addition of the magnesia solution after the addition of a little tartaric acid and ammonia, or after the addition of nitric acid by ammonium molybdate.

382. Estimation of Phosphoric Acid in Muck and Peat Soils.—The amount of phosphoric acid obtained by extraction with hydrochloric or sulfuric acid is markedly less in these soils than that obtained after the incineration of the sample, as pointed out by Schmoeger.^[247] This is due to the fact that the phosphoric acid is ordinarily combined in the form of nuclein. Extraction of the soils with ether shows that it is not present in the form of lecithin. The nuclein products, as is well known, are decomposed by heating in presence of a liquid at a high temperature for some time. The heating can either take place in an autoclave or in sealed glass tubes. The method is as follows:

The sample of soil is thoroughly rubbed up in a mortar with water, and then hydrochloric acid added until one gram of the water-free peat is suspended in about ten cubic centimeters of twelve per cent hydrochloric acid. The sample is placed in a glass or porcelain vessel in an autoclave and heated to 140°–160° for ten hours. The phosphoric acid is then determined in the extract in the usual way. The percentage of phosphoric acid determined in this way is found to correspond to the amount determined by the incineration of the substance.

The total phosphoric acid is determined in peats by the incineration of the sample and the estimation of the phosphoric acid in the ash. The phosphoric acid soluble in hydrochloric acid solution is determined by extracting a sample of the soil with twelve per cent hydrochloric acid in the usual way. The difference between this and the total is calculated as phosphoric acid in organic compounds.

Or the total phosphoric acid is determined by treating the soil with twelve per cent hydrochloric acid, in the proportion of one gram of soil to ten cubic centimeters of the acid, and the solution is placed in an autoclave and

heated for ten hours to 140°–160°, as above described. The phosphoric acid is then determined by the usual method. The difference between the total phosphoric acid as thus determined and the phosphoric acid soluble in hydrochloric acid is calculated as phosphoric acid in organic compounds.

383. Method of Goss.—On account of the length of time required to determine the phosphoric acid in soils by the usual methods, Goss^[248] has proposed the following modification which in his hands has given satisfactory results:

Weigh ten grams of the air-dried soil, which has been sifted through a one millimeter mesh sieve, and transfer to a pear-shaped, straight necked, kjeldahl digestion flask, which has been marked to hold 250 cubic centimeters. Add approximately seven-tenths gram of yellow mercuric oxid and twenty to thirty cubic centimeters of concentrated sulfuric acid, as for the determination of nitrogen. Twenty cubic centimeters of acid are nearly always sufficient, but in the case of unusually finely divided clay soils containing little or no sand it is necessary to use thirty cubic centimeters to prevent caking of contents of flask. In doubtful cases twenty cubic centimeters of acid should first be added and at the end of five or ten minutes, if contents show a tendency to cake, ten cubic centimeters more should be introduced. Thoroughly mix the contents of the flask by shaking, place on a suitable support over a burner, boil for one hour, cool, add about 100 cubic centimeters of water, five cubic centimeters of concentrated hydrochloric acid, and two cubic centimeters of concentrated nitric acid, boil gently for two minutes to oxidize iron, cool, make up to volume, and filter through a dry folded paper until perfectly clear. In order to secure a clear filtrate it will usually be found necessary to pour the first portion of the filtrate back through the paper three or four times. Transfer 100 cubic centimeters of the filtrate to an ordinary flask of about 450 cubic centimeters capacity, add strong ammonia until a permanent precipitate forms, then six or eight cubic centimeters of nitric acid to dissolve the precipitate, and boil until clear. In the case of many soils it is not absolutely necessary to oxidize with hydrochloric and nitric acids, as a clear solution can be secured at this point without further oxidation. In the case of some soils, however, and especially in subsoils, the solution cannot be cleared up even by prolonged boiling with nitric acid, but if the solution have been previously oxidized, a clear solution can be secured without any difficulty whatever. Remove the flask from the lamp and after two minutes add

seventy-five cubic centimeters of molybdate solution, place the unstoppered flask in an open water-bath kept at a temperature of 80° for fifteen minutes, shaking vigorously four or five times while in the bath; then remove, let stand ten minutes to allow precipitate to settle, filter through a nine centimeter filter avoiding too strong a pressure at first, wash the flask and precipitate thoroughly with ammonium nitrate solution, place the flask in which the precipitation was made under the funnel, shut off pump and close all valves to filtering jar to form an air-cushion and prevent too rapid filtration, fill paper two-thirds full of hot water, add a few cubic centimeters of strong ammonia, aid solution, if necessary, by stirring precipitate with a small glass rod.

As pointed out by Hilgard, aluminum is sometimes carried down with the phosphoric acid upon precipitating with molybdate solution, in which case some of the phosphoric acid will not be dissolved in the treatment with ammonia. This will be indicated, first, by the appearance of a white precipitate upon dissolving the yellow precipitate in ammonia; and, second, by the difficulty experienced afterward in washing. If such a precipitate be present in any appreciable quantity, proceed as follows:

After washing out all the ammoniacal solution in the usual manner, place a small beaker under the funnel, close all valves, fill the filter one-third full of hot water, add the same amount of concentrated hydrochloric acid, proceed as if dissolving phosphomolybdate in ammonia, and receive final solution and washings in flask used. As soon as the yellow precipitate is dissolved open the valve to filtering jar but do not turn on the pump; after the solution has all passed through rinse the filter once with a small amount of hot water; after the last portion has passed through remove the flask and place a No. 1 lipped beaker under the funnel and heat the solution in the flask to boiling. If the solutions have not been oxidized, a blue color is sometimes present upon dissolving the yellow precipitate in ammonia. This can be discharged by boiling the ammoniacal solution for a minute or two and shaking at the same time. Again pour the solution through the filter, avoiding use of pump at first, otherwise loss from spattering is likely to ensue, wash out flask and filter with a small amount of hot water, (the total filtrate should not exceed fifty cubic centimeters), add hydrochloric acid to contents of beaker while hot, until yellow color appears, then add a few drops of ammonia until solution clears, cool, add fifty cubic centimeters of filtered magnesia mixture from a burette, a drop at a time with constant

stirring, let stand fifteen minutes, add twenty cubic centimeters of strong ammonia specific gravity nine-tenths, let stand over night, filter, wash precipitate with dilute ammonia, dry, ignite intensely over blast-lamp for ten minutes, cool in desiccator and weigh $\text{Mg}_2\text{P}_2\text{O}_7$ secured.

Time of Digestion.—Experience has shown that very little phosphoric acid is extracted from the sample by digestion with sulfuric acid after the first thirty minutes.

Time Required to Precipitate Phosphomolybdate.— When the yellow precipitate is obtained according to the method of Goss practically the whole of it will be thrown down in five minutes.

Agreement with Standard Methods.—Comparative tests of the Goss method against standard methods have shown that it gives almost identical results with them. The variations were never more than 0.02 to 0.03 per cent.

While this method has not been sufficiently tried to receive unconditional recommendation it possesses merits which entitle it to the attention of analysts.

384. Estimation of the Sulfuric Acid.—Sulfuric acid is generally present in small proportions in soils. Since the plants have need of sulfur it is proper to inquire into the presence of the compound which is its principal source. It is in combination with lime that sulfuric acid exists almost always. In addition to this there is also some sulfur combined with the organic matter of the soil.

By digesting a soil for six hours with hot, concentrated nitric acid the sulfates are dissolved, and there is transformed into sulfuric acid an important part of the sulfur which is combined with the humic substances. The quantity of soil to be operated upon should be about fifty grams.

After filtering and washing with hot water the filtered liquor is collected, in the French Commission method, ^[249] in a flask and carried to boiling, and five cubic centimeters of a saturated solution of barium chlorid or sufficient to be in slight excess are added. The boiling is continued for some minutes and the flask is allowed to stand for twenty-four hours. The filtrate is received upon a filter and washed with boiling water. The filter is dried and incinerated, allowed to cool, and as there may have been a slight reduction of the sulfate a few drops of nitric acid are added and a drop of sulfuric acid. It is now evaporated to dryness on a water-bath, heated to redness for

a few moments, cooled and weighed. The weight of the barium sulfate obtained multiplied by 0.3433, gives the quantity of sulfuric acid obtained from the fifty grams of soil.

If it is desired to estimate only the sulfur which exists in the form of sulfate it is necessary to treat the soil with hydrochloric acid in a very dilute state, heating for a few moments only and afterwards precipitate by barium nitrate. If, on the other hand, it is desired to estimate the total sulfur which is sometimes of great interest, it is necessary to employ the process of Berthelot and André.

385. Method of Berthelot and André.—Sulfur may exist in the soil in three forms; viz.,

1. Mineral compounds, consisting generally of sulfates and sometimes of sulfids.
2. Sulfur, existing in ethereal compounds or their analogues, as in urine.
3. Organic compounds containing sulfur.

Estimation of Total Sulfur.—The principle on which this operation, as described by Berthelot and André, rests is that already described for phosphorus; viz., oxidation in a current of oxygen and passing the vapors over a column of alkaline carbonate at or near a red heat.^[250]

The ordinary methods of oxidation in the wet way give generally inexact results.

Estimation of Sulfur Pre-existing as Sulfates.—The sample is treated with cold, dilute hydrochloric acid. The filtrate is treated with barium chlorid, the precipitate collected, dried, ignited, washed with a mixture of sulfuric and hydrofluoric acids to remove silica, and afterwards weighed as barium sulfate.

Estimation of Sulfur as Sulfids.—The sample is distilled with dilute hydrochloric acid, and the hydrogen sulfid produced is made to pass through an acidulated solution of copper sulfate in such a way as to transform the sulfur in the hydrogen sulfid into a sulfid, which is afterwards collected and weighed in the usual way. The use of a titrated solution of iodine is not advisable on account of the organic matter which may be present.

Estimation of Sulfur in Ethereal Compounds.—These compounds can be decomposed by boiling with a solution of potash or concentrated

hydrochloric acid. The resulting sulfuric acid is precipitated with barium chlorid. Subtract from the sulfates thus obtained those pre-existing as sulfates; the difference represents the sulfur present in ethers.

Estimation of Sulfur in Other Organic Compounds.—This is estimated indirectly by subtracting from the total sulfur that present as sulfates, sulfids, and ethers.

386. Method of Von Bemmelen.—As Von Bemmelen^[251] observes, the estimation of sulfuric acid in soils presents a number of difficulties. A small part of it can be present as sulfate insoluble in water. In addition to this, there is always some sulfur in the organic bodies present. If the soil is extracted with water then the sulfuric acid can be estimated therein when only a trace of humus substance has gone into solution. On the contrary, if there is much humus substance in solution, and also iron oxid, as is the case when the extraction is made with hydrochloric acid, then both of these must be removed, otherwise the estimation is very inexact. By fusing the residue of the solution with sodium carbonate and a little potassium nitrate the organic substance is destroyed, and after treatment with water the iron oxid is separated. If any sulfur has been dissolved in the organic substance present, this is then oxidized to sulfuric acid. The estimation of the sulfuric acid and of the sulfur, therefore, remains unsatisfactory.

In a sample of clay from Java, which was rich in calcium carbonate, but which contained no basic iron sulfate, there was found the following percentages of sulfuric acid:

Exhausted in the cold with very weak hydrochloric acid, 0.04 per cent; the residue treated in the cold with concentrated hydrochloric acid, the solution evaporated and fused with sodium carbonate and potassium nitrate, 0.07 per cent; again, the residue treated with aqua regia to oxidize the sulfur, the solution evaporated to dryness, fused with sodium carbonate and potassium nitrate, 0.14 per cent; in all 0.25 percent. A sample of the same soil treated directly with aqua regia, and then evaporated and fused as above, gave two-tenths per cent sulfuric acid. A sample of the same soil ignited in a crucible with sodium carbonate and potassium nitrate gave 0.16 per cent of sulfuric acid. The difference between 0.04 and 0.07 per cent can be attributed to the sulfur in the organic substance which was dissolved by the concentrated hydrochloric acid; the quantity, however, is too small to draw any safe conclusion. Possibly it might have been that the very dilute

hydrochloric acid did not dissolve all of the sulfate. The quantity of sulfur combined in the organic substance in the above soil may be derived from the following equation; viz., $\frac{0.2-0.07}{80} \times 32 = 0.05$ per cent of sulfur.

The estimation of the sulfur in a sample of soil from Deli was carried on with still greater exactness by three different methods.

The quantities of hydrochloric acid, nitric acid, and sodium carbonate employed were measured or weighed, and the minute content of sulfuric acid therein estimated and subtracted from the final results. The methods employed were as follows:

(A) Extraction with water and afterwards with very dilute hydrochloric acid.

(B) Extraction with cold hydrochloric acid, one part to three of water.

(C) Extraction with aqua regia.

(D) Ignition with sodium carbonate and potassium nitrate.

(F) Ignition in a combustion tube with sodium carbonate in a stream of oxygen.

The percentages of sulfuric acid obtained by the different methods were as follows:

(A)	0.058	per	cent.
(B)	0.070	„	„
(C)	0.140	„	„
(D)	0.125	„	„
(F)	0.106	„	„

387. Method of Wolff.—In regard to the sulfuric acid Wolff calls attention to the fact that in soils which have been ignited, a larger quantity of this acid is found than in soils containing humus.^[252] This, doubtless, arises from the oxidation of the organic sulfur. The following special method for determining the sulfuric acid is therefore proposed:

Fifty grams of fresh air-dried soil are placed in a platinum dish with a concentrated solution of pure sodium nitrate. After drying, the heat is raised gradually to redness. In this way the complete ignition of the humus present takes place. After cooling, the mass is diluted with hydrochloric acid, with the addition of a little nitric acid, and boiled. In the solution, the silicic acid

is first separated and the sulfuric acid estimated in the usual way with barium sulfate.

388. Method of the Italian Chemists.—The determination of the sulfuric acid is conducted as follows by the Italian chemists:^[253] The soil is completely extracted by diluted hydrochloric acid and the sulfuric acid precipitated in the solution with barium chlorid. If a soil is very rich in calcium sulfate it should first be treated with a warm solution of sodium carbonate to decompose the calcium sulfate, and the sulfuric acid determined in the solution after having added hydrochloric acid.

389. Estimation of the Chlorin.—The estimation of the chlorin is of great importance in certain cases. When this element is lacking in the soil, which, however, is rare, certain plants appear to suffer from its absence. The quality of the forage plants in particular is influenced by it; but when the chlorids are too abundant, which is a frequent case, they prevent or arrest completely the progress of vegetation. Salty soils are, in general, completely sterile. In the proportion of one pound in a thousand in the earth, sodium chlorid is to be regarded as injurious. It is necessary, therefore, in analysis to take account of two cases; viz., those of soils poor in chlorids and those of soils rich in chlorids.

For soils poor in chlorids the French method directs that^[254] 200 grams of the earth are to be washed on a filter with boiling water. The liquor is evaporated to dryness and gently heated to a temperature inferior to redness in order to destroy the organic matter. The residue is taken up by small quantities of water and to the filtered liquor the volume of which should not exceed forty to fifty cubic centimeters are added ten cubic centimeters of pure nitric acid and a sufficient quantity of silver nitrate to produce a complete precipitation. The precipitate is vigorously shaken and allowed to stand for a few hours in a darkened locality. The precipitate is collected upon a double filter and the silver chlorid, after proper desiccation, is weighed.

When the soil is rich in chlorids it is washed as has just been described upon a filter. The wash-waters are made up to one liter and fifty cubic centimeters, equivalent to ten grams of the soil, are taken for analysis. This quantity is treated exactly as described above.

390. Wolff's Method of Estimating Chlorin in Soils.^[255]—Three hundred grams of the soil are treated with 900 cubic centimeters of pure

water containing a little nitric acid for forty-eight hours with frequent shaking. Four hundred and fifty cubic centimeters are then filtered and the clear liquid evaporated to 200 cubic centimeters. The chlorin is then precipitated with silver nitrate. The quantity obtained, corresponds to that found in 150 grams of the air-dried soil.

A second method, Mohr's, is as follows: Fifty grams of the soil are placed in a platinum dish and moistened with a concentrated solution of potassium nitrate, free from chlorin. The mass is evaporated to dryness and gradually heated to a red heat. After cooling it is moistened with water and washed into a beaker and the solid mass quickly separated. The clear liquid is poured off and the residue again washed with water. The clear liquid obtained is saturated with acetic acid, carefully evaporated to dryness and after solution in water, filtration and the addition of a little nitric acid, the chlorin therein is precipitated by a silver nitrate solution, and the precipitate collected and weighed as usual.

391. Method of Petermann.^[256]—Chlorin in the soil is estimated at the Gembloux station by digesting 1,000 grams of the sample with two liters of distilled water with frequent shaking for thirty-six hours. After allowing to stand for twelve hours with the addition of one gram of powdered magnesium sulfate to facilitate the deposition of suspended matter one liter of liquid is siphoned and evaporated in a platinum dish with the addition of a few drops of a solution of potassium carbonate free from chlorin and nitric acid. The concentrated solution is filtered, washed, and made up to 250 cubic centimeters. Take 100 cubic centimeters of the solution add some nitric acid and precipitate the chlorin with silver nitrate. The rest of the solution is reserved for the estimation of nitrate.

392. Estimation of Silicic Acid.—*Direct Estimation.*—The sample of soil in the method of Berthelot and André^[257] is mixed with two or three times its weight of pure sodium carbonate and fused in a silver crucible until complete decomposition has taken place. The residue is dissolved in water and dilute hydrochloric acid. The silicates are decomposed by this treatment and the solution is evaporated to dryness on the water-bath, and when dry slightly heated. The silicic acid (silica) is by this treatment rendered insoluble. It is collected on a filter, washed, ignited, and weighed. The resulting compound should be mixed with ammonium fluorid and sulfuric acid, and after the disappearance of the silica the residue should be

dried and weighed. The loss in weight represents the true silica. The loss in weight should be corrected by calculating the sulfates of the alkalies back to oxids. This correction can be neglected when the work has been carefully done, and the washing of the original silica has been well performed.

Indirect Estimation.—The total silica may be estimated indirectly by subtracting from the total weight of the sample the sum of the weights of the other constituents resulting from the separate estimation of each of them after decomposing the sample with hydrofluoric acid.

393. Simultaneous Estimation of Different Elements.—The operations and processes for the estimation of each of the elements have been described, but it is often best to carry on an operation in such a way as to gain time by making a single decomposition upon a quantity of soil of some considerable magnitude, and using the results of the solution for the determination of the different substances. From the operations already described it will be easy to make a combination of methods by which all or nearly all the important constituents in a soil may be determined in a single sample. Of the various methods proposed, that of the commission of the French agricultural chemists may be taken as a type.^[258] In the case of the estimation of lime, potash, magnesia, and sulfuric acid, in which the operation is carried on in a soil which is not incinerated, time may be saved by digesting a considerable quantity of the soil with concentrated nitric acid for a period of five hours. It is best to take 100 grams of the soil and increase proportionally the nitric acid. The filtrate, after washing, is made up to one liter and thoroughly shaken. From this amount of liquid, portions are taken corresponding to the weights of soil upon which the operation for the determination of each of the constituents would be conducted. For example, for the estimation of lime in the case of a very calcareous earth, ten cubic centimeters representing one gram of the original sample, in the case of a soil poor in carbonates 100 cubic centimeters representing ten grams, and for the estimation of potash, magnesia, and sulfuric acid 200 cubic centimeters, representing twenty grams of the soil, should be taken. This method avoids frequent weighings of the earth and separate treatments thereof by the acid.

On the other hand, in the same portion of the solution, the different elements can be estimated. For example, for the estimation of the potash as has been indicated, in the place of precipitating as a whole the sulfuric acid,

lime, etc., and of afterwards separating the magnesia in the sole aim of eliminating these bodies, they can be collected separately and weighed, thus securing at a single operation several determinations.

At the end, some barium chlorid is added and if the barium sulfate is then collected and weighed, the estimation of the sulfuric acid is effected. To the filtrate there are afterwards added some ammonia and ammonium carbonate to precipitate, at once, the excess of barium and the iron and aluminum oxids, the lime and the phosphoric acid. This separation being effected the filtrate contains still the magnesia and the alkalies. The first can be separated by carbonation by means of oxalic acid, collected, and weighed. Finally the potash itself can be estimated in the state of perchlorate. It has thus been possible in the same suite of operations to estimate in a given quantity of the liquid, the sulfuric acid, the magnesia, the lime, and the potash.

394. Estimation of Kaolin in Soils.—True kaolin is a hydrated aluminum silicate, having the formula $H_4Al_2Si_2O_9$. This substance is, even in concentrated hydrochloric acid, almost completely insoluble. It contains, theoretically, 13.94 per cent of water of combination. The following methods, due to Sachsse and Becker,^[259] can be used for its determination.

Estimation of the Water of Combination.—Heat from one to two grams of kaolin, dried at 100° , for half an hour in a covered platinum crucible to a temperature which shows an incipient red heat when the crucible is partly protected from the daylight with the hand. This treatment does not quite give the whole of the water of combination but nearly all of it. A kaolin is changed by this treatment into a substance which is easily soluble in dilute hydrochloric acid.

Estimation of the Kaolin in Impure Kaolins.—Mineral kaolin, or the kaolin obtained by silt analysis, is dried at 100° to constant weight. It is then heated with strong hydrochloric acid until all the matters which will pass into solution have been dissolved. The residual kaolin is then washed thoroughly with water and ignited for half an hour at a low red heat. The residual mass is a second time extracted with hydrochloric acid until silica no longer passes into solution. The soluble silica is then estimated in the usual way and calculated to kaolin. The result will give the pure kaolin in the sample examined.

The estimation may also be made as follows: Two samples of the impure kaolin are taken and dried to constant weight at 100°. One is extracted with hydrochloric acid in the manner described above and the amount of silica determined. The second is treated directly by ignition to low redness for half an hour, dissolved in hydrochloric acid and the amount of silica determined. The difference in the two percentages of silica corresponds to the silica equivalent to the pure kaolin.

Statement of Results.—It is convenient to incorporate the data obtained by the above methods with the complete mass analysis of the silicate examined. In the sample given below the analysis was made on a clay silt obtained with a velocity of two-tenths millimeter per second.

The mass analysis gave the following data:

Loss on ignition	10.04
SiO ₂	51.52
Al ₂ O ₃	17.93
Fe ₂ O ₃	7.42
CaO	1.57
MgO	6.27
K ₂ O	4.1
Na ₂ O	1.61

The loss on ignition was made up of the combined water and a trace of humus. On gentle ignition only 7.52 per cent of water came off.

The examination of the non-ignited and the gently ignited silica by means of dilute hydrochloric acid, gave the following data:

	Non-ignited.	Gently ignited.	Difference.
Water	10.04	10.04	
Insoluble residue	40.	34.54	−5.46
Al ₂ O ₃	9.04	10.	+0.96
Fe ₂ O ₃	5.96	7.27	+1.31
SiO ₂	25.27	28.	+2.73
Alkalies and alkaline earths	9.69	10.15	+0.46

By a comparison of these data with those obtained by the mass analysis, the following representation of the distribution of the various components

in the clay is obtained:

23.52 per cent SiO_2 in the form of quartz and undecomposed silicates.

2.73 per cent SiO_2 in the form of kaolin.

25.27 per cent in the form of easily decomposable silicates and of the hydrates of SiO_2 .

7.93 per cent Al_2O_3 in the form of undecomposed silicates.

0.96 per cent Al_2O_3 in the form of kaolin.

9.04 per cent Al_2O_3 in the form of easily decomposed silicates and of hydrates.

0.15 per cent Fe_2O_3 in the form of undecomposed silicates.

1.31 per cent Fe_2O_3 in the form of kaolin.

5.96 per cent Fe_2O_3 in the form of easily decomposable silicates and hydrates.

3.40 per cent of alkalies and alkaline earths in the form of undecomposed silicates.

9.69 per cent of alkalies and alkaline earths in the form of easily decomposable silicates.

10.04 per cent of water, including a trace of humus.

Collecting these results the following statement is obtained.

The clay analyzed contained:

10.04 per cent of water, a trace of humus.

35.15 per cent of undecomposed silicates and quartz.

5.00 per cent of kaolin.

50.27 per cent of easily decomposable silicates, hydrates of SiO_2 and hydroxids.

ESTIMATION OF NITROGEN IN SOILS.

395. Introductory Considerations.—The great economic and biologic value of nitrogen as a plant food renders its estimation in soils of especial importance. It is necessary, first of all, to remember that the nitrogen present in soils may be found in three forms; viz., first, in organic compounds, second, as ammonia, and third, as nitric or nitrous acid. Further than this each of these classes of nitrogen may be subdivided. The organic nitrogen may be in a form easily nitrified and rendered available for plant food, or it may be inert and resistant to nitrification, as in humus, or exist in an amid state. The ammoniacal nitrogen may exist in small quantities as gaseous ammonia, or be combined with mineral or organic acids. As nitric or nitrous acid the nitrogen will be found combined with bases, or perhaps in minute quantities as free acid, in passing under the influence of the nitrifying ferment from the organic to the inorganic state. To the latter state it must finally come before it is suited to absorption by plants.

In general, far the largest part of soil nitrogen, excluding the atmosphere diffused in the pores of the soil, is found in the organic state and is derived from the débris of animal and vegetable life and from added fertilizers. As ammonia, the nitrogen can only be regarded as in a transition state, arising from the processes of decay, or incomplete nitrification. As nitric acid, it is found as a completed product of nitrification, or as the result of electrical action. The processes of nitrification and the isolation and determination of the nitrifying organisms will be considered in a special chapter of this manual. By reason of the great solubility of the nitrates, and the inability of the soil to retain them, there can never be a great accumulation of nitric acid in the soil save in localities deficient in rain-fall or in specially protected spots, such as caves. The nitric acid, therefore, produced in the soil passes at once into growing vegetation, or is found eventually in the drainage waters.

The formation of ammonia in soil containing much vegetable matter is thought by Berthelot and André^[260] to be due to the progressive decomposition of amid principles under the influence of dilute acids or alkalies, either in the cold or at an elevated temperature. Soils of the above description, of themselves, contain neither free ammonia nor ammoniacal

salts, and the ammonia which is found in the analysis of these soils comes from the reaction above indicated. The ammonia which comes from these soils, in place of what is given off to the surrounding atmosphere, comes from the same class of decompositions, and these decompositions, in this case, are effected by the water itself, and by the alkaline carbonates of the soil. The amid principles which are thus decomposed belong either to the class of amids proper, derived by the displacement of hydrogen in ammonia by acids, or to the class of alkalamids derived from nitrogenous bases, both volatile and fixed. Among these alkalamids some are soluble in water and some insoluble, and the decomposition of these last by acids or by alkalies may furnish bodies which themselves are either soluble or insoluble in water.

To determine the nature of the nitrogenous principles in a soil rather rich in humus, Berthelot and André applied the following treatment:

A soil containing 19.1 grams of carbon and 1.67 grams of nitrogen per kilogram was first subjected to treatment, at room temperature, with a concentrated solution of potash. By this treatment 17.4 per cent of the nitrogen content was set free under the form of ammonia. One-quarter of this was obtained during the first three days; one-eighth during the next three days. Afterward the action became much more feeble and was continued during forty days longer, and the evolution of the gas was diminished almost proportionately to the time. It appears from the above observations that the amid principles of the soil, decomposable by potash, belong to two distinct groups, which are broken up with very unequal velocities. The soil, treated on the water-bath for three hours at 100° with strong potash, showed the following behavior in respect of its nitrogenous constituents: Nitrogen eliminated in the form of ammonia, sixteen per cent; nitrogen remaining in the part soluble in potash, ten per cent; nitrogen remaining in the part insoluble in potash, seventy-four per cent.

Treatment with Acid.—The nitrogenous compounds of the soil are also decomposed by dilute acids, and often more rapidly than by the alkalies. The method of treatment is substantially the same as that set forth above. The decompositions effected either by alkalies or by acids tend in general to lower the molecular weights of the resulting products. The prolonged action of alkalies at the temperature of boiling water rendered soluble, after twenty-four hours of treatment, 93.6 per cent of the organic nitrogen found

in the vegetable mould. By treating the earth successively with alkalies and acids 95.5 per cent of the total nitrogen were decomposed. These experiments show how the insoluble nitrogen in humic compounds can be gradually rendered assimilable. The action of vegetables is not assuredly identical with those which acids and alkalies exercise. However, both present certain degrees of comparison from the point of view of the mechanisms set in play by the earthy carbonates and carbon dioxide, as well as by the acids formed by vegetation. The reactions which take place naturally, while they are not so violent as those produced in the laboratory, make up by their duration what they lack in intensity.

For a more detailed study of the nature of the nitrogenous elements in soil the following method of treatment, due to Berthelot and André, is recommended:

Treatment of the Soil with Alkalies.—1. Reaction with cold, dilute solution of potash. Take fifty grams of the sample, dried at 110°, and mix with a large excess of ten per cent potash solution and place under a bell-jar containing standard sulfuric acid. The mixture is left for a long time in order to secure as fully as possible the ammonia set free.

Example: Fifty grams of a soil contained 0.0814 gram of nitrogen. Treated as above it gave the following quantities of nitrogen as ammonia:

Nitrogen as Ammonia.				
After	3	days	0.0034	gram.
„	6	„	0.0054	„
„	11	„	0.0065	„
„	17	„	0.0078	„
„	25	„	0.0093	„
„	41	„	0.0107	„
„	46	„	0.0141	„

It is seen that the action still continued after forty days. In the space of forty days 17.4 per cent of the total nitrogen contained in the soil had been converted into ammonia by dilute potash. According to the above observations the amid principles transformed into ammonia under the influence of dilute potash, exist in groups which are acted on with very unequal rapidity.

2. Reaction with hot dilute solution of potash. Take 200 grams of the soil sample, mix with one and one-half liters of dilute potash solution containing fifty grams of potash. Place in a flask and heat on boiling water-bath for six hours. The flask is furnished with a stopper and tubes, and a current of pure hydrogen is made to pass through the liquid, having the double object of preventing any oxidizing effect from the air and of carrying away the ammonia which may be formed. The escaping hydrogen and ammonia are passed into a bulb apparatus containing titrated sulfuric acid.

The sample of soil employed contained in 200 grams, 0.3256 gram of nitrogen. There was obtained at the end of six hours' heating, 0.0366 gram of nitrogen. In other words, 11.24 per cent of the total nitrogen in the sample appeared as ammonia.

Examination of Residue.—After the separation of the ammonia as above described, pour the residue in the flask on a filter, wash with hot water, and determine nitrogen in filtrate and in solid matter on the filter by combustion with soda-lime. The filtrate is, of course, first evaporated to dryness after being neutralized with sulfuric acid.

The insoluble part contained 0.041 gram of nitrogen, *i. e.*, 12.84 per cent of the entire amount.

The soluble part contained 0.2411 gram of nitrogen, *i. e.*, 74.05 per cent of the whole.

Summary of Data.—In the sample analyzed the following data were obtained:

		Of the whole.		
Nitrogen	as ammonia	11.24	per	cent.
„	in insoluble part	12.84	„	„
„	„ soluble part	74.05	„	„
„	not determined	1.87	„	„
	Sum	100.00	„	„

The same experiment in which the heating on the water-bath was continued for thirteen hours gave the following data:

		Of the whole.		
Nitrogen	as ammonia	16.03	per	cent.
„	in insoluble part	9.98	„	„
„	„ soluble part	74.01	„	„
	Sum	100.00	„	„

Further Treatment of Matter Insoluble in Hot Dilute Potash.—A portion of the insoluble portion from the last experiment was treated for thirteen hours longer under the same conditions with dilute hot potash. The soluble and insoluble portions were determined as already described. Of the nitrogen insoluble after thirteen hours, 64.21 per cent remained insoluble after the second thirteen hours. This fact shows that slow and progressive decomposition of the alkalamids in the soil occurs under the influence of hot dilute potash.

Treatment of Matter Insoluble in Hot Dilute Potash with Hydrochloric Acid.—A part of the material insoluble in hot potash after thirteen hours is mixed with dilute hydrochloric acid, in such proportion as to have one-fifth the weight of pure hydrochloric acid to the dry solid matter. Heat in flask on a boiling water-bath for thirteen hours. Determine the nitrogen in the insoluble residue.

Example: In the case given it was found that 54.91 per cent of the nitrogen insoluble in dilute hot potash were dissolved by the hot hydrochloric acid.

This fact shows that insoluble nitrogen compounds contained in the soil are dissolved by dilute acids even more readily than by dilute alkalies at the temperature of boiling water.

Several reactions appear to take place simultaneously when potash is brought into contact with the nitrogenous principles of arable earth. Some of these principles, during the first period of the action become soluble and even form compounds which are not precipitable by acids. When, however, the action of the potash is prolonged, the dissolved bodies lose little by little a part of their nitrogen as ammonia or as soluble alkalamids. They become thus changed either to compounds no longer soluble in the potash, or to those insoluble in the solution when acidified. These compounds, it is true, contain nitrogen, but are poorer in this element and have a higher molecular

weight, or, in other words, are condensation products. These last principles are not absolutely stable in the presence of potash, but are decomposed much more slowly than the original principles from which they were derived.

In general, it may be said that under the influence of alkalies on the nitrogenous principles of the soil there is a tendency to form two classes of bodies, the one more soluble with a lower molecular weight, the other less soluble with a higher molecular weight. The inverse relation between solubility and condensation is in agreement with what is observed in similar reactions with organic bodies in general. It certainly plays an important rôle in the transformations which an arable soil undergoes, either through the mild influences of the air and natural waters, or the more energetic action of vegetables themselves.

The methods of estimating nitric nitrogen will be made the theme of a special study in connection with the chapter on nitrification. There will be considered first, therefore, the methods of determining organic and ammoniacal nitrogen with only such incidental treatment of the methods for nitric nitrogen as the processes applicable to the other forms may contain.

396. Provisional Methods of the Association of Official Agricultural Chemists.^[261]—The nitrogen compounds in the soil are usually placed in three classes.

1. The nitrogen combined with oxygen as nitrates, or nitrites, existing as soluble salts in the soil.
2. The nitrogen combined with hydrogen as ammonia, or organic nitrogen easily convertible into ammonia. The ammonia may exist as salts, or may be occluded by hydrated ferric or aluminum oxids and organic matter in the soil.
3. The inert nitrogen of the soil or the humus nitrogen.

Active Soil Nitrogen.—The material proposed for reducing the nitrates to ammonia, and at the same time to bring ammonia salts and organic nitrogen into a condition for separation by distillation, is sodium amalgam. Liquid sodium amalgam may be readily prepared by placing 100 cubic centimeters of mercury in a flask of half a liter capacity, covering the warmed mercury with melted paraffin, and dropping into the flask at short intervals pieces of metallic sodium, the size of a large pea (taking care that the violence of the reaction does not project the contents from the flask), till 6.75 grams of

sodium have combined with the mercury. The amalgam contains one-half per cent of sodium and may be preserved indefinitely under the covering of paraffin. To estimate the active soil nitrogen, weigh fifty grams of air-dried soil and place it in a clean mortar. Take 200 cubic centimeters of ammonia-free distilled water, rub up the soil with a part of the water to a smooth paste, transfer this to a flask of one liter capacity, washing the last traces of the soil into the flask with the rest of the water. Add twenty-five cubic centimeters of the liquid sodium amalgam and shake the flask so as to break the sodium amalgam into small globules distributed through the soil. Insert a stopper with a valve and set aside in a cool place for twenty-four hours. Pour into the flask fifty cubic centimeters of milk of lime, and distill, on a sand-bath, 100 cubic centimeters into a flask containing twenty cubic centimeters of decinormal sulfuric acid, and titrate with decinormal soda solution, using dimethyl-orange as indicator. Estimate the nitrogen of the ammonia found as active soil nitrogen.

If the ammonia produced is too small in amount to be readily estimated volumetrically, determine the ammonia by nesslerizing the distillate.

Estimation of Nitrates in the Soil.—When it is desired to estimate separately the nitrates in the soil the following method may be used: Evaporate 100 cubic centimeters of the soil extract to dryness on the water-bath, dissolve the soluble portion, of the residue in 100 cubic centimeters of ammonia-free distilled water, filtering out any insoluble residue, place the solution in a flask and add ten cubic centimeters of liquid sodium amalgam, insert stopper with valve, set it aside to digest in a cool place for twenty-four hours, add fifty cubic centimeters of milk of lime, distill and titrate as above, and estimate the nitrogen as N_2O_5 .

Nesslerizing may be substituted for titration when the amount of nitrates is small.

An approximate estimation of the amount of nitrates will be of value in determining which method of estimation to use. This may be done by evaporating a measured quantity of the soil extract, say five cubic centimeters, on a porcelain crucible cover on a steam-bath or radiator, having first dissolved a minute fragment of pure brucin sulfate in the soil extract. When dry pour over the residue concentrated sulfuric acid, free from nitrates, and observe the color reactions produced.

If the nitrate (reckoned as KNO_3) left upon evaporating the quantity of water taken does not exceed the two-thousandth part of a milligram, only a pink color will be developed by adding the sulfuric acid; with the three-thousandth part of a milligram, a pink with faint reddish lines; with the four-thousandth part, a reddish color; with the five-thousandth part, a red color.

By increasing or diminishing the amount of soil extract evaporated to secure a color reaction of a certain intensity, an approximate estimate may be made of the amount of nitrates present.

Blank experiments to test the acid and the brucin sulfate will be required before confidence can be placed in such estimations.

Total Nitrogen of Soils.—The total nitrogen of soils may be determined by the usual combustion with soda-lime, but this process is often unsatisfactory because of the large amount of material required when the organic matter or humus is small in amount.

A modification of the kjeldahl method is more easy to carry out and gives results equally satisfactory. Place twenty grams of soil in a kjeldahl flask, and add twenty cubic centimeters of sulfuric acid (free from ammonia) holding in solution one gram of salicylic acid. If the soil contain much lime or magnesia in the form of carbonate, enough more sulfuric acid must be added to secure a strongly acid condition of the contents of the flask. Add gradually two grams of zinc dust, shaking the contents of the flask to secure intimate mixture. Place the flask in a sand-bath and heat till the acid boils, and maintain the boiling for ten minutes. Add one gram of mercury and continue the boiling for one hour, adding ten cubic centimeters of sulfuric acid if the contents of the flask are likely to become solid. Cool the flask and wash out the soluble materials with 200 cubic centimeters of pure water, leaving the heavy earthy materials. Rinse the residue with 100 cubic centimeters of water, and add this to the first washings. Place this soluble acid extract in a liter digestion flask, add thirty-five cubic centimeters of a solution of potassium sulfid, and shake the flask to secure intimate mixture of the contents. Introduce a few fragments of granulated zinc, pour in seventy-five cubic centimeters of a saturated solution of caustic soda, connect the flask with a condenser and distill 150 cubic centimeters into a flask containing twenty cubic centimeters of acid, using the same acid and alkali for titration used in the kjeldahl method under fertilizers.

Enter the nitrogen found in this operation as total soil nitrogen.

The difference between the total soil nitrogen and the active soil nitrogen will express the inert nitrogen of the soil.

397. Hilgard's Method.^[262]—The humus determination will, in the case of virgin soils, usually indicate approximately the store of nitrogen in the soil, which must be gradually made available by nitrification. Ordinarily (outside of the arid regions) the determination of ammonia and nitrates present in the soil is of little interest for general purposes, since these factors will vary with the season and from day to day. Kedzie proposes to estimate the active soil nitrogen (ammonia plus nitrates and nitrites) by treatment of the whole soil with sodium amalgam and distillation with lime. The objection to this process is that the formation of ammonia by the reaction of the alkali and lime upon the humus amids would greatly exaggerate the active nitrogen and lead to a serious overestimate of the soil's immediate resources.

The usual content of nitrogen in black soil-humus is from six to eight per cent in the regions of summer rains. From late determinations it would seem that in the arid regions the usually small amount of humus (often less than two-tenths per cent) is materially compensated by a higher nitrogen percentage. It thus becomes necessary to determine the humus nitrogen directly; and this is easily done by substituting in the grandeaun process of humus extraction potash or soda-lye for ammonia water, and determining the nitrogen by the kjeldahl method in the filtrate.

The lye used should have the strength of four per cent in the case of potassium hydroxid, three per cent in that of sodium hydroxid. The black humus filtrate is carefully neutralized with sulfuric acid, evaporated to a small bulk in a beaker or evaporating basin, and the reduced liquid finally evaporated to dryness in the kjeldahl flask itself by means of a current of air. The beaker or basin is washed either with some of the alkaline lye, or, after evaporation, with warm concentrated sulfuric acid, which is then used in the nitrogen determination in the usual way.

For the determination of nitrates in the soil it is, of course, usually necessary to use large amounts of material, say not less than 100 grams, and, according to circumstances, five or more times that amount. In the evaporated solution the nitric acid is best determined by the reduction method, as ammonia.

Usually the soil filtrate is clear and contains no appreciable amount of organic matter that would interfere with the determination; yet in the case of alkaline soils (impregnated with sodium carbonate) a very dark colored solution may be obtained. In that case the soil may advantageously be mixed with a few per cent of powdered gypsum before leaching; or the gypsum may be used in the filtrate to discolor it by the decomposition of sodium carbonate and the precipitation of calcium humate. The evaporated filtrate can then be used for the nitrate determination by either the kjeldahl, griess, or the nessler process, which will, of course, include such portions of the ammoniacal salts as may have been leached out.

For the separate determination of these and of the occluded ammonia, when desired, it is probably best to mix the wetted soil intimately with about ten per cent of magnesium oxid and distill into titrated hydrochloric acid. For general purposes, however, this determination is usually of little interest.

398. Müller's Modified Kjeldahl Method.—Numerous difficulties, as stated by Müller,^[263] have attended the attempts to apply the kjeldahl method for the estimation of nitrogen to samples of soil, and he has modified the method to some extent and made comparisons of the quantity of nitrogen by this modified method and by the soda-lime method.

The principal difficulty encountered by him has been in the regular heating of the mixture of fuming sulfuric acid and soil. The particles of soil are deposited at the bottom of the flask and the result is that the bottom layers become overheated, and, being poor conductors of heat, fail to transmit a sufficient quantity of heat to penetrate to the upper layers of the liquid to complete the reaction. In order to avoid this difficulty Müller heats his flask in a small stove formed with a straight vertical cylinder of iron or copper, the upper end of which is covered with a sheet of iron pierced with a hole which allows the neck of the flask to pass through, while the lower end is closed with a piece of sheet iron furnished on its upper surface with a layer of asbestos. This cylinder of metal is surrounded with a second one, concentric with the first through which passes a current of heated gases furnished by an ordinary bunsen. By heating the flask in this stove or furnace an even distribution of the heat is secured to all parts of the mixture, but the little drops of sulfuric acid, which are condensed on the cold part of the neck, sometimes lead to the fracture of the glass as they run down the

sides of the flask to the hot portions. To prevent the reflux of this condensed acid, which only needs to be done near the end of the reaction, when it is necessary to heat to a very high temperature, the neck of the flask is bent at the point immediately above its emergence at the upper surface of the furnace, and carried into a flask of about seventy-five cubic centimeters capacity, which will receive the drops of sulfuric acid condensed during the operation. The furnace has the following dimensions; height, twelve centimeters; diameter of interior cylinder, five and one-half centimeters; diameter of exterior cylinder, seven and one-half centimeters.

It is supported on a triangle of large iron wire and is heated by an ordinary bunsen, or by a concentric bunsen, according to the temperature which it is necessary to obtain. The proportions which should be observed between the amount of earth employed and the sulfuric acid are about as follows:

Of the dry earth, fifteen grams; of the fuming sulfuric acid, thirty cubic centimeters. There should also be added to the mixture about three-tenths of a gram of pure stearic acid, or better, benzoic acid. When the soil to be analyzed does not contain carbonate, the sulfuric acid should be added in two portions. At first add about twenty cubic centimeters of the acid, and after shaking it, the other ten cubic centimeters, running it in from a burette or a pipette in such a manner as to wash thoroughly the neck and sides of the flask. If the earth contain carbonate, however, it is necessary to add the fuming acid in small portions of about five cubic centimeters at a time, waiting each time until the disengagement of the gas caused by the previous addition has ceased. A soil which contains from thirty to forty per cent of calcium carbonate should be carefully treated in a porcelain capsule with a slight excess of sulfuric acid, pure and dilute. The mixture is afterward to be evaporated to dryness upon a sand-bath and the residue heated in a drying oven to 110° . The mass is then pulverized, introduced into the flask, treated with three-tenths of a gram of benzoic acid and thirty cubic centimeters of fuming sulfuric acid, and treated as indicated above. In all cases it is necessary to continue the heating until the contents of the flask are colorless.

With soils containing considerable quantities of iron, however, a slight red color will probably be observed which will not interfere with the accuracy of the tests.

The heating should at first be gentle and the temperature afterward elevated little by little, and finally the heat should be sufficiently great to distill about one and one-half cubic centimeters of sulfuric acid. The operation lasts from twelve to thirteen hours. As the reaction is terminated the cooled mass is taken up with water absolutely free from ammonia. It is filtered into a flask, and washed upon the filter until the volume of the filtered liquid is about 350 cubic centimeters. Afterward an excess of soda-lye, at 50° baumé is added, then a few pieces of quartz to facilitate boiling. The flask is then connected with a condenser, the liquid distilled and received in a conical flask closed by a cork having two holes, of which one permits the entrance of the end of the condenser, and the other a glass tube which is connected with a small flask containing water, the neck of the receiving flask being inclined toward the condenser to avoid the entrainment of any of the alkaline liquid which may be distilled. The receiving flask rests upon two or three pieces of sheet iron and is heated with an ordinary burner, and ebullition is perfectly regular. From 170 to 180 cubic centimeters of the liquid are distilled in from three and one-half to four hours. The distilled liquid, treated with a few drops of litmus, is titrated by a solution of sulfuric or hydrochloric acid, of which one cubic centimeter corresponds to 0.001 cubic centimeter of nitrogen.

399. Modification of the Kjeldahl Method by Arnold and Wedemeyer.^[264]—For the oxidizing liquid a mixture of three grams of benzoic acid with forty cubic centimeters of H_2SO_4 is employed. After placing in the digestion flask with the nitrogenous body the whole is gently shaken for a few minutes to prevent clotting. The temperature is then raised until acid vapors begin to come off, when one gram of copper sulfate and one gram of mercuric oxid are added; and after ten to fifteen minutes, to avoid foaming, ten to twenty grams of potassium sulfate. The sublimate noticed on the walls of the flask is benzoic acid and does not interfere with the accuracy of the determination.

This method has given good results with the alkaline nitrates, the nitrates of barium, mercury, silver, lead, and with strychnia, ammonia, pyridin, azobenzol, dinitrobenzol, and picric acid.

400. Prevention of Bumping During Distillation.—Daffert has employed the modified kjeldahl method, but found considerable difficulty in using the same owing to the violent bumping of the liquid in the

distillation. This was especially the case where the sample contained a large proportion of sand. To overcome this annoyance and danger he devised the following process:^[265]

Fit into the mouth of a large-mouthed distillation flask a stopper having two perforations. Through one of the perforations pass the usual distillation tube, through the other a similar tube connected with a supply of steam. Bring the contents to a brisk boil, after which a small current of steam is turned on, allowing the same to pass in a small stream throughout the distillation. By this means, not only is all danger from bumping avoided, but the time required for the distillation shortened. By the old method it usually requires from fifteen to twenty minutes, whereas the former requires from six to ten minutes.

It is advisable to filter all samples of soils having a large proportion of sand.

401. Determination of Organic Nitrogen by the Soda-Lime Method.

—In the description of the method following, the directions of the French Commission of Agricultural Chemists have been taken as the basis of the analytical process.^[266] This method is, in this country, almost superseded by the moist combustion process with sulfuric acid. By reason of its long use, however, and because it is still regarded as the best method by the agricultural chemists of France, Italy, and England, it merits a full description. It is recommended also by Berthelot and André,^[267] by the International Congress of Chemists, held in Paris in 1889, by the Italian chemists, and by the official Belgian method,^[268] in all cases where nitrates are not present in notable quantities.

The nitrogen which is found in soils in the organic state is transformed into ammonia when it is heated with soda-lime. This reaction is the base of the process of analysis which has so long been used for this class of bodies. The analytical process is conducted as follows:

A well-cleaned glass combustion tube, closed at one end, is used. The length of the tube is from thirty-five to forty centimeters. It is filled first to a depth of two centimeters with calcium oxalate; afterwards to a depth of five centimeters with soda-lime in small fragments; afterwards with the mixture to be analyzed; viz., of ten grams of the sample of soil, or twenty grams if poor in nitrogen and organic matters, with soda-lime reduced to a coarse powder. This mixture should occupy a length of about twenty centimeters in

the tube. The soil and soda-lime are mixed in a mortar. Afterwards the mortar is rubbed with small quantities of soda-lime, and this, together with the copper boat which has been used in introducing the mixture, is thoroughly washed with the soda-lime, which is poured into the tube until it is filled to within four centimeters of its open extremity. The open end of the tube is then closed with a wad of asbestos packed sufficiently tight to prevent the carrying off of the soda-lime by the gas which may be generated during the combustion. The combustion should be commenced by heating the tube near the open extremity until it is red and carrying the heat progressively towards the part containing the soil mixed with the soda-lime. An ordinary gas combustion furnace should be used and the heat graduated in such a way that the bubbles of gas pass off regularly and not too rapidly. The gas is conducted into a bulb tube containing a decinormal standard sulfuric acid colored with litmus. The combustion is continued until the whole of the organic material is decomposed, care being taken not to raise the combustion tube above a low redness in order to avoid its softening. At the end, however, the temperature of the combustion tube should be raised to a bright red, and the part containing the calcium oxalate should be heated little by little for the purpose of evolving hydrogen, which is used to drive out the last traces of ammonia. After the combustion is completed, and the last traces of ammonia driven out, the standard acid which has received the evolved ammonia is removed, the tube leading to it washed, the wash-water collected with the rest of the liquid and titrated with a standard solution of lime-water, the strength of which has previously been determined against standard sulfuric acid.

402. Preparation of the Standard Sulfuric Acid.—The sulfuric acid to be used in making the standard solutions should be previously boiled for half an hour in a platinum dish and allowed to cool in a desiccator. It should contain 61.25 grams of sulfuric acid in one liter. It is recommended that the flask which holds the sulfuric acid should be one which has been used for a long time for holding concentrated sulfuric acid, in order to avoid any action of the alkali in the glass upon the acid after its strength has been determined. The solution before described is of such strength as to have each cubic centimeter equivalent to one milligram of nitrogen.

For the estimation of the nitrogen in the soil a tenth normal solution should be used, which is prepared by taking 100 cubic centimeters of the normal solution, described above, and diluting to one liter.

Preparation of the Lime-Water.—From 200 to 300 grams of slaked lime are placed in a closed flask of about five liters capacity. This is filled with water and shaken frequently, and left to deposit the matter in suspension. The water which contains the saline particles which may have been present in the lime is then poured off. Fresh water is then poured on and the flask shaken from time to time. To use this lime-water the clear part of it is decanted into a flask, avoiding, as much as possible, access to the air. The flask is closed with a cork carrying two tubes drawn out and bent at a right angle. One of these serves for pouring off the water and the other serves for the entrance of the air. These two tubes are themselves closed by means of a rubber tube carrying a pinch-cock. The strength of the lime-water is fixed by titration with the decinormal standard sulfuric acid.

Preparation of the Soda-Lime.—Six hundred grams of slaked lime in fine powder are saturated with 300 grams of caustic soda dissolved in 300 cubic centimeters of water. The whole is rubbed into a paste and introduced into a crucible which is heated to redness. The contents of the crucible, still hot, are poured out, and rapidly reduced to fragments in a copper mortar in such a manner as to have the pieces about the size of a pea, and without having too much finely powdered soda-lime mixed with it. While the matter is still hot it is placed in a flask and well-stoppered. In order that this reagent should contain no nitrogen it is indispensable to use in its preparation materials which contain no trace of nitrates.

Preparation of the Calcium Oxalate.—In a small copper vessel place 100 grams of oxalic acid and add gradually, bringing it to boiling, enough water to dissolve it. Afterwards place in the solution small portions of slaked lime in a state of powder, constantly testing it until turmeric paper indicates that there is a little lime in excess. It is then evaporated, stirring vigorously on the open fire, and the evaporation is finally finished on a steam-bath. The dried material is placed in a flask and well-stoppered. The oxalic acid which is used in this preparation should be free from every trace of nitrogen.

Preparation of the Litmus Solution.—Five grams of litmus are placed in a flask with a flat bottom. Afterwards a few cubic centimeters of ammonia are added, twenty five grams of crystallized sodium carbonate, and ten cubic centimeters of water. This mixture is left to digest for sometime, with frequent stirring, at a temperature of from 60°–80°. The digestion is finished in about four or five days, during which time, at intervals, a few

drops of ammonia are added, sufficient to maintain always the ammoniacal odor. At the end of this time 200 cubic centimeters of water are added and the digestion allowed to continue several days more, still maintaining the solution alkaline with ammonia. A slight excess of hydrochloric acid is added, and the matter which is precipitated is received upon a filter where it is washed several times with cold water and allowed to dry at a low temperature.

For use, from one to two grams of this dry precipitate are dissolved in 100 cubic centimeters of alcohol, and there is thus obtained a litmus solution of extreme sensibility.

403. Treatment of Soil Containing Nitrates.—Nitrates exist in small quantities in all arable soils. When treated for nitrogen by the soda-lime method above described, a part of the nitric nitrogen is changed to the state of ammonia, while another part escapes estimation altogether, causing an error which it is important to point out. When the soils contain only small quantities of nitrates this error is insignificant and does not affect sensibly the results, but in the case of earths rich in nitrates it is necessary first to eliminate them before the determination of the nitrogen by the soda-lime method. The operation is carried on as follows:

Twenty grams of the soil are washed on a small funnel, furnished with a plug of asbestos, with small quantities of pure water, in such a way as to cause thirty to forty cubic centimeters of water to pass through. The whole of the nitrate is thus removed. The soil is now dried and submitted to analysis by the soda-lime method as just described. There are removed with the nitrate only small traces of organic nitrogen, too small to influence the results of the analysis. If, however, it is desired to remove altogether this slight cause of error, evaporate the wash-waters, above described, to two or three cubic centimeters; add a few drops of a concentrated solution of ferrous chlorid and as much hydrochloric acid, and boil some minutes in order to drive off, in the state of nitrogen dioxid, all the nitric acid. The residue is evaporated to dryness and contains the traces of organic nitrogen. This is added to the soil which is to be treated by the soda-lime method.

404. Müller's Method.—The determination of nitrogen in the soil by soda-lime is carried on as follows by Müller:^[269]

Fifteen grams of fine earth, dried and mixed with a little sugar, are mixed with thirty grams of soda-lime in powder. The bottom of the combustion

tube contains a little moist soda-lime, which is heated at the end of the operation at the same time that a current of pure hydrogen is made to pass through it, and the temperature of the tube is raised, little by little, to a distinct redness. The contents of the receiving bulbs are distilled, after the addition of water and soda, in the same apparatus which served in the estimation of nitrogen, by the kjeldahl method; the determinations and titrations are made also under the same conditions.

Blank determinations are also made under the same conditions to determine the amount of correction to be made by the two methods. Soda-lime, heated with pure sugar, gave 0.0002 gram of nitrogen for a total weight of fifty-five grams of the soda-lime contained in the tube. The fuming sulfuric acid gave 0.0011 cubic centimeters of ammoniacal nitrogen for the volume of thirty cubic centimeters.

The numbers obtained by the kjeldahl method in general, are lower than those obtained by the soda-lime method when no stearic or benzoic acid is used. The numbers obtained when stearic acid alone was used were sometimes inferior to those obtained by the soda-lime method. The numbers obtained when benzoic acid is used are, in general, about the same as those obtained by the soda-lime method.

It would seem that the double distillation, outlined above, for the kjeldahl method, would not be necessary if due care were exercised in the first distillation. This variation, therefore, seems to be unnecessary.

In the soda-lime method, time would be saved by the reception of the ammonia in standard acid, and its titration in the usual way, unless a further purification of the nitrogenous products of the combustion by the final distillation be desired.

405. Volumetric Determination of the Nitrogen.—Instead of separating the nitrates, the total nitrogen in the soil can be determined directly by the classic method of Dumas, which consists in bringing the whole of the nitrogen into a gaseous state and afterwards measuring its volume.

The following method illustrates the general principles of the determination:

A glass combustion tube closed at one end, about one meter in length, is selected. In the bottom of this tube is placed some potassium bicarbonate in a crystalline form, in small pieces, filling the tube to a distance of about twenty centimeters. Afterwards copper oxid is placed to the depth of ten

centimeters and finally a mixture of from twenty to thirty grams of the earth with thirty to forty grams of copper oxid in a fine state of subdivision, and about ten grams of metallic copper obtained by reducing the copper oxid by hydrogen. Next the tube is filled with copper oxid to a depth of from twenty to twenty-five centimeters, and afterwards with reduced copper to the depth of at least twenty-five centimeters, and after this another layer of copper oxid of about five centimeters, and finally a plug of asbestos. The combustion tube is closed with a stopper carrying a glass tube of about ninety centimeters in length, of which the extremity, bent into the form of a J, extends to a mercury trough. The glass combustion tube is surrounded with brass gauze, except that part which contains the potassium bicarbonate. The beginning of the operation consists in heating the tube to decompose a part of the potassium bicarbonate, until the whole of the apparatus is filled with carbon dioxid. In order to determine that the whole of the air has been expelled and that the apparatus is entirely filled with carbon dioxid, a part of the gas which is disengaged, is received into a jar filled with mercury, in which a little potash-lye has been placed. If the gas is entirely absorbed by the potash, so that there remain only unappreciable particles, the tube can be regarded as completely free of air. When assurance is given that the air is all out of the apparatus, a jar of about 300 cubic centimeters capacity, filled with mercury and containing from thirty to forty cubic centimeters of a solution of potash of a density of 42° baumé, is placed over the outlet tube. The combustion is commenced by heating the anterior part of the tube, avoiding the heating of the part containing the earth. When the first part of the tube has reached the red stage the part containing the earth is gradually heated in order to obtain a gentle evolution of gas. The temperature of the tube is carried to redness and the heating gradually carried back toward the closed extremity, but avoiding raising the temperature of the part containing the potassium bicarbonate. The red heat is continued as long as bubbles of gas are discharged into the reservoir. When the evolution of gas has ceased the apparatus is again filled with carbon dioxid for the purpose of driving out the last traces of nitrogen, by heating again the part of the tube containing the potassium bicarbonate. The evolution of the carbon dioxid should be maintained for about fifteen minutes. At the end of this time all the nitrogen will be found in the receiving jar. Sometimes a small quantity of nitrogen dioxid is formed incidentally in the operation. After waiting for a quarter of an hour, in order

to permit all the carbon dioxide which may have escaped into the reservoir to be completely absorbed, the receiving jar is carried to a water-basin and the mercury allowed gradually to escape; its place being taken by the water. The gas is then transferred into an azotometer where its volume and temperature are read in the usual way.

In order to absorb any nitrogen dioxide which may be admixed with the nitrogen itself, a little crystal of ferrous sulfate is introduced. The reservoir containing the nitrogen is carried to the mercury trough, and the water which it contains is nearly all run out in such a way as to be replaced with mercury, great care being exercised to avoid any escape of gas. Afterwards there is introduced over the mercury a crystal of ferrous sulfate and the azotometer is shaken until this crystal is dissolved by the water which it still contains. It is then allowed to remain for twenty hours. At the end of this time the nitrogen dioxide is absorbed and the volume of the gas is again read as before. One-half only of the total loss should be subtracted, since the volume of the nitrogen dioxide is twice the volume of the nitrogen itself. For the practice of this method, in connection with the use of a mercury pump, the directions which will be given under fertilizers may be consulted.

406. Estimation of Ammonia.—Ammonia exists ordinarily only in very small quantities in the soil, since it is incessantly transformed into nitrate or diffused in the air. Nevertheless, it is sometimes interesting to determine its quantity.

The method of determining the ammonia in soils is one of extreme delicacy on account of the small proportion therein, and the difficulty of expelling it without at the same time converting some of the organic nitrogen into ammoniacal compounds. The various methods employed for this purpose may be classified as follows:

1. Treatment of the soil with soda-lye in the cold, and the absorption of the ammonia given off by standard sulfuric acid.
2. The method of Boussingault, which consists in replacing the soda-lye with magnesia and distilling the ammonia at a boiling temperature, absorbing the distillate in a standard acid.
3. A modification of the above method, due to Schloesing, which consists first in extracting the ammonia by hydrochloric acid and subjecting the extract to distillation with magnesia.

4. The method of Knop consists in treating the soil in a closed cylinder with soda-lye containing bromin. The ammonia set free by the lye is decomposed in the presence of bromin into free nitrogen and hydrochloric acid. The nitrogen is collected and measured in an azotometer. The brom-soda-lye is prepared by dissolving 100 grams of sodium hydroxid in 1,200 cubic centimeters of water and adding twenty-five cubic centimeters of bromin.

5. The process described under 4, as shown by Baumann,^[270] does not give accurate results and it has been modified by him as follows: Two hundred grams of soil are treated with 100 cubic centimeters of dilute hydrochloric acid (one part acid and four of water) free of ammonia; 300 cubic centimeters of ammonia-free distilled water are added and the whole digested for two hours with frequent stirring. If a soil contain much calcium carbonate larger quantities of acid must be used. Two hundred cubic centimeters of the filtrate are placed in an evolution flask, connected with an azotometer, with five grams of freshly burned magnesia. The mixture is then oxidized as follows: Ozone is generated by adding three parts by weight of sulfuric acid to one part of dry and powdered potassium permanganate. A stream of air is drawn through the ozone generator by an aspirator, and the ozone is conducted into a flask containing the hydrochloric acid extract of the soil and magnesia. The oxidation is completed in about ten minutes. The mixture is then brought into the evolution flask of the azotometer and the nitrogen set free and measured in the usual way.

It has been shown that if asparagin or glutamin be present in the soil they are decomposed by the soda-lye and the results obtained are too high. It has been further proved that soils which contain a notable quantity of humus give, with soda-lye in the cold, a practically continuous evolution of ammonia. Moreover, soils which are rich in humus and which have been treated by distillation with magnesia give, on subsequent treatment with soda-lye, considerable additional quantities of ammonia.

Comparison of Methods of Estimating Ammonia.—Baumann has determined the ammonia-nitrogen in various soils by the soda-lime method; distillation of the hydrochloric acid extract with magnesia, and the azotometric method modified as indicated above. These methods will be designated as 1, 2, 3, respectively in the following table.

	METHOD.		
	1.	2.	3.
	Ammonia-nitrogen in one kilogram of soil.		
No. of sample.	Gram.	Gram.	Gram.
1	0.0448	0.02227	0.02781
2	0.0168	0.01105	0.01326
3	0.0336	0.01771	0.02214
4	0.0056	0.00443	0.00443
5	0.0280	0.02337	0.02894
6	0.0196	0.01243	0.01672

From the above figures it is seen that the method usually attributed to Schloesing gives uniformly higher numbers than either of the other processes, while the third gives slightly higher values than the second.

407. The Magnesia Distillation Process.—If a sample of soil be distilled directly with magnesia and water, there is danger on the one side of not extracting all the ammonia, by reason of the absorbing power of these bodies, and on the other, of transforming into ammonia the nitrogen of the organic matters. It is therefore preferable to separate the ammonia from the soil in the form of chlorid, and to subject this extract to distillation.

In fifty grams of the soil the humidity is determined by drying at 100° until there is no further loss of weight. The quantity of moisture being known, 200 grams of soil are taken and moistened with water, and then there is added, in small portions, some dilute hydrochloric acid, shaking frequently until the whole of the calcium carbonate present is decomposed. The liquor should remain acid at the end of the operation, but without containing a notable excess of acidity. Knowing beforehand the quantity of moisture contained in the 200 grams, water is added until the total quantity shall be equal to 500 cubic centimeters. The whole is then shaken and allowed to repose, and filtered rapidly, covering the funnel with a glass vessel and receiving the liquid which runs through in a flask with a narrow opening. Two hundred and fifty cubic centimeters of this liquor, or mixture, represent 100 grams of earth of known humidity. This quantity is introduced into a flask for determining the ammonia and five grams of calcined magnesia added. Before commencing the distillation, assurance should be

had that the magnesia has completely saturated the acid in excess, and that the liquor is alkaline.

If, by chance, the liquor should be still acid it would be necessary to add sufficient magnesia in order that the reaction should be manifestly alkaline. Afterwards the distillation is begun and the ammonia is received in an appropriate vessel containing one-tenth normal sulfuric acid and titrated in the usual way, or nesslerized.

Inasmuch as the quantities of ammonia contained in the earth are generally very small it is necessary to be very particular in order to avoid errors. The distilled water which is employed should be deprived of all traces of ammonia by prolonged ebullition, and the hydrochloric acid should be distilled in the presence of a little sulfuric acid. The treatment with hydrochloric acid is for the purpose of destroying the absorbing properties of the soil for ammonia, and to permit this last to enter into solution as chlorid. When there is need of very great precision it is convenient to make a blank operation with the hydrochloric acid and water which are employed, in order to make a correction for the traces of ammonia which these reagents may contain.

408. Estimation of Ammoniacal and Amid Nitrogen by the Method of Berthelot and André.^[271]—Heat one hundred grams of earth for thirty hours on a steam-bath with about .500 cubic centimeters of dilute hydrochloric acid (fifteen grams of hydrochloric acid to 500 cubic centimeters of water). At the end of this time throw the contents of the flask on a filter and wash with hot water until acid reaction has ceased. Determine both the ammoniacal and amid nitrogen in the soluble, and the total nitrogen in the insoluble portion, the ammoniacal by distillation with magnesia, and the amid and total with soda-lime.

Example. A soil contained 0.1669 per cent total nitrogen. Of this there were obtained:

As ammoniacal nitrogen	13.7	per	cent
In the soluble part as amid nitrogen	56.2	„	„
In the insoluble part, total nitrogen	29.7	„	„
Sum	99.6	„	„

Treatment of the Insoluble Portion.—Treat the part insoluble in hydrochloric acid with a three per cent solution of potash on a steam-bath for thirty hours. Estimate the nitrogen remaining insoluble, from which the part dissolved can be determined by difference. The potash will dissolve usually about two-thirds of the remaining nitrogen.

About ninety per cent of the total nitrogen present in an arable soil will be rendered soluble by successive treatment with acid and alkali. The reverse treatment will give practically the same result. It is therefore immaterial, from an analytical standpoint, whether the acid or alkali be used first.

409. Estimation of Volatile Nitrogenous Compounds Emitted by Arable Soil.—The following method, due to Berthelot and André,^[272] may be practiced:

Porcelain pots, containing one kilogram of soil, are placed under bell-jars of fifty liters capacity adjusted to glass dishes designed to receive the waters of condensation.

During the first period the pots are to be sprinkled from time to time, during the duration of the experiment, through the upper tubulature, so as to prevent the soil from becoming dry. The water is partly condensed on the sides of the bell-jar. It is removed each week through the inferior tubulature, treated with a little dilute sulfuric acid, and preserved for further study. A small vessel containing dilute sulfuric acid is placed near the porcelain pot for the purpose of collecting, as far as possible, the evolved ammonia.

During the second period the pots are not sprinkled, the soil becomes dry and there is no longer any condensation of water on the walls of the bell-jar. The two periods should include about five months, from May to October.

At the end of the second period the following determinations are to be made:

1. The ammonia absorbed by the dilute sulfuric acid.
2. The ammonia set free by distillation with magnesia, such as may have accumulated in the condensed water.
3. The organic nitrogen contained in the latter after elimination of the ammonia. This is determined by adding a slight excess of acid, evaporation to dryness, and combustion with soda-lime, or by moist combustion with sulfuric acid.

Example:

Earth Employed.—One kilogram of sandy clay containing total nitrogen, 0.09 gram. Nitrogen in sprinkling water, 0.000048 gram.

Nitrogen in Exhaled Products.—

FIRST PERIOD. SPRINKLING.

Ammoniacal nitrogen collected in the dilute sulfuric acid	0.00012	gram.
Ammoniacal nitrogen collected in the condensation waters	0.00012	„
Organic nitrogen in condensation waters	0.00220	„
Sum	0.00244	„

SECOND PERIOD. NO SPRINKLING.

Ammoniacal nitrogen in dilute sulfuric acid	0.000007	gram.
„ „ „ condensed water	0.000007	„
Organic „ „ „ „	0.000040	„
Sum	0.000054	„

Conclusions.—The exhalation of nitrogenous compounds takes place with a certain relative activity, about two milligrams in two months and a half, as long as the soil is kept moist by sprinkling.

In the second period, without sprinkling, the exhalation is reduced to a mere trace.

The vessel containing the dilute sulfuric acid placed near the porcelain pot absorbs only about one-half of the ammoniacal nitrogen set free. The nitrogen emitted under other forms than ammonia is, in every instance, greatly superior in quantity, and this is the most important of the observed phenomena. This is true at least with the kind of soil with which the experiment was made. With arable soil containing twenty times as much nitrogen as the soil described above this order is reversed,^[273] the ammoniacal prevailing over the non-ammoniacal nitrogen volatilized.

These phenomena are doubtless greatly influenced in soil under culture by microbes, and the lowest orders of vegetation to which are doubtless due the traces of non-ammoniacal volatile nitrogenous compounds, a sort of vegetable ptomaines.

410. General Conclusions.—In the light of our present knowledge concerning the methods of nitrogen determination in the soil in the form of organic compounds and ammonia, moist combustion with sulfuric acid is to be preferred to the older soda-lime process. For the nitrogen combined as ammonia, the extraction of the sample with hydrochloric acid and subsequent distillation with an excess of freshly calcined magnesia, are recommended. For the study of the progressive decomposition of the nitrogenous compounds, the various processes devised by Berthelot and André are the best.

The origin of the nitric acid in the soil, the methods of studying the various nitrifying organisms, and of estimating the nitric acid produced, will form the subject of the next part.

NOTE.—At the Eleventh Annual Convention of the Association of Official Agricultural Chemists, held in Washington, August 23, 24 and 25, 1894, the following process of soil extraction was adopted as the official method:

Preparation of the Sample.—500 grams or more, of the air-dried soil, which may be either the original soil or that which has been passed through a sieve of coarser mesh, are sifted upon a sieve with circular openings one-half millimeter in diameter, rubbing, if necessary, with a rubber pestle in a mortar, until the fine earth has been separated as completely as possible from the particles that are too coarse to pass through the sieve. The fine earths thoroughly mixed and preserved in a tightly stoppered bottle from which the portions for analysis are weighed out.

The coarse part is weighed and may be subjected to further examination, (as in *Bulletin 38, Div. of Chem.*, pp. 65, 75 and 200.) It may sometimes be necessary to wash the soil through the one-half millimeter sieve with water, in which case proceed as directed on pp. 65 and 75 of the above *Bulletin*. The use of water is to be avoided whenever possible.

Determination of Moisture.—Heat two to five grams of the air-dried soil in a flat-bottomed, tared platinum dish; heat for five hours in a water-oven kept briskly boiling; cover the dish, cool in a desiccator, and weigh.

Repeat the heating, cooling, and weighing at intervals of two hours till constant weight is found, and estimate the moisture by the loss of weight. Weigh rapidly to avoid absorption of moisture from the air. An air-bath must not be used in this determination.

Determination of Volatile Matter.—The platinum dish and soil used to determine moisture are used also to determine volatile matter. Heat the dish and dried soil to full redness until all organic matter is burned away. If the soil contain appreciable quantities of carbonates, the contents of the dish, after cooling, are to be moistened with a few drops of a saturated solution of ammonium carbonate, dried and heated to dull redness to expel ammonium salts, cooled in the desiccator and weighed.

The loss in weight represents the organic matter, water of combination, ammonium salts, etc.

Extraction of Acid-Soluble Materials.—In the following scheme for soil analysis it is intended to use the air-dried soil from the sample bottle for each separate investigation. The determination of moisture, made once for all on a separate portion of air-dried soil, will afford the datum for calculating the results of analysis upon the soil dried at the temperature of boiling water. It is not desirable to ignite the soil before analysis or to heat it so as to change its chemical properties.

The acid digestion is to be performed in a flask so arranged that the evaporation of acid shall be reduced to a minimum, but to take place under atmospheric pressure and at the temperature of boiling water. Any flask resistant to acids is suitable, but it is not necessary to use a condenser, as a simple bohemian glass tube eighteen inches in length will answer the purpose of preventing loss of acid. Where it is not desired to determine sulfur trioxid, an erlenmeyer fitted with a rubber stopper and hard glass tube will answer. The flask must be immersed in the water-bath up to the neck or at least to the level of the acid and the water must be kept boiling continuously during the digestion.

In the following scheme, ten grams of soil are taken, this being a convenient quantity in most soils, in which the insoluble matter is about eighty per cent. If desired, a larger quantity of such soil may be taken, using a proportionately larger quantity of acid and making up the soil solution to a proportionately larger volume. In very sandy soils, where the proportion of insoluble matter is ninety per cent or more, twenty grams of soil are to be digested with 100 cubic centimeters of acid and the solution made up to 500 cubic centimeters or a larger quantity may be used, preserving the same proportions. It is very important that the analyst assure himself of the purity

of all the reagents to be used in the analysis of soils before beginning the work.

Acid Digestion of the Soil.—Place ten grams of the air-dried soil in a 150 to 200 cubic centimeter bohemian flask, add 100 cubic centimeters of pure hydrochloric acid of specific gravity 1.115, insert the stopper with condensing tube, place in a water or steam-bath and digest for ten hours continuously at the temperature of boiling water, shaking once each hour. Pour the clear liquid from the flask into a small beaker, wash the residue out of the flask with distilled water on a filter adding the washings to the contents of the beaker. The residue after washing until free of acid, is to be dried and ignited as directed below. Add one or two cubic centimeters of nitric acid to the filtrate, and evaporate to dryness on the water-bath, finishing on a sand or air-bath to complete dryness; take up with hot water and a few cubic centimeters of hydrochloric acid, and again evaporate to complete dryness. Take up as before, filter and wash thoroughly with cold water or with hot water slightly acidified at first with hydrochloric acid. Cool and make up to 500 cubic centimeters. This is solution “A.” The residue is to be added to the main residue and the whole ignited and weighed, giving the insoluble matter.

The determination of the various components of the solution remains essentially as described in the provisional methods of the Association which have already been given.

It is directed that all results of soil analysis be calculated on the basis of the sample dried to constant weight at the temperature of boiling water.

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PART SEVENTH.

THE ORIGIN AND ESTIMATION OF OXIDIZED NITROGEN IN SOILS, RAIN AND DRAINAGE WATERS.

411. Introductory Considerations.—The estimation of oxidized nitrogen in the soil would properly find a place in the preceding part; but on account of the late progress in our knowledge of the source of this indispensable and costly plant food it has become necessary to give it especial attention. The present part will, therefore, be devoted to a brief statement of our present knowledge in respect of the origin of oxidized nitrogen, a description of the nitrifying ferments and methods for their isolation and determination and finally the most approved methods of estimating the ammonia, nitrous, and nitric acids formed thereby both in the soil and the waters pertaining thereto or proceeding therefrom. It is scarcely necessary to caution the reader not to consider this part in any sense a treatise on the bacteria active in soil chemistry. Its object is rather to place in the hands of the soil analyst data which will enable him to intelligently study the soil phenomena depending on these organisms and to determine the extent and character of their biological and chemical functions. These are matters which, up to the present time, have found no place in manuals dedicated to agricultural analysis.

412. Organic Nitrogen in the Soil.—With the exception of the small quantities of nitric acid added to the soil directly by rain water, the whole of the supply of this substance is derived from the products of the oxidation of nitrogenous bodies. These products are either stored as the results of past nitrification or are formed synchronously with their consumption by the

growing plant. Nitrogenous compounds are present as organic vegetable or animal remains and as humus. All vegetable and animal material deposited in or on the soil contains more or less of these proteid or nitrogenous matters while the amount of nitric acid supplied in this way is probably represented entirely by the quantity in the organism of the plant or animal and unabsorbed at the time of its death. In other words it is not demonstrated that nitrates or nitrites are in any sense a special product of plant growth save in the case of nitrifying organisms themselves which are supposed to be of a vegetable nature. Animal organisms do not in any sense assimilate nitric nitrogen.

With most plants, the quantity of proteid nitrogen which they can deliver to the soil is in no case greater than the sum of organic and nitric nitrogen supplied in their food and they can therefore be regarded only as the carriers and conservers of this substance. On the other hand there are some plants notably those belonging to the leguminous family which permit of the development on their rootlets of colonies of bacteria which have the faculty of rendering atmospheric nitrogen available for plant growth. Whether or not there exist plants other than the micro-organisms mentioned which are capable of directly oxidizing and fixing atmospheric nitrogen is still an unanswered question. It is not probable, however, that the difficult task of oxidizing atmospheric or free nitrogen would be accomplished in nature in only one way. In fact it has already been established that organisms do exist which are capable of oxidizing free nitrogen in a manner wholly independent of other plant life and to produce weighable quantities of nitric acid when developed in media of mineral matters and pure carbohydrates to which free nitrogen has access. It is, therefore, fair to assume that the fixation of free nitrogen is a function of chemical activity quite independent of ordinary plant life and that the leguminous plants take no further part in this process than that of providing in their radical development a favorable nidus for the growth of the nitrifying organism.

By the action of denitrifying organisms a portion of the nitrogen of nitric acid is constantly restored to a free state, a far larger portion, perhaps, than is fixed in the atmosphere itself by the action of electricity. Were it not, therefore, for the activity of the nitrifying ferments the stores of nitrogen available for growing plants would constantly become less. Instead of this being the case, however, it is probable that the contrary is true and that, by a

wise system of agriculture, the total nitrogen at the disposal of plants may become greater and greater in quantity.

413. Development of Nitric and Nitrous Acids in Soils.—Owing to the solubility of nitrates there can be but little accumulation of them in soils in those countries where there is any considerable amount of rain-fall. On the other hand in arid regions there may be found extensive deposits of nitrates. The occurrence of a certain quantity of nitrates in the soil, however, is essential to the growth of plants. Until within a few years little was known of the origin of nitric acid in the soil. The presence of nitrates in drainage waters was well established, likewise the consumption of nitric acid by the growing plant, but the method of its supply was unknown. In a general way it was said that the nitric acid came from electrical action and the oxidation of the albuminous bodies in the soil, but without specifying the manner in which this change takes place. The researches of Schloesing and Müntz, of Springer, Winogradsky, Frankland, Warington and others have demonstrated the fact that this oxidation is caused by means of bacteria and that the nitrates formed can be consumed and destroyed by other species of this organism. In the one case the process has been called nitrification and in the other denitrification.^[274]

The influence of these low organisms both in producing fertility in a soil and maintaining it in a state of fertility is of the highest importance.

414. Conditions Necessary for Nitrification.—In order to properly understand the reasons for many of the steps in investigating a soil for nitrifying organisms, it will be useful to state the general conditions on which nitrification depends.

The nitrifying organism, like every other one, first of all feels the necessity for food. In general, food which is given to microbes of all kinds consists of some organic matter together with the addition of mineral substances necessary to growth. These substances in general are phosphoric acid, potash, and lime. Of these articles of bacterial food phosphoric acid seems to be the most important. With the nitrifying organisms, however, it has been found that the organic matter can be omitted. In fact, as will be seen further on, the omission of organic matter supplies the best condition for the proper isolation of the organisms. In other words some forms of the nitrifying organisms have the property of subsisting wholly on mineral substances, *i. e.*, are true vegetables.

The presence of oxygen is also necessary to the growth of the common nitro-organisms. In an atmosphere deprived of oxygen or in which the oxygen is reduced to a very low percentage, the process of nitrification is retarded or stopped as the oxygen diminishes or disappears.

The presence of a base with which the nitrous or nitric acid formed may unite is also essential to the proper conduct of the process. For this reason the nitrification should take place in a solution which is feebly alkaline or in the presence of a base which can be easily decomposed so that no acidity can take place. Calcium carbonate is a base well suited to favor the nitrifying process and its presence in a soil favors the rapid oxidation of proteid matter. The mistake must not be made, however, of supposing that an excess of alkali would favor nitrification. The contrary is true. A slight excess of alkali may prevent nitrification altogether when it is due to the common organisms present in an arable soil. It may be that in soils charged with alkali a different organism exists which is capable of exercising its functions when the alkali is in excess.

The temperature to which the nitrifying body is subjected is also a matter of importance. The nitrifying organisms have the property of remaining active at lower temperatures than most bodies of their class. On the contrary their action is retarded and destroyed by high temperatures. The most favorable temperature for nitrification is about that of blood heat; viz., 37°. At 50° the organism shows very little activity and at 55° its activity ceases altogether. Nitrification, however, according to Warington, cannot be started in a solution if the initial temperature is 40°.

Desiccation has the same retarding influence on nitrification that a high temperature has. Even thoroughly air-drying a soil may destroy its nitrifying qualities.

Darkness is also necessary to the proper progress of nitrification. In a strong light, the activity of the organism is very much diminished or destroyed altogether. A bright light like sunshine may even stop nitrification which has set in.

415. Effect of Potassium Salts on Rate of Nitrification.—Dumont and Crochetelle have described some experiments to determine the effect of potassium salts alone and in combination with lime on nitrification.^[275]

Soil rich in vegetable mold (18.5 per cent of humus and 0.29 per cent of lime) was treated with varying amounts of potassium sulfate and carbonate

and kept for twenty days at 25°. In the untreated soil the amount of nitric acid produced was twenty-five parts per million. When potassium carbonate was applied in quantities of from one-tenth to six per cent the amount of nitric acid increased from forty-seven parts per million to 438 parts when four and one-half per cent of the potassium salt were used. Larger quantities caused a decrease in the amount of nitric acid produced. Very little effect, on the contrary, was produced by the action of potassium sulfate. When one-half per cent was employed the quantity of nitric acid formed rose to fifty parts per million, while with quantities as high as five per cent it fell below the normal; viz., twenty-five parts per million.

When calcium carbonate was added to the soil in conjunction with potassium sulfate there was a marked increase in the amount of nitrogen oxidized. The activity of potassium sulfate in promoting nitrification is therefore increased by the presence of the calcium salt, potassium carbonate and calcium sulfate being formed.

416. Production of Nitrous and Nitric Acids.—In the following pages the study of the methods of isolating the nitrous and nitric ferments will be considered as one process, the final isolation of the two classes of bodies being the result of their synchronous cultivation in appropriate media. The special process of the production of ammonia by oxidation is not so well-known, and will therefore be described in brief.

It is now generally conceded that the action of the nitrous organism is precedent to that of the nitric, but the two processes go on so nearly together as to prevent the accumulation of any large quantities of the lower salt in the soil.

Whether or not the formation of ammonia precedes that of nitrous acid is still a subject for experimental demonstration. Chemically, both nitrous acid and ammonia may be produced by the reduction of nitric acid. In nature, the reverse of this process may be the customary method.

417. Production of Ammonia in the Soil by the Action of Microbes.—It is highly probable that organic nitrogen in the soil in passing into the form of nitric acid exists at some period of the process in the form of ammonia.

Marchal has isolated and studied some of these ammonia-making bacteria.^[276] *Bacillus mycoides* is the most active of these organisms. It occurs constantly in surface soils and is present in the air and in natural

waters. In decomposing albumen it produces a strongly alkaline solution due to ammonium carbonate. Organic carbon, during this process, is converted chiefly into carbon dioxide, but small quantities of formic, propionic, and butyric acids are also produced. Any organic sulfur which is present is converted into acid. No hydrogen or nitrogen is eliminated in a free state. While slight alkalinity is favorable to the development of this bacterium, yet it may be propagated in a feeble sulfuric acid solution when the acid is less than one per cent.

The greatest activity of this organism is manifested at 30°. Below 5° and above 42° no ammonia is produced. The bacillus will not develop in an atmosphere of hydrogen or carbon dioxide, except in solutions of organic matter and nitrate. In addition to its action on egg albumen it decomposes other proteid bodies as well as leucin, tyrosin, creatin, and asparagin. It, however, does not oxidize urea, nor does it develop in solutions of ammonium salts and nitrates, except as mentioned above. When soluble carbohydrates are present, acids are formed. It is concluded from these experiments that the final oxidation of organic nitrogenous matter is preceded by its conversion into ammonium carbonate.

418. Summary of Statements.—All nitrogenous matters which would be naturally present in the soil may become subject to nitrification when the proper conditions are supplied. Munro has also succeeded in nitrifying ethylamin, thiocyanates, and gelatin, urea, asparagin, and the albuminoids of milk and rapeseed.

The products of nitrification are ammonia, nitrous or nitric acid, carbon dioxide, and water. The ammonia and nitrous acid may not appear in soils as the final products of nitrification, as the nitric organism attacks the latter at once and converts it into nitric acid. Nitrous acid and ammonia may also be produced in soils as one of the retrograde steps in denitrification.

To summarize the conditions necessary for nitrification it may be said that first, the proper material must be supplied; *viz.*, an organic or inorganic nitrogenous compound capable of oxidation. In the second place, the medium must be faintly alkaline, the temperature must not be too high, the nitrifying organisms must have abundant food, and the process must take place in the dark.

419. Order of Oxidation.—It is quite definitely determined that activity of the ammoniacal and nitrous organisms is the first step in the process,

since the nitric organism appears to have no power whatever to oxidize proteid compounds; while, on the other hand, the nitrous organism can not, in any case, complete the conversion of nitrous into nitric acid.

The conditions which permit certain organisms to oxidize free nitrogen have not been definitely determined. The presence of such bodies in the tubercles attached to the rootlets of certain leguminous plants has been established. Lately, Winogradsky has isolated from the soil a nitrifying organism which is capable of converting free nitrogen into forms suited to nourish plant growth. This organism is cultivated in dextrose with careful exclusion of all nitrogen, save that which exists in the air carefully freed of every trace of ammonia or oxidized nitrogen.

Under the influence of the growth of this organism the sugar undergoes a butyric fermentation, and nitrogen in an oxidized form is assimilated in an amount apparently equal to about one five-hundredth of the sugar consumed.

This result leads Warington^[277] to remark that it is a fact of extraordinary interest, both to the physiologist and chemist, that a vegetable organism should be able to acquire from the air all the nitrogen it needs.

420. The Nitrification of Ammonia.—The same organism which converts organic nitrogen into nitrous acid acts also on ammonia and its compounds with a similar result. In fact, the formation of ammonia may be regarded as one of the stages on the road from albuminoid to nitric nitrogen.

Data have been collected by Schloesing on the nitrification of ammonia taking place in arable soil, tending to show that this phenomenon is accomplished without appreciable loss of nitrogen in the gaseous state.^[278] This, however, does not hold good when the quantity of ammonium carbonate introduced into the earth is largely increased. In two experiments, conducted by Schloesing, with a larger quantity of ammonium carbonate, the loss of nitrogen was very notable. In certain conditions the production of nitrous acid may take place, and it is interesting to know whether the appearance of nitrites has any influence on the disengagement of free nitrogen. In order to determine this question a solution of calcium nitrite was prepared by decomposing silver nitrite with calcium chlorid. From the results of the experiments made it was seen that the nitrites were only the results of a retarded and partially incomplete nitrification. They are, moreover, thus an obstacle to the normal work of the nitrifying organisms.

It is also established that when they are present a disengagement of gaseous nitrogen takes place, whether the nitrites are formed during the progress of the experiment, or whether they were originally present. However, it is not best to say that the nitrites themselves have been the cause of the disengagement of the nitrogen. It may happen that the disengagement of the nitrogen and the presence of nitrites are simply simultaneous and due to one and the same cause. The destruction of nitrates in the midst of reducing agents furnishes, according to the nature of these bodies and the circumstances, nitrous acid, nitrogen dioxid, nitrogen protoxid, free nitrogen, and even ammonia.

This destruction of nitrates and the appearance of oxids of nitrogen and of free nitrogen are more likely to be due to the presence of a separate denitrifying ferment as pointed out by Springer than to have arisen in the manner mentioned above by Schloesing. In the present state of our knowledge, moreover, we can hardly regard the presence of nitrites as an obstacle to complete nitrification. On the other hand, it seems to be well established that the production of nitrites or ammonia is a necessary step between organic nitrogen and nitric acid.

421. Occurrence of Nitrifying Organisms.—According to the observations of Schloesing and Müntz the nitrifying organisms are widely distributed.^[279] Arable soil containing considerable humus seems to be the medium in which they grow most freely and in which they accomplish their most important functions. Sewage waters are also rich in nitrifying ferments, and, in fact, all waters containing organic matter. They are also found in running waters but not in great numbers. They affect chiefly the surface of bodies, and especially are found on the bottom of culture-flasks.

These authors have not found the nitrifying organisms in normal air. They could not seed sterilized flasks by admitting air freely. The absence of these ferments from the air is explained by reason of their sensitiveness to desiccation.

The method used by Schloesing and Müntz for the separation of the organism consisted in the preparation of original and subcultures in sterilized solutions containing nitrifiable matters. The proof of isolation was assumed when a given subculture contained only one kind of organism as seen with the microscope. The appearance of this organism, as described by

the authors, was that of the later isolations by Warington and Winogradsky, but the method used could hardly now be regarded as decisive.

422. Determination of Nitrifying Power of Soils.—In studying the distribution of the nitrifying organisms in a soil the general method of procedure is based on the production of nitrification in a convenient solution by the organisms present in a given sample of soil. If the solution seeded with the given portion of soil remain unaffected, it will show that there were no nitrifying organisms present in the seed used. On the other hand, the vigor of the nitrifying process when once it is started, may be taken as an evidence of the number and activity of the organisms in the soil, a sample of which was used for seed.

423. Composition of the Culture Medium.—The solution recommended by Warington for the culture and isolation of the nitrifying ferments has the following composition:

Ammonium chlorid	80	milligrams.
Sodium potassium tartrate	80	„
Potassium phosphate	40	„
Magnesium sulfate	20	„
calcium carbonate	about 200	„
Pure bacteria-free water to make one liter.		

424. Apparatus and Manipulation.—The experiments are conducted in short, wide-mouthed bottles. The initial volume of the solution in each bottle is 100 cubic centimeters, and the bottle should be of such size as to give a depth of liquid of from three to five centimeters.

The neck of the bottle is closed with a plug of cotton and this is protected from dust by tying over it a cap of filter paper. Arranged in this way, filtered air has free access to the solution. The bottle with the solution thus protected is placed in a water-oven and kept near the temperature of boiling water for six to eight hours to destroy any organisms present. When cool, the solution is ready for use.

The calcium carbonate used should be prepared by precipitation and added in a moist state. The calcium carbonate solution should be added after the sterilization of the liquid, the precipitated carbonate being boiled just before it is added.

Preparation of Seed.—The seed employed to start the nitrification should be a small quantity of fresh soil, usually about one-tenth of a gram. If a previously nitrified solution be used for seed it should be thoroughly shaken and about one cubic centimeter of the solution removed for seeding the new bottle.

In introducing the nitrifying liquor into the bottle the plug should be lifted slightly and a small pipette inserted by means of which the liquor is added. The operation should be carried on in a room perfectly free from dust and to which no one but the operator has access. The greatest care should be exercised to prevent any particles of matter entering the solution except that which is purposely added. In withdrawing the liquor from the nitrifying solution cotton wool should be pressed around the top of the pipette so that the entering air may be filtered before admission to the interior of the bottle. The pipette which is used should be kept in boiling water until it is required for use. After use it should be washed and replaced in boiling water until again required.

After seeding, the bottles should be placed in a dark cupboard and exposed to the ordinary temperature of the laboratory. If a higher or stated temperature be desired, the bottle should be placed in a metal box the temperature of which can be regulated to any degree.

Test of the Commencement of Nitrification.—The beginning of the nitrification can be determined in a solution by testing it with diphenylamin. One cubic centimeter of the solution withdrawn as above indicated, is placed in a small beaker, a drop of solution of diphenylamin sulfate in sulfuric acid added, and then two cubic centimeters of concentrated sulfuric acid and the contents of the beaker well shaken. The development of a violet-blue color shows the presence of nitric or nitrous acid. This test will detect one part of nitric nitrogen in twenty million of water.

Determining the Progress of Nitrification.—The progress of nitrification is determined by repeated examinations for ammonia by nesslerizing, and for nitrous acid with metaphenylenediamin. Each experiment is made with five cubic centimeters of the solution withdrawn as above indicated and placed in test-tubes, always of the same size. The reaction with the nessler solution is then made by adding it in the usual way. The colorations are recorded as, trace, small, moderate, considerable, large, and abundant.

If the change produced by the organism consisted in the formation of nitrites only, the ammonia in the original solution would fall from *large* to *trace*, while the nitrous acid would increase from *trace* to *large*. If the nitrification consisted in the production of nitrates only, the ammonia would diminish without any corresponding production of nitrous acid. In mother solutions which contain ammonium carbonate instead of sulfate, it should not be forgotten that the ammonia might gradually disappear owing to the volatilization of the carbonate without any corresponding production of free nitrites or nitrates. The complete disappearance of the ammonia in the above experiments shows the completion of the process.

425. To Determine the Distribution of the Nitrifying Organism in the Soil.—The principle on which the determination of the distribution of the nitrifying organism in the soil depends, rests upon seeding the growth solutions with samples of soil taken at different depths and carefully protected from the time of sampling until the time of seeding from any admixture of accidental organisms.

The method of Warington is the simplest and best to follow.^[280] The samples of soil are taken by digging a pit of convenient depth usually from eight to ten feet. A fresh surface is then cut on one of the sides of the pit at the spot selected for sampling. This surface is scraped with a freshly ignited platinum spatula. The spatula should then be washed, re-ignited, and cooled, and a small portion of the soil, at the depth required, detached with the spatula and transferred at once into one of the growth bottles already described.

The growth solution best suited for the purpose contains four cubic centimeters of urine per liter. Each bottle should also contain some freshly precipitated calcium carbonate. In sterilizing urine solutions the calcium carbonate should be added before the heating instead of afterwards. The quantity of soil taken for each seeding should be about one-tenth of a gram.

Inasmuch as the cotton stopper has to be lifted to introduce the soil, opportunity is given for the entrance of any organisms floating in the air. Experience, however, has shown that air free from soil dust very seldom contains nitrifying organisms. The seeded bottles are placed in a dark cupboard of moderate temperature as already described.

426. Sterilized Urine Solution.—The sterilized urine solution used for the determination of the distribution of the nitrifying organisms in the soil,

is made by taking four cubic centimeters of healthy urine, diluting to one liter, adding some freshly precipitated calcium carbonate, stoppering with cotton wool and heating for several hours at the boiling temperature of water.

As a result of Warington's experiments it was shown that the nitrifying organism in the soil did not exist, at least in portions of one-tenth of a gram, to a greater depth than eighteen inches. In only one case was nitrification produced from a sample of soil taken at a greater depth and this may have been due to the accidental introduction of organisms from other sources. It may be assumed that any long delay in the commencement of nitrification under favorable conditions, implies the presence of a very limited quantity of organisms in the solution. Thus a comparative study of the period of incubation and the progress of nitrification in solutions seeded with soils taken at different depths or at different places, becomes a fair index of the number and vitality of the nitrifying organisms contained therein.

427. Depth to Which Micro-Organisms are Found.—Koch states that at the depth of about one meter, the soil is nearly free from every kind of bacteria.^[281] These observations have been corroborated by Pumpelly and Smyth who find that no infection of a bacterial nature is produced in a sterilized solution from samples of clay taken at the depth of nine feet below the surface.^[282]

It is evident from the nature of the experiments above described that the nitrifying processes go on almost exclusively in those portions of the soil which are subject to cultivation, while in the subsoil and below the processes of nitrification are either retarded or arrested. Any stores of nitrogenous matter, therefore, in an insoluble state, resting in the subsoil, are preserved from oxidation and consequent waste until such time as they may be removed to near the surface.

428. Isolation of the Nitrous and Nitric Organisms in the Soil.—The action of the organisms which produce nitrification either in form of nitrites or nitrates, having been thoroughly established, and the method of testing the soil therefor given, it remains to describe a method by means of which these organisms in the soil may be isolated and obtained in a state of purity. The difficulties attending this process are extremely great on account of the similarity of the two organisms. All earlier attempts to make pure cultures of the two separate organisms were attended with but little success.

According to Winogradsky the method of culture on gelatin so long practiced is not to be relied upon.^[283] It is very difficult to eliminate by this process the organisms which grow rapidly in gelatin and which mature their colonies in two or three days, but where they require eight or ten days to produce a colony the method is successful. In fact, by the gelatin process as it was at first practiced, a good deal was owing to chance, but sometimes by a happy accident a pure nitro-bacterium might be isolated.

Formerly it was considered that a liquid could be regarded as sterile if it gave no growth upon gelatin. It has, however, now been demonstrated that a liquid may contain large numbers of nitro-bacteria and still produce no growth upon gelatin. However, for the organisms which accompany the nitro-bacteria in soils, it is regarded as certain that if no growth on gelatin is produced by them they are absent. Therefore in the case of a solution which has been seeded with a soil, if it can be brought to such a state as to produce no growth on gelatin, it may be safely assumed that it contains no bacterial organisms save those which are capable of producing nitrites or nitrates. Therefore if such a solution produce nitrification and at the same time no growth upon gelatin, it may be considered as a proof of the isolation of the nitro-organisms from all others.

This method was also worked out independently by Mr. and Mrs. Frankland.^[284]

Winogradsky says further he confesses that he has advanced these views only provisionally and without being convinced of their infallibility. Strictly speaking, the proof of seeding gelatin is not sufficient alone because the absence of growth can not be regarded as the exclusive privilege of the nitro-bacteria. Such might be the case sometimes for an accidental mixture of microbes, introduced with any given sample of soil into the cultures, but the criterion is not absolute. Microbes, for example, of a sulfurous or ferruginous nature may be cited, for which the gelatin layer is not only unfavorable but even fatal. It may thus happen that there may be eliminated from the solution all that will grow upon gelatin without freeing it from some special kinds of cultures, refractory like the nitro-bacteria, but which might reappear if they should be resown in some favorable nutritive solution. On account of this fault in the process, Winogradsky has been impressed with the necessity of bringing out a better method.

In using the gelatin media it is necessary to find the one that is suited to nourish these organisms, which would evidently be the way promising the greatest success. This having been found, and those organisms which produce colonies being easily recognizable, a great step towards the solution of the problem will have been made and the more so as the medium would be at the same time absolutely unfavorable to other forms of microbes. On account of the slow degree of development of the nitro-organisms, all others would probably have opportunity to grow and strengthen to their exclusion, unless these interfering organisms could be completely removed.

429. The Culture Solution.—The culture-solution, first proposed by Winogradsky, had the following composition:

To ten grams of gelatin or one part of agar-agar in 100 cubic centimeters of water add potassium phosphate, one-tenth of a gram; magnesium sulfate, five-hundredths of a gram; calcium chlorid, trace; and sodium carbonate, half a gram. The solution being sterilized in the usual way by heating, there are added to it a few cubic centimeters of a sterilized solution containing two-tenths per cent of ammonium sulfate. Such a solution has been proved to be very favorable to nitro-organisms. Nevertheless the experiments with such solutions gave no definite results and they were abandoned.

The non-success of this method led Winogradsky to adopt a nitrifying solution which absolutely excluded all organic substances. Instead of using an animal or vegetable gelatinous substance he used one of a mineral nature, first proposed by Graham and Kühne.^[285] Two of these gelatinous mineral substances were considered; viz., the aluminum hydroxid and the hydrate of silica. The latter was chosen.

430. Preparation of the Mineral Gelatinous Solution.—The soluble glass which is found in commerce is generally of a thick, sirupy consistence. It is first diluted with three times its volume of water. One hundred cubic centimeters of this liquid are poured with constant stirring into fifty cubic centimeters of dilute hydrochloric acid and the mixture placed in a dialyzer. It is useless to employ a standard solution of silica. All that is necessary is to submit to dialysis a liquid with an excess of acid and sufficiently dilute not to be exposed to the danger of being spontaneously gelatinized in the dialyzer. The dialyzer is left for one day in running water and two days in distilled water, often renewed. The solution is then ready

for use. This is the case when it is no longer rendered turbid on the addition of silver nitrate, showing that the hydrochloric acid has been entirely extracted. The solution is then to be sterilized by boiling, and preserved in a glass flask closed with a plug of cotton.

More recent instructions by Winogradsky for preparing the gelatinous silica recommend dialyzing the soluble glass after treatment with hydrochloric acid in a parchment tube.^[286] The proportions of silicate and acid are 100 cubic centimeters of the silicate solution (1.06 specific gravity) and 100 cubic centimeters of hydrochloric acid (1.1 specific gravity). With a dialyzing tube placed two days in running water and one day in distilled water frequently changed it will be found that the acid is completely removed. One hundred cubic centimeters of the residual liquor giving no reaction for hydrochloric acid are concentrated to twenty cubic centimeters. When cold there is added one cubic centimeter each of a solution of ammonium sulfate and of sodium carbonate, together with corresponding quantities of the other nutrient salts commonly employed. The ammonium sulfate should never exceed two to two and a half, and the sodium carbonate four parts per thousand. To the flask containing the above substances is added one drop of the seed-liquor, which may be a soil water or a drop from some previous culture. The flask is shaken and the mixture poured into a low circular glass dish which is covered by one slightly larger in diameter (Petri double dish). To the liquid in the dish is added a drop of a cold saturated solution of common salt, and it is then stirred with a platinum spatula. The addition of the salt greatly favors the setting of the jelly. The jelly may set in from two to three hours, but a longer time secures better results in the end.

In employing these preparations as seed, after the organisms have grown, it is absolutely necessary to use the isolated cellules and not the aggregated masses (*zoöglææ*). The latter are rarely free of foreign germs which adhere to their gelatinous envelope. Since the *zoöglææ* can not be broken up by any artificial means it is necessary to await their spontaneous disintegration in order to separate the mobile monads. The opalescence of the culture-liquid is a sure index of this separation.

The particles of mineral gelatin to be used as seed for nitrifying are best taken as follows:

A glass tube is drawn out immediately preceding the operation, until the end is as fine as a hair. The surface of the mineral gelatin is magnified by means of a dissecting microscope magnifying 80 to 100, to the proper degree and the preparation table is so arranged as to give a perfect support to the right hand which should hold the filament of glass. The smallest colony is then pricked with the needle and the end of the glass is broken and dropped into the flask which is to be seeded. The seed is thus selected in as small a particle as may be desired, only a few cells, but it can always be ascertained with certainty that some of the particles have been obtained by this operation.

The method of cultivation on mineral jelly is considered by Winogradsky an important resource in the study of the nitrifying organisms. It removes the chief difficulties heretofore existing in discovering and characterizing these organisms among the innumerable micro-organisms of the soil. The long series of cultures necessary to separate the organisms are rendered nugatory. By directly introducing a little of the earth into the silicic jelly the active organisms in nitrification can be at once discovered. It is preferable, however, as indicated below, to previously produce a nitrification in an aqueous solution by a trace of earth and to take from it the seed for impregnating the solid medium. In order to show at once a proof of its nitrifying character, it is only necessary to take a small bit of the mineral jelly, the size of a grain of rye, and to throw it into a little sulfuric acid which has been treated with diphenylamin. There is at once formed a blue spot equal in intensity to a saturated solution of anilin blue.

In regard to the growths which nitro-organisms make in a medium of the kind described, they are far from being so marked as are those produced by ordinary micro-organisms.

A nitro-bacterium is not capable of the energy of growth which is recognized for the greater number of microbes. The colonies contained in the gelatin always remain small. The largest among them are just visible to the naked eye like white points. Along the striae, on the contrary, there is formed quite a thick white crust. To the naked eye, in general, there is nothing very characteristic in the formation of colonies in a medium of this nature. But this impression changes altogether when the plaques are examined with a low magnifying power. The colonies, especially those of

the interior surface, reveal then an aspect so curious as to be well remembered when once seen.

This mineral gelatin, as has already been noticed, is very unfavorable to the growth of microbes other than nitro-bacteria and becomes altered only under the action of the air. If the plaques be carefully preserved from desiccation the culture of these organisms can be continued for several weeks. Although they do not seem to increase, the colonies, as well as the jelly, are still in a good condition at the end of that time. Nevertheless the expectation that this medium would prevent the formation of any foreign organism has not been realized. Some of the organisms which accompany the nitro-bacteria in soil, also grow upon the silicic jelly; but they do not form colonies, properly so-called, and their growth is extremely slow. They generally make their appearance before the nitro-bacteria and spread exclusively upon the surface in form of white spots, so transparent that without careful examination they would not be discovered. Having reached a certain size the spots do not change during entire weeks. This circumstance renders the operations of isolation somewhat delicate, but does not prevent them.

431. Preparation and Treatment of the Solution to be Nitrified.—The organisms having been grown on the siliceous gelatin in the manner described they are tested for their nitrifying power as follows:

The mineral solution which is to be nitrified with the above preparation is composed of ammonium sulfate, four-tenths gram; magnesium sulfate, half a gram; potassium phosphate, one-tenth gram; calcium chlorid, trace; sodium carbonate, six-tenths to nine-tenths gram; and distilled water, 100 cubic centimeters. The sulfates with the calcium chlorid on the one hand, and the phosphate and carbonate on the other, are dissolved separately and the two solutions sterilized separately and mixed after cooling. The seeding is then done as described above.

432. Isolation of the Nitrous and Nitric Organisms.—Instead of proceeding immediately to the isolation of special organisms in the soil, the preliminary period of purification is prolonged by Winogradsky by allowing the free growth to take place of all the organisms which can be maintained in the ordinary medium.^[287]

The composition of the culture solution employed is as follows: Distilled water, 1,000 parts; potassium phosphate, one part; magnesium sulfate, half

a part; calcium chlorid, trace. Each flask receives besides this some magnesium carbonate, freshly washed with boiling water and added in slight excess.

The flasks thus charged are sterilized, and after sterilization there are added two cubic centimeters of a solution of two per cent of ammonium sulfate, which, when added to fifteen or twenty cubic centimeters of liquid give from two to two and a half parts per thousand.

They are then seeded with soil. The reasons for this preliminary treatment are as follows: First, all the observations upon the enfeeblement of the oxidizing power of these organisms have been made upon cultures seeded simply by the fresh soil, and in cultures derived therefrom. In the second place, the existence of the two forms, one nitrous and the other nitric, prevents at once the isolation of a single organism.

Samples of soil from Europe, Africa, Asia, Australia, and America, were used for seed for the experiments. First, the cultures were made by seeding with a small quantity of each of these samples of soil, and each one of these cultures served as a point of departure for a series of subcultures. The temperature of the cultures should be kept constantly at 30°.

The method of following the nitrification adopted by Winogradsky is essentially that of Warington, the percentage of ammonia remaining at any time being determined by nesslerizing. To detect the presence of nitric acid the nitrous acid is decomposed by boiling with ammonium chlorid in excess, or with urea, and then diphenylamin is used as a reagent. By treatment with ammonium chlorid and boiling, the ammonium nitrite is resolved into free nitrogen and water as indicated by the equation $\text{NH}_4\text{NO}_2 = \text{N}_2 + 2\text{H}_2\text{O}$. Or the total oxidized nitrogen may be estimated by the Schloesing method or by any of the standard methods hereafter given. The nitrous acid is then determined by potassium permanganate and the nitric acid by difference.

A great difference is to be noted between freshly taken earth and that which has been kept for a long while, especially when sealed. With fresh earth taken near the surface a mere trace is sufficient to produce nitrification. With samples of earth which have been kept for a long while and thoroughly dried, several grams must be added in order to secure perfect nitrification. The period of incubation with the samples of earth ranges from three to twenty days. The beginning of the phenomenon is

revealed by the appearance of nitrous acid, of which the quantity is increased very rapidly, but in the end it disappears and is transformed into nitric acid.

433. Statement of the Results.—The method of stating the results of examination of soils for nitrifying organisms is illustrated by the following example:

Soil from Zurich. The culture was seeded on the 11th of October, one gram of soil being taken. On the 20th of October the nitrous acid had reached its maximum of intensity and there was no ammonia left. On the 29th of October the nitrous acid remained almost stationary and there was hardly any nitric acid present. On the 1st of November the reaction for nitrous acid began to decrease. On the 5th of November the reaction for nitric acid was very intense. On the 11th of November the nitrous acid had all disappeared except a mere trace.

The above order of phenomena was observed with all the samples of soil tried, from which it is concluded with certainty that nitrifying organisms transplanted directly from their natural medium in the soil into a liquid easily nitrifiable produce at once nitrous acid in abundance. The phenomenon of nitrification is divided into two periods therefore, of which the first is devoted to the production of nitrites, and the second consists in the oxidation of the nitrites, and this does not commence until the total disappearance of the ammonia. Occasionally the formation and oxidation of the nitrites practically go on together, but never equally, the oxidation of the nitrites being always sensibly behind their formation.

434. Method for Subcultures.—From the mother cultures described above, Winogradsky makes subcultures as follows:

The solution to be nitrified is prepared as in the mother cultures. The seeding is accomplished by adding a small quantity of the liquor of the mother culture after shaking. Subcultures can be made in this way to the seventh generation.

In respect of the oxidation of the nitrites the results may be entered as negative if they have not disappeared at the end of two months.

To determine whether the process of oxidizing the nitrites is in progress or not the total nitrous acid is estimated, and the process repeated at the end of eight or ten days. Should there be no diminution of the nitrous acid

within this time it may be considered that the further oxidizing action is not taking place.

435. Use of a Solid Medium.—It may be justly claimed that the action of nitrifying organisms in a liquid is not to be compared with their action in a solid medium, such as a soil which is their natural habitat. It might be, therefore, that the inability of the nitrous organism to produce nitrates is due to the nature of the medium in which it is cultivated. Winogradsky in order to determine this question cultivated the organism in a solid medium of two kinds, first a silicate gelatin impregnated with an ammonium salt and second in sterilized earth. The silicate jelly is prepared as follows:

Mix a jelly of silica containing some ammonium sulfate with sterilized soil. The seeding is done with one of the subcultures which no longer has the power of producing nitrates.

In the case of the jelly the seeding is accomplished as follows:

A minute drop of a culture liquid is taken with a capillary glass tube and applied in striae to different parts of the solid jelly; or a minute drop of the culture liquid may be mixed with the jelly before solidification. The Petri dishes in which these cultures are made can be preserved in a moist atmosphere and thus the desiccation be easily prevented for a long time. From time to time small pieces of the jelly as large as a pea can be taken and tested for the progress of nitrification.

Results.—The nitrous reaction, both in the prepared jelly and in sterilized soil, will appear in a few days. At the end of from seven to twelve days it will have attained its maximum intensity and will then remain stationary indefinitely. Sterilized soil has no power to generate the nitric from the nitrous ferment. The two organisms are, therefore, of different species.

After a few generations the power of producing nitrates seems to be lost although the nitrous ferment may still be active. This suppression of the power to oxidize the nitrites is not due to any pernicious influence of the culture-medium but to the condition of the successive solutions at the time of taking the seeding samples.

436. Microscopic Examination.—A small particle of the deposit in the culture-liquid is spread on a glass slide and dried. There is then added a drop of very dilute perfectly transparent malachite green solution. Malachite green is Bittermandelölgrün, or tetramethyldiamidotriphenylcarbinol. Use the zinc chlorid double salt or

oxalate. In about half a minute it is washed and colored by a very dilute solution of gentian violet which is left to act for some time. The cells then appear distinctly colored on a colorless background.

In examining in this way nitrous cultures under a moderate enlargement there are seen particles of material covered with scattered groups and massive zoöglææ composed of cells which are, doubtless, identical.

By their round or roundish forms, by their relative size and especially by their numbers and uniformity they are at once distinguished from the other vegetations which are generally of a purely bacillus shape.

With the exception of some shreds of mycelium coming from some oidium in the soil the microscope reveals nothing but the organisms described. The microscopic appearance^{[\[288\]](#)} of the nitrous ferment is shown in Fig. 69.

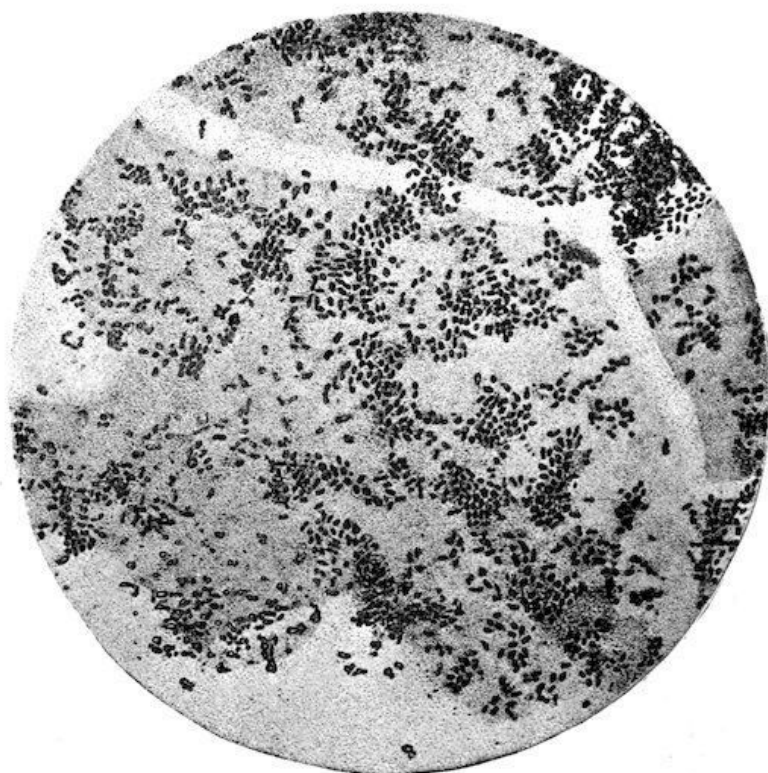
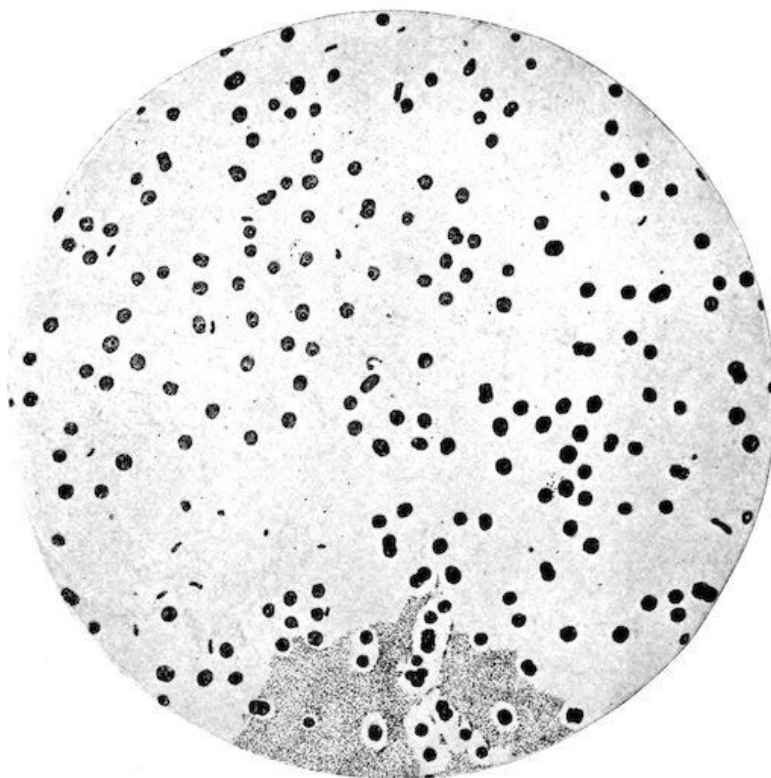


Figure 69. (Upper figure.) Nitrous ferment prepared by Winogradsky from soil from Cito.

Figure 70. (Lower figure.) Nitric ferment prepared by Winogradsky from soil from Cito.

The general conclusions of Winogradsky are:

1. Each soil possesses but one organism capable of oxidizing ammonia.
2. Soils from one locality have always the same kind of nitrifying ferment.
3. Soils from different and distant countries contain nitrifying organisms which differ from one another in some respects so much so that it may be necessary to distinguish a few species or even genera in these bodies.

437. Isolation of the Nitric Ferment in Soils.—The principle of the separation of this ferment as described by Winogradsky rests upon the fact that in culture solutions of a mineral nature free from ammonia the nitrous ferment will not grow, whereas if nitrite or nitrous acid be present the nitric ferment will grow.^[289] In a few generations, therefore, the nitrous ferment will be entirely eliminated.

Solution employed:

Distilled water	1,000	grams.
Potassium phosphate	1	gram.
Magnesium sulfate	0.5	„
Calcium chlorid	trace.	
Potassium nitrite	0.22	gram.

To culture-flasks containing 100 cubic centimeters of the above mixture after sterilization about one-tenth gram of fresh soil is added. In favorable conditions the nitrous acid will disappear in about fifteen days.

Subcultures are made by seeding fresh portions of the sterilized solution with one or two cubic centimeters of the mother culture. The operation is continued until the nitrous ferment is eliminated.

The organisms in the deposit in the culture-flasks are then subjected to microscopic examination in the manner already described for the nitrous ferment; or proceed as follows:

438. Culture on Solid Media.—Take a liquid which has been employed in the culture of a nitrous ferment and evaporate to one third of its bulk. Gelatinize the residue by adding double its volume of the silicic acid solution prepared as already directed.

The jelly is placed in the glass vessels usually employed. The seeding may be done with a few drops of a culture-liquid containing the nitric ferment as obtained above. The first reaction will appear in from eight to ten days. In about forty-five days the nitrous acid in the jelly will have entirely disappeared. Two classes of colonies are noticed under the microscope. The first to appear are small colonies which never extend beneath the surface of the jelly. In cultures seeded with these colonies there is no oxidation of nitrous acid. The second class of colonies extends into the interior of the jelly. They are much larger than the first, of a yellowish-gray color and not spherical but rather lenticular in shape. Cultures seeded with these colonies will lose their nitrous acid in about ten days or two weeks.

The growth of these organisms in a liquid scarcely merit the name of cultures. The naked eye can usually distinguish no form of vegetation. The liquid remains clear, the surface is free from any film, no flocks are deposited. Colored and examined in the microscope the organisms found are so puny as to make doubtful their oxidizing power. There is an apparent contradiction between the powerful chemical action that these organisms can produce and their apparent deficiency in physical properties.

These organisms are best found by cultivating them in a very limpid solution. The bottoms of the culture boxes will be found covered with an extremely tenuous gelatinous deposit communicating to the glass a feeble grayish-blue tint. The culture bottle is inclined and the bottom scratched with a recently drawn-out capillary tube. The colonies rise in the tube together with a little of the liquid. The colonies are dried, mounted, and colored as already described and when examined with the microscope are found to be composed exclusively of masses of an organism of extreme minuteness.

The organism remains attached so firmly to the bottom of the culture bottle that it can be washed several times with pure water without danger of detachment and thus rendered more pure.

In old cultures which are sustained by new additions of nitrite an extremely transparent pellicle on the bottom of the flask can be

distinguished. By shaking the liquid some fragments may be detached and made to float through the fluid. With a little care and patience these flocks can be captured, mounted, and colored. Since they show the nitric organism in its natural state their preparations are of the greatest interest.

The best preparations are made by coloring with malachite green and gentian violet and then coloring again hot with magenta. Afterwards the preparation is washed with warm water at 50°–60° which takes almost the whole of the color from the gelatinous matter. The cells are then clearly presented colored a reddish violet on a rose background. These organisms^[290] are shown in figure 70.

The figure shows the cells united by a gelatinous membrane and grouped in small dense masses composed often of a single layer of organisms. The cells are generally elongated, rarely regularly spherical or oval. Their mean length does not exceed half a micromillimeter and their thickness is from two to three times less.

The difference in form of the nitrous and nitric ferments is very marked and leaves no doubt of the existence of these two forms which are as distinct as could be desired in microbic discrimination.

439. Dilution Method of Warington.—The method pursued by Warington in preparing pure cultures of the nitrifying ferment is based on the well-known principle of dilution which may be expressed as follows:^[291] In a liquid containing bacterial ferments dilution may be practiced until a drop of the liquid may be taken which will contain no more than a single organism of any one kind. If now proper solutions be seeded with single drops of this solution, some of them may give colonies of pure cultures of any given organism. The solution to be nitrified employed by Warington had the following composition:

Water	1000	parts.
Ammonium carbonate	0.25	„
Ammonium chlorid	0.50	„
Potassium phosphate	0.04	„
Magnesium sulphate	0.02	„
Calcium sulphate	0.02	„

The ammonium chlorid is added to prevent the precipitation of magnesium and calcium phosphates. The solution is kept in wide-mouthed,

stoppered bottles to prevent the loss of ammonium carbonate, the bottles being only half full. About 100 cubic centimeters are taken for each experiment. These bottles are sterilized and seeded with fresh soil in the ordinary way. They are then covered with paper caps and placed in a dark cupboard at a constant temperature of 22°.

Special Media.—A quantity of arable soil is exhausted of nitrates by washing with cold water under pressure. The soil is then boiled with water and filtered. The clear amber-colored solution obtained may be used instead of water in the above formula.

Solid Media.—(1) Ordinary ten per cent gelatin made with beef broth and peptone. (P)

(2) A ten per cent urine solution solidified with six per cent of gelatin. (U)

(3) A solution of one gram of asparagin, one-half gram sodium acetate, one-half gram potassium phosphate, two-tenths gram magnesium sulfate, two-tenths gram calcium sulfate, and one liter of water solidified with six per cent gelatin. (As)

Other solid media may also be employed for the purpose of favoring, as much as possible, the growth of the nitrifying organisms.

The first culture in the ammonium carbonate solution given above, is always made by seeding with a little unmanured arable soil. Subcultures are seeded from this mother culture by seeding new solutions with a few drops of the original. In all cases tried by Warington the subcultures produced only nitrous fermentation while the original cultures produced the nitric fermentation.

440. Microscopic Examination.—The microscopic examination of the organisms formed is conducted as follows:

The cover glasses for microscopic objects are placed at the bottom of the culture-flask, the cover glasses being previously sterilized. At the end of the nitrification the liquid is removed with a pipette and the flask containing the cover glasses dried at 35°. The cover glasses are then removed and stained. The microscopic appearance of the organisms obtained by the previous cultures showed masses of corpuscles usually of oval shape and having a length generally exceeding one micromillimeter. An immersion objective giving a magnification of 800 diameters is suitable for this work.

Other forms of organisms are also met, the whole series being characterized as follows:

(1) The corpuscles already mentioned. Larger ones are frequently rough in outline resembling masses of siliceous sea-sand. The smaller oval corpuscles are regular in form.

(2) Some very small circular organisms often appearing as mere points and staining much more plainly than the preceding.

(3) A few slender bacilli, staining faintly.

All the cultures obtained by the above method give abundant growth on gelatin.

441. Trials with the Dilution Method.—One part of the third subculture in the ammonium carbonate solution described above, is mixed with 500 parts of thoroughly boiled water and one drop from a sterilized capillary tube is added to each of five bottles containing the sterilized ammonium carbonate solution. In Warington's experiments one of the five bottles was found to have nitrified after forty-one days. After ninety-one days two more were nitrified. Two bottles did not nitrify at all. All three solutions which nitrified gave growths on gelatin. The growths took place more speedily on gelatin U and As than on P.

The organisms obtained on gelatin were seeded in appropriate liquid media but no nitrification was obtained.

A subculture from solution No. 2 of the first dilution mentioned above, was diluted to one one-thousandth, one ten-thousandth, one one-hundred-thousandth, and one one-millionth. Each of these dilutions was used for seeding with five sterilized solutions of ammonium carbonate, using the method of seeding above described. At the end of 190 days not one of these solutions had nitrified.

Warington supposed that the cause of failure in the method just mentioned might be due to the alkalinity of the ammonium carbonate. While this solution could be seeded in the ordinary way with fresh earth it might be that the faint alkalinity which it presented might prevent it altogether from action when the nitrifying agent was reduced to a few organisms.

He therefore changed the solution to one of the following composition:

Water	1,000	parts.
Ammonium chlorid	0.02	part.
Potassium phosphate	0.06	„
Magnesium sulfate	0.03	„
Calcium sulfate	0.03	„

The solution was divided in twenty stoppered bottles which were half filled. The bottles were divided into four series, A, B, C, D, each one consisting of five bottles, and these were respectively seeded with one drop from dilutions to one one-thousandth, one ten-thousandth, one one-hundred-thousandth, and one one-millionth of a second subculture of No. 3 in the first dilution series.

After 115 days, nitrification had occurred in ten of the bottles. The other ten did not nitrify at all. Each of the nitrifying solutions was spread on gelatin, P and U being employed. Growth took place far more easily on gelatin U than on gelatin P. Of the ten nitrified solutions there were three which gave no growth on gelatin U, either when spread on the surface or introduced into the substance of the jelly. There were therefore secured nitrifying solutions which did not contain organisms capable of growing on gelatin. The supposition is therefore fair that they were pure nitrifying organisms. These fresh, pure organisms had the faculty of converting ammonia into nitrous acid only and not into nitric acid.

With the organisms thus prepared a number of solutions of potassium nitrite containing phosphates and other mineral ingredients were seeded. In no case was any loss of nitrite found, which is proof that the solution contained no organisms capable of oxidizing nitrous acid. The organisms prepared as above, have the power of nitrifying organic substances containing nitrogenous bodies.

The organism isolated as described and examined under the microscope is seen to contain two forms. The first one is nearly spherical in shape, the corpuscles varying in size from mere points to a diameter of one micromillimeter. The form is very striking and easily stained. The second form is oval-shaped and attains a length distinctly exceeding one micromillimeter. Sometimes it is a regular oval and sometimes it is egg-shaped. This form is stained less easily than the preceding or spherical form.

442. Method of Staining.—The method of staining employed is as follows:

A drop of the culture-liquid is placed on a glass slide and mixed with the filtered stain by means of a wire. A cover glass is placed on the drop and allowed to stand for half an hour. It is then pressed down on the slide and the liquid which exudes wiped off and hollis glue run around the cover glass. In this way the organism is stained in its own culture-fluid and can be seen in its true form without any possibility of the destruction of its shape by drying. The plate is bright and clear though colored.

If the preparation is to be mounted in balsam a drop of the culture is dried in the center of a cover glass. It is then placed for some minutes in dilute acetic acid to remove matter which would cause turbidity. The cover glass with its contents is then washed, dried, and stained for some hours in methyl violet.

443. Classification of Nitrifying Organisms.—The names proposed by Winogradsky for the various organisms are the following:

For the general group of microbes transforming ammonia into nitric acid, *Nitro-bacteria*.

For the nitrous ferments of the Old World

Genus, *Nitrosomonas*:

Species, *Nitrosomonas europaea*.

Nitrosomonas javanensis.

For the nitrous microbes of the New World:

Genus, *Nitrosococcus*.

Species, not determined.

For the nitric ferment:

Genus, *Nitrobacter*.

444. Nitrification in Sterilized Soil.—The process of nitrification in sterilized soil, when seeded with pure cultures, is determined as follows:

Preparation of Sample.—The fresh sample of arable soil is freed from pebbles and vegetable debris and reduced to as fine a state of subdivision as is possible in the fresh state. It is placed in quantities of about 800 grams in large crystallizing dishes.

One dish is set aside for use in the natural state, and the other, hermetically closed, is placed in a sterilizing apparatus and subjected to the action of steam for two and a half hours. This treatment is repeated three times on as many successive days.

Seeding of Sample.—Each of the two dishes is moistened with fifty cubic centimeters of pure water containing 500 milligrams of ammonium sulfate. The sterilized portion is then seeded with a preparation of the pure nitrous ferment, produced as before described. The seed is prepared by filtering a few cubic centimeters of the nitrous culture liquid through asbestos. The asbestos is well washed and then thrown into a flask containing a few cubic centimeters of sterilized water and well shaken. The water carrying the filaments of asbestos is poured drop by drop on the surface of the soil in as many places as possible. The two dishes of soil are kept at an even temperature of 20° in a dark place. Winogradsky found that, treated in this way, the unsterilized soil produced only nitrates, while the sterilized portions produced only nitrites.^[292]

445. Variation of the Determinations.—To vary the conditions of the experiment Winogradsky uses twelve flasks of the erlenmeyer shape, four having bottoms twelve centimeters in diameter, and eight of them five centimeters in diameter. In each of the four large flasks are placed 100 grams of fresh soil, and in each of the eight small flasks twenty-five grams. The eight small flasks are designated a, b, c, d, and a', b', c', d', and the four large flasks A, B, C, D.

The flasks a, b, c, d, and a', b', c', d', are placed in a stove at 30° for several days before use, while A, B, C, and D, are kept at 22°–23° for the same length of time. The soil in the small flasks is, therefore, somewhat drier than that in the large ones.

The flasks are treated as follows:

a, a', A, contain the soil as prepared above for control.

b, b', B, are sterilized at 135° and seeded with a drop of the pure nitrous culture.

c, c', C, sterilized as above and seeded with a little of the unsterilized earth.

d, d', D, sterilized as above and seeded with pure nitrous and pure nitric cultures.

After sterilization there was added to the small flasks two cubic centimeters of a twenty per cent sterilized ammonium sulfate solution, and to the large ones six cubic centimeters. At the end of a month or six weeks the contents of the flasks are thrown on a filter and washed with cold water until a drop of the filtrate gives no blue color with diphenylamin. The respective quantities of nitrite and nitrate are then determined in the filtrates by the usual processes, which will be fully described further along.

446. Sterilization.—One of the chief requisites for success in the bacteriological investigation of soils is found in the thoroughness of the sterilizing processes. The value of cultures depends chiefly on the care with which the introduction of foreign germs is prevented. In the following description a mere outline of the method of sterilization is presented, while those who wish to study more carefully the details of the process are referred to the standard works on bacteriology.

447. Sterilization of the Hands.—It is important that the hands of the operator handling apparatus and materials for bacteriological work should be sterilized. The sterilization may be accomplished in the following way:

The nails should be cut short and thoroughly cleaned with soap and brush. The hands are thoroughly washed in hot water with soap. After washing in hot water the hands should be washed in alcohol and ether. They are then dipped in the sterilizing solution.

This liquid may consist of a three per cent solution of carbolic acid, which is the one most commonly employed. A solution of corrosive sublimate, however, is perhaps the best disinfectant. It should contain from one to two parts of the crystallized salt to 1,000 parts of water. It has lately been advised to use the sublimate in an acid solution. Acetic acid or citric acid may be employed, but hydrochloric acid is recommended as the best, in a preparation of one-half part per 1,000. For stronger solutions of sublimate containing more than a half per cent, equal quantities of common salt should be added. The solution should be made with sterilized water.

After dipping the hands in the sterilizing solution they should be dried with a napkin taken directly from a sterilizing oven, where it has been kept for some time at the temperature of boiling water. Where only ordinary work in bacteriology is contemplated this sterilization of the hands is not necessary. It is practiced chiefly in antiseptic surgery.

448. Sterilizing Apparatus.—With platinum instruments the most effective and easiest way for sterilizing is to hold them in the flame of a bunsen until they are red hot. Steel and copper instruments can not be treated in this way without injury. They are best sterilized by submitting them to dry heat in a drying oven at a temperature of 150°–160° for two hours. Glass and porcelain apparatus can be sterilized best in the same way.

All apparatus and materials employed should be used in a space as free as possible from dust, so that any germs which might be carried in the dust can be excluded from the apparatus in transferring it from one place to another.

449. Methods of Applying Heat.—Sterilization by means of heat may take place in several ways.

First. Submitting the Materials to Dry Heat Without Pressure.—The temperature in sterilization of this kind may vary from the temperature of boiling water at sea-level to 160° obtained by an oil-bath or by an air-oven.

Second. Sterilization in a Liquid Under Pressure.—This form of sterilization may be effected by sealing the liquid in a strong vessel and submitting it to the required temperature. If the temperature required be greater than that of boiling water the vessel can be immersed in a solution of some mineral salt which will raise the boiling-point.

Third. Sterilization in Steam Under Pressure.—This method of sterilization consists in placing the body in a proper receptacle in vessels to which the steam can have access and then admitting steam from a boiler at any required pressure. In the case of small apparatus, such as the autoclave, the steam can be generated in the apparatus itself. The variety of apparatus used in the above method of sterilization is very great, but all the forms of apparatus employed depend upon the principles indicated.

450. The Sterilizing Oven.—The apparatus for sterilization by means of hot, dry air usually consists of a double-walled vessel made of sheet-iron, usually with a copper bottom. The apparatus is shown in Fig. 71.

The temperature is controlled by means of a thermometer, T, and the gas-regulator, R. This is one of the ordinary gas-regulators by means of which the amount of gas supplied to the lamp is increased if the temperature should fall, and diminished if it should rise above the required degree. The best form of the sterilizing ovens is provided with a means for circulating the hot air so that the temperature may be made uniform throughout the

mass. This can be accomplished by introducing a mechanical stirrer, or by the movement of the air itself.

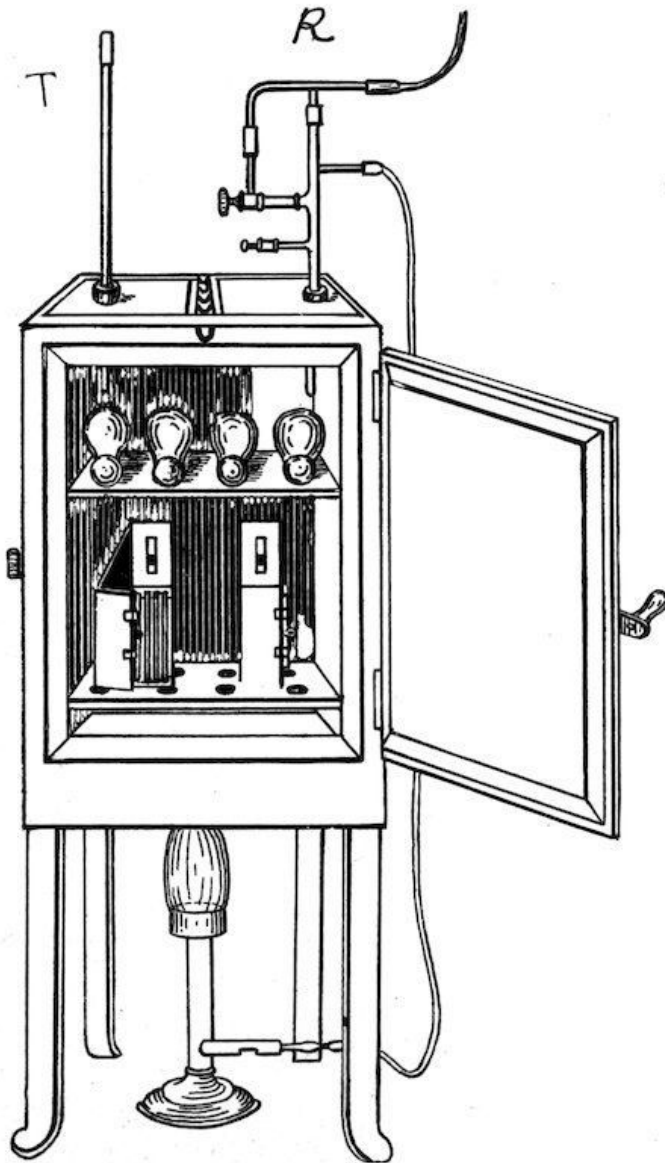


FIGURE 71. STERILIZING OVEN.

and is shown in Fig. 72.

The top is movable and held in place by the clamp, *a*, which is fixed by the screw worked by the lever, *b*. The vessel itself is double-jacketed and the pressure is obtained from water in the vessel heated by means of the lamp, *c*. The actual steam pressure is indicated by the index *d*. The safety-valve, *e*, allows any excess of steam to escape above the amount required

Between the walls of the vessel may be placed water, provided the temperature of sterilization be that of boiling water. If it should require a higher temperature than boiling water, a solution of salt can be added until the required temperature is reached, or the space between the two walls may be left vacant and hot air made to circulate around the oven.

The exterior of the oven, except at the bottom where the lamp strikes the copper surface, should be protected by thick layers of asbestos or other non-conducting material. To avoid danger of flying filaments, this covering should be coated with some smooth paint which will leave an even surface not easily abraded.

451. Sterilization with Steam at High Pressure.—

The apparatus used for this is commonly called an autoclave

for the maintenance of the pressure. This, however, is best regulated by the lamp. The outer jacket permits the heat from the lamp to circulate around the inner pressure vessel, and the holes near the top, *oo*, are for the escape of the heated gases. Enough water is placed in the bottom of the inner pressure vessel to supply all the aqueous vapor necessary to produce the required pressure and still leave some water in excess.

The materials to be sterilized are held on the shelves of the stand and the vessels may be of various kinds according to the nature of the material to be sterilized. The vessels containing the material being covered, the steam does not come in actual contact with it. At the end of the operation the safety-valve must not be opened to allow the escape of the steam, otherwise the remaining water would be rapidly converted into vapor and would be projected over the materials on the shelves. The lamp should be extinguished and the apparatus allowed to cool. The autoclave is not only useful for sterilizing purposes but can be made of general use in the laboratory where heat under pressure, as in the estimation of starch, etc., is required.

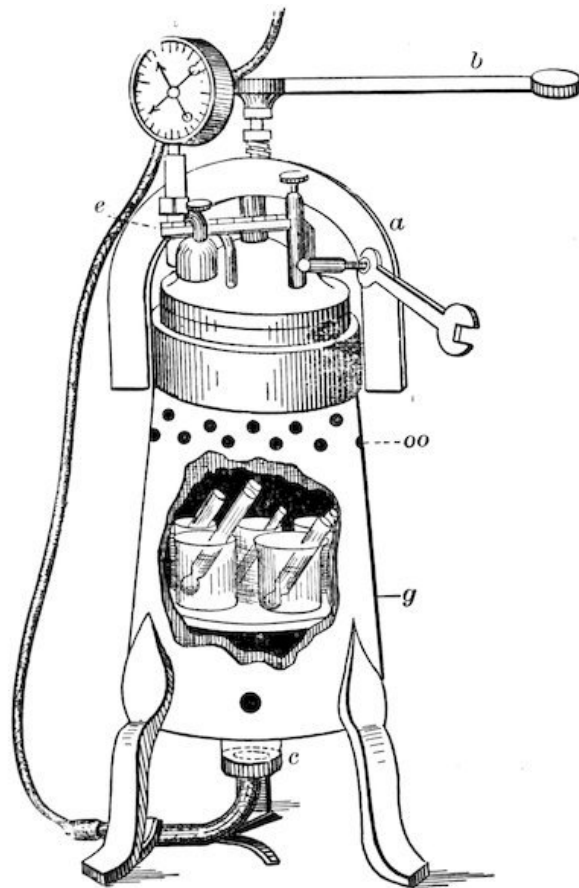


FIGURE 72. AUTOCLAVE STERILIZER.

These two forms of apparatus are sufficient to illustrate the general principles of sterilization by hot air and steam. There are, however, many variations of these forms designed for special use in certain kinds of work. For full descriptions of these, reference is made to catalogues of bacteriological apparatus.

452. Arnold's Sterilizing Apparatus.—A very simple and cheap steam sterilizer has been devised by Arnold.

Water is poured into the pan or reservoir, *B*, Fig. 73, whence it passes through three small apertures into the shallow copper vessel, *A*. It is there

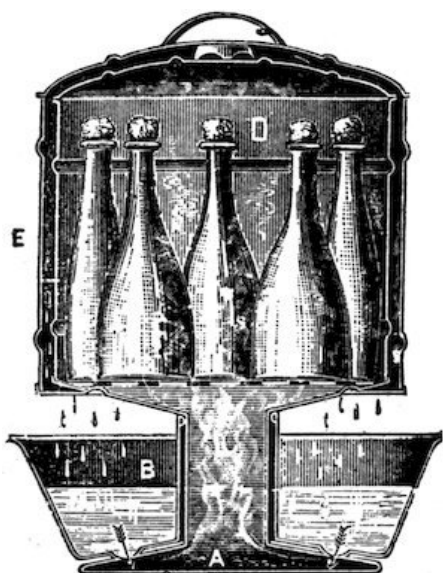


FIGURE 73. ARNOLD'S STERILIZER.

converted into steam by being heated with any convenient lamp, and rises through the funnel in the center to the sterilizing chamber. Here it accumulates under moderate pressure at a temperature of 100° . The excess of steam escapes about the cover, becomes imprisoned under the hood, E, and serves to form a steam-jacket between the wall of the sterilizing chamber and the hood. As the steam is forced down from above and meets the air it condenses and drips back into the reservoir. Such an apparatus as this is better suited to commercial purposes, as the sterilizing of milk, than for scientific uses.

453. Thermostats for Culture Apparatus.

—It is important in the culture of micro-organisms that the temperature should be kept constant during the entire time of growth. Inasmuch as some operations continue for as much as three months it is necessary to have special forms of apparatus by means of which a given temperature, during the time specified, can be maintained. This is secured by means of an oven with an automatic temperature regulator, practically built on the principle of the hot air sterilizing oven already described.

The essential principles of construction are, however, that the regulator for the temperature should be delicate and that the non-conducting medium surrounding the apparatus should be as perfect as possible, so that the variations in temperature from changes in the exterior temperature, are reduced to a minimum. This delicacy is secured by introducing a drop of chloroform-ether into a confined space over the mercury of the regulating apparatus. The doors of the chamber are double, the interior one being of glass so that the exterior door can be opened for inspection of the progress of the bacterial growth without materially interfering with the interior temperature. A convenient form is shown in Fig. 74.

The apparatus figured, is oval in shape, although circular or other forms are equally as effective. The arrangement of the lamp, *a*, thermometers, *t t t*,

and gas-regulator, *g*, and the double doors, *d d*, is shown in the engraving and does not require further description.

The usual temperatures for cultures range from 22° to 35°, and the apparatus once set at any temperature will remain fixed with extremely minute variations for an indefinite time. The apparatus possesses a heat zone which, by the arrangement of the regulator, is kept absolutely constant. The space between the walls of the apparatus being filled with water, the temperature is maintained even in every part. The apparatus, as constructed, is independent not only of the surrounding temperature within ordinary variations, but also of the pressure of the barometer. Three thermometers are employed to determine the temperature of the heating zone, the water space and the inner space. The arrangement of the gas-regulator is of an especial kind, as mentioned above, by means of which the consumption of gas is reduced to a minimum. This apparatus can be regulated to suit the character of the work.

454. Microscopic Apparatus Required.—Any good microscope, capable of accurate observation, of high power, may be used for the bacteriological observations necessary to soil analysis.

Preference should be given to the patterns adapted to receive any additional accessories which may be subsequently required for advanced work. The stage, in addition to being fitted with a sliding bar, should have a large circular or horseshoe opening to facilitate focusing operations. A mechanical stage is a desirable acquisition if really well made, but a plain stage is preferable for many purposes. A rackwork, centering sub-stage is essential for advanced work, and in the absence of the more complete form,

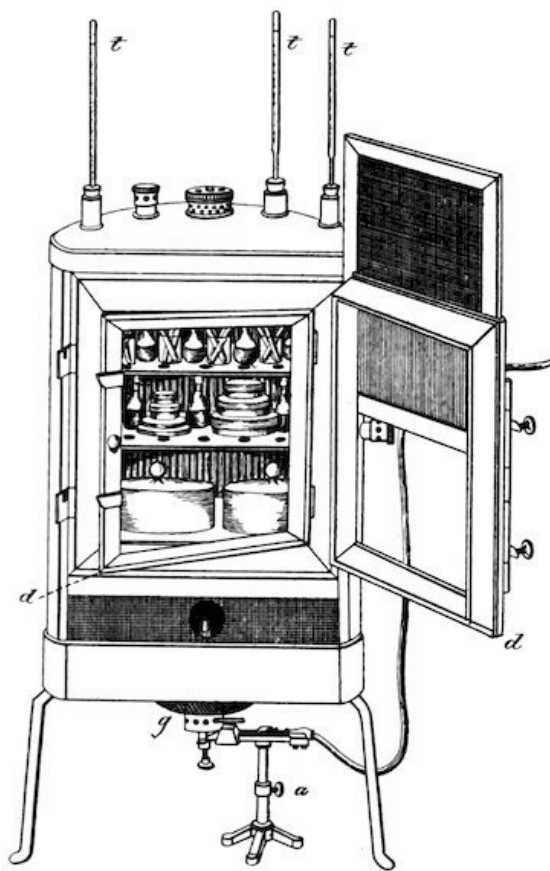


FIGURE 74. LAUTENSCHLÄGER'S THERMOSTAT.

there should at least be a fitting beneath the stage to take the diaphragm and condenser. An iris diaphragm will be found more useful than any other kind in practice, since the size of the opening can be increased very gradually at will.

One of the best lamps is known as the paraffin lamp and is fitted with a half-inch wick. This will give even more light than is actually required, and a steady flame, perfectly under control, may be obtained. For the minute details to be observed in high-grade microscopic work, such as is required in the bacteriological examination of soils, reference must be had to the standard works on bacteriology and microscopy.

455. General Conclusions.—The nitrogenous food of plants is provided in several ways; viz., (1) By the nitrogen brought to soil in rain and snow. This nitrogen is chiefly in the form of ammonia and nitric acid. The nitrogen gas in solution in rain water has no significance as a plant food. (2) By the action of certain anaerobic organisms herding in the rootlets of leguminous plants, free nitrogen may be oxidized and put into form for assimilation. (3) By the action of certain organisms on nitrogenous compounds pre-existing in the soil, ammonia, nitrous acid, and finally, nitric acid, are produced. It is believed that the plant organism, unaided by the activity of a micro-organism, is unable to assimilate nitrogen unless it be fully oxidized to nitric acid. (4) There exist micro-organisms capable of acting directly on free nitrogen independent of other plant growth, but the significance of this possible source of plant food is, at the present time, unknown. (5) The micro-organisms of importance to agriculture may be isolated and developed to the exclusion of other organisms of a similar character. This isolation is best accomplished in culture-media consisting essentially of a mineral gelatin to which is added only pure carbohydrates and the necessary mineral nourishment. (6) The nitrifying ferments consist probably of several species, of different geographic distribution. Different types of soils probably have nitrifying organisms of different properties. This is illustrated by the fact that nitrification is accomplished in dry alkaline soils under conditions in which the ordinary nitrifying organisms would fail to develop. (7) The study of typical soils in respect of the kind, activity, and vigor of their nitrifying organisms has become as important a factor in soil analysis as the usual determination of physical and chemical composition.

DETERMINATION OF NITRIC AND NITROUS ACIDS IN SOILS.

456. Classification of Methods.—The minute quantities in which highly oxidized nitrogen exists in soils render the operations of its quantitative estimation extremely delicate. On the other hand, the easy solubility of these forms of combination and the absence of absorptive powers therefor, in the soil, render the separation of them from the soil a matter of great ease. It is possible, therefore, to secure all the nitrates and nitrites present in a large quantity of earth in a solution which can be concentrated under proper precautions to a volume convenient for manipulation. The method of this extraction is the same for all the processes of determination. The methods of analysis suited to soil extracts, as a rule, may also be used in the determination of the same compounds in rain, drainage, and sewage waters, and for the qualitative and quantitative control of the progress of nitrification. The various processes employed may be classified as follows:

1. The conversion of the nitrogen into the gaseous state and the determination of its volume directly. This is accomplished by combustion with copper oxid and metallic copper.
2. The conversion of the nitrogen into nitric oxid and the volumetric determination thereof. The decomposition of a nitrate with ferrous chlorid in the presence of free hydrochloric acid is an instance of this type of methods.
3. The oxidation of colored organic solutions and the consequent disappearance of the characteristic color, or its change into a different tint. The indigo and indigotin processes are examples of this method.
4. The production of color, in a colorless or practically colorless solution, by the treatment thereof with the nitrate in presence of an acid which decomposes it with the liberation of oxidizing compounds. The depth of color produced is compared with that formed by a known quantity of a pure nitrate solution until the two colorations are alike. The methods depending on the use of carbazol or acid phenol sulfate illustrate this class of reactions.
5. The conversion of the nitrogen into ammonia by moist combustion with sulfuric acid in the presence of certain organic compounds, *e. g.*,

salicylic acid, and the collection of the ammonia in standard acid, the excess of which, is determined by titration.

6. The reduction of nitrates to ammonia by nascent hydrogen and the recovery of the ammonia produced by distillation and collection in standard acid.

7. The reduction of nitrates by electrolytic action and the collection of the ammonia as above.

8. The decomposition of nitrates with the quantitative evolution of a different element, and the direct or indirect measurement of the evolved substance. The quantitative evolution of chlorine on treating a nitrate with hydrochloric acid, the collection of the chlorine in potassium iodide, and the determination of the iodine set free, form a process belonging here.

457. Relative Merit of Methods.—The processes mentioned in the classifications embraced under numbers (1) and (5) of the preceding schedule are sufficiently described in the paragraphs devoted thereto, under soil and fertilizers. In practice at the present time it is rare to determine the nitrogen in nitrates by the copper oxide method. The more rapid and equally exact processes of colorimetric comparison or reduction by nascent hydrogen are in all respects to be preferred.

The indigo methods among the colorimetric processes are not so much in use now as those which depend on the development of a color. Lawes and Gilbert considered them far inferior to the Schloesing method. The developed color methods are especially delicate and are to be preferred in all cases where the detection of the merest traces of nitrates is desired. Where nitrates are present in considerable quantities the reduction method with nascent hydrogen is to be preferred over all others. In all these cases the judgment of the analyst must be exercised. The particular method to be employed in any given case can not be determined save by the intelligent discrimination of the operator.

458. The Extraction of Nitric Acid from the Soil.—The easy solubility of nitric acid and of nitrates in water is taken advantage of in the separation of these bodies from the soil. A convenient quantity, usually about 1,000 grams of the fine soil, is taken for the extraction. Instead of freeing the soil entirely from water, it is better to determine the amount of water in the air-dried or prepared sample, and base the calculation on 1,000 grams of the water-free soil.

All samples of soil, when taken for the purpose of examining for nitrates, should be rapidly dried to prevent the process of nitrification from continuing after the sample is taken. For this purpose the soil should be placed in a thin layer in a warm place, 50°–60°, until air-dried. It still contains in this case a little moisture but not enough to permit nitrification to go on.

One thousand grams of soil prepared as above are treated with 2,000 cubic centimeters of distilled water, free of nitric acid, and allowed to stand for forty-eight hours with frequent shaking. One thousand cubic centimeters of the extract are then filtered, corresponding to 500 grams of the dry soil. A small quantity of pure sodium carbonate should be added to the filtrate which is then evaporated on a water-bath to a volume of about 100 cubic centimeters. Should a precipitate be formed during evaporation it should be separated by filtration, the filter washed thoroughly, and the filtrate again evaporated to a volume of 100 cubic centimeters.

In taking a soil for the determination of nitrates, it is well to extend the sampling to a considerable depth. If the sample be taken only to the depth of nine inches, it should be in dry weather when the nitrates are near the surface.

The temperature at which a soil is dried has also an influence on the quality of nitric nitrogen remaining after desiccation.

If a wet soil be dried at 100°, the nitrates present will suffer partial decomposition. This is probably due to deoxidation by organic matter present. On the other hand, ordinary air-drying affords opportunity for continued nitrification, thus increasing the residuum of oxidized nitrogen. The above method is essentially that followed by Warington at Rothamstead.

The method of drying practiced at Rothamstead, in order to secure results as nearly accurate as possible is the following:[\[293\]](#)

The soil is broken up directly after it is taken from the field, and spread on trays in layers one inch deep. The trays are then placed in a room at 55°. The drying is completed in twenty-four hours. After drying, stones and roots are removed, and the soil is finely powdered and placed in bottles.

For extracting the nitrates, a funnel is prepared by cutting off the bottom from a bottle four and a half inches in diameter. A nicely fitting disk of copper gauze is placed in the bottom of this funnel, and this is covered with

two filter papers, the upper one having a slightly greater diameter than the lower. The paper is first moistened, and then from 200 to 500 grams of the dry powdered soil introduced. The funnel is connected with the receiving flask of a filter pump, and pure water poured on the soil until it is thoroughly saturated. The water is then added in small quantities. When the filtrate amounts to 100 cubic centimeters the process may be discontinued, since all the nitrates in the soil will be found in this part of the filtrate.

The extract obtained above is evaporated to convenient bulk for the determination of nitric nitrogen.

THE NITRIC OXID PROCESS.

459. Method of Schloesing.—The processes for estimating nitrogen by combustion with copper oxid and by moist combustion with sulfuric acid have both been used for the determination of the quantity of nitrogen existing in a highly oxidized state. These processes will be fully discussed under the head of fertilizers. In the case of soil extracts, drainage waters, etc., it will be sufficient to discuss, for the present, only those processes adapted especially to a quick and accurate estimation of oxidized nitrogen.

The principle of the method of Schloesing depends on the decomposition of nitrates in the presence of a ferrous salt and a strong mineral acid.^[294] The nitrogen in the process appears as nitric oxid, the volume of which may be directly measured, or it may be converted into nitric acid and titrated by an alkali.

The typical reactions which take place are represented in the following equation:



460. Schloesing's Modified Method.—The Schloesing method as now practiced by the French chemists is conducted in the apparatus shown in Fig. 75.^[295] The carbon dioxid is generated by the action of the hydrochloric acid in F on the fragments of marble in A. After passing the wash-bottle the gas enters the small tubulated retort, C, which contains the nitrate in solution. For ordinary soils 100 grams are placed in an extraction flask, plugged with cotton, and a layer of the same material is placed over the soil for the purpose of securing an even distribution of the extracting liquid. This liquid is distilled water containing in each liter one gram of calcium chlorid. The purpose of using the calcium chlorid is to prevent the soil from becoming compacted which would render the extraction of the nitrate difficult. The extracting liquid is allowed to fall drop by drop from a mariotte bottle until the filtrate amounts to 500 cubic centimeters. This volume is concentrated on a sand-bath until it is reduced to ten or fifteen cubic centimeters when it is transferred to a flat-bottomed dish and the evaporation finished over steam, care being taken not to allow the temperature to exceed 100°.

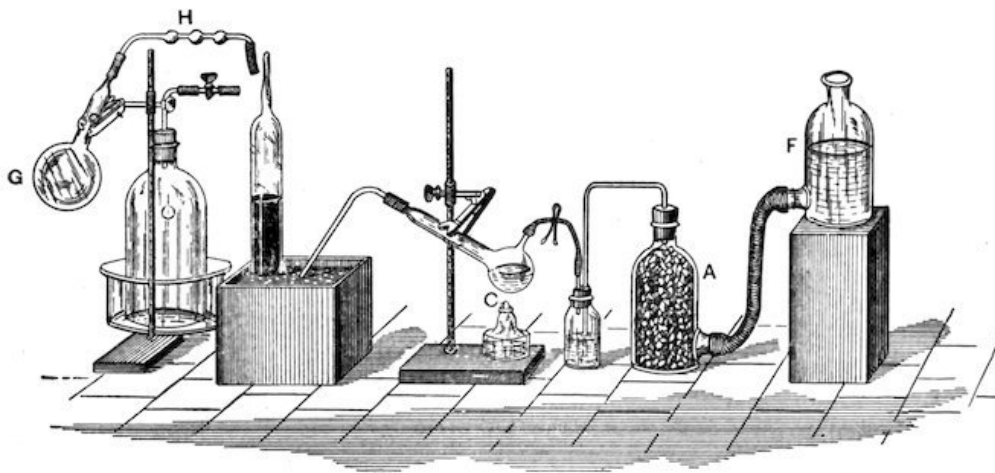


FIGURE 75. SCHLOESING'S APPARATUS FOR NITRIC ACID.

Another and more rapid method for dissolving the nitrate, may also be practiced. In a flask holding about one liter, place 220 grams of the soil and 660 cubic centimeters of distilled water and shake vigorously, or enough water to make 660 cubic centimeters together with the moisture remaining in the air-dried sample taken. All the nitrates pass into solution. Throw the contents of the flask into a filter and take 600 cubic centimeters of the filtrate which will contain all the nitrates in 200 grams of the sample taken. This filtrate is evaporated as described above.

In the flat dish containing the dried nitrates, pour three or four cubic centimeters of ferrous chlorid solution and stir with a small glass rod until complete solution of the nitrate takes place. By means of a small funnel the solution is poured into C, and the capsule and funnel are well rinsed with two cubic centimeters of hydrochloric acid. The washing is repeated three times as above described, and once with one cubic centimeter of water, which is added cautiously so as to form a layer over the surface of the heavier liquid. The tubulated flask is then connected with the carbon dioxid apparatus, previously freed from air, and the gas allowed to flow evenly until the whole of the apparatus is completely air-free. The other details of the method are essentially the same as those adopted by the Commission of French Agricultural Chemists which will be given below.

461. The French Agricultural Method.—The Schloesing method as practiced by the French agricultural chemists is very slightly different from

the procedure just described.^[296] The process with soils is carried on as follows:

Five hundred grams of the soil are taken and introduced into a flask of about two liters capacity and shaken thoroughly with a liter of distilled water. The whole of the nitrates of the soil is thus passed into solution. The solution is filtered and 400 cubic centimeters of the filtrate are taken, which correspond to 200 grams of the soil. This liquid is evaporated in a flask, adding a fragment of paraffin to prevent foaming, until its volume is reduced to fifteen or twenty cubic centimeters. It is afterwards transferred through a filter into a capsule with a flat bottom in which the evaporation is finished on a steam-bath, taking care that the temperature does not exceed 100°. An important precaution is, not to allow the contact of the water with the soil to be too prolonged, to avoid the reduction of the nitrates which could take place under the influence of the denitrifying organisms which are developed with so great a rapidity in moist earth. The apparatus in which the transformation of the nitrates into nitric oxid takes place is essentially that already described (Fig. 75). The carbon dioxid generator is connected by means of a rubber tube and a small wash-bottle to the small retort in which the reaction takes place, and from which the exit tube leads to a mercury trough. The gas which is disengaged is received under a jar drawn out to a fine point in its upper part, which carries about fifteen cubic centimeters of potash solution containing two parts of water to one of potash.

The operation is conducted as follows:

Into the small capsule which contains the dried matter, three or four cubic centimeters of ferrous chlorid are poured. By means of a stirring rod the residue sticking to the sides of the capsule is detached with care and all the matter is thus collected in the bottom. By means of a small funnel the contents of the capsule are introduced into the retort. About two cubic centimeters of hydrochloric acid are used for washing out the materials and this acid is also introduced into the retort. The washing with hydrochloric acid is repeated three or four times, and finally the apparatus is washed with one cubic centimeter of water, which is also poured in by the small funnel with great care, so that this water may form a layer over the surface of the liquid. The apparatus is now connected and filled completely with carbon dioxid. Since it is necessary that this gas should be completely free of air,

the flask, which generates it, is first filled with the acidulated water from the acid flask, and the air is thus almost totally displaced by the liquid. The evolution of carbon dioxide gas which follows, completely frees the apparatus from air. When this is accomplished the retort is connected with the rest of the apparatus and the gas allowed to pass for about two minutes until the air is completely driven out of all the connections. The current is arrested for a moment by pinching the rubber tube which conducts the carbon dioxide into the retort, and the vessel which is to receive the gas is then placed over the delivery-tube, this vessel being filled with mercury and a strong solution of potash. The communication between the retort and the carbon dioxide flask is broken and the flask is heated slightly by means of a small lamp. The first bubbles of gas evolved should be entirely absorbed by the potash. This will be an indication of the complete absence of the air. When the liquid is in a state of ebullition the nitrogen dioxide is set free. The boiling is regulated in such a way that the evolution is regular and the liquid of the retort may not, by a too violent boiling, pass into the receiver. The boiling is continued until the larger part of the liquid is distilled and only three or four cubic centimeters remain in the retort. At this time a few bubbles of carbon dioxide are allowed to flow through in order to cause to pass into the receiver the last traces of nitric oxide. The gas received is left for some minutes in contact with the potash.

Afterward in a small flask, G, the neck of which is drawn out to a fine point, and carrying a bulb-tube, H, and a piece of rubber tubing, there are boiled twenty-five or thirty cubic centimeters of water for five or six minutes in order to drive all the air out of the flask, and while the boiling is continued the rubber tubing is fastened to the drawn-out part of the jar containing the nitric oxide. Within the rubber tubing the drawn-out point is broken and the vapor of water is forced into the jar and drives before it the solution of potash which has filled the capillary part of the drawn-out tube. As soon as the point is broken, the boiling of the flask is stopped and by its cooling the nitric oxide passes into it. It is necessary to press the rubber tubing with the fingers in order that the passage of the gas into the flask be not too rapid. As the solution of potash rises in the bell-jar which contains the nitric oxide near to the point where the rubber tubing covers its drawn-out portion, the fingers are removed and a clamp put in their place. There still remains a little nitric oxide in the flask and to drive this out it is necessary to introduce five or six cubic centimeters of pure hydrogen,

which are allowed to pass over into the receiving flask, by releasing the clamp in the same way as the nitric oxid. The hydrogen being introduced in successive portions, finally carries all the nitric oxid into the flask without allowing any of the potash to enter.

The flask containing the nitric oxid is now connected with a reservoir of oxygen. The oxygen is allowed to enter, bubble by bubble, by means of cooling the flask by immersion in water. The transformation of nitric oxid into nitric acid is not entirely complete for twenty-four hours. It is necessary, therefore, to wait that long after the introduction of the oxygen before determining the amount of nitric acid produced.

The contents of the flask are placed in a titration-jar, the flask being washed two or three times and a few drops of tincture of litmus being added. The nitric acid is then determined by a standard solution of calcium hydroxid or some other standard alkali. From the titration the content of nitric acid is calculated.

The French Committee further suggests that this method may be modified in the way of making it more rapid by collecting the nitric acid in a graduated tube filled with mercury and containing some potash. The volume of the gas is determined and the pressure of the barometer and the temperature observed, and the usual calculations made to reduce the volume to zero and to a pressure of 760 millimeters of mercury. Each cubic centimeter of nitric oxid thus measured corresponds to 2.417 milligrams of nitric acid. The presence of organic matter does not interfere with the determination of nitric acid by either of the methods given above.

462. Modification of Warington.—The method of procedure and description of apparatus used, as employed by Warington, are as follows:

The vessel in which the reaction takes place is a small tubulated receiver, A (Fig. 76), about four centimeters in diameter, mounted and connected as shown in the illustration. The delivery-tube dips into a jar of mercury in a trough containing the same liquid. The long supply funnel-tube *a* is of small bore, holding in all only one-half cubic centimeter. The connecting tube F, carrying a clamp, is also of small diameter and serves to connect the apparatus with a supply of carbon dioxid.

In practice, the supply-tube *a* is first filled with strong hydrochloric acid and carbon dioxid passed through the apparatus until the air is all expelled.

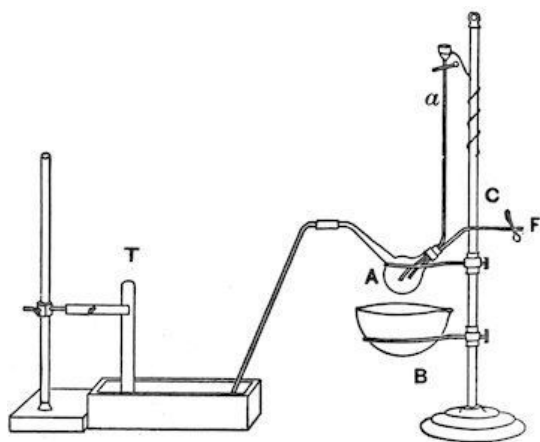


FIGURE 76. WARINGTON'S APPARATUS FOR NITRIC ACID.

This is indicated when a portion of the gas collected over the mercury, is entirely absorbed by caustic alkali.

At this point the current of carbon dioxid is stopped by the clamp C, and a bath of calcium chlorid, B, heated to 140° is brought under the bulb A, until the latter is half immersed therein. The temperature of the bath is maintained by a lamp. By allowing a few drops of hydrochloric acid to enter the receiver, the carbon dioxid is almost wholly expelled. The end of the delivery-tube is

then connected with the tube, T, filled with mercury, and the apparatus is ready for use.

The nitrate, in which the nitric acid is to be determined, in a dry state, is dissolved in two cubic centimeters of the ferrous chlorid solution (one gram of iron in ten cubic centimeters), one cubic centimeter of strong hydrochloric acid is added, and the whole is then introduced into the receiver through the supply-tube, being followed by successive rinsings with hydrochloric acid, each rinsing not exceeding one-half cubic centimeter. The contents of the receiver are, in a few moments, boiled to dryness; a little carbon dioxid is admitted before dryness is reached, and again afterwards to drive over all remains of nitric oxid. In the recovered gas the carbon dioxid is first absorbed by caustic potash, and afterwards the nitric oxid by ferrous chlorid. All measurements of the gas are made in Frankland's modification of Regnault's apparatus. The carbon dioxid should be as free as possible from oxygen. The carbon dioxid generator is formed of two vessels, the lower one consisting of a bottle with a tubule in the side near the bottom; this bottle is supported in an inverted position and contains the marble from which the gas is generated. The upper vessel consists of a similar bottle standing upright and containing the hydrochloric acid required to act on the marble. The two vessels are connected by a glass tube passing from the side tubule of the upper vessel to the inverted mouth of the lower vessel. The acid from the upper vessel thus enters below the marble. Carbon dioxid is generated and removed at pleasure by opening a stop-cock attached to the side tubule of the lower vessel thus allowing hydrochloric

acid to descend and come in contact with the marble. A good Kipp's generator of any approved form may also be used instead of the simple apparatus, above described.

The fragments of marble used are previously boiled in water in a strong flask. After boiling has proceeded for some time, a rubber stopper is fixed in the neck of the flask and the flame removed. Boiling will then continue for some time in a partial vacuum.

The hydrochloric acid is also well boiled and has dissolved in it a moderate quantity of cuprous chlorid. As soon as the acid has been placed in the upper reservoir, it is covered by a layer of oil. The apparatus being thus charged is at once set in active work by opening the stop-cock of the marble reservoir; the acid descends, enters the marble reservoir, and the carbon dioxid produced drives out the air. As the acid reservoir is kept on a higher level than the marble reservoir, the latter is always under internal pressure, and leakage of air from without, into the apparatus, cannot occur.

The presence of the cuprous chlorid in the hydrochloric acid not only insures the removal of dissolved oxygen, but affords an indication to the eye of the maintenance of this condition. While the acid remains of an olive tint, oxygen is absent; but should the color change to a blue-green, more cuprous chlorid must be added. All the reagents employed should be previously boiled.

In order to secure absolute freedom from air, the following modifications on the above process have been adopted by Warington: The apparatus having been mounted as described, the funnel-tube attached to the bulb retort is filled with water, and the apparatus connected with the carbon dioxid generator. Carbon dioxid is then passed through the apparatus until a moderate stream of bubbles rises in the mercury trough. The stop-cock is left in this position, and the admission of gas is controlled by the pinch-cock. The bath of calcium chlorid is so adjusted as to cause the bulb retort to be almost entirely submerged, and the temperature of the bath is kept at 130° to 140°. Small quantities of water are next admitted into the bulb and expelled as steam in the current of carbon dioxid, the supply of this gas being shut off before the evaporation is entirely completed, so as to leave as little carbon dioxid as possible in the apparatus. Previous to very delicate experiments it is advisable to introduce through the funnel-tube a small quantity of potassium nitrate, ferrous chlorid, and hydrochloric acid, rinsing

the tube with the latter reagent. Any trace of oxygen remaining in the apparatus is then consumed by the nitric oxid formed; and after boiling to dryness and driving out the nitric acid with carbon dioxid, the apparatus is in a perfect condition for a quantitative experiment.

463. Preparation of the Materials to be Analyzed.—According to Warrington, soil extracts may be used without other preparation than concentration.

Vegetable juices which coagulate when heated, require to be boiled and filtered or else evaporated to a thin sirup, treated with alcohol, and filtered. A clear solution being thus obtained, it is concentrated over a water-bath to a minimum volume in a beaker of small size. As soon as cool, it is mixed with one cubic centimeter of a cold saturated solution of ferrous chlorid and one cubic centimeter of hydrochloric acid, both reagents having been boiled and cooled immediately before use.

In mixing with the reagents, care must be taken that bubbles of air are not entangled, which is apt to occur with viscid extracts.

The quantity of ferrous chlorid mentioned is amply sufficient for most soil extracts, but it is well to use two cubic centimeters in the first experiment, the presence of a considerable excess of ferrous chlorid in the retort being thus insured. With bulky vegetable extracts more ferrous chlorid should be employed. To the sirup from twenty grams of mangel-wurzel sap, five cubic centimeters of ferrous chlorid and two cubic centimeters of hydrochloric acid are usually added.

464. Measurement of the Gas.—The measurement of the gas was for some time conducted by the use of concentrated potash for absorbing the carbon dioxid, and ferrous chlorid for absorbing the nitric oxid. The use of the ferrous chlorid, however, was found to introduce a source of error. The treatment of the gas with oxygen and pyrogallol over potash has therefore been substituted by Warrington for its absorption by ferrous chlorid.

The chief source of error attending the oxygen process lies in the small quantity of carbon monoxid produced during the absorption with pyrogallol; this error becomes negligible if the oxygen be only used in small excess. The amount of oxygen employed can be regulated by the use of Bischof's gas delivery-tube. This may be made of a test-tube having a small perforation half an inch from the mouth. The tube is partly filled with oxygen over mercury, and its mouth is then closed by a finely perforated

stopper made from a piece of wide tube and fitted tightly into the test-tube by means of a covering of rubber. When this tube is inclined, the side perforation being downwards, the oxygen is discharged in small bubbles from the perforated stopper, while mercury enters through the opening. Using this tube, the supply of oxygen is perfectly under control and can be stopped as soon as a fresh bubble ceases to produce a red tinge on entering. Warington concludes his description by stating that in the reaction proposed by Schloesing the analyst has a means of determining a very small quantity of nitric acid with considerable accuracy, even in the presence of organic matter; but to accomplish this, the various simplifications consisting in the omission of the stream of carbon dioxid, and the collection of the gas over caustic soda must be abandoned, and special precautions must be taken to exclude all traces of oxygen from the apparatus.

465. Spiegel's Modification.—Spiegel noticed inaccuracies in the results of the ferrous chlorid method of estimating nitric acid when carbon dioxid is used, which sometimes amounted to three per cent of the nitric acid present in the sample. The following suggestions are made by him for the improvement of the process:^[297]

As regards the use of carbon dioxid in the operation, the first difficulty consists in obtaining it entirely free from air. By the use of small pieces of marble, which, before being placed in the Kipp apparatus, are kept for a long while in boiling water, a product is obtained which, after thirty minutes of moderate evolution, leaves only a trace of unabsorbed gas in contact with potash-lye. The apparatus used is illustrated in Fig. 77.

A is a round flask of about 150 cubic centimeters capacity, furnished with a well-fitting rubber stopper provided with two holes, one for the entrance of the funnel-tube B and the other for the delivery-tube C. The tube B ends about two centimeters above the bottom of A and carries a bulb-shaped funnel at its top capable of holding about fifty cubic centimeters. The gas-tube D is ground into the bulb of B as shown in the figure.

After the flask had been filled with the solution to be examined, carbon dioxid is conducted through D and the flask is heated to boiling until the gas which escapes through C no longer contains any air. The measuring tube is brought over the end of the delivery-tube C, in the usual manner, but not shown in the figure. In the funnel of B are placed twenty cubic centimeters of previously prepared and boiled ferrous chlorid solution and

this liquid is allowed to flow partly into A by lifting slightly the gas-tube, D. About forty cubic centimeters of concentrated, boiled hydrochloric acid are afterwards added to it in the same way. As soon as the liquid in the flask A is again boiling, the stream of carbon dioxide is shut off and allowed to flow again only towards the end of the operation, when the contents of the flask are reduced almost to dryness. As will be seen from the above directions no unboiled liquids of any kind are to be used as reagents in the apparatus described. If the flask A were made much smaller the efficiency of this apparatus would be increased. It appears to have few, if any, advantages over Warington's process.

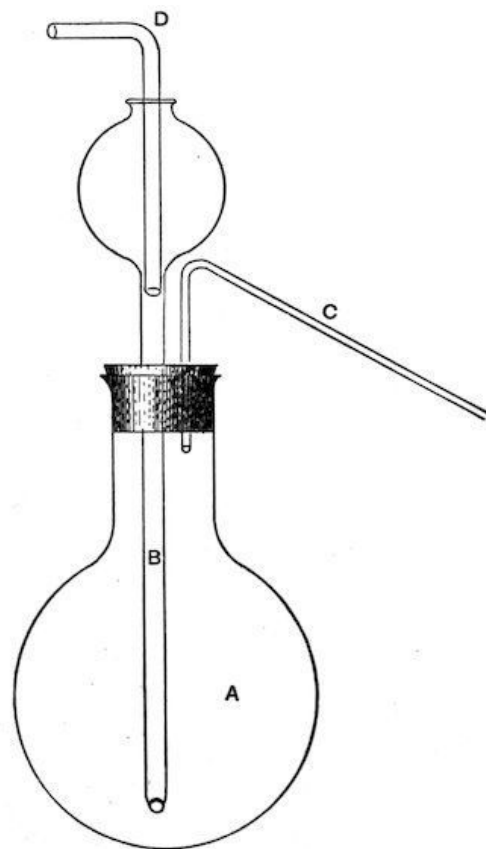


FIGURE 77. SPIEGEL'S APPARATUS FOR NITRIC ACID.

466. Schulze-Tiemann Method.—The modification of Schulze-Tiemann in the ferrous salt method consists chiefly in the omission of the use of carbon dioxide, and in the simplified form of apparatus, which permits rapid work and gives, also, according to some authorities, very exact and reliable results.^[298] The extract, representing 500 grams of the fine soil, is reduced by evaporation to 100 cubic centimeters and placed in a glass flask, A (Fig. 78), of 500 cubic centimeters capacity. The flask is closed with a rubber stopper, carrying two bent glass tubes which pass through it. The tube *a b c* is drawn out into a point at *a* and reaches about two centimeters below the surface of the rubber stopper. The tube *e f g* passes just to the lower surface of the rubber stopper. The two tubes mentioned are connected, by means of rubber tubes and pinch-cocks, with the tubes *d* and *h*. The pinch-cocks at *c* and *g* must be capable of closing the tubes air-tight. The end of the tube *g h* passes into a crystallizing dish, B, and is bent upward to a point passing two to three centimeters into the measuring tube C. The point within the tube is covered with a piece of rubber tubing. The measuring tube C is divided into tenths of a cubic centimeter, and together with the crystallizing dish B, is filled

with a ten per cent solution of boiled soda-lye, which is obtained by dissolving 12.9 parts of sodium hydroxid in 100 parts of water.

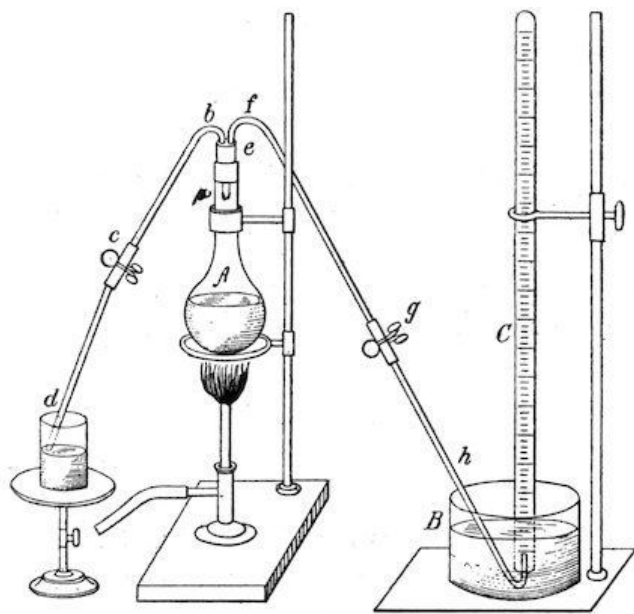


FIGURE 78. SCHULZE-TIEMANN'S NITRIC ACID APPARATUS.

The liquid which is to be examined for nitric acid, the pinch-cocks being opened and the tube *g h* not dipping into the crystallizing dish, is boiled for one hour in order to drive the air out of the flask *A*. The end of the tube *e f g h* is then brought into the crystallizing dish containing the sodium hydroxid solution so that the steam escaping from the flask *A*, escapes partly through the tube *b c d*, and partly through the tube *f g h*, not allowing, however, the bubbles to enter the measuring tube *C*. To determine whether the air is all expelled, the

pinch-cock at *g* is closed and the soda-lye will thereupon rise to *g* in case no air interferes. It is best to close the tube at *g* first with the thumb and finger and then the rise of the soda-lye to that point can be determined by the impulse felt. The tube is then firmly closed by means of the pinch-cock *g*. The rest of the steam is allowed to escape through the tube *a b c d*, and the evaporation is continued until the contents of the flask are evaporated to about ten cubic centimeters. The flask into which the tube *c d* dips, is filled with freshly boiled water. The lamp is removed from the flask *A*, the pinch-cock is closed, whereupon the tube *c d* becomes filled with the freshly boiled water. The measuring tube *C*, filled with freshly boiled soda-lye is closed with the thumb and brought into the dish *B*, care being taken that no bubble of air enters. It is placed over the end of the tube *g h*.

The pressure of the external air will now flatten the rubber tubes at *c* and *g*. The flask at the end of *c d* holding freshly boiled water is then replaced with one filled with a nearly saturated solution of ferrous chlorid containing some hydrochloric acid. The flask containing the ferrous chlorid solution should be graduated so that the amount which is sucked into the flask *A* can be determined. The pinch-cock *c* is opened and from fifteen to twenty cubic

centimeters of the ferrous chlorid solution allowed to flow into *A*. The end of the tube *c d* is then placed in another flask containing strong hydrochloric acid, and the latter allowed to flow into the tube in small quantities at a time until all the ferrous chlorid is washed out of the tube *b c d* into *A*. At the point *b* there is sometimes formed a little bubble of hydrochloric acid in the state of gas, which by heating the flask *A* completely disappears.

The flask *A* is next warmed gently until the rubber tubes at the pinch-cocks begin to assume their normal condition. The pinch-cock at *g* is now replaced by the thumb and finger, and as soon as the pressure within the flask *A* is somewhat stronger, caused by the nitric oxid gas evolved from the mixture, it is allowed to pass through the tube *e f g h* and escape into the measuring cylinder *C*. By a manipulation of the finger and thumb at *g*, it is possible to prevent regurgitation of the sodium hydroxid into *A*, and at the same time to relieve the pressure of the nitric oxid in *A*, which would be difficult to do by means of the pinch-cock alone.

The boiling of the liquid is continued until there is no longer any increase of the volume of gas in the measuring cylinder *C*. After the end of the operation the tube *g h* is removed from the dish *B* and the measuring tube *C* is closed by means of the thumb while its end is still beneath the surface of the soda-lye, and it is shaken until all traces of any hydrochloric acid, which may have escaped absorption, are removed. It is then placed in a large glass cylinder filled with water at the temperature at which the volume of gas is to be read. After being kept at this constant temperature for about half an hour the volume of the nitric oxid can be read. For this purpose the measuring cylinder *C* is sunk into the water of the large cylinder until the level of the liquids within and without the tube is the same. The usual correction for pressure of the atmosphere, as determined by the barometer, and for the tension of the aqueous vapor at the temperature at which the reading is made, is applied. The correction is made by means of the following formula:

$$V' = (V \times 273 \times (B - f)) \div ((273 + t) \times 760)$$

In this formula *V'* denotes the volume of the gas at the temperature of zero, and at 760 millimeters barometric pressure; *V* the volume of the gas as read at the barometric pressure observed, *B*, and the temperature observed, *t*, while *f* denotes the tension of the aqueous vapor in millimeters of mercury pressure at the observed temperature *t*. The tension of the aqueous

vapor at temperatures from zero to 26°, expressed in millimeters of mercury, is given in the following table:

Temp. °	Tension in mm. mercury.
0	4.6
1	4.9
2	5.3
3	5.7
4	6.1
5	6.5
6	6.9
7	7.4
8	8.0
9	8.5
10	9.1
11	9.7
12	10.4
13	11.1
14	11.9
15	12.7
16	13.5
17	14.4
18	15.3
19	16.3
20	17.4
21	18.5
22	19.6
23	20.9
24	22.2
25	23.5
26	25.0

From the gas volume reduced by the above formula the nitric acid is calculated as follows:

One cubic centimeter of nitric oxid weighs at 0° and 760 millimeters barometric pressure 1.343 milligrams.

Since two molecules of NO (molecular weight sixty) correspond to one molecule of N_2O_5 (108) we have the following equation: $60 : 108 = 1.343 : x$. Whence $x = 2.417$ milligrams, the weight of nitric acid corresponding to one cubic centimeter of nitric oxid.

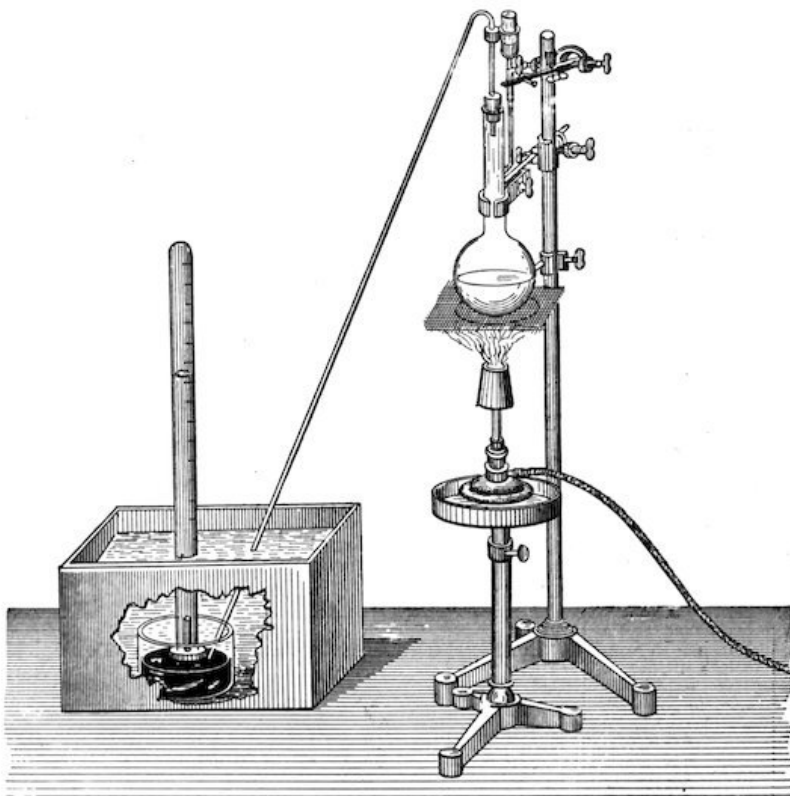


FIGURE 79. DE KONICK'S APPARATUS.

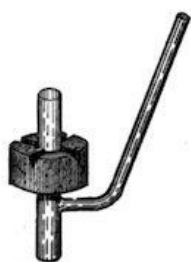


FIGURE 80. END OF DELIVERY-TUBE.

467. DeKonick's Modification of Schloesing's Method.

—This modification consists in an arrangement of the gas delivery-tube, whereby the regurgitation of the water in the measuring burette into the evolution flask is prevented by a device for sealing the delivery-tube with mercury.^[299] The apparatus is arranged as shown in Fig. 79. The flask in which the decomposition takes place is provided with a long neck, into which a side tube is sealed and bent upwards, carrying a small funnel attached to it by rubber tubing. The piece of rubber tubing carries a pinch-cock, by means of which the solution containing the nitrate and hydrochloric acid can be introduced into the flask. The small gas delivery-tube is arranged as shown in the figure, and carries at the end next the burette a device shown in Fig. 80. The cork represented in this device has radial notches cut in it, so as to permit of a free communication between the water in the burette and in the pneumatic trough. The open end of the burette, when the apparatus is mounted ready for use, rests on the notched surface of the cork, and the end of the delivery-

tube is placed in the crystallizing dish resting on the bottom of the pneumatic trough.

The end of the delivery-tube, as indicated, has fused onto it a vertical tube open at both ends and six to seven centimeters in length, and carrying the notched cork already described. The crystallizing dish in the bottom of the pneumatic trough is filled with mercury until the point of union of the delivery-tube with the vertical end is sealed to the depth of a few millimeters. As the gas is evolved it bubbles up through the mercury into the measuring tube and the displaced water passes out through the notches in the cork. Should any back pressure supervene the mercury at once rises in the delivery-tube which is of such a length as to prevent its entrance into the flask. The operation can then be carried on with absolute safety.

To make an estimation there are placed in the flask about forty cubic centimeters of ferrous chlorid solution containing about 200 grams of iron to the liter, and also an equal volume of hydrochloric acid of one and one-tenth specific gravity. The side tube is also filled up to the funnel with the acid. The contents of the flask are boiled until all air is expelled, which can be determined by holding a test-tube filled with water over the end of the delivery-tube. The solution containing the nitrate is next placed in the funnel, the pinch-cock opened and the liquid allowed to run into the flask by means of the partial vacuum produced by stopping the boiling and allowing the mercury to rise in the delivery-tube. All the solution is washed into the flask by successive rinsings of the funnel with hydrochloric acid, being careful to allow no bubble of air to enter. The contents of the flask are again raised to the boiling-point and the nitric oxid evolved collected in the nitrometer. The solution examined should contain enough nitrate to afford from sixty to eighty cubic centimeters of gas. Without refilling the flask, from eight to nine determinations can be made by regenerating the ferrous chlorid by treatment with zinc chlorid. Care must be exercised not to add the zinc chlorid in excess, otherwise ammonia and not nitric oxid will be produced. The side tube and funnel must also be carefully freed from zinc chlorid by washing with hydrochloric acid.

468. Schmidt's Process.—In the case of a water, or the aqueous extract of a soil, according to the content of nitric acid, from fifty to one hundred cubic centimeters are evaporated to thirty cubic centimeters, and the residue sucked into the generating flask of the apparatus, Fig. 81, and, with the

rinsings with distilled water, evaporated again to from twenty to thirty cubic centimeters, and the flask then connected, as shown in the figure, to a Schliff measuring apparatus, B.^[300] This apparatus is previously filled to *i* with mercury, and the bulb *g* connected with *k* by a rubber tube.

The apparatus is then filled with a twenty per cent, previously boiled and still warm, caustic soda solution until the bulb *g* is partially filled when raised a little above the cock *h*. Then *h* is closed and *g* held, by an appropriate support, on about the same level with *h*. The cock at *b* is then closed and *e* opened. Meanwhile the ebullition in the flask is continued, and the air bubbles rising in the Schliff apparatus are removed, from time to time, by carefully opening *h* and raising *g*. When bubbles no longer come over, the cock at *e* is closed and at *b* opened, and the steam issuing at *a* is conducted through a mixture of ferrous chlorid and strong hydrochloric acid to free it, as far as possible, from air. When the contents of the flask have been evaporated to about five cubic centimeters, *b* is closed and the lamp at once removed.

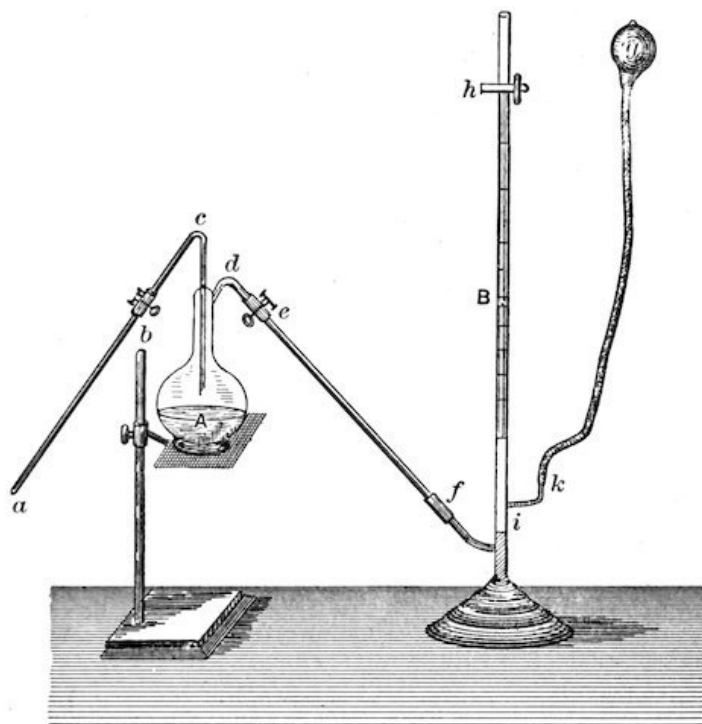


FIGURE 81. SCHMIDT'S APPARATUS.

By carefully opening *b* about ten cubic centimeters of a mixture of ferrous chlorid and hydrochloric acid are allowed to enter the flask, when *b* is closed and the flask slowly heated until the positive pressure is restored. The pinch-cock *e* is then opened and the contents of the flask evaporated nearly to dryness. The cock *e* is again closed and the flame removed. Another quantity (fifteen cubic centimeters) of ferrous chlorid and hydrochloric acid solution is sucked into the flask and the process of distillation repeated, whereby the whole of the nitric oxid is collected in *h*. The nitric oxid evolved is measured in the usual way and calculated to nitric

acid, one cubic centimeter of nitrogen dioxid being equal to 2.417 milligrams of nitric acid.

469. Merits of the Ferrous Chlorid Process.—The possibility of an accurate determination of nitrates; by decomposition with a ferrous salt in presence of an excess of acid, has been established by many years of experience and by the testimony of many analysts. The method is applicable especially where the quantity of nitrate is not too small and when organic matter is present. In the case of minute quantities of nitrate, however, the process is inapplicable and must give way to some of the colorimetric methods to be hereafter described.

In respect of the apparatus modern practice has led to the preference of that form which does not require the use of carbon dioxid for displacing the air. Steam appears to be quite as effective as carbon dioxid and is much more easily employed. That form of apparatus should be used which is the simplest in construction and has the least cubical content.

The measurement of the evolved gas is most simply made by collecting over lye in an azotometer, reading the volume, noting the reading of the barometer and thermometer and then reducing to standard conditions of pressure and temperature by the customary calculations. Where a very strong lye is used the tension of the aqueous vapor may be neglected. While every analyst should have a thorough knowledge of the ferrous chlorid method and the principles on which it is based it can not be compared in simplicity to the later methods with pure nitrates which are based on the conversion of the nitric acid into ammonia by the action of nascent hydrogen. In accuracy, moreover, it does not appear to have any marked advantage over the reduction methods.

470. Mercury and Sulfuric Acid Method.—This simple and accurate method of determining nitric acid in the absence of organic matter is known as the Crum-Frankland process.^{[\[301\]](#)}

The method rests on the principle of converting nitric acid into nitric oxid by the action of mercury in the presence of sulfuric acid. The operation as at first described is conducted in a glass jar eight inches long by one and a half inches in diameter filled with mercury and inverted in a trough containing the same liquid. The nitrate to be examined, in a solid form, is passed into the tube together with three cubic centimeters of water and five of sulfuric

acid. With occasional shaking, two hours are allowed for the disengagement of the gas, which is then measured.

471. Warington's Modification.—A graduated shaking tube is employed which allows the nitrate solution and oil of vitriol to be brought to a definite volume. The nitrate solution, with rinsings, is always two cubic centimeters and enough sulfuric acid is added to increase the volume to five cubic centimeters. The sulfuric acid should give no gas when shaken with distilled water. Any gas given off in the apparatus before shaking, is not expelled but is included in the final result. The persistent froth sometimes noticed where some kinds of organic matter are present, is reduced by the addition of a few drops of hot water through the stop-cock of the apparatus. The nitric oxid is finally measured in Frankland's modification of Regnault's apparatus.

This method, accurate for pure nitrates, unfortunately fails in the presence of any considerable amount of organic matter.

According to Warington's observations the presence of chlorids is no hindrance to the accurate determination of both nitric and nitrous acids by the mercury method. This simplifies the operation as carried on by Frankland who directs that any chlorin present, be removed before the determination of the nitric acid is commenced.

472. Noyes' Method.—In the analyses made by Noyes for the National Board of Health, the Crum-Frankland method was employed.^[302] The apparatus used was essentially that which is now known as Lunge's nitrometer and it will be described in the next paragraph. No correction is made by Noyes for the tension of aqueous vapor in the measurement of the nitric oxid because of the moderate dilution of the sulfuric acid by the liquid holding the nitric compounds in solution. The chlorin was not removed from the dry residue of the evaporated water as its presence in moderate quantity does not interfere with the accuracy of the process. In order to obtain the amount of nitrogen in the form of nitrates, the total volume of nitric oxid must be diminished by that due to nitrites present, which must be determined in a separate analysis. The method of manipulation is given in the following paragraph.

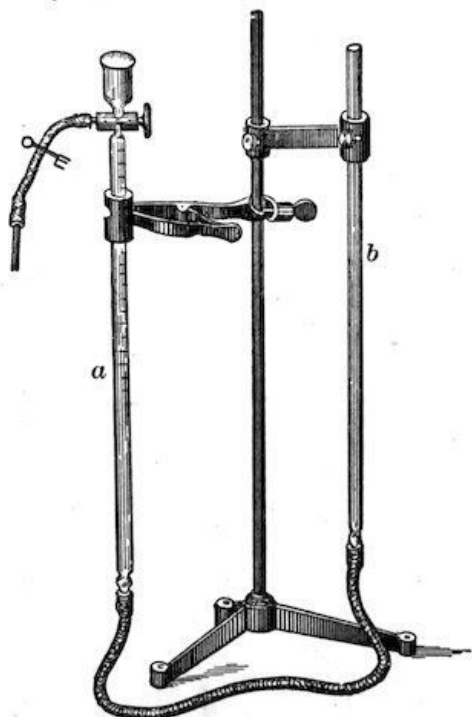


FIGURE 82. LUNGE'S NITROMETER.

473. Lunge's Nitrometer.—The apparatus employed by Noyes, in a somewhat more elaborate form, is known as Lunge's nitrometer.^[303] This apparatus is shown in Fig. 82. It consists of a burette, *a*, divided into one-fifth cubic centimeters. At its upper end it is expanded into a cup-shaped funnel attached by a three-way glass stop-cock. Below, the burette is joined to a plain tube, *b*, of similar size, by means of rubber tubing. The apparatus is first filled with mercury through the tube *b*, the stop-cock being so adjusted as to allow the mercury to fill the cup at the top of *a*. The cock is then turned until the mercury in the cup flows out through the side tube carrying the rubber tube and clamp. The three-way cock is closed, and the solution containing the nitrate placed in the cup. By lowering the tube *b* and opening the cock the liquid is carefully passed into *a*, being careful to close the cock before all the liquid has passed out of the cup. By repeated rinsings with pure concentrated sulfuric acid, every particle of the nitric compound is finally introduced into *a*, together with a large excess of sulfuric acid. The total volume of the introduced liquid should not exceed ten cubic centimeters. The mixture

of the mercury, nitric compound, and sulfuric acid is effected by detaching *a* from its support, compressing the rubber connection between *a* and *b*, placing *a* nearly in a horizontal position, and quickly bringing it into a vertical position with vigorous shaking.

After about five minutes the reaction is complete, and the level of the liquids in the two tubes is so adjusted as to compensate for the difference in specific gravity between the acid mixture in *a* and the mercury in *b*; in other words, the mercury column in *b* should stand above the mercury column in *a* one-seventh of the length of the acid mixture in *a*. This secures atmospheric pressure on the nitric oxide which has been collected in *a*. The measured volume of nitric oxide should be reduced to 0° and 760 millimeters barometric pressure. Each cubic centimeter of nitric oxide thus obtained corresponds to 1.343 milligrams NO; 2.417 milligrams N₂O₅; 4.521 milligrams KNO₃; 1.701 milligrams N₂O₃; 2.820 milligrams HNO₃; and 3.805 milligrams NaNO₃.

474. Lunge's Improved Apparatus.—Lunge has lately improved his apparatus for generating and measuring gases and extended its applicability.^[304] The part of it designed to measure the volume of a gas is the same in all cases. For generating the gas, the apparatus varies according to the character of the substance under examination.

The measuring apparatus is shown in Fig. 83. It is composed essentially of three tubes, conveniently mounted on a wooden holder with a box base for securing any spilled mercury. The support is not shown in the illustration.

The tubes A, B, C, are mutually connected by means of a three-way tube and rubber tubing with very thick walls to safely hold the mercury without expansion. In the middle of the measuring tube A, is found a bulb of seventy cubic centimeters capacity. Above and below the

bulb the tube is divided into tenths of a cubic centimeter, and its diameter is such, viz., 11.3 millimeters, that each cubic centimeter occupies a length of one centimeter. The upper end of A is closed with a glass cock with two oblique perforations, by means of which communication can be established at will, either through *e* with the apparatus for generating the gas, or through *d* with the absorption apparatus, or the opening be completely closed.

The volume of air under the observed conditions which would measure exactly 100 cubic centimeters at 0° and 760 millimeters pressure of mercury, is calculated by the formula

$$V = \frac{100(273 + t)760}{273(b - f)},$$

where *t* equals observed temperature, *b* the barometric pressure less the correction noted above and *f* the tension of the vapor of water under existing conditions. For example:

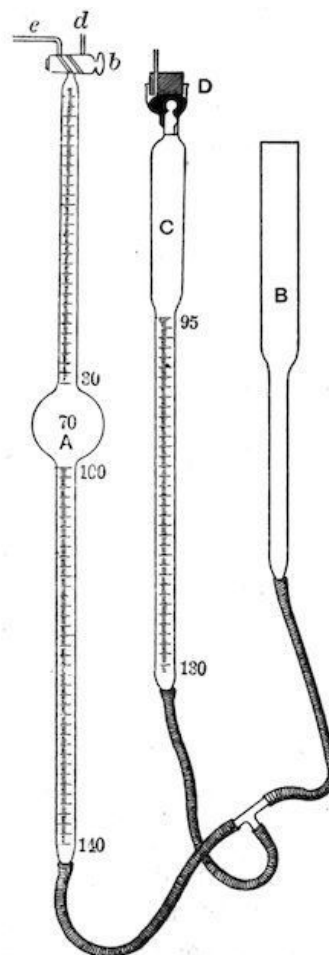


FIGURE. 83.

LUNGE'S IMPROVED APPARATUS.

Let the temperature be	18°
Barometric reading	755
Correction for <i>t</i>	2
Corrected barometer	753
Vapor of water tension	16

$$\text{Then } V = \frac{100(273 + 18)760}{273(753 - 16)} = 109.9.$$

This indicates that 109.9 cubic centimeters of air would occupy a volume of 100 cubic centimeters when subjected to standard conditions.

The tubes A, B, and C are filled with mercury of which about two and a half kilograms will be required. By means of the leveling tube B, the stopper in C being opened, the mercury in C is brought exactly to 109.9 cubic centimeters. The stopper in C is then closed, mercury poured into D, which is then closed with a rubber stopper, carrying a small glass tube as indicated in the figure.

The leveling tube B serves to regulate the pressure on the gas in A and this is secured by depressing or elevating it as the case may require.

The tube for reducing the volume to standard conditions of temperature and pressure, viz., 0° and 760 millimeters of mercury, is shown in C. In its narrow part which has the same internal diameter as A it is graduated into tenths of a cubic centimeter. The upper end of C is furnished with a heavy glass neck D surmounted by a glass cup. In the neck is placed a ground-glass stopper, carrying a groove below, which corresponds to a similar groove above in the side of the neck whereby communication can be established at will between the interior of C and the exterior. The joint is also sealed by pouring mercury into D as is shown in the figure. When the stopper is well ground and greased the reduction tube may be raised or lowered as much as may be necessary without any danger of escape or entrance of gas. To determine the position of the reduction tube C the reading of the barometer and thermometer at room temperature is taken. From the reading of the barometer subtract one millimeter if the temperature be below 12°, two millimeters at a temperature from 12° to 19°, three from 20° to 25°, and four above 25°.

When a gas has been introduced into the measuring tube A it is brought to the volume which it would assume under standard conditions by adjusting the tube C in such a way as to bring the level of mercury in C and A to the same point and the level of the mercury in C is exactly at 100 cubic centimeters. The gas in A is then at the volume which it would occupy under standard conditions and this volume can be directly read. This adjustment is secured by moving the tubes B and C up or down. If gases are to be measured wet, a drop of water should be put on the side of the upper part of C, and, if dry, of sulfuric acid, before the adjustment for temperature and pressure.

475. Method of Manipulation.—By the action of mercury in the presence of sulfuric acid, the nitrogen in nitrates, nitrites, nitrosulfates, nitroses, nitrocellulose, nitroglycerol, and the greater number of explosives, may be obtained and measured as nitric oxid. The nitrogen compounds are decomposed in the apparatus shown in Fig. 84.

To make an analysis, the apparatus is filled with mercury, through F, until the two openings in the cock and *i* are entirely occupied with that liquid. The cock *h* is then closed, and the nitrogen compound, in solution, introduced through *g*, care being taken that no air enters *g* when F is depressed and *h* opened to admit the sample. The funnel *g* is washed several times with a few drops of sulfuric acid, which are successively introduced into G. The total liquid introduced should not exceed ten to fifteen cubic centimeters, of which the greater part should be sulfuric acid. The rubber tube connecting G and F is carefully closed with a clamp and G violently shaken for a few minutes until no further evolution of nitric oxid takes place. In shaking, the apparatus should be so held as to prevent the escape of the mercury from the small tube *i* by keeping it closed with the finger or drawing over it a rubber cap.

After the evolution of the gas has ceased, the tube *e*, Fig. 83, is brought into contact with *i*, Fig. 84, and the two are joined by a tight-fitting piece of rubber tubing in such a way as to exclude any particle of air. The tube F, Fig. 84, is lifted and B and C, Fig. 83, depressed. On

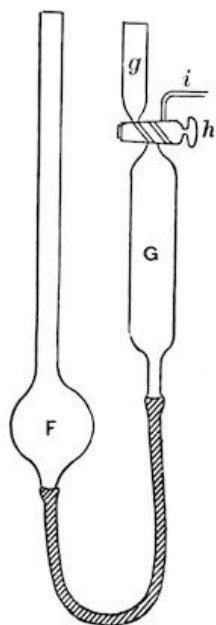


FIGURE 84.

LUNGE'S ANALYTIC
APPARATUS.

carefully opening the cocks *h* and *b* and bringing *i* and *e* into union, the gas is passed from *G* into *A*. When all the gas has entered *A* and the acid mixture from *G* has reached *b* the latter is closed, and also *h*. The apparatus *G* is disconnected and removed. The gas in *A* is then reduced to normal conditions by manipulating the reduction tube *C* in the manner already described.

The gas in *A* is measured dry by reason of having been generated in presence of rather strong sulfuric acid. Consequently, for this operation the adjustment of the volume of gas in *C* should be made in contact with a drop of strong sulfuric acid. In order to make the readings, a quantity of material must be taken which will give less than thirty or from 100 to 140 cubic centimeters of nitric oxid.

The quantities of the different compounds of nitric acid corresponding to the number of cubic centimeters of nitric oxid, measured under standard conditions, are shown in the following table:

Cubic centimeters of NO.	Weight in milligrams.	CORRESPONDING TO		
		N ₂ O ₃ in milligrams.	HNO ₃ in milligrams.	NaNO ₃ in milligrams.
1	1.343	1.701	2.820	3.805
2	2.682	3.402	5.640	7.610
3	4.029	5.103	8.460	11.415
4	5.372	6.804	11.280	15.220
5	6.715	8.506	14.100	19.025
6	8.058	10.206	16.920	22.830
7	9.401	11.907	19.740	26.635
8	10.744	13.608	22.560	30.440
9	12.087	15.309	25.380	34.245

476. Utility of the Method.—Where it is desirable that the nitric oxid method be used, and at the same time heating be avoided, the decomposition of a nitrate by means of metallic mercury and sulfuric acid affords a convenient and accurate procedure. But, as a rule, there is no objection to the application of the lamp, and in such cases the mercury method appears to have no advantage over the ferrous chlorid process. Nevertheless, in the hands of a skilled worker the results are reliable, and the process is a quicker one, on the whole, than by distillation with ferrous chlorid and hydrochloric acid. This method, however, can not be recommended as in any way superior to the reduction methods to be hereinafter described.

ESTIMATION OF NITRIC ACID BY OXIDATION OF A COLORED SOLUTION.

477. Method of Boussingault.—The process for the estimation of nitric acid by the decoloration of a solution of indigo is due originally to Boussingault.^[305] In this method the extract, obtained by washing slowly 200 grams of soil until the filtrate amounts to 300 cubic centimeters, is evaporated until its volume is no greater than two or three cubic centimeters, and it is transferred to a test-tube, with washings, and again evaporated in the tube until the volume is not greater than that last mentioned. A few drops of solution of indigo are added, and then two cubic centimeters of pure hydrochloric acid; the whole is then heated. As the color of the indigo disappears more is added. When the color ceases to fade, the liquid in the test-tube is concentrated by boiling. If concentration fail to destroy the blue or green color, another one-half cubic centimeter of hydrochloric acid is introduced. The reaction is completed when neither concentration nor fresh addition of hydrochloric acid destroys the excess of indigo present. The color produced by a small excess of indigo is a bright sap-green; this tint is the final reaction sought. The small excess of indigo necessary to produce a green color is deducted in every experiment.

When more than mere traces of organic matter are present, Boussingault advises that the nitric acid be first separated by distillation and then reduced by the indigo solution. For this purpose the concentrated solution of the nitrate, two or three cubic centimeters, is placed in a small tubulated retort with two grams of manganese dioxid in fine powder. The retort is next half filled with fragments of broken glass, over which is poured one cubic centimeter of concentrated sulfuric acid. The retort is heated carefully by means of a small flame, which is kept in motion so as to successively come in contact with all parts of the bottom of the retort. The distillate is received in a graduated test-tube which is kept cool. The distillation is continued until the vapors of sulfuric acid begin to appear. The apparatus is allowed to cool, the stopper of the retort removed, two cubic centimeters of water introduced, and the distillation again made until fumes of sulfuric acid are again seen. The distillation with water is made twice in order to remove every trace of nitric acid from the retort. The distillate is neutralized with a solution of potassium hydroxid and concentrated to two cubic centimeters, and the nitric acid estimated in the manner already described. The manganese dioxid used should be previously well washed and the sulfuric must be free of nitric acid.

Preparation of the Indigo Solution.—Fifty grams of indigo in fine powder are digested for twenty-four hours, at 40°, in a liter of distilled water. The water is then poured off and replaced with a fresh supply. After the second decantation the residue is treated with 750 cubic centimeters of equal parts of water and pure concentrated hydrochloric acid and boiled for an hour. After cooling, the undissolved portion is collected on a filter and washed at first with hot, and afterwards with cold water, until the filtrate is no longer colored and is free of acid. The dried residue is treated with ether under a bell-jar, or in a continuous extraction apparatus, until the ether is only of a faint blue tint. The fifty grams of indigo at first taken will give about twenty-five grams of the purified article, which, however, will still leave a little ash on combustion.

Solution in Sulfuric Acid.—Five grams of the purified indigo are placed in a flask having a ground-glass stopper, treated with twenty-five grams of fuming sulfuric acid, and allowed to digest two or three days at a temperature of from 50° to 60°. From seventy to 200 drops of the solution thus made are placed in 100 cubic centimeters of water for use in the process.

Standardization of the Indigo Solution.—The solution as prepared above is standardized by a solution of one gram of pure potassium nitrate in 1,000 cubic centimeters of distilled water. The oxidation of the indigo solution is accomplished as described above. For this strength of standard nitrate solution two cubic centimeters are taken corresponding to two milligrams of potassium nitrate. The indigo solution for this strength should have only twenty drops of the sulfuric acid solution of indigo to 100 cubic centimeters of water. If twenty grams of potassium nitrate are taken for 1,000 cubic centimeters of the standard solution then 200 drops of the sulfindigotic acid should be used to 100 cubic centimeters of water.

478. Method of Marx.—As usually practiced, the indigo method is conducted according to the variation described by Marx.^[306] There are required for the process the following reagents and apparatus:

- a. A solution of pure potassium nitrate containing 1.8724 grams per liter. One cubic centimeter of the solution is equivalent to one milligram of nitric anhydrid (N_2O_5).
- b. A solution of the best indigo carmine in water which should be approximately standardized by solution in the manner described hereafter, and then diluted so that six to eight cubic centimeters equal one milligram of nitric acid.
- c. Chemically pure sulfuric acid of specific gravity 1.842, perfectly free from sulfurous and arsenious acids and nitrogen oxids.
- d. Several thin flasks of about 200 cubic centimeters capacity.
- e. A small cylindrical measure holding fifty cubic centimeters and divided into cubic centimeters.
- f. A Mohr's burette divided into tenths of a cubic centimeter.
- g. A twenty-five cubic centimeter pipette or another burette.
- h. A five cubic centimeter pipette divided into cubic centimeters or half cubic centimeters.
- i. A measuring flask of 250 cubic centimeters capacity.

Preliminary Trial.—Twenty-five cubic centimeters of the sample are transferred to a flask; the fifty cubic centimeter measure is filled with sulfuric acid and the burette with indigo solution. The sulfuric acid is added to the sample all at once, shaken for a moment, and the indigo run in as quickly as possible with shaking until a permanent greenish tint is produced. If the sample do not require more than twenty cubic centimeters of indigo solution of the above strength, it can be titrated directly, otherwise it must be diluted with a proper quantity of pure water, and subjected again to the preliminary trial.

The Actual Titration.—(1) Twenty-five cubic centimeters of the sample properly diluted if necessary, are measured and poured into a flask, and as much indigo as was used in the preliminary trial, is added; a quantity of sulfuric acid, equal in volume to the liquid in the flask, is added all at once, the mixture shaken, and indigo solution run in quickly out of the burette until the liquid remains permanently of a greenish tint.

(2) The last experiment is repeated as often as may be necessary adding to the water at first half a cubic centimeter less indigo than the total quantity used previously, afterwards

proceeding as in (1) until the final test shows too little indigo used.

(3) From the rough titration of the indigo, calculate the amount of potassium nitrate solution corresponding with the indigo solution used in (2), multiply the result by ten, transfer this quantity of the standard nitrate solution to a 250 cubic centimeter flask, fill with pure water to the mark, and titrate twenty-five cubic centimeters of this fluid with indigo as in (2). If the quantity of indigo solution used is nearly the same as that required in (2), its exact value may be calculated, but if it is not, another nitrate solution may be made up in the 250 cubic centimeter flask, more closely resembling the sample in strength, and the titration with the indigo solution must be repeated.

(4) If the water contain any considerable amount of organic matter, it must first be destroyed by potassium permanganate. In this case, the estimation of the organic matter and nitric acid may be conveniently combined.

The use of permanganate in the above case is likely to introduce an error as has been shown by Warington. The method therefore can not be recommended in the presence of organic matter.

479. Method of Warington.—The modification of the indigo method as used by Warington, applicable only in absence of organic matter, is the one chiefly employed in England.^[307]

Instead of the ordinary indigo of commerce, indigotin is used. The normal solution of the coloring matter is made of such a strength as to be equivalent to a solution of potassium nitrate containing 0.14 gram of nitrogen per liter. Where large quantities of the coloring matter are to be used it is advisable to prepare it about four times the strength given above and then dilute it as required. Four grams of sublimed indigotin will furnish more than two liters of the color solution.

The solution is prepared as follows:

Four grams of indigotin are digested for a few hours with five times that weight of Nordhausen sulfuric acid, diluted with water, filtered, and made up to a volume of two liters. The strength of the indigotin solution is determined with a solution of potassium nitrate of the strength mentioned above. The process is performed as follows:

From ten to twenty cubic centimeters of the standard nitrate solution are placed in a wide-mouthed flask of about 150 cubic centimeters capacity. A portion of the indigotin solution is next added, such as will be deemed sufficient for the process, and the whole is well mixed. Strong sulfuric acid is next measured out from a burette into a test-tube, in volume equal to the united volumes of the nitrate solution and indigotin. The whole of the sulfuric acid is then poured as quickly as possible, into the solution in the flask and rapidly mixed, and the flask transferred to a calcium chlorid bath, the temperature of which should be maintained at 140°. It is essential to the success of the operation that the sulfuric acid should be mixed with the greatest rapidity. It should be poured in at once and the whole well shaken without waiting for the test-tube containing the acid, to drain. The flask should be covered by a watch-glass while it is held in the bath. As soon as the larger part of the indigotin is oxidized the flask in the bath should be gently rotated. With very weak solutions of nitrate it may be necessary sometimes to keep the flask in the bath for five minutes. When the indigo color is quickly discharged it shows the presence of nitric acid in considerable excess and a considerably larger quantity of indigo must be taken in the next experiment. The experiments are continued until just the quantity of indigo necessary to consume the nitric acid is taken, the amount of indigo being in very slight excess, not exceeding one-tenth cubic centimeter of the indigo solution used. The

tint produced by the small excess of indigo remaining is best seen by filling the flask with water. On substances of approximately known strength about four experiments are usually necessary to determine the amount of indigo to be taken, but with unknown substances a larger number may be necessary.

Usually in determinations of this kind it is directed to use double the volume of sulfuric acid mentioned above. In this case not only is the quantity of indigo oxidized much greater than with a smaller portion of acid, but the prejudicial effect of organic matter is also greater than when the smaller quantity of acid is employed.

An indigo solution standardized as above is strictly to be used for a solution of nitrate of the strength employed during the standardization. The quantity of indigo oxidized in proportion to the nitric acid present diminishes as the nitrate solution becomes more dilute. Instead of determining this during each series of experiments it may be estimated once for all and a table of corrections used.

The following table is based upon experimental determinations:

Strength of niter solution used.		Indigo required, cubic centimeters.	Difference between amounts of indigo, cubic centimeters.	Nitrogen corresponding to one cubic centimeter of indigo, gram.	Difference between the nitrogen values, gram.	Difference in the nitrogen values for a difference of one cubic centimeter in the amount of indigo, gram.
$\frac{8}{64}$	Normal	10.00		0.000035000		
$\frac{7}{64}$	„	8.71	1.29	0.000035161	0.000000161	0.000000125
$\frac{6}{64}$	„	7.43	1.28	0.000035330	0.000000169	0.000000132
$\frac{5}{64}$	„	6.14	1.29	0.000035627	0.000000298	0.000000231
$\frac{4}{64}$	„	4.86	1.28	0.000036008	0.000000381	0.000000298
$\frac{3}{64}$	„	3.57	1.29	0.000036763	0.000000756	0.000000586
$\frac{2}{64}$	„	2.29	1.28	0.000038209	0.000001445	0.000001129
$\frac{1}{64}$	„	1.00	1.29	0.000043750	0.000005541	0.000004295

The table is used as follows:

Suppose that twenty cubic centimeters of water under examination have required 5.36 cubic centimeters of indigo solution for the oxidation of the nitric acid contained therein. By inspection of the table it is seen that this number is five-tenths cubic centimeter above the nearest quantity given; viz., 4.86 cubic centimeters. From the last column in the table it is found that the correction for five-tenths cubic centimeter of indigo solution is 0.000000149 cubic centimeter, being half that for the one cubic centimeter given in the table. This is to be subtracted from the unit value in nitrogen given in the first “gram” column of the table; viz., 0.000036008. It is thus seen that the 5.86 cubic centimeters of indigo solution are equivalent to 0.000035859 gram of nitrogen per cubic centimeter. The water under examination, therefore, contains nine and six-tenths parts of nitrogen as nitric acid per million.

Attention must also be paid in standardizing indigo solutions to the initial temperature of the solutions. A rise in the initial temperature will be attended by a diminution in the quantity of indigo oxidized. Experiments with a room temperature of 10° and a room temperature of 20°, being the initial temperatures of the experiments, showed that at the higher temperature the amount of indigo consumed was about five per cent less when the strong solutions of nitrate

were employed. The indigo solution should, therefore, be standardized at the same temperature at which the determinations are made.

If twenty cubic centimeters of the standard nitrate solution employed be used in setting the indigo solution, this standard will enable the operator to determine nitric acid up to 17.5 parts of nitrogen per million in water or soil extracts.

The presence of an abundance of chlorids in the water under examination tends to diminish the content of nitric acid found, and also tends to introduce an error, which is sometimes of a plus and sometimes of a minus quantity, according to the strength of the nitric acid present. The reaction is shortened in weak solutions by the presence of chlorids, and the quantity of indigo consumed is consequently increased. The error introduced by chlorids is usually of an insignificant nature.

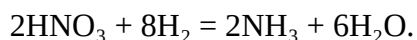
On account of the interference of organic matters with the reaction of indigo it is not of much use in the examination of nitrates washed out of soils, although in some cases the results may be quite accurate. This method must, therefore, be considered as applicable, in general, to waters or soil extracts which contain little or no organic matter.

In analytical work pertaining particularly to agriculture, the use of the indigo method for determining nitric acid has been largely employed, both in the analyses of soil extracts and drainage and irrigation waters. The method, however, can hardly survive as an important one in such work in competition with more modern and speedy processes of analysis.

DETERMINATION OF NITRIC NITROGEN BY REDUCTION TO AMMONIA.

480. Classification of Methods.—When nitrogen is present in a highly oxidized state, *e. g.*, as nitric acid, it may be quickly and accurately estimated by reduction to ammonia. This action is effected by the reducing power of nascent hydrogen, and this substance may be secured in the active state by the action of an acid or alkali on a metal, or by means of an electric current. The processes depending on the use of a finely divided metal in the presence of an acid or alkali have come into general use within a few years, and are now employed generally instead of the more elaborate estimations depending on the use of copper oxid or indigo.

The typical reaction which takes place in all cases is represented by the following equation:



The method will be considered under three heads; *viz.*, 1. Reduction in an alkaline solution; 2. Reduction in an acid solution; 3. Reduction by means of an electric current.

In the first class of processes the reduction and distillation may go on together. In the second class the reduction is accomplished first and the distillation effected afterwards, with the addition of an alkali. In the third class of operations the reduction is accomplished by means of an electric current and the ammonia subsequently obtained by distillation, or determined by nesslerizing. These processes may be applied to rain and drainage waters, and to soil extracts. On account of the ease with which the analyses are accomplished, the short time required and the accuracy of the results, the reduction methods for nitrates have already commended themselves to analysts, and are quite likely to supersede all others for practical use where weighable quantities of nitrates are present. For the minute traces of nitrates found in rain and drainage waters, and in some soil extracts, the reduction method may also be applied, but in these cases the ammonia which is formed must be determined by colorimetry (nesslerizing) and not by distillation. The processes about to be described are especially applicable to the examination of soils and waters rich in nitrates.

REDUCTION IN ALKALINE SOLUTIONS.

481. Provisional Method of the Association of Official Agricultural Chemists.^[308]—

Extraction of the Nitrates.—Place one kilogram of the dried soil, calculated to water-free substance, on a percolator of glass or tin. Moisten the soil thoroughly with pure distilled water, and allow to stand for half an hour. Add fresh portions of pure distilled water until the filtrate secured amounts to one liter. If the first filtrate be cloudy before use it may be refiltered.

Qualitative Test for Nitrates.—Evaporate five cubic centimeters of the soil extract in a porcelain crucible, having first dissolved a small quantity of pure brucin sulfate therein. When dry, add to the residue a drop of concentrated sulfuric acid free of nitrates. If the nitrate calculated as potassium nitrate does not exceed the two-thousandth part of a milligram only a pink color will be developed; with the three-thousandth part of a milligram a pink color with reddish lines; with the four-thousandth part of a milligram a reddish color; with the five-thousandth part of a milligram a distinct red color.

Estimation of the Nitrates.—Evaporate 100 cubic centimeters of the soil extract to dryness on a steam-bath. Dissolve the soluble portions of the residue in 100 cubic centimeters of ammonia-free distilled water, filtering out any insoluble residue. Place the solution in a flask, add ten cubic centimeters of sodium amalgam, stopper the flask with a valve which will permit the escape of hydrogen, and allow to stand in a cool room for twenty-four hours. Add fifty cubic centimeters of milk of lime and titrate the ammonia produced by distillation, with standard acid and estimate as nitrogen pentoxid. Where the amount of ammonia is small, nesslerizing may be substituted for titration.

Preparation of Sodium Amalgam.—Place 100 cubic centimeters of mercury in a flask of half a liter capacity; warm until paraffin will remain melted over the surface; drop successively in the paraffin-covered mercury, pieces of metallic sodium of the size of a pea until 6.75 grams have united with the mercury. The amalgam contains then 0.5 per cent of metallic sodium and may be preserved indefinitely under the covering of paraffin.

482. Method of the Experiment Station at Möckern.^[309]—The principle of this reaction is based on the reducing action exercised by nascent hydrogen on a nitrate, the hydrogen being generated by the action of soda-lye on a mixture of zinc dust and finely divided iron.

Ten grams of nitrate are dissolved in 500 cubic centimeters of water. Of this solution twenty-five cubic centimeters, corresponding to one-half gram, are placed in a distillation flask of about 400 cubic centimeters capacity, 120 cubic centimeters of water added, and about five grams of well-washed and dried zinc dust and an equal weight of reduced iron. To the solution are added eighty cubic centimeters of soda-lye of 32° B. The flask is then connected with the condensing apparatus and the distillation carried on synchronously with the reduction, the ammonia being collected in twenty cubic centimeters of titrated sulfuric acid. The distillation is continued from one to two hours, or until 100 cubic centimeters have been distilled, and the remaining sulfuric acid is titrated in the usual way. Soil extracts and sewage waters should be concentrated until they have approximately the proportion of nitrates given above.

483. Method of Devarda.—The inconvenience due to slow action and other causes, arising from the use of pure metals in the reduction of nitrates to ammonia, has been overcome, to some extent, by Devarda, by use of an alloy, in a state of fine powder, consisting of aluminum,

copper, and zinc.^[310] The alloy consists of forty-five per cent of aluminum, fifty per cent of copper, and five per cent of zinc. In dissolving, the copper is left in a finely divided state, which is a great help in distillation in preventing bumping.

The analytical process is carried out as follows: The solution containing the nitrate, in quantity equivalent to about one-half gram of potassium nitrate, is placed in a flask having a capacity of about one liter, and diluted with sixty cubic centimeters of water and five cubic centimeters of alcohol, and then forty cubic centimeters of caustic potash solution added of specific gravity one and three-tenths. From two to two and one-half grams of the alloy, described above, are introduced, and the flask attached to a condenser with a receiver containing standard acid. The connection between the flask and the condenser is made by means of a tube having on the limb next the flask a bulb filled with glass beads to prevent the contents of the flask splashing over into the receiver, and on the other limb another bulb to prevent the acid in the receiver finding its way into the distillation flask, should regurgitation occur. When the flask has been thus connected with the condenser it is gently heated for half an hour, at the end of which time the evolution of hydrogen will have slackened or ceased, and then the distillation is begun, at first cautiously, until the zinc of the alloy has completely dissolved, and then more vigorously, the time necessary being about twenty minutes from the time when the contents of the flask begin to boil. The distillate is caught in standard acid and the ammonia determined by backward titration in the ordinary way. It is to be noted that the strength of the alkali used is of importance, as if it be too strong, the action on the alloy is unduly vigorous at the beginning of the operation, and if too weak, the contents of the flask have to be heated overmuch, the result in both cases being the formation of a fine spray of caustic solution, which is very difficult to stop, even with complicated washing attachments to the distilling flask. The test analyses on pure nitrates are satisfactory. This method has been used with satisfaction in the laboratory of the Department of Agriculture, but does not appear to have any special advantage over the process of Ulsch, to be described further on.

484. Variation of Stoklassa.—Stoklassa has subjected the method of Devarda to a comparative test with the following methods:^[311]

1. Wagner's Schloesing-Grandeau method.
2. Lunge's nitrometer method.
3. Stutzer's method.

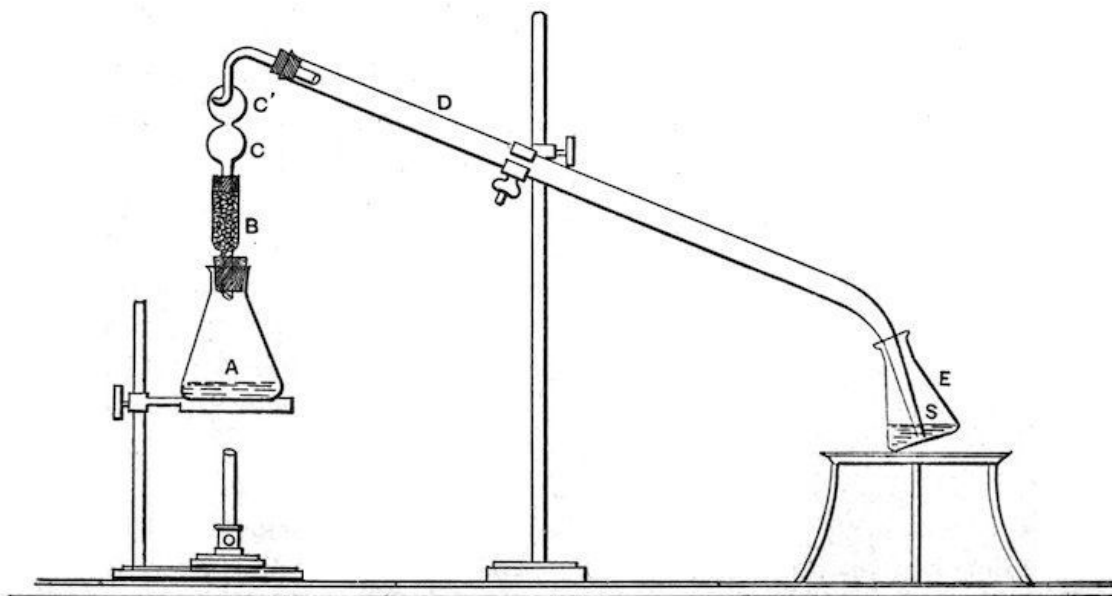


FIGURE 85. STOKLASSA'S NITRIC ACID APPARATUS.

The reduction takes place in a copper erlenmeyer, A, Fig. 85, in which, in addition to the solution containing the nitrate, are placed 200 cubic centimeters of water, forty cubic centimeters of potassium hydroxid solution of 33° B., five cubic centimeters of alcohol, and finally two and one-half grams of the finely powdered Devarda alloy. The distillate passes through a tube, B, filled with glass pearls and into the condenser D, through the bulbs, C C'.

After the flask is connected with the distilling apparatus, it is gently warmed and the reduction is ended in about twenty minutes. The ammonia which is formed is then distilled into E, containing the standard acid, S, requiring about twenty minutes more. The comparative results given, show that the Devarda method is equally as accurate as any of the other methods mentioned, giving practically theoretical results.

In so far, however, as speed of an analysis, is concerned the first place is awarded to the Lunge nitrometer method, with which a complete analysis can be made in from thirty to forty minutes. In the second rank, so far as speed is concerned, the Devarda method is recommended. All the methods give accurate results.

485. Method of Sievert.^[312]—Two grams of potassium or sodium nitrate are dissolved and made up to 1,000 cubic centimeters. Fifty cubic centimeters of the solution are placed in a 600 cubic centimeter flask and diluted with fifty cubic centimeters of water, and from eighteen to twenty grams of caustic alkali added. After the alkali is dissolved, seventy-five cubic centimeters of ninety-six per cent alcohol are added and a few pieces of bone-black to prevent foaming. From ten to fifteen grams of zinc or iron dust are then added to the flask which is closed and connected with a tube holding about 200 cubic centimeters, which contains about ten cubic centimeters of normal sulfuric acid. This tube is kept cool by being immersed in water. The whole mixture is now allowed to stand for three or four hours and then the alcohol is distilled slowly and the ammonia formed by the reduction of the nitrates is carried over with it. The distillation lasts for about two hours. The contents of the tube are carefully rinsed into a dish and the excess of sulfuric acid titrated with one-fourth normal soda-lye.

For soil extracts and substances containing unknown quantities of nitric acid, a preliminary test will indicate approximately the amount thereof, and this will be an indication for the quantity to be used in the analysis.

The method of Stutzer differs from the foregoing in the substitution of aluminum dust instead of iron or zinc.^[313] The reducing power of aluminum, however, varies greatly according to the method in which the metal has been prepared. Pure aluminum prepared by the electric method, reduces the nitric acid much less vigorously than the metal prepared by the older methods of fusion with sodium. For this reason the method of Stutzer is not to be preferred to that of Sievert.

REDUCTION IN AN ACID SOLUTION.

486. Variation of the Sodium Amalgam Process.—This method is described by Monnier and Auriol.^[314]

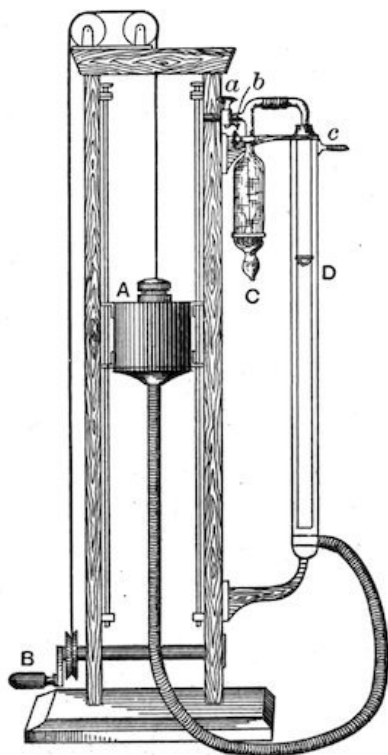


FIGURE 86. VARIATION OF THE SODIUM AMALGAM PROCESS.

The principle of the operation depends on the reduction of the dissolved nitrate by titrated sodium amalgam in presence of an acid, and the estimation of the quantity of nitric acid present from the deficit in the volume of hydrogen. The apparatus employed is conveniently mounted as shown in Fig. 86. The brass vessel A is movable by means of the cord on the pulley B, in such a way as to be fixed at any required altitude. It is filled with water and connected by a rubber tube to the cooling tube D. Within the cooling tube there is a graduated cylinder open at its lower end. Its upper end is connected directly with the apparatus C. The cooling tube D has a small side tube, *c*, near its upper end, by means of which the air can enter or escape when the position of A is changed. The apparatus C, in which the reaction takes place, is a glass cylinder. Its upper end is continuous with the T tube provided with the stop-cocks *a* and *b*. One arm of the T permits connection with the graduated measuring tube by means of a rubber union. The lower end of C is closed with a large hollow ground-glass stopper, carrying a small receptacle within, so that it forms two separate water-tight compartments, open at the top.

The sodium amalgam is prepared as follows:

In a clay crucible are heated 400 grams of mercury, and, little by little, with constant stirring, four grams of dry sodium are added. When cold, the amalgam is placed in a burette, having a ground-glass stopper, and covered with petroleum. The strength of the amalgam is established in the following manner. A small glass thimble, ground even at the top, is filled with the amalgam and struck off even with a ground-glass straight edge. In this way the same quantity of amalgam is taken for each test. This measured portion of the amalgam is placed in the inner vessel of the glass stopper to C. Ten cubic centimeters of water, containing sixty centigrams of tartaric acid, are placed on the outer ring of the glass stopper, which is then inserted, well oiled, in C, closing it air- and water-tight. The tartaric acid solution also carries a piece of litmus paper, so that its constant acidity may be insured. The vessel A is then fixed in a position which brings the water in the graduated burette and tube D exactly to the 0 mark. The cock *a* is next closed, *b* opened, and C is inverted until all the amalgam is poured into the solution of tartaric acid. The evolved hydrogen mixed with the air contained in the apparatus, is passed into the graduated burette. After fifteen minutes, the reaction is ended. The water level within and without the graduated tube is restored and the volume of gas evolved noted and reduced by the usual tables to 0° and 760 millimeters pressure of the barometer.

An amalgam prepared as above will give about three cubic centimeters of hydrogen for each gram. The thimble should hold from twelve to fifteen grams.

The estimation of nitric acid should be made in a solution containing about one-tenth per cent of nitrate. Ten cubic centimeters are taken, to which six-tenths gram of tartaric acid is added, and placed in the outer part of the glass stopper. The rest of the process is conducted exactly as described above. The deficit in hydrogen is calculated to nitrogen pentoxid.

The reduction by sodium amalgam is not so convenient a form of estimating nitric acid as many of the other forms of using nascent hydrogen. As practiced by calculating from the deficit of hydrogen, however, it has some advantages by reason of the fact that no heating is required. The presence of organic neutral bodies, or even those of an acid nature, like humus, does not, therefore, interfere with the work. Likewise, mineral bodies in solution, which are not reduced by nascent hydrogen, do not interfere with the accuracy of the reaction.

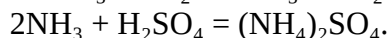
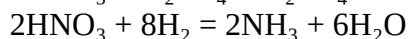
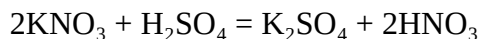
487. Method of Schmitt.—In the method of Schmitt forty cubic centimeters of glacial acetic acid are placed in a flask of 600 cubic centimeters content, and fifteen grams of a mixture of zinc and iron dust added.^[315] To this a quantity of the solution containing the nitrate, representing about half a gram of the pure nitrate, is added with constant shaking, in portions which do not evolve hydrogen too rapidly. After about fifteen minutes when the evolution of nitrogen has somewhat diminished, an additional fifteen grams of the metal dust are added. If the contents of the flask should become thick they can be diluted with thirty cubic centimeters of water. The reduction is complete in from thirty to forty minutes. The contents of the flask are now saturated with enough soda-lye not only to neutralize the excess of acetic acid, but to keep the zinc hydroxid also in solution. For this purpose about 200 cubic centimeters of soda-lye of 1.25 specific gravity are necessary. The ammonia is obtained by distillation into standard acid in the usual way.

488. Method of Ulsch.—In practice the method of Ulsch has come into general use.^[316]

For the determination of nitrogen by this method half a gram of saltpeter or four-tenths gram of sodium nitrate is taken and dissolved in twenty five cubic centimeters of water, in a flask with a content of about 600 cubic centimeters. Five grams of iron reduced by hydrogen, and ten cubic centimeters of sulfuric acid diluted with two volumes of water are then added to the flask. To avoid mechanical losses during the evolution of hydrogen a pear-shaped glass stopper is hung in the neck of the flask. After the first violent evolution of hydrogen has passed, the flask is slowly heated until in about four minutes it is brought to a gentle boil. The boiling is continued for about six minutes when the reduction is complete. About fifty cubic centimeters of water are then added; also an excess of soda-lye and a few particles of zinc and the ammonia is distilled and collected in standard acid in the usual way.

The method of Ulsch can also be applied, according to Fricke, to the analysis of nitrates contained in drinking and drainage waters, and it is regarded by him as one of the best methods to be employed in such investigations.^[317]

The method of Ulsch in this laboratory has given entirely satisfactory results, and is generally used in preference to other methods in cases where a considerable quantity of nitrates is present. It is based on the following reactions:



REDUCTION BY THE ELECTRIC CURRENT.

489. Method of Williams-Warington.—From the losses which naturally occur during the evaporation of water, even with all the precautions noted, Warington was led to try some method for the determination of nitrates and nitrites in waters without previous concentration. ^[318] The reduction of these bodies by the copper-zinc couple formed the basis of these experiments, and they resulted in the following method of manipulation, which is based on a process devised by Williams. ^[319]

The method consists in boiling rapidly one liter of the rain water in a retort, with a little magnesia previously raised to a low red heat and then washed, until 250 cubic centimeters have distilled over. The residue is then made up to 800 cubic centimeters, transferred to a wide-mouthed, stoppered bottle supplied with strips of copper and zinc forming electric couples, and set aside, at a constant temperature of from 21°–24°, for three days. A measured portion of the solution is then distilled, and the ammonia determined in the distillate by nesslerizing.

This plan has two advantages: First, the ammonia, as well as the nitrogen as nitrates and nitrites, can be determined in the course of the same operation and in the same sample of water. For this purpose it is only necessary to fit the retort to an efficient condenser and to remove all ammonia from the apparatus by boiling distilled water in the retort before introducing the rain water. The distillate of 250 cubic centimeters from the rain water, as described above, is well mixed and the ammonia determined, in from twenty-five to one hundred cubic centimeters thereof, diluted to 150 cubic centimeters with ammonia-free water. Second, the nitrogen, as nitrates and nitrites, is determined directly and alone; the error of the determination is as small as nesslerizing admits of, since it is possible, if necessary, to distill 600 cubic centimeters of the boiled rain water corresponding to 750 cubic centimeters of the original, and thus obtain a full amount of ammonia for determination, even when the rain water has been poor in nitrates.

The determination of nitric nitrogen, in a given sample, by the above method gave a mean quantity of product of 0.162 part per million, while the determination, in the same lot of samples, by the modified Schloesing method gave 0.125 part per million. This result confirms the supposition that in the complete evaporation necessary to the manipulation of the Schloesing method there is a loss of nitrogen. The amount of nitrogen as nitrates and nitrites in the rain water at Rothamstead, for the twelve months ending April 1, 1888, was found, by the Schloesing method, to be 0.614 pound per acre, the total rain-fall being 21.96 inches. For the year ending April 1, 1889, by the copper-zinc method, it amounted to 0.917 pound per acre, the total rain-fall being 29.27 inches.

The amounts found in other localities are quite different from the above, as for instance, the mean of seven stations in Germany for thirteen years, beginning in 1864, showed 10.18 pounds of nitrogen per acre. The average amount for ten years at the observatory of Mont Sauris, near Paris, showed 12.36 pounds of nitrogen per acre. The average for three years at Lincoln, as determined by Professor G. Gray, shows one and six-tenths pounds of nitrogen per acre per annum. At Tokio, in Japan, Kellner found, for one year, 1.02 pounds per acre.

490. Determination of the Ammonia.—The method used at Rothamstead is to make one determination of ammonia in the whole of the distillate obtained, the strength of which is regulated by varying the amount introduced into the retort, so that it shall be equal to about two

cubic centimeters of the standard ammonia solution. A 150 cubic centimeter cylinder is first filled with the rain water, and fifty cubic centimeters of nessler reagent added. The depth of tint indicates what quantity of rain water will be required for distillation. This having been determined, the appropriate volume of the rain water, provided it do not exceed 600 cubic centimeters, is placed in the retort described above, and the distillation continued until the 150 cubic centimeter cylinder is filled. The titration is made in the usual way.

491. Preparation of the Copper-Zinc Couple.—For 800 cubic centimeters of boiled rain water, prepared as described, six strips of zinc foil, four inches long by one and a quarter inches wide, are taken and bent at right angles along their center to obtain stiffness. The couple is cleansed and coated by washing in a series of five beakers containing, respectively, dilute solution of sodium hydroxid, very dilute sulfuric acid, a three per cent solution of copper sulfate, ordinary distilled water, and distilled water free from ammonia. Through these five beakers the zinc foil is successively passed. It is rinsed both after the alkali and the acid. But after the copper has been deposited, the strips are simply drained and carefully placed in the distilled water, it being difficult to rinse without removing the copper. The couples should be entirely submerged when placed in the rain water. The strips should remain in the copper sulfate solution long enough to be well covered with copper.

492. Substitution of an Aluminum-Mercury Couple for Copper-Zinc.—Ormandy and Cohen have proposed to use an aluminum-mercury couple for the copper-zinc in the process described above.^[320]

This couple acts more quickly than the copper-zinc, and the results are equally as accurate. Nitrites are reduced in about one hour by this apparatus, while the zinc-copper couple of Gladstone and Tribe requires about six times as long. Aluminum foil, free of grease, should be used. The foil should be heated over a bunsen just before amalgamation. The clean, very thin foil is coated with mercury by shaking with a concentrated solution of mercuric chlorid. It should be prepared immediately before use.

The amalgamated foil is introduced into the sample of water to be analyzed, and left until all the aluminum is converted into oxid. The presence of the oxid favors the prevention of bumping during the subsequent distillation. The distilled ammonia, collected in dilute acid, is determined by nesslerizing, the free ammonia in the sample having been previously determined. The increase in ammonia is due to nitrates or nitrites reduced by the couple.

IODOMETRIC ESTIMATION OF NITRIC ACID IN NITRATES.

493. Method of De Koninck and Nihoul.—This process is applicable only in the absence of organic bodies and other reducing agents.

The principle on which it rests, as applied by McGowan, is as follows:^[321]

When a fairly concentrated solution of a nitrate is warmed with an excess of pure, strong hydrochloric acid, the nitrate is completely decomposed, and the production of nitrosyl chlorid and chlorin is quantitative. The reaction, as shown by Tilden, is represented by the following equation:^[322]



One molecule of nitric acid thus yields two atoms of chlorin and one molecule of nitrosyl chlorid capable of setting free three atoms of iodine. The iodine can be estimated in the usual manner by titration with sodium thiosulfate. The nitrosyl chlorid is decomposed by the potassium iodide, nitric oxide escaping.

The apparatus employed is shown in Fig. 87.

A is a small, round-bottomed flask, into the neck of which a glass stopper, *x*, is accurately ground (with fine emery and oil). The capacity of the bulb is about forty-six cubic centimeters, and the length of the neck, from *x* to *y*, ninety millimeters. The first condenser is a simple tube, slightly enlarged at the foot into two small bulbs.^[323] The length from *a* to *b* is 300 millimeters, from *b* to *c* 180 millimeters, and from *e* to *f* thirty millimeters. The capacity of the bulb B is twenty-five cubic centimeters, and the total capacity of the two bulbs and tube, up to the top of C, forty-one cubic centimeters. This condenser is immersed, up to the level of *c*, in a beaker full of water. D is a geissler bulb apparatus, E is a calcium chlorid tube, filled with broken glass, which acts as a tower and *g* is a small funnel, attached by rubber and clip to the branch T tube *h*. Between the T tube *i* and the wash-bottle for the carbon dioxide is placed a short piece of glass tubing, *s*, containing a strip of filter paper, slightly moistened with iodide of starch solution. This tube *s* is really hardly necessary, as no chlorin escapes backwards if a moderate current of carbon dioxide is kept passing, but it serves as a check. A glance at the joints *o*, *p*, and *q*, which are of narrow india-rubber tubing, is sufficient to show that, by using this arrangement, practically no rubber is exposed to the action of the chlorin. The tiny piece of rubber tubing at the joint *o* may be done away with, the narrower tube there being accurately ground into the wider one; this makes the condensing apparatus practically perfect.

The actual operation is performed in the following manner:

The evolution flask is washed and thoroughly dried, and the nitrate (say, about 0.25 gram of potassium nitrate) is tapped into it from the weighing tube. Two cubic centimeters of water are now added, and the bulb is gently warmed, so as to bring the nitrate into solution, after which the stopper of the flask is firmly inserted. About fifteen cubic centimeters of a solution of potassium iodide (one in four) are run into the first condensing tube, any iodide adhering to the upper portion of the tube being washed down with a little water, and five cubic centimeters of the same solution, mixed with eight to ten cubic centimeters of water, are sucked into the geissler bulbs whilst the glass in the tower E is also thoroughly moistened with the iodide. The

geissler bulbs should be so arranged that gas only bubbles through the last of them, the liquid in the others remaining quiescent.

All the joints having been made tight the carbon dioxide is turned on briskly and passed through the apparatus until a small tubeful collected at *l*, over caustic potash solution, shows that no appreciable amount of air is left in it. The small outlet tube *l*, is now replaced by a calcium chlorid tube, filled with broken glass which has been moistened with the above-mentioned iodid solution, and closed by a cork through which an outlet tube passes, the object of this trap tube being to prevent any air getting back into the apparatus.

The brisk current of carbon dioxide is continued for a minute or two longer, so as to practically expel all the air from this last tube. The stream of gas is now stopped for an instant, and about fifteen cubic centimeters of pure concentrated hydrochloric acid, free from chlorin, run into A through the funnel *g* (into the tube of which it is well to have run a few drops of water before beginning to expel the air from the apparatus), and A is shaken so as to mix its contents thoroughly. A slow current of carbon dioxide is now again turned on (one to two bubbles through the wash-bottle per second), and A is gently warmed over a burner. It is a distinct advantage that the reaction does not begin until the mixed solutions are warmed, when the liquid becomes orange-colored, the color again disappearing after the nitrosyl chlorid and chlorin have been expelled. The warming should be very gentle at first in order to make sure of the conversion of all the nitric acid, and also because the first escaping vapors are relatively very rich in chlorin; afterwards the liquid in A is briskly boiled. A very little practice enables the operator to judge as to the proper rate of warming. When the volume of liquid in A has been reduced to about seven cubic centimeters (by which time it is again colorless) the stream of carbon dioxide is slightly quickened and the apparatus allowed to cool a little. The burner is now set aside for a few minutes, and two cubic centimeters more of hydrochloric acid, previously warmed in a test-tube, run in gently through *g*; there is no fear either of the iodid solution running back, or of any bubbles of air escaping through *y* if this is done carefully. This is a precautionary measure, in case a trace of the liberated chlorin might have lodged in the comparatively cool liquid in the tube *h*. The carbon dioxide is once more turned on slowly and the liquid in A is boiled again until it is reduced to about five cubic centimeters. It is now only necessary to allow the apparatus to cool, passing carbon dioxide all the time, after which the contents of the condensers are transferred to a flask and titrated with thiosulfate. At the end of a properly conducted experiment, the glass in the upper part of tower E should be quite colorless and there should be only a mere trace of iodine showing in the lower part of the tower, while the liquid in the last bulb of the geissler apparatus ought to be pale yellow. During the operation, the stopper of A and the various joints can be tested from time to time by means of a piece of

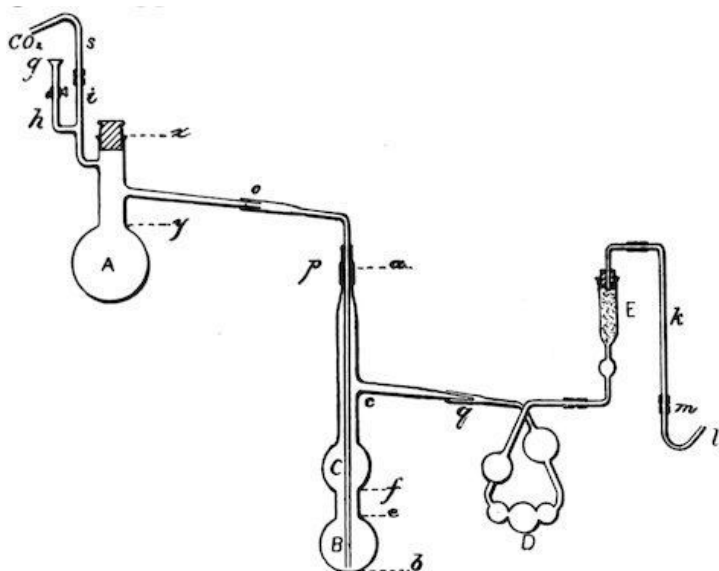


FIGURE 87. MCGOWAN'S APPARATUS FOR THE IODOMETRIC ESTIMATION OF NITRIC ACID.

iodid of starch paper, and before disjoining it is well to test the escaping gas (say at *m*) in the same way, to make sure that all nitric oxid has been thoroughly expelled.

The method is capable of giving accurate results, but it can not be preferred to the reduction or colorimetric processes.

494. Method of Gooch and Gruener.—The principle on which this method rests depends on the decomposition of a nitrate in presence of a hot saturated solution of manganous chlorid and hydrochloric acid in an atmosphere of carbon dioxid.^[324] The products of decomposition are passed into a solution of potassium iodid and the liberated iodin is titrated with standard sodium thiosulfate. The products of the reaction are chlorin, nitric oxid, and possibly nitrosyl chlorid, and under proper precautions the iodin set free is quantitatively proportional to the weight of nitrate decomposed. The manganous mixture is acted on slowly at ordinary temperatures, but on heating, the nitrate is decomposed with the formation of a higher manganese chlorid and nitric oxid. When the heat is continued a sufficient length of time the chlorin from the higher chlorids is evolved and only manganous chlorid remains. During the heating the color of the solution passes from green to black and at the end the green color is restored. The apparatus employed is shown in Fig. 88.

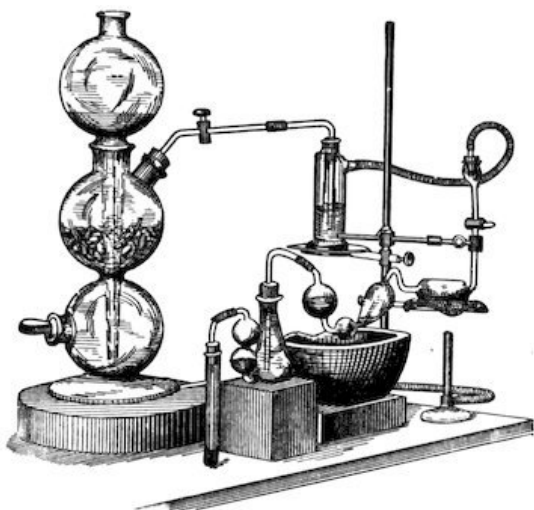


FIGURE 88. APPARATUS OF GOOCH AND GRUENER.

A plain pipette bent as is shown in the figure serves as the generating flask and for the attachment on the one hand to the carbon dioxid apparatus and on the other to the system of absorption bulbs for containing the potassium iodid. The latter should be glass, sealed to the evolution bulb of the pipette to prevent the action of the evolved gases on organic materials. The point of the potassium iodid apparatus is drawn out so as to be pushed well into the second receiver, being held in place by a piece of rubber tubing. The third tube acts simply as a trap to exclude the air from the absorption apparatus. The first receiver contains in solution three grams, the second two, and the third one gram of potassium iodid. During the reaction the first receiver is kept cool by immersion in water. Before connecting the

apparatus with the carbon dioxid generator the solution of manganous chlorid and afterwards the nitrate solution are drawn into the bulb of the pipette by gentle suction. After connecting the apparatus the current of carbon dioxid is started and kept up until all the air is expelled. Heat is then applied to the bulb of the pipette and the distillation continued until all the liquid has passed over. At the end of the reaction the contents of the receivers are united by disconnecting the apparatus from the carbon dioxid generator and passing water through the pipette. The introduction of the manganous chlorid into the mixture does not interfere with the titration of the iodine. This is accomplished in the usual way with sodium thiosulfate using starch as an indicator. The quantity of material used should contain about the amount of nitric acid that is found in two-tenths of a gram of potassium nitrate. This method, so similar to the preceding, is somewhat less complex, and, to that extent, preferable to it.

ESTIMATION OF NITRIC ACID BY COLORIMETRIC COMPARISON.

495. Delicacy of the Method.—The remarkable delicacy of those methods of chemical analysis, which depend on the production of a pronounced color, which can be compared with that produced by a known quantity of a given substance, has been long illustrated by the nesslerizing process for the estimation of ammonia. By such methods minute qualities of substances can be quantitatively determined with great accuracy, when they would escape all effort for their estimation by gravimetric methods. Processes based on this principle are, therefore, peculiarly applicable to the detection and estimation of oxidized nitrogen in waters and soil extracts, whether they be present as nitric, nitrous, or ammoniacal compounds.

In the following paragraphs will be given with sufficient detail for the needs of the analyst, the principles and practice of the colorimetric comparison methods which have been approved as best by the experience of analysts. These methods are applicable especially to cases in which only minute quantities of the substances looked for are present, and where celerity of determination is especially desirable. They are, therefore, of especial value in the analysis of rain, drainage, and irrigation waters, and of soil extracts poor in oxidized nitrogen.

496. Hooker's Method.—The quantitative action depends upon the deep green coloration given by nitric acid, when dissolved in sulfuric acid and carbazol.^[325] Other oxidizing bodies, such as iron, chlorine, bromine, chromic acid, etc., give the same reaction, but not in such a prominent manner. Such bodies with the exception of chlorine and iron, are not often found in waters. In the application of the process, iron, if present in quantities greater than one-tenth part per one hundred thousand, must be removed. Chlorides also, even when present in very small quantities, interfere with the delicacy of the reaction and must be removed. Easily destructible organic matter tends to lower the result, but not materially, unless present in large excess. Calcium carbonate and sulfate, soda, and other alkalies, in the quantities in which they are usually present in water, do not affect the result. The following reagents are required:

1. Concentrated sulfuric acid.
2. An acetic acid solution of carbazol; diphenylimid, ($C_6H_4
C_6H_4/ NH$.)
3. A sulfuric acid solution of carbazol.
4. Standard solutions of potassium nitrate.
5. A solution of aluminum sulfate.
6. A solution of silver sulfate.

1. The sulfuric acid, used for all purposes in the process, should be entirely free from nitrogen oxides. It may be readily tested by dissolving in it a small quantity of carbazol. If the solution be at first golden-yellow or brown, the acid is sufficiently pure; if it be green or greenish, another and better sample must be taken. It is essential also that the specific gravity of the acid be fully 1.84, and it is well to ascertain that this is really the case.

2. The acetic acid solution of carbazol is prepared by dissolving six-tenths gram in about ninety cubic centimeters of strongest acetic acid, by the aid of gentle heat. It is allowed to cool, and is then made up to 100 cubic centimeters by the further addition of acetic acid. The exact

strength of this solution, is of no material importance to the success of the process, and the above proportions have been selected principally because they are convenient. The solution will remain unchanged for several months. The use of this solution merely facilitates the preparation of that next described, which will not keep, and has, consequently, to be freshly prepared for each series of determinations.

3. The sulfuric acid solution of carbazol is easily made in a few seconds, but it is advisable to allow it to stand from one and one-half to two hours before using. It is prepared by rapidly adding fifteen cubic centimeters of sulfuric acid, to one cubic centimeter of the above-described acetic acid solution. This quantity usually suffices for from two to three nitrate estimations. When freshly prepared it is golden-yellow or brown; it changes gradually, however, and in the course of one and one-half or two hours it becomes olive-green. This change is probably due to traces of oxidizing agents, which occur in the sulfuric and acetic acids, and which, although not present in sufficient quantity to act immediately, gradually bring about the reaction described. The greenish color does not interfere with the process, as might at first be supposed; on the contrary, the solution is not sensitive to small quantities of nitric acid until it has undergone the change to olive-green, and it is for this reason, that it should be prepared about two hours before required for use. This solution may be thoroughly depended on for six hours after preparation. The intensities of color produced by the more concentrated solutions of nitrates after this time, gradually approach each other and become ultimately the same.

4. The standard solutions of potassium nitrate are very readily prepared. The solutions which are to be compared directly with the waters examined, may be prepared as required, but if many determinations are to be made with a variety of waters, it will be found best to prepare a complete series, differing from each other by 0.02 part nitrogen in 100,000. This series may include solutions containing quantities of nitrogen in 100,000 parts, represented by all the odd numbers from 0.03 up to 0.39. It will be found convenient to prepare them in quantities of 100 cubic centimeters at a time, from a stock solution of potassium nitrate which contains 0.00001 gram nitrogen, or 0.000045 nitric acid in one cubic centimeter. Each cubic centimeter of this solution, when diluted to 100 cubic centimeters, represents 0.01 nitrogen in 100,000, and consequently if it is desired to make a solution containing 0.35 part nitrogen in 100,000, thirty-five cubic centimeters are taken and made up to 100 cubic centimeters, and so on. The solution of potassium nitrate (b) is best prepared from a stronger one (a) containing 0.0001 gram nitrogen to the cubic centimeter, or 0.7214 gram potassium nitrate to the liter; 100 cubic centimeters of (a) made up to one liter give the solution (b). It is obvious that the series of solutions, above described, could be made directly from (a), but by first making (b), greater accuracy is secured.

5. For purposes which will be presently described, a solution of aluminum sulfate is required, containing five grams to the liter. The salt used must be free from chlorin and iron; and the solution should give no reaction when tested with carbazol.

6. The solution of silver sulfate is required for the removal of chlorin from the water or soil extract to be examined. It is prepared by dissolving 4.3943 grams of the salt in pure distilled water and making up to one liter. The sulfate is preferably obtained by dissolving metallic silver in pure sulfuric acid. The solution should be tested with carbazol in the same way as will be presently described for water; if perfectly pure, no reaction will be obtained. As silver sulfate is often prepared by precipitation from the nitrate, it is very apt to contain nitric acid,

and consequently, if the source of the salt be unknown, this test should on no account be omitted.

The analytical process is carried on as follows:

Two cubic centimeters of the water are carefully delivered by means of a pipette into the bottom of a test-tube; four cubic centimeters of sulfuric acid are added, and the solution thoroughly mixed by the help of a glass rod. The test-tube is then immersed in cold water, and when well cooled, one cubic centimeter of the sulfuric acid solution of carbazol is added, and the whole again mixed as before. The intensity of the color is now observed, and a little experience enables a fairly good opinion to be formed of the quantity of nitric acid present. Suppose that the water be roughly estimated to contain about 0.15 part nitrogen per 100,000; in such a case solutions of potassium nitrate containing 0.11, 0.15, 0.19 part nitrogen are selected from the series. Two cubic centimeters are taken from each, and treated, side by side, with a fresh quantity of the water, precisely as described for the preliminary experiment, the various operations being performed as nearly simultaneously as possible with each of the samples, and under precisely similar conditions. Two or three minutes after the carbazol has been added, the intensity of the color of each is observed. If that given by the water is matched by any of the standard solutions, the estimation is at an end. Similarly, if it falls between two of these, the mean may be taken as representing the nitrogen present in cases in which great accuracy is not required. If this be done, the maximum error will be 0.02 part nitrogen, or 0.09 part nitric acid per 100,000. If greater exactness be required, or it be found that the color given by the water is either darker or lighter than that given by all the standard solutions, a new trial must be made. In such a case the water must be again tested simultaneously with the solutions with which it is to be compared. This is rendered necessary principally for the reason that the shade of the solutions to which the carbazol has been added is apt to change on standing. Hence it is desirable that the water, and the standard potassium nitrate with which it is to be compared, should have the carbazol added at as nearly the same time as possible. When finally the color falls between that given by any two consecutive members of the standard potassium nitrate series, the estimation may be considered at an end, and the mean of these solutions taken as representing the nitrogen present.

The greatest neatness should be observed in all steps of the analysis. The quantity of water operated upon is so small that if the greatest care be not exercised throughout, sources of error may be readily introduced. The test-tubes should be rinsed out with nitrate-free water before being used and then dried. The tint should be determined by looking through the tube and not through the length of the column of liquid.

Influence of Nitrites.—If the quantity of nitrous acid in the water is known a correction can be applied for nitrates by deducting one-fifth of the number found for nitrites when estimated as nitrates.

Influence of Iron.—Although ferrous salts give no reaction with carbazol, nitrates are apt to be overestimated in their presence. On the other hand, ferric compounds, like other oxidizing agents, may give a characteristic green color with carbazol. In all cases when iron is present in any considerable quantity it is best to remove it by rendering the water slightly alkaline, evaporating to dryness, and redissolving the soluble residue until the solution reaches the original volume.

Influence of Chlorids.—The presence of chlorids furnishes by far the most serious source of error in the process by intensifying the action of the nitric acid. If, however, nitrates be absent

chlorids give no reaction with carbazol. The chlorids are removed by a standard silver sulfate solution, the quantity of chlorids present having been first determined by a standard silver nitrate solution. For this purpose an ordinary sugar flask can be employed marked at 100 and 110 cubic centimeters. This flask is filled to the 100 cubic centimeter mark with the water to be examined; the necessary quantity of silver sulfate is added and then two cubic centimeters of the solution of aluminum sulfate, previously described, and the contents of the flask brought up to 110 cubic centimeters by the addition of pure distilled water. The whole is shaken up and filtered, the first portion of the filtrate being rejected. The aluminum sulfate by reacting with the carbonates usually present in the water and producing the precipitation of alumina, facilitates the removal of the precipitated silver chlorid.

The above-described method on account of its delicacy is not well suited to aqueous solutions of soils except where the quantity of nitric nitrogen present is extremely minute.

Hooker also first suggested the use of diphenylamin for detecting the presence of nitrates,^[326] a method afterwards worked out by Spiegel.^[327]

In the variation of the method as practiced by Rideal the standard potassium nitrate and the pure sulfuric acid mentioned below are required, and in addition, the following reagents:^[328]

- (a) Silver sulfate solution containing 4.3945 grams per liter.
- (b) Aluminum sulfate solution free from chlorids and iron, five grams per liter.
- (c) Carbazol solution; six-tenths gram carbazol dissolved in glacial acetic acid and made up to 100 cubic centimeters with the glacial acid. For use, one cubic centimeter of this solution is withdrawn by a pipette and mixed with fifteen cubic centimeters of pure redistilled sulfuric acid.

The process is carried out as follows: To 100 cubic centimeters of water the amount of chlorin which has been previously ascertained is removed by the silver sulfate solution. Two cubic centimeters of the aluminum sulfate are added and the whole made up to a convenient volume, say about 110 cubic centimeters. The liquid is filtered and two cubic centimeters of the filtrate taken for an estimation of nitrates. To the two cubic centimeters are added four cubic centimeters of concentrated sulfuric acid and the mixture cooled.

One cubic centimeter of the carbazol solution in sulfuric acid is added and a bright green color appears in a few moments, if nitrates are present. Comparison is made with solutions of standard potassium nitrate.

497. Phenylsulfuric Acid Method.—Rideal also proposes a variation of the method described by Hooker, which consists in the substitution of phenylsulfuric acid for carbazol.^[329]

The solutions required are:

- (a) A standard solution of potassium nitrate containing 0.7215 gram of the pure crystallized salt in a liter of water.
- (b) Phenylsulfuric acid, (acid phenyl sulfate,) prepared by dissolving fifteen grams of pure crystallized phenol in 92.5 cubic centimeters of pure, redistilled sulfuric acid free from nitrates and diluted with seven and one-half cubic centimeters of water.

The process is conducted as follows:

A known volume of water, from twenty-five to one hundred cubic centimeters, according to its richness in nitrates, is evaporated to dryness in a porcelain dish, one cubic centimeter of phenylsulfuric acid added then one cubic centimeter of pure water and three drops of strong

sulfuric acid and the mixture gently warmed. A yellow color shows the presence of nitrates. Dilute to about twenty-five cubic centimeters with water and add ammonia in slight excess. Pour into a narrow nessler tube and add the washings and make up to 100 cubic centimeters. Imitate the color of the solution with the standard potassium nitrate treated with the same reagents.

The phenylsulfuric acid should be prepared some time before use, as the fresh solution imparts a greenish tint to the yellow of the ammonium picrate formed.

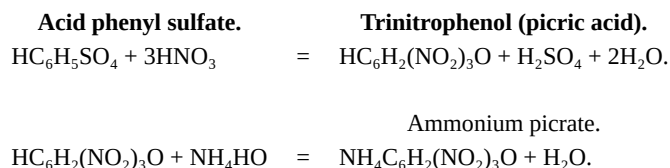
498. Variation of Leffmann and Beam.—The phenyl sulfate process, as described by Leffmann and Beam, is conducted as follows.^[330]

Solutions Required.—*Acid phenyl sulfate:* 18.5 cubic centimeters of strong sulfuric acid are added to one and one-half cubic centimeters of water and three grams of pure phenol. Preserve in a tightly-stoppered bottle.

Standard potassium nitrate: 0.722 gram of potassium nitrate, previously heated to a temperature just sufficient to fuse it, is dissolved in water, and the solution made up to 1000 cubic centimeters. One cubic centimeter of this solution will contain 0.0001 gram of nitrogen.

Analytical Process.—A measured volume of the water is evaporated just to dryness in a platinum or porcelain basin. One cubic centimeter of the acid phenyl sulfate is added and thoroughly mixed with the residue by means of a glass rod. One cubic centimeter of water, and three drops of strong sulfuric acid are added, and the dish gently warmed. The liquid is then diluted with about twenty-five cubic centimeters of water, ammonium hydroxid added in excess, and the solution made up to 100 cubic centimeters.

The reactions are:



The ammonium picrate imparts to the solution a yellow color, the intensity of which is proportional to the amount present.

Five cubic centimeters of the standard solution of potassium nitrate are similarly evaporated in a platinum dish, treated as above, and made up to 100 cubic centimeters. The color produced is compared to that given by the water, and one or the other of the solutions diluted until the tints of the two agree. The comparative volumes of the liquids furnish the necessary data for determining the amount of nitrate present, as the following example will show:

Five cubic centimeters of standard nitrate are treated as above, and made up to 100 cubic centimeters, representing 0.0005 gram nitrogen.

Suppose 100 cubic centimeters of water similarly treated are found to require dilution to 150 cubic centimeters before the tint will match that of the standard; then

$$100 : 150 :: 0.005 : 0.0075$$

i. e., the water contains seven and one-half milligrams of nitrogen as nitrate per liter.

The ammonium picrate solution keeps very well, especially in the dark. A good plan, therefore, is to make up a standard solution equivalent to, say, ten milligrams of nitrogen as

nitrate per liter, to which the color obtained from the water may be directly compared.

The results obtained by this method are quite accurate. Care should be taken that the same quantity of acid phenyl sulfate be used for the water and for the comparison liquid, otherwise different tints instead of depths of tints are produced.

With subsoil and other waters probably containing much nitrates, ten cubic centimeters of the sample will be sufficient for the test, but with river and spring waters, twenty-five to one hundred cubic centimeters may be used. When the organic matter is sufficient to color the residue, it will be well to purify the water by addition of alum and subsequent filtration, before evaporating. The method may also be used to determine small quantities of nitrates in aqueous extracts of soils when the quantity is too small for estimation by the ferrous chlorid or reduction processes.

499. Variation of Johnson.—The ammonium picrate method has given very satisfactory results as practiced by Johnson, who varies the process as described below.^[331]

The standard solution of potassium nitrate is prepared by dissolving 0.7215 gram of the pure salt in a liter of distilled water. Dilute 100 cubic centimeters of this solution to one liter with distilled water. Ten cubic centimeters of this dilute solution contain nitrogen equivalent to one part as nitrates in 100,000.

The Solution of Acid Phenyl Sulfate.—This is prepared by pouring two parts by measure of pure crystallized phenol liquefied by hot water into five parts by measure of pure concentrated sulfuric acid and digesting the whole in the water-bath for eight hours. After cooling, add one and one-half volumes of distilled water and one-half volume strong hydrochloric acid to each volume of the above mixture.

The analytical processes are carried on as follows: Ten cubic centimeters of the water under examination and ten cubic centimeters of the standard potassium nitrate are placed in small beakers and put near the edge of a hot plate. When nearly evaporated they are put on the top of the water-bath and left there until completely dry. The residue, in each case, is then treated with one cubic centimeter of the acid phenyl sulfate and the beakers placed on the top of the water-bath. In good water, a red color ought not to appear for about ten minutes.

After standing about fifteen minutes, the beakers are removed, the contents of each washed successively into 100 cubic centimeter flasks, about twenty cubic centimeters of 0.96 per cent. ammonia added, and the 100 cubic centimeters made up by the addition of water and the yellow liquid transferred to the nessler glass and the tints appropriately compared.

500. Estimation of Nitric in Presence of Nitrous Acid.—The detection of nitrous in presence of nitric acid can be accomplished by the method proposed by Griess, as described further on, through the development of azocolors, with metaphenylenediamin and other bodies, which are not produced under similar conditions by nitric acid. The detection and estimation of nitric in the presence of nitrous acid, however, is not so easy. Lunge and Lwoff propose brucin for this purpose, which, contrary to most authorities, does not give the red-yellow color with nitrous acid.^[332] The reagent is prepared by dissolving two-tenths gram of brucin in 100 cubic centimeters of sulfuric acid, pure and concentrated. It is almost impossible to prepare a sulfuric acid which does not give a trace of color with brucin; but with the purest acids this trace may be neglected.

A solution of nitrate is also prepared containing 0.01 milligram of nitrogen as nitric acid in one cubic centimeter. It is made by dissolving 0.0721 gram of pure potassium nitrate in 100

cubic centimeters of distilled water, and diluting ten cubic centimeters thereof with pure concentrated sulfuric acid to 100 cubic centimeters. Both solutions are conveniently preserved in burettes with glass stop-cocks. The liquid to be tested for nitric acid should be mixed with sulfuric acid in such a way that the mixture will have a specific gravity of one and seven-tenths. If the liquid to be tested is water, this concentration is reached by adding three times its volume of the strong acid. For the comparison of colors, cylinders of colorless glass are employed, marked at fifty cubic centimeters. They should be about twenty-four centimeters high and extend about ten centimeters above the mark. There is placed in the cylinder one cubic centimeter of the solution of nitrate in sulfuric acid, and the same quantity of the brucin mixture, and it is filled to the mark with pure sulfuric acid. The contents of the cylinder are poured into a flask and warmed at from 70°–80°, until the final yellow tint is secured, and then poured into the cylinder again. The liquid to be tested is treated in exactly the same way. The tints are then equalized by pouring out a part of the contents of the deeper colored cylinder, taking account of the volume, and filling up with pure concentrated sulfuric acid.

In this manner the content of nitric acid in the liquid under examination can be compared directly with the solution of potassium nitrate of known strength. The coloration is distinctly produced with 0.01 milligram in fifty cubic centimeters of liquid, at least three-fourths of which must be sulfuric acid.

501. Piccini Process.—The method proposed by Piccini may also be used.^[333]

About five cubic centimeters of the nitrite solution are placed in a small beaker, some pure urea dissolved therein and a few drops of sulfuric acid, and then held in boiling water for three minutes. The nitrous acid is thus completely destroyed. Ammonium chlorid may be substituted for urea. The reaction is given on page 478. The nitric acid present is then determined by diphenylamin or other suitable reagents. Diphenylamin reacts both with nitrous and nitric acids, producing a violet tint. Warrington calls attention to a slight difference, however, in its deportment with these two acids. When the solution of the reagent is not too strong a drop of it produces but little turbidity when added to water or to a solution containing nitric acid. When, however, nitrous acid is present, a cream-colored turbidity is produced. The violet color also appears at once on adding sulfuric acid when a nitrite is present, while in the case of nitrates, more sulfuric acid is required, except when the solution is very strong. In this connection, it must not be forgotten that in heating nitrites with urea or ammonium chlorid in the presence of a slight excess of sulfuric acid a trace of nitric acid may be formed.

ESTIMATION OF NITROUS ACID BY COLORIMETRIC COMPARISON.

502. Application of the Method.—The most minute traces of nitrous acid may be detected by colorimetric methods and the determination of the quantity present may be approximated with great exactness by comparison with a solution of a nitrite of known strength. Especially in following the progress of nitrification is this method, in some of its forms, of essential importance. In delicacy and celerity it has the same advantages as the colorimetric methods for the determination of nitric acid.

503. Metaphenylenediamin Method.—This process depends upon the development of a yellow color in water containing nitrous acid on the addition of a reagent containing metaphenylenediamin; $m\text{-C}_6\text{H}_4(\text{NH}_2)_2$. This variety of the phenylenediamins is readily obtained from common dinitrobenzene. It melts at 63° and boils at 287° . In order to preserve the reagent in shape for use it should be prepared in the following manner:

Dissolve two grams of the chlorid in ten cubic centimeters of ammonia, and place the solution in a glass-stoppered flask. To this solution are added five grams of powdered animal-black, and the whole vigorously shaken. After allowing to settle, the shaking is repeated at intervals of an hour, three or four times, and the flask then allowed to remain at rest for twenty-four hours.

The supernatant liquid is generally sufficiently decolorized by this treatment. If not, the shaking and subsidence must be repeated until a completely colorless liquid is obtained. The solution can be kept, indefinitely, in contact with the animal-black. Aqueous and alcoholic solutions of the salt can not be kept.

The test is applied by mixing five drops of the reagent with five cubic centimeters of sulfuric acid. The mixture must be colorless. To the mixture add 100 cubic centimeters of the water to be tested, and heat on the water-bath for five minutes. A yellow coloration indicates the presence of nitrous acid.

The metaphenylenediamin test is fairly satisfactory in perfectly colorless waters and aqueous extracts. Many waters and soil extracts, however, have a yellowish tint, and this interferes in a marked way with a proper judgment of the yellow triaminazobenzol developed in the application of the above test.

The decoloration of such waters by means of sodium carbonate or hydroxid and alum, is a matter of some difficulty and not wholly without action on the nitrites which may be present. The method, therefore, is inferior to the one next described.

504. Sulfanilic Acid and Naphthylamin Test for Nitrous Acid.—A very delicate test for the presence of nitrous acid, first described by Griess, is the coloration produced thereby in an acid solution of sulfanilic acid and naphthylamin.^[334]

Sulfuric or acetic acid may be used as the acidifying agent, preferably the latter. The solutions are prepared as follows:

- (1) Dissolve one-half gram of sulfanilic acid in 150 cubic centimeters of dilute acetic acid.
- (2) Boil one-tenth gram of naphthylamin with twenty cubic centimeters of water, decant the colorless solution from the residue and acidify it with 150 cubic centimeters of dilute acetic acid.

The two solutions may at once be mixed and preserved in a well-stoppered flask. The action of light on the mixture is not hurtful, but air should be carefully excluded because of the traces of nitrous acid which it may contain. Whenever the mixed solutions show a red tint it is an indication that they have absorbed some nitrous acid. The red color may be discharged and the solution again fitted for use by the introduction of a little zinc dust, and shaking.

The water, or aqueous solution of a soil, to be tested for nitrites, in portions of about twenty cubic centimeters, is treated with a few cubic centimeters of the mixed reagent and warmed to 70°–80°. If nitrous acid, in the proportion of one part to one million be present, the red color will appear in a few minutes. If the content of nitrous acid be greater, *e. g.*, one part in one thousand, only a yellow color will be produced, unless a greater quantity of the reagent be used.

Leffmann and Beam recommend the following method of conducting the determinations.^[335]

Solutions required:

Naphthylammonium Chlorid.—Saturated solution in water free from nitrites. It should be colorless; a small quantity of animal charcoal allowed to remain in the bottle will keep it in this condition.

Paraamidobenzene Sulfonic Acid (Sulfanilic Acid).—Saturated solution in water, free from nitrites.

Hydrochloric Acid.—Twenty-five cubic centimeters of concentrated pure hydrochloric acid added to seventy-five cubic centimeters of water, free from nitrites.

Standard Sodium Nitrite.—0.275 gram pure silver nitrite is dissolved in pure water, and a dilute solution of pure sodium chlorid added until the precipitate ceases to form. It is then diluted with pure water to 250 cubic centimeters and allowed to stand until clear. For use, ten cubic centimeters of this solution are diluted to 100. It is to be kept in the dark.

One cubic centimeter of the dilute solution is equivalent to 0.00001 gram of nitrogen.

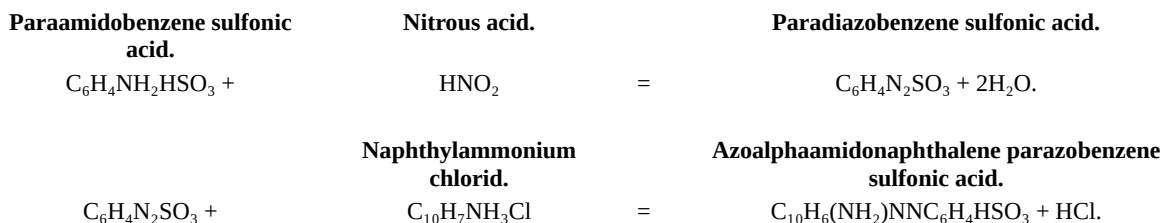
The silver nitrite is prepared in the following manner: A hot concentrated solution of silver nitrate is added to a concentrated solution of the purest sodium or potassium nitrite available, filtered while hot and allowed to cool. The silver nitrite will separate in fine needle-like crystals, which are freed from the mother-liquor by filtration with the aid of a filter pump. The crystals are dissolved in the smallest possible quantity of hot water, allowed to cool and crystallize, and again separated by means of the pump. They are then thoroughly dried in the water-bath, and preserved in a tightly-stoppered bottle away from the light. Their purity may be tested by heating a weighed quantity to redness in a tared, porcelain crucible and noting the weight of the metallic silver. One hundred and fifty-four parts of silver nitrite leave a residue of 108 parts of silver.

Analytical Process.—One hundred cubic centimeters of the water are placed in one of the color-comparison cylinders, the measuring vessel and cylinder having previously been rinsed with the water to be tested. By means of a pipette, one cubic centimeter each of the solutions of sulfanilic acid, dilute hydrochloric acid, and naphthylammonium chlorid is dropped into the water in the order named. It is convenient to have three pipettes for this test, and to use them for no other purpose. In any case the pipette must be rinsed out thoroughly with nitrite-free water each time before using, as nitrites, in quantity sufficient to give a distinct reaction, may be taken up from the air.

One cubic centimeter of the standard nitrite solution is placed in another clean cylinder, made up with nitrite-free water to 100 cubic centimeters and treated with the reagents, as above.

In the presence of nitrites a pink color is produced, which, in dilute solutions, may require half an hour for complete development. At the end of this time the two solutions are compared, the colors equalized by diluting the darker, and the calculation made as explained under the estimation of nitrates.

The following are the reactions:



The last named body gives the color to the liquid.

The method pursued by Tanner, in the preparation of the reagent, is as follows:

Sulfanilic acid is prepared by mixing thirty grams of anilin slowly, with sixty grams of fuming sulfuric acid, in a porcelain dish. The brown, sirupy liquid formed is carefully heated until quite dark in color, and until the evolution of sulfurous fumes is noticed. After cooling, the thick, semi-fluid mass is poured into half a liter of cold water and allowed to stand for some hours. The liquid portion is then decanted from the nearly black undissolved crystalline mass. To the residue half a liter of hot water is added and allowed to stand until cold, and the liquid again decanted. The undissolved portion is then treated with one liter of hot water and filtered. The filtrate is treated with animal charcoal to decolorize it, and allowed to stand for twenty-four hours and again filtered, the filtrate diluted to 1,500 cubic centimeters and used as required. This solution tends to turn pink on keeping, and thus its color interferes with the delicacy of the test, and a small amount of animal-char is kept in a small bottle containing the portion for immediate use, and this bottle is filled, from time to time, from the larger one.

The solution of naphthylamin hydrochlorate is made with one gram of the salt dissolved in 100 cubic centimeters of water. The solution is to be occasionally filtered, and not more than 100 cubic centimeters should be prepared at a time.

The analytical operations are carried on as follows:

A standard solution of pure potassium nitrite, made from the silver salt in distilled water perfectly free from nitrites, is placed in a color-glass, similar to those used in the nessler reaction, together with a second glass containing the water to be tested. These glasses should be marked to hold 100 cubic centimeters at the same depth. To each of the tubes a few drops of pure hydrochloric acid are added and two cubic centimeters of the sulfanilic solution. Afterwards, to each tube are added two cubic centimeters of the solution of naphthylamin hydrochlorate, and it is allowed to stand for twenty minutes, at the end of which time the color should be fully developed. Each tube is covered by a piece of glass in order to prevent access of air. It is unnecessary to add that the standard solutions of nitrite of different strength should be employed until the one is found which resembles, as nearly as possible, the color developed in the sample of water under examination.

505. Lunge and Lwoff's Process for Nitrous Acid.—The reaction of nitrous acid with α naphthylamin, first described by Griess, may be made reliable, quantitatively, by proceeding as below:^[336]

Boil 0.100 gram of pure white α naphthylamin for fifteen minutes with 100 cubic centimeters of water, add five cubic centimeters of glacial acetic acid, or its equivalent of dilute acid, and afterwards one gram of sulfanilic acid dissolved in 100 cubic centimeters of hot water. The mixture is kept in a well-closed flask. A slight red tint in the mixture is of no significance, inasmuch as this completely disappears when one part of it is mixed with fifty parts of the liquid to be examined. If the coloration be very strong it can be removed by adding a little zinc dust. One cubic centimeter of this reagent will give a distinct coloration with 0.001 milligram of nitrous nitrogen in 100 cubic centimeters of water.

The analysis is conducted in cylinders of white glass marked at fifty cubic centimeters. One cubic centimeter of the above reagent is placed in each of two cylinders with forty cubic centimeters of water and five grams of solid sodium acetate. In one of the cylinders is placed one cubic centimeter of a normal solution of a nitrite prepared by dissolving 0.0493 gram of pure sodium nitrite corresponding to ten milligrams of nitrogen in 100 cubic centimeters of water, and adding ten cubic centimeters of this solution to ninety cubic centimeters of pure sulfuric acid. This secures a normal solution of nitrosylsulfuric acid, of which each cubic centimeter corresponds to 0.01 milligram of nitrogen.

In the other cylinder is placed one cubic centimeter of the solution to be examined, and the contents of both cylinders are well mixed so that the nitrous acid in a nascent state may act on the reagent. The colors are compared after any convenient period, but, as a rule, after five minutes.

The chief improvement made by Lunge and Lwoff on the method of Griess is in keeping the reagent in a mixed state ready for use, by means of which any nitrous impurities in the components thereof are surely indicated. Its advantage over the method of Ilosvay^[337] consists in using the comparative normal nitrite solution as nitrosylsulfuric acid, in which state it is much more stable.

506. Estimation of Nitrous Acid with Starch as Indicator.—The method of procedure, depending on the blue color produced in a solution of starch in presence of a nitrite and zinc iodid when treated with sulfuric acid, is not of wide application on account of the interference produced by organic matter. The soil extract or water is treated in a test-tube, with a few drops of starch solution and some zinc iodid, to which is added some sulfuric acid. The decomposition of the nitrite is attended with the setting free of an equivalent amount of iodine which gives a blue coloration to the starch solution. The depth of the tint is imitated by treating a standard solution of nitrite in a similar way until the proper quantity is found, which gives at once the proportion of nitrite in the sample examined. This process, however, is scarcely more than a qualitative one.

507. Estimation of Nitrites by the Method of Chabrier.—In order to make the estimation of the evolved nitrous acid more definite by the iodine method, Chabrier has elaborated a plan for titrating it with a reducing agent.^[338]

The substance chosen for this purpose is sodium hyposulfite. In point of fact, it is not the nitrous acid which is attacked by the hyposulfite, but the equivalent amount of free iodine representing it. In the case of a soil where the quantity of nitrites is usually very small, it is well to take as much as one kilogram. The extraction should be made rapidly, with water, free

of nitrites, in order to avoid any reducing action on the nitrates which may be present. In the case of water, from five to ten liters should be evaporated to a small volume. The concentration should take place in a large flask, rather than in an open dish, in order to avoid any possibility of the absorption of nitrites produced by combustion. When the volume has been reduced to about 100 cubic centimeters it is transferred to a small flask and the concentration continued until only ten or fifteen cubic centimeters are left. The residue is filtered into a woulff bottle, F, Fig. 89, of about 100 cubic centimeters capacity.

One of the side tubulures carries a burette, B, containing five per cent sulfuric acid, the other one filled with a hyposulfite solution of known strength. The middle tubule serves to introduce a glass tube through which carbon dioxid or illuminating gas passes for the purpose of driving out the air from the solution and the flask. If carbon dioxid be used it should be generated by the action of sulfuric acid on marble. The cork holding this is furnished with a slot or valve to permit the exit of the air and the excess of the gas.

Before inserting the middle stopper, a few cubic centimeters of potassium iodid solution and a few drops of thin starch paste are added, the potassium salt being always used in excess of the nitrite supposed to be present.

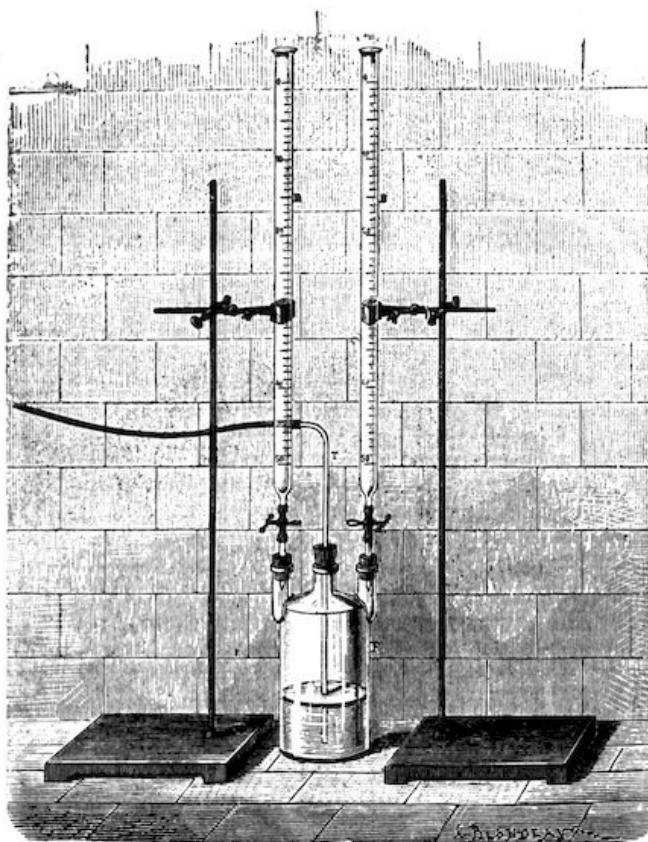


FIGURE 89. METHOD OF CHABRIER.

After the air has all been expelled from the flask the analytical process is commenced, the carbon dioxid current being slowly continued. At first, a few drops of the dilute sulfuric acid are allowed to flow into the flask. As soon as the liquid is colored blue a sufficient quantity of

the thiosulfate solution is added to discharge the color. The successive addition of acid and thiosulfate is continued until another portion of the acid fails to develop the blue color, thus indicating that all the nitrite has been decomposed. From the volume of thiosulfate used the quantity of nitrite is calculated.

The Thiosulfate Solution.—The thiosulfate solution is conveniently prepared, when a large number of analyses is to be made, by dissolving twenty-five grams of pure crystallized sodium thiosulfate in 100 cubic centimeters of water and diluting any convenient part thereof to 100 or 1,000 cubic centimeters, according to the supposed strength of nitrite solution under examination.

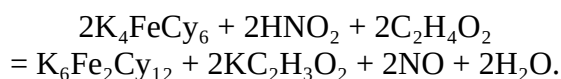
For fixing the strength of the solution dissolve 3 348 grams of pure iodine in a solution of potassium iodide and make the volume up to one liter. Each cubic centimeter of this solution corresponds to one milligram of nitrous acid. A given volume of the iodine solution is titrated against the thiosulfate, but it is best not to add the starch paste until the greater part of the iodine has been removed. The starch paste is then added and the titration continued until the blue color has been discharged. Ten cubic centimeters of the iodine solution is a convenient quantity for the titration and the thiosulfate should be diluted by adding to ten cubic centimeters of the solution mentioned above, 990 cubic centimeters of water. Each liter of this dilute solution contains two and a half grams of the sodium thiosulphate.

Example.—Let us suppose that it has required 21.3 cubic centimeters of thiosulfate to absorb ten cubic centimeters of the iodine solution; further that ten liters of water have been evaporated and titrated as described above, and that the volume of thiosulfate employed was 13.8 cubic centimeters. From this is derived the following formula: $\frac{13.8 \times 10}{21.3} = 6.48$ milligrams of nitrous acid; or 0.648 milligram per liter.

508. Estimation of Nitrous Acid By Coloration of Solution of Ferrous Salt.—This method, due to Picini is based on the production of the well-known brown color formed by the action of nitric oxide on a ferrous salt.^[339] The nitrite is decomposed by heating with acetic acid while nitrates thus treated do not develop the reaction. The tint produced is imitated as above by testing against a standard solution of nitrite. Ferrous chloride is to be preferred to other ferrous salts for the above purpose. The process should be carried on in solutions free of air.

509. Estimation of Nitrous Acid By Decomposition with Potassium Ferrocyanide.—The method of Schaeffer was first described in 1851, but little attention has been paid to it since. The method has lately been brought into notice again by Deventer.^[340]

The reaction depends upon the decomposition of nitrous acid by potassium ferrocyanide in the presence of acetic acid with the formation of potassium ferricyanide and acetate, and nitric oxide. The reaction is expressed by the following equation:



A eudiometer with a glass stop-cock is arranged as shown in Fig. 90. The lower part of the eudiometer is closed with a rubber stopper carrying a glass tube which ends in the pan *f* as shown at *e*. The eudiometer is filled to the stop-cock with a solution of potassium ferrocyanide of about fourteen per cent strength. The dish *f* is also filled up to the height indicated in the figure with the same solution. The solution of nitrite is used in such quantities that the nitric oxide evolved will occupy a space of about twenty cubic centimeters. The whole eudiometer should contain about fifty-seven cubic centimeters. The nitrite solution is added to the

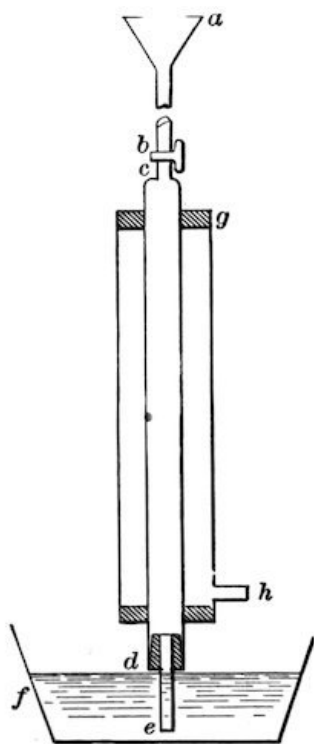


FIGURE 90.

SCHAEFFER'S NITROUS ACID
METHOD.

eudiometer by means of a funnel, *a*. The vessel containing it is washed out with a little water and then with acetic acid and finally with a few cubic centimeters of strong potassium ferrocyanid solution. The last fluid flows through the solution of nitrite and acetic acid and thus mixes it with the solution already in the eudiometer. The liquids reacting on each other float together on the strong ferrocyanid solution and each one of them is at once pressed downward by the gases which are evolved. When the evolution of gas becomes slower the apparatus should be shaken for about twenty minutes, moving it back and forth without taking the bottom of it out of the dish. When there is no longer any evolution of gas, water is added through a slowly, until the heavy potassium ferrocyanid solution is almost completely driven out of the eudiometer. The opening of the tube at *e* is then closed with the thumb, the apparatus is taken out of the dish, shaken for some time in a vertical direction and again placed in the dish. Water of any required temperature is now allowed to flow through the jacket, *g*, *h*, until the temperature is constant, when the volume of nitric oxid is read. The whole experiment can be performed in less than an hour. Operating in this way, at the end there is in the eudiometer a liquid which is not very different from water and one whose coefficient of solubility for nitric oxid is practically the same as that of water. The gas volume read is to be corrected for temperature, pressure, tension of the aqueous vapor, height of the water column in the eudiometer, and, after the end of the

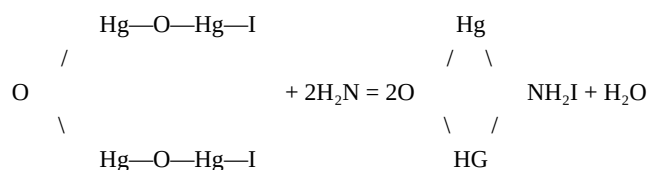
calculation, five per cent of the volume of water remaining in the eudiometer is to be added to the volume of gas obtained. This is to compensate for the volume of the gas absorbed by the water. The method gives good quantitative results.

510. Method of Collecting Samples of Rain Water for Analysis.—Warington collects rain water in a large leaden gauge having an area of 0.001 of an acre.^[341] Of the daily collection of rain, dew, and snow water, an aliquot part amounting to a gallon for each inch of precipitation is placed in a carboy; at the end of each month the contents of the carboy are mixed, and a sample taken for analysis. In the carboy receiving the rain for nitric acid estimation a little mercuric chlorid is placed each month with the view of preventing any change of ammonia into nitric acid. It may be doubted, however, if this precaution is necessary, as the rain water thus collected always contains a very appreciable amount of lead; and experiments have shown that on the whole, rain water more frequently gains than loses ammonia by keeping.

Preparation of the Sample.—The method first employed by Warington was to concentrate ten pounds of the rain water in a retort, a little magnesia being used to decompose any ammonium nitrite or nitrate present. Concentration by evaporation in the open air, and especially over gas, results in a distinct addition to nitrites present. When concentrated to a small bulk, the water is filtered and evaporated to dryness in a very small beaker. The nitrogen as nitrates and nitrites is then determined by means of the methods already described.

DETERMINATION OF FREE AND ALBUMINOID AMMONIA IN RAIN AND DRAINAGE WATERS AND SOIL EXTRACTS.

511. Nessler Process.—The quantities of free ammonia in rain and most drainage waters are minute, but may reach considerable magnitude in some sewages. By reason of these minute proportions, gravi- and volumetric methods are not suitable for its quantitative determination. Recourse is therefore had to the delicate colorimetric reaction first proposed by Nessler. This reaction is based on the yellowish-brown coloration produced by ammonia in a solution of mercuric iodid in potassium iodid. The coloration is due to the formation of oxydimercuric ammonium iodid, $\text{NH}_2\text{Hg}_2\text{OI}$, and takes place between the molecule of free ammonia and the mercuric iodid dissolved in the alkaline potassium iodid as represented by the following equation:



Nessler Reagent.—Dissolve thirty-five grams of potassium iodid in 100 cubic centimeters of water. Add to this solution gradually a solution of seventeen grams of mercuric chlorid in 300 cubic centimeters of water until a permanent precipitate of mercuric iodid is formed. Add now enough of a twenty per cent solution of sodium hydroxid to make 1000 cubic centimeters.

The mixed solutions, at room temperature, are treated with additional mercuric chlorid until the precipitate formed, after thorough stirring, remains undissolved. This precipitate is then allowed to subside, and when the supernatant liquid is perfectly clear, it is decanted or filtered through asbestos and kept in a well-stoppered bottle in a dark place. The part in use should be transferred to a smaller bottle as required. The solution should be made for a few days before using, since its delicacy is increased by keeping. The nessler reagent should show a faint yellow tint. If colorless it is not delicate, and shows the addition of an insufficient quantity of mercuric chlorid. When properly prepared, two cubic centimeters of the reagent poured into fifty cubic centimeters of water containing 0.05 milligram of ammonia will at once develop a yellowish-brown tint.

Preparation of Ammonia-Free Water.—To pure distilled water add pure, recently-ignited sodium carbonate, from one to two grams per one liter, and distill. When one-fourth of the whole has passed over, the distillate may be regarded as free from ammonia; fifty cubic centimeters of the following distillate should give no reaction with the nessler reagent. The distillation should be continued until the residual volume in the retort is about one-fourth of the original, and the distillate free of ammonia is carefully preserved in close glass-stoppered bottles previously washed with ammonia-free water. Pure water, free of ammonia may also be obtained by distilling with sulfuric acid.

Comparative Solution of Ammonium Chlorid containing 0.00001 gram Ammonia in one cubic centimeter.—Dissolve 3.15 grams H_4NCl in ammonia-free water and make the volume

up to one liter. Take ten cubic centimeters of the above solution and dilute to 1000 with water, free from ammonia.

Solution containing 0.00001 gram Nitrogen in one cubic centimeter.—Dissolve 3.82 grams H_4NCl in water, free from ammonia and dilute with same to 1000 cubic centimeters. Dilute ten cubic centimeters of the above solution to 1000.

The Distillation.—Any kind of suitable retort or flask connected with a good condenser may be used. The capacity of the retort should be from 700 to 1,000 cubic centimeters. The retort and condenser preferred by Leffmann and Beam are shown in Fig. 91. Any good lamp may be used in which the flame is under complete control. The gauze burner shown in the figure is easily controlled and distributes the heat evenly over the surface of the retort thus diminishing the danger of fracture. The apparatus having been previously rinsed with distilled water receives 500 cubic centimeters of the liquid to be tested for ammonia, together with a few pieces of recently ignited pumice stone to prevent bumping and five cubic centimeters of the twenty percent sodium carbonate solution to render its contents alkaline. The water is raised to the boiling-point and with gentle ebullition fifty cubic centimeters of distillate collected. The distillate is conveniently collected in a color-comparison cylinder of thin white glass and flat bottom, about two and a half centimeters in diameter, and marked at fifty and one hundred cubic centimeters. Two cubic centimeters of the nessler reagent are added and if ammonia be present a yellowish-brown color will be developed, the intensity of which is matched by taking portions of the ammonium chlorid solution, diluting to fifty cubic centimeters with pure water and treating with the same quantity of the nessler reagent. The process is repeated until a distillate is obtained which gives no reaction for ammonia. The sum of the quantities obtained in the several distillates gives the total amount of ammonia in the 500 cubic centimeters of the water taken. In most cases practically all the ammonia is obtained in three or four portions of the distillate.

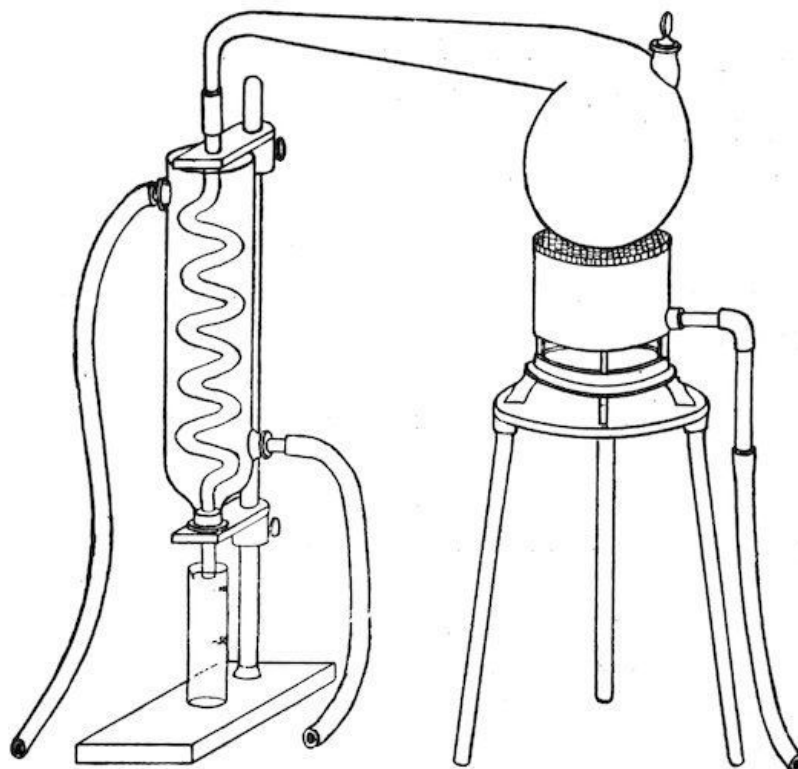


FIGURE 91. RETORT FOR DISTILLING AMMONIA.

Albuminoid Ammonia.—The residue from the process just described is employed for the purpose of determining the albuminoid ammonia. Two hundred grams of potassium hydroxid and eight grams of potassium permanganate are dissolved in 1,000 parts of distilled water. Fifty cubic centimeters of the solution are placed in a porcelain dish with 100 cubic centimeters of distilled water and evaporated to fifty cubic centimeters. This liquid is placed in the retort and the distillation resumed and continued until an ammonia-free distillate is obtained. The total albuminoid ammonia is determined by taking the sum of the quantities in the several distillates.

512. Nessler Reagent of Ilosvay.—To secure greater delicacy in nesslerizing, Ilosvay uses a reagent prepared as follows:^[342]

Dissolve two grams of potassium iodid in five cubic centimeters of water, heat the solution gently, and add three grams of mercuric iodid. After the solution is cooled, add an additional portion of three grams of the mercury salt, and then twenty cubic centimeters of water, and wait until the precipitation is complete. After filtering, there are added to the filtrate from twenty to thirty cubic centimeters of a twenty per cent solution of potassium hydroxid. Only the limpid supernatant liquid is used in the analytical work. With this reagent, Ilosvay has been able to detect 0.02 milligram of ammonia in 110 cubic centimeters of water.

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NOTE.—On page 158, paragraph 172, third line, insert, “and determining matters dissolved therein,” after “flow.”

PART EIGHTH.

SPECIAL EXAMINATION OF WATERS, VEGETABLE SOILS, AND UNUSUAL SOIL CONSTITUENTS.

513. Further Examination of Waters.—Having described in the preceding part the approved methods of determining the oxidized nitrogen in waters and soil extracts there remains to be considered the examination of waters for other substances of importance to agriculture. Rain waters add practically nothing to the soil but nitric acid and ammonia, and, therefore, demand no further discussion here. In drainage and sewage waters, in addition to the oxidized nitrogen, there may be sufficient quantities of phosphoric acid and potash to make their further analysis of interest. But by far the most practical point to be considered is in the case of waters used for irrigation purposes where the continued addition to the soil of mineral matters may eventually convert fertile fields into barren wastes. In irrigated lands there is practically no drainage and the whole of the water is removed by superficial evaporation. It is easily seen how these mineral matters tend to accumulate in that part of the soil in which the rootlets of plants seek their nourishment.

514. Estimation of Total Solid Matter.—The total solid contents of a sample of water are determined by evaporating a known volume or weight to dryness and weighing the residue. For comparative purposes a given volume of water may be taken if the solid contents do not exceed four grams in a United States gallon. The water should be measured at a temperature of about 15°. Where the content of mineral matter is greater it is best to weigh the water and calculate the solid contents to parts per one

hundred thousand. For practical purposes in the United States it is customary to state the content of solid matter in grains per gallon. Since, however, the gallon has so many different values it is always necessary to indicate what particular measure is meant.

In ordinary spring and well waters the volume to be used is conveniently taken at 100 cubic centimeters. To avoid calculation a volume in cubic centimeters corresponding to some decimal part of a gallon in grains may be taken and the weight in milligrams will then be equivalent to the grains per gallon. Thus in the imperial gallon which contains 70,000 grains of distilled water at 15°.5, seventy cubic centimeters may be taken. If the residue weigh twenty-five milligrams the water contains twenty-five grains of solid matter per gallon. The United States gallon at 15°.5 contains 58,304 grains of distilled water. In this case 58.3 cubic centimeters should be used, or double this amount and the weight in milligrams be divided by two.

The evaporation may be made in a platinum, porcelain, or aluminum dish, preferably with a flat bottom; The dish does not need to hold the whole volume at once, but the water may be added from time to time as the evaporation continues. The dish, however, should, as a rule, hold not less than 100 cubic centimeters. The evaporation is best conducted over a steam-bath, and after the complete disappearance of the liquid the heating should be continued until the residue is perfectly dry.

In the case of mineral waters highly impregnated with inorganic salts, a smaller volume or weight may be taken, and greater care must be exercised in drying the residue. For the purpose of qualitatively determining the percentage of special ingredients, quantities of the water should be taken inversely corresponding to the content of the ingredient desired. In general, it will not be necessary to evaporate the sample to complete dryness, but only to concentrate it to a volume convenient for the application of the analytical process. Where a complete quantitative analysis of the solid residue is desired, a sufficient quantity of the water is evaporated to give a weighable amount of the least abundant ingredient. The total solid content of the water having been previously determined, the actual weight or volume of the water taken to obtain the above residue is of no importance.

515. Estimation of the Chlorin.—The chlorin in the solid residue from a sample of water may be determined directly by dissolving the soluble salts in distilled water, to which enough nitric acid is added to preserve the

solution slightly acid. After filtering and washing, silver chlorid is added, little by little, with constant shaking until a further addition of the reagent produces no further precipitate. The beaker or flask should be placed in a dark place, on a shaking apparatus which is kept in motion until the precipitate has entirely settled in a granular state. The silver chlorid is then collected on a gooch, washed free of all soluble matter, dried at 150° and weighed. If the precipitate be ignited to incipient fusion, a porcelain gooch should be used.

A more convenient method is to determine the chlorin directly in the water, or, where the quantity is too minute, after proper concentration, volumetrically by means of a titrated solution of silver nitrate, using potassium chromate as indicator. As soon as the chlorin has all united with the silver, any additional quantity of the silver nitrate will form red silver chromate, the permanent appearance of which indicates the end of the reaction. This process is especially applicable to water, which in a neutral state contains no other acids capable of precipitating silver. The chromate indicator does not work well in an acid solution.

516. Solutions Employed.—A quantity of pure silver nitrate, about five grams, is dissolved in pure water and made up to a volume of one liter. For determining the actual strength of the solution, 0.824 gram of pure sodium chlorid is dissolved in water and the volume made up to half a liter. Twenty-five cubic centimeters of this solution are placed in a porcelain dish, and a few drops of the solution of potassium chromate added. The silver nitrate solution is allowed to flow into the porcelain dish from a burette graduated to tenths of a cubic centimeter. The red color produced as each drop falls, disappears on stirring as long as there is any undecomposed chlorid. Finally a point is reached when the red color becomes permanent, a single drop in excess of the silver nitrate being sufficient to impart a faint red tint to the contents of the dish.

The solution of potassium chromate is prepared by dissolving five grams of the salt in 100 cubic centimeters of water. Silver nitrate solution is added until a permanent red precipitate is produced, which is removed by filtration, and the filtrate is employed as the indicator as above described. Water with any considerable quantity of chlorin can be treated directly with the reagents; when the percentage of chlorin is low, previous concentration to a convenient volume is advisable.

In waters containing bromids and iodids these halogens would be included with the chlorin estimated as above. For agricultural purposes such waters have little importance. In the case of soluble carbonates capable of precipitating silver this action can be prevented by acidifying the water with nitric acid and afterwards removing the excess of acid with precipitated calcium carbonate. In this reaction McElroy recommends the use of Congo paper, which is not affected by the carbon dioxid but is turned blue as soon as an excess of nitric acid is added. After the addition of the calcium carbonate the mixture should be boiled to expel carbon dioxid.^[343]

Irrigation waters from natural sources or derived from sewage rarely contain enough chlorin to make their use objectionable. On the other hand, when water is obtained for this purpose from artesian wells it may often contain a quantity of chlorin which will eventually do more harm to the arable soil than the water will do good.

517. Carbon Dioxid.—Free carbon dioxid in water has no significance in respect of its use for irrigation purposes. Such waters, however, are usually of a highly mineral nature and thus are justly open to suspicion when used for farm animals and on the field. The presence of free carbon dioxid as has already been pointed out in paragraph 42, gives to water, one of its chief sources of power as an agent for dissolving rocks and ultimately forming soil. The estimation of the total free carbon dioxid in a sample of water issuing from a spring or well is a matter of some delicacy by reason of the tendency of this gas to escape as soon as the water reaches the open air and is relieved from the natural pressure to which it has been subjected. The actual quantity of the gas remaining in solution at any given time is determined as follows: 100 cubic centimeters of the water are placed in a flask with three cubic centimeters of a saturated solution of calcium and two of ammonium chlorid. To this mixture is added forty-five cubic centimeters of a titrated solution of calcium hydroxid. The flask is stoppered, well shaken, and set aside for twelve hours to allow the complete separation of the calcium carbonate formed.

When the supernatant liquid is perfectly clear an aliquot part thereof, from fifty to one hundred cubic centimeters, is removed and titrated with decinormal acid with phenacetolin or lacmoid as an indicator. From the quantity of calcium hydroxid remaining unprecipitated the amount which has been converted into carbonate can be determined by difference. The

difference between the quantity of calcium hydroxid originally present in the solution and that remaining after the above treatment multiplied by the factor 0.0022 will give the weight of carbon dioxid present in the water in a free state or in excess of that present as normal carbonates.

UNUSUAL CONSTITUENTS OF SOIL.

518. Boric Acid.—Boron, while not regarded as an essential plant food, is yet found quite uniformly in the ashes of a large number of plants. It may, therefore, be of some interest to the agricultural analyst to determine the amount of it which may be present in a soil extract or mineral water. For this purpose the following method due to Gooch may be employed.^[344] To one liter of the water supposed to contain boric acid add enough sodium carbonate to produce distinct alkalinity. After evaporation to dryness acidify the residue with hydrochloric acid, apply a piece of turmeric paper and dry at a moderate heat. The usual brown-red tint will reveal the presence of boric acid.

The quantitative estimation of the acid is accomplished as follows: One or more liters of the water rendered alkaline as above are evaporated to dryness. With the aid of as small a quantity as possible of acetic acid the dry residue is transferred to a distillation flask and condenser arranged as shown in Fig. 92. About one gram of recently ignited pure lime, cooled in a desiccator and weighed accurately, is introduced into the flask at the bottom of the condenser and slaked by a few cubic centimeters of water. When the flask is attached, the terminal tube of the condensing apparatus should dip into the lime-water in the flask. The heating-bath is partly filled with paraffin at a temperature of about 120°. The paraffin-bath is raised so that the entire bulb of the flask is immersed therein and the distillation continued until all the liquid has been distilled. The bath is removed and after cooling somewhat, ten cubic centimeters of methyl alcohol are introduced by means of the stoppered funnel-tube and the process of distillation repeated. This operation with methyl alcohol is repeated five times. The boric acid passes off with the distillate and is found in the flask below the condenser as calcium borate. The contents of the distillation flask are evaporated to dryness and ignited conveniently in the same crucible in which the lime was burned. The increase in weight represents the quantity of boric anhydrid, B_2O_2 obtained.

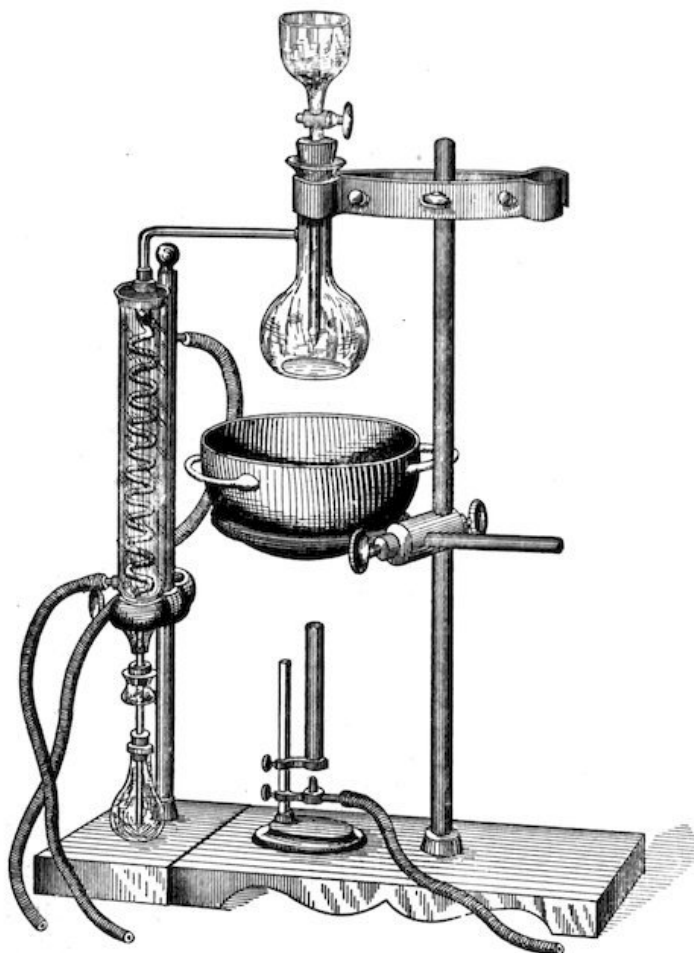


FIGURE 92. GOOCH'S APPARATUS FOR BORIC ACID.

519. Method Of Moissan.—The principle of the method of Gooch, which has just been described, is applied by Moissan in a slightly modified manner.^[345]

In this method the generating flask is made smaller than in the Gooch apparatus, and the funnel at the top is oval and provided with a ground-glass stopper. It is closed at the bottom with a glass stop-cock, and the slender funnel-tube enters through a rubber stopper and ends about the middle of the bulb of the flask. The delivery-tube is longer than in Fig. 91, and is bent upward at its middle part in the form of an obtuse angle. The receiving flask is connected with the condenser by means of a tube-shaped funnel, which prevents any regurgitation into the generating flask. The receiving flask also has attached to it a three-bulb potash absorption tube, through which all vapors escaping from the receiving flask must pass. The

bulbs contain a five per cent solution of ammonia. The receiving flask should be placed in a crystallizing dish and kept surrounded with ice-water.

The boron which is to be estimated should be in the form of boric acid. This can readily be accomplished by treating the residue to be analyzed with nitric acid in a sealed tube. The mixture is introduced into the generating flask, washing with a little nitric acid, and evaporated to dryness. The heat is removed, and, by means of the funnel, ten cubic centimeters of methyl alcohol added, and distillation is renewed. This operation with methyl alcohol is repeated four times, taking care to distill to dryness in each case before the addition of a fresh quantity of alcohol. Afterwards, there is introduced into the apparatus one cubic centimeter each of distilled water and nitric acid and the distillation again carried to dryness. The treatment with methyl alcohol, as described above, is then repeated three times. To determine whether all the boric acid have passed over, the receiving flask at the bottom of the condenser is disconnected and a drop of the alcohol taken from the end of the condensing tube by means of a filament of filter paper. On burning, the flame should not show any trace of green. In case a green color is observed, the distillation with nitric acid and methyl alcohol must be repeated.

The ammonia in the potash bulbs serves to arrest any of the vapors carrying boric acid which might escape from the receiving flask. The contents of the bulbs are to be mixed with the liquid in the receiving dish, and the whole poured onto a known weight of recently ignited calcium oxid contained in a platinum dish, and the mixture briskly stirred. If the liquid be very acid the platinum dish should be kept in ice-water to prevent heating. After fifteen minutes the liquid usually becomes alkaline, and it is then evaporated at a temperature below the boiling-point of methyl alcohol (66°). The mass, after the methyl alcohol has disappeared, is dried at a gradually increasing temperature, and finally, the dish is ignited over a blast, at first covered and afterwards open. The dish is covered and weighed and again ignited until constant weight is obtained.

The lime used should be specially prepared by igniting calcium nitrate incompletely, and reigniting a portion of this to constant weight just before beginning each analysis. The calcium oxid is then obtained in a perfectly fresh state. It should be employed in considerable excess, for each half

gram of boric acid at least eight grams of the lime. The operation is tedious but the results are quite accurate.

SPECIAL TREATMENT OF MUCK SOILS.

520. General Considerations.—Deposits of muck which are to be used as soil for cultural purposes, or marsh lands, containing large quantities of organic matter, require a special treatment in addition to the general principles of examination illustrated in the previous pages. These soils, essentially of an organic origin, do not permit of the same treatment either chemical or physical as is practiced with soils of a mineral nature. For instance, it would be useless to attempt a silt analysis with organic soils, and the extraction of them with hydrochloric acid for the purpose of determining the materials passing into solution would prove of little utility. The object of the examination is not only to obtain knowledge of the ultimate constitution of the sample, but also, and this is the practical point, to gain some idea of its stores of plant food and of the proper steps necessary to secure a supply of the deficient nutrients. The final analytical processes for the estimation of the constituents of a muck or vegetable soil are the same as those already given, but the preliminary treatment is radically different.

521. Sampling.—First of all the geologic and meteorologic conditions of the muck formations must be determined as nearly as possible. It is fair to presume that these formations are of comparatively recent origin, in fact that they are still in progress. The geologic formation in the vicinity of the deposit should be noted. Information should be given in respect of the character of the water, whether running or stagnant, fresh, salt, or brackish, and changes of level to which it is subject, should be noted. It should be particularly stated whether the vegetable growth contributing to the formation be subject to frost or freezing. The character of the growth is to be carefully noted, and observation made of any changes in vegetation due to drainage preparatory to cultivation. It is to the original vegetation that the chief vegetable accretions in the muck must be accredited. In all cases, for purposes of comparison, some samples must be taken from parts of the field which have not been under cultivation or fertilization. The original properties of the muck can thus be determined and compared with the portions which have been changed under cultivation. If the vegetation in different parts of the field vary it is an indication that the muck is not

homogeneous, and in such cases all the different kinds must be separately sampled. Any alluvial deposit should be carefully separated from the muck found *in situ*, for the two layers are radically different in nature.

The sampling should be made by digging a pit, if possible to the bottom of the muck formation, and taking the samples at depths of one foot from one or all of the sides. The samples from sections of even depth are to be mixed and about five kilograms of the well-mixed sample preserved. Blocks of the unbroken and unshattered material should also be taken from each section for the purpose of determining permeability to water and air. All living vegetable matter should be as fully as possible removed before the sampling begins. The nature of the subsoil must be observed, and it should be stated whether it be sand, clay, limestone, etc. Fresh samples should be taken at various depths for the purpose of determining the content of moisture in the manner described in paragraph 65. The tubes used are made sharp at the end to be inserted in the soil, and so arranged as to cut cylinders of soil a trifle smaller than their interior diameter. By this means the sample slips easily into its place. The same care and judgment must be used in taking these samples as are required in the case of common soils.

Illustration.—Samples of muck soil taken at Runnymede, Florida.

(a) Formation. Littoral fresh-water lacustrine deposits, varying from a few inches to four feet in depth, and from a few feet to half a mile in width.

(b) Vegetation before drainage. Saw grass (*Cladium Mariscus* or *effusum*).

(c) Principal present vegetation (see pages 59–60).

(d) *Kinds of Soil.*—The muck shows two distinct colors, black and brown. The vegetation, however, seems to be the same in both cases. The black muck has the appearance of being more thoroughly decomposed.

(e) *Geologic Formation.*—This portion of the Florida peninsula is covered generally with sand due to marine submergence during recent geologic periods. The forest growth is pine. The drainage from the pine land is towards the muck deposits. The pine land lies from four to ten feet higher than the surface of the muck and is much less subject to frost.

522. Water Content.—The capacity of a muck soil for retaining water is very great. In a moist state these soils are heavy and apparently quite firm. When dry they are light and fluffy and unsuited to hold the rootlets of

plants. Saturated to their greatest capacity they hold considerably more than their own weight of water. Attention has already been called to the danger of drying such samples at a high temperature. As in most cases of drying exposure at the temperature of boiling water until a constant weight is obtained is a perfectly safe way. It is hard to say what comes off in addition to water at a higher temperature. All that comes off even at the temperature of boiling water is not water.

The method of determination usually employed in this laboratory is the following:

From four to five grams of the material are spread as evenly as possible over the flat bottom of a circular aluminum dish, about seven centimeters in diameter. The dish is exposed for three hours at the temperature of boiling water and then kept for two hours in an air-bath at 110°. At the end of this time constant weight is obtained. Additional drying at 110° for five hours, usually gives no further loss of volatile matter. The dish should be covered during weighing on account of the hygroscopicity of the residue. When well sampled the dry matter thus obtained serves as the basis of calculation for the general analytical data.

Results.—Samples of muck soil taken in brass tubes in March during the dry season had the following contents of moisture:

					Matter volatile at 110°, per cent.
Taken	near the	surface			61.60
„	one foot	below	the	surface	84.35
„	two feet	„	„	„	81.52

It is thus seen that the normal content of moisture in such a soil during the dry season, exclusive of the top layer, is about eighty per cent.

523. Organic Carbon and Hydrogen.—The organic carbon and hydrogen in muck soils are determined on the carefully dried sample by combustion with copper oxid. This process gives not only the quantities of these bodies combined as humus, but also those in a less advanced state of decomposition and present as fatty bodies or resins. The method employed is given on pages 319–20.

Results.—The data obtained on a sample of muck soil from Florida are as follow:

				Per cent. carbon.	Per cent. hydrogen.
One	foot	from	surface	57.67	4.48
Two	feet	„	„	47.07	5.15
Three	„	„	„	8.52	0.53

The last sample was largely mixed with sand, the muck at the point when it was taken not being quite three feet deep.

524. Total Volatile and Organic Matter and Ash.—The ignition of the sample should be very carefully conducted at the lowest possible temperature. About five grams of the air-dried sample or double that amount of the moist sample should be taken. In the latter case the calculations should be made on the basis of the dry material. The ignition should be continued with frequent stirring with a platinum wire until all organic matter is destroyed. At the same time in a large dish one or more kilograms of the sample should be ignited in order to secure an ash for analysis. The ash should be quickly weighed to avoid absorption of moisture.

525. Sulfur.—The sulfur present in muck is combined either in an organic form or with iron. It may be estimated by the method of Fleischer.
[\[346\]](#)

From five to ten grams of the sample are ignited carefully in a hard glass tube in a stream of air or better of oxygen. The sulfur compounds escape as sulfuric or sulfurous acid.

The combustion is carried on in the apparatus shown in Fig. 93.

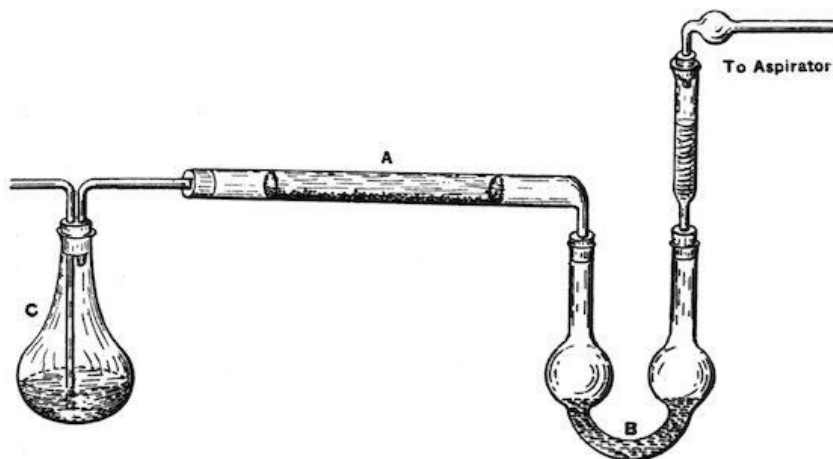


FIGURE 93. APPARATUS FOR DETERMINING SULFUR.

The end of the tube A, next to B, is lightly stopped with a plug of glass wool, the substance introduced and held in place by a second plug of glass wool next to C. A is connected to the working flask C, containing water, as is shown in the illustration. The chief object of the flask is to control the rate of aspiration of the air or oxygen. A is also connected with the bulb-tube B, as shown in the figure. B contains potash-lye, free of sulfur. On the top of B is placed a drying tube filled with glass pearls, moistened with potash-lye. This is connected with the aspirator by a small bulb-tube bent at right angles, as indicated. The bulb of this tube contains a little neutral litmus solution, which must suffer no change of color during the progress of this analysis. The tube, thus arranged, is placed in a combustion furnace and gradually heated to redness, beginning with the part next to B. A moderate stream of air or oxygen is passed through the tube during the operation. Any product of the combustion collecting in the tube before reaching B, is driven into B by careful heating. At the end of the combustion the contents of B are acidified with hydrochloric acid, and treated with bromin to convert the sulfurous into sulfuric acid. The excess of bromin is afterwards removed by boiling, and the sulfuric acid precipitated by barium chlorid and estimated in the usual way.

The total sulfuric acid having thus been determined, the sample is extracted with water and the sulfuric acid estimated in the residue.

The sulfuric acid in a muck which is injurious to vegetation is classified by Fleischer, as follows:

(1) Free sulfuric acid. (The residue which is obtained by calculation as sulfates of the bases in the water extract.)

(2) The sulfuric acid contained as copperas (calculated from the iron oxid content of the aqueous extract).

(3) Sulfuric acid arising from the oxidation of pyrites (calculated from the sulfuric acid obtained by treatment of the water-extracted sample).

A better idea of the distribution of the sulfur in the sample can be obtained by estimating it according to the method given in paragraph 385.

526. Phosphoric Acid.—The method for determining the phosphorus in muck is given in paragraph 382. The process given in 378 may also be used.

The method of extraction with hydrochloric acid is wholly unreliable as a means of determining the available phosphoric acid in muck.

There are some vegetable soils which contain so much iron and lime that the whole of the acid ordinarily used would be consumed thereby. This fact has been clearly pointed out by Wiklund in determining the phosphoric acid in a large number of peaty soils.^[347] His experiments, were made with acid of only four per cent strength. In some cases, however, it may be found useful to determine the quantity of phosphoric acid which can be extracted with hydrochloric acid, and afterwards to separate the humus and determine the content of phosphoric acid therein.

527. Humus.—In this laboratory the humus is estimated by the method of Huston and McBride, as given in paragraph 312. In samples so rich in organic matter the method of Grandeau does not give as good results.

Often more than half the weight of the dry substance is soluble in ammonia after treatment with acid. The nitrogen in the original sample and the separated humus should be estimated by moist combustion with sulfuric acid in the usual manner.

528. The Mineral Contents of Humus.—The material obtained by precipitating the alkaline extract of a vegetable earth with an acid does not consist alone of oxygen, carbon, hydrogen, and nitrogen. The complex molecules which make up this mixture contain certain quantities of iron, sulfur, and phosphorus in an organic state. These bodies are left as inorganic compounds on ignition, provided there is enough of base present to combine with all the acid elements. Much of the sulfur and phosphorus, however, in these compounds might be lost by simple ignition. In such

cases moist oxidation of these bodies must be practiced, or the gases of combustion passed over bodies capable of absorbing the oxidized materials in order to detect and determine them. The proper methods of accomplishing this have already been pointed out for vegetable soils, and the same processes are applicable in the case of extracted and precipitated humus.

Another proof that both phosphorus and sulfur are present in humus in an organic state is found in the fact pointed out by Eggertz and Nilson, that the ash of muck soils is always richer in sulfuric and phosphoric acids than the solution obtained therefrom by hydrochloric acid.^[348]

In a sample of muck examined by them there was found in the ash 1.46 per cent SO_3 , and in the acid extract only 0.05 per cent SO_3 ; and in the ash 0.3 per cent P_2O_5 , while in the extract only 0.04 per cent P_2O_5 .

529. Combustion of the Humus.—The percentage composition of the extracted humus can be determined, after drying to constant weight, by combustion with copper oxid. There is little use in trying to assign definite chemical formulas to any of the components of the complex which we call humus. Some of the supposed formulas have been given on pages 61 and 62.

530. Ether Extract.—Most peaty soils, when very dry, are not easily moistened with water. This is due to a superficial coating of fatty or resinous bodies which prevents the water from reaching the muck particles. In such cases water will pass between the particles and percolate to a considerable depth, but without wetting. This oily matter can be removed by treating the dry material with ether in any approved extraction apparatus. For the separation of the more purely fatty bodies, light petroleum may be used, while the total of such matters is extractable with sulfuric ether. The extracted bodies should be examined to determine their nature, whether fatty, resinous, or of other materials soluble in ether. The quantity of this material in some muck soils is remarkably high. In a Florida muck, examined in this laboratory, 18.95 per cent in the air-dried substance, which contained still 41.83 per cent of water, or about 32.5 per cent of the water-free material were found to be soluble in ether.

The color of the ether extract may be almost black, showing the extraction of a part of the humus or coloring matter in the muck. This

extractive coloring matter may also be a partial oxidation product of the original chlorophyll of the plant.

531. Further Examination of the Ether Extract.—The ether extract should be first treated with petroleum ether, unless this substance be used first in extraction. Afterwards, it is to be exhausted with strong alcohol, and the quantities of material soluble in the three reagents separately determined.

The nitrogen is further to be determined in the several extracts, and, for control, in the residue of the muck.

The method of procedure practiced in this laboratory is to first extract the sample with petroleum ether, which will yield any free fat acids, fats, or oils, waxes, and possibly some resinous matter. A weighed portion of the sample, about two grams, is extracted quantitatively by one of the methods which will be described in the second volume of this manual.

From two to five kilograms of the sample are then extracted in bulk for the purpose of securing a sufficient quantity of the material to use for further analysis.^[349]

In each case the petroleum is followed by pure ether, and in this way the chlorophyll, resins, etc., are obtained. This extract is examined also for its several proximate constituents.^[350]

The treatment with ether is followed by extraction with absolute alcohol for the removal of tannins and other glucosides, resins insoluble in ether, etc., and the extract subjected to the usual examination.^[351] Instead of absolute alcohol a spirit of ninety-five per cent strength, or even of eighty per cent, may be used. The final residue should be subjected to the usual determination for nitrogen, volatile matter, ash, etc., in the manner already described. The large amount of resinous and other matters soluble in petroleum and ether, which is found in the Florida muck soils, is probably due to the proximity of pine forests, the débris of which, sooner or later, find their way to these littoral deposits. Considerable portions of organic humic acids and even humus itself, may also be removed by ether and alcohol and in every case nitrogen should be determined in these extracts.

RARE CONSTITUENTS OF SOILS.

532. Estimation of Copper.—The natural occurrence of copper in many vegetables has acquired additional significance by reason of its relation to added copper in canned peas and other preserves. Formerly, copper was not regarded, in any sense, as a plant food. Even now it can scarcely be considered as more than an accidental and non-essential constituent of vegetable matter. It is by no means certain, however, that copper may not be, in some sense, in organic combination, as phosphorus and sulfur often are. It is said, also, to be found in certain animal organisms, notably in the oyster. In the estimation of copper in soils, there is first made a hydrochloric acid solution of the sample. The solution is treated with well-washed hydrogen sulfid until well saturated. The precipitate is collected at once on a gooch, and washed with water containing the precipitating reagent. The filtrate is dried, gently ignited or roasted, and dissolved in aqua regia. After evaporating to dryness on a steam-bath, water and hydrochloric acid are added, and the copper reprecipitated in the manner described above.

If zinc be present in the sample the solution should be made very strongly acid with hydrochloric before the treatment with hydrogen sulfid, otherwise some zinc may be carried down with the copper.^[352] If lead be present it is also precipitated with the copper and can be separated as described below. In the filtrate from the solution in nitric acid after the second precipitation the copper is precipitated as hydroxid by potash, collected in a porcelain gooch, dried, ignited, and weighed as CuO . Or the copper may be secured as sulfate and estimated electrolytically in the manner described in volume second for the gravimetric estimation of sugar.

533. Estimation of Lead.—If the soil contain lead this metal will be thrown down with the copper as sulfid in the manner described above. In this case the mixed sulfids are dissolved in nitric acid, diluted with water, filtered, and washed. The filtrate is treated with sulfuric acid in considerable excess, and evaporated until all the nitric acid has passed off and the sulfuric acid begins to escape. After cooling, water is added and the lead sulfate collected on a porcelain gooch and washed with water containing sulfuric acid. Finally it is washed with alcohol, dried, ignited, and the lead weighed as PbSO_4 .

534. Estimation of Zinc.—If zinc be present in the hydrochloric acid extract of a soil it may be estimated as carbonate after freeing it carefully of iron. The principal part of the iron should first be separated in the usual way by sodium acetate. In the warm solution (acid with acetic) the zinc is precipitated by hydrogen sulfid in excess. The beaker in which the precipitation takes place should be left covered in a warm place at least twelve hours. After collecting the zinc sulfid on a filter it is washed with water saturated with hydrogen sulfid. In order to free the zinc from every trace of iron it is better to dissolve the precipitate in hot dilute hydrochloric acid and reprecipitate as above, and, after boiling with some potassium chlorate, saturate it with ammonia. Any remaining trace of iron is precipitated as ferric hydroxid while the zinc remains in solution. The ferric hydroxid is separated by filtration and the filtrate, after acidifying with acetic, is treated with hydrogen sulfid as above. The zinc sulfid is dissolved again in hot hydrochloric acid, oxidized with potassium chlorate, the acid almost neutralized with soda and the zinc precipitated as carbonate with the sodium salt. After precipitation, the contents of the beaker are boiled until all free carbon dioxid is expelled, the carbonate collected on a filter, washed with hot water, dried, ignited, and weighed as ZnO.

535. Estimation of Boron.—Boron has been found in the ashes of many plants and agricultural products. Whether or not it be an essential or only accidental constituent of plants has not been determined. Its occurrence in the soil, nevertheless, is a matter which the agricultural chemist can not overlook. The boron should be dissolved from the soil by gently heating with dilute nitric acid followed by washing with hot water. Boiling should be avoided on account of the volatility of boric acid. In the solution thus obtained, concentrated on a bath at a moderate temperature to a convenient volume, the boron is to be estimated by the method given in paragraphs 518 and 519.

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NOTE.—On page 557, paragraph **500**, ninth line, read “red-yellow” instead of “blue.”

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TRANSCRIBER'S NOTES

1. Silently corrected palpable typographical errors; retained non-standard spellings and dialect.
2. Corrected the items listed on p. [viii](#).
3. Reindexed chapter footnotes using numbers and table footnotes using letters.

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SOILS ***

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